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Volume 1

University Physics

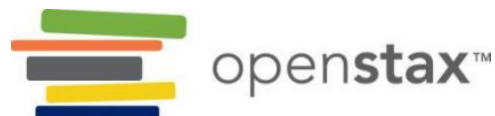
Volume 1

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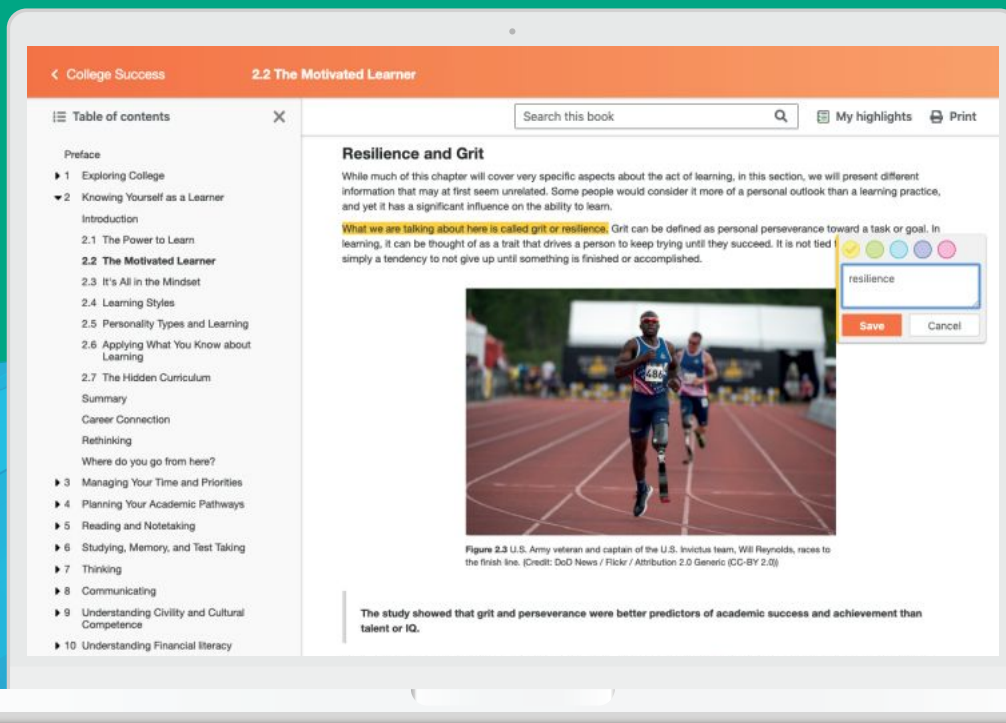
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PREFACE

Welcome to *University Physics*, an OpenStax resource. This textbook was written to increase student access to high-quality learning materials, maintaining highest standards of academic rigor at little to no cost.

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Format

You can access this textbook for free in web view or PDF through OpenStax.org, and for a low cost in print.

About *University Physics*

University Physics is designed for the two- or three-semester calculus-based physics course. The text has been developed to meet the scope and sequence of most university physics courses and provides a foundation for a career in mathematics, science, or engineering. The book provides an important opportunity for students to learn the core concepts of physics and understand how those concepts apply to their lives and to the world around them.

Due to the comprehensive nature of the material, we are offering the book in three volumes for flexibility and efficiency.

Coverage and scope

Our *University Physics* textbook adheres to the scope and sequence of most two- and three-semester physics courses nationwide. We have worked to make physics interesting and accessible to students while maintaining the mathematical rigor inherent in the subject. With this objective in mind, the content of this textbook has been developed and arranged to provide a logical progression from fundamental to more advanced concepts, building upon what students have already learned and emphasizing connections between topics and between theory and applications. The goal of each section is to enable students not just to recognize concepts, but to work with them in ways that will be useful in later courses and future careers. The organization and pedagogical features were developed and vetted with feedback from science educators dedicated to the project.

VOLUME I

Unit 1: Mechanics

Chapter 1: Units and Measurement
 Chapter 2: Vectors
 Chapter 3: Motion Along a Straight Line
 Chapter 4: Motion in Two and Three Dimensions
 Chapter 5: Newton's Laws of Motion
 Chapter 6: Applications of Newton's Laws
 Chapter 7: Work and Kinetic Energy
 Chapter 8: Potential Energy and Conservation of Energy
 Chapter 9: Linear Momentum and Collisions
 Chapter 10: Fixed-Axis Rotation
 Chapter 11: Angular Momentum
 Chapter 12: Static Equilibrium and Elasticity
 Chapter 13: Gravitation
 Chapter 14: Fluid Mechanics

Unit 2: Waves and Acoustics

Chapter 15: Oscillations
 Chapter 16: Waves
 Chapter 17: Sound

VOLUME II

Unit 1: Thermodynamics

Chapter 1: Temperature and Heat
 Chapter 2: The Kinetic Theory of Gases
 Chapter 3: The First Law of Thermodynamics
 Chapter 4: The Second Law of Thermodynamics

Unit 2: Electricity and Magnetism

Chapter 5: Electric Charges and Fields
 Chapter 6: Gauss's Law
 Chapter 7: Electric Potential
 Chapter 8: Capacitance
 Chapter 9: Current and Resistance
 Chapter 10: Direct-Current Circuits
 Chapter 11: Magnetic Forces and Fields
 Chapter 12: Sources of Magnetic Fields
 Chapter 13: Electromagnetic Induction
 Chapter 14: Inductance
 Chapter 15: Alternating-Current Circuits
 Chapter 16: Electromagnetic Waves

VOLUME III

Unit 1: Optics

Chapter 1: The Nature of Light
 Chapter 2: Geometric Optics and Image Formation
 Chapter 3: Interference
 Chapter 4: Diffraction

Unit 2: Modern Physics

Chapter 5: Relativity

Chapter 6: Photons and Matter Waves
 Chapter 7: Quantum Mechanics
 Chapter 8: Atomic Structure
 Chapter 9: Condensed Matter Physics
 Chapter 10: Nuclear Physics
 Chapter 11: Particle Physics and Cosmology

Pedagogical foundation

Throughout *University Physics* you will find derivations of concepts that present classical ideas and techniques, as well as modern applications and methods. Most chapters start with observations or experiments that place the material in a context of physical experience. Presentations and explanations rely on years of classroom experience on the part of long-time physics professors, striving for a balance of clarity and rigor that has proven successful with their students. Throughout the text, links enable students to review earlier material and then return to the present discussion, reinforcing connections between topics. Key historical figures and experiments are discussed in the main text (rather than in boxes or sidebars), maintaining a focus on the development of physical intuition. Key ideas, definitions, and equations are highlighted in the text and listed in summary form at the end of each chapter. Examples and chapter-opening images often include contemporary applications from daily life or modern science and engineering that students can relate to, from smart phones to the internet to GPS devices.

Assessments that reinforce key concepts

In-chapter **Examples** generally follow a three-part format of Strategy, Solution, and Significance to emphasize how to approach a problem, how to work with the equations, and how to check and generalize the result. Examples are often followed by questions and answers to help reinforce for students the important ideas of the examples. **Problem-Solving Strategies** in each chapter break down methods of approaching various types of problems into steps students can follow for guidance. The book also includes exercises at the end of each chapter so students can practice what they've learned.

Conceptual questions do not require calculation but test student learning of the key concepts.

Problems categorized by section test student problem-solving skills and the ability to apply ideas to practical situations.

Additional Problems apply knowledge across the chapter, forcing students to identify what concepts and equations are appropriate for solving given problems. Randomly located throughout the problems are **Unreasonable**

Results exercises that ask students to evaluate the answer to a problem and explain why it is not reasonable and what assumptions made might not be correct.

Challenge Problems extend text ideas to interesting but difficult situations.

Answers for selected exercises are available in an **Answer Key** at the end of the book.

Answers to Questions in the Book

Answers to Unreasonable Results are not provided. All answers to Check Your Understanding are provided in the Answer Key in the book. Odd-numbered answers to Conceptual Questions, Problems, Additional Problems, and Challenge Problems are also provided in the Answer Key. Students can find additional answer information in the Student Solution Guide via the Student Resources page. Answers to even-numbered Conceptual Questions, Problems, Additional Problems, and Challenge Problems are provided only to instructors in the Instructor Answer Guide via the Instructor Resources page.

Additional resources

Student and instructor resources

We've compiled additional resources for both students and instructors, including Getting Started Guides, PowerPoint slides, and answer and solution guides for instructors and students. Instructor resources require a verified instructor account, which you can apply for when you log in or create your account on OpenStax.org. Take advantage of these resources to supplement your OpenStax book.

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CHAPTER 1

Units and Measurement



FIGURE 1.1 This image might be showing any number of things. It might be a whirlpool in a tank of water or perhaps a collage of paint and shiny beads done for art class. Without knowing the size of the object in units we all recognize, such as meters or inches, it is difficult to know what we're looking at. In fact, this image shows the Whirlpool Galaxy (and its companion galaxy), which is about 60,000 light-years in diameter (about 6×10^{17} km across). (credit: modification of work by S. Beckwith (STScI) Hubble Heritage Team, (STScI/AURA), ESA, NASA)

CHAPTER OUTLINE

- 1.1 The Scope and Scale of Physics**
- 1.2 Units and Standards**
- 1.3 Unit Conversion**
- 1.4 Dimensional Analysis**
- 1.5 Estimates and Fermi Calculations**
- 1.6 Significant Figures**
- 1.7 Solving Problems in Physics**

INTRODUCTION As noted in the figure caption, the chapter-opening image is of the Whirlpool Galaxy, which we examine in the first section of this chapter. Galaxies are as immense as atoms are small, yet the same laws of physics describe both, along with all the rest of nature—an indication of the underlying unity in the universe. The laws of physics are surprisingly few, implying an underlying simplicity to nature's apparent complexity. In this text, you learn about the laws of physics. Galaxies and atoms may seem far removed from your daily life, but as you begin to explore this broad-ranging subject, you may soon come to realize that physics plays a much larger role in your life than you first thought, no matter your life goals or career choice.

1.1 The Scope and Scale of Physics

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Describe the scope of physics.
- Calculate the order of magnitude of a quantity.
- Compare measurable length, mass, and timescales quantitatively.
- Describe the relationships among models, theories, and laws.

Physics is devoted to the understanding of all natural phenomena. In physics, we try to understand physical phenomena at all scales—from the world of subatomic particles to the entire universe. Despite the breadth of the subject, the various subfields of physics share a common core. The same basic training in physics will prepare you to work in any area of physics and the related areas of science and engineering. In this section, we investigate the scope of physics; the scales of length, mass, and time over which the laws of physics have been shown to be applicable; and the process by which science in general, and physics in particular, operates.

The Scope of Physics

Take another look at the chapter-opening image. The Whirlpool Galaxy contains billions of individual stars as well as huge clouds of gas and dust. Its companion galaxy is also visible to the right. This pair of galaxies lies a staggering billion trillion miles (1.4×10^{21} mi) from our own galaxy (which is called the *Milky Way*). The stars and planets that make up the Whirlpool Galaxy might seem to be the furthest thing from most people's everyday lives, but the Whirlpool is a great starting point to think about the forces that hold the universe together. The forces that cause the Whirlpool Galaxy to act as it does are thought to be the same forces we contend with here on Earth, whether we are planning to send a rocket into space or simply planning to raise the walls for a new home. The gravity that causes the stars of the Whirlpool Galaxy to rotate and revolve is thought to be the same as what causes water to flow over hydroelectric dams here on Earth. When you look up at the stars, realize the forces out there are the same as the ones here on Earth. Through a study of physics, you may gain a greater understanding of the interconnectedness of everything we can see and know in this universe.

Humans have created and manufactured millions of different objects. Successive technological periods (often referred to as the Stone Age, the Bronze Age, the Iron Age, and so on) are marked by knowledge of the physical properties of certain materials and the ability to manipulate them. This knowledge all stems from physics, whether it's the way a rock would flake when constructing a spear point, the effect of integrating carbon with iron in South Indian and Sri Lankan furnaces to create the earliest high-quality steel, or the proper way to combine perfectly ground and polished pieces of glass to create optical instruments. Our current technological age, the Information Age, can be traced to critical innovations made by people from all backgrounds working together. Mohamed M. Atalla and Dawon Kahng, for example, invented the MOSFET (metal-oxide-semiconductor field-effect transistor). Although unknown to most people, this tiny device, created in 1959 by an Egyptian-born scientist and a Korean-born scientist working in a lab in New Jersey (U.S.), is the basis for modern electronics. More MOSFETs have been produced than any other object in human history. They are used in computers, smartphones, microwave ovens, automotive controls, medical instruments, and nearly every other electronic device.

Next, think about the most exciting modern technologies that you have heard about in the news, such as trains that levitate above tracks, “invisibility cloaks” that bend light around them, and microscopic robots that fight cancer cells in our bodies. All these groundbreaking advances, commonplace or unbelievable, rely on the principles of physics. Aside from playing a significant role in technology, professionals such as engineers, pilots, physicians, physical therapists, electricians, and computer programmers apply physics concepts in their daily work. For example, a pilot must understand how wind forces affect a flight path; a physical therapist must understand how the muscles in the body experience forces as they move and bend. As you will learn in this text, the principles of physics are propelling new, exciting technologies, and these principles are applied in a wide range of careers.

The underlying order of nature makes science in general, and physics in particular, interesting and enjoyable to study. For example, what do a bag of chips and a car battery have in common? Both contain energy that can be converted to other forms. The law of conservation of energy (which says that energy can change form but is never lost) ties together such topics as food calories, batteries, heat, light, and watch springs. Understanding this law

makes it easier to learn about the various forms energy takes and how they relate to one another. Apparently unrelated topics are connected through broadly applicable physical laws, permitting an understanding beyond just the memorization of lists of facts.

Science consists of theories and laws that are the general truths of nature, as well as the body of knowledge they encompass. Scientists are continuously trying to expand this body of knowledge and to perfect the expression of the laws that describe it. **Physics**, which comes from the Greek *phúsis*, meaning “nature,” is concerned with describing the interactions of energy, matter, space, and time to uncover the fundamental mechanisms that underlie every phenomenon. This concern for describing the basic phenomena in nature essentially defines the *scope of physics*.

Physics aims to understand the world around us at the most basic level. It emphasizes the use of a small number of quantitative laws to do this, which can be useful to other fields pushing the performance boundaries of existing technologies. Consider a smartphone (Figure 1.2). Physics describes how electricity interacts with the various circuits inside the device. This knowledge helps engineers select the appropriate materials and circuit layout when building a smartphone. Knowledge of the physics underlying these devices is required to shrink their size or increase their processing speed. Or, think about a GPS. Physics describes the relationship between the speed of an object, the distance over which it travels, and the time it takes to travel that distance. GPS relies on precise calculations that account for variations in the Earth's landscapes, the exact distance between orbiting satellites, and even the effect of a complex occurrence of time dilation. Most of these calculations are founded on algorithms developed by Gladys West, a mathematician and computer scientist who programmed the first computers capable of highly accurate remote sensing and positioning. When you use a GPS device, it utilizes these algorithms to recognize where you are and how your position relates to other objects on Earth.

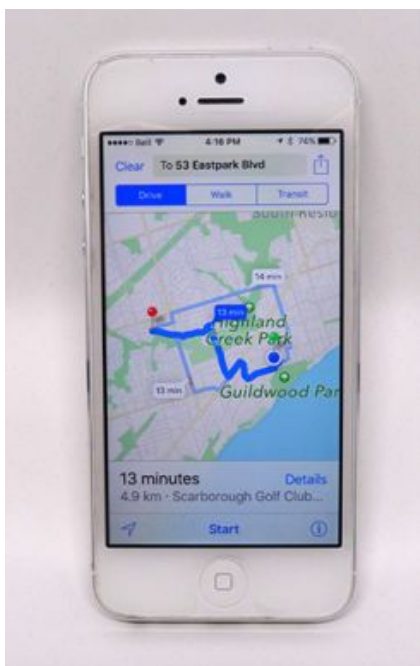


FIGURE 1.2 The Apple iPhone is a common smartphone with a GPS function. Physics describes the way that electricity flows through the circuits of this device. Engineers use their knowledge of physics to construct an iPhone with features that consumers will enjoy. One specific feature of an iPhone is the GPS function. A GPS uses physics equations to determine the drive time between two locations on a map. (credit: Jane Whitney)

Knowledge of physics is useful in everyday situations as well as in nonscientific professions. It can help you understand how microwave ovens work, why metals should not be put into them, and why they might affect pacemakers. Physics allows you to understand the hazards of radiation and to evaluate these hazards rationally and more easily. Physics also explains the reason why a black car radiator helps remove heat in a car engine, and it explains why a white roof helps keep the inside of a house cool. Similarly, the operation of a car's ignition system as well as the transmission of electrical signals throughout our body's nervous system are much easier to understand when you think about them in terms of basic physics.

Physics is a key element of many important disciplines and contributes directly to others. Chemistry, for

example—since it deals with the interactions of atoms and molecules—has close ties to atomic and molecular physics. Most branches of engineering are concerned with designing new technologies, processes, or structures within the constraints set by the laws of physics. In architecture, physics is at the heart of structural stability and is involved in the acoustics, heating, lighting, and cooling of buildings. Parts of geology rely heavily on physics, such as radioactive dating of rocks, earthquake analysis, and heat transfer within Earth. Some disciplines, such as biophysics and geophysics, are hybrids of physics and other disciplines.

Physics has many applications in the biological sciences. On the microscopic level, it helps describe the properties of cells and their environments. On the macroscopic level, it explains the heat, work, and power associated with the human body and its various organ systems. Physics is involved in medical diagnostics, such as radiographs, magnetic resonance imaging, and ultrasonic blood flow measurements. Medical therapy sometimes involves physics directly; for example, cancer radiotherapy uses ionizing radiation. Physics also explains sensory phenomena, such as how musical instruments make sound, how the eye detects color, and how lasers transmit information.

It is not necessary to study all applications of physics formally. What is most useful is knowing the basic laws of physics and developing skills in the analytical methods for applying them. The study of physics also can improve your problem-solving skills. Furthermore, physics retains the most basic aspects of science, so it is used by all the sciences, and the study of physics makes other sciences easier to understand.

The Scale of Physics

From the discussion so far, it should be clear that to accomplish your goals in any of the various fields within the natural sciences and engineering, a thorough grounding in the laws of physics is necessary. The reason for this is simply that the laws of physics govern everything in the observable universe at all measurable scales of length, mass, and time. Now, that is easy enough to say, but to come to grips with what it really means, we need to get a little bit quantitative. So, before surveying the various scales that physics allows us to explore, let's first look at the concept of “order of magnitude,” which we use to come to terms with the vast ranges of length, mass, and time that we consider in this text ([Figure 1.3](#)).

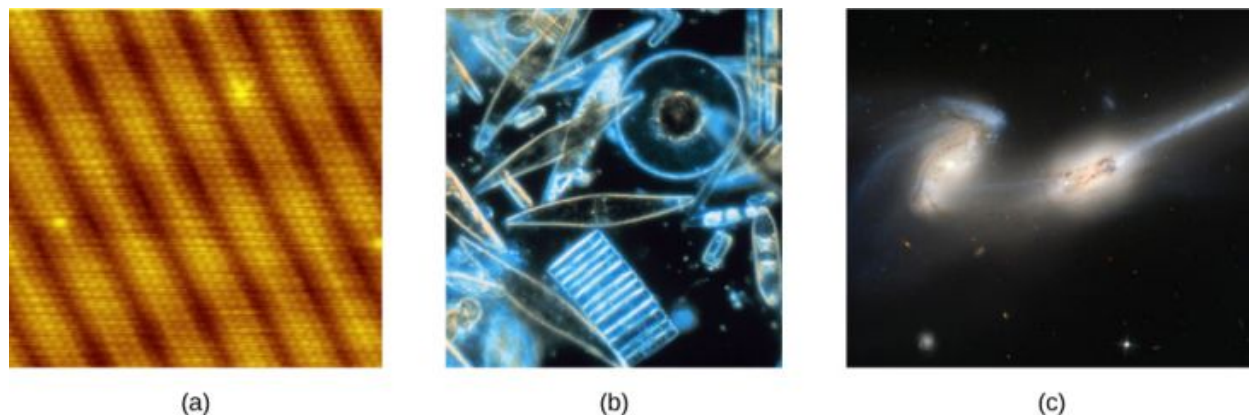


FIGURE 1.3 (a) Using a scanning tunneling microscope, scientists can see the individual atoms (diameters around 10^{-10} m) that compose this sheet of gold. (b) Tiny phytoplankton swim among crystals of ice in the Antarctic Sea. They range from a few micrometers ($1\ \mu\text{m}$ is 10^{-6} m) to as much as 2 mm ($1\ \text{mm}$ is 10^{-3} m) in length. (c) These two colliding galaxies, known as NGC 4676A (right) and NGC 4676B (left), are nicknamed “The Mice” because of the tail of gas emanating from each one. They are located 300 million light-years from Earth in the constellation Coma Berenices. Eventually, these two galaxies will merge into one. (credit a: modification of work by “Erwinrossen”/Wikimedia Commons; credit b: modification of work by Prof. Gordon T. Taylor, Stony Brook University; NOAA Corps Collections; credit c: modification of work by NASA, H. Ford (JHU), G. Illingworth (UCSC/LO), M. Clampin (STScI), G. Hartig (STScI), the ACS Science Team, and ESA)

Order of magnitude

The **order of magnitude** of a number is the power of 10 that most closely approximates it. Thus, the order of magnitude refers to the scale (or size) of a value. Each power of 10 represents a different order of magnitude. For example, 10^1 , 10^2 , 10^3 , and so forth, are all different orders of magnitude, as are $10^0 = 1$, 10^{-1} , 10^{-2} , and 10^{-3} . To find the order of magnitude of a number, take the base-10 logarithm of the number and round it to the nearest integer, then the order of magnitude of the number is simply the resulting power of 10. For example, the order of magnitude of 800 is 10^3 because $\log_{10} 800 \approx 2.903$, which rounds to 3. Similarly, the order of magnitude of 450 is

10^3 because $\log_{10} 450 \approx 2.653$, which rounds to 3 as well. Thus, we say the numbers 800 and 450 are of the same order of magnitude: 10^3 . However, the order of magnitude of 250 is 10^2 because $\log_{10} 250 \approx 2.397$, which rounds to 2.

An equivalent but quicker way to find the order of magnitude of a number is first to write it in scientific notation and then check to see whether the first factor is greater than or less than $\sqrt{10} = 10^{0.5} \approx 3$. The idea is that $\sqrt{10} = 10^{0.5}$ is halfway between $1 = 10^0$ and $10 = 10^1$ on a log base-10 scale. Thus, if the first factor is less than $\sqrt{10}$, then we round it down to 1 and the order of magnitude is simply whatever power of 10 is required to write the number in scientific notation. On the other hand, if the first factor is greater than $\sqrt{10}$, then we round it up to 10 and the order of magnitude is one power of 10 higher than the power needed to write the number in scientific notation. For example, the number 800 can be written in scientific notation as 8×10^2 . Because 8 is bigger than $\sqrt{10} \approx 3$, we say the order of magnitude of 800 is $10^{2+1} = 10^3$. The number 450 can be written as 4.5×10^2 , so its order of magnitude is also 10^3 because 4.5 is greater than 3. However, 250 written in scientific notation is 2.5×10^2 and 2.5 is less than 3, so its order of magnitude is 10^2 .

The order of magnitude of a number is designed to be a ballpark estimate for the scale (or size) of its value. It is simply a way of rounding numbers consistently to the nearest (on a logarithmic scale) power of 10. In other words, if numbers differ by one order of magnitude, the one is larger than the other by about ten. This makes doing rough mental math with very big and very small numbers easier. For example, the diameter of a hydrogen atom is on the order of 10^{-10} m, whereas the diameter of the Sun is on the order of 10^9 m, so it would take roughly $10^9/10^{-10} = 10^{19}$ hydrogen atoms to stretch across the diameter of the Sun. This is much easier to do in your head than using the more precise values of 1.06×10^{-10} m for a hydrogen atom diameter and 1.39×10^9 m for the Sun's diameter, to find that it would take 1.31×10^{19} hydrogen atoms to stretch across the Sun's diameter. In addition to being easier, the rough estimate is also nearly as informative as the precise calculation.

Known ranges of length, mass, and time

The vastness of the universe and the breadth over which physics applies are illustrated by the wide range of examples of known lengths, masses, and times (given as orders of magnitude) in [Figure 1.4](#). Examining this table will give you a feeling for the range of possible topics in physics and numerical values. A good way to appreciate the vastness of the ranges of values in [Figure 1.4](#) is to try to answer some simple comparative questions, such as the following:

- How many hydrogen atoms does it take to stretch across the diameter of the Sun?
(Answer: $10^9 \text{ m}/10^{-10} \text{ m} = 10^{19}$ hydrogen atoms)
- How many protons are there in a bacterium?
(Answer: $10^{-15} \text{ kg}/10^{-27} \text{ kg} = 10^{12}$ protons)
- How many floating-point operations can a supercomputer do in 1 day?
(Answer: $10^5 \text{ s}/10^{-17} \text{ s} = 10^{22}$ floating-point operations)

In studying [Figure 1.4](#), take some time to come up with similar questions that interest you and then try answering them. Doing this can breathe some life into almost any table of numbers.

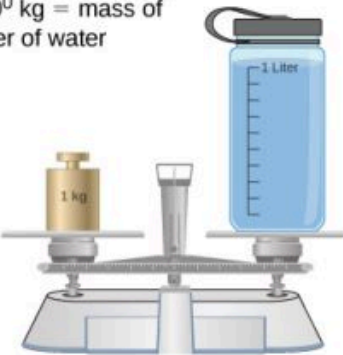
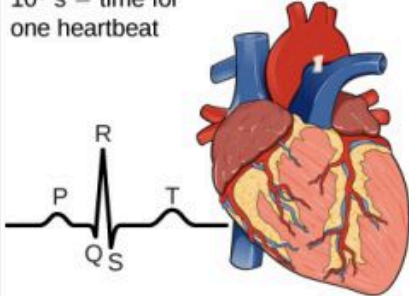
Length in Meters (m)	Masses in Kilograms (kg)	Time in Seconds (s)
10^{-15} m = diameter of proton	10^{-30} kg = mass of electron	10^{-22} s = mean lifetime of very unstable nucleus
10^{-14} m = diameter of large nucleus	10^{-27} kg = mass of proton	10^{-17} s = time for single floating-point operation in a supercomputer
10^{-10} m = diameter of hydrogen atom	10^{-15} kg = mass of bacterium	10^{-15} s = time for one oscillation of visible light
10^{-7} m = diameter of typical virus	10^{-5} kg = mass of mosquito	10^{-13} s = time for one vibration of an atom in a solid
10^{-2} m = pinky fingernail width	10^{-2} kg = mass of hummingbird	10^{-3} s = duration of a nerve impulse
10^0 m = height of 4 year old child 	10^0 kg = mass of liter of water 	10^0 s = time for one heartbeat 
10^2 m = length of football field	10^2 kg = mass of person	10^5 s = one day
10^7 m = diameter of Earth	10^{19} kg = mass of atmosphere	10^7 s = one year
10^{13} m = diameter of solar system	10^{22} kg = mass of Moon	10^9 s = human lifetime
10^{16} m = distance light travels in a year (one light-year)	10^{25} kg = mass of Earth	10^{11} s = recorded human history
10^{21} m = Milky Way diameter	10^{30} kg = mass of Sun	10^{17} s = age of Earth
10^{26} m = distance to edge of observable universe	10^{53} kg = upper limit on mass of known universe	10^{18} s = age of the universe

FIGURE 1.4 This table shows the orders of magnitude of length, mass, and time.

INTERACTIVE

Visit [this site \(https://openstax.org/l/21scaleuniv\)](https://openstax.org/l/21scaleuniv) to explore interactively the vast range of length scales in our universe. Scroll down and up the scale to view hundreds of organisms and objects, and click on the individual objects to learn more about each one.

Building Models

How did we come to know the laws governing natural phenomena? What we refer to as the laws of nature are concise descriptions of the universe around us. They are human statements of the underlying laws or rules that all natural processes follow. Such laws are intrinsic to the universe; humans did not create them and cannot change them. We can only discover and understand them. Their discovery is a very human endeavor, with all the elements of mystery, imagination, struggle, triumph, and disappointment inherent in any creative effort (Figure 1.5). The cornerstone of discovering natural laws is observation; scientists must describe the universe as it is, not as we imagine it to be.

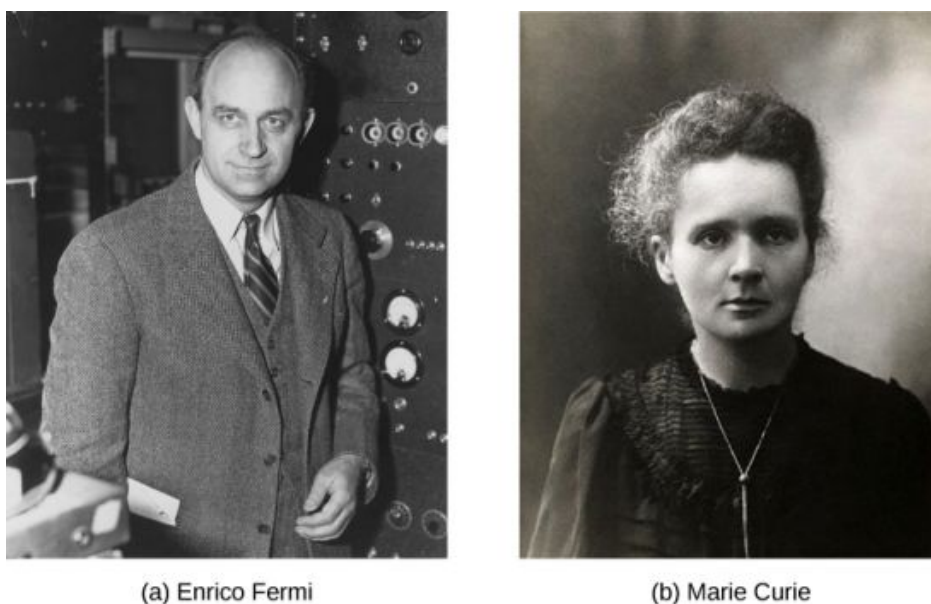


FIGURE 1.5 (a) Enrico Fermi (1901–1954) was born in Italy. On accepting the Nobel Prize in Stockholm in 1938 for his work on artificial radioactivity produced by neutrons, he took his family to America rather than return home to the government in power at the time. He became an American citizen and was a leading participant in the Manhattan Project. (b) Marie Curie (1867–1934) sacrificed monetary assets to help finance her early research and damaged her physical well-being with radiation exposure. She is the only person to win Nobel prizes in both physics and chemistry. One of her daughters also won a Nobel Prize. (credit a: modification of work by United States Department of Energy)

A **model** is a representation of something that is often too difficult (or impossible) to display directly. Although a model is justified by experimental tests, it is only accurate in describing certain aspects of a physical system. An example is the Bohr model of single-electron atoms, in which the electron is pictured as orbiting the nucleus, analogous to the way planets orbit the Sun (Figure 1.6). We cannot observe electron orbits directly, but the mental image helps explain some of the observations we can make, such as the emission of light from hot gases (atomic spectra). However, other observations show that the picture in the Bohr model is not really what atoms look like. The model is “wrong,” but is still useful for some purposes. Physicists use models for a variety of purposes. For example, models can help physicists analyze a scenario and perform a calculation or models can be used to represent a situation in the form of a computer simulation. Ultimately, however, the results of these calculations and simulations need to be double-checked by other means—namely, observation and experimentation.

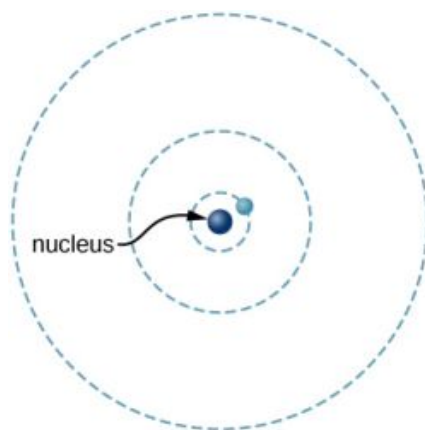


FIGURE 1.6 What is a model? The Bohr model of a single-electron atom shows the electron orbiting the nucleus in one of several possible circular orbits. Like all models, it captures some, but not all, aspects of the physical system.

The word *theory* means something different to scientists than what is often meant when the word is used in everyday conversation. In particular, to a scientist a theory is not the same as a “guess” or an “idea” or even a “hypothesis.” The phrase “it’s just a theory” seems meaningless and silly to scientists because science is founded on the notion of theories. To a scientist, a **theory** is a testable explanation for patterns in nature supported by scientific evidence and verified multiple times by various groups of researchers. Some theories include models to

help visualize phenomena whereas others do not. Newton's theory of gravity, for example, does not require a model or mental image, because we can observe the objects directly with our own senses. Although models are meant only to describe certain aspects of a physical system accurately, a theory should describe all aspects of any system that falls within its domain of applicability. In particular, any experimentally testable implication of a theory should be verified. If an experiment ever shows an implication of a theory to be false, then the theory is either thrown out or modified suitably (for example, by limiting its domain of applicability).

A **law** uses concise language to describe a generalized pattern in nature supported by scientific evidence and repeated experiments. Often, a law can be expressed in the form of a single mathematical equation. Laws and theories are similar in that they are both scientific statements that result from a tested hypothesis and are supported by scientific evidence. However, the designation *law* is usually reserved for a concise and very general statement that describes phenomena in nature, such as the law that energy is conserved during any process, or Newton's second law of motion, which relates force (F), mass (m), and acceleration (a) by the simple equation $F = ma$. A theory, in contrast, is a less concise statement of observed behavior. For example, the theory of evolution and the theory of relativity cannot be expressed concisely enough to be considered laws. The biggest difference between a law and a theory is that a theory is much more complex and dynamic. A law describes a single action whereas a theory explains an entire group of related phenomena. Less broadly applicable statements are usually called principles (such as Pascal's principle, which is applicable only in fluids), but the distinction between laws and principles often is not made carefully.

The models, theories, and laws we devise sometimes imply the existence of objects or phenomena that are as yet unobserved. These predictions are remarkable triumphs and tributes to the power of science. It is the underlying order in the universe that enables scientists to make such spectacular predictions. However, if experimentation does not verify our predictions, then the theory or law is wrong, no matter how elegant or convenient it is. Laws can never be known with absolute certainty because it is impossible to perform every imaginable experiment to confirm a law for every possible scenario. Physicists operate under the assumption that all scientific laws and theories are valid until a counterexample is observed. If a good-quality, verifiable experiment contradicts a well-established law or theory, then the law or theory must be modified or overthrown completely.

The study of science in general, and physics in particular, is an adventure much like the exploration of an uncharted ocean. Discoveries are made; models, theories, and laws are formulated; and the beauty of the physical universe is made more sublime for the insights gained.

1.2 Units and Standards

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Describe how SI base units are defined.
- Describe how derived units are created from base units.
- Express quantities given in SI units using metric prefixes.

As we saw previously, the range of objects and phenomena studied in physics is immense. From the incredibly short lifetime of a nucleus to the age of Earth, from the tiny sizes of subnuclear particles to the vast distance to the edges of the known universe, from the force exerted by a jumping flea to the force between Earth and the Sun, there are enough factors of 10 to challenge the imagination of even the most experienced scientist. Giving numerical values for physical quantities and equations for physical principles allows us to understand nature much more deeply than qualitative descriptions alone. To comprehend these vast ranges, we must also have accepted units in which to express them. We shall find that even in the potentially mundane discussion of meters, kilograms, and seconds, a profound simplicity of nature appears: all physical quantities can be expressed as combinations of only seven base physical units.

We define a **physical quantity** either by specifying how it is measured or by stating how it is calculated from other measurements. For example, we might define distance and time by specifying methods for measuring them, such as using a meter stick and a stopwatch. Then, we could define *average speed* by stating that it is calculated as the total distance traveled divided by time of travel.

Measurements of physical quantities are expressed in terms of **units**, which are standardized values. For example,

the length of a race, which is a physical quantity, can be expressed in units of meters (for sprinters) or kilometers (for distance runners). Without standardized units, it would be extremely difficult for scientists to express and compare measured values in a meaningful way (Figure 1.7).

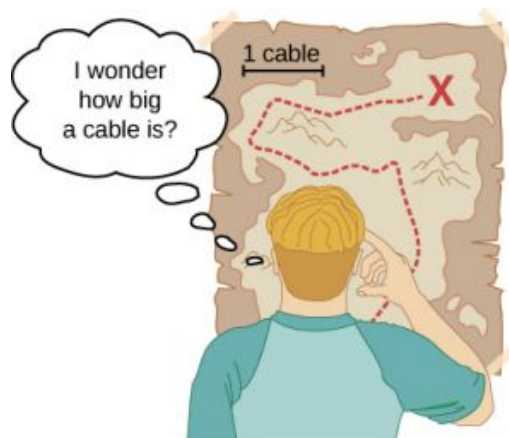


FIGURE 1.7 Distances given in unknown units are maddeningly useless.

Two major systems of units are used in the world: **SI units** (for the French *Système International d'Unités*), also known as the *metric system*, and **English units** (also known as the *customary* or *imperial system*). English units were historically used in nations once ruled by the British Empire and are still widely used in the United States. English units may also be referred to as the *foot–pound–second* (fps) system, as opposed to the *centimeter–gram–second* (cgs) system. You may also encounter the term *SAE units*, named after the Society of Automotive Engineers. Products such as fasteners and automotive tools (for example, wrenches) that are measured in inches rather than metric units are referred to as *SAE fasteners* or *SAE wrenches*.

Virtually every other country in the world (except the United States) now uses SI units as the standard. The metric system is also the standard system agreed on by scientists and mathematicians.

SI Units: Base and Derived Units

In any system of units, the units for some physical quantities must be defined through a measurement process. These are called the **base quantities** for that system and their units are the system's **base units**. All other physical quantities can then be expressed as algebraic combinations of the base quantities. Each of these physical quantities is then known as a **derived quantity** and each unit is called a **derived unit**. The choice of base quantities is somewhat arbitrary, as long as they are independent of each other and all other quantities can be derived from them. Typically, the goal is to choose physical quantities that can be measured accurately to a high precision as the base quantities. The reason for this is simple. Since the derived units can be expressed as algebraic combinations of the base units, they can only be as accurate and precise as the base units from which they are derived.

Based on such considerations, the International Standards Organization recommends using seven base quantities, which form the International System of Quantities (ISQ). These are the base quantities used to define the SI base units. [Table 1.1](#) lists these seven ISQ base quantities and the corresponding SI base units.

ISQ Base Quantity	SI Base Unit
Length	meter (m)
Mass	kilogram (kg)
Time	second (s)
Electrical current	ampere (A)

TABLE 1.1 ISQ Base Quantities and Their SI Units

ISQ Base Quantity	SI Base Unit
Thermodynamic temperature	kelvin (K)
Amount of substance	mole (mol)
Luminous intensity	candela (cd)

TABLE 1.1 ISQ Base Quantities and Their SI Units

You are probably already familiar with some derived quantities that can be formed from the base quantities in [Table 1.1](#). For example, the geometric concept of area is always calculated as the product of two lengths. Thus, area is a derived quantity that can be expressed in terms of SI base units using square meters ($\text{m} \times \text{m} = \text{m}^2$). Similarly, volume is a derived quantity that can be expressed in cubic meters (m^3). Speed is length per time; so in terms of SI base units, we could measure it in meters per second (m/s). Volume mass density (or just density) is mass per volume, which is expressed in terms of SI base units such as kilograms per cubic meter (kg/m^3). Angles can also be thought of as derived quantities because they can be defined as the ratio of the arc length subtended by two radii of a circle to the radius of the circle. This is how the radian is defined. Depending on your background and interests, you may be able to come up with other derived quantities, such as the mass flow rate (kg/s) or volume flow rate (m^3/s) of a fluid, electric charge ($\text{A} \cdot \text{s}$), mass flux density [$\text{kg}/(\text{m}^2 \cdot \text{s})$], and so on. We will see many more examples throughout this text. For now, the point is that every physical quantity can be derived from the seven base quantities in [Table 1.1](#), and the units of every physical quantity can be derived from the seven SI base units.

For the most part, we use SI units in this text. Non-SI units are used in a few applications in which they are in very common use, such as the measurement of temperature in degrees Celsius ($^{\circ}\text{C}$), the measurement of fluid volume in liters (L), and the measurement of energies of elementary particles in electron-volts (eV). Whenever non-SI units are discussed, they are tied to SI units through conversions. For example, 1 L is 10^{-3} m^3 .

INTERACTIVE

Check out a comprehensive source of information on [SI units \(https://openstax.org/l/21SIUnits\)](https://openstax.org/l/21SIUnits) at the National Institute of Standards and Technology (NIST) Reference on Constants, Units, and Uncertainty.

Units of Time, Length, and Mass: The Second, Meter, and Kilogram

The initial chapters in this textbook are concerned with mechanics, fluids, and waves. In these subjects all pertinent physical quantities can be expressed in terms of the base units of length, mass, and time. Therefore, we now turn to a discussion of these three base units, leaving discussion of the others until they are needed later.

The second

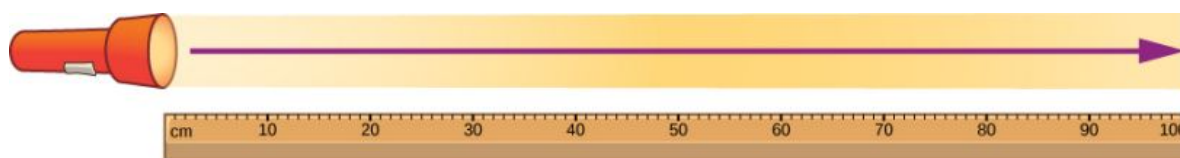
The SI unit for time, the **second** (abbreviated s), has a long history. For many years it was defined as $1/86,400$ of a mean solar day. More recently, a new standard was adopted to gain greater accuracy and to define the second in terms of a nonvarying or constant physical phenomenon (because the solar day is getting longer as a result of the very gradual slowing of Earth's rotation). Cesium atoms can be made to vibrate in a very steady way, and these vibrations can be readily observed and counted. In 1967, the second was redefined as the time required for 9,192,631,770 of these vibrations to occur ([Figure 1.8](#)). Note that this may seem like more precision than you would ever need, but it isn't—GPSs rely on the precision of atomic clocks to be able to give you turn-by-turn directions on the surface of Earth, far from the satellites broadcasting their location.



FIGURE 1.8 An atomic clock such as this one uses the vibrations of cesium atoms to keep time to a precision of better than a microsecond per year. The fundamental unit of time, the second, is based on such clocks. This image looks down from the top of an atomic fountain nearly 30 feet tall. (credit: Steve Jurvetson)

The meter

The SI unit for length is the **meter** (abbreviated m); its definition has also changed over time to become more precise. The meter was first defined in 1791 as $1/10,000,000$ of the distance from the equator to the North Pole. This measurement was improved in 1889 by redefining the meter to be the distance between two engraved lines on a platinum–iridium bar now kept near Paris. By 1960, it had become possible to define the meter even more accurately in terms of the wavelength of light, so it was again redefined as 1,650,763.73 wavelengths of orange light emitted by krypton atoms. In 1983, the meter was given its current definition (in part for greater accuracy) as the distance light travels in a vacuum in $1/299,792,458$ of a second ([Figure 1.9](#)). This change came after knowing the speed of light to be exactly 299,792,458 m/s.



Light travels a distance of 1 meter
in $1/299,792,458$ of a second

FIGURE 1.9 The meter is defined to be the distance light travels in $1/299,792,458$ of a second in a vacuum. Distance traveled is speed multiplied by time.

The kilogram

The SI unit for mass is the **kilogram** (abbreviated kg); From 1795–2018 it was defined to be the mass of a platinum–iridium cylinder kept with the old meter standard at the International Bureau of Weights and Measures near Paris. However, this cylinder has lost roughly 50 micrograms since it was created. Because this is the standard, this has shifted how we defined a kilogram. Therefore, a new definition was adopted in May 2019 based on the Planck constant and other constants which will never change in value. We will study Planck’s constant in quantum mechanics, which is an area of physics that describes how the smallest pieces of the universe work. The kilogram is measured on a Kibble balance (see [Figure 1.10](#)). When a weight is placed on a Kibble balance, an electrical current is produced that is proportional to Planck’s constant. Since Planck’s constant is defined, the exact current measurements in the balance define the kilogram.



FIGURE 1.10 Redefining the SI unit of mass. The U.S. National Institute of Standards and Technology's Kibble balance is a machine that balances the weight of a test mass with the resulting electrical current needed for a force to balance it.

Metric Prefixes

SI units are part of the **metric system**, which is convenient for scientific and engineering calculations because the units are categorized by factors of 10. [Table 1.2](#) lists the metric prefixes and symbols used to denote various factors of 10 in SI units. For example, a centimeter is one-hundredth of a meter (in symbols, $1 \text{ cm} = 10^{-2} \text{ m}$) and a kilometer is a thousand meters ($1 \text{ km} = 10^3 \text{ m}$). Similarly, a megagram is a million grams ($1 \text{ Mg} = 10^6 \text{ g}$), a nanosecond is a billionth of a second ($1 \text{ ns} = 10^{-9} \text{ s}$), and a terameter is a trillion meters ($1 \text{ Tm} = 10^{12} \text{ m}$).

Prefix	Symbol	Meaning	Prefix	Symbol	Meaning
yotta-	Y	10^{24}	yocto-	y	10^{-24}
zetta-	Z	10^{21}	zepto-	z	10^{-21}
exa-	E	10^{18}	atto-	a	10^{-18}
peta-	P	10^{15}	femto-	f	10^{-15}
tera-	T	10^{12}	pico-	p	10^{-12}
giga-	G	10^9	nano-	n	10^{-9}
mega-	M	10^6	micro-	μ	10^{-6}
kilo-	k	10^3	milli-	m	10^{-3}
hecto-	h	10^2	centi-	c	10^{-2}
deka-	da	10^1	deci-	d	10^{-1}

TABLE 1.2 Metric Prefixes for Powers of 10 and Their Symbols

The only rule when using metric prefixes is that you cannot “double them up.” For example, if you have

measurements in petameters ($1 \text{ Pm} = 10^{15} \text{ m}$), it is not proper to talk about megagigameters, although $10^6 \times 10^9 = 10^{15}$. In practice, the only time this becomes a bit confusing is when discussing masses. As we have seen, the base SI unit of mass is the kilogram (kg), but metric prefixes need to be applied to the gram (g), because we are not allowed to “double-up” prefixes. Thus, a thousand kilograms (10^3 kg) is written as a megagram (1 Mg) since

$$10^3 \text{ kg} = 10^3 \times 10^3 \text{ g} = 10^6 \text{ g} = 1 \text{ Mg}.$$

Incidentally, 10^3 kg is also called a *metric ton*, abbreviated t. This is one of the units outside the SI system considered acceptable for use with SI units.

As we see in the next section, metric systems have the advantage that conversions of units involve only powers of 10. There are 100 cm in 1 m, 1000 m in 1 km, and so on. In nonmetric systems, such as the English system of units, the relationships are not as simple—there are 12 in. in 1 ft, 5280 ft in 1 mi, and so on.

Another advantage of metric systems is that the same unit can be used over extremely large ranges of values simply by scaling it with an appropriate metric prefix. The prefix is chosen by the order of magnitude of physical quantities commonly found in the task at hand. For example, distances in meters are suitable in construction, whereas distances in kilometers are appropriate for air travel, and nanometers are convenient in optical design. With the metric system there is no need to invent new units for particular applications. Instead, we rescale the units with which we are already familiar.



EXAMPLE 1.1

Using Metric Prefixes

Restate the mass $1.93 \times 10^{13} \text{ kg}$ using a metric prefix such that the resulting numerical value is bigger than one but less than 1000.

Strategy

Since we are not allowed to “double-up” prefixes, we first need to restate the mass in grams by replacing the prefix symbol k with a factor of 10^3 (see [Table 1.2](#)). Then, we should see which two prefixes in [Table 1.2](#) are closest to the resulting power of 10 when the number is written in scientific notation. We use whichever of these two prefixes gives us a number between one and 1000.

Solution

Replacing the k in kilogram with a factor of 10^3 , we find that

$$1.93 \times 10^{13} \text{ kg} = 1.93 \times 10^{13} \times 10^3 \text{ g} = 1.93 \times 10^{16} \text{ g}.$$

From [Table 1.2](#), we see that 10^{16} is between “peta-” (10^{15}) and “exa-” (10^{18}). If we use the “peta-” prefix, then we find that $1.93 \times 10^{16} \text{ g} = 1.93 \times 10^1 \text{ Pg}$, since $16 = 1 + 15$. Alternatively, if we use the “exa-” prefix we find that $1.93 \times 10^{16} \text{ g} = 1.93 \times 10^{-2} \text{ Eg}$, since $16 = -2 + 18$. Because the problem asks for the numerical value between one and 1000, we use the “peta-” prefix and the answer is 19.3 Pg.

Significance

It is easy to make silly arithmetic errors when switching from one prefix to another, so it is always a good idea to check that our final answer matches the number we started with. An easy way to do this is to put both numbers in scientific notation and count powers of 10, including the ones hidden in prefixes. If we did not make a mistake, the powers of 10 should match up. In this problem, we started with $1.93 \times 10^{13} \text{ kg}$, so we have $13 + 3 = 16$ powers of 10. Our final answer in scientific notation is $1.93 \times 10^1 \text{ Pg}$, so we have $1 + 15 = 16$ powers of 10. So, everything checks out.

If this mass arose from a calculation, we would also want to check to determine whether a mass this large makes any sense in the context of the problem. For this, [Figure 1.4](#) might be helpful.

✓ CHECK YOUR UNDERSTANDING 1.1

Restate $4.79 \times 10^5 \text{ kg}$ using a metric prefix such that the resulting number is bigger than one but less than 1000.

1.3 Unit Conversion

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Use conversion factors to express the value of a given quantity in different units.

It is often necessary to convert from one unit to another. For example, if you are reading a European cookbook, some quantities may be expressed in units of liters and you need to convert them to cups. Or perhaps you are reading walking directions from one location to another and you are interested in how many miles you will be walking. In this case, you may need to convert units of feet or meters to miles.

Let's consider a simple example of how to convert units. Suppose we want to convert 80 m to kilometers. The first thing to do is to list the units you have and the units to which you want to convert. In this case, we have units in *meters* and we want to convert to *kilometers*. Next, we need to determine a conversion factor relating meters to kilometers. A **conversion factor** is a ratio that expresses how many of one unit are equal to another unit. For example, there are 12 in. in 1 ft, 1609 m in 1 mi, 100 cm in 1 m, 60 s in 1 min, and so on. Refer to [Appendix B](#) for a more complete list of conversion factors. In this case, we know that there are 1000 m in 1 km. Now we can set up our unit conversion. We write the units we have and then multiply them by the conversion factor so the units cancel out, as shown:

$$80 \cancel{\text{m}} \times \frac{1 \text{ km}}{1000 \cancel{\text{m}}} = 0.080 \text{ km}.$$

Note that the unwanted meter unit cancels, leaving only the desired kilometer unit. You can use this method to convert between any type of unit. Now, the conversion of 80 m to kilometers is simply the use of a metric prefix, as we saw in the preceding section, so we can get the same answer just as easily by noting that

$$80 \text{ m} = 8.0 \times 10^1 \text{ m} = 8.0 \times 10^{-2} \text{ km} = 0.080 \text{ km},$$

since “kilo-” means 10^3 (see [Table 1.2](#)) and $1 = -2 + 3$. However, using conversion factors is handy when converting between units that are not metric or when converting between derived units, as the following examples illustrate.

EXAMPLE 1.2

Converting Nonmetric Units to Metric

The distance from the university to home is 10 mi and it usually takes 20 min to drive this distance. Calculate the average speed in meters per second (m/s). (*Note:* Average speed is distance traveled divided by time of travel.)

Strategy

First we calculate the average speed using the given units, then we can get the average speed into the desired units by picking the correct conversion factors and multiplying by them. The correct conversion factors are those that cancel the unwanted units and leave the desired units in their place. In this case, we want to convert miles to meters, so we need to know the fact that there are 1609 m in 1 mi. We also want to convert minutes to seconds, so we use the conversion of 60 s in 1 min.

Solution

1. Calculate average speed. Average speed is distance traveled divided by time of travel. (Take this definition as a given for now. Average speed and other motion concepts are covered in later chapters.) In equation form,

$$\text{Average speed} = \frac{\text{Distance}}{\text{Time}}.$$

2. Substitute the given values for distance and time:

$$\text{Average speed} = \frac{10 \text{ mi}}{20 \text{ min}} = 0.50 \frac{\text{mi}}{\text{min}}.$$

3. Convert miles per minute to meters per second by multiplying by the conversion factor that cancels miles and leave meters, and also by the conversion factor that cancels minutes and leave seconds:

$$0.50 \frac{\cancel{\text{mile}}}{\cancel{\text{min}}} \times \frac{1609 \text{ m}}{1 \cancel{\text{ mile}}} \times \frac{1 \cancel{\text{ min}}}{60 \text{ s}} = \frac{(0.50)(1609)}{60} \text{ m/s} = 13 \text{ m/s}.$$

Significance

Check the answer in the following ways:

1. Be sure the units in the unit conversion cancel correctly. If the unit conversion factor was written upside down, the units do not cancel correctly in the equation. We see the “miles” in the numerator in 0.50 mi/min cancels the “mile” in the denominator in the first conversion factor. Also, the “min” in the denominator in 0.50 mi/min cancels the “min” in the numerator in the second conversion factor.
2. Check that the units of the final answer are the desired units. The problem asked us to solve for average speed in units of meters per second and, after the cancellations, the only units left are a meter (m) in the numerator and a second (s) in the denominator, so we have indeed obtained these units.

CHECK YOUR UNDERSTANDING 1.2

Light travels about 9 Pm in a year. Given that a year is about 3×10^7 s, what is the speed of light in meters per second?

EXAMPLE 1.3

Converting between Metric Units

The density of iron is 7.86 g/cm^3 under standard conditions. Convert this to kg/m^3 .

Strategy

We need to convert grams to kilograms and cubic centimeters to cubic meters. The conversion factors we need are $1 \text{ kg} = 10^3 \text{ g}$ and $1 \text{ cm} = 10^{-2} \text{ m}$. However, we are dealing with cubic centimeters ($\text{cm}^3 = \text{cm} \times \text{cm} \times \text{cm}$), so we have to use the second conversion factor three times (that is, we need to cube it). The idea is still to multiply by the conversion factors in such a way that they cancel the units we want to get rid of and introduce the units we want to keep.

Solution

$$7.86 \frac{\cancel{\text{g}}}{\cancel{\text{cm}}^3} \times \frac{\text{kg}}{10^3 \cancel{\text{g}}} \times \left(\frac{\cancel{\text{cm}}}{10^{-2} \text{ m}} \right)^3 = \frac{7.86}{(10^3)(10^{-6})} \text{ kg/m}^3 = 7.86 \times 10^3 \text{ kg/m}^3$$

Significance

Remember, it's always important to check the answer.

1. Be sure to cancel the units in the unit conversion correctly. We see that the gram (“g”) in the numerator in 7.86 g/cm^3 cancels the “g” in the denominator in the first conversion factor. Also, the three factors of “cm” in the denominator in 7.86 g/cm^3 cancel with the three factors of “cm” in the numerator that we get by cubing the second conversion factor.
2. Check that the units of the final answer are the desired units. The problem asked for us to convert to kilograms per cubic meter. After the cancellations just described, we see the only units we have left are “kg” in the numerator and three factors of “m” in the denominator (that is, one factor of “m” cubed, or “m³”). Therefore, the units on the final answer are correct.

✓ CHECK YOUR UNDERSTANDING 1.3

We know from [Figure 1.4](#) that the diameter of Earth is on the order of 10^7 m, so the order of magnitude of its surface area is 10^{14} m². What is that in square kilometers (that is, km²)? (Try doing this both by converting 10^7 m to km and then squaring it and then by converting 10^{14} m² directly to square kilometers. You should get the same answer both ways.)

Unit conversions may not seem very interesting, but not doing them can be costly. One famous example of this situation was seen with the *Mars Climate Orbiter*. This probe was launched by NASA on December 11, 1998. On September 23, 1999, while attempting to guide the probe into its planned orbit around Mars, NASA lost contact with it. Subsequent investigations showed a piece of software called SM_FORCES (or “small forces”) was recording thruster performance data in the English units of pound-force-seconds (lbf-s). However, other pieces of software that used these values for course corrections expected them to be recorded in the SI units of newton-seconds (N-s), as dictated in the software interface protocols. This error caused the probe to follow a very different trajectory from what NASA thought it was following, which most likely caused the probe either to burn up in the Martian atmosphere or to shoot out into space. This failure to pay attention to unit conversions cost hundreds of millions of dollars, not to mention all the time invested by the scientists and engineers who worked on the project.

✓ CHECK YOUR UNDERSTANDING 1.4

Given that 1 lbf (pound-force) is 4.45 N, were the numbers being output by SM_FORCES too big or too small?

1.4 Dimensional Analysis

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Find the dimensions of a mathematical expression involving physical quantities.
- Determine whether an equation involving physical quantities is dimensionally consistent.

The **dimension** of any physical quantity expresses its dependence on the base quantities as a product of symbols (or powers of symbols) representing the base quantities. [Table 1.3](#) lists the base quantities and the symbols used for their dimension. For example, a measurement of length is said to have dimension L or L¹, a measurement of mass has dimension M or M¹, and a measurement of time has dimension T or T¹. Like units, dimensions obey the rules of algebra. Thus, area is the product of two lengths and so has dimension L², or length squared. Similarly, volume is the product of three lengths and has dimension L³, or length cubed. Speed has dimension length over time, L/T or LT⁻¹. Volumetric mass density has dimension M/L³ or ML⁻³, or mass over length cubed. In general, the dimension of any physical quantity can be written as L^aM^bT^cI^dΘ^eN^fJ^g for some powers *a*, *b*, *c*, *d*, *e*, *f*, and *g*. We can write the dimensions of a length in this form with *a* = 1 and the remaining six powers all set equal to zero: L¹ = L¹M⁰T⁰I⁰Θ⁰N⁰J⁰. Any quantity with a dimension that can be written so that all seven powers are zero (that is, its dimension is L⁰M⁰T⁰I⁰Θ⁰N⁰J⁰) is called **dimensionless** (or sometimes “of dimension 1,” because anything raised to the zero power is one). Physicists often call dimensionless quantities *pure numbers*.

Base Quantity	Symbol for Dimension
Length	L
Mass	M
Time	T
Current	I

TABLE 1.3 Base Quantities and Their Dimensions

Base Quantity	Symbol for Dimension
Thermodynamic temperature	Θ
Amount of substance	N
Luminous intensity	J

TABLE 1.3 Base Quantities and Their Dimensions

Physicists often use square brackets around the symbol for a physical quantity to represent the dimensions of that quantity. For example, if r is the radius of a cylinder and h is its height, then we write $[r] = L$ and $[h] = L$ to indicate the dimensions of the radius and height are both those of length, or L . Similarly, if we use the symbol A for the surface area of a cylinder and V for its volume, then $[A] = L^2$ and $[V] = L^3$. If we use the symbol m for the mass of the cylinder and ρ for the density of the material from which the cylinder is made, then $[m] = M$ and $[\rho] = ML^{-3}$.

The importance of the concept of dimension arises from the fact that any mathematical equation relating physical quantities must be **dimensionally consistent**, which means the equation must obey the following rules:

- Every term in an expression must have the same dimensions; it does not make sense to add or subtract quantities of differing dimension (think of the old saying: “You can’t add apples and oranges”). In particular, the expressions on each side of the equality in an equation must have the same dimensions.
- The arguments of any of the standard mathematical functions such as trigonometric functions (such as sine and cosine), logarithms, or exponential functions that appear in the equation must be dimensionless. These functions require pure numbers as inputs and give pure numbers as outputs.

If either of these rules is violated, an equation is not dimensionally consistent and cannot possibly be a correct statement of physical law. This simple fact can be used to check for typos or algebra mistakes, to help remember the various laws of physics, and even to suggest the form that new laws of physics might take. This last use of dimensions is beyond the scope of this text, but is something you may learn later in your academic career.



EXAMPLE 1.4

Using Dimensions to Remember an Equation

Suppose we need the formula for the area of a circle for some computation. Like many people who learned geometry too long ago to recall with any certainty, two expressions may pop into our mind when we think of circles: πr^2 and $2\pi r$. One expression is the circumference of a circle of radius r and the other is its area. But which is which?

Strategy

One natural strategy is to look it up, but this could take time to find information from a reputable source. Besides, even if we think the source is reputable, we shouldn’t trust everything we read. It is nice to have a way to double-check just by thinking about it. Also, we might be in a situation in which we cannot look things up (such as during a test). Thus, the strategy is to find the dimensions of both expressions by making use of the fact that dimensions follow the rules of algebra. If either expression does not have the same dimensions as area, then it cannot possibly be the correct equation for the area of a circle.

Solution

We know the dimension of area is L^2 . Now, the dimension of the expression πr^2 is

$$[\pi r^2] = [\pi] \cdot [r]^2 = 1 \cdot L^2 = L^2,$$

since the constant π is a pure number and the radius r is a length. Therefore, πr^2 has the dimension of area. Similarly, the dimension of the expression $2\pi r$ is

$$[2\pi r] = [2] \cdot [\pi] \cdot [r] = 1 \cdot 1 \cdot L = L,$$

since the constants 2 and π are both dimensionless and the radius r is a length. We see that $2\pi r$ has the dimension

of length, which means it cannot possibly be an area.

We rule out $2\pi r$ because it is not dimensionally consistent with being an area. We see that πr^2 is dimensionally consistent with being an area, so if we have to choose between these two expressions, πr^2 is the one to choose.

Significance

This may seem like kind of a silly example, but the ideas are very general. As long as we know the dimensions of the individual physical quantities that appear in an equation, we can check to see whether the equation is dimensionally consistent. On the other hand, knowing that true equations are dimensionally consistent, we can match expressions from our imperfect memories to the quantities for which they might be expressions. Doing this will not help us remember dimensionless factors that appear in the equations (for example, if you had accidentally conflated the two expressions from the example into $2\pi r^2$, then dimensional analysis is no help), but it does help us remember the correct basic form of equations.

CHECK YOUR UNDERSTANDING 1.5

Suppose we want the formula for the volume of a sphere. The two expressions commonly mentioned in elementary discussions of spheres are $4\pi r^2$ and $4\pi r^3/3$. One is the volume of a sphere of radius r and the other is its surface area. Which one is the volume?

EXAMPLE 1.5

Checking Equations for Dimensional Consistency

Consider the physical quantities s , v , a , and t with dimensions $[s] = L$, $[v] = LT^{-1}$, $[a] = LT^{-2}$, and $[t] = T$. Determine whether each of the following equations is dimensionally consistent: (a) $s = vt + 0.5at^2$; (b) $s = vt^2 + 0.5at$; and (c) $v = \sin(at^2/s)$.

Strategy

By the definition of dimensional consistency, we need to check that each term in a given equation has the same dimensions as the other terms in that equation and that the arguments of any standard mathematical functions are dimensionless.

Solution

- a. There are no trigonometric, logarithmic, or exponential functions to worry about in this equation, so we need only look at the dimensions of each term appearing in the equation. There are three terms, one in the left expression and two in the expression on the right, so we look at each in turn:

$$\begin{aligned}[s] &= L \\ [vt] &= [v] \cdot [t] = LT^{-1} \cdot T = LT^0 = L \\ [0.5at^2] &= [a] \cdot [t]^2 = LT^{-2} \cdot T^2 = LT^0 = L.\end{aligned}$$

All three terms have the same dimension, so this equation is dimensionally consistent.

- b. Again, there are no trigonometric, exponential, or logarithmic functions, so we only need to look at the dimensions of each of the three terms appearing in the equation:

$$\begin{aligned}[s] &= L \\ [vt^2] &= [v] \cdot [t]^2 = LT^{-1} \cdot T^2 = LT \\ [at] &= [a] \cdot [t] = LT^{-2} \cdot T = LT^{-1}.\end{aligned}$$

None of the three terms has the same dimension as any other, so this is about as far from being dimensionally consistent as you can get. The technical term for an equation like this is *nonsense*.

- c. This equation has a trigonometric function in it, so first we should check that the argument of the sine function is dimensionless:

$$\left[\frac{at^2}{s} \right] = \frac{[a] \cdot [t]^2}{[s]} = \frac{LT^{-2} \cdot T^2}{L} = \frac{L}{L} = 1.$$

The argument is dimensionless. So far, so good. Now we need to check the dimensions of each of the two terms (that is, the left expression and the right expression) in the equation:

$$[v] = LT^{-1}$$

$$\left[\sin \left(\frac{at^2}{s} \right) \right] = 1.$$

The two terms have different dimensions—meaning, the equation is not dimensionally consistent. This equation is another example of “nonsense.”

Significance

If we are trusting people, these types of dimensional checks might seem unnecessary. But, rest assured, any textbook on a quantitative subject such as physics (including this one) almost certainly contains some equations with typos. Checking equations routinely by dimensional analysis save us the embarrassment of using an incorrect equation. Also, checking the dimensions of an equation we obtain through algebraic manipulation is a great way to make sure we did not make a mistake (or to spot a mistake, if we made one).

✓ CHECK YOUR UNDERSTANDING 1.6

Is the equation $v = at$ dimensionally consistent?

One further point that needs to be mentioned is the effect of the operations of calculus on dimensions. We have seen that dimensions obey the rules of algebra, just like units, but what happens when we take the derivative of one physical quantity with respect to another or integrate a physical quantity over another? The derivative of a function is just the slope of the line tangent to its graph and slopes are ratios, so for physical quantities v and t , we have that the dimension of the derivative of v with respect to t is just the ratio of the dimension of v over that of t :

$$\left[\frac{dv}{dt} \right] = \frac{[v]}{[t]}.$$

Similarly, since integrals are just sums of products, the dimension of the integral of v with respect to t is simply the dimension of v times the dimension of t :

$$\left[\int v dt \right] = [v] \cdot [t].$$

By the same reasoning, analogous rules hold for the units of physical quantities derived from other quantities by integration or differentiation.

1.5 Estimates and Fermi Calculations

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Estimate the values of physical quantities.

On many occasions, physicists, other scientists, and engineers need to make *estimates* for a particular quantity. Other terms sometimes used are *guesstimates*, *order-of-magnitude approximations*, *back-of-the-envelope calculations*, or *Fermi calculations*. (The physicist Enrico Fermi mentioned earlier was famous for his ability to estimate various kinds of data with surprising precision.) Will that piece of equipment fit in the back of the car or do we need to rent a truck? How long will this download take? About how large a current will there be in this circuit when it is turned on? How many houses could a proposed power plant actually power if it is built? Note that estimating does not mean guessing a number or a formula at random. Rather, **estimation** means using prior experience and sound physical reasoning to arrive at a rough idea of a quantity's value. Because the process of determining a reliable approximation usually involves the identification of correct physical principles and a good

guess about the relevant variables, estimating is very useful in developing physical intuition. Estimates also allow us to perform “sanity checks” on calculations or policy proposals by helping us rule out certain scenarios or unrealistic numbers. They allow us to challenge others (as well as ourselves) in our efforts to learn truths about the world.

Many estimates are based on formulas in which the input quantities are known only to a limited precision. As you develop physics problem-solving skills (which are applicable to a wide variety of fields), you also will develop skills at estimating. You develop these skills by thinking quantitatively and being willing to take risks. As with any skill, experience helps. Familiarity with dimensions (see [Table 1.3](#)) and units (see [Table 1.1](#) and [Table 1.2](#)), and the scales of base quantities (see [Figure 1.4](#)) also helps.

To make some progress in estimating, you need to have some definite ideas about how variables may be related. The following strategies may help you in practicing the art of estimation:

- *Get big lengths from smaller lengths.* When estimating lengths, remember that anything can be a ruler. Thus, imagine breaking a big thing into smaller things, estimate the length of one of the smaller things, and multiply to get the length of the big thing. For example, to estimate the height of a building, first count how many floors it has. Then, estimate how big a single floor is by imagining how many people would have to stand on each other's shoulders to reach the ceiling. Last, estimate the height of a person. The product of these three estimates is your estimate of the height of the building. It helps to have memorized a few length scales relevant to the sorts of problems you find yourself solving. For example, knowing some of the length scales in [Figure 1.4](#) might come in handy. Sometimes it also helps to do this in reverse—that is, to estimate the length of a small thing, imagine a bunch of them making up a bigger thing. For example, to estimate the thickness of a sheet of paper, estimate the thickness of a stack of paper and then divide by the number of pages in the stack. These same strategies of breaking big things into smaller things or aggregating smaller things into a bigger thing can sometimes be used to estimate other physical quantities, such as masses and times.
- *Get areas and volumes from lengths.* When dealing with an area or a volume of a complex object, introduce a simple model of the object such as a sphere or a box. Then, estimate the linear dimensions (such as the radius of the sphere or the length, width, and height of the box) first, and use your estimates to obtain the volume or area from standard geometric formulas. If you happen to have an estimate of an object's area or volume, you can also do the reverse; that is, use standard geometric formulas to get an estimate of its linear dimensions.
- *Get masses from volumes and densities.* When estimating masses of objects, it can help first to estimate its volume and then to estimate its mass from a rough estimate of its average density (recall, density has dimension mass over length cubed, so mass is density times volume). For this, it helps to remember that the density of air is around 1 kg/m^3 , the density of water is 10^3 kg/m^3 , and the densest everyday solids max out at around 10^4 kg/m^3 . Asking yourself whether an object floats or sinks in either air or water gets you a ballpark estimate of its density. You can also do this the other way around; if you have an estimate of an object's mass and its density, you can use them to get an estimate of its volume.
- *If all else fails, bound it.* For physical quantities for which you do not have a lot of intuition, sometimes the best you can do is think something like: Well, it must be bigger than this and smaller than that. For example, suppose you need to estimate the mass of a moose. Maybe you have a lot of experience with moose and know their average mass offhand. If so, great. But for most people, the best they can do is to think something like: It must be bigger than a person (of order 10^2 kg) and less than a car (of order 10^3 kg). If you need a single number for a subsequent calculation, you can take the geometric mean of the upper and lower bound—that is, you multiply them together and then take the square root. For the moose mass example, this would be

$$(10^2 \times 10^3)^{0.5} = 10^{2.5} = 10^{0.5} \times 10^2 \approx 3 \times 10^2 \text{ kg}.$$

The tighter the bounds, the better. Also, no rules are unbreakable when it comes to estimation. If you think the value of the quantity is likely to be closer to the upper bound than the lower bound, then you may want to bump up your estimate from the geometric mean by an order or two of magnitude.

- *One “sig. fig.” is fine.* There is no need to go beyond one significant figure, or one digit in the coefficient of an expression in scientific notation, when doing calculations to obtain an estimate. In most cases, the order of magnitude is good enough. The goal is just to get in the ballpark figure, so keep the arithmetic as simple as possible.
- *Ask yourself: Does this make any sense?* Last, check to see whether your answer is reasonable. How does it compare with the values of other quantities with the same dimensions that you already know or can look up

easily? If you get some wacky answer (for example, if you estimate the mass of the Atlantic Ocean to be bigger than the mass of Earth, or some time span to be longer than the age of the universe), first check to see whether your units are correct. Then, check for arithmetic errors. Then, rethink the logic you used to arrive at your answer. If everything checks out, you may have just proved that some slick new idea is actually bogus.



EXAMPLE 1.6

Mass of Earth's Oceans

Estimate the total mass of the oceans on Earth.

Strategy

We know the density of water is about 10^3 kg/m^3 , so we start with the advice to “get masses from densities and volumes.” Thus, we need to estimate the volume of the planet’s oceans. Using the advice to “get areas and volumes from lengths,” we can estimate the volume of the oceans as surface area times average depth, or $V = AD$. We know the diameter of Earth from [Figure 1.4](#) and we know that most of Earth’s surface is covered in water, so we can estimate the surface area of the oceans as being roughly equal to the surface area of the planet. By following the advice to “get areas and volumes from lengths” again, we can approximate Earth as a sphere and use the formula for the surface area of a sphere of diameter d —that is, $A = \pi d^2$, to estimate the surface area of the oceans. Now we just need to estimate the average depth of the oceans. For this, we use the advice: “If all else fails, bound it.” We happen to know the deepest points in the ocean are around 10 km and that it is not uncommon for the ocean to be deeper than 1 km, so we take the average depth to be around $(10^3 \times 10^4)^{0.5} \approx 3 \times 10^3 \text{ m}$. Now we just need to put it all together, heeding the advice that “one ‘sig. fig.’ is fine.”

Solution

We estimate the surface area of Earth (and hence the surface area of Earth’s oceans) to be roughly

$$A = \pi d^2 = \pi(10^7 \text{ m})^2 \approx 3 \times 10^{14} \text{ m}^2.$$

Next, using our average depth estimate of $D = 3 \times 10^3 \text{ m}$, which was obtained by bounding, we estimate the volume of Earth’s oceans to be

$$V = AD = (3 \times 10^{14} \text{ m}^2)(3 \times 10^3 \text{ m}) = 9 \times 10^{17} \text{ m}^3.$$

Last, we estimate the mass of the world’s oceans to be

$$M = \rho V = (10^3 \text{ kg/m}^3)(9 \times 10^{17} \text{ m}^3) = 9 \times 10^{20} \text{ kg}.$$

Thus, we estimate that the order of magnitude of the mass of the planet’s oceans is 10^{21} kg .

Significance

To verify our answer to the best of our ability, we first need to answer the question: Does this make any sense? From [Figure 1.4](#), we see the mass of Earth’s atmosphere is on the order of 10^{19} kg and the mass of Earth is on the order of 10^{25} kg . It is reassuring that our estimate of 10^{21} kg for the mass of Earth’s oceans falls somewhere between these two. So, yes, it does seem to make sense. It just so happens that we did a search on the Web for “mass of oceans” and the top search results all said $1.4 \times 10^{21} \text{ kg}$, which is the same order of magnitude as our estimate. Now, rather than having to trust blindly whoever first put that number up on a website (most of the other sites probably just copied it from them, after all), we can have a little more confidence in it.



CHECK YOUR UNDERSTANDING 1.7

[Figure 1.4](#) says the mass of the atmosphere is 10^{19} kg . Assuming the density of the atmosphere is 1 kg/m^3 , estimate the height of Earth’s atmosphere. Do you think your answer is an underestimate or an overestimate? Explain why.

How many piano tuners are there in New York City? How many leaves are on that tree? If you are studying

photosynthesis or thinking of writing a smartphone app for piano tuners, then the answers to these questions might be of great interest to you. Otherwise, you probably couldn't care less what the answers are. However, these are exactly the sorts of estimation problems that people in various tech industries have been asking potential employees to evaluate their quantitative reasoning skills. If building physical intuition and evaluating quantitative claims do not seem like sufficient reasons for you to practice estimation problems, how about the fact that being good at them just might land you a high-paying job?

INTERACTIVE

For practice estimating relative lengths, areas, and volumes, check out this [PhET \(https://openstax.org//21lengthgame\)](https://openstax.org//21lengthgame) simulation, titled “Estimation.”

1.6 Significant Figures

LEARNING OBJECTIVES

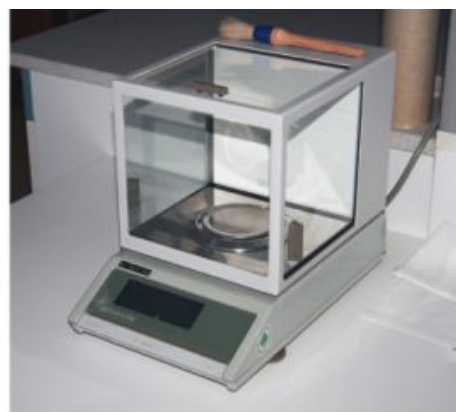
By the end of this section, you will be able to:

- Determine the correct number of significant figures for the result of a computation.
- Describe the relationship between the concepts of accuracy, precision, uncertainty, and discrepancy.
- Calculate the percent uncertainty of a measurement, given its value and its uncertainty.
- Determine the uncertainty of the result of a computation involving quantities with given uncertainties.

[Figure 1.11](#) shows two instruments used to measure the mass of an object. The digital scale has mostly replaced the double-pan balance in physics labs because it gives more accurate and precise measurements. But what exactly do we mean by *accurate* and *precise*? Aren't they the same thing? In this section we examine in detail the process of making and reporting a measurement.



(a)



(b)

FIGURE 1.11 (a) A double-pan mechanical balance is used to compare different masses. Usually an object with unknown mass is placed in one pan and objects of known mass are placed in the other pan. When the bar that connects the two pans is horizontal, then the masses in both pans are equal. The “known masses” are typically metal cylinders of standard mass such as 1 g, 10 g, and 100 g. (b) Many mechanical balances, such as double-pan balances, have been replaced by digital scales, which can typically measure the mass of an object more precisely. A mechanical balance may read only the mass of an object to the nearest tenth of a gram, but many digital scales can measure the mass of an object up to the nearest thousandth of a gram. (credit a: modification of work by Serge Melki; credit b: modification of work by Karel Jakubec)

Accuracy and Precision of a Measurement

Science is based on observation and experiment—that is, on measurements. **Accuracy** is how close a measurement is to the accepted reference value for that measurement. For example, let's say we want to measure the length of standard printer paper. The packaging in which we purchased the paper states that it is 11.0 in. long. We then measure the length of the paper three times and obtain the following measurements: 11.1 in., 11.2 in., and 10.9 in. These measurements are quite accurate because they are very close to the reference value of 11.0 in. In contrast, if we had obtained a measurement of 12 in., our measurement would not be very accurate. Notice that the concept of accuracy requires that an accepted reference value be given.

The **precision** of measurements refers to how close the agreement is between repeated independent measurements (which are repeated under the same conditions). Consider the example of the paper measurements. The precision of the measurements refers to the spread of the measured values. One way to analyze the precision of the measurements is to determine the range, or difference, between the lowest and the highest measured values. In this case, the lowest value was 10.9 in. and the highest value was 11.2 in. Thus, the measured values deviated from each other by, at most, 0.3 in. These measurements were relatively precise because they did not vary too much in value. However, if the measured values had been 10.9 in., 11.1 in., and 11.9 in., then the measurements would not be very precise because there would be significant variation from one measurement to another. Notice that the concept of precision depends only on the actual measurements acquired and does not depend on an accepted reference value.

The measurements in the paper example are both accurate and precise, but in some cases, measurements are accurate but not precise, or they are precise but not accurate. Let's consider an example of a GPS attempting to locate the position of a restaurant in a city. Think of the restaurant location as existing at the center of a bull's-eye target and think of each GPS attempt to locate the restaurant as a black dot. In [Figure 1.12\(a\)](#), we see the GPS measurements are spread out far apart from each other, but they are all relatively close to the actual location of the restaurant at the center of the target. This indicates a low-precision, high-accuracy measuring system. However, in [Figure 1.12\(b\)](#), the GPS measurements are concentrated quite closely to one another, but they are far away from the target location. This indicates a high-precision, low-accuracy measuring system.

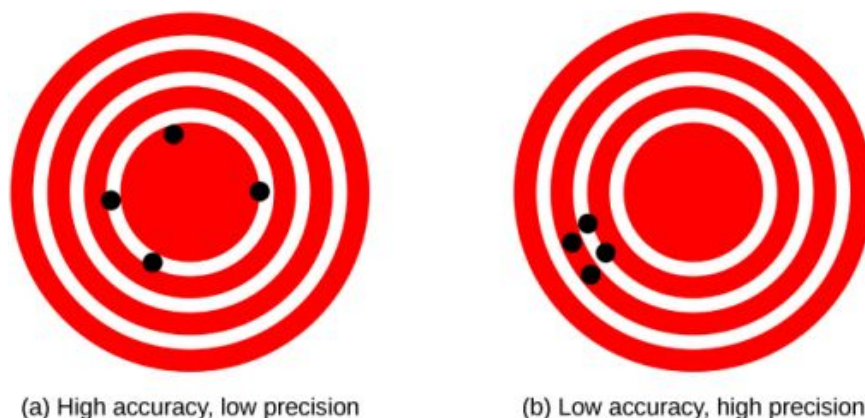


FIGURE 1.12 A GPS attempts to locate a restaurant at the center of the bull's-eye. The black dots represent each attempt to pinpoint the location of the restaurant. (a) The dots are spread out quite far apart from one another, indicating low precision, but they are each rather close to the actual location of the restaurant, indicating high accuracy. (b) The dots are concentrated rather closely to one another, indicating high precision, but they are rather far away from the actual location of the restaurant, indicating low accuracy. (credit a and credit b: modification of works by "DarkEvil"/Wikimedia Commons)

Accuracy, Precision, Uncertainty, and Discrepancy

The precision of a measuring system is related to the **uncertainty** in the measurements whereas the accuracy is related to the **discrepancy** from the accepted reference value. Uncertainty is a quantitative measure of how much your measured values deviate from one another. There are many different methods of calculating uncertainty, each of which is appropriate to different situations. Some examples include taking the range (that is, the largest minus the smallest) or finding the standard deviation of the measurements. Discrepancy (or "measurement error") is the difference between the measured value and a given standard or expected value. If the measurements are not very precise, then the uncertainty of the values is high. If the measurements are not very accurate, then the discrepancy of the values is high.

Recall our example of measuring paper length; we obtained measurements of 11.1 in., 11.2 in., and 10.9 in., and the accepted value was 11.0 in. We might average the three measurements to say our best guess is 11.1 in.; in this case, our discrepancy is $11.1 - 11.0 = 0.1$ in., which provides a quantitative measure of accuracy. We might calculate the uncertainty in our best guess by using half of the range of our measured values: 0.15 in. Then we would say the length of the paper is 11.1 in. plus or minus 0.15 in. The uncertainty in a measurement, A , is often denoted as δA (read "delta A "), so the measurement result would be recorded as $A \pm \delta A$. Returning to our paper example, the measured length of the paper could be expressed as 11.1 ± 0.15 in. Since the discrepancy of 0.1 in. is less than the uncertainty of 0.15 in., we might say the measured value agrees with the accepted reference value to

within experimental uncertainty.

Some factors that contribute to uncertainty in a measurement include the following:

- Limitations of the measuring device
- The skill of the person taking the measurement
- Irregularities in the object being measured
- Any other factors that affect the outcome (highly dependent on the situation)

In our example, such factors contributing to the uncertainty could be the smallest division on the ruler is 1/16 in., the person using the ruler has bad eyesight, the ruler is worn down on one end, or one side of the paper is slightly longer than the other. At any rate, the uncertainty in a measurement must be calculated to quantify its precision. If a reference value is known, it makes sense to calculate the discrepancy as well to quantify its accuracy.

Percent uncertainty

Another method of expressing uncertainty is as a percent of the measured value. If a measurement A is expressed with uncertainty δA , the **percent uncertainty** is defined as

$$\text{Percent uncertainty} = \frac{\delta A}{A} \times 100\%.$$



EXAMPLE 1.7

Calculating Percent Uncertainty: A Bag of Apples

A grocery store sells 5-lb bags of apples. Let's say we purchase four bags during the course of a month and weigh the bags each time. We obtain the following measurements:

- Week 1 weight: 4.8 lb
- Week 2 weight: 5.3 lb
- Week 3 weight: 4.9 lb
- Week 4 weight: 5.4 lb

We then determine the average weight of the 5-lb bag of apples is 5.1 ± 0.3 lb from using half of the range. What is the percent uncertainty of the bag's weight?

Strategy

First, observe that the average value of the bag's weight, A , is 5.1 lb. The uncertainty in this value, δA , is 0.3 lb. We can use the following equation to determine the percent uncertainty of the weight:

$$\text{Percent uncertainty} = \frac{\delta A}{A} \times 100\%.$$

1.1

Solution

Substitute the values into the equation:

$$\text{Percent uncertainty} = \frac{\delta A}{A} \times 100\% = \frac{0.3 \text{ lb}}{5.1 \text{ lb}} \times 100\% = 5.9\% \approx 6\%.$$

Significance

We can conclude the average weight of a bag of apples from this store is $5.1 \text{ lb} \pm 6\%$. Notice the percent uncertainty is dimensionless because the units of weight in $\delta A = 0.2 \text{ lb}$ canceled those in $A = 5.1 \text{ lb}$ when we took the ratio.



CHECK YOUR UNDERSTANDING 1.8

A high school track coach has just purchased a new stopwatch. The stopwatch manual states the stopwatch has an uncertainty of ± 0.05 s. Runners on the track coach's team regularly clock 100-m sprints of 11.49 s to 15.01 s. At the

school's last track meet, the first-place sprinter came in at 12.04 s and the second-place sprinter came in at 12.07 s. Will the coach's new stopwatch be helpful in timing the sprint team? Why or why not?

Uncertainties in calculations

Uncertainty exists in anything calculated from measured quantities. For example, the area of a floor calculated from measurements of its length and width has an uncertainty because the length and width have uncertainties. How big is the uncertainty in something you calculate by multiplication or division? If the measurements going into the calculation have small uncertainties (a few percent or less), then the **method of adding percents** can be used for multiplication or division. This method states *the percent uncertainty in a quantity calculated by multiplication or division is the sum of the percent uncertainties in the items used to make the calculation*. For example, if a floor has a length of 4.00 m and a width of 3.00 m, with uncertainties of 2% and 1%, respectively, then the area of the floor is 12.0 m^2 and has an uncertainty of 3%. (Expressed as an area, this is $0.36 \text{ m}^2 [12.0 \text{ m}^2 \times 0.03]$, which we round to 0.4 m^2 since the area of the floor is given to a tenth of a square meter.)

Precision of Measuring Tools and Significant Figures

An important factor in the precision of measurements involves the precision of the measuring tool. In general, a precise measuring tool is one that can measure values in very small increments. For example, a standard ruler can measure length to the nearest millimeter whereas a caliper can measure length to the nearest 0.01 mm. The caliper is a more precise measuring tool because it can measure extremely small differences in length. The more precise the measuring tool, the more precise the measurements.

When we express measured values, we can only list as many digits as we measured initially with our measuring tool. For example, if we use a standard ruler to measure the length of a stick, we may measure it to be 36.7 cm. We can't express this value as 36.71 cm because our measuring tool is not precise enough to measure a hundredth of a centimeter. It should be noted that the last digit in a measured value has been estimated in some way by the person performing the measurement. For example, the person measuring the length of a stick with a ruler notices the stick length seems to be somewhere in between 36.6 cm and 36.7 cm, and he or she must estimate the value of the last digit. Using the method of **significant figures**, the rule is that *the last digit written down in a measurement is the first digit with some uncertainty*. To determine the number of significant digits in a value, start with the first measured value at the left and count the number of digits through the last digit written on the right. For example, the measured value 36.7 cm has three digits, or three significant figures. Significant figures indicate the precision of the measuring tool used to measure a value.

Zeros

Special consideration is given to zeros when counting significant figures. The zeros in 0.053 are not significant because they are placeholders that locate the decimal point. There are two significant figures in 0.053. The zeros in 10.053 are not placeholders; they are significant. This number has five significant figures. The zeros in 1300 may or may not be significant, depending on the style of writing numbers. They could mean the number is known to the last digit or they could be placeholders. So 1300 could have two, three, or four significant figures. To avoid this ambiguity, we should write 1300 in scientific notation as 1.3×10^3 , 1.30×10^3 , or 1.300×10^3 , depending on whether it has two, three, or four significant figures. *Zeros are significant except when they serve only as placeholders.*

Significant figures in calculations

When combining measurements with different degrees of precision with the mathematical operations of addition, subtraction, multiplication, or division, then *the number of significant digits in the final answer can be no greater than the number of significant digits in the least-precise measured value*. There are two different rules, one for multiplication and division and the other for addition and subtraction.

1. *For multiplication and division, the result should have the same number of significant figures as the quantity with the least number of significant figures entering into the calculation.* For example, the area of a circle can be calculated from its radius using $A = \pi r^2$. Let's see how many significant figures the area has if the radius has only two—say, $r = 1.2 \text{ m}$. Using a calculator with an eight-digit output, we would calculate

$$A = \pi r^2 = (3.1415927\ldots) \times (1.2 \text{ m})^2 = 4.5238934 \text{ m}^2.$$

But because the radius has only two significant figures, it limits the calculated quantity to two significant figures, or

$$A = 4.5 \text{ m}^2,$$

although π is good to at least eight digits.

2. *For addition and subtraction, the answer can contain no more decimal places than the least-precise measurement.* Suppose we buy 7.56 kg of potatoes in a grocery store as measured with a scale with precision 0.01 kg, then we drop off 6.052 kg of potatoes at your laboratory as measured by a scale with precision 0.001 kg. Then, we go home and add 13.7 kg of potatoes as measured by a bathroom scale with precision 0.1 kg. How many kilograms of potatoes do we now have and how many significant figures are appropriate in the answer? The mass is found by simple addition and subtraction:

$$7.56 \text{ kg} - 6.052 \text{ kg} + 13.7 \text{ kg} = 15.208 \text{ kg}$$

Next, we identify the least-precise measurement: 13.7 kg. This measurement is expressed to the 0.1 decimal place, so our final answer must also be expressed to the 0.1 decimal place. Thus, the answer is rounded to the tenths place, giving us 15.2 kg.

Significant figures in this text

In this text, most numbers are assumed to have three significant figures. Furthermore, consistent numbers of significant figures are used in all worked examples. An answer given to three digits is based on input good to at least three digits, for example. If the input has fewer significant figures, the answer will also have fewer significant figures. Care is also taken that the number of significant figures is reasonable for the situation posed. In some topics, particularly in optics, more accurate numbers are needed and we use more than three significant figures. Finally, if a number is *exact*, such as the two in the formula for the circumference of a circle, $C = 2\pi r$, it does not affect the number of significant figures in a calculation. Likewise, conversion factors such as 100 cm/1 m are considered exact and do not affect the number of significant figures in a calculation.

1.7 Solving Problems in Physics

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Describe the process for developing a problem-solving strategy.
- Explain how to find the numerical solution to a problem.
- Summarize the process for assessing the significance of the numerical solution to a problem.



FIGURE 1.13 Problem-solving skills are essential to your success in physics. (credit: “scui3asteveo”/Flickr)

Problem-solving skills are clearly essential to success in a quantitative course in physics. More important, the ability

to apply broad physical principles—usually represented by equations—to specific situations is a very powerful form of knowledge. It is much more powerful than memorizing a list of facts. Analytical skills and problem-solving abilities can be applied to new situations whereas a list of facts cannot be made long enough to contain every possible circumstance. Such analytical skills are useful both for solving problems in this text and for applying physics in everyday life.

As you are probably well aware, a certain amount of creativity and insight is required to solve problems. No rigid procedure works every time. Creativity and insight grow with experience. With practice, the basics of problem solving become almost automatic. One way to get practice is to work out the text's examples for yourself as you read. Another is to work as many end-of-section problems as possible, starting with the easiest to build confidence and then progressing to the more difficult. After you become involved in physics, you will see it all around you, and you can begin to apply it to situations you encounter outside the classroom, just as is done in many of the applications in this text.

Although there is no simple step-by-step method that works for every problem, the following three-stage process facilitates problem solving and makes it more meaningful. The three stages are strategy, solution, and significance. This process is used in examples throughout the book. Here, we look at each stage of the process in turn.

Strategy

Strategy is the beginning stage of solving a problem. The idea is to figure out exactly what the problem is and then develop a strategy for solving it. Some general advice for this stage is as follows:

- *Examine the situation to determine which physical principles are involved.* It often helps to *draw a simple sketch* at the outset. You often need to decide which direction is positive and note that on your sketch. When you have identified the physical principles, it is much easier to find and apply the equations representing those principles. Although finding the correct equation is essential, keep in mind that equations represent physical principles, laws of nature, and relationships among physical quantities. Without a conceptual understanding of a problem, a numerical solution is meaningless.
- *Make a list of what is given or can be inferred from the problem as stated (identify the “knowns”).* Many problems are stated very succinctly and require some inspection to determine what is known. Drawing a sketch can be very useful at this point as well. Formally identifying the knowns is of particular importance in applying physics to real-world situations. For example, the word *stopped* means the velocity is zero at that instant. Also, we can often take initial time and position as zero by the appropriate choice of coordinate system.
- *Identify exactly what needs to be determined in the problem (identify the unknowns).* In complex problems, especially, it is not always obvious what needs to be found or in what sequence. Making a list can help identify the unknowns.
- *Determine which physical principles can help you solve the problem.* Since physical principles tend to be expressed in the form of mathematical equations, a list of knowns and unknowns can help here. It is easiest if you can find equations that contain only one unknown—that is, all the other variables are known—so you can solve for the unknown easily. If the equation contains more than one unknown, then additional equations are needed to solve the problem. In some problems, several unknowns must be determined to get at the one needed most. In such problems it is especially important to keep physical principles in mind to avoid going astray in a sea of equations. You may have to use two (or more) different equations to get the final answer.

Solution

The solution stage is when you do the math. *Substitute the knowns (along with their units) into the appropriate equation and obtain numerical solutions complete with units.* That is, do the algebra, calculus, geometry, or arithmetic necessary to find the unknown from the knowns, being sure to carry the units through the calculations. This step is clearly important because it produces the numerical answer, along with its units. Notice, however, that this stage is only one-third of the overall problem-solving process.

Significance

After having done the math in the solution stage of problem solving, it is tempting to think you are done. But, always remember that physics is not math. Rather, in doing physics, we use mathematics as a tool to help us understand

nature. So, after you obtain a numerical answer, you should always assess its significance:

- *Check your units.* If the units of the answer are incorrect, then an error has been made and you should go back over your previous steps to find it. One way to find the mistake is to check all the equations you derived for dimensional consistency. However, be warned that correct units do not guarantee the numerical part of the answer is also correct.
- *Check the answer to see whether it is reasonable. Does it make sense?* This step is extremely important: –the goal of physics is to describe nature accurately. To determine whether the answer is reasonable, check both its magnitude and its sign, in addition to its units. The magnitude should be consistent with a rough estimate of what it should be. It should also compare reasonably with magnitudes of other quantities of the same type. The sign usually tells you about direction and should be consistent with your prior expectations. Your judgment will improve as you solve more physics problems, and it will become possible for you to make finer judgments regarding whether nature is described adequately by the answer to a problem. This step brings the problem back to its conceptual meaning. If you can judge whether the answer is reasonable, you have a deeper understanding of physics than just being able to solve a problem mechanically.
- *Check to see whether the answer tells you something interesting. What does it mean?* This is the flip side of the question: Does it make sense? Ultimately, physics is about understanding nature, and we solve physics problems to learn a little something about how nature operates. Therefore, assuming the answer does make sense, you should always take a moment to see if it tells you something about the world that you find interesting. Even if the answer to this particular problem is not very interesting to you, what about the method you used to solve it? Could the method be adapted to answer a question that you do find interesting? In many ways, it is in answering questions such as these that science progresses.

Chapter Review

Key Terms

accuracy the degree to which a measured value agrees with an accepted reference value for that measurement

base quantity physical quantity chosen by convention and practical considerations such that all other physical quantities can be expressed as algebraic combinations of them

base unit standard for expressing the measurement of a base quantity within a particular system of units; defined by a particular procedure used to measure the corresponding base quantity

conversion factor a ratio that expresses how many of one unit are equal to another unit

derived quantity physical quantity defined using algebraic combinations of base quantities

derived units units that can be calculated using algebraic combinations of the fundamental units

dimension expression of the dependence of a physical quantity on the base quantities as a product of powers of symbols representing the base quantities; in general, the dimension of a quantity has the form $L^a M^b T^c I^d \Theta^e N^f J^g$ for some powers a , b , c , d , e , f , and g .

dimensionally consistent equation in which every term has the same dimensions and the arguments of any mathematical functions appearing in the equation are dimensionless

dimensionless quantity with a dimension of $L^0 M^0 T^0 I^0 \Theta^0 N^0 J^0 = 1$; also called quantity of dimension 1 or a pure number

discrepancy the difference between the measured value and a given standard or expected value

English units system of measurement used in the United States; includes units of measure such as feet, gallons, and pounds

estimation using prior experience and sound physical reasoning to arrive at a rough idea of a quantity's value; sometimes called an "order-of-magnitude approximation," a "guesstimate," a "back-of-the-envelope calculation," or a "Fermi calculation"

kilogram SI unit for mass, abbreviated kg

law description, using concise language or a mathematical formula, of a generalized pattern in nature supported by scientific evidence and repeated experiments

meter SI unit for length, abbreviated m

method of adding percents the percent uncertainty in a quantity calculated by multiplication or division is the sum of the percent uncertainties in the items used to make the calculation.

metric system system in which values can be calculated in factors of 10

model representation of something often too difficult (or impossible) to display directly

order of magnitude the size of a quantity as it relates to a power of 10

percent uncertainty the ratio of the uncertainty of a measurement to the measured value, expressed as a percentage

physical quantity characteristic or property of an object that can be measured or calculated from other measurements

physics science concerned with describing the interactions of energy, matter, space, and time; especially interested in what fundamental mechanisms underlie every phenomenon

precision the degree to which repeated measurements agree with each other

second the SI unit for time, abbreviated s

SI units the international system of units that scientists in most countries have agreed to use; includes units such as meters, liters, and grams

significant figures used to express the precision of a measuring tool used to measure a value

theory testable explanation for patterns in nature supported by scientific evidence and verified multiple times by various groups of researchers

uncertainty a quantitative measure of how much measured values deviate from one another

units standards used for expressing and comparing measurements

Key Equations

$$\text{Percent uncertainty} \quad \text{Percent uncertainty} = \frac{\delta A}{A} \times 100\%$$

Summary

1.1 The Scope and Scale of Physics

- Physics is about trying to find the simple laws that describe all natural phenomena.
- Physics operates on a vast range of scales of length, mass, and time. Scientists use the concept of the order of magnitude of a number to

track which phenomena occur on which scales. They also use orders of magnitude to compare the various scales.

- Scientists attempt to describe the world by formulating models, theories, and laws.

1.2 Units and Standards

- Systems of units are built up from a small number of base units, which are defined by accurate and precise measurements of conventionally chosen base quantities. Other units are then derived as algebraic combinations of the base units.
- Two commonly used systems of units are English units and SI units. All scientists and most of the other people in the world use SI, whereas nonscientists in the United States still tend to use English units.
- The SI base units of length, mass, and time are the meter (m), kilogram (kg), and second (s), respectively.
- SI units are a metric system of units, meaning values can be calculated by factors of 10. Metric prefixes may be used with metric units to scale the base units to sizes appropriate for almost any application.

1.3 Unit Conversion

- To convert a quantity from one unit to another, multiply by conversion factors in such a way that you cancel the units you want to get rid of and introduce the units you want to end up with.
- Be careful with areas and volumes. Units obey the rules of algebra so, for example, if a unit is squared we need two factors to cancel it.

1.4 Dimensional Analysis

- The dimension of a physical quantity is just an expression of the base quantities from which it is derived.
- All equations expressing physical laws or principles must be dimensionally consistent. This fact can be used as an aid in remembering physical laws, as a way to check whether claimed relationships between physical quantities are possible, and even to derive new physical laws.

1.5 Estimates and Fermi Calculations

- An estimate is a rough educated guess at the

value of a physical quantity based on prior experience and sound physical reasoning. Some strategies that may help when making an estimate are as follows:

- Get big lengths from smaller lengths.
- Get areas and volumes from lengths.
- Get masses from volumes and densities.
- If all else fails, bound it.
- One “sig. fig.” is fine.
- Ask yourself: Does this make any sense?

1.6 Significant Figures

- Accuracy of a measured value refers to how close a measurement is to an accepted reference value. The discrepancy in a measurement is the amount by which the measurement result differs from this value.
- Precision of measured values refers to how close the agreement is between repeated measurements. The uncertainty of a measurement is a quantification of this.
- The precision of a measuring tool is related to the size of its measurement increments. The smaller the measurement increment, the more precise the tool.
- Significant figures express the precision of a measuring tool.
- When multiplying or dividing measured values, the final answer can contain only as many significant figures as the value with the least number of significant figures.
- When adding or subtracting measured values, the final answer cannot contain more decimal places than the least-precise value.

1.7 Solving Problems in Physics

The three stages of the process for solving physics problems used in this book are as follows:

- *Strategy*: Determine which physical principles are involved and develop a strategy for using them to solve the problem.
- *Solution*: Do the math necessary to obtain a numerical solution complete with units.
- *Significance*: Check the solution to make sure it makes sense (correct units, reasonable magnitude and sign) and assess its significance.

Conceptual Questions

1.1 The Scope and Scale of Physics

1. What is physics?
2. Some have described physics as a “search for simplicity.” Explain why this might be an appropriate description.
3. If two different theories describe experimental observations equally well, can one be said to be more valid than the other (assuming both use accepted rules of logic)?
4. What determines the validity of a theory?
5. Certain criteria must be satisfied if a measurement or observation is to be believed. Will the criteria necessarily be as strict for an expected result as for an unexpected result?
6. Can the validity of a model be limited or must it be universally valid? How does this compare with the required validity of a theory or a law?

1.2 Units and Standards

7. Identify some advantages of metric units.
8. What are the SI base units of length, mass, and time?
9. What is the difference between a base unit and a derived unit? (b) What is the difference between a base quantity and a derived quantity? (c) What

is the difference between a base quantity and a base unit?

10. For each of the following scenarios, refer to [Figure 1.4](#) and [Table 1.2](#) to determine which metric prefix on the meter is most appropriate for each of the following scenarios. (a) You want to tabulate the mean distance from the Sun for each planet in the solar system. (b) You want to compare the sizes of some common viruses to design a mechanical filter capable of blocking the pathogenic ones. (c) You want to list the diameters of all the elements on the periodic table. (d) You want to list the distances to all the stars that have now received any radio broadcasts sent from Earth 10 years ago.

1.6 Significant Figures

11. (a) What is the relationship between the precision and the uncertainty of a measurement? (b) What is the relationship between the accuracy and the discrepancy of a measurement?

1.7 Solving Problems in Physics

12. What information do you need to choose which equation or equations to use to solve a problem?
13. What should you do after obtaining a numerical answer when solving a problem?

Problems

1.1 The Scope and Scale of Physics

14. Find the order of magnitude of the following physical quantities. (a) The mass of Earth’s atmosphere: 5.1×10^{18} kg; (b) The mass of the Moon’s atmosphere: 25,000 kg; (c) The mass of Earth’s hydrosphere: 1.4×10^{21} kg; (d) The mass of Earth: 5.97×10^{24} kg; (e) The mass of the Moon: 7.34×10^{22} kg; (f) The Earth–Moon distance (semimajor axis): 3.84×10^8 m; (g) The mean Earth–Sun distance: 1.5×10^{11} m; (h) The equatorial radius of Earth: 6.38×10^6 m; (i) The mass of an electron: 9.11×10^{-31} kg; (j) The mass of a proton: 1.67×10^{-27} kg; (k) The mass of the Sun: 1.99×10^{30} kg.
15. Use the orders of magnitude you found in the previous problem to answer the following questions to within an order of magnitude. (a) How many electrons would it take to equal the

mass of a proton? (b) How many Earths would it take to equal the mass of the Sun? (c) How many Earth–Moon distances would it take to cover the distance from Earth to the Sun? (d) How many Moon atmospheres would it take to equal the mass of Earth’s atmosphere? (e) How many moons would it take to equal the mass of Earth? (f) How many protons would it take to equal the mass of the Sun?

For the remaining questions, you need to use [Figure 1.4](#) to obtain the necessary orders of magnitude of lengths, masses, and times.

16. Roughly how many heartbeats are there in a lifetime?
17. A generation is about one-third of a lifetime. Approximately how many generations have passed since the year 0?
18. Roughly how many times longer than the mean

life of an extremely unstable atomic nucleus is the lifetime of a human?

19. Calculate the approximate number of atoms in a bacterium. Assume the average mass of an atom in the bacterium is 10 times the mass of a proton.
20. (a) Calculate the number of cells in a hummingbird assuming the mass of an average cell is 10 times the mass of a bacterium. (b) Making the same assumption, how many cells are there in a human?
21. Assuming one nerve impulse must end before another can begin, what is the maximum firing rate of a nerve in impulses per second?
22. About how many floating-point operations can a supercomputer perform each year?
23. Roughly how many floating-point operations can a supercomputer perform in a human lifetime?

1.2 Units and Standards

24. The following times are given using metric prefixes on the base SI unit of time: the second. Rewrite them in scientific notation without the prefix. For example, 47 Ts would be rewritten as 4.7×10^{13} s. (a) 980 Ps; (b) 980 fs; (c) 17 ns; (d) 577 μ s.
25. The following times are given in seconds. Use metric prefixes to rewrite them so the numerical value is greater than one but less than 1000. For example, 7.9×10^{-2} s could be written as either 7.9 cs or 79 ms. (a) 9.57×10^5 s; (b) 0.045 s; (c) 5.5×10^{-7} s; (d) 3.16×10^7 s.
26. The following lengths are given using metric prefixes on the base SI unit of length: the meter. Rewrite them in scientific notation without the prefix. For example, 4.2 Pm would be rewritten as 4.2×10^{15} m. (a) 89 Tm; (b) 89 pm; (c) 711 mm; (d) 0.45 μ m.
27. The following lengths are given in meters. Use metric prefixes to rewrite them so the numerical value is bigger than one but less than 1000. For example, 7.9×10^{-2} m could be written either as 7.9 cm or 79 mm. (a) 7.59×10^7 m; (b) 0.0074 m; (c) 8.8×10^{-11} m; (d) 1.63×10^{13} m.
28. The following masses are written using metric prefixes on the gram. Rewrite them in scientific notation in terms of the SI base unit of mass: the kilogram. For example, 40 Mg would be written as 4×10^4 kg. (a) 23 mg; (b) 320 Tg; (c) 42 ng; (d) 7 g; (e) 9 Pg.
29. The following masses are given in kilograms. Use metric prefixes on the gram to rewrite them so the numerical value is bigger than one but less than 1000. For example, 7×10^{-4} kg could be written as 70 cg or 700 mg. (a) 3.8×10^{-5} kg; (b) 2.3×10^{17} kg; (c) 2.4×10^{-11} kg; (d) 8×10^{15} kg; (e) 4.2×10^{-3} kg.

1.3 Unit Conversion

30. The volume of Earth is on the order of 10^{21} m³. (a) What is this in cubic kilometers (km³)? (b) What is it in cubic miles (mi³)? (c) What is it in cubic centimeters (cm³)?
31. The speed limit on some interstate highways is roughly 100 km/h. (a) What is this in meters per second? (b) How many miles per hour is this?
32. A car is traveling at a speed of 33 m/s. (a) What is its speed in kilometers per hour? (b) Is it exceeding the 90 km/h speed limit?
33. In SI units, speeds are measured in meters per second (m/s). But, depending on where you live, you're probably more comfortable of thinking of speeds in terms of either kilometers per hour (km/h) or miles per hour (mi/h). In this problem, you will see that 1 m/s is roughly 4 km/h or 2 mi/h, which is handy to use when developing your physical intuition. More precisely, show that (a) $1.0 \text{ m/s} = 3.6 \text{ km/h}$ and (b) $1.0 \text{ m/s} = 2.2 \text{ mi/h}$.
34. American football is played on a 100-yd-long field, excluding the end zones. How long is the field in meters? (Assume that 1 m = 3.281 ft.)
35. Soccer fields vary in size. A large soccer field is 115 m long and 85.0 m wide. What is its area in square feet? (Assume that 1 m = 3.281 ft.)
36. What is the height in meters of a person who is 6 ft 1.0 in. tall?
37. Mount Everest, at 29,028 ft, is the tallest mountain on Earth. What is its height in kilometers? (Assume that 1 m = 3.281 ft.)
38. The speed of sound is measured to be 342 m/s on a certain day. What is this measurement in kilometers per hour?
39. Tectonic plates are large segments of Earth's crust that move slowly. Suppose one such plate has an average speed of 4.0 cm/yr. (a) What

distance does it move in 1.0 s at this speed? (b) What is its speed in kilometers per million years?

40. The average distance between Earth and the Sun is 1.5×10^{11} m. (a) Calculate the average speed of Earth in its orbit (assumed to be circular) in meters per second. (b) What is this speed in miles per hour?
41. The density of nuclear matter is about 10^{18} kg/m³. Given that 1 mL is equal in volume to cm³, what is the density of nuclear matter in megagrams per microliter (that is, Mg/ μ L)?
42. The density of aluminum is 2.7 g/cm³. What is the density in kilograms per cubic meter?
43. A commonly used unit of mass in the English system is the pound-mass, abbreviated lbm, where 1 lbm = 0.454 kg. What is the density of water in pound-mass per cubic foot?
44. A furlong is 220 yd. A fortnight is 2 weeks. Convert a speed of one furlong per fortnight to millimeters per second.
45. It takes 2π radians (rad) to get around a circle, which is the same as 360°. How many radians are in 1°?
46. Light travels a distance of about 3×10^8 m/s. A light-minute is the distance light travels in 1 min. If the Sun is 1.5×10^{11} m from Earth, how far away is it in light-minutes?
47. A light-nanosecond is the distance light travels in 1 ns. Convert 1 ft to light-nanoseconds.
48. An electron has a mass of 9.11×10^{-31} kg. A proton has a mass of 1.67×10^{-27} kg. What is the mass of a proton in electron-masses?
49. A fluid ounce is about 30 mL. What is the volume of a 12 fl-oz can of soda pop in cubic meters?

1.4 Dimensional Analysis

50. A student is trying to remember some formulas from geometry. In what follows, assume A is area, V is volume, and all other variables are lengths. Determine which formulas are dimensionally consistent. (a) $V = \pi r^2 h$; (b) $A = 2\pi r^2 + 2\pi rh$; (c) $V = 0.5bh$; (d) $V = \pi d^2$; (e) $V = \pi d^3/6$.
51. Consider the physical quantities s , v , a , and t with dimensions $[s] = L$, $[v] = LT^{-1}$, $[a] = LT^{-2}$, and $[t] = T$. Determine whether each of the following equations is dimensionally

consistent. (a) $v^2 = 2as$; (b) $s = vt^2 + 0.5at^2$; (c) $v = s/t$; (d) $a = v/t$.

52. Consider the physical quantities m , s , v , a , t , and r with dimensions $[m] = M$, $[s] = L$, $[v] = LT^{-1}$, $[a] = LT^{-2}$, $[t] = T$, and $[r] = L$. Assuming each of the following equations is dimensionally consistent, find the dimension of the quantity on the left-hand side of the equation: (a) $F = ma$; (b) $K = 0.5mv^2$; (c) $p = mv$; (d) $W = mas$; (e) $L = mvr$.
53. Suppose quantity s is a length and quantity t is a time. Suppose the quantities v and a are defined by $v = ds/dt$ and $a = dv/dt$. (a) What is the dimension of v ? (b) What is the dimension of the quantity a ? What are the dimensions of (c) $\int v dt$, (d) $\int a dt$, and (e) da/dt ?
54. Suppose $[V] = L^3$, $[\rho] = ML^{-3}$, and $[t] = T$. (a) What is the dimension of $\int \rho dV$? (b) What is the dimension of dV/dt ? (c) What is the dimension of $\rho(dV/dt)$?
55. The arc length formula says the length s of arc subtended by angle θ in a circle of radius r is given by the equation $s = r\theta$. What are the dimensions of (a) s , (b) r , and (c) θ ?

1.5 Estimates and Fermi Calculations

56. Assuming the human body is made primarily of water, estimate the volume of a person.
57. Assuming the human body is primarily made of water, estimate the number of molecules in it. (Note that water has a molecular mass of 18 g/mol and there are roughly 10^{24} atoms in a mole.)
58. Estimate the mass of air in a classroom.
59. Estimate the number of molecules that make up Earth, assuming an average molecular mass of 30 g/mol. (Note there are on the order of 10^{24} objects per mole.)
60. Estimate the surface area of a person.
61. Roughly how many solar systems would it take to tile the disk of the Milky Way?
62. (a) Estimate the density of the Moon. (b) Estimate the diameter of the Moon. (c) Given that the Moon subtends at an angle of about half a degree in the sky, estimate its distance from

Earth.

- 63.** The average density of the Sun is on the order 10^3 kg/m^3 . (a) Estimate the diameter of the Sun. (b) Given that the Sun subtends at an angle of about half a degree in the sky, estimate its distance from Earth.
- 64.** Estimate the mass of a virus.
- 65.** A floating-point operation is a single arithmetic operation such as addition, subtraction, multiplication, or division. (a) Estimate the maximum number of floating-point operations a human being could possibly perform in a lifetime. (b) How long would it take a supercomputer to perform that many floating-point operations?

1.6 Significant Figures

- 66.** Consider the equation $4000/400 = 10.0$. Assuming the number of significant figures in the answer is correct, what can you say about the number of significant figures in 4000 and 400?
- 67.** Suppose your bathroom scale reads your mass as 65 kg with a 3% uncertainty. What is the uncertainty in your mass (in kilograms)?
- 68.** A good-quality measuring tape can be off by 0.50 cm over a distance of 20 m. What is its percent uncertainty?
- 69.** An infant's pulse rate is measured to be 130 ± 5 beats/min. What is the percent uncertainty in this measurement?
- 70.** (a) Suppose that a person has an average heart rate of 72.0 beats/min. How many beats do they have in 2.0 years? (b) In 2.00 years? (c) In 2.000 years?
- 71.** A can contains 375 mL of soda. How much is left after 308 mL is removed?
- 72.** State how many significant figures are proper in the results of the following calculations: (a) $(106.7)(98.2)/(46.210)(1.01)$; (b) $(18.7)^2$; (c)

$$(1.60 \times 10^{-19})(3712)$$

- 73.** (a) How many significant figures are in the numbers 99 and 100.? (b) If the uncertainty in each number is 1, what is the percent uncertainty in each? (c) Which is a more meaningful way to express the accuracy of these two numbers: significant figures or percent uncertainties?
- 74.** (a) If your speedometer has an uncertainty of 2.0 km/h at a speed of 90 km/h, what is the percent uncertainty? (b) If it has the same percent uncertainty when it reads 60 km/h, what is the range of speeds you could be going?
- 75.** (a) A person's blood pressure is measured to be 120 ± 2 mm Hg. What is its percent uncertainty? (b) Assuming the same percent uncertainty, what is the uncertainty in a blood pressure measurement of 80 mm Hg?
- 76.** A person measures their heart rate by counting the number of beats in 30 s. If 40 ± 1 beats are counted in 30.0 ± 0.5 s, what is the heart rate and its uncertainty in beats per minute?
- 77.** What is the area of a circle 3.102 cm in diameter?
- 78.** Determine the number of significant figures in the following measurements: (a) 0.0009, (b) 15,450.0, (c) 6×10^3 , (d) 87.990, and (e) 30.42.
- 79.** Perform the following calculations and express your answer using the correct number of significant digits. (a) A woman has two bags weighing 13.5 lb and one bag with a weight of 10.2 lb. What is the total weight of the bags? (b) The force F on an object is equal to its mass m multiplied by its acceleration a . If a wagon with mass 55 kg accelerates at a rate of 0.0255 m/s^2 , what is the force on the wagon? (The unit of force is called the *newton* and it is expressed with the symbol N.)

Additional Problems

- 80.** Consider the equation $y = mt + b$, where the dimension of y is length and the dimension of t is time, and m and b are constants. What are the dimensions and SI units of (a) m and (b) b ?
- 81.** Consider the equation $s = s_0 + v_0 t + a_0 t^2/2 + j_0 t^3/6 + S_0 t^4/24 + ct^5/120$, where s is a length and t is a time. What are the

dimensions and SI units of (a) s_0 , (b) v_0 , (c) a_0 , (d) j_0 , (e) S_0 , and (f) c ?

- 82.** (a) A car speedometer has a 5% uncertainty. What is the range of possible speeds when it reads 90 km/h? (b) Convert this range to miles per hour. Note 1 km = 0.6214 mi.
- 83.** A marathon runner completes a 42.188-km

course in 2 h, 30 min, and 12 s. There is an uncertainty of 25 m in the distance traveled and an uncertainty of 1 s in the elapsed time. (a) Calculate the percent uncertainty in the distance. (b) Calculate the percent uncertainty in the elapsed time. (c) What is the average speed in meters per second? (d) What is the uncertainty in the average speed?

- 84.** The sides of a small rectangular box are measured to be 1.80 ± 0.1 cm, 2.05 ± 0.02 cm, and 3.1 ± 0.1 cm long. Calculate its volume and uncertainty in cubic centimeters.
- 85.** When nonmetric units were used in the United Kingdom, a unit of mass called the pound-mass (lbm) was used, where $1 \text{ lbm} = 0.4539 \text{ kg}$. (a) If there is an uncertainty of 0.0001 kg in the

pound-mass unit, what is its percent uncertainty? (b) Based on that percent uncertainty, what mass in pound-mass has an uncertainty of 1 kg when converted to kilograms?

- 86.** The length and width of a rectangular room are measured to be 3.955 ± 0.005 m and 3.050 ± 0.005 m. Calculate the area of the room and its uncertainty in square meters.
- 87.** A car engine moves a piston with a circular cross-section of 7.500 ± 0.002 cm in diameter a distance of 3.250 ± 0.001 cm to compress the gas in the cylinder. (a) By what amount is the gas decreased in volume in cubic centimeters? (b) Find the uncertainty in this volume.

Challenge Problems

- 88.** The first atomic bomb was detonated on July 16, 1945, at the Trinity test site about 200 mi south of Los Alamos. In 1947, the U.S. government declassified a film reel of the explosion. From this film reel, British physicist G. I. Taylor was able to determine the rate at which the radius of the fireball from the blast grew. Using dimensional analysis, he was then able to deduce the amount of energy released in the explosion, which was a closely guarded secret at the time. Because of this, Taylor did not publish his results until 1950. This problem challenges you to recreate this famous calculation. (a) Using keen physical insight developed from years of experience, Taylor decided the radius r of the fireball should depend only on the following variables:

t , the time since the explosion,

ρ , and the density of the air,

E , the energy of the initial explosion.

Thus, he made the educated guess that

$r = kE^a \rho^b t^c$ for some dimensionless constant k and some unknown exponents a , b , and c . Given that $[E] = \text{ML}^2\text{T}^{-2}$, determine the values of the exponents necessary to make this equation dimensionally consistent. (*Hint:* Notice the equation implies that $k = rE^{-a} \rho^{-b} t^{-c}$ and that $[k] = 1$.) (b) By analyzing data from high-energy conventional explosives, Taylor found the formula he derived seemed to be valid as long as the constant k had the value 1.03. From the film reel, he was able to determine many values of r and the corresponding values of t . For example, he found that after 25.0 ms, the fireball had a

radius of 130.0 m. Use these values, along with an average air density of 1.25 kg/m^3 , to calculate the initial energy release of the Trinity detonation in joules (J). (*Hint:* To get energy in joules, you need to make sure all the numbers you substitute in are expressed in terms of SI base units.) (c) The energy released in large explosions is often cited in units of “tons of TNT” (abbreviated “t TNT”), where 1 t TNT is about 4.2 GJ. Convert your answer to (b) into kilotons of TNT (that is, kt TNT). Compare your answer with the quick-and-dirty estimate of 10 kt TNT made by physicist Enrico Fermi shortly after witnessing the explosion from what was thought to be a safe distance. (Reportedly, Fermi made his estimate by dropping some shredded bits of paper right before the remnants of the shock wave hit him and looked to see how far they were carried by it.)

- 89.** The purpose of this problem is to show the entire concept of dimensional consistency can be summarized by the old saying “You can’t add apples and oranges.” If you have studied power series expansions in a calculus course, you know the standard mathematical functions such as trigonometric functions, logarithms, and exponential functions can be expressed as infinite sums of the form

$$\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots,$$

where the a_n are dimensionless constants for all $n = 0, 1, 2, \dots$ and x is the argument of the

function. (If you have not studied power series in calculus yet, just trust us.) Use this fact to explain why the requirement that all terms in an equation have the same dimensions is sufficient as a definition of dimensional consistency. That is, it actually implies the arguments of standard

mathematical functions must be dimensionless, so it is not really necessary to make this latter condition a separate requirement of the definition of dimensional consistency as we have done in this section.

CHAPTER 2

Vectors



FIGURE 2.1 A signpost gives information about distances and directions to towns or to other locations relative to the location of the signpost. Distance is a scalar quantity. Knowing the distance alone is not enough to get to the town; we must also know the direction from the signpost to the town. The direction, together with the distance, is a vector quantity commonly called the displacement vector. A signpost, therefore, gives information about displacement vectors from the signpost to towns. (credit: modification of work by "studio tdes"/Flickr, thedailyenglishshow.com)

CHAPTER OUTLINE

2.1 Scalars and Vectors

2.2 Coordinate Systems and Components of a Vector

2.3 Algebra of Vectors

2.4 Products of Vectors

INTRODUCTION Vectors are essential to physics and engineering. Many fundamental physical quantities are vectors, including displacement, velocity, force, and electric and magnetic vector fields. Scalar products of vectors define other fundamental scalar physical quantities, such as energy. Vector products of vectors define still other fundamental vector physical quantities, such as torque and angular momentum. In other words, vectors are a component part of physics in much the same way as sentences are a component part of literature.

In introductory physics, vectors are Euclidean quantities that have geometric representations as arrows in one dimension (in a line), in two dimensions (in a plane), or in three dimensions (in space). They can be added, subtracted, or multiplied. In this chapter, we explore elements of vector algebra for applications in mechanics and in electricity and magnetism. Vector operations also have numerous generalizations in other branches of physics.

2.1 Scalars and Vectors

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Describe the difference between vector and scalar quantities.
- Identify the magnitude and direction of a vector.
- Explain the effect of multiplying a vector quantity by a scalar.
- Describe how one-dimensional vector quantities are added or subtracted.
- Explain the geometric construction for the addition or subtraction of vectors in a plane.
- Distinguish between a vector equation and a scalar equation.

Many familiar physical quantities can be specified completely by giving a single number and the appropriate unit. For example, “a class period lasts 50 min” or “the gas tank in my car holds 65 L” or “the distance between two posts is 100 m.” A physical quantity that can be specified completely in this manner is called a **scalar quantity**. Scalar is a synonym of “number.” Time, mass, distance, length, volume, temperature, and energy are examples of **scalar** quantities.

Scalar quantities that have the same physical units can be added or subtracted according to the usual rules of algebra for numbers. For example, a class ending 10 min earlier than 50 min lasts $50 \text{ min} - 10 \text{ min} = 40 \text{ min}$. Similarly, a 60-cal serving of corn followed by a 200-cal serving of donuts gives $60 \text{ cal} + 200 \text{ cal} = 260 \text{ cal}$ of energy. When we multiply a scalar quantity by a number, we obtain the same scalar quantity but with a larger (or smaller) value. For example, if yesterday’s breakfast had 200 cal of energy and today’s breakfast has four times as much energy as it had yesterday, then today’s breakfast has $4(200 \text{ cal}) = 800 \text{ cal}$ of energy. Two scalar quantities can also be multiplied or divided by each other to form a derived scalar quantity. For example, if a train covers a distance of 100 km in 1.0 h, its speed is $100.0 \text{ km}/1.0 \text{ h} = 27.8 \text{ m/s}$, where the speed is a derived scalar quantity obtained by dividing distance by time.

Many physical quantities, however, cannot be described completely by just a single number of physical units. For example, when the U.S. Coast Guard dispatches a ship or a helicopter for a rescue mission, the rescue team must know not only the distance to the distress signal, but also the direction from which the signal is coming so they can get to its origin as quickly as possible. Physical quantities specified completely by giving a number of units (magnitude) and a direction are called **vector quantities**. Examples of vector quantities include displacement, velocity, position, force, and torque. In the language of mathematics, physical vector quantities are represented by mathematical objects called **vectors** (Figure 2.2). We can add or subtract two vectors, and we can multiply a vector by a scalar or by another vector, but we cannot divide by a vector. The operation of division by a vector is not defined.

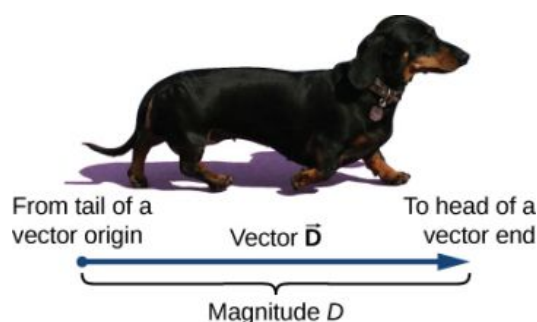


FIGURE 2.2 We draw a vector from the initial point or origin (called the “tail” of a vector) to the end or terminal point (called the “head” of a vector), marked by an arrowhead. Magnitude is the length of a vector and is always a positive scalar quantity. (credit “photo”: modification of work by Cate Sevilla)

Let’s examine vector algebra using a graphical method to be aware of basic terms and to develop a qualitative understanding. In practice, however, when it comes to solving physics problems, we use analytical methods, which we’ll see in the next section. Analytical methods are more simple computationally and more accurate than graphical methods. From now on, to distinguish between a vector and a scalar quantity, we adopt the common convention that a letter in bold type with an arrow above it denotes a vector, and a letter without an arrow denotes a scalar. For example, a distance of 2.0 km, which is a scalar quantity, is denoted by $d = 2.0 \text{ km}$, whereas a displacement of 2.0

km in some direction, which is a vector quantity, is denoted by $\vec{d}$.

Suppose you tell a friend on a camping trip that you have discovered a terrific fishing hole 6 km from your tent. It is unlikely your friend would be able to find the hole easily unless you also communicate the direction in which it can be found with respect to your campsite. You may say, for example, “Walk about 6 km northeast from my tent.” The key concept here is that you have to give not one but *two* pieces of information—namely, the distance or magnitude (6 km) *and* the direction (northeast).

Displacement is a general term used to describe a *change in position*, such as during a trip from the tent to the fishing hole. Displacement is an example of a vector quantity. If you walk from the tent (location A) to the hole (location B), as shown in [Figure 2.3](#), the vector $\vec{D}$, representing your **displacement**, is drawn as the arrow that originates at point A and ends at point B . The arrowhead marks the end of the vector. The direction of the displacement vector $\vec{D}$ is the direction of the arrow. The length of the arrow represents the **magnitude** D of vector $\vec{D}$. Here, $D = 6$ km. Since the magnitude of a vector is its length, which is a positive number, the magnitude is also indicated by placing the absolute value notation around the symbol that denotes the vector; so, we can write equivalently that $D \equiv |\vec{D}|$. To solve a vector problem graphically, we need to draw the vector $\vec{D}$ to scale. For example, if we assume 1 unit of distance (1 km) is represented in the drawing by a line segment of length $u = 2$ cm, then the total displacement in this example is represented by a vector of length $d = 6u = 6(2 \text{ cm}) = 12$ cm, as shown in [Figure 2.4](#). Notice that here, to avoid confusion, we used $D = 6$ km to denote the magnitude of the actual displacement and $d = 12$ cm to denote the length of its representation in the drawing.

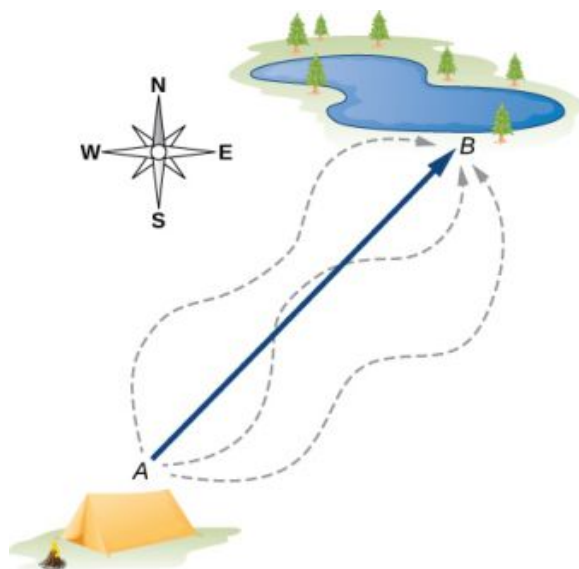


FIGURE 2.3 The displacement vector from point A (the initial position at the campsite) to point B (the final position at the fishing hole) is indicated by an arrow with origin at point A and end at point B . The displacement is the same for any of the actual paths (dashed curves) that may be taken between points A and B .

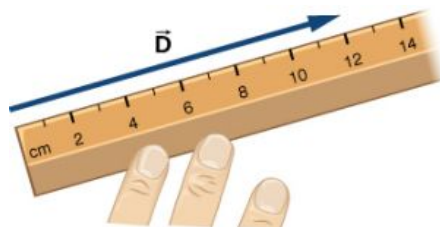


FIGURE 2.4 A displacement $\vec{D}$ of magnitude 6 km is drawn to scale as a vector of length 12 cm when the length of 2 cm represents 1 unit of displacement (which in this case is 1 km).

Suppose your friend walks from the campsite at A to the fishing pond at B and then walks back: from the fishing pond at B to the campsite at A . The magnitude of the displacement vector $\vec{D}_{AB}$ from A to B is the same as the magnitude of the displacement vector $\vec{D}_{BA}$ from B to A (it equals 6 km in both cases), so we can write

$D_{AB} = D_{BA}$. However, vector $\vec{D}_{AB}$ is *not* equal to vector $\vec{D}_{BA}$ because these two vectors have different directions: $\vec{D}_{AB} \neq \vec{D}_{BA}$. In [Figure 2.3](#), vector $\vec{D}_{BA}$ would be represented by a vector with an origin at point B and an end at point A , indicating vector $\vec{D}_{BA}$ points to the southwest, which is exactly 180° opposite to the direction of vector $\vec{D}_{AB}$. We say that vector $\vec{D}_{BA}$ is **antiparallel** to vector $\vec{D}_{AB}$ and write $\vec{D}_{AB} = -\vec{D}_{BA}$, where the minus sign indicates the antiparallel direction.

Two vectors that have identical directions are said to be **parallel vectors**—meaning, they are *parallel* to each other. Two parallel vectors $\vec{A}$ and $\vec{B}$ are equal, denoted by $\vec{A} = \vec{B}$, if and only if they have equal magnitudes $|\vec{A}| = |\vec{B}|$. Two vectors with directions perpendicular to each other are said to be **orthogonal vectors**. These relations between vectors are illustrated in [Figure 2.5](#).

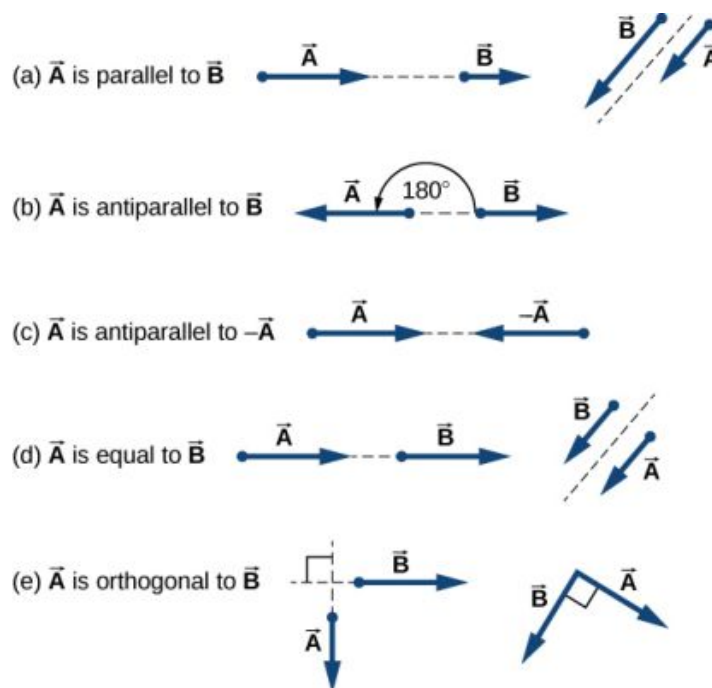


FIGURE 2.5 Various relations between two vectors $\vec{A}$ and $\vec{B}$. (a) $\vec{A} \neq \vec{B}$ because $A \neq B$. (b) $\vec{A} \neq \vec{B}$ because they are not parallel and $A \neq B$. (c) $\vec{A} \neq -\vec{A}$ because they have different directions (even though $|\vec{A}| = |-\vec{A}| = A$). (d) $\vec{A} = \vec{B}$ because they are parallel *and* have identical magnitudes $A = B$. (e) $\vec{A} \neq \vec{B}$ because they have different directions (are not parallel); here, their directions differ by 90° —meaning, they are orthogonal.

✓ CHECK YOUR UNDERSTANDING 2.1

Two motorboats named *Alice* and *Bob* are moving on a lake. Given the information about their velocity vectors in each of the following situations, indicate whether their velocity vectors are equal or otherwise. (a) *Alice* moves north at 6 knots and *Bob* moves west at 6 knots. (b) *Alice* moves west at 6 knots and *Bob* moves west at 3 knots. (c) *Alice* moves northeast at 6 knots and *Bob* moves south at 3 knots. (d) *Alice* moves northeast at 6 knots and *Bob* moves southwest at 6 knots. (e) *Alice* moves northeast at 2 knots and *Bob* moves closer to the shore northeast at 2 knots.

Algebra of Vectors in One Dimension

Vectors can be multiplied by scalars, added to other vectors, or subtracted from other vectors. We can illustrate these vector concepts using an example of the fishing trip seen in [Figure 2.6](#).

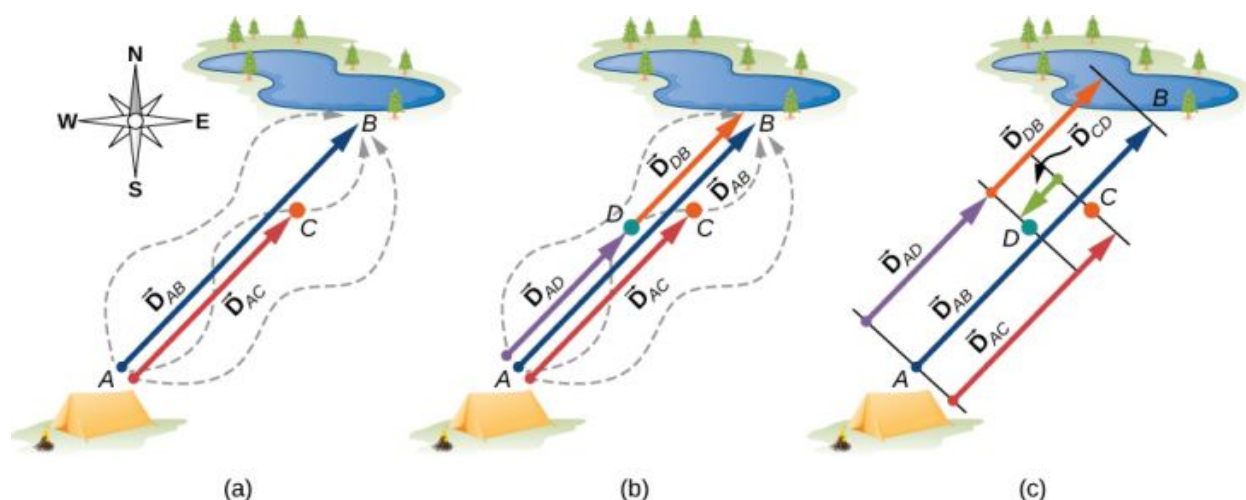


FIGURE 2.6 Displacement vectors for a fishing trip. (a) Stopping to rest at point C while walking from camp (point A) to the pond (point B). (b) Going back for the dropped tackle box (point D). (c) Finishing up at the fishing pond.

Suppose your friend departs from point A (the campsite) and walks in the direction to point B (the fishing pond), but, along the way, stops to rest at some point C located three-quarters of the distance between A and B, beginning from point A (Figure 2.6(a)). What is his displacement vector $\vec{D}_{AC}$ when he reaches point C? We know that if he walks all the way to B, his displacement vector relative to A is $\vec{D}_{AB}$, which has magnitude $D_{AB} = 6$ km and a direction of northeast. If he walks only a 0.75 fraction of the total distance, maintaining the northeasterly direction, at point C he must be $0.75D_{AB} = 4.5$ km away from the campsite at A. So, his displacement vector at the rest point C has magnitude $D_{AC} = 4.5$ km $= 0.75D_{AB}$ and is parallel to the displacement vector $\vec{D}_{AB}$. All of this can be stated succinctly in the form of the following **vector equation**:

$$\vec{D}_{AC} = 0.75\vec{D}_{AB}.$$

In a vector equation, both sides of the equation are vectors. The previous equation is an example of a vector multiplied by a positive scalar (number) $\alpha = 0.75$. The result, $\vec{D}_{AC}$, of such a multiplication is a new vector with a direction parallel to the direction of the original vector $\vec{D}_{AB}$.

In general, when a vector $\vec{A}$ is multiplied by a *positive* scalar α , the result is a new vector $\vec{B}$ that is *parallel* to $\vec{A}$:

$$\vec{B} = \alpha\vec{A}. \quad 2.1$$

The magnitude $|\vec{B}|$ of this new vector is obtained by multiplying the magnitude $|\vec{A}|$ of the original vector, as expressed by the **scalar equation**:

$$B = |\alpha|A. \quad 2.2$$

In a scalar equation, both sides of the equation are numbers. Equation 2.2 is a scalar equation because the magnitudes of vectors are scalar quantities (and positive numbers). If the scalar α is *negative* in the vector equation Equation 2.1, then the magnitude $|\vec{B}|$ of the new vector is still given by Equation 2.2, but the direction of the new vector $\vec{B}$ is *antiparallel* to the direction of $\vec{A}$. These principles are illustrated in Figure 2.7(a) by two examples where the length of vector $\vec{A}$ is 1.5 units. When $\alpha = 2$, the new vector $\vec{B} = 2\vec{A}$ has length $B = 2A = 3.0$ units (twice as long as the original vector) and is parallel to the original vector. When $\alpha = -2$, the new vector $\vec{C} = -2\vec{A}$ has length $C = |-2|A = 3.0$ units (twice as long as the original vector) and is antiparallel to the original vector.

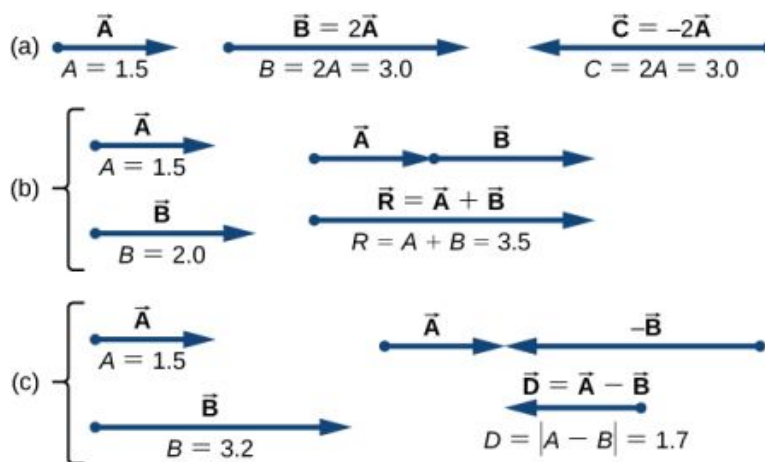


FIGURE 2.7 Algebra of vectors in one dimension. (a) Multiplication by a scalar. (b) Addition of two vectors ($\vec{R}$ is called the *resultant* of vectors $\vec{A}$ and $\vec{B}$). (c) Subtraction of two vectors ($\vec{D}$ is the difference of vectors $\vec{A}$ and $\vec{B}$).

Now suppose your fishing buddy departs from point A (the campsite), walking in the direction to point B (the fishing hole), but he realizes he lost his tackle box when he stopped to rest at point C (located three-quarters of the distance between A and B , beginning from point A). So, he turns back and retraces his steps in the direction toward the campsite and finds the box lying on the path at some point D only 1.2 km away from point C (see [Figure 2.6\(b\)](#)). What is his displacement vector $\vec{D}_{AD}$ when he finds the box at point D ? What is his displacement vector $\vec{D}_{DB}$ from point D to the hole? We have already established that at rest point C his displacement vector is $\vec{D}_{AC} = 0.75\vec{D}_{AB}$. Starting at point C , he walks southwest (toward the campsite), which means his new displacement vector $\vec{D}_{CD}$ from point C to point D is antiparallel to $\vec{D}_{AB}$. Its magnitude $|\vec{D}_{CD}|$ is $D_{CD} = 1.2 \text{ km} = 0.2D_{AB}$, so his second displacement vector is $\vec{D}_{CD} = -0.2\vec{D}_{AB}$. His total displacement $\vec{D}_{AD}$ relative to the campsite is the **vector sum** of the two displacement vectors: vector $\vec{D}_{AC}$ (from the campsite to the rest point) and vector $\vec{D}_{CD}$ (from the rest point to the point where he finds his box):

$$\vec{D}_{AD} = \vec{D}_{AC} + \vec{D}_{CD}. \quad 2.3$$

The vector sum of two (or more) vectors is called the **resultant vector** or, for short, the *resultant*. When the vectors on the right-hand-side of [Equation 2.3](#) are known, we can find the resultant $\vec{D}_{AD}$ as follows:

$$\vec{D}_{AD} = \vec{D}_{AC} + \vec{D}_{CD} = 0.75\vec{D}_{AB} - 0.2\vec{D}_{AB} = (0.75 - 0.2)\vec{D}_{AB} = 0.55\vec{D}_{AB}. \quad 2.4$$

When your friend finally reaches the pond at B , his displacement vector $\vec{D}_{AB}$ from point A is the vector sum of his displacement vector $\vec{D}_{AD}$ from point A to point D and his displacement vector $\vec{D}_{DB}$ from point D to the fishing hole: $\vec{D}_{AB} = \vec{D}_{AD} + \vec{D}_{DB}$ (see [Figure 2.6\(c\)](#)). This means his displacement vector $\vec{D}_{DB}$ is the **difference of two vectors**:

$$\vec{D}_{DB} = \vec{D}_{AB} - \vec{D}_{AD} = \vec{D}_{AB} + (-\vec{D}_{AD}). \quad 2.5$$

Notice that a difference of two vectors is nothing more than a vector sum of two vectors because the second term in [Equation 2.5](#) is vector $-\vec{D}_{AD}$ (which is antiparallel to $\vec{D}_{AD}$). When we substitute [Equation 2.4](#) into [Equation 2.5](#), we obtain the second displacement vector:

$$\vec{D}_{DB} = \vec{D}_{AB} - \vec{D}_{AD} = \vec{D}_{AB} - 0.55\vec{D}_{AB} = (1.0 - 0.55)\vec{D}_{AB} = 0.45\vec{D}_{AB}. \quad 2.6$$

This result means your friend walked $D_{DB} = 0.45D_{AB} = 0.45(6.0 \text{ km}) = 2.7 \text{ km}$ from the point where he finds his tackle box to the fishing hole.

When vectors $\vec{A}$ and $\vec{B}$ lie along a line (that is, in one dimension), such as in the camping example, their resultant $\vec{R} = \vec{A} + \vec{B}$ and their difference $\vec{D} = \vec{A} - \vec{B}$ both lie along the same direction. We can illustrate the addition or

subtraction of vectors by drawing the corresponding vectors to scale in one dimension, as shown in [Figure 2.7](#).

To illustrate the resultant when $\vec{A}$ and $\vec{B}$ are two parallel vectors, we draw them along one line by placing the origin of one vector at the end of the other vector in head-to-tail fashion (see [Figure 2.7\(b\)](#)). The magnitude of this resultant is the sum of their magnitudes: $R = A + B$. The direction of the resultant is parallel to both vectors. When vector $\vec{A}$ is antiparallel to vector $\vec{B}$, we draw them along one line in either head-to-head fashion ([Figure 2.7\(c\)](#)) or tail-to-tail fashion. The magnitude of the vector difference, then, is the *absolute value* $D = |A - B|$ of the difference of their magnitudes. The direction of the difference vector $\vec{D}$ is parallel to the direction of the longer vector.

In general, in one dimension—as well as in higher dimensions, such as in a plane or in space—we can add any number of vectors and we can do so in any order because the addition of vectors is **commutative**,

$$\vec{A} + \vec{B} = \vec{B} + \vec{A}, \quad 2.7$$

and **associative**,

$$(\vec{A} + \vec{B}) + \vec{C} = \vec{A} + (\vec{B} + \vec{C}). \quad 2.8$$

Moreover, multiplication by a scalar is **distributive**:

$$\alpha_1 \vec{A} + \alpha_2 \vec{A} = (\alpha_1 + \alpha_2) \vec{A}. \quad 2.9$$

We used the distributive property in [Equation 2.4](#) and [Equation 2.6](#).

When adding many vectors in one dimension, it is convenient to use the concept of a **unit vector**. A unit vector, which is denoted by a letter symbol with a hat, such as $\hat{u}$, has a magnitude of one and does not have any physical unit so that $|\hat{u}| \equiv u = 1$. The only role of a unit vector is to specify direction. For example, instead of saying vector $\vec{D}_{AB}$ has a magnitude of 6.0 km and a direction of northeast, we can introduce a unit vector $\hat{u}$ that points to the northeast and say succinctly that $\vec{D}_{AB} = (6.0 \text{ km})\hat{u}$. Then the southwesterly direction is simply given by the unit vector $-\hat{u}$. In this way, the displacement of 6.0 km in the southwesterly direction is expressed by the vector

$$\vec{D}_{BA} = (-6.0 \text{ km})\hat{u}.$$



EXAMPLE 2.1

A Ladybug Walker

A long measuring stick rests against a wall in a physics laboratory with its 200-cm end at the floor. A ladybug lands on the 100-cm mark and crawls randomly along the stick. It first walks 15 cm toward the floor, then it walks 56 cm toward the wall, then it walks 3 cm toward the floor again. Then, after a brief stop, it continues for 25 cm toward the floor and then, again, it crawls up 19 cm toward the wall before coming to a complete rest ([Figure 2.8](#)). Find the vector of its total displacement and its final resting position on the stick.

Strategy

If we choose the direction along the stick toward the floor as the direction of unit vector $\hat{u}$, then the direction toward the wall is $+\hat{u}$ and the direction toward the wall is $-\hat{u}$. The ladybug makes a total of five displacements:

$$\begin{aligned} \vec{D}_1 &= (15 \text{ cm})(+\hat{u}), \\ \vec{D}_2 &= (56 \text{ cm})(-\hat{u}), \\ \vec{D}_3 &= (3 \text{ cm})(+\hat{u}), \\ \vec{D}_4 &= (25 \text{ cm})(+\hat{u}), \text{ and} \\ \vec{D}_5 &= (19 \text{ cm})(-\hat{u}). \end{aligned}$$

The total displacement $\vec{D}$ is the resultant of all its displacement vectors.

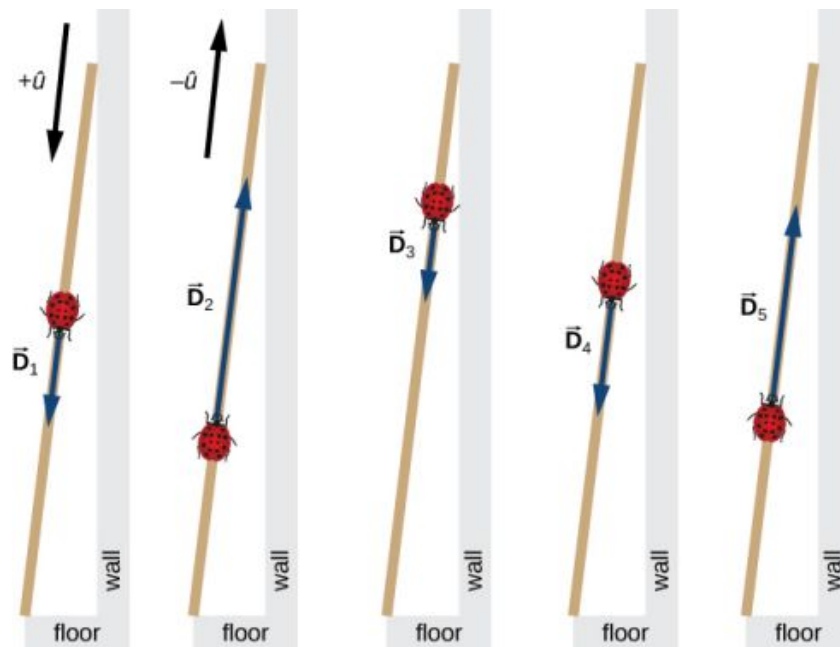


FIGURE 2.8 Five displacements of the ladybug. Note that in this schematic drawing, magnitudes of displacements are not drawn to scale. (credit "ladybug": modification of work by "Persian Poet Gal"/Wikimedia Commons)

Solution

The resultant of all the displacement vectors is

$$\begin{aligned}\vec{D} &= \vec{D}_1 + \vec{D}_2 + \vec{D}_3 + \vec{D}_4 + \vec{D}_5 \\ &= (15 \text{ cm})(+\hat{u}) + (56 \text{ cm})(-\hat{u}) + (3 \text{ cm})(+\hat{u}) + (25 \text{ cm})(+\hat{u}) + (19 \text{ cm})(-\hat{u}) \\ &= (15 - 56 + 3 + 25 - 19)\text{cm}\hat{u} \\ &= -32 \text{ cm}\hat{u}.\end{aligned}$$

In this calculation, we use the distributive law given by [Equation 2.9](#). The result reads that the total displacement vector points away from the 100-cm mark (initial landing site) toward the end of the meter stick that touches the wall. The end that touches the wall is marked 0 cm, so the final position of the ladybug is at the $(100 - 32)\text{cm} = 68\text{-cm}$ mark.

✓ CHECK YOUR UNDERSTANDING 2.2

A cave diver enters a long underwater tunnel. When her displacement with respect to the entry point is 20 m, she accidentally drops her camera, but she doesn't notice it missing until she is some 6 m farther into the tunnel. She swims back 10 m but cannot find the camera, so she decides to end the dive. How far from the entry point is she? Taking the positive direction out of the tunnel, what is her displacement vector relative to the entry point?

Algebra of Vectors in Two Dimensions

When vectors lie in a plane—that is, when they are in two dimensions—they can be multiplied by scalars, added to other vectors, or subtracted from other vectors in accordance with the general laws expressed by [Equation 2.1](#), [Equation 2.2](#), [Equation 2.7](#), and [Equation 2.8](#). However, the addition rule for two vectors in a plane becomes more complicated than the rule for vector addition in one dimension. We have to use the laws of geometry to construct resultant vectors, followed by trigonometry to find vector magnitudes and directions. This geometric approach is commonly used in navigation ([Figure 2.9](#)). In this section, we need to have at hand two rulers, a triangle, a protractor, a pencil, and an eraser for drawing vectors to scale by geometric constructions.



FIGURE 2.9 In navigation, the laws of geometry are used to draw resultant displacements on nautical maps.

For a geometric construction of the sum of two vectors in a plane, we follow **the parallelogram rule**. Suppose two vectors $\vec{A}$ and $\vec{B}$ are at the arbitrary positions shown in [Figure 2.10](#). Translate either one of them in parallel to the beginning of the other vector, so that after the translation, both vectors have their origins at the same point. Now, at the end of vector $\vec{A}$ we draw a line parallel to vector $\vec{B}$ and at the end of vector $\vec{B}$ we draw a line parallel to vector $\vec{A}$ (the dashed lines in [Figure 2.10](#)). In this way, we obtain a parallelogram. From the origin of the two vectors we draw a diagonal that is the resultant $\vec{R}$ of the two vectors: $\vec{R} = \vec{A} + \vec{B}$ ([Figure 2.10\(a\)](#)). The other diagonal of this parallelogram is the vector difference of the two vectors $\vec{D} = \vec{A} - \vec{B}$, as shown in [Figure 2.10\(b\)](#). Notice that the end of the difference vector is placed at the end of vector $\vec{A}$.

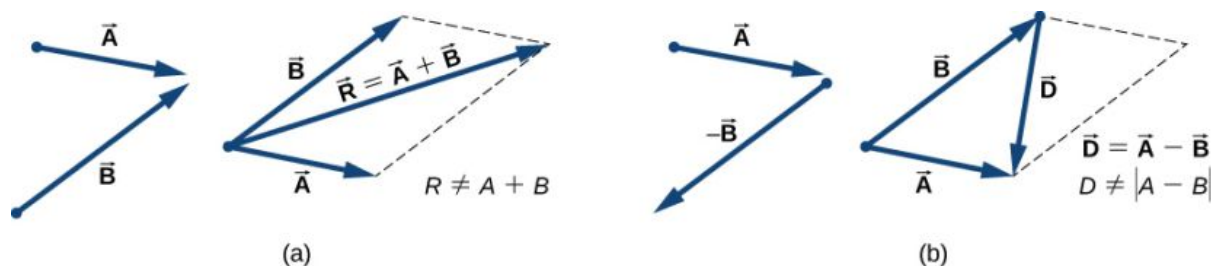


FIGURE 2.10 The parallelogram rule for the addition of two vectors. Make the parallel translation of each vector to a point where their origins (marked by the dot) coincide and construct a parallelogram with two sides on the vectors and the other two sides (indicated by dashed lines) parallel to the vectors. (a) Draw the resultant vector $\vec{R}$ along the diagonal of the parallelogram from the common point to the opposite corner. Length R of the resultant vector is *not* equal to the sum of the magnitudes of the two vectors. (b) Draw the difference vector $\vec{D} = \vec{A} - \vec{B}$ along the diagonal connecting the ends of the vectors. Place the origin of vector $\vec{D}$ at the end of vector $\vec{B}$ and the end (arrowhead) of vector $\vec{D}$ at the end of vector $\vec{A}$. Length D of the difference vector is *not* equal to the difference of magnitudes of the two vectors.

It follows from the parallelogram rule that neither the magnitude of the resultant vector nor the magnitude of the difference vector can be expressed as a simple sum or difference of magnitudes A and B , because the length of a diagonal cannot be expressed as a simple sum of side lengths. When using a geometric construction to find magnitudes $|\vec{R}|$ and $|\vec{D}|$, we have to use trigonometry laws for triangles, which may lead to complicated algebra.

There are two ways to circumvent this algebraic complexity. One way is to use the method of components, which we examine in the next section. The other way is to draw the vectors to scale, as is done in navigation, and read approximate vector lengths and angles (directions) from the graphs. In this section we examine the second approach.

If we need to add three or more vectors, we repeat the parallelogram rule for the pairs of vectors until we find the resultant of all of the resultants. For three vectors, for example, we first find the resultant of vector 1 and vector 2, and then we find the resultant of this resultant and vector 3. The order in which we select the pairs of vectors does not matter because the operation of vector addition is commutative and associative (see [Equation 2.7](#) and [Equation 2.8](#)). Before we state a general rule that follows from repetitive applications of the parallelogram rule, let's look at the following example.

Suppose you plan a vacation trip in Florida. Departing from Tallahassee, the state capital, you plan to visit your uncle Joe in Jacksonville, see your cousin Vinny in Daytona Beach, stop for a little fun in Orlando, see a circus performance in Tampa, and visit the University of Florida in Gainesville. Your route may be represented by five displacement vectors $\vec{A}$, $\vec{B}$, $\vec{C}$, $\vec{D}$, and $\vec{E}$, which are indicated by the red vectors in [Figure 2.11](#). What is your total displacement when you reach Gainesville? The total displacement is the vector sum of all five displacement vectors, which may be found by using the parallelogram rule four times. Alternatively, recall that the displacement vector has its beginning at the initial position (Tallahassee) and its end at the final position (Gainesville), so the total displacement vector can be drawn directly as an arrow connecting Tallahassee with Gainesville (see the green vector in [Figure 2.11](#)). When we use the parallelogram rule four times, the resultant $\vec{R}$ we obtain is exactly this green vector connecting Tallahassee with Gainesville: $\vec{R} = \vec{A} + \vec{B} + \vec{C} + \vec{D} + \vec{E}$.

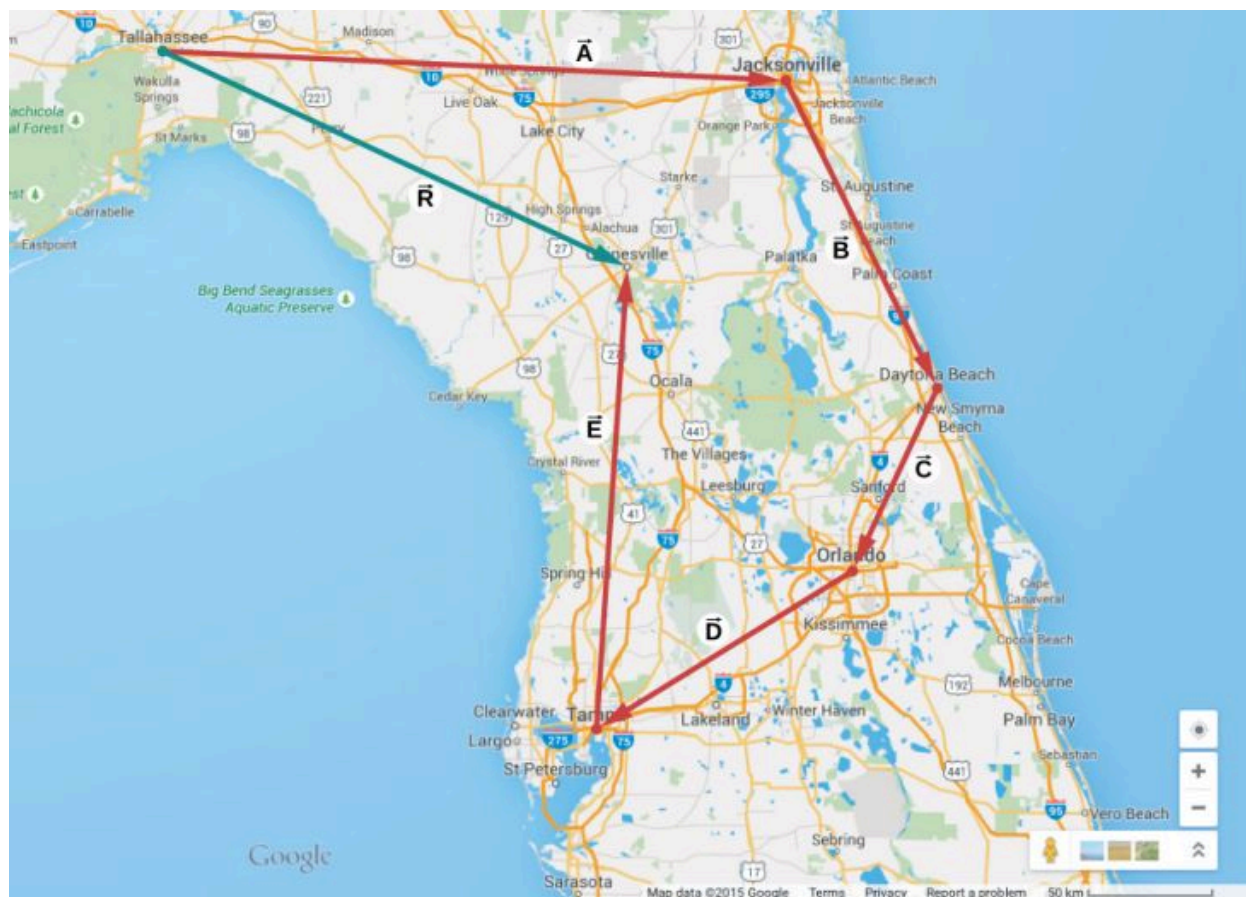


FIGURE 2.11 When we use the parallelogram rule four times, we obtain the resultant vector $\vec{R} = \vec{A} + \vec{B} + \vec{C} + \vec{D} + \vec{E}$, which is the green vector connecting Tallahassee with Gainesville.

Drawing the resultant vector of many vectors can be generalized by using the following **tail-to-head geometric construction**. Suppose we want to draw the resultant vector $\vec{R}$ of four vectors $\vec{A}$, $\vec{B}$, $\vec{C}$, and $\vec{D}$ ([Figure 2.12\(a\)](#)). We select any one of the vectors as the first vector and make a parallel translation of a second vector to a position where the origin (“tail”) of the second vector coincides with the end (“head”) of the first vector. Then, we select a third vector and make a parallel translation of the third vector to a position where the origin of the third vector coincides with the end of the second vector. We repeat this procedure until all the vectors are in a head-to-tail

arrangement like the one shown in Figure 2.12. We draw the resultant vector $\vec{R}$ by connecting the origin (“tail”) of the first vector with the end (“head”) of the last vector. The end of the resultant vector is at the end of the last vector. Because the addition of vectors is associative and commutative, we obtain the same resultant vector regardless of which vector we choose to be first, second, third, or fourth in this construction.

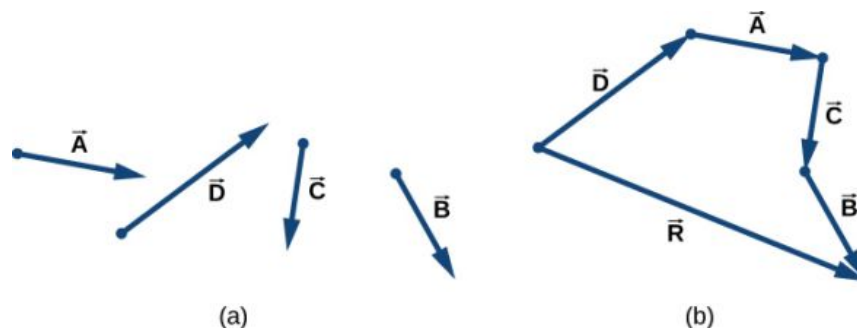


FIGURE 2.12 Tail-to-head method for drawing the resultant vector $\vec{R} = \vec{A} + \vec{B} + \vec{C} + \vec{D}$. (a) Four vectors of different magnitudes and directions. (b) Vectors in (a) are translated to new positions where the origin (“tail”) of one vector is at the end (“head”) of another vector. The resultant vector is drawn from the origin (“tail”) of the first vector to the end (“head”) of the last vector in this arrangement.

EXAMPLE 2.2

Geometric Construction of the Resultant

The three displacement vectors $\vec{A}$, $\vec{B}$, and $\vec{C}$ in Figure 2.13 are specified by their magnitudes $A = 10.0$, $B = 7.0$, and $C = 8.0$, respectively, and by their respective direction angles with the horizontal direction $\alpha = 35^\circ$, $\beta = -110^\circ$, and $\gamma = 30^\circ$. The physical units of the magnitudes are centimeters. Choose a convenient scale and use a ruler and a protractor to find the following vector sums: (a) $\vec{R} = \vec{A} + \vec{B}$, (b) $\vec{D} = \vec{A} - \vec{B}$, and (c) $\vec{S} = \vec{A} - 3\vec{B} + \vec{C}$.

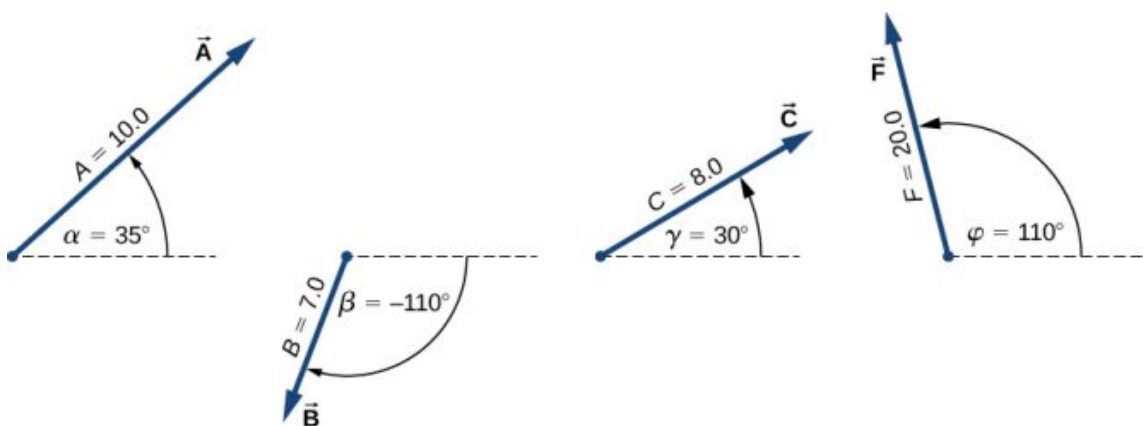


FIGURE 2.13 Vectors used in Example 2.2 and in the Check Your Understanding feature that follows.

Strategy

In geometric construction, to find a vector means to find its magnitude and its direction angle with the horizontal direction. The strategy is to draw to scale the vectors that appear on the right-hand side of the equation and construct the resultant vector. Then, use a ruler and a protractor to read the magnitude of the resultant and the direction angle. For parts (a) and (b) we use the parallelogram rule. For (c) we use the tail-to-head method.

Solution

For parts (a) and (b), we attach the origin of vector $\vec{B}$ to the origin of vector $\vec{A}$, as shown in Figure 2.14, and construct a parallelogram. The shorter diagonal of this parallelogram is the sum $\vec{A} + \vec{B}$. The longer of the diagonals is the difference $\vec{A} - \vec{B}$. We use a ruler to measure the lengths of the diagonals, and a protractor to measure the angles with the horizontal. For the resultant $\vec{R}$, we obtain $R = 5.8$ cm and $\theta_R \approx 0^\circ$. For the difference $\vec{D}$, we obtain $D = 16.2$ cm and $\theta_D = 49.3^\circ$, which are shown in Figure 2.14.

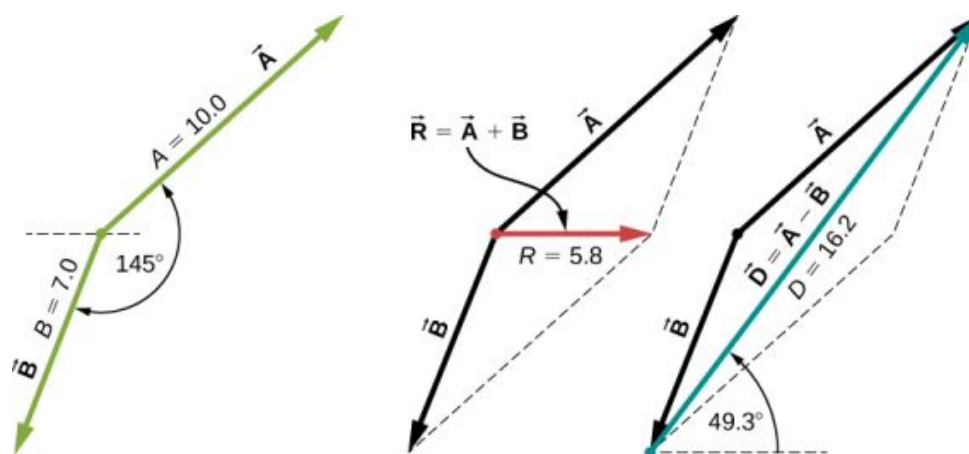


FIGURE 2.14 Using the parallelogram rule to solve (a) (finding the resultant, red) and (b) (finding the difference, blue).

For (c), we can start with vector $-3\vec{B}$ and draw the remaining vectors tail-to-head as shown in [Figure 2.15](#). In vector addition, the order in which we draw the vectors is unimportant, but drawing the vectors to scale is very important. Next, we draw vector $\vec{S}$ from the origin of the first vector to the end of the last vector and place the arrowhead at the end of $\vec{S}$. We use a ruler to measure the length of $\vec{S}$, and find that its magnitude is $S = 36.9$ cm. We use a protractor and find that its direction angle is $\theta_S = 52.9^\circ$. This solution is shown in [Figure 2.15](#).

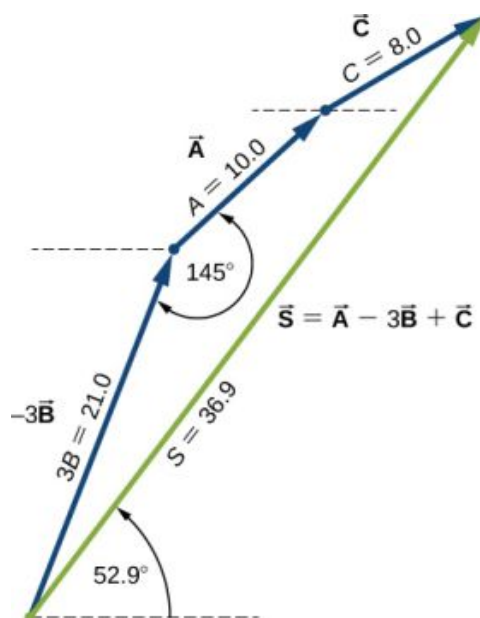


FIGURE 2.15 Using the tail-to-head method to solve (c) (finding vector $\vec{S}$, green).

✓ CHECK YOUR UNDERSTANDING 2.3

Using the three displacement vectors $\vec{A}$, $\vec{B}$, and $\vec{F}$ in [Figure 2.13](#), choose a convenient scale, and use a ruler and a protractor to find vector $\vec{G}$ given by the vector equation $\vec{G} = \vec{A} + 2\vec{B} - \vec{F}$.

🔗 INTERACTIVE

Observe the addition of vectors in a plane by visiting this [vector calculator \(https://openstax.org/l/21compveccalc\)](https://openstax.org/l/21compveccalc),

and by engaging the Phet simulation below.

Access multimedia content (<https://openstax.org/books/university-physics-volume-1/pages/2-1-scalars-and-vectors>)

2.2 Coordinate Systems and Components of a Vector

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Describe vectors in two and three dimensions in terms of their components, using unit vectors along the axes.
- Distinguish between the vector components of a vector and the scalar components of a vector.
- Explain how the magnitude of a vector is defined in terms of the components of a vector.
- Identify the direction angle of a vector in a plane.
- Explain the connection between polar coordinates and Cartesian coordinates in a plane.

Vectors are usually described in terms of their components in a coordinate system. Even in everyday life we naturally invoke the concept of vector components in a rectangular coordinate system. For example, if you ask someone for directions to a particular location, you will more likely be told to go 40 km east and 30 km north than 50 km in the direction 37° north of east.

In a rectangular (Cartesian) xy -coordinate system in a plane, a point in a plane is described by a pair of coordinates (x, y) . In a similar fashion, a vector $\vec{A}$ in a plane is described by a pair of its *vector* coordinates. The x -coordinate of vector $\vec{A}$ is called its x -component and the y -coordinate of vector $\vec{A}$ is called its y -component. The vector x -component is a vector denoted by $\vec{A}_x$. The vector y -component is a vector denoted by $\vec{A}_y$. In the Cartesian system, the x and y **vector components** of a vector are the orthogonal projections, as illustrated in Figure 2.16, of this vector onto the x - and y -axes, respectively. In this way, following the parallelogram rule for vector addition, each vector on a Cartesian plane can be expressed as the vector sum of its vector components:

$$\vec{A} = \vec{A}_x + \vec{A}_y. \quad 2.10$$

As illustrated in Figure 2.16, vector $\vec{A}$ is the diagonal of the rectangle where the x -component $\vec{A}_x$ is the side parallel to the x -axis and the y -component $\vec{A}_y$ is the side parallel to the y -axis. Vector component $\vec{A}_x$ is orthogonal to vector component $\vec{A}_y$.

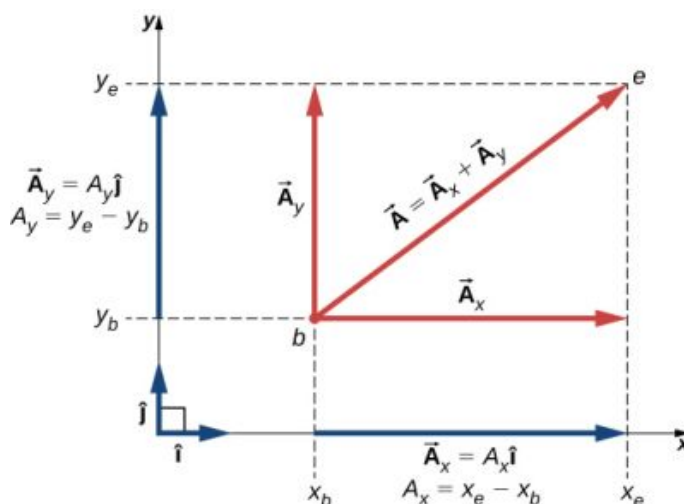


FIGURE 2.16 Vector $\vec{A}$ in a plane in the Cartesian coordinate system is the vector sum of its vector x - and y -components. The x -vector component $\vec{A}_x$ is the orthogonal projection of vector $\vec{A}$ onto the x -axis. The y -vector component $\vec{A}_y$ is the orthogonal projection of vector $\vec{A}$ onto the y -axis. The numbers A_x and A_y that multiply the unit vectors are the scalar components of the vector.

It is customary to denote the positive direction on the x -axis by the unit vector $\hat{i}$ and the positive direction on the

y-axis by the unit vector $\hat{j}$. **Unit vectors of the axes**, $\hat{i}$ and $\hat{j}$, define two orthogonal directions in the plane. As shown in [Figure 2.16](#), the x- and y- components of a vector can now be written in terms of the unit vectors of the axes:

$$\begin{cases} \vec{A}_x = A_x \hat{i} \\ \vec{A}_y = A_y \hat{j} \end{cases} \quad 2.11$$

The vectors $\vec{A}_x$ and $\vec{A}_y$ defined by [Equation 2.11](#) are the *vector components* of vector $\vec{A}$. The numbers A_x and A_y that define the vector components in [Equation 2.11](#) are the **scalar components** of vector $\vec{A}$. Combining [Equation 2.10](#) with [Equation 2.11](#), we obtain **the component form of a vector**:

$$\vec{A} = A_x \hat{i} + A_y \hat{j}. \quad 2.12$$

If we know the coordinates $b(x_b, y_b)$ of the origin point of a vector (where b stands for “beginning”) and the coordinates $e(x_e, y_e)$ of the end point of a vector (where e stands for “end”), we can obtain the scalar components of a vector simply by subtracting the origin point coordinates from the end point coordinates:

$$\begin{cases} A_x = x_e - x_b \\ A_y = y_e - y_b \end{cases} \quad 2.13$$



EXAMPLE 2.3

Displacement of a Mouse Pointer

A mouse pointer on the display monitor of a computer at its initial position is at point $b(6.0 \text{ cm}, 1.6 \text{ cm})$ with respect to the lower left-side corner. If you move the pointer to an icon located at point $e(2.0 \text{ cm}, 4.5 \text{ cm})$, what is the displacement vector of the pointer?

Strategy

The origin of the xy -coordinate system is the lower left-side corner of the computer monitor. Therefore, the unit vector $\hat{i}$ on the x -axis points horizontally to the right and the unit vector $\hat{j}$ on the y -axis points vertically upward. The origin of the displacement vector is located at point $b(6.0, 1.6)$ and the end of the displacement vector is located at point $e(2.0, 4.5)$. Substitute the coordinates of these points into [Equation 2.13](#) to find the scalar components D_x and D_y of the displacement vector $\vec{D}$. Finally, substitute the coordinates into [Equation 2.12](#) to write the displacement vector in the vector component form.

Solution

We identify $x_b = 6.0$, $x_e = 2.0$, $y_b = 1.6$, and $y_e = 4.5$, where the physical unit is 1 cm. The scalar x - and y -components of the displacement vector are

$$\begin{aligned} D_x &= x_e - x_b = (2.0 - 6.0)\text{cm} = -4.0 \text{ cm}, \\ D_y &= y_e - y_b = (4.5 - 1.6)\text{cm} = +2.9 \text{ cm}. \end{aligned}$$

The vector component form of the displacement vector is

$$\vec{D} = D_x \hat{i} + D_y \hat{j} = (-4.0 \text{ cm})\hat{i} + (2.9 \text{ cm})\hat{j} = (-4.0\hat{i} + 2.9\hat{j})\text{cm}. \quad 2.14$$

This solution is shown in [Figure 2.17](#).

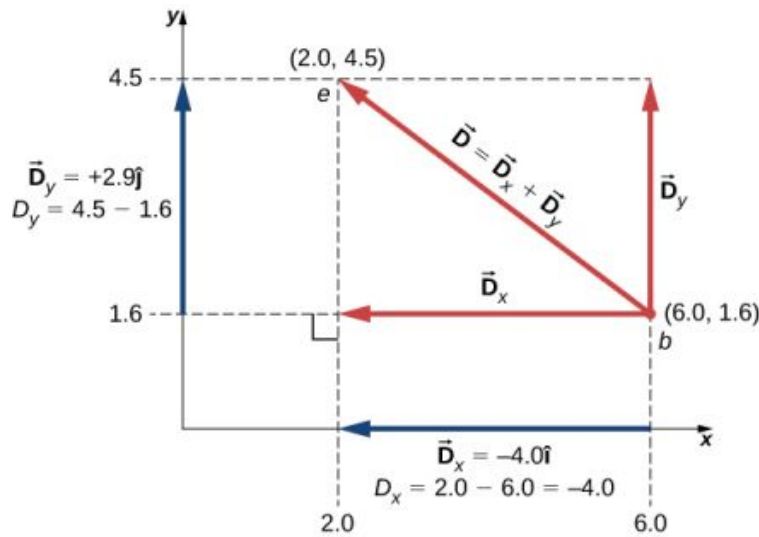


FIGURE 2.17 The graph of the displacement vector. The vector points from the origin point at b to the end point at e .

Significance

Notice that the physical unit—here, 1 cm—can be placed either with each component immediately before the unit vector or globally for both components, as in Equation 2.14. Often, the latter way is more convenient because it is simpler.

The vector x -component $\vec{D}_x = -4.0\hat{i} = 4.0(-\hat{i})$ of the displacement vector has the magnitude $|\vec{D}_x| = |-4.0||\hat{i}| = 4.0$ because the magnitude of the unit vector is $|\hat{i}| = 1$. Notice, too, that the direction of the x -component is $-\hat{i}$, which is antiparallel to the direction of the $+x$ -axis; hence, the x -component vector $\vec{D}_x$ points to the left, as shown in Figure 2.17. The scalar x -component of vector $\vec{D}$ is $D_x = -4.0$.

Similarly, the vector y -component $\vec{D}_y = +2.9\hat{j}$ of the displacement vector has magnitude $|\vec{D}_y| = |2.9||\hat{j}| = 2.9$ because the magnitude of the unit vector is $|\hat{j}| = 1$. The direction of the y -component is $+\hat{j}$, which is parallel to the direction of the $+y$ -axis. Therefore, the y -component vector $\vec{D}_y$ points up, as seen in Figure 2.17. The scalar y -component of vector $\vec{D}$ is $D_y = +2.9$. The displacement vector $\vec{D}$ is the resultant of its two vector components.

The vector component form of the displacement vector Equation 2.14 tells us that the mouse pointer has been moved on the monitor 4.0 cm to the left and 2.9 cm upward from its initial position.

✓ CHECK YOUR UNDERSTANDING 2.4

A blue fly lands on a sheet of graph paper at a point located 10.0 cm to the right of its left edge and 8.0 cm above its bottom edge and walks slowly to a point located 5.0 cm from the left edge and 5.0 cm from the bottom edge. Choose the rectangular coordinate system with the origin at the lower left-side corner of the paper and find the displacement vector of the fly. Illustrate your solution by graphing.

When we know the scalar components A_x and A_y of a vector $\vec{A}$, we can find its magnitude A and its direction angle θ_A . The **direction angle**—or **direction**, for short—is the angle the vector forms with the positive direction on the x -axis. The angle θ_A is measured in the *counterclockwise direction* from the $+x$ -axis to the vector (Figure 2.18). Because the lengths A , A_x , and A_y form a right triangle, they are related by the Pythagorean theorem:

$$A^2 = A_x^2 + A_y^2 \Leftrightarrow A = \sqrt{A_x^2 + A_y^2}. \quad 2.15$$

This equation works even if the scalar components of a vector are negative. The direction angle θ_A of a vector is

defined via the tangent function of angle θ_A in the triangle shown in [Figure 2.18](#):

$$\tan \theta = \frac{A_y}{A_x}$$

2.16

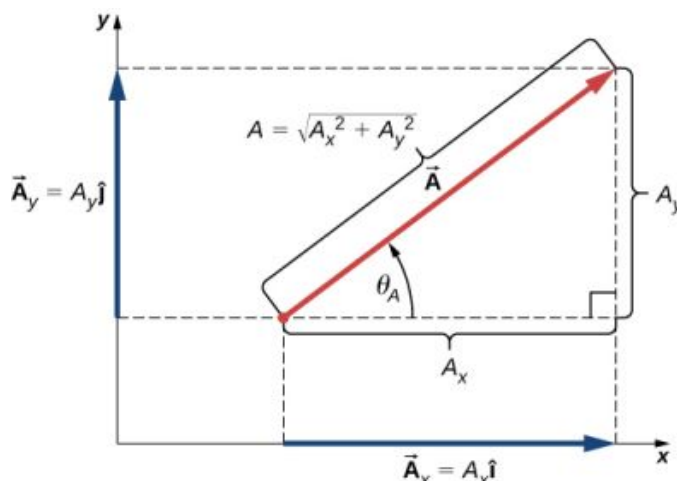


FIGURE 2.18 When the vector lies either in the first quadrant or in the fourth quadrant, where component A_x is positive ([Figure 2.19](#)), the direction angle θ_A in [Equation 2.16](#) is identical to the angle θ .

When the vector lies either in the first quadrant or in the fourth quadrant, where component A_x is positive ([Figure 2.19](#)), the angle θ in [Equation 2.16](#) is identical to the direction angle θ_A . For vectors in the fourth quadrant, angle θ is negative, which means that for these vectors, direction angle θ_A is measured *clockwise* from the positive x-axis. Similarly, for vectors in the second quadrant, angle θ is negative. When the vector lies in either the second or third quadrant, where component A_x is negative, the direction angle is $\theta_A = \theta + 180^\circ$ ([Figure 2.19](#)).

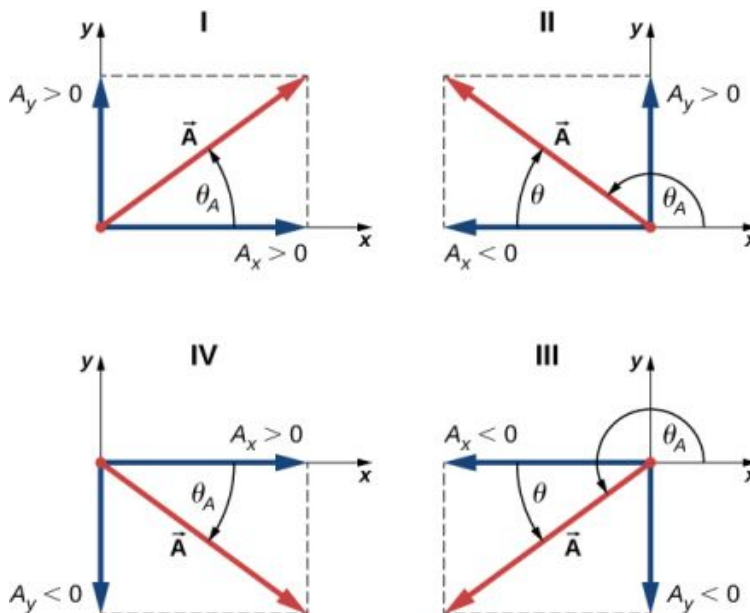


FIGURE 2.19 Scalar components of a vector may be positive or negative. Vectors in the first quadrant (I) have both scalar components positive and vectors in the third quadrant have both scalar components negative. For vectors in quadrants II and III, the direction angle of a vector is $\theta_A = \theta + 180^\circ$.



EXAMPLE 2.4

Magnitude and Direction of the Displacement Vector

You move a mouse pointer on the display monitor from its initial position at point (6.0 cm, 1.6 cm) to an icon located

at point (2.0 cm, 4.5 cm). What are the magnitude and direction of the displacement vector of the pointer?

Strategy

In [Example 2.3](#), we found the displacement vector $\vec{D}$ of the mouse pointer (see [Equation 2.14](#)). We identify its scalar components $D_x = -4.0$ cm and $D_y = +2.9$ cm and substitute into [Equation 2.15](#) and [Equation 2.16](#) to find the magnitude D and direction θ_D , respectively.

Solution

The magnitude of vector $\vec{D}$ is

$$D = \sqrt{D_x^2 + D_y^2} = \sqrt{(-4.0 \text{ cm})^2 + (2.9 \text{ cm})^2} = \sqrt{(4.0)^2 + (2.9)^2} \text{ cm} = 4.9 \text{ cm}.$$

The direction angle is

$$\tan \theta = \frac{D_y}{D_x} = \frac{+2.9 \text{ cm}}{-4.0 \text{ cm}} = -0.725 \Rightarrow \theta = \tan^{-1}(-0.725) = -35.9^\circ.$$

Vector $\vec{D}$ lies in the second quadrant, so its direction angle is

$$\theta_D = \theta + 180^\circ = -35.9^\circ + 180^\circ = 144.1^\circ.$$

CHECK YOUR UNDERSTANDING 2.5

If the displacement vector of a blue fly walking on a sheet of graph paper is $\vec{D} = (-5.00\hat{i} - 3.00\hat{j})\text{cm}$, find its magnitude and direction.

In many applications, the magnitudes and directions of vector quantities are known and we need to find the resultant of many vectors. For example, imagine 400 cars moving on the Golden Gate Bridge in San Francisco in a strong wind. Each car gives the bridge a different push in various directions and we would like to know how big the resultant push can possibly be. We have already gained some experience with the geometric construction of vector sums, so we know the task of finding the resultant by drawing the vectors and measuring their lengths and angles may become intractable pretty quickly, leading to huge errors. Worries like this do not appear when we use analytical methods. The very first step in an analytical approach is to find vector components when the direction and magnitude of a vector are known.

Let us return to the right triangle in [Figure 2.18](#). The quotient of the adjacent side A_x to the hypotenuse A is the cosine function of direction angle θ_A , $A_x/A = \cos \theta_A$, and the quotient of the opposite side A_y to the hypotenuse A is the sine function of θ_A , $A_y/A = \sin \theta_A$. When magnitude A and direction θ_A are known, we can solve these relations for the scalar components:

$$\begin{cases} A_x = A \cos \theta_A \\ A_y = A \sin \theta_A \end{cases} \quad \text{2.17}$$

When calculating vector components with [Equation 2.17](#), care must be taken with the angle. The direction angle θ_A of a vector is the angle measured *counterclockwise* from the positive direction on the x -axis to the vector. The clockwise measurement gives a negative angle.



EXAMPLE 2.5

Components of Displacement Vectors

A rescue party for a missing child follows a search dog named Trooper. Trooper wanders a lot and makes many trial sniffs along many different paths. Trooper eventually finds the child and the story has a happy ending, but his displacements on various legs seem to be truly convoluted. On one of the legs he walks 200.0 m southeast, then he

runs north some 300.0 m. On the third leg, he examines the scents carefully for 50.0 m in the direction 30° west of north. On the fourth leg, Trooper goes directly south for 80.0 m, picks up a fresh scent and turns 23° west of south for 150.0 m. Find the scalar components of Trooper's displacement vectors and his displacement vectors in vector component form for each leg.

Strategy

Let's adopt a rectangular coordinate system with the positive x -axis in the direction of geographic east, with the positive y -direction pointed to geographic north. Explicitly, the unit vector $\hat{i}$ of the x -axis points east and the unit vector $\hat{j}$ of the y -axis points north. Trooper makes five legs, so there are five displacement vectors. We start by identifying their magnitudes and direction angles, then we use [Equation 2.17](#) to find the scalar components of the displacements and [Equation 2.12](#) for the displacement vectors.

Solution

On the first leg, the displacement magnitude is $L_1 = 200.0$ m and the direction is southeast. For direction angle θ_1 we can take either 45° measured clockwise from the east direction or $45^\circ + 270^\circ$ measured counterclockwise from the east direction. With the first choice, $\theta_1 = -45^\circ$. With the second choice, $\theta_1 = +315^\circ$. We can use either one of these two angles. The components are

$$\begin{aligned} L_{1x} &= L_1 \cos \theta_1 = (200.0 \text{ m}) \cos 315^\circ = 141.4 \text{ m}, \\ L_{1y} &= L_1 \sin \theta_1 = (200.0 \text{ m}) \sin 315^\circ = -141.4 \text{ m}. \end{aligned}$$

The displacement vector of the first leg is

$$\vec{L}_1 = L_{1x}\hat{i} + L_{1y}\hat{j} = (141.4\hat{i} - 141.4\hat{j}) \text{ m}.$$

On the second leg of Trooper's wanderings, the magnitude of the displacement is $L_2 = 300.0$ m and the direction is north. The direction angle is $\theta_2 = +90^\circ$. We obtain the following results:

$$\begin{aligned} L_{2x} &= L_2 \cos \theta_2 = (300.0 \text{ m}) \cos 90^\circ = 0.0, \\ L_{2y} &= L_2 \sin \theta_2 = (300.0 \text{ m}) \sin 90^\circ = 300.0 \text{ m}, \\ \vec{L}_2 &= L_{2x}\hat{i} + L_{2y}\hat{j} = (300.0 \text{ m})\hat{j}. \end{aligned}$$

On the third leg, the displacement magnitude is $L_3 = 50.0$ m and the direction is 30° west of north. The direction angle measured counterclockwise from the eastern direction is $\theta_3 = 30^\circ + 90^\circ = +120^\circ$. This gives the following answers:

$$\begin{aligned} L_{3x} &= L_3 \cos \theta_3 = (50.0 \text{ m}) \cos 120^\circ = -25.0 \text{ m}, \\ L_{3y} &= L_3 \sin \theta_3 = (50.0 \text{ m}) \sin 120^\circ = +43.3 \text{ m}, \\ \vec{L}_3 &= L_{3x}\hat{i} + L_{3y}\hat{j} = (-25.0\hat{i} + 43.3\hat{j}) \text{ m}. \end{aligned}$$

On the fourth leg of the excursion, the displacement magnitude is $L_4 = 80.0$ m and the direction is south. The direction angle can be taken as either $\theta_4 = -90^\circ$ or $\theta_4 = +270^\circ$. We obtain

$$\begin{aligned} L_{4x} &= L_4 \cos \theta_4 = (80.0 \text{ m}) \cos (-90^\circ) = 0, \\ L_{4y} &= L_4 \sin \theta_4 = (80.0 \text{ m}) \sin (-90^\circ) = -80.0 \text{ m}, \\ \vec{L}_4 &= L_{4x}\hat{i} + L_{4y}\hat{j} = (-80.0 \text{ m})\hat{j}. \end{aligned}$$

On the last leg, the magnitude is $L_5 = 150.0$ m and the angle is $\theta_5 = -23^\circ + 270^\circ = +247^\circ$ (23° west of south), which gives

$$\begin{aligned} L_{5x} &= L_5 \cos \theta_5 = (150.0 \text{ m}) \cos 247^\circ = -58.6 \text{ m}, \\ L_{5y} &= L_5 \sin \theta_5 = (150.0 \text{ m}) \sin 247^\circ = -138.1 \text{ m}, \\ \vec{L}_5 &= L_{5x}\hat{i} + L_{5y}\hat{j} = (-58.6\hat{i} - 138.1\hat{j}) \text{ m}. \end{aligned}$$

✓ CHECK YOUR UNDERSTANDING 2.6

If Trooper runs 20 m west before taking a rest, what is his displacement vector?

Polar Coordinates

To describe locations of points or vectors in a plane, we need two orthogonal directions. In the Cartesian coordinate system these directions are given by unit vectors $\hat{i}$ and $\hat{j}$ along the x -axis and the y -axis, respectively. The Cartesian coordinate system is very convenient to use in describing displacements and velocities of objects and the forces acting on them. However, it becomes cumbersome when we need to describe the rotation of objects. When describing rotation, we usually work in the **polar coordinate system**.

In the polar coordinate system, the location of point P in a plane is given by two **polar coordinates** (Figure 2.20). The first polar coordinate is the **radial coordinate** r , which is the distance of point P from the origin. The second polar coordinate is an angle φ that the radial vector makes with some chosen direction, usually the positive x -direction. In polar coordinates, angles are measured in radians, or **rads**. The radial vector is attached at the origin and points away from the origin to point P . This radial direction is described by a unit radial vector $\hat{r}$. The second unit vector $\hat{\theta}$ is a vector orthogonal to the radial direction $\hat{r}$. The positive $+\hat{\theta}$ direction indicates how the angle φ changes in the counterclockwise direction. In this way, a point P that has coordinates (x, y) in the rectangular system can be described equivalently in the polar coordinate system by the two polar coordinates (r, φ) . Equation 2.17 is valid for any vector, so we can use it to express the x - and y -coordinates of vector $\vec{r}$. In this way, we obtain the connection between the polar coordinates and rectangular coordinates of point P :

$$\begin{cases} x = r \cos \varphi \\ y = r \sin \varphi \end{cases} \quad 2.18$$

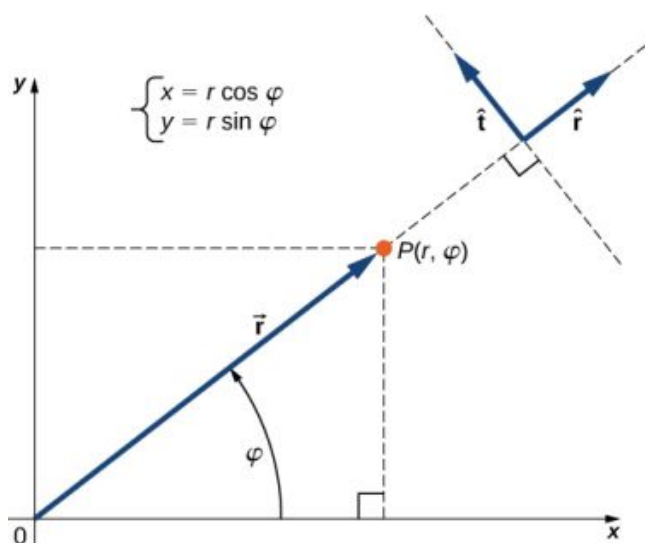


FIGURE 2.20 Using polar coordinates, the unit vector $\hat{r}$ defines the positive direction along the radius r (radial direction) and, orthogonal to it, the unit vector $\hat{\theta}$ defines the positive direction of rotation by the angle φ .

✿ EXAMPLE 2.6

Polar Coordinates

A treasure hunter finds one silver coin at a location 20.0 m away from a dry well in the direction 20° north of east and finds one gold coin at a location 10.0 m away from the well in the direction 20° north of west. What are the polar and rectangular coordinates of these findings with respect to the well?

Strategy

The well marks the origin of the coordinate system and east is the $+x$ -direction. We identify radial distances from the locations to the origin, which are $r_S = 20.0$ m (for the silver coin) and $r_G = 10.0$ m (for the gold coin). To find the angular coordinates, we convert 20° to radians: $20^\circ = \pi 20/180 = \pi/9$. We use [Equation 2.18](#) to find the x - and y -coordinates of the coins.

Solution

The angular coordinate of the silver coin is $\varphi_S = \pi/9$, whereas the angular coordinate of the gold coin is $\varphi_G = \pi - \pi/9 = 8\pi/9$. Hence, the polar coordinates of the silver coin are $(r_S, \varphi_S) = (20.0 \text{ m}, \pi/9)$ and those of the gold coin are $(r_G, \varphi_G) = (10.0 \text{ m}, 8\pi/9)$. We substitute these coordinates into [Equation 2.18](#) to obtain rectangular coordinates. For the gold coin, the coordinates are

$$\begin{cases} x_G = r_G \cos \varphi_G = (10.0 \text{ m}) \cos 8\pi/9 = -9.4 \text{ m} \\ y_G = r_G \sin \varphi_G = (10.0 \text{ m}) \sin 8\pi/9 = 3.4 \text{ m} \end{cases} \Rightarrow (x_G, y_G) = (-9.4 \text{ m}, 3.4 \text{ m}).$$

For the silver coin, the coordinates are

$$\begin{cases} x_S = r_S \cos \varphi_S = (20.0 \text{ m}) \cos \pi/9 = 18.9 \text{ m} \\ y_S = r_S \sin \varphi_S = (20.0 \text{ m}) \sin \pi/9 = 6.8 \text{ m} \end{cases} \Rightarrow (x_S, y_S) = (18.9 \text{ m}, 6.8 \text{ m}).$$

Vectors in Three Dimensions

To specify the location of a point in space, we need three coordinates (x, y, z) , where coordinates x and y specify locations in a plane, and coordinate z gives a vertical position above or below the plane. Three-dimensional space has three orthogonal directions, so we need not two but *three* unit vectors to define a three-dimensional coordinate system. In the Cartesian coordinate system, the first two unit vectors are the unit vector of the x -axis $\hat{i}$ and the unit vector of the y -axis $\hat{j}$. The third unit vector $\hat{k}$ is the direction of the z -axis ([Figure 2.21](#)). The order in which the axes are labeled, which is the order in which the three unit vectors appear, is important because it defines the orientation of the coordinate system. The order x - y - z , which is equivalent to the order $\hat{i}$ - $\hat{j}$ - $\hat{k}$, defines the standard right-handed coordinate system (positive orientation). Notice that while previously the y -axis was used for the upward direction, in three dimensions we usually choose z to represent the vertical. A right-handed coordinate system with z up and with one of the remaining axes pointing to the right and the other out of the page must have the y -axis pointing right and x out of the page, as illustrated here.

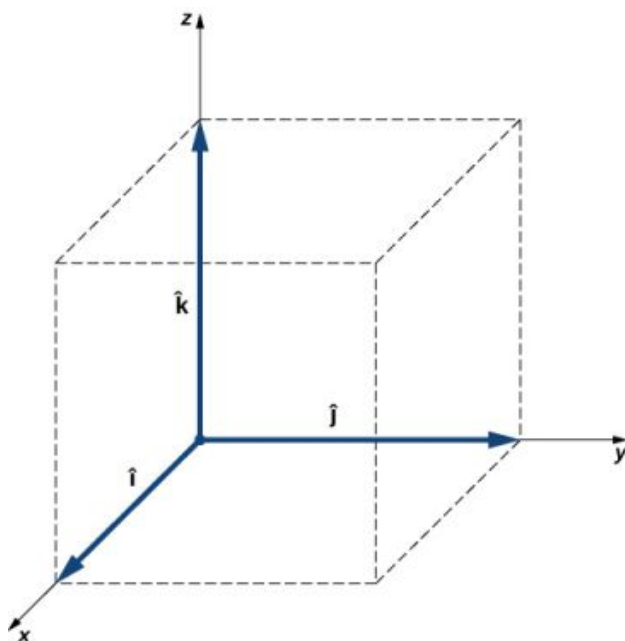


FIGURE 2.21 Three unit vectors define a Cartesian system in three-dimensional space. The order in which these unit vectors appear defines the orientation of the coordinate system. The order shown here defines the right-handed orientation.

In three-dimensional space, vector $\vec{A}$ has three vector components: the x-component $\vec{A}_x = A_x \hat{i}$, which is the part of vector $\vec{A}$ along the x-axis; the y-component $\vec{A}_y = A_y \hat{j}$, which is the part of $\vec{A}$ along the y-axis; and the z-component $\vec{A}_z = A_z \hat{k}$, which is the part of the vector along the z-axis. A vector in three-dimensional space is the vector sum of its three vector components (Figure 2.22):

$$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}. \quad 2.19$$

If we know the coordinates of its origin $b(x_b, y_b, z_b)$ and of its end $e(x_e, y_e, z_e)$, its scalar components are obtained by taking their differences: A_x and A_y are given by Equation 2.13 and the z-component is given by

$$A_z = z_e - z_b. \quad 2.20$$

Magnitude A is obtained by generalizing Equation 2.15 to three dimensions:

$$A = \sqrt{A_x^2 + A_y^2 + A_z^2}. \quad 2.21$$

This expression for the vector magnitude comes from applying the Pythagorean theorem twice. As seen in Figure 2.22, the diagonal in the xy-plane has length $\sqrt{A_x^2 + A_y^2}$ and its square adds to the square A_z^2 to give A^2 . Note that when the z-component is zero, the vector lies entirely in the xy-plane and its description is reduced to two dimensions.

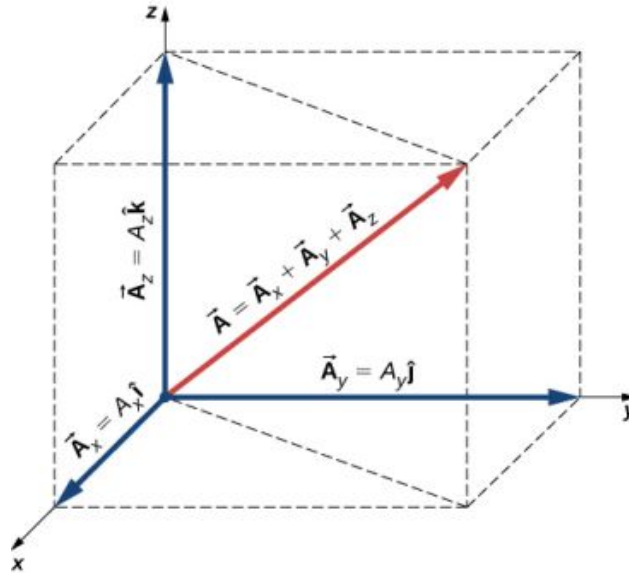


FIGURE 2.22 A vector in three-dimensional space is the vector sum of its three vector components.



EXAMPLE 2.7

Takeoff of a Drone

During a takeoff of IAI Heron (Figure 2.23), its position with respect to a control tower is 100 m above the ground, 300 m to the east, and 200 m to the north. One minute later, its position is 250 m above the ground, 1200 m to the east, and 2100 m to the north. What is the drone's displacement vector with respect to the control tower? What is the magnitude of its displacement vector?



FIGURE 2.23 The drone IAI Heron in flight. (credit: SSgt Reynaldo Ramon, USAF)

Strategy

We take the origin of the Cartesian coordinate system as the control tower. The direction of the $+x$ -axis is given by unit vector $\hat{i}$ to the east, the direction of the $+y$ -axis is given by unit vector $\hat{j}$ to the north, and the direction of the $+z$ -axis is given by unit vector $\hat{k}$, which points up from the ground. The drone's first position is the origin (or, equivalently, the beginning) of the displacement vector and its second position is the end of the displacement vector.

Solution

We identify $b(300.0 \text{ m}, 200.0 \text{ m}, 100.0 \text{ m})$ and $e(1200 \text{ m}, 2100 \text{ m}, 250 \text{ m})$, and use [Equation 2.13](#) and [Equation 2.20](#) to find the scalar components of the drone's displacement vector:

$$\begin{cases} D_x = x_e - x_b = 1200.0 \text{ m} - 300.0 \text{ m} = 900.0 \text{ m}, \\ D_y = y_e - y_b = 2100.0 \text{ m} - 200.0 \text{ m} = 1900.0 \text{ m}, \\ D_z = z_e - z_b = 250.0 \text{ m} - 100.0 \text{ m} = 150.0 \text{ m}. \end{cases}$$

We substitute these components into [Equation 2.19](#) to find the displacement vector:

$$\vec{D} = D_x \hat{i} + D_y \hat{j} + D_z \hat{k} = 900.0 \text{ m} \hat{i} + 1900.0 \text{ m} \hat{j} + 150.0 \text{ m} \hat{k} = (0.90 \hat{i} + 1.90 \hat{j} + 0.15 \hat{k}) \text{ km}.$$

We substitute into [Equation 2.21](#) to find the magnitude of the displacement:

$$D = \sqrt{D_x^2 + D_y^2 + D_z^2} = \sqrt{(0.90 \text{ km})^2 + (1.90 \text{ km})^2 + (0.15 \text{ km})^2} = 2.11 \text{ km}.$$

✓ CHECK YOUR UNDERSTANDING 2.7

If the average velocity vector of the drone in the displacement in [Example 2.7](#) is $\vec{u} = (15.0 \hat{i} + 31.7 \hat{j} + 2.5 \hat{k}) \text{ m/s}$, what is the magnitude of the drone's velocity vector?

2.3 Algebra of Vectors

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Apply analytical methods of vector algebra to find resultant vectors and to solve vector equations for unknown vectors.
- Interpret physical situations in terms of vector expressions.

Vectors can be added together and multiplied by scalars. Vector addition is associative ([Equation 2.8](#)) and commutative ([Equation 2.7](#)), and vector multiplication by a sum of scalars is distributive ([Equation 2.9](#)). Also, scalar multiplication by a sum of vectors is distributive:

$$\alpha(\vec{A} + \vec{B}) = \alpha\vec{A} + \alpha\vec{B}.$$

2.22

In this equation, α is any number (a scalar). For example, a vector antiparallel to vector $\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$ can be expressed simply by multiplying $\vec{A}$ by the scalar $\alpha = -1$:

$$-\vec{A} = -A_x\hat{i} - A_y\hat{j} - A_z\hat{k}. \quad 2.23$$



EXAMPLE 2.8

Direction of Motion

In a Cartesian coordinate system where $\hat{i}$ denotes geographic east, $\hat{j}$ denotes geographic north, and $\hat{k}$ denotes altitude above sea level, a military convoy advances its position through unknown territory with velocity $\vec{v} = (4.0\hat{i} + 3.0\hat{j} + 0.1\hat{k})\text{km/h}$. If the convoy had to retreat, in what geographic direction would it be moving?

Solution

The velocity vector has the third component $\vec{v}_z = (+0.1\text{km/h})\hat{k}$, which says the convoy is climbing at a rate of 100 m/h through mountainous terrain. At the same time, its velocity is 4.0 km/h to the east and 3.0 km/h to the north, so it moves on the ground in direction $\tan^{-1}(3/4) \approx 37^\circ$ north of east. If the convoy had to retreat, its new velocity vector $\vec{u}$ would have to be antiparallel to $\vec{v}$ and be in the form $\vec{u} = -\alpha\vec{v}$, where α is a positive number. Thus, the velocity of the retreat would be $\vec{u} = \alpha(-4.0\hat{i} - 3.0\hat{j} - 0.1\hat{k})\text{km/h}$. The negative sign of the third component indicates the convoy would be descending. The direction angle of the retreat velocity is $\tan^{-1}(-3\alpha/-4\alpha) \approx 37^\circ$ south of west. Therefore, the convoy would be moving on the ground in direction 37° south of west while descending on its way back.

The generalization of the number zero to vector algebra is called the **null vector**, denoted by $\vec{0}$. All components of the null vector are zero, $\vec{0} = 0\hat{i} + 0\hat{j} + 0\hat{k}$, so the null vector has no length and no direction.

Two vectors $\vec{A}$ and $\vec{B}$ are **equal vectors** if and only if their difference is the null vector:

$$\vec{0} = \vec{A} - \vec{B} = (A_x\hat{i} + A_y\hat{j} + A_z\hat{k}) - (B_x\hat{i} + B_y\hat{j} + B_z\hat{k}) = (A_x - B_x)\hat{i} + (A_y - B_y)\hat{j} + (A_z - B_z)\hat{k}.$$

This vector equation means we must have simultaneously $A_x - B_x = 0$, $A_y - B_y = 0$, and $A_z - B_z = 0$. Hence, we can write $\vec{A} = \vec{B}$ if and only if the corresponding components of vectors $\vec{A}$ and $\vec{B}$ are equal:

$$\vec{A} = \vec{B} \Leftrightarrow \begin{cases} A_x = B_x \\ A_y = B_y \\ A_z = B_z \end{cases} \quad 2.24$$

Two vectors are equal when their corresponding scalar components are equal.

Resolving vectors into their scalar components (i.e., finding their scalar components) and expressing them analytically in vector component form (given by Equation 2.19) allows us to use vector algebra to find sums or differences of many vectors *analytically* (i.e., without using graphical methods). For example, to find the resultant of two vectors $\vec{A}$ and $\vec{B}$, we simply add them component by component, as follows:

$$\vec{R} = \vec{A} + \vec{B} = (A_x\hat{i} + A_y\hat{j} + A_z\hat{k}) + (B_x\hat{i} + B_y\hat{j} + B_z\hat{k}) = (A_x + B_x)\hat{i} + (A_y + B_y)\hat{j} + (A_z + B_z)\hat{k}.$$

In this way, using Equation 2.24, scalar components of the resultant vector $\vec{R} = R_x\hat{i} + R_y\hat{j} + R_z\hat{k}$ are the sums of corresponding scalar components of vectors $\vec{A}$ and $\vec{B}$:

$$\begin{cases} R_x = A_x + B_x, \\ R_y = A_y + B_y, \\ R_z = A_z + B_z. \end{cases}$$

Analytical methods can be used to find components of a resultant of many vectors. For example, if we are to sum up N vectors $\vec{F}_1, \vec{F}_2, \vec{F}_3, \dots, \vec{F}_N$, where each vector is $\vec{F}_k = F_{kx}\hat{i} + F_{ky}\hat{j} + F_{kz}\hat{k}$, the resultant vector $\vec{F}_R$ is

$$\begin{aligned}\vec{F}_R &= \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \dots + \vec{F}_N = \sum_{k=1}^N \vec{F}_k = \sum_{k=1}^N (F_{kx}\hat{i} + F_{ky}\hat{j} + F_{kz}\hat{k}) \\ &= \left(\sum_{k=1}^N F_{kx} \right) \hat{i} + \left(\sum_{k=1}^N F_{ky} \right) \hat{j} + \left(\sum_{k=1}^N F_{kz} \right) \hat{k}.\end{aligned}$$

Therefore, scalar components of the resultant vector are

$$\begin{cases} F_{Rx} = \sum_{k=1}^N F_{kx} = F_{1x} + F_{2x} + \dots + F_{Nx} \\ F_{Ry} = \sum_{k=1}^N F_{ky} = F_{1y} + F_{2y} + \dots + F_{Ny} \\ F_{Rz} = \sum_{k=1}^N F_{kz} = F_{1z} + F_{2z} + \dots + F_{Nz}. \end{cases} \quad 2.25$$

Having found the scalar components, we can write the resultant in vector component form:

$$\vec{F}_R = F_{Rx}\hat{i} + F_{Ry}\hat{j} + F_{Rz}\hat{k}.$$

Analytical methods for finding the resultant and, in general, for solving vector equations are very important in physics because many physical quantities are vectors. For example, we use this method in kinematics to find resultant displacement vectors and resultant velocity vectors, in mechanics to find resultant force vectors and the resultants of many derived vector quantities, and in electricity and magnetism to find resultant electric or magnetic vector fields.



EXAMPLE 2.9

Analytical Computation of a Resultant

Three displacement vectors $\vec{A}$, $\vec{B}$, and $\vec{C}$ in a plane (Figure 2.13) are specified by their magnitudes $A = 10.0$, $B = 7.0$, and $C = 8.0$, respectively, and by their respective direction angles with the horizontal direction $\alpha = 35^\circ$, $\beta = -110^\circ$, and $\gamma = 30^\circ$. The physical units of the magnitudes are centimeters. Resolve the vectors to their scalar components and find the following vector sums: (a) $\vec{R} = \vec{A} + \vec{B} + \vec{C}$, (b) $\vec{D} = \vec{A} - \vec{B}$, and (c) $\vec{S} = \vec{A} - 3\vec{B} + \vec{C}$.

Strategy

First, we use Equation 2.17 to find the scalar components of each vector and then we express each vector in its vector component form given by Equation 2.12. Then, we use analytical methods of vector algebra to find the resultants.

Solution

We resolve the given vectors to their scalar components:

$$\begin{cases} A_x = A \cos \alpha = (10.0 \text{ cm}) \cos 35^\circ = 8.19 \text{ cm} \\ A_y = A \sin \alpha = (10.0 \text{ cm}) \sin 35^\circ = 5.73 \text{ cm} \\ B_x = B \cos \beta = (7.0 \text{ cm}) \cos (-110^\circ) = -2.39 \text{ cm} \\ B_y = B \sin \beta = (7.0 \text{ cm}) \sin (-110^\circ) = -6.58 \text{ cm} \\ C_x = C \cos \gamma = (8.0 \text{ cm}) \cos 30^\circ = 6.93 \text{ cm} \\ C_y = C \sin \gamma = (8.0 \text{ cm}) \sin 30^\circ = 4.00 \text{ cm} \end{cases}$$

For (a) we may substitute directly into [Equation 2.25](#) to find the scalar components of the resultant:

$$\begin{cases} R_x = A_x + B_x + C_x = 8.19 \text{ cm} - 2.39 \text{ cm} + 6.93 \text{ cm} = 12.73 \text{ cm} \\ R_y = A_y + B_y + C_y = 5.73 \text{ cm} - 6.58 \text{ cm} + 4.00 \text{ cm} = 3.15 \text{ cm} \end{cases}$$

Therefore, the resultant vector is $\vec{R} = R_x \hat{i} + R_y \hat{j} = (12.7\hat{i} + 3.1\hat{j})\text{cm}$.

For (b), we may want to write the vector difference as

$$\vec{D} = \vec{A} - \vec{B} = (A_x \hat{i} + A_y \hat{j}) - (B_x \hat{i} + B_y \hat{j}) = (A_x - B_x) \hat{i} + (A_y - B_y) \hat{j}.$$

Then, the scalar components of the vector difference are

$$\begin{cases} D_x = A_x - B_x = 8.19 \text{ cm} - (-2.39 \text{ cm}) = 10.58 \text{ cm} \\ D_y = A_y - B_y = 5.73 \text{ cm} - (-6.58 \text{ cm}) = 12.31 \text{ cm} \end{cases}$$

Hence, the difference vector is $\vec{D} = D_x \hat{i} + D_y \hat{j} = (10.6\hat{i} + 12.3\hat{j})\text{cm}$.

For (c), we can write vector $\vec{S}$ in the following explicit form:

$$\begin{aligned} \vec{S} &= \vec{A} - 3\vec{B} + \vec{C} = (A_x \hat{i} + A_y \hat{j}) - 3(B_x \hat{i} + B_y \hat{j}) + (C_x \hat{i} + C_y \hat{j}) \\ &= (A_x - 3B_x + C_x) \hat{i} + (A_y - 3B_y + C_y) \hat{j}. \end{aligned}$$

Then, the scalar components of $\vec{S}$ are

$$\begin{cases} S_x = A_x - 3B_x + C_x = 8.19 \text{ cm} - 3(-2.39 \text{ cm}) + 6.93 \text{ cm} = 22.29 \text{ cm} \\ S_y = A_y - 3B_y + C_y = 5.73 \text{ cm} - 3(-6.58 \text{ cm}) + 4.00 \text{ cm} = 29.47 \text{ cm} \end{cases}$$

The vector is $\vec{S} = S_x \hat{i} + S_y \hat{j} = (22.3\hat{i} + 29.5\hat{j})\text{cm}$.

Significance

Having found the vector components, we can illustrate the vectors by graphing or we can compute magnitudes and direction angles, as shown in [Figure 2.24](#). Results for the magnitudes in (b) and (c) can be compared with results for the same problems obtained with the graphical method, shown in [Figure 2.14](#) and [Figure 2.15](#). Notice that the analytical method produces exact results and its accuracy is not limited by the resolution of a ruler or a protractor, as it was with the graphical method used in [Example 2.2](#) for finding this same resultant.

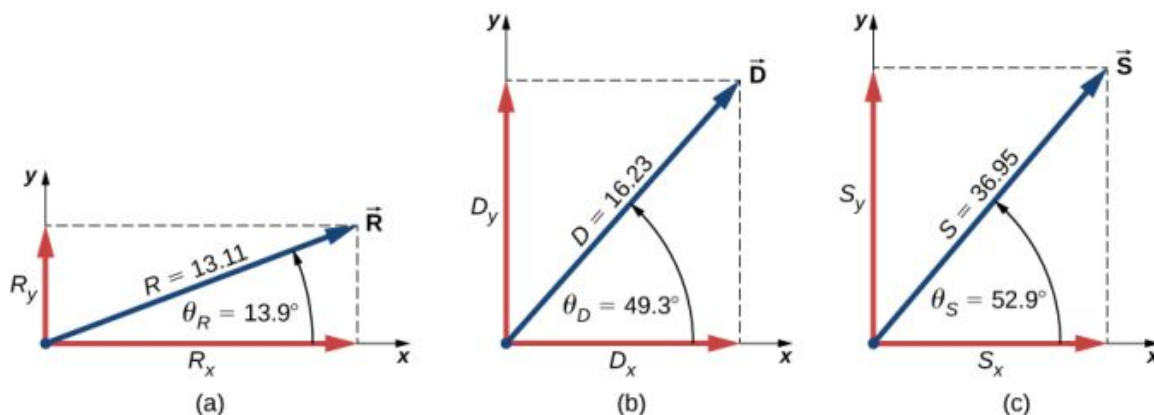


FIGURE 2.24 Graphical illustration of the solutions obtained analytically in [Example 2.9](#).

✓ CHECK YOUR UNDERSTANDING 2.8

Three displacement vectors $\vec{A}$, $\vec{B}$, and $\vec{F}$ ([Figure 2.13](#)) are specified by their magnitudes $A = 10.00$, $B = 7.00$, and $F = 20.00$, respectively, and by their respective direction angles with the horizontal direction $\alpha = 35^\circ$, $\beta = -110^\circ$, and

$\varphi = 110^\circ$. The physical units of the magnitudes are centimeters. Use the analytical method to find vector $\vec{G} = \vec{A} + 2\vec{B} - \vec{F}$. Verify that $G = 28.15$ cm and that $\theta_G = -68.65^\circ$.

EXAMPLE 2.10

The Tug-of-War Game

Four dogs named Astro, Balto, Clifford, and Dug play a tug-of-war game with a toy (Figure 2.25). Astro pulls on the toy in direction $\alpha = 55^\circ$ south of east, Balto pulls in direction $\beta = 60^\circ$ east of north, and Clifford pulls in direction $\gamma = 55^\circ$ west of north. Astro pulls strongly with 160.0 units of force (N), which we abbreviate as $A = 160.0$ N. Balto pulls even stronger than Astro with a force of magnitude $B = 200.0$ N, and Clifford pulls with a force of magnitude $C = 140.0$ N. When Dug pulls on the toy in such a way that his force balances out the resultant of the other three forces, the toy does not move in any direction. With how big a force and in what direction must Dug pull on the toy for this to happen?

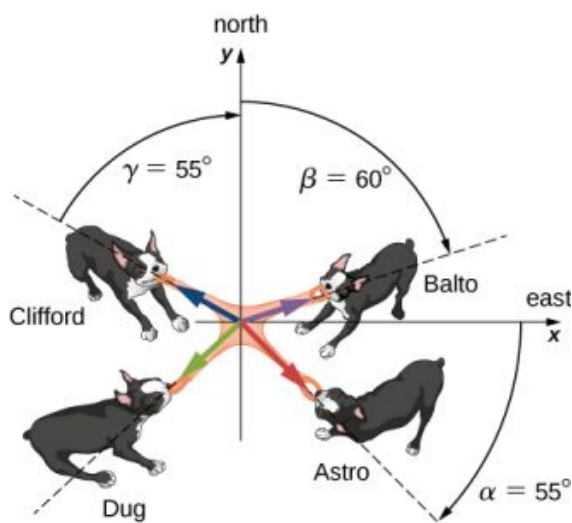


FIGURE 2.25 Four dogs play a tug-of-war game with a toy.

Strategy

We assume that east is the direction of the positive x-axis and north is the direction of the positive y-axis. As in Example 2.9, we have to resolve the three given forces— $\vec{A}$ (the pull from Astro), $\vec{B}$ (the pull from Balto), and $\vec{C}$ (the pull from Clifford)—into their scalar components and then find the scalar components of the resultant vector $\vec{R} = \vec{A} + \vec{B} + \vec{C}$. When the pulling force $\vec{D}$ from Dug balances out this resultant, the sum of $\vec{D}$ and $\vec{R}$ must give the null vector $\vec{D} + \vec{R} = \vec{0}$. This means that $\vec{D} = -\vec{R}$, so the pull from Dug must be antiparallel to $\vec{R}$.

Solution

The direction angles are $\theta_A = -\alpha = -55^\circ$, $\theta_B = 90^\circ - \beta = 30^\circ$, and $\theta_C = 90^\circ + \gamma = 145^\circ$, and substituting them into Equation 2.17 gives the scalar components of the three given forces:

$$\begin{cases} A_x = A \cos \theta_A = (160.0 \text{ N}) \cos (-55^\circ) = +91.8 \text{ N} \\ A_y = A \sin \theta_A = (160.0 \text{ N}) \sin (-55^\circ) = -131.1 \text{ N} \\ B_x = B \cos \theta_B = (200.0 \text{ N}) \cos 30^\circ = +173.2 \text{ N} \\ B_y = B \sin \theta_B = (200.0 \text{ N}) \sin 30^\circ = +100.0 \text{ N} \\ C_x = C \cos \theta_C = (140.0 \text{ N}) \cos 145^\circ = -114.7 \text{ N} \\ C_y = C \sin \theta_C = (140.0 \text{ N}) \sin 145^\circ = +80.3 \text{ N} \end{cases}$$

Now we compute scalar components of the resultant vector $\vec{R} = \vec{A} + \vec{B} + \vec{C}$:

$$\begin{cases} R_x = A_x + B_x + C_x = +91.8 \text{ N} + 173.2 \text{ N} - 114.7 \text{ N} = +150.3 \text{ N} \\ R_y = A_y + B_y + C_y = -131.1 \text{ N} + 100.0 \text{ N} + 80.3 \text{ N} = +49.2 \text{ N} \end{cases}$$

The antiparallel vector to the resultant $\vec{R}$ is

$$\vec{D} = -\vec{R} = -R_x \hat{i} - R_y \hat{j} = (-150.3 \hat{i} - 49.2 \hat{j}) \text{ N}.$$

The magnitude of Dug's pulling force is

$$D = \sqrt{D_x^2 + D_y^2} = \sqrt{(-150.3)^2 + (-49.2)^2} \text{ N} = 158.1 \text{ N}.$$

The direction of Dug's pulling force is

$$\theta = \tan^{-1} \left(\frac{D_y}{D_x} \right) = \tan^{-1} \left(\frac{-49.2 \text{ N}}{-150.3 \text{ N}} \right) = \tan^{-1} \left(\frac{49.2}{150.3} \right) = 18.1^\circ.$$

Dug pulls in the direction 18.1° south of west because both components are negative, which means the pull vector lies in the third quadrant ([Figure 2.19](#)).

CHECK YOUR UNDERSTANDING 2.9

Suppose that Balto in [Example 2.10](#) leaves the game to attend to more important matters, but Astro, Clifford, and Dug continue playing. Astro and Clifford's pull on the toy does not change, but Dug runs around and bites on the toy in a different place. With how big a force and in what direction must Dug pull on the toy now to balance out the combined pulls from Clifford and Astro? Illustrate this situation by drawing a vector diagram indicating all forces involved.

EXAMPLE 2.11

Vector Algebra

Find the magnitude of the vector $\vec{C}$ that satisfies the equation $2\vec{A} - 6\vec{B} + 3\vec{C} = 2\hat{j}$, where $\vec{A} = \hat{i} - 2\hat{k}$ and $\vec{B} = -\hat{j} + \hat{k}/2$.

Strategy

We first solve the given equation for the unknown vector $\vec{C}$. Then we substitute $\vec{A}$ and $\vec{B}$; group the terms along each of the three directions $\hat{i}$, $\hat{j}$, and $\hat{k}$; and identify the scalar components C_x , C_y , and C_z . Finally, we substitute into [Equation 2.21](#) to find magnitude C .

Solution

$$\begin{aligned} 2\vec{A} - 6\vec{B} + 3\vec{C} &= 2\hat{j} \\ 3\vec{C} &= 2\hat{j} - 2\vec{A} + 6\vec{B} \\ \vec{C} &= \frac{2}{3}\hat{j} - \frac{2}{3}\vec{A} + 2\vec{B} \\ &= \frac{2}{3}\hat{j} - \frac{2}{3}(\hat{i} - 2\hat{k}) + 2\left(-\hat{j} + \frac{\hat{k}}{2}\right) = \frac{2}{3}\hat{j} - \frac{2}{3}\hat{i} + \frac{4}{3}\hat{k} - 2\hat{j} + \hat{k} \\ &= -\frac{2}{3}\hat{i} + \left(\frac{2}{3} - 2\right)\hat{j} + \left(\frac{4}{3} + 1\right)\hat{k} \\ &= -\frac{2}{3}\hat{i} - \frac{4}{3}\hat{j} + \frac{7}{3}\hat{k}. \end{aligned}$$

The components are $C_x = -2/3$, $C_y = -4/3$, and $C_z = 7/3$, and substituting into [Equation 2.21](#) gives

$$C = \sqrt{C_x^2 + C_y^2 + C_z^2} = \sqrt{(-2/3)^2 + (-4/3)^2 + (7/3)^2} = \sqrt{23/3}.$$

EXAMPLE 2.12

Displacement of a Skier

Starting at a ski lodge, a cross-country skier goes 5.0 km north, then 3.0 km west, and finally 4.0 km southwest before taking a rest. Find their total displacement vector relative to the lodge when they are at the rest point. How far and in what direction must they ski from the rest point to return directly to the lodge?

Strategy

We assume a rectangular coordinate system with the origin at the ski lodge and with the unit vector $\hat{i}$ pointing east and the unit vector $\hat{j}$ pointing north. There are three displacements: $\vec{D}_1$, $\vec{D}_2$, and $\vec{D}_3$. We identify their magnitudes as $D_1 = 5.0$ km, $D_2 = 3.0$ km, and $D_3 = 4.0$ km. We identify their directions as the angles $\theta_1 = 90^\circ$, $\theta_2 = 180^\circ$, and $\theta_3 = 180^\circ + 45^\circ = 225^\circ$. We resolve each displacement vector to its scalar components and substitute the components into [Equation 2.25](#) to obtain the scalar components of the resultant displacement $\vec{D}$ from the lodge to the rest point. On the way back from the rest point to the lodge, the displacement is $\vec{B} = -\vec{D}$. Finally, we find the magnitude and direction of $\vec{B}$.

Solution

Scalar components of the displacement vectors are

$$\begin{cases} D_{1x} = D_1 \cos \theta_1 = (5.0 \text{ km}) \cos 90^\circ = 0 \\ D_{1y} = D_1 \sin \theta_1 = (5.0 \text{ km}) \sin 90^\circ = 5.0 \text{ km} \\ D_{2x} = D_2 \cos \theta_2 = (3.0 \text{ km}) \cos 180^\circ = -3.0 \text{ km} \\ D_{2y} = D_2 \sin \theta_2 = (3.0 \text{ km}) \sin 180^\circ = 0 \\ D_{3x} = D_3 \cos \theta_3 = (4.0 \text{ km}) \cos 225^\circ = -2.8 \text{ km} \\ D_{3y} = D_3 \sin \theta_3 = (4.0 \text{ km}) \sin 225^\circ = -2.8 \text{ km} \end{cases}$$

Scalar components of the net displacement vector are

$$\begin{cases} D_x = D_{1x} + D_{2x} + D_{3x} = (0 - 3.0 - 2.8) \text{ km} = -5.8 \text{ km} \\ D_y = D_{1y} + D_{2y} + D_{3y} = (5.0 + 0 - 2.8) \text{ km} = +2.2 \text{ km} \end{cases}$$

Hence, the skier's net displacement vector is $\vec{D} = D_x \hat{i} + D_y \hat{j} = (-5.8 \hat{i} + 2.2 \hat{j}) \text{ km}$. On the way back to the lodge, their displacement is $\vec{B} = -\vec{D} = -(-5.8 \hat{i} + 2.2 \hat{j}) \text{ km} = (5.8 \hat{i} - 2.2 \hat{j}) \text{ km}$. Its magnitude is

$$B = \sqrt{B_x^2 + B_y^2} = \sqrt{(5.8)^2 + (-2.2)^2} \text{ km} = 6.2 \text{ km} \text{ and its direction angle is } \theta = \tan^{-1}(-2.2/5.8) = -20.8^\circ.$$

Therefore, to return to the lodge, they must go 6.2 km in a direction about 21° south of east.

Significance

Notice that no figure is needed to solve this problem by the analytical method. Figures are required when using a graphical method; however, we can check if our solution makes sense by sketching it, which is a useful final step in solving any vector problem.

EXAMPLE 2.13

Displacement of a Jogger

A jogger runs up a flight of 200 identical steps to the top of a hill and then runs along the top of the hill 50.0 m before he stops at a drinking fountain ([Figure 2.26](#)). His displacement vector from point A at the bottom of the steps to point B at the fountain is $\vec{D}_{AB} = (-90.0 \hat{i} + 30.0 \hat{j}) \text{ m}$. What is the height and width of each step in the flight? What is the actual distance the jogger covers? If he makes a loop and returns to point A, what is his net displacement vector?

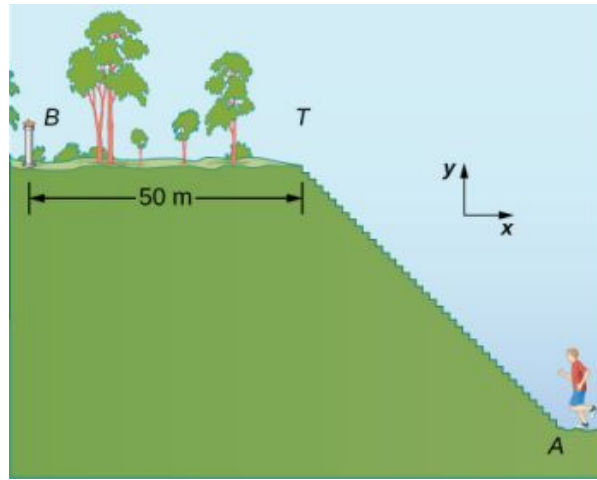


FIGURE 2.26 A jogger runs up a flight of steps.

Strategy

The displacement vector $\vec{D}_{AB}$ is the vector sum of the jogger's displacement vector $\vec{D}_{AT}$ along the stairs (from point A at the bottom of the stairs to point T at the top of the stairs) and his displacement vector $\vec{D}_{TB}$ on the top of the hill (from point T at the top of the stairs to the fountain at point B). We must find the horizontal and the vertical components of $\vec{D}_{AT}$. If each step has width w and height h , the horizontal component of $\vec{D}_{AT}$ must have a length of $200w$ and the vertical component must have a length of $200h$. The actual distance the jogger covers is the sum of the distance he runs up the stairs and the distance of 50.0 m that he runs along the top of the hill.

Solution

In the coordinate system indicated in Figure 2.26, the jogger's displacement vector on the top of the hill is

$\vec{D}_{TB} = (-50.0 \text{ m})\hat{i}$. His net displacement vector is

$$\vec{D}_{AB} = \vec{D}_{AT} + \vec{D}_{TB}.$$

Therefore, his displacement vector $\vec{D}_{AT}$ along the stairs is

$$\begin{aligned}\vec{D}_{AT} &= \vec{D}_{AB} - \vec{D}_{TB} = (-90.0\hat{i} + 30.0\hat{j})\text{m} - (-50.0\text{m})\hat{i} = [(-90.0 + 50.0)\hat{i} + 30.0\hat{j}]\text{m} \\ &= (-40.0\hat{i} + 30.0\hat{j})\text{m}.\end{aligned}$$

Its scalar components are $D_{ATx} = -40.0 \text{ m}$ and $D_{ATy} = 30.0 \text{ m}$. Therefore, we must have

$$200w = |-40.0|\text{m} \text{ and } 200h = 30.0 \text{ m}.$$

Hence, the step width is $w = 40.0 \text{ m}/200 = 0.2 \text{ m} = 20 \text{ cm}$, and the step height is $h = 30.0 \text{ m}/200 = 0.15 \text{ m} = 15 \text{ cm}$. The distance that the jogger covers along the stairs is

$$D_{AT} = \sqrt{D_{ATx}^2 + D_{ATy}^2} = \sqrt{(-40.0)^2 + (30.0)^2} \text{ m} = 50.0 \text{ m}.$$

Thus, the actual distance he runs is $D_{AT} + D_{TB} = 50.0 \text{ m} + 50.0 \text{ m} = 100.0 \text{ m}$. When he makes a loop and comes back from the fountain to his initial position at point A, the total distance he covers is twice this distance, or 200.0 m. However, his net displacement vector is zero, because when his final position is the same as his initial position, the scalar components of his net displacement vector are zero (Equation 2.13).

In many physical situations, we often need to know the direction of a vector. For example, we may want to know the direction of a magnetic field vector at some point or the direction of motion of an object. We have already said direction is given by a unit vector, which is a dimensionless entity—that is, it has no physical units associated with it. When the vector in question lies along one of the axes in a Cartesian system of coordinates, the answer is simple, because then its unit vector of direction is either parallel or antiparallel to the direction of the unit vector of an axis. For example, the direction of vector $\vec{d} = -5 \text{ m}\hat{i}$ is unit vector $\hat{d} = -\hat{i}$. The general rule of finding the unit vector $\hat{V}$ of

direction for any vector $\vec{V}$ is to divide it by its magnitude V :

$$\hat{V} = \frac{\vec{V}}{V}. \quad 2.26$$

We see from this expression that the unit vector of direction is indeed dimensionless because the numerator and the denominator in Equation 2.26 have the same physical unit. In this way, Equation 2.26 allows us to express the unit vector of direction in terms of unit vectors of the axes. The following example illustrates this principle.



EXAMPLE 2.14

The Unit Vector of Direction

If the velocity vector of the military convoy in Example 2.8 is $\vec{v} = (4.000\hat{i} + 3.000\hat{j} + 0.100\hat{k})\text{km/h}$, what is the unit vector of its direction of motion?

Strategy

The unit vector of the convoy's direction of motion is the unit vector $\hat{v}$ that is parallel to the velocity vector. The unit vector is obtained by dividing a vector by its magnitude, in accordance with Equation 2.26.

Solution

The magnitude of the vector $\vec{v}$ is

$$v = \sqrt{v_x^2 + v_y^2 + v_z^2} = \sqrt{4.000^2 + 3.000^2 + 0.100^2}\text{km/h} = 5.001\text{km/h}.$$

To obtain the unit vector $\hat{v}$, divide $\vec{v}$ by its magnitude:

$$\begin{aligned} \hat{v} &= \frac{\vec{v}}{v} = \frac{(4.000\hat{i} + 3.000\hat{j} + 0.100\hat{k})\text{km/h}}{5.001\text{km/h}} \\ &= \frac{(4.000\hat{i} + 3.000\hat{j} + 0.100\hat{k})}{5.001} \\ &= \frac{4.000}{5.001}\hat{i} + \frac{3.000}{5.001}\hat{j} + \frac{0.100}{5.001}\hat{k} \\ &= (79.98\hat{i} + 59.99\hat{j} + 2.00\hat{k}) \times 10^{-2}. \end{aligned}$$

Significance

Note that when using the analytical method with a calculator, it is advisable to carry out your calculations to at least three decimal places and then round off the final answer to the required number of significant figures, which is the way we performed calculations in this example. If you round off your partial answer too early, you risk your final answer having a huge numerical error, and it may be far off from the exact answer or from a value measured in an experiment.



CHECK YOUR UNDERSTANDING 2.10

Verify that vector $\hat{v}$ obtained in Example 2.14 is indeed a unit vector by computing its magnitude. If the convoy in Example 2.8 was moving across a desert flatland—that is, if the third component of its velocity was zero—what is the unit vector of its direction of motion? Which geographic direction does it represent?

2.4 Products of Vectors

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Explain the difference between the scalar product and the vector product of two vectors.
- Determine the scalar product of two vectors.
- Determine the vector product of two vectors.
- Describe how the products of vectors are used in physics.

A vector can be multiplied by another vector but may not be divided by another vector. There are two kinds of products of vectors used broadly in physics and engineering. One kind of multiplication is a *scalar multiplication of two vectors*. Taking a scalar product of two vectors results in a number (a scalar), as its name indicates. Scalar products are used to define work and energy relations. For example, the work that a force (a vector) performs on an object while causing its displacement (a vector) is defined as a scalar product of the force vector with the displacement vector. A quite different kind of multiplication is a *vector multiplication of vectors*. Taking a vector product of two vectors returns as a result a vector, as its name suggests. Vector products are used to define other derived vector quantities. For example, in describing rotations, a vector quantity called *torque* is defined as a vector product of an applied force (a vector) and its distance from pivot to force (a vector). It is important to distinguish between these two kinds of vector multiplications because the scalar product is a scalar quantity and a vector product is a vector quantity.

The Scalar Product of Two Vectors (the Dot Product)

Scalar multiplication of two vectors yields a scalar product.

SCALAR PRODUCT (DOT PRODUCT)

The **scalar product** $\vec{A} \cdot \vec{B}$ of two vectors $\vec{A}$ and $\vec{B}$ is a number defined by the equation

$$\vec{A} \cdot \vec{B} = AB \cos \varphi, \quad 2.27$$

where φ is the angle between the vectors (shown in [Figure 2.27](#)). The scalar product is also called the **dot product** because of the dot notation that indicates it.

In the definition of the dot product, the direction of angle φ does not matter, and φ can be measured from either of the two vectors to the other because $\cos \varphi = \cos (-\varphi) = \cos (2\pi - \varphi)$. The dot product is a negative number when $90^\circ < \varphi \leq 180^\circ$ and is a positive number when $0^\circ \leq \varphi < 90^\circ$. Moreover, the dot product of two parallel vectors is $\vec{A} \cdot \vec{B} = AB \cos 0^\circ = AB$, and the dot product of two antiparallel vectors is $\vec{A} \cdot \vec{B} = AB \cos 180^\circ = -AB$. The scalar product of two *orthogonal vectors* vanishes: $\vec{A} \cdot \vec{B} = AB \cos 90^\circ = 0$. The scalar product of a vector with itself is the square of its magnitude:

$$\vec{A}^2 \equiv \vec{A} \cdot \vec{A} = AA \cos 0^\circ = A^2. \quad 2.28$$

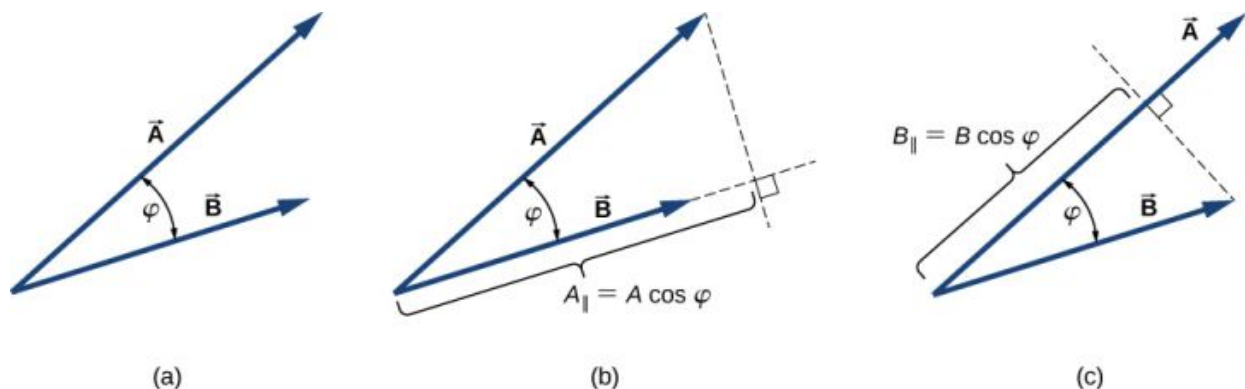


FIGURE 2.27 The scalar product of two vectors. (a) The angle between the two vectors. (b) The orthogonal projection $A_{\parallel}$ of vector $\vec{A}$ onto

the direction of vector $\vec{B}$. (c) The orthogonal projection $B_{\parallel}$ of vector $\vec{B}$ onto the direction of vector $\vec{A}$.



EXAMPLE 2.15

The Scalar Product

For the vectors shown in [Figure 2.13](#), find the scalar product $\vec{A} \cdot \vec{F}$.

Strategy

From [Figure 2.13](#), the magnitudes of vectors $\vec{A}$ and $\vec{F}$ are $A = 10.0$ and $F = 20.0$. Angle θ , between them, is the difference: $\theta = \varphi - \alpha = 110^\circ - 35^\circ = 75^\circ$. Substituting these values into [Equation 2.27](#) gives the scalar product.

Solution

A straightforward calculation gives us

$$\vec{A} \cdot \vec{F} = AF \cos \theta = (10.0)(20.0) \cos 75^\circ = 51.76.$$



CHECK YOUR UNDERSTANDING 2.11

For the vectors given in [Figure 2.13](#), find the scalar products $\vec{A} \cdot \vec{B}$ and $\vec{F} \cdot \vec{C}$.

In the Cartesian coordinate system, scalar products of the unit vector of an axis with other unit vectors of axes always vanish because these unit vectors are orthogonal:

$$\begin{aligned}\hat{i} \cdot \hat{j} &= |\hat{i}| |\hat{j}| \cos 90^\circ = (1)(1)(0) = 0, \\ \hat{i} \cdot \hat{k} &= |\hat{i}| |\hat{k}| \cos 90^\circ = (1)(1)(0) = 0, \\ \hat{k} \cdot \hat{j} &= |\hat{k}| |\hat{j}| \cos 90^\circ = (1)(1)(0) = 0.\end{aligned}\tag{2.29}$$

In these equations, we use the fact that the magnitudes of all unit vectors are one: $|\hat{i}| = |\hat{j}| = |\hat{k}| = 1$. For unit vectors of the axes, [Equation 2.28](#) gives the following identities:

$$\hat{i} \cdot \hat{i} = i^2 = \hat{j} \cdot \hat{j} = j^2 = \hat{k} \cdot \hat{k} = k^2 = 1.\tag{2.30}$$

The scalar product $\vec{A} \cdot \vec{B}$ can also be interpreted as either the product of B with the projection $A_{\parallel}$ of vector $\vec{A}$ onto the direction of vector $\vec{B}$ ([Figure 2.27\(b\)](#)) or the product of A with the projection $B_{\parallel}$ of vector $\vec{B}$ onto the direction of vector $\vec{A}$ ([Figure 2.27\(c\)](#)):

$$\begin{aligned}\vec{A} \cdot \vec{B} &= AB \cos \varphi \\ &= B(A \cos \varphi) = BA_{\parallel} \\ &= A(B \cos \varphi) = AB_{\parallel}.\end{aligned}$$

For example, in the rectangular coordinate system in a plane, the scalar x-component of a vector is its dot product with the unit vector $\hat{i}$, and the scalar y-component of a vector is its dot product with the unit vector $\hat{j}$:

$$\begin{cases} \vec{A} \cdot \hat{i} = |\vec{A}| |\hat{i}| \cos \theta_A = A \cos \theta_A = A_x \\ \vec{A} \cdot \hat{j} = |\vec{A}| |\hat{j}| \cos (90^\circ - \theta_A) = A \sin \theta_A = A_y \end{cases}.$$

Scalar multiplication of vectors is commutative,

$$\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A},\tag{2.31}$$

and obeys the distributive law:

$$\vec{A} \cdot (\vec{B} + \vec{C}) = \vec{A} \cdot \vec{B} + \vec{A} \cdot \vec{C}. \quad 2.32$$

We can use the commutative and distributive laws to derive various relations for vectors, such as expressing the dot product of two vectors in terms of their scalar components.

✓ CHECK YOUR UNDERSTANDING 2.12

For vector $\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$ in a rectangular coordinate system, use [Equation 2.29](#) through [Equation 2.32](#) to show that $\vec{A} \cdot \hat{i} = A_x$, $\vec{A} \cdot \hat{j} = A_y$, and $\vec{A} \cdot \hat{k} = A_z$.

When the vectors in [Equation 2.27](#) are given in their vector component forms,

$$\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k} \text{ and } \vec{B} = B_x\hat{i} + B_y\hat{j} + B_z\hat{k},$$

we can compute their scalar product as follows:

$$\begin{aligned} \vec{A} \cdot \vec{B} &= (A_x\hat{i} + A_y\hat{j} + A_z\hat{k}) \cdot (B_x\hat{i} + B_y\hat{j} + B_z\hat{k}) \\ &= A_xB_x\hat{i} \cdot \hat{i} + A_xB_y\hat{i} \cdot \hat{j} + A_xB_z\hat{i} \cdot \hat{k} \\ &\quad + A_yB_x\hat{j} \cdot \hat{i} + A_yB_y\hat{j} \cdot \hat{j} + A_yB_z\hat{j} \cdot \hat{k} \\ &\quad + A_zB_x\hat{k} \cdot \hat{i} + A_zB_y\hat{k} \cdot \hat{j} + A_zB_z\hat{k} \cdot \hat{k}. \end{aligned}$$

Since scalar products of two different unit vectors of axes give zero, and scalar products of unit vectors with themselves give one (see [Equation 2.29](#) and [Equation 2.30](#)), there are only three nonzero terms in this expression. Thus, the scalar product simplifies to

$$\vec{A} \cdot \vec{B} = A_xB_x + A_yB_y + A_zB_z. \quad 2.33$$

We can use [Equation 2.33](#) for the scalar product in terms of scalar components of vectors to find the angle between two vectors. When we divide [Equation 2.27](#) by AB , we obtain the equation for $\cos \varphi$, into which we substitute [Equation 2.33](#):

$$\cos \varphi = \frac{\vec{A} \cdot \vec{B}}{AB} = \frac{A_xB_x + A_yB_y + A_zB_z}{AB}. \quad 2.34$$

Angle φ between vectors $\vec{A}$ and $\vec{B}$ is obtained by taking the inverse cosine of the expression in [Equation 2.34](#).



EXAMPLE 2.16

Angle between Two Forces

Three dogs are pulling on a stick in different directions, as shown in [Figure 2.28](#). The first dog pulls with force $\vec{F}_1 = (10.0\hat{i} - 20.4\hat{j} + 2.0\hat{k})\text{N}$, the second dog pulls with force $\vec{F}_2 = (-15.0\hat{i} - 6.2\hat{k})\text{N}$, and the third dog pulls with force $\vec{F}_3 = (5.0\hat{i} + 12.5\hat{j})\text{N}$. What is the angle between forces $\vec{F}_1$ and $\vec{F}_2$?

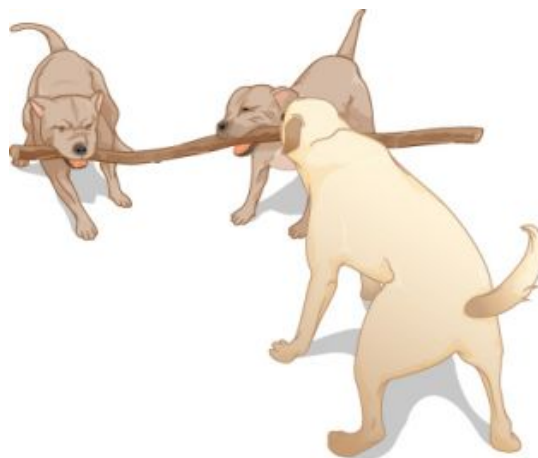


FIGURE 2.28 Three dogs are playing with a stick.

Strategy

The components of force vector $\vec{F}_1$ are $F_{1x} = 10.0 \text{ N}$, $F_{1y} = -20.4 \text{ N}$, and $F_{1z} = 2.0 \text{ N}$, whereas those of force vector $\vec{F}_2$ are $F_{2x} = -15.0 \text{ N}$, $F_{2y} = 0.0 \text{ N}$, and $F_{2z} = -6.2 \text{ N}$. Computing the scalar product of these vectors and their magnitudes, and substituting into [Equation 2.34](#) gives the angle of interest.

Solution

The magnitudes of forces $\vec{F}_1$ and $\vec{F}_2$ are

$$F_1 = \sqrt{F_{1x}^2 + F_{1y}^2 + F_{1z}^2} = \sqrt{10.0^2 + 20.4^2 + 2.0^2} \text{ N} = 22.8 \text{ N}$$

and

$$F_2 = \sqrt{F_{2x}^2 + F_{2y}^2 + F_{2z}^2} = \sqrt{15.0^2 + 6.2^2} \text{ N} = 16.2 \text{ N}.$$

Substituting the scalar components into [Equation 2.33](#) yields the scalar product

$$\begin{aligned} \vec{F}_1 \cdot \vec{F}_2 &= F_{1x}F_{2x} + F_{1y}F_{2y} + F_{1z}F_{2z} \\ &= (10.0 \text{ N})(-15.0 \text{ N}) + (-20.4 \text{ N})(0.0 \text{ N}) + (2.0 \text{ N})(-6.2 \text{ N}) \\ &= -162.4 \text{ N}^2. \end{aligned}$$

Finally, substituting everything into [Equation 2.34](#) gives the angle

$$\cos \varphi = \frac{\vec{F}_1 \cdot \vec{F}_2}{F_1 F_2} = \frac{-162.4 \text{ N}^2}{(22.8 \text{ N})(16.2 \text{ N})} = -0.439 \Rightarrow \varphi = \cos^{-1}(-0.439) = 116.0^\circ.$$

Significance

Notice that when vectors are given in terms of the unit vectors of axes, we can find the angle between them without knowing the specifics about the geographic directions the unit vectors represent. Here, for example, the $+x$ -direction might be to the east and the $+y$ -direction might be to the north. But, the angle between the forces in the problem is the same if the $+x$ -direction is to the west and the $+y$ -direction is to the south.

CHECK YOUR UNDERSTANDING 2.13

Find the angle between forces $\vec{F}_1$ and $\vec{F}_3$ in [Example 2.16](#).

EXAMPLE 2.17

The Work of a Force

When force $\vec{F}$ pulls on an object and when it causes its displacement $\vec{D}$, we say the force performs work. The amount of work the force does is the scalar product $\vec{F} \cdot \vec{D}$. If the stick in [Example 2.16](#) moves momentarily and gets displaced by vector $\vec{D} = (-7.9\hat{j} - 4.2\hat{k})$ cm, how much work is done by the third dog in [Example 2.16](#)?

Strategy

We compute the scalar product of displacement vector $\vec{D}$ with force vector $\vec{F}_3 = (5.0\hat{i} + 12.5\hat{j})$ N, which is the pull from the third dog. Let's use W_3 to denote the work done by force $\vec{F}_3$ on displacement $\vec{D}$.

Solution

Calculating the work is a straightforward application of the dot product:

$$\begin{aligned} W_3 &= \vec{F}_3 \cdot \vec{D} = F_{3x}D_x + F_{3y}D_y + F_{3z}D_z \\ &= (5.0 \text{ N})(0.0 \text{ cm}) + (12.5 \text{ N})(-7.9 \text{ cm}) + (0.0 \text{ N})(-4.2 \text{ cm}) \\ &= -98.7 \text{ N} \cdot \text{cm}. \end{aligned}$$

Significance

The SI unit of work is called the joule (J), where $1 \text{ J} = 1 \text{ N} \cdot \text{m}$. The unit $\text{cm} \cdot \text{N}$ can be written as $10^{-2} \text{ m} \cdot \text{N} = 10^{-2} \text{ J}$, so the answer can be expressed as $W_3 = -0.9875 \text{ J} \approx -1.0 \text{ J}$.

CHECK YOUR UNDERSTANDING 2.14

How much work is done by the first dog and by the second dog in [Example 2.16](#) on the displacement in [Example 2.17](#)?

The Vector Product of Two Vectors (the Cross Product)

Vector multiplication of two vectors yields a vector product.

VECTOR PRODUCT (CROSS PRODUCT)

The **vector product** of two vectors $\vec{A}$ and $\vec{B}$ is denoted by $\vec{A} \times \vec{B}$ and is often referred to as a **cross product**. The vector product is a vector that has its direction perpendicular to both vectors $\vec{A}$ and $\vec{B}$. In other words, vector $\vec{A} \times \vec{B}$ is perpendicular to the plane that contains vectors $\vec{A}$ and $\vec{B}$, as shown in [Figure 2.29](#). The magnitude of the vector product is defined as

$$|\vec{A} \times \vec{B}| = AB \sin \varphi, \quad 2.35$$

where angle φ , between the two vectors, is measured from vector $\vec{A}$ (first vector in the product) to vector $\vec{B}$ (second vector in the product), as indicated in [Figure 2.29](#), and is between 0° and 180° .

According to [Equation 2.35](#), the vector product vanishes for pairs of vectors that are either parallel ($\varphi = 0^\circ$) or antiparallel ($\varphi = 180^\circ$) because $\sin 0^\circ = \sin 180^\circ = 0$.

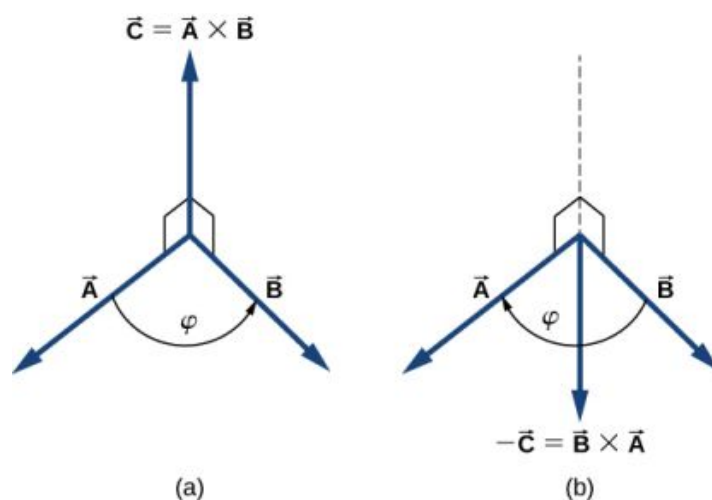


FIGURE 2.29 The vector product of two vectors is drawn in three-dimensional space. (a) The vector product $\vec{A} \times \vec{B}$ is a vector perpendicular to the plane that contains vectors $\vec{A}$ and $\vec{B}$. Small squares drawn in perspective mark right angles between $\vec{A}$ and $\vec{C}$, and between $\vec{B}$ and $\vec{C}$ so that if $\vec{A}$ and $\vec{B}$ lie on the floor, vector $\vec{C}$ points vertically upward to the ceiling. (b) The vector product $\vec{B} \times \vec{A}$ is a vector antiparallel to vector $\vec{A} \times \vec{B}$.

On the line perpendicular to the plane that contains vectors $\vec{A}$ and $\vec{B}$ there are two alternative directions—either up or down, as shown in [Figure 2.29](#)—and the direction of the vector product may be either one of them. In the standard right-handed orientation, where the angle between vectors is measured counterclockwise from the first vector, vector $\vec{A} \times \vec{B}$ points *upward*, as seen in [Figure 2.29\(a\)](#). If we reverse the order of multiplication, so that now $\vec{B}$ comes first in the product, then vector $\vec{B} \times \vec{A}$ must point *downward*, as seen in [Figure 2.29\(b\)](#). This means that vectors $\vec{A} \times \vec{B}$ and $\vec{B} \times \vec{A}$ are *antiparallel* to each other and that vector multiplication is *not* commutative but *anticommutative*. The **anticommutative property** means the vector product reverses the sign when the order of multiplication is reversed:

$$\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}.$$

2.36

The **corkscrew right-hand rule** is a common mnemonic used to determine the direction of the vector product. As shown in [Figure 2.30](#), a corkscrew is placed in a direction perpendicular to the plane that contains vectors $\vec{A}$ and $\vec{B}$, and its handle is turned in the direction from the first to the second vector in the product. The direction of the cross product is given by the progression of the corkscrew.

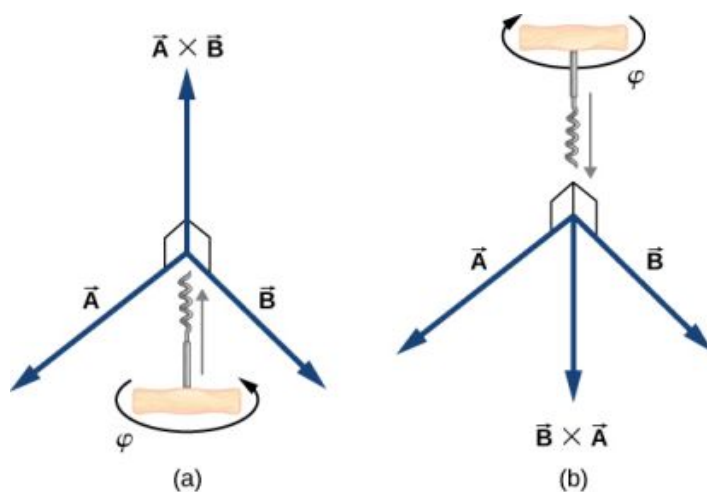


FIGURE 2.30 The corkscrew right-hand rule can be used to determine the direction of the cross product $\vec{A} \times \vec{B}$. Place a corkscrew in the direction perpendicular to the plane that contains vectors $\vec{A}$ and $\vec{B}$, and turn it in the direction from the first to the second vector in the product. The direction of the cross product is given by the progression of the corkscrew. (a) Upward movement means the cross-product vector points up. (b) Downward movement means the cross-product vector points downward.

EXAMPLE 2.18

The Torque of a Force

The mechanical advantage that a familiar tool called a *wrench* provides (Figure 2.31) depends on magnitude F of the applied force, on its direction with respect to the wrench handle, and on how far from the nut this force is applied.

The distance R from the nut to the point where force vector $\vec{F}$ is attached is represented by the radial vector $\vec{R}$. The physical vector quantity that makes the nut turn is called *torque* (denoted by $\vec{\tau}$), and it is the vector product of the distance between the pivot to force with the force: $\vec{\tau} = \vec{R} \times \vec{F}$.

To loosen a rusty nut, a 20.00-N force is applied to the wrench handle at angle $\varphi = 40^\circ$ and at a distance of 0.25 m from the nut, as shown in Figure 2.31(a). Find the magnitude and direction of the torque applied to the nut. What would the magnitude and direction of the torque be if the force were applied at angle $\varphi = 45^\circ$, as shown in Figure 2.31(b)? For what value of angle φ does the torque have the largest magnitude?

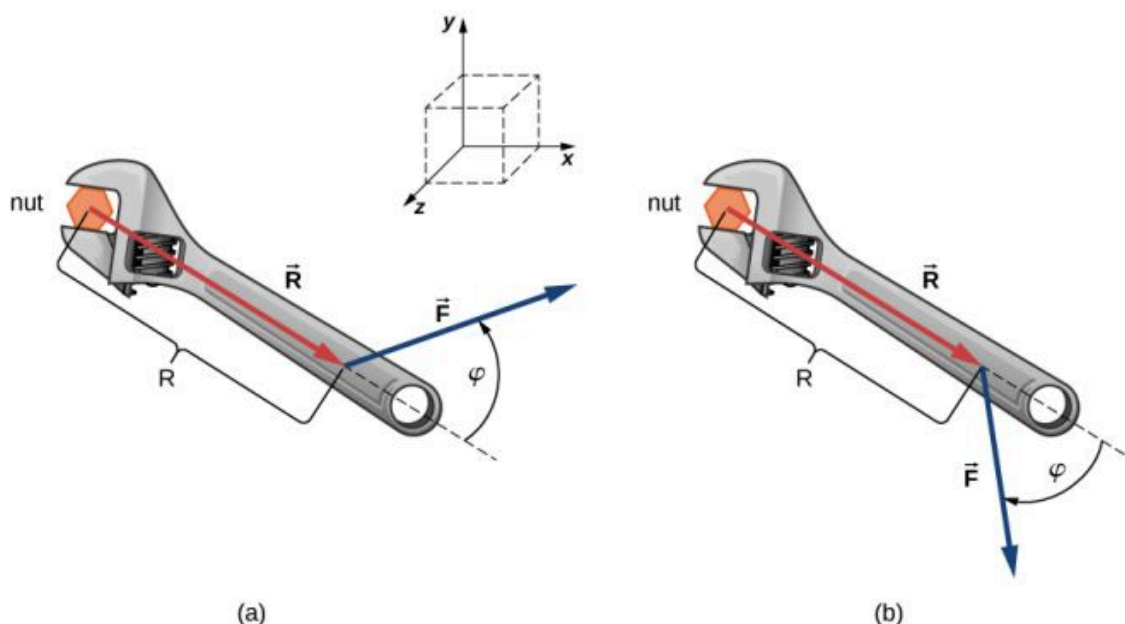


FIGURE 2.31 A wrench provides grip and mechanical advantage in applying torque to turn a nut. (a) Turn counterclockwise to loosen the nut. (b) Turn clockwise to tighten the nut.

Strategy

We adopt the frame of reference shown in Figure 2.31, where vectors $\vec{R}$ and $\vec{F}$ lie in the xy -plane and the origin is at the position of the nut. The radial direction along vector $\vec{R}$ (pointing away from the origin) is the reference direction for measuring the angle φ because $\vec{R}$ is the first vector in the vector product $\vec{\tau} = \vec{R} \times \vec{F}$. Vector $\vec{\tau}$ must lie along the z -axis because this is the axis that is perpendicular to the xy -plane, where both $\vec{R}$ and $\vec{F}$ lie. To compute the magnitude τ , we use Equation 2.35. To find the direction of $\vec{\tau}$, we use the corkscrew right-hand rule (Figure 2.30).

Solution

For the situation in (a), the corkscrew rule gives the direction of $\vec{R} \times \vec{F}$ in the positive direction of the z -axis. Physically, it means the torque vector $\vec{\tau}$ points out of the page, perpendicular to the wrench handle. We identify $F = 20.00$ N and $R = 0.25$ m, and compute the magnitude using Equation 2.35:

$$\tau = |\vec{R} \times \vec{F}| = RF \sin \varphi = (0.25 \text{ m})(20.00 \text{ N}) \sin 40^\circ = 3.21 \text{ N} \cdot \text{m}.$$

For the situation in (b), the corkscrew rule gives the direction of $\vec{R} \times \vec{F}$ in the negative direction of the z -axis. Physically, it means the vector $\vec{\tau}$ points into the page, perpendicular to the wrench handle. The magnitude of this torque is

$$\tau = |\vec{\mathbf{R}} \times \vec{\mathbf{F}}| = RF \sin \varphi = (0.25 \text{ m})(20.00 \text{ N}) \sin 45^\circ = 3.53 \text{ N} \cdot \text{m}.$$

The torque has the largest value when $\sin \varphi = 1$, which happens when $\varphi = 90^\circ$. Physically, it means the wrench is most effective—giving us the best mechanical advantage—when we apply the force perpendicular to the wrench handle. For the situation in this example, this best-torque value is $\tau_{\text{best}} = RF = (0.25 \text{ m})(20.00 \text{ N}) = 5.00 \text{ N} \cdot \text{m}$.

Significance

When solving mechanics problems, we often do not need to use the corkscrew rule at all, as we'll see now in the following equivalent solution. Notice that once we have identified that vector $\vec{\mathbf{R}} \times \vec{\mathbf{F}}$ lies along the z-axis, we can write this vector in terms of the unit vector $\hat{\mathbf{k}}$ of the z-axis:

$$\vec{\mathbf{R}} \times \vec{\mathbf{F}} = RF \sin \varphi \hat{\mathbf{k}}.$$

In this equation, the number that multiplies $\hat{\mathbf{k}}$ is the scalar z-component of the vector $\vec{\mathbf{R}} \times \vec{\mathbf{F}}$. In the computation of this component, care must be taken that the angle φ is measured *counterclockwise* from $\vec{\mathbf{R}}$ (first vector) to $\vec{\mathbf{F}}$ (second vector). Following this principle for the angles, we obtain $RF \sin (+40^\circ) = +3.2 \text{ N} \cdot \text{m}$ for the situation in (a), and we obtain $RF \sin (-45^\circ) = -3.5 \text{ N} \cdot \text{m}$ for the situation in (b). In the latter case, the angle is negative because the graph in [Figure 2.31](#) indicates the angle is measured clockwise; but, the same result is obtained when this angle is measured counterclockwise because $+(360^\circ - 45^\circ) = +315^\circ$ and $\sin (+315^\circ) = \sin (-45^\circ)$. In this way, we obtain the solution without reference to the corkscrew rule. For the situation in (a), the solution is $\vec{\mathbf{R}} \times \vec{\mathbf{F}} = +3.2 \text{ N} \cdot \text{m} \hat{\mathbf{k}}$; for the situation in (b), the solution is $\vec{\mathbf{R}} \times \vec{\mathbf{F}} = -3.5 \text{ N} \cdot \text{m} \hat{\mathbf{k}}$.

CHECK YOUR UNDERSTANDING 2.15

For the vectors given in [Figure 2.13](#), find the vector products $\vec{\mathbf{A}} \times \vec{\mathbf{B}}$ and $\vec{\mathbf{C}} \times \vec{\mathbf{F}}$.

Similar to the dot product ([Equation 2.32](#)), the cross product has the following distributive property:

$$\vec{\mathbf{A}} \times (\vec{\mathbf{B}} + \vec{\mathbf{C}}) = \vec{\mathbf{A}} \times \vec{\mathbf{B}} + \vec{\mathbf{A}} \times \vec{\mathbf{C}}. \quad 2.37$$

The distributive property is applied frequently when vectors are expressed in their component forms, in terms of unit vectors of Cartesian axes.

When we apply the definition of the cross product, [Equation 2.35](#), to unit vectors $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, and $\hat{\mathbf{k}}$ that define the positive x-, y-, and z-directions in space, we find that

$$\hat{\mathbf{i}} \times \hat{\mathbf{i}} = \hat{\mathbf{j}} \times \hat{\mathbf{j}} = \hat{\mathbf{k}} \times \hat{\mathbf{k}} = 0. \quad 2.38$$

All other cross products of these three unit vectors must be vectors of unit magnitudes because $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, and $\hat{\mathbf{k}}$ are orthogonal. For example, for the pair $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$, the magnitude is $|\hat{\mathbf{i}} \times \hat{\mathbf{j}}| = ij \sin 90^\circ = (1)(1)(1) = 1$. The direction of the vector product $\hat{\mathbf{i}} \times \hat{\mathbf{j}}$ must be orthogonal to the xy-plane, which means it must be along the z-axis. The only unit vectors along the z-axis are $-\hat{\mathbf{k}}$ or $+\hat{\mathbf{k}}$. By the corkscrew rule, the direction of vector $\hat{\mathbf{i}} \times \hat{\mathbf{j}}$ must be parallel to the positive z-axis. Therefore, the result of the multiplication $\hat{\mathbf{i}} \times \hat{\mathbf{j}}$ is identical to $+\hat{\mathbf{k}}$. We can repeat similar reasoning for the remaining pairs of unit vectors. The results of these multiplications are

$$\begin{cases} \hat{\mathbf{i}} \times \hat{\mathbf{j}} = +\hat{\mathbf{k}}, \\ \hat{\mathbf{j}} \times \hat{\mathbf{k}} = +\hat{\mathbf{i}}, \\ \hat{\mathbf{k}} \times \hat{\mathbf{i}} = +\hat{\mathbf{j}}. \end{cases} \quad 2.39$$

Notice that in [Equation 2.39](#), the three unit vectors $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, and $\hat{\mathbf{k}}$ appear in the *cyclic order* shown in a diagram in [Figure 2.32](#)(a). The cyclic order means that in the product formula, $\hat{\mathbf{i}}$ follows $\hat{\mathbf{k}}$ and comes before $\hat{\mathbf{j}}$, or $\hat{\mathbf{k}}$ follows $\hat{\mathbf{j}}$ and comes before $\hat{\mathbf{i}}$, or $\hat{\mathbf{j}}$ follows $\hat{\mathbf{i}}$ and comes before $\hat{\mathbf{k}}$. The cross product of two different unit vectors is always a

third unit vector. When two unit vectors in the cross product appear in the cyclic order, the result of such a multiplication is the remaining unit vector, as illustrated in [Figure 2.32\(b\)](#). When unit vectors in the cross product appear in a different order, the result is a unit vector that is antiparallel to the remaining unit vector (i.e., the result is with the minus sign, as shown by the examples in [Figure 2.32\(c\)](#) and [Figure 2.32\(d\)](#)). In practice, when the task is to find cross products of vectors that are given in vector component form, this rule for the cross-multiplication of unit vectors is very useful.

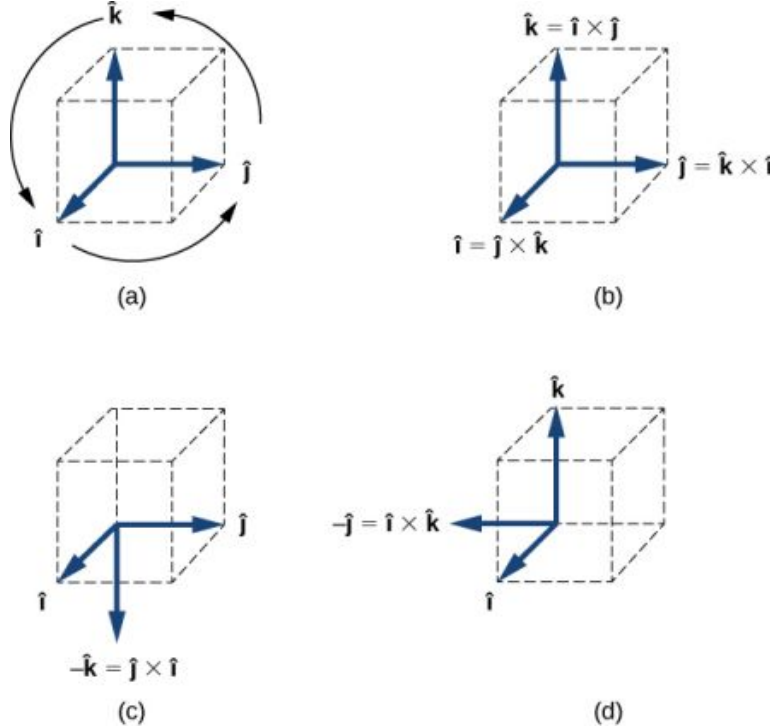


FIGURE 2.32 (a) The diagram of the cyclic order of the unit vectors of the axes. (b) The only cross products where the unit vectors appear in the cyclic order. These products have the positive sign. (c, d) Two examples of cross products where the unit vectors do not appear in the cyclic order. These products have the negative sign.

Suppose we want to find the cross product $\vec{A} \times \vec{B}$ for vectors $\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$ and $\vec{B} = B_x \hat{i} + B_y \hat{j} + B_z \hat{k}$. We can use the distributive property ([Equation 2.37](#)), the anticommutative property ([Equation 2.36](#)), and the results in [Equation 2.38](#) and [Equation 2.39](#) for unit vectors to perform the following algebra:

$$\begin{aligned}
 \vec{A} \times \vec{B} &= (A_x \hat{i} + A_y \hat{j} + A_z \hat{k}) \times (B_x \hat{i} + B_y \hat{j} + B_z \hat{k}) \\
 &= A_x \hat{i} \times (B_x \hat{i} + B_y \hat{j} + B_z \hat{k}) + A_y \hat{j} \times (B_x \hat{i} + B_y \hat{j} + B_z \hat{k}) + A_z \hat{k} \times (B_x \hat{i} + B_y \hat{j} + B_z \hat{k}) \\
 &= A_x B_x \hat{i} \times \hat{i} + A_x B_y \hat{i} \times \hat{j} + A_x B_z \hat{i} \times \hat{k} \\
 &\quad + A_y B_x \hat{j} \times \hat{i} + A_y B_y \hat{j} \times \hat{j} + A_y B_z \hat{j} \times \hat{k} \\
 &\quad + A_z B_x \hat{k} \times \hat{i} + A_z B_y \hat{k} \times \hat{j} + A_z B_z \hat{k} \times \hat{k} \\
 &= A_x B_x (0) + A_x B_y (+\hat{k}) + A_x B_z (-\hat{j}) \\
 &\quad + A_y B_x (-\hat{k}) + A_y B_y (0) + A_y B_z (+\hat{i}) \\
 &\quad + A_z B_x (+\hat{j}) + A_z B_y (-\hat{i}) + A_z B_z (0).
 \end{aligned}$$

When performing algebraic operations involving the cross product, be very careful about keeping the correct order of multiplication because the cross product is anticommutative. The last two steps that we still have to do to complete our task are, first, grouping the terms that contain a common unit vector and, second, factoring. In this way we obtain the following very useful expression for the computation of the cross product:

$$\vec{C} = \vec{A} \times \vec{B} = (A_y B_z - A_z B_y) \hat{i} + (A_z B_x - A_x B_z) \hat{j} + (A_x B_y - A_y B_x) \hat{k}.$$

2.40

In this expression, the scalar components of the cross-product vector are

$$\begin{cases} C_x = A_y B_z - A_z B_y, \\ C_y = A_z B_x - A_x B_z, \\ C_z = A_x B_y - A_y B_x. \end{cases} \quad 2.41$$

When finding the cross product, in practice, we can use either [Equation 2.35](#) or [Equation 2.40](#), depending on which one of them seems to be less complex computationally. They both lead to the same final result. One way to make sure if the final result is correct is to use them both.

EXAMPLE 2.19

A Particle in a Magnetic Field

When moving in a magnetic field, some particles may experience a magnetic force. Without going into details—a detailed study of magnetic phenomena comes in later chapters—let’s acknowledge that the magnetic field $\vec{\mathbf{B}}$ is a vector, the magnetic force $\vec{\mathbf{F}}$ is a vector, and the velocity $\vec{\mathbf{u}}$ of the particle is a vector. The magnetic force vector is proportional to the vector product of the velocity vector with the magnetic field vector, which we express as $\vec{\mathbf{F}} = \zeta \vec{\mathbf{u}} \times \vec{\mathbf{B}}$. In this equation, a constant ζ takes care of the consistency in physical units, so we can omit physical units on vectors $\vec{\mathbf{u}}$ and $\vec{\mathbf{B}}$. In this example, let’s assume the constant ζ is positive.

A particle moving in space with velocity vector $\vec{\mathbf{u}} = -5.0\hat{\mathbf{i}} - 2.0\hat{\mathbf{j}} + 3.5\hat{\mathbf{k}}$ enters a region with a magnetic field and experiences a magnetic force. Find the magnetic force $\vec{\mathbf{F}}$ on this particle at the entry point to the region where the magnetic field vector is (a) $\vec{\mathbf{B}} = 7.2\hat{\mathbf{i}} - \hat{\mathbf{j}} - 2.4\hat{\mathbf{k}}$ and (b) $\vec{\mathbf{B}} = 4.5\hat{\mathbf{k}}$. In each case, find magnitude F of the magnetic force and angle θ the force vector $\vec{\mathbf{F}}$ makes with the given magnetic field vector $\vec{\mathbf{B}}$.

Strategy

First, we want to find the vector product $\vec{\mathbf{u}} \times \vec{\mathbf{B}}$, because then we can determine the magnetic force using $\vec{\mathbf{F}} = \zeta \vec{\mathbf{u}} \times \vec{\mathbf{B}}$. Magnitude F can be found either by using components, $F = \sqrt{F_x^2 + F_y^2 + F_z^2}$, or by computing the magnitude $|\vec{\mathbf{u}} \times \vec{\mathbf{B}}|$ directly using [Equation 2.35](#). In the latter approach, we would have to find the angle between vectors $\vec{\mathbf{u}}$ and $\vec{\mathbf{B}}$. When we have $\vec{\mathbf{F}}$, the general method for finding the direction angle θ involves the computation of the scalar product $\vec{\mathbf{F}} \cdot \vec{\mathbf{B}}$ and substitution into [Equation 2.34](#). To compute the vector product we can either use [Equation 2.40](#) or compute the product directly, whichever way is simpler.

Solution

The components of the velocity vector are $u_x = -5.0$, $u_y = -2.0$, and $u_z = 3.5$.

(a) The components of the magnetic field vector are $B_x = 7.2$, $B_y = -1.0$, and $B_z = -2.4$. Substituting them into [Equation 2.41](#) gives the scalar components of vector $\vec{\mathbf{F}} = \zeta \vec{\mathbf{u}} \times \vec{\mathbf{B}}$:

$$\begin{cases} F_x = \zeta(u_y B_z - u_z B_y) = \zeta[(-2.0)(-2.4) - (3.5)(-1.0)] = 8.3\zeta \\ F_y = \zeta(u_z B_x - u_x B_z) = \zeta[(3.5)(7.2) - (-5.0)(-2.4)] = 13.2\zeta \\ F_z = \zeta(u_x B_y - u_y B_x) = \zeta[(-5.0)(-1.0) - (-2.0)(7.2)] = 19.4\zeta \end{cases}$$

Thus, the magnetic force is $\vec{\mathbf{F}} = \zeta(8.3\hat{\mathbf{i}} + 13.2\hat{\mathbf{j}} + 19.4\hat{\mathbf{k}})$ and its magnitude is

$$F = \sqrt{F_x^2 + F_y^2 + F_z^2} = \zeta \sqrt{(8.3)^2 + (13.2)^2 + (19.4)^2} = 24.9\zeta.$$

To compute angle θ , we may need to find the magnitude of the magnetic field vector,

$$B = \sqrt{B_x^2 + B_y^2 + B_z^2} = \sqrt{(7.2)^2 + (-1.0)^2 + (-2.4)^2} = 7.6,$$

and the scalar product $\vec{\mathbf{F}} \cdot \vec{\mathbf{B}}$:

$$\vec{\mathbf{F}} \cdot \vec{\mathbf{B}} = F_x B_x + F_y B_y + F_z B_z = (8.3\zeta)(7.2) + (13.2\zeta)(-1.0) + (19.4\zeta)(-2.4) = 0.$$

Now, substituting into [Equation 2.34](#) gives angle θ :

$$\cos \theta = \frac{\vec{\mathbf{F}} \cdot \vec{\mathbf{B}}}{FB} = \frac{0}{(24.9\zeta)(7.6)} = 0 \Rightarrow \theta = 90^\circ.$$

Hence, the magnetic force vector is perpendicular to the magnetic field vector. (We could have saved some time if we had computed the scalar product earlier.)

(b) Because vector $\vec{\mathbf{B}} = 4.5\hat{\mathbf{k}}$ has only one component, we can perform the algebra quickly and find the vector product directly:

$$\begin{aligned}\vec{\mathbf{F}} &= \zeta \vec{\mathbf{u}} \times \vec{\mathbf{B}} = \zeta(-5.0\hat{\mathbf{i}} - 2.0\hat{\mathbf{j}} + 3.5\hat{\mathbf{k}}) \times (4.5\hat{\mathbf{k}}) \\ &= \zeta[(-5.0)(4.5)\hat{\mathbf{i}} \times \hat{\mathbf{k}} + (-2.0)(4.5)\hat{\mathbf{j}} \times \hat{\mathbf{k}} + (3.5)(4.5)\hat{\mathbf{k}} \times \hat{\mathbf{k}}] \\ &= \zeta[-22.5(-\hat{\mathbf{j}}) - 9.0(+\hat{\mathbf{i}}) + 0] = \zeta(-9.0\hat{\mathbf{i}} + 22.5\hat{\mathbf{j}}).\end{aligned}$$

The magnitude of the magnetic force is

$$F = \sqrt{F_x^2 + F_y^2 + F_z^2} = \zeta \sqrt{(-9.0)^2 + (22.5)^2 + (0.0)^2} = 24.2\zeta.$$

Because the scalar product is

$$\vec{\mathbf{F}} \cdot \vec{\mathbf{B}} = F_x B_x + F_y B_y + F_z B_z = (-9.0\zeta)(0) + (22.5\zeta)(0) + (0)(4.5) = 0,$$

the magnetic force vector $\vec{\mathbf{F}}$ is perpendicular to the magnetic field vector $\vec{\mathbf{B}}$.

Significance

Even without actually computing the scalar product, we can predict that the magnetic force vector must always be perpendicular to the magnetic field vector because of the way this vector is constructed. Namely, the magnetic force vector is the vector product $\vec{\mathbf{F}} = \zeta \vec{\mathbf{u}} \times \vec{\mathbf{B}}$ and, by the definition of the vector product (see [Figure 2.29](#)), vector $\vec{\mathbf{F}}$ must be perpendicular to both vectors $\vec{\mathbf{u}}$ and $\vec{\mathbf{B}}$.

✓ CHECK YOUR UNDERSTANDING 2.16

Given two vectors $\vec{\mathbf{A}} = -\hat{\mathbf{i}} + \hat{\mathbf{j}}$ and $\vec{\mathbf{B}} = 3\hat{\mathbf{i}} - \hat{\mathbf{j}}$, find (a) $\vec{\mathbf{A}} \times \vec{\mathbf{B}}$, (b) $|\vec{\mathbf{A}} \times \vec{\mathbf{B}}|$, (c) the angle between $\vec{\mathbf{A}}$ and $\vec{\mathbf{B}}$, and (d) the angle between $\vec{\mathbf{A}} \times \vec{\mathbf{B}}$ and vector $\vec{\mathbf{C}} = \hat{\mathbf{i}} + \hat{\mathbf{k}}$.

In conclusion to this section, we want to stress that “dot product” and “cross product” are entirely different mathematical objects that have different meanings. The dot product is a scalar; the cross product is a vector. Later chapters use the terms *dot product* and *scalar product* interchangeably. Similarly, the terms *cross product* and *vector product* are used interchangeably.

Chapter Review

Key Terms

anticommutative property change in the order of operation introduces the minus sign

antiparallel vectors two vectors with directions that differ by 180°

associative terms can be grouped in any fashion

commutative operations can be performed in any order

component form of a vector a vector written as the vector sum of its components in terms of unit vectors

corkscrew right-hand rule a rule used to determine the direction of the vector product

cross product the result of the vector multiplication of vectors is a vector called a cross product; also called a vector product

difference of two vectors vector sum of the first vector with the vector antiparallel to the second

direction angle in a plane, an angle between the positive direction of the x-axis and the vector, measured counterclockwise from the axis to the vector

displacement change in position

distributive multiplication can be distributed over terms in summation

dot product the result of the scalar multiplication of two vectors is a scalar called a dot product; also called a scalar product

equal vectors two vectors are equal if and only if all their corresponding components are equal; alternately, two parallel vectors of equal magnitudes

magnitude length of a vector

null vector a vector with all its components equal to zero

orthogonal vectors two vectors with directions that differ by exactly 90° , synonymous with perpendicular vectors

parallel vectors two vectors with exactly the same direction angles

parallelogram rule geometric construction of the vector sum in a plane

polar coordinate system an orthogonal coordinate

system where location in a plane is given by polar coordinates

polar coordinates a radial coordinate and an angle

radial coordinate distance to the origin in a polar coordinate system

resultant vector vector sum of two (or more) vectors

scalar a number, synonymous with a scalar quantity in physics

scalar component a number that multiplies a unit vector in a vector component of a vector

scalar equation equation in which the left-hand and right-hand sides are numbers

scalar product the result of the scalar multiplication of two vectors is a scalar called a scalar product; also called a dot product

scalar quantity quantity that can be specified completely by a single number with an appropriate physical unit

tail-to-head geometric construction geometric construction for drawing the resultant vector of many vectors

unit vector vector of a unit magnitude that specifies direction; has no physical unit

unit vectors of the axes unit vectors that define orthogonal directions in a plane or in space

vector mathematical object with magnitude and direction

vector components orthogonal components of a vector; a vector is the vector sum of its vector components.

vector equation equation in which the left-hand and right-hand sides are vectors

vector product the result of the vector multiplication of vectors is a vector called a vector product; also called a cross product

vector quantity physical quantity described by a mathematical vector—that is, by specifying both its magnitude and its direction; synonymous with a vector in physics

vector sum resultant of the combination of two (or more) vectors

Key Equations

Multiplication by a scalar (vector equation)

$$\vec{B} = \alpha \vec{A}$$

Multiplication by a scalar (scalar equation for magnitudes)

$$B = |\alpha|A$$

Resultant of two vectors

$$\vec{D}_{AD} = \vec{D}_{AC} + \vec{D}_{CD}$$

Commutative law	$\vec{A} + \vec{B} = \vec{B} + \vec{A}$
Associative law	$(\vec{A} + \vec{B}) + \vec{C} = \vec{A} + (\vec{B} + \vec{C})$
Distributive law	$\alpha_1 \vec{A} + \alpha_2 \vec{A} = (\alpha_1 + \alpha_2) \vec{A}$
The component form of a vector in two dimensions	$\vec{A} = A_x \hat{i} + A_y \hat{j}$
Scalar components of a vector in two dimensions	$\begin{cases} A_x = x_e - x_b \\ A_y = y_e - y_b \end{cases}$
Magnitude of a vector in a plane	$A = \sqrt{A_x^2 + A_y^2}$
The direction angle of a vector in a plane	$\theta_A = \tan^{-1} \left(\frac{A_y}{A_x} \right)$
Scalar components of a vector in a plane	$\begin{cases} A_x = A \cos \theta_A \\ A_y = A \sin \theta_A \end{cases}$
Polar coordinates in a plane	$\begin{cases} x = r \cos \varphi \\ y = r \sin \varphi \end{cases}$
The component form of a vector in three dimensions	$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$
The scalar z-component of a vector in three dimensions	$A_z = z_e - z_b$
Magnitude of a vector in three dimensions	$A = \sqrt{A_x^2 + A_y^2 + A_z^2}$
Distributive property	$\alpha(\vec{A} + \vec{B}) = \alpha\vec{A} + \alpha\vec{B}$
Antiparallel vector to $\vec{A}$	$-\vec{A} = -A_x \hat{i} - A_y \hat{j} - A_z \hat{k}$
Equal vectors	$\vec{A} = \vec{B} \Leftrightarrow \begin{cases} A_x = B_x \\ A_y = B_y \\ A_z = B_z \end{cases}$
Components of the resultant of N vectors	$\begin{cases} F_{Rx} = \sum_{k=1}^N F_{kx} = F_{1x} + F_{2x} + \dots + F_{Nx} \\ F_{Ry} = \sum_{k=1}^N F_{ky} = F_{1y} + F_{2y} + \dots + F_{Ny} \\ F_{Rz} = \sum_{k=1}^N F_{kz} = F_{1z} + F_{2z} + \dots + F_{Nz} \end{cases}$
General unit vector	$\hat{V} = \frac{\vec{V}}{V}$
Definition of the scalar product	$\vec{A} \cdot \vec{B} = AB \cos \varphi$
Commutative property of the scalar product	$\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$
Distributive property of the scalar product	$\vec{A} \cdot (\vec{B} + \vec{C}) = \vec{A} \cdot \vec{B} + \vec{A} \cdot \vec{C}$
Scalar product in terms of scalar components of vectors	$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$
Cosine of the angle between two vectors	$\cos \varphi = \frac{\vec{A} \cdot \vec{B}}{AB}$
Dot products of unit vectors	$\hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0$
Magnitude of the vector product (definition)	$ \vec{A} \times \vec{B} = AB \sin \varphi$

Anticommutative property of the vector product

$$\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$$

Distributive property of the vector product

$$\vec{A} \times (\vec{B} + \vec{C}) = \vec{A} \times \vec{B} + \vec{A} \times \vec{C}$$

Cross products of unit vectors

$$\begin{cases} \hat{i} \times \hat{j} = +\hat{k}, \\ \hat{j} \times \hat{k} = +\hat{i}, \\ \hat{k} \times \hat{i} = +\hat{j}. \end{cases}$$

The cross product in terms of scalar components of vectors

$$\vec{A} \times \vec{B} = (A_y B_z - A_z B_y)\hat{i} + (A_z B_x - A_x B_z)\hat{j} + (A_x B_y - A_y B_x)\hat{k}$$

Summary

2.1 Scalars and Vectors

- A vector quantity is any quantity that has magnitude and direction, such as displacement or velocity. Vector quantities are represented by mathematical objects called vectors.
- Geometrically, vectors are represented by arrows, with the end marked by an arrowhead. The length of the vector is its magnitude, which is a positive scalar. On a plane, the direction of a vector is given by the angle the vector makes with a reference direction, often an angle with the horizontal. The direction angle of a vector is a scalar.
- Two vectors are equal if and only if they have the same magnitudes and directions. Parallel vectors have the same direction angles but may have different magnitudes. Antiparallel vectors have direction angles that differ by 180° . Orthogonal vectors have direction angles that differ by 90° .
- When a vector is multiplied by a scalar, the result is another vector of a different length than the length of the original vector. Multiplication by a positive scalar does not change the original direction; only the magnitude is affected. Multiplication by a negative scalar reverses the original direction. The resulting vector is antiparallel to the original vector. Multiplication by a scalar is distributive. Vectors can be divided by nonzero scalars but cannot be divided by vectors.
- Two or more vectors can be added to form another vector. The vector sum is called the resultant vector. We can add vectors to vectors or scalars to scalars, but we cannot add scalars to vectors. Vector addition is commutative and associative.
- To construct a resultant vector of two vectors in a plane geometrically, we use the parallelogram rule. To construct a resultant vector of many vectors in a plane geometrically, we use the tail-

to-head method.

2.2 Coordinate Systems and Components of a Vector

- Vectors are described in terms of their components in a coordinate system. In two dimensions (in a plane), vectors have two components. In three dimensions (in space), vectors have three components.
- A vector component of a vector is its part in an axis direction. The vector component is the product of the unit vector of an axis with its scalar component along this axis. A vector is the resultant of its vector components.
- Scalar components of a vector are differences of coordinates, where coordinates of the origin are subtracted from end point coordinates of a vector. In a rectangular system, the magnitude of a vector is the square root of the sum of the squares of its components.
- In a plane, the direction of a vector is given by an angle the vector has with the positive x-axis. This direction angle is measured counterclockwise. The scalar x-component of a vector can be expressed as the product of its magnitude with the cosine of its direction angle, and the scalar y-component can be expressed as the product of its magnitude with the sine of its direction angle.
- In a plane, there are two equivalent coordinate systems. The Cartesian coordinate system is defined by unit vectors $\hat{i}$ and $\hat{j}$ along the x-axis and the y-axis, respectively. The polar coordinate system is defined by the radial unit vector $\hat{r}$, which gives the direction from the origin, and a unit vector $\hat{t}$, which is perpendicular (orthogonal) to the radial direction.

2.3 Algebra of Vectors

- Analytical methods of vector algebra allow us to find resultants of sums or differences of vectors without having to draw them. Analytical methods of vector addition are exact, contrary to graphical

methods, which are approximate.

- Analytical methods of vector algebra are used routinely in mechanics, electricity, and magnetism. They are important mathematical tools of physics.

2.4 Products of Vectors

- There are two kinds of multiplication for vectors. One kind of multiplication is the scalar product, also known as the dot product. The other kind of multiplication is the vector product, also known as the cross product. The scalar product of vectors is a number (scalar). The vector product of vectors is a vector.
- Both kinds of multiplication have the distributive property, but only the scalar product has the commutative property. The vector product has the anticommutative property, which means that when we change the order in which two vectors are multiplied, the result acquires a minus sign.
- The scalar product of two vectors is obtained by

multiplying their magnitudes with the cosine of the angle between them. The scalar product of orthogonal vectors vanishes; the scalar product of antiparallel vectors is negative.

- The vector product of two vectors is a vector perpendicular to both of them. Its magnitude is obtained by multiplying their magnitudes by the sine of the angle between them. The direction of the vector product can be determined by the corkscrew right-hand rule. The vector product of two either parallel or antiparallel vectors vanishes. The magnitude of the vector product is largest for orthogonal vectors.
- The scalar product of vectors is used to find angles between vectors and in the definitions of derived scalar physical quantities such as work or energy.
- The cross product of vectors is used in definitions of derived vector physical quantities such as torque or magnetic force, and in describing rotations.

Conceptual Questions

2.1 Scalars and Vectors

- A weather forecast states the temperature is predicted to be -5°C the following day. Is this temperature a vector or a scalar quantity? Explain.
- Which of the following is a vector: a person's height, the altitude on Mt. Everest, the velocity of a fly, the age of Earth, the boiling point of water, the cost of a book, Earth's population, or the acceleration of gravity?
- Give a specific example of a vector, stating its magnitude, units, and direction.
- What do vectors and scalars have in common? How do they differ?
- Suppose you add two vectors $\vec{A}$ and $\vec{B}$. What relative direction between them produces the resultant with the greatest magnitude? What is the maximum magnitude? What relative direction between them produces the resultant with the smallest magnitude? What is the minimum magnitude?
- Is it possible to add a scalar quantity to a vector quantity?
- Is it possible for two vectors of different magnitudes to add to zero? Is it possible for three vectors of different magnitudes to add to

zero? Explain.

- Does the odometer in an automobile indicate a scalar or a vector quantity?
- When a 10,000-m runner competing on a 400-m track crosses the finish line, what is the runner's net displacement? Can this displacement be zero? Explain.
- A vector has zero magnitude. Is it necessary to specify its direction? Explain.
- Can a magnitude of a vector be negative?
- Can the magnitude of a particle's displacement be greater than the distance traveled?
- If two vectors are equal, what can you say about their components? What can you say about their magnitudes? What can you say about their directions?
- If three vectors sum up to zero, what geometric condition do they satisfy?

2.2 Coordinate Systems and Components of a Vector

- Give an example of a nonzero vector that has a component of zero.
- Explain why a vector cannot have a component greater than its own magnitude.
- If two vectors are equal, what can you say about their components?

18. If vectors $\vec{A}$ and $\vec{B}$ are orthogonal, what is the component of $\vec{B}$ along the direction of $\vec{A}$? What is the component of $\vec{A}$ along the direction of $\vec{B}$?
19. If one of the two components of a vector is not zero, can the magnitude of the other vector component of this vector be zero?
20. If two vectors have the same magnitude, do their components have to be the same?

2.4 Products of Vectors

21. What is wrong with the following expressions? How can you correct them? (a) $C = \vec{A}\vec{B}$, (b)

Problems

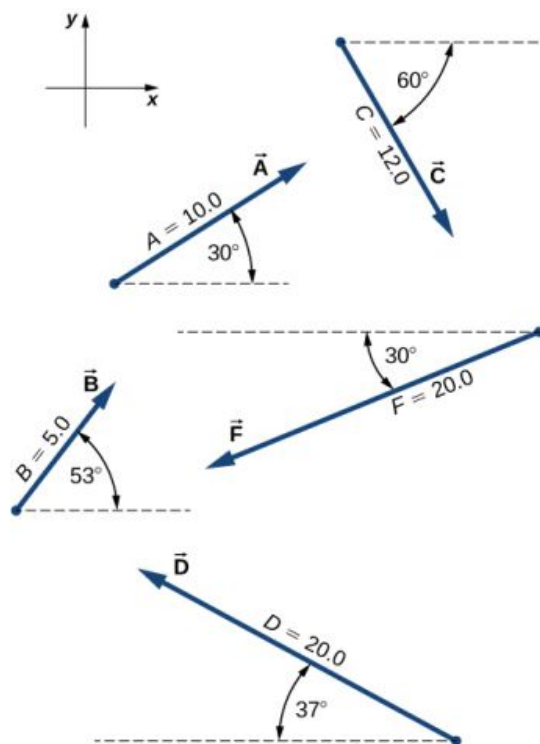
2.1 Scalars and Vectors

25. A scuba diver makes a slow descent into the depths of the ocean. His vertical position with respect to a boat on the surface changes several times. He makes the first stop 9.0 m from the boat but has a problem with equalizing the pressure, so he ascends 3.0 m and then continues descending for another 12.0 m to the second stop. From there, he ascends 4 m and then descends for 18.0 m, ascends again for 7 m and descends again for 24.0 m, where he makes a stop, waiting for his buddy. Assuming the positive direction up to the surface, express his net vertical displacement vector in terms of the unit vector. What is his distance to the boat?
26. In a tug-of-war game on one campus, 15 students pull on a rope at both ends in an effort to displace the central knot to one side or the other. Two students pull with force 196 N each to the right, four students pull with force 98 N each to the left, five students pull with force 62 N each to the left, three students pull with force 150 N each to the right, and one student pulls with force 250 N to the left. Assuming the positive direction to the right, express the net pull on the knot in terms of the unit vector. How big is the net pull on the knot? In what direction?
27. Suppose you walk 18.0 m straight west and then 25.0 m straight north. How far are you from your starting point and what is the compass direction of a line connecting your starting point to your final position? Use a graphical method.
28. For the vectors given in the following figure, use a graphical method to find the following resultants: (a) $\vec{A} + \vec{B}$, (b) $\vec{C} + \vec{B}$, (c) $\vec{D} + \vec{F}$, (d) $\vec{A} - \vec{B}$, (e) $\vec{D} - \vec{F}$,

$\vec{C} = \vec{A}\vec{B}$, (c) $C = \vec{A} \times \vec{B}$, (d) $C = A\vec{B}$, (e) $C + 2\vec{A} = B$, (f) $\vec{C} = A \times \vec{B}$, (g) $\vec{A} \cdot \vec{B} = \vec{A} \times \vec{B}$, (h) $\vec{C} = 2\vec{A} \cdot \vec{B}$, (i) $C = \vec{A}/\vec{B}$, and (j) $C = \vec{A}/B$.

22. If the cross product of two vectors vanishes, what can you say about their directions?
23. If the dot product of two vectors vanishes, what can you say about their directions?
24. What is the dot product of a vector with the cross product that this vector has with another vector?

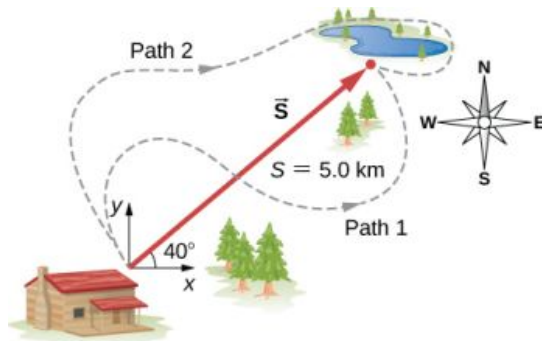
(f) $\vec{A} + 2\vec{F}$, (g) $\vec{C} - 2\vec{D} + 3\vec{F}$; and (h) $\vec{A} - 4\vec{D} + 2\vec{F}$.



29. A delivery driver starts at the post office, drives 40 km north, then 20 km west, then 60 km northeast, and finally 50 km north to stop for lunch. Use a graphical method to find their net displacement vector.
30. An adventurous dog strays from home, runs three blocks east, two blocks north, one block east, one block north, and two blocks west. Assuming that each block is about 100 m, how far from home and in what direction is the dog? Use a graphical method.
31. In an attempt to escape a desert island, a

castaway builds a raft and sets out to sea. The wind shifts a great deal during the day and he is blown along the following directions: 2.50 km and 45.0° north of west, then 4.70 km and 60.0° south of east, then 1.30 km and 25.0° south of west, then 5.10 km straight east, then 1.70 km and 5.00° east of north, then 7.20 km and 55.0° south of west, and finally 2.80 km and 10.0° north of east. Use a graphical method to find the castaway's final position relative to the island.

32. A small plane flies 40.0 km in a direction 60° north of east and then flies 30.0 km in a direction 15° north of east. Use a graphical method to find the total distance the plane covers from the starting point and the direction of the path to the final position.
33. A trapper walks a 5.0-km straight-line distance from his cabin to the lake, as shown in the following figure. Use a graphical method (the parallelogram rule) to determine the trapper's displacement directly to the east and displacement directly to the north that sum up to his resultant displacement vector. If the trapper walked only in directions east and north, zigzagging his way to the lake, how many kilometers would he have to walk to get to the lake?

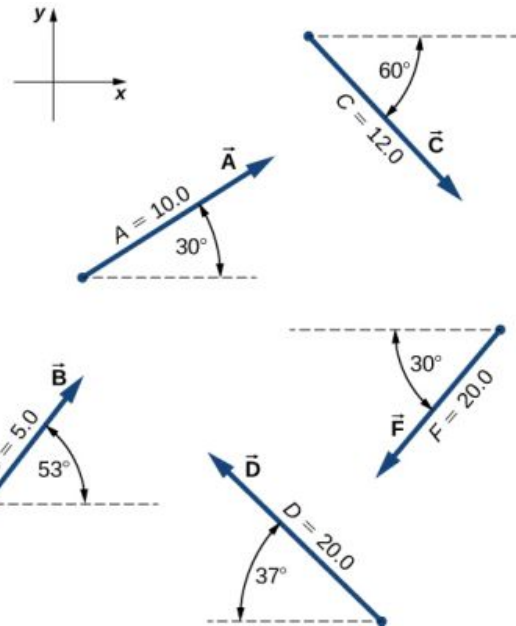


34. A surveyor measures the distance across a river that flows straight north by the following method. Starting directly across from a tree on the opposite bank, the surveyor walks 100 m along the river to establish a baseline. She then sights across to the tree and reads that the angle from the baseline to the tree is 35° . How wide is the river?
35. A pedestrian walks 6.0 km east and then 13.0 km north. Use a graphical method to find the pedestrian's resultant displacement and geographic direction.
36. The magnitudes of two displacement vectors are $A = 20$ m and $B = 6$ m. What are the largest and the smallest values of the magnitude of the

resultant $\vec{R} = \vec{A} + \vec{B}$?

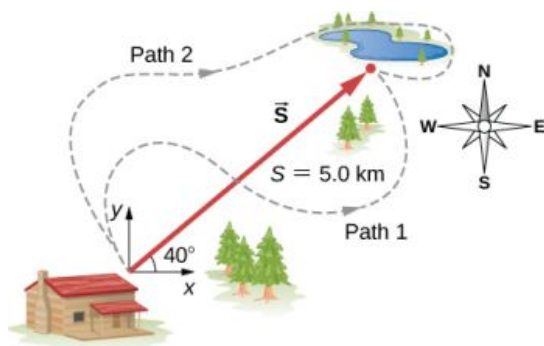
2.2 Coordinate Systems and Components of a Vector

37. Assuming the $+x$ -axis is horizontal and points to the right, resolve the vectors given in the following figure to their scalar components and express them in vector component form.



38. Suppose you walk 18.0 m straight west and then 25.0 m straight north. How far are you from your starting point? What is your displacement vector? What is the direction of your displacement? Assume the $+x$ -axis is to the east.
39. You drive 7.50 km in a straight line in a direction 15° east of north. (a) Find the distances you would have to drive straight east and then straight north to arrive at the same point. (b) Show that you still arrive at the same point if the east and north legs are reversed in order. Assume the $+x$ -axis is to the east.
40. A sledge is being pulled by two horses on a flat terrain. The net force on the sledge can be expressed in the Cartesian coordinate system as vector $\vec{F} = (-2980.0\hat{i} + 8200.0\hat{j})\text{N}$, where $\hat{i}$ and $\hat{j}$ denote directions to the east and north, respectively. Find the magnitude and direction of the pull.
41. A trapper walks a 5.0-km straight-line distance from her cabin to the lake, as shown in the following figure. Determine the east and north components of her displacement vector. How many more

kilometers would she have to walk if she walked along the component displacements? What is her displacement vector?



42. The polar coordinates of a point are $4\pi/3$ and 5.50 m. What are its Cartesian coordinates?
43. Two points in a plane have polar coordinates $P_1(2.500 \text{ m}, \pi/6)$ and $P_2(3.800 \text{ m}, 2\pi/3)$. Determine their Cartesian coordinates and the distance between them in the Cartesian coordinate system. Round the distance to a nearest centimeter.
44. A chameleon is resting quietly on a lanai screen, waiting for an insect to come by. Assume the origin of a Cartesian coordinate system at the lower left-hand corner of the screen and the horizontal direction to the right as the $+x$ -direction. If its coordinates are (2.000 m, 1.000 m), (a) how far is it from the corner of the screen? (b) What is its location in polar coordinates?
45. Two points in the Cartesian plane are $A(2.00 \text{ m}, -4.00 \text{ m})$ and $B(-3.00 \text{ m}, 3.00 \text{ m})$. Find the distance between them and their polar coordinates.
46. A fly enters through an open window and zooms around the room. In a Cartesian coordinate system with three axes along three edges of the room, the fly changes its position from point $b(4.0 \text{ m}, 1.5 \text{ m}, 2.5 \text{ m})$ to point $e(1.0 \text{ m}, 4.5 \text{ m}, 0.5 \text{ m})$. Find the scalar components of the fly's displacement vector and express its displacement vector in vector component form. What is its magnitude?
48. A particle undergoes three consecutive displacements given by vectors $\vec{D}_1 = (3.0\hat{i} - 4.0\hat{j} - 2.0\hat{k})\text{mm}$, $\vec{D}_2 = (1.0\hat{i} - 7.0\hat{j} + 4.0\hat{k})\text{mm}$, and $\vec{D}_3 = (-7.0\hat{i} + 4.0\hat{j} + 1.0\hat{k})\text{mm}$. (a) Find the resultant displacement vector of the particle. (b) What is the magnitude of the resultant displacement? (c) If all displacements were along one line, how far would the particle travel?
49. Given two displacement vectors $\vec{A} = (3.00\hat{i} - 4.00\hat{j} + 4.00\hat{k})\text{m}$ and $\vec{B} = (2.00\hat{i} + 3.00\hat{j} - 7.00\hat{k})\text{m}$, find the displacements and their magnitudes for (a) $\vec{C} = \vec{A} + \vec{B}$ and (b) $\vec{D} = 2\vec{A} - \vec{B}$.
50. A small plane flies 40.0 km in a direction 60° north of east and then flies 30.0 km in a direction 15° north of east. Use the analytical method to find the total distance the plane covers from the starting point, and the geographic direction of its displacement vector. What is its displacement vector?
51. In an attempt to escape a desert island, a castaway builds a raft and sets out to sea. The wind shifts a great deal during the day, and she is blown along the following straight lines: 2.50 km and 45.0° north of west, then 4.70 km and 60.0° south of east, then 1.30 km and 25.0° south of west, then 5.10 km due east, then 1.70 km and 5.00° east of north, then 7.20 km and 55.0° south of west, and finally 2.80 km and 10.0° north of east. Use the analytical method to find the resultant vector of all her displacement vectors. What is its magnitude and direction?
52. Assuming the $+x$ -axis is horizontal to the right for the vectors given in the following figure, use the analytical method to find the following resultants: (a) $11.67\hat{i} + 8.99\hat{j}$, (b) $9.01\hat{i} - 6.40\hat{j}$, (c) $\vec{D} + \vec{F}$, (d) $5.65\hat{i} + 1.01\hat{j}$, (e) $\vec{D} - \vec{F}$, (f) $\vec{A} + 2\vec{F}$, (g) $\vec{C} - 2\vec{D} + 3\vec{F}$, and (h) $\vec{A} - 4\vec{D} + 2\vec{F}$.

2.3 Algebra of Vectors

47. For vectors $\vec{B} = -\hat{i} - 4\hat{j}$ and $\vec{A} = -3\hat{i} - 2\hat{j}$, calculate (a) $\vec{A} + \vec{B}$ and its magnitude and direction angle, and (b) $\vec{A} - \vec{B}$ and its magnitude and direction angle.

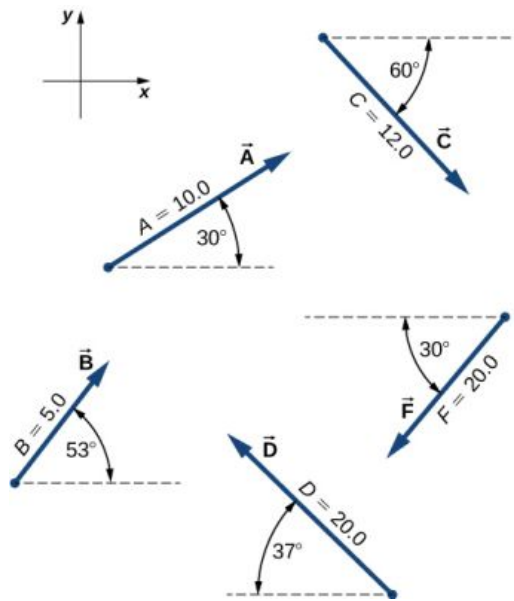


FIGURE 2.33

53. Given the vectors in the preceding figure, find vector $\vec{R}$ that solves equations (a) $\vec{D} + \vec{R} = \vec{F}$ and (b) $\vec{C} - 2\vec{D} + 5\vec{R} = 3\vec{F}$. Assume the $+x$ -axis is horizontal to the right.
54. A delivery driver starts at the post office, drives 40 km north, then 20 km west, then 60 km northeast, and finally 50 km north to stop for lunch. Use the analytical method to determine the following: (a) Find his net displacement vector. (b) How far is the restaurant from the post office? (c) If he returns directly from the restaurant to the post office, what is his displacement vector on the return trip? (d) What is his compass heading on the return trip? Assume the $+x$ -axis is to the east.
55. An adventurous dog strays from home, runs three blocks east, two blocks north, and one block east, one block north, and two blocks west. Assuming that each block is about a 100 yd, use the analytical method to find the dog's net displacement vector, its magnitude, and its direction. Assume the $+x$ -axis is to the east. How would your answer be affected if each block was about 100 m?
56. If $\vec{D} = (6.00\hat{i} - 8.00\hat{j})\text{m}$, $\vec{B} = (-8.00\hat{i} + 3.00\hat{j})\text{m}$, and $\vec{A} = (26.0\hat{i} + 19.0\hat{j})\text{m}$, find the unknown constants a and b such that $a\vec{D} + b\vec{B} + \vec{A} = \vec{0}$.
57. Given the displacement vector $\vec{D} = (3\hat{i} - 4\hat{j})\text{m}$, find the displacement vector $\vec{R}$ so that

$$\vec{D} + \vec{R} = -4D\hat{j}.$$

58. Find the unit vector of direction for the following vector quantities: (a) Force $\vec{F} = (3.0\hat{i} - 2.0\hat{j})\text{N}$, (b) displacement $\vec{D} = (-3.0\hat{i} - 4.0\hat{j})\text{m}$, and (c) velocity $\vec{v} = (-5.00\hat{i} + 4.00\hat{j})\text{m/s}$.
59. At one point in space, the direction of the electric field vector is given in the Cartesian system by the unit vector $\hat{E} = 1/\sqrt{5}\hat{i} - 2/\sqrt{5}\hat{j}$. If the magnitude of the electric field vector is $E = 400.0\text{ V/m}$, what are the scalar components E_x , E_y , and E_z of the electric field vector $\vec{E}$ at this point? What is the direction angle θ_E of the electric field vector at this point?
60. A barge is pulled by the two tugboats shown in the following figure. One tugboat pulls on the barge with a force of magnitude 4000 units of force at 15° above the line AB (see the figure) and the other tugboat pulls on the barge with a force of magnitude 5000 units of force at 12° below the line AB. Resolve the pulling forces to their scalar components and find the components of the resultant force pulling on the barge. What is the magnitude of the resultant pull? What is its direction relative to the line AB?

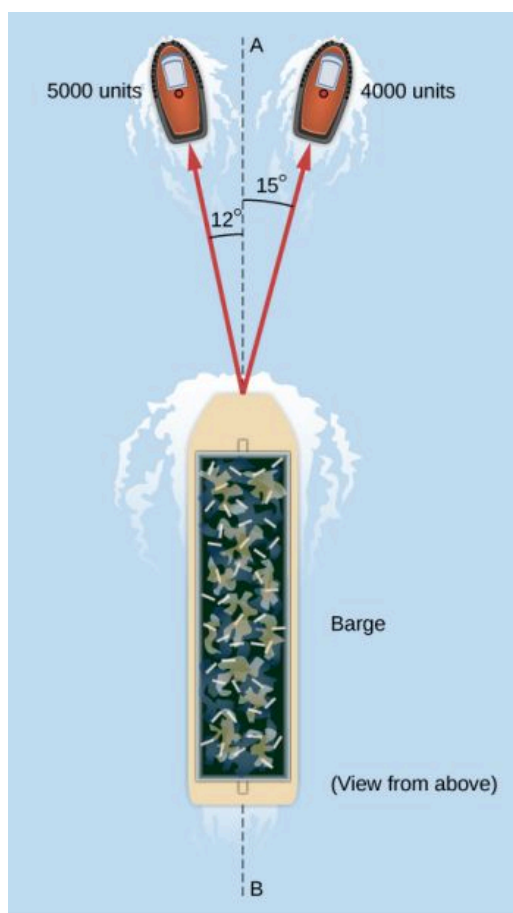
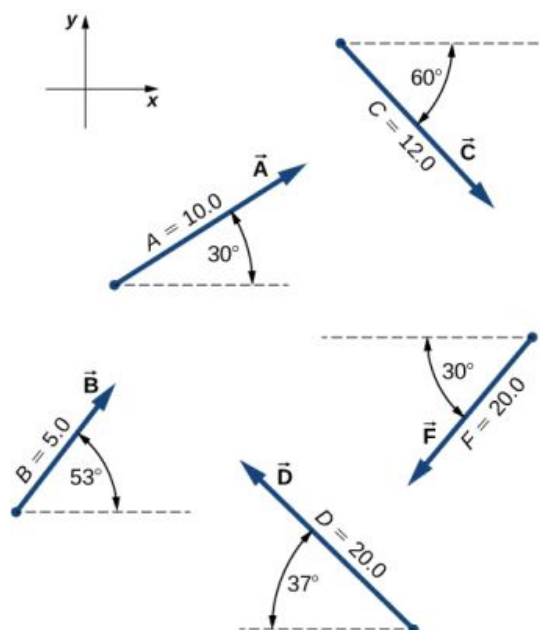


FIGURE 2.34

61. In the control tower at a regional airport, an air traffic controller monitors two aircraft that travel in straight paths directly away from the control tower. One plane is a cargo carrier Boeing 747 climbing at 10° above the horizontal, and moving 30° north of west. At some moment in time, the controller notes that the Boeing is at an altitude of 2500 m and the DC-3 is at an altitude of 3000 m. The other plane is a Douglas DC-3 climbing at 5° above the horizontal, and cruising directly west. (a) Find the position vectors of the planes relative to the control tower at this time. (b) What is the distance between the planes at this time?

2.4 Products of Vectors

62. Assuming the $+x$ -axis is horizontal to the right for the vectors in the following figure, find the following scalar products: (a) $\vec{A} \cdot \vec{C}$, (b) $\vec{A} \cdot \vec{F}$, (c) $\vec{D} \cdot \vec{C}$, (d) $\vec{A} \cdot (\vec{F} + 2\vec{C})$, (e) $\hat{i} \cdot \vec{B}$, (f) $\hat{j} \cdot \vec{B}$, (g) $(3\hat{i} - \hat{j}) \cdot \vec{B}$, and (h) $\vec{B} \cdot \vec{B}$.



63. Assuming the $+x$ -axis is horizontal to the right for the vectors in the preceding figure, find (a) the component of vector $\vec{A}$ along vector $\vec{C}$, (b) the component of vector $\vec{C}$ along vector $\vec{A}$, (c) the component of vector $\hat{i}$ along vector $\vec{F}$, and (d) the component of vector $\vec{F}$ along vector $\hat{i}$.
64. Find the angle between vectors for (a) $\vec{D} = (-3.0\hat{i} - 4.0\hat{j})\text{m}$ and $\vec{A} = (-3.0\hat{i} + 4.0\hat{j})\text{m}$ and (b) $\vec{D} = (2.0\hat{i} - 4.0\hat{j} + \hat{k})\text{m}$ and $\vec{B} = (-2.0\hat{i} + 3.0\hat{j} + 2.0\hat{k})\text{m}$.
65. Find the angles that vector $\vec{D} = (2.0\hat{i} - 4.0\hat{j} + \hat{k})\text{m}$ makes with the x -, y -, and z - axes.
66. Show that the force vector $\vec{D} = (2.0\hat{i} - 4.0\hat{j} + \hat{k})\text{N}$ is orthogonal to the force vector $\vec{G} = (3.0\hat{i} + 4.0\hat{j} + 10.0\hat{k})\text{N}$.
67. Assuming the $+x$ -axis is horizontal to the right for the vectors in the previous figure, find the following vector products: (a) $\vec{A} \times \vec{C}$, (b) $\vec{A} \times \vec{F}$, (c) $\vec{D} \times \vec{C}$, (d) $\vec{A} \times (\vec{F} + 2\vec{C})$, (e) $\hat{i} \times \vec{B}$, (f) $\hat{j} \times \vec{B}$, (g) $(3\hat{i} - \hat{j}) \times \vec{B}$, and (h) $\vec{B} \times \vec{B}$.
68. Find the cross product $\vec{A} \times \vec{C}$ for (a) $\vec{A} = 2.0\hat{i} - 4.0\hat{j} + \hat{k}$ and $\vec{C} = 3.0\hat{i} + 4.0\hat{j} + 10.0\hat{k}$, (b) $\vec{A} = 3.0\hat{i} + 4.0\hat{j} + 10.0\hat{k}$ and

$\vec{C} = 2.0\hat{i} - 4.0\hat{j} + \hat{k}$, (c) $\vec{A} = -3.0\hat{i} - 4.0\hat{j}$ and $\vec{C} = -3.0\hat{i} + 4.0\hat{j}$, and (d) $\vec{C} = -2.0\hat{i} + 3.0\hat{j} + 2.0\hat{k}$ and $\vec{A} = -9.0\hat{j}$.

69. For the vectors in the earlier figure, find (a) $(\vec{A} \times \vec{F}) \cdot \vec{D}$, (b) $(\vec{A} \times \vec{F}) \cdot (\vec{D} \times \vec{B})$, and (c)

$$(\vec{A} \cdot \vec{F})(\vec{D} \times \vec{B}).$$

70. (a) If $\vec{A} \times \vec{F} = \vec{B} \times \vec{F}$, can we conclude $\vec{A} = \vec{B}$? (b) If $\vec{A} \cdot \vec{F} = \vec{B} \cdot \vec{F}$, can we conclude $\vec{A} = \vec{B}$? (c) If $F\vec{A} = \vec{B}F$, can we conclude $\vec{A} = \vec{B}$? Why or why not?

Additional Problems

71. You fly 32.0 km in a straight line in still air in the direction 35.0° south of west. (a) Find the distances you would have to fly due south and then due west to arrive at the same point. (b) Find the distances you would have to fly first in a direction 45.0° south of west and then in a direction 45.0° west of north. Note these are the components of the displacement along a different set of axes—namely, the one rotated by 45° with respect to the axes in (a).

72. Rectangular coordinates of a point are given by $(2, y)$ and its polar coordinates are given by $(r, \pi/6)$. Find y and r .

73. If the polar coordinates of a point are (r, φ) and its rectangular coordinates are (x, y) , determine the polar coordinates of the following points: (a) $(-x, y)$, (b) $(-2x, -2y)$, and (c) $(3x, -3y)$.

74. Vectors $\vec{A}$ and $\vec{B}$ have identical magnitudes of 5.0 units. Find the angle between them if $\vec{A} + \vec{B} = 5\sqrt{2}\hat{j}$.

75. Starting at the island of Moi in an unknown archipelago, a fishing boat makes a round trip with two stops at the islands of Noi and Poi. It sails from Moi for 4.76 nautical miles (nmi) in a direction 37° north of east to Noi. From Noi, it sails 69° west of north to Poi. On its return leg from Poi, it sails 28° east of south. What distance does the boat sail between Noi and Poi? What distance does it sail between Moi and Poi? Express your answer both in nautical miles and in kilometers. Note: 1 nmi = 1852 m.

76. An air traffic controller notices two signals from two planes on the radar monitor. One plane is at altitude 800 m and in a 19.2-km horizontal distance to the tower in a direction 25° south of west. The second plane is at altitude 1100 m and its horizontal distance is 17.6 km and 20° south of west. What is the distance between these planes?

77. Show that when $\vec{A} + \vec{B} = \vec{C}$, then $C^2 = A^2 + B^2 + 2AB \cos \varphi$, where φ is the

angle between vectors $\vec{A}$ and $\vec{B}$.

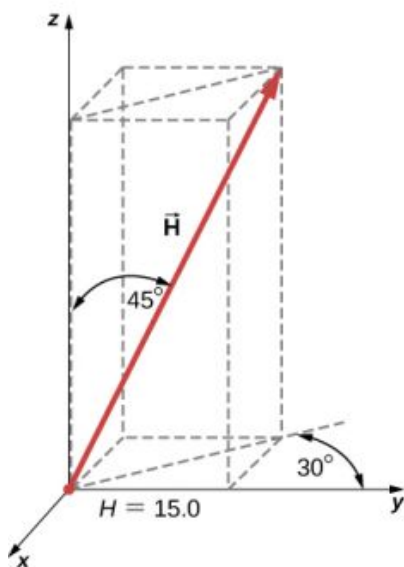
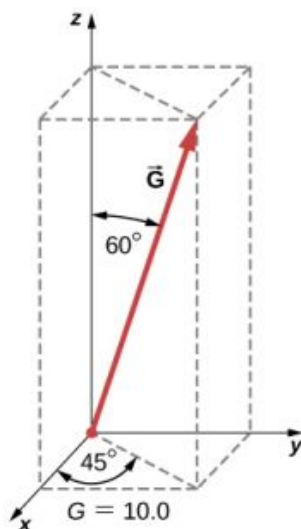
78. Four force vectors each have the same magnitude f . What is the largest magnitude the resultant force vector may have when these forces are added? What is the smallest magnitude of the resultant? Make a graph of both situations.

79. A skater glides along a circular path of radius 5.00 m in clockwise direction. When he coasts around one-half of the circle, starting from the west point, find (a) the magnitude of his displacement vector and (b) how far he actually skated. (c) What is the magnitude of his displacement vector when he skates all the way around the circle and comes back to the west point?

80. A stubborn dog is being walked on a leash by its owner. At one point, the dog encounters an interesting scent at some spot on the ground and wants to explore it in detail, but the owner gets impatient and pulls on the leash with force $\vec{F} = (98.0\hat{i} + 132.0\hat{j} + 32.0\hat{k})\text{N}$ along the leash. (a) What is the magnitude of the pulling force? (b) What angle does the leash make with the vertical?

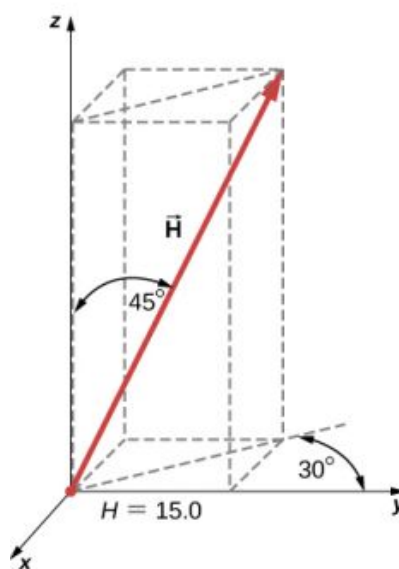
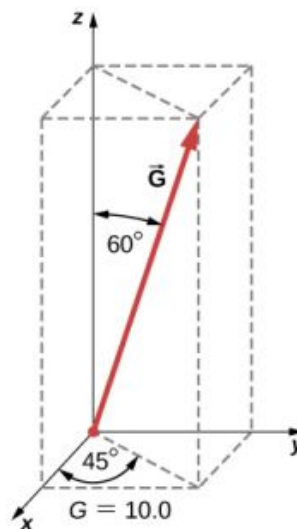
81. If the velocity vector of a polar bear is $\vec{u} = (-18.0\hat{i} - 13.0\hat{j})\text{km/h}$, how fast and in what geographic direction is it heading? Here, $\hat{i}$ and $\hat{j}$ are directions to geographic east and north, respectively.

82. Find the scalar components of three-dimensional vectors $\vec{G}$ and $\vec{H}$ in the following figure and write the vectors in vector component form in terms of the unit vectors of the axes.

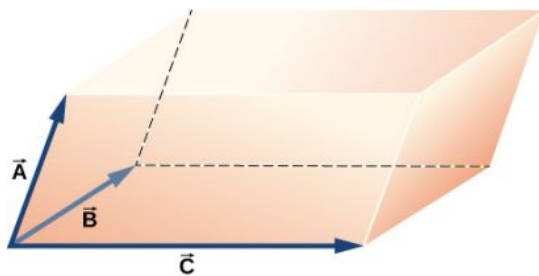


83. A diver explores a shallow reef off the coast of Belize. She initially swims 90.0 m north, makes a turn to the east and continues for 200.0 m, then follows a big grouper for 80.0 m in the direction 30° north of east. In the meantime, a local current displaces her by 150.0 m south. Assuming the current is no longer present, in what direction and how far should she now swim to come back to the point where she started?
84. A force vector $\vec{A}$ has x- and y-components, respectively, of -8.80 units of force and 15.00 units of force. The x- and y-components of force vector $\vec{B}$ are, respectively, 13.20 units of force and -6.60 units of force. Find the components of force vector $\vec{C}$ that satisfies the vector equation $\vec{A} - \vec{B} + 3\vec{C} = 0$.

85. Vectors $\vec{A}$ and $\vec{B}$ are two orthogonal vectors in the xy-plane and they have identical magnitudes. If $\vec{A} = 3.0\hat{i} + 4.0\hat{j}$, find $\vec{B}$.
86. For the three-dimensional vectors in the following figure, find (a) $\vec{G} \times \vec{H}$, (b) $|\vec{G} \times \vec{H}|$, and (c) $\vec{G} \cdot \vec{H}$.

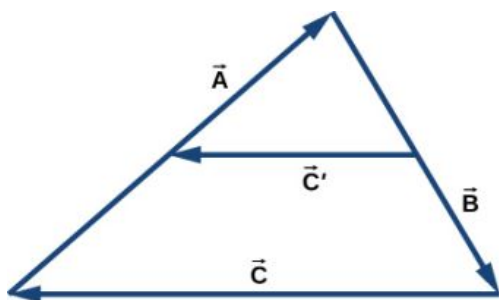


87. Show that $(\vec{B} \times \vec{C}) \cdot \vec{A}$ is the volume of the parallelepiped, with edges formed by the three vectors in the following figure.



Challenge Problems

88. Vector $\vec{B}$ is 5.0 cm long and vector $\vec{A}$ is 4.0 cm long. Find the angle between these two vectors when $|\vec{A} + \vec{B}| = 3.0$ cm.
89. What is the component of the force vector $\vec{G} = (3.0\hat{i} + 4.0\hat{j} + 10.0\hat{k})\text{N}$ along the force vector $\vec{H} = (1.0\hat{i} + 4.0\hat{j})\text{N}$?
90. The following figure shows a triangle formed by the three vectors $\vec{A}$, $\vec{B}$, and $\vec{C}$. If vector $\vec{C}'$ is drawn between the midpoints of vectors $\vec{A}$ and $\vec{B}$, show that $\vec{C}' = \vec{C}/2$.



91. Distances between points in a plane do not change when a coordinate system is rotated. In other words, the magnitude of a vector is *invariant* under rotations of the coordinate system. Suppose a coordinate system S is rotated about its origin by angle φ to become a new coordinate system S' , as shown in the following figure. A point in a plane has coordinates (x, y) in S and coordinates (x', y') in S' .

(a) Show that, during the transformation of rotation, the coordinates in S' are expressed in terms of the coordinates in S by the following relations:

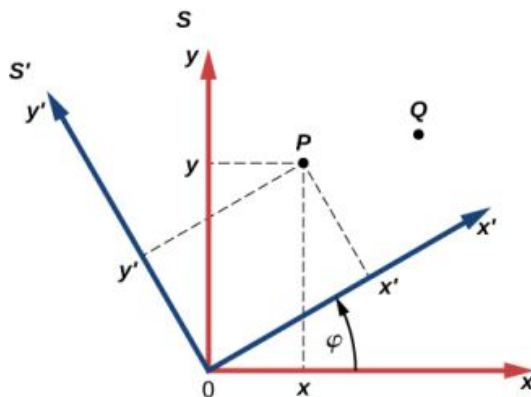
$$\begin{cases} x' = x \cos \varphi + y \sin \varphi \\ y' = -x \sin \varphi + y \cos \varphi \end{cases}$$

(b) Show that the distance of point P to the origin is invariant under rotations of the coordinate system. Here, you have to show that

$$\sqrt{x^2 + y^2} = \sqrt{x'^2 + y'^2}.$$

(c) Show that the distance between points P and Q is invariant under rotations of the coordinate system. Here, you have to show that

$$\sqrt{(x_P - x_Q)^2 + (y_P - y_Q)^2} = \sqrt{(x'_P - x'_Q)^2 + (y'_P - y'_Q)^2}.$$



CHAPTER 3

Motion Along a Straight Line



FIGURE 3.1 A JR Central L0 series five-car maglev (magnetic levitation) train undergoing a test run on the Yamanashi Test Track. The maglev train's motion can be described using kinematics, the subject of this chapter. (credit: modification of work by "Maryland GovPics"/Flickr)

CHAPTER OUTLINE

3.1 Position, Displacement, and Average Velocity

3.2 Instantaneous Velocity and Speed

3.3 Average and Instantaneous Acceleration

3.4 Motion with Constant Acceleration

3.5 Free Fall

3.6 Finding Velocity and Displacement from Acceleration

INTRODUCTION Our universe is full of objects in motion. From the stars, planets, and galaxies; to the motion of people and animals; down to the microscopic scale of atoms and molecules—everything in our universe is in motion. We can describe motion using the two disciplines of kinematics and dynamics. We study dynamics, which is concerned with the causes of motion, in [Newton's Laws of Motion](#); but, there is much to be learned about motion without referring to what causes it, and this is the study of kinematics. Kinematics involves describing motion through properties such as position, time, velocity, and acceleration.

A full treatment of **kinematics** considers motion in two and three dimensions. For now, we discuss motion in one dimension, which provides us with the tools necessary to study multidimensional motion. A good example of an object undergoing one-dimensional motion is the maglev (magnetic levitation) train depicted at the beginning of this chapter. As it travels, say, from Tokyo to Kyoto, it is at different positions along the track at various times in its journey, and therefore has displacements, or changes in position. It also has a variety of velocities along its path and it undergoes accelerations (changes in velocity). With the skills learned in this chapter we can calculate these quantities and average velocity. All these quantities can be described using kinematics, without knowing the train's mass or the forces involved.

3.1 Position, Displacement, and Average Velocity

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Define position, displacement, and distance traveled.
- Calculate the total displacement given the position as a function of time.
- Determine the total distance traveled.
- Calculate the average velocity given the displacement and elapsed time.

When you're in motion, the basic questions to ask are: Where are you? Where are you going? How fast are you getting there? The answers to these questions require that you specify your position, your displacement, and your average velocity—the terms we define in this section.

Position

To describe the motion of an object, you must first be able to describe its **position** (x): *where it is at any particular time*. More precisely, we need to specify its position relative to a convenient frame of reference. A frame of reference is an arbitrary set of axes from which the position and motion of an object are described. Earth is often used as a frame of reference, and we often describe the position of an object as it relates to stationary objects on Earth. For example, a rocket launch could be described in terms of the position of the rocket with respect to Earth as a whole, whereas a cyclist's position could be described in terms of where she is in relation to the buildings she passes [Figure 3.2](#). In other cases, we use reference frames that are not stationary but are in motion relative to Earth. To describe the position of a person in an airplane, for example, we use the airplane, not Earth, as the reference frame. To describe the position of an object undergoing one-dimensional motion, we often use the variable x . Later in the chapter, during the discussion of free fall, we use the variable y .



FIGURE 3.2 These cyclists in Vietnam can be described by their position relative to buildings or a canal. Their motion can be described by their change in position, or displacement, in a frame of reference. (credit: modification of work by Suzan Black)

Displacement

If an object moves relative to a frame of reference—for example, if a professor moves to the right relative to a whiteboard [Figure 3.3](#)—then the object's position changes. This change in position is called **displacement**. The word *displacement* implies that an object has moved, or has been displaced. Although position is the numerical value of x along a straight line where an object might be located, displacement gives the *change* in position along this line. Since displacement indicates direction, it is a vector and can be either positive or negative, depending on the choice of positive direction. Also, an analysis of motion can have many displacements embedded in it. If right is positive and an object moves 2 m to the right, then 4 m to the left, the individual displacements are 2 m and -4 m, respectively.

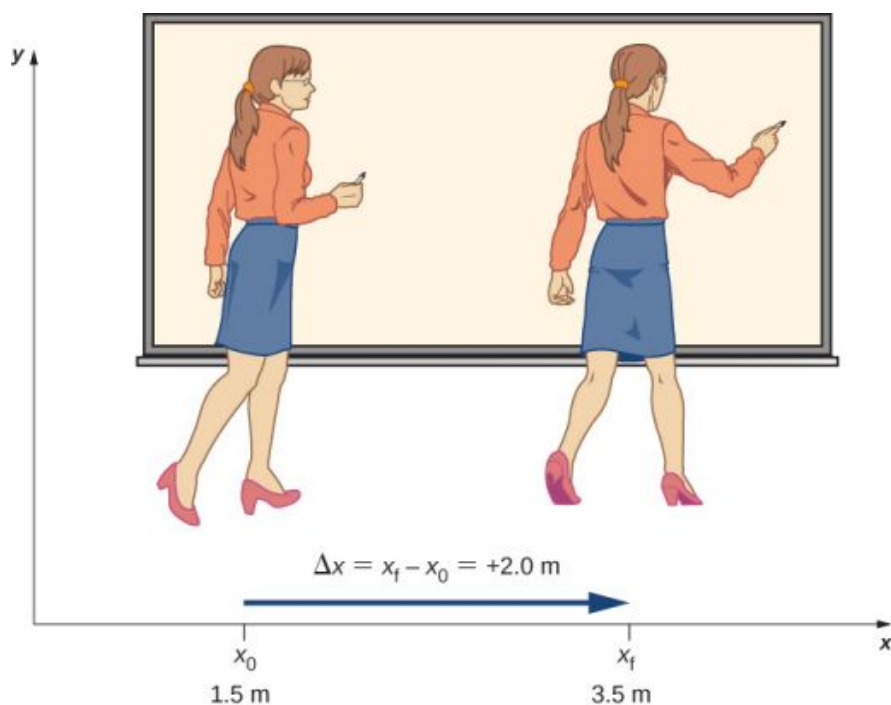


FIGURE 3.3 A professor paces left and right while lecturing. Her position relative to Earth is given by x . The $+2.0\text{-m}$ displacement of the professor relative to Earth is represented by an arrow pointing to the right.

DISPLACEMENT

Displacement Δx is the change in position of an object:

$$\Delta x = x_f - x_0,$$

3.1

where Δx is displacement, x_f is the final position, and x_0 is the initial position.

We use the uppercase Greek letter delta (Δ) to mean “change in” whatever quantity follows it; thus, Δx means *change in position* (final position less initial position). We always solve for displacement by subtracting initial position x_0 from final position x_f . Note that the SI unit for displacement is the meter, but sometimes we use kilometers or other units of length. Keep in mind that when units other than meters are used in a problem, you may need to convert them to meters to complete the calculation (see [Appendix B](#)).

Objects in motion can also have a series of displacements. In the previous example of the pacing professor, the individual displacements are 2 m and -4 m , giving a total displacement of -2 m . We define **total displacement** Δx_{Total} , as *the sum of the individual displacements*, and express this mathematically with the equation

$$\Delta x_{\text{Total}} = \sum \Delta x_i,$$

3.2

where Δx_i are the individual displacements. In the earlier example,

$$\Delta x_1 = x_1 - x_0 = 2 - 0 = 2\text{ m}.$$

Similarly,

$$\Delta x_2 = x_2 - x_1 = -2 - (2) = -4\text{ m}.$$

Thus,

$$\Delta x_{\text{Total}} = \Delta x_1 + \Delta x_2 = 2 - 4 = -2\text{ m}.$$

The total displacement is $2 - 4 = -2\text{ m}$ along the x -axis. It is also useful to calculate the magnitude of the displacement, or its size. The magnitude of the displacement is always positive. This is the absolute value of the

displacement, because displacement is a vector and cannot have a negative value of magnitude. In our example, the magnitude of the total displacement is 2 m, whereas the magnitudes of the individual displacements are 2 m and 4 m.

The magnitude of the total displacement should not be confused with the distance traveled. Distance traveled x_{Total} , is the total length of the path traveled between two positions. In the previous problem, the **distance traveled** is the sum of the magnitudes of the individual displacements:

$$x_{\text{Total}} = |\Delta x_1| + |\Delta x_2| = 2 + 4 = 6 \text{ m.}$$

Average Velocity

To calculate the other physical quantities in kinematics we must introduce the time variable. The time variable allows us not only to state where the object is (its position) during its motion, but also how fast it is moving. How fast an object is moving is given by the rate at which the position changes with time.

For each position x_i , we assign a particular time t_i . If the details of the motion at each instant are not important, the rate is usually expressed as the **average velocity** $\bar{v}$. This vector quantity is simply the total displacement between two points divided by the time taken to travel between them. The time taken to travel between two points is called the **elapsed time** Δt .

AVERAGE VELOCITY

If x_1 and x_2 are the positions of an object at times t_1 and t_2 , respectively, then

$$\begin{aligned} \text{Average velocity} = \bar{v} &= \frac{\text{Displacement between two points}}{\text{Time needed to make the displacement}} \\ \bar{v} &= \frac{\Delta x}{\Delta t} = \frac{x_2 - x_1}{t_2 - t_1}. \end{aligned}$$

3.3

It is important to note that the average velocity is a vector and can be negative, depending on positions x_1 and x_2 .



EXAMPLE 3.1

Delivering Flyers

Jill sets out from her home to deliver flyers for her yard sale, traveling due east along her street lined with houses. At 0.5 km and 9 minutes later she runs out of flyers and has to retrace her steps back to her house to get more. This takes an additional 9 minutes. After picking up more flyers, she sets out again on the same path, continuing where she left off, and ends up 1.0 km from her house. This third leg of her trip takes 15 minutes. At this point she turns back toward her house, heading west. After 1.75 km and 25 minutes she stops to rest.

- What is Jill's total displacement to the point where she stops to rest?
- What is the magnitude of the final displacement?
- What is the average velocity during her entire trip?
- What is the total distance traveled?
- Make a graph of position versus time.

A sketch of Jill's movements is shown in [Figure 3.4](#).

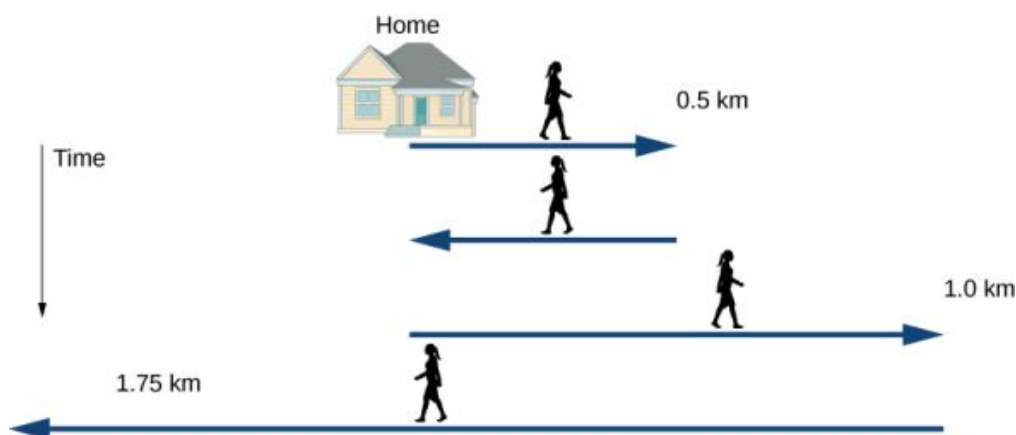


FIGURE 3.4 Timeline of Jill's movements.

Strategy

The problem contains data on the various legs of Jill's trip, so it would be useful to make a table of the physical quantities. We are given position and time in the wording of the problem so we can calculate the displacements and the elapsed time. We take east to be the positive direction. From this information we can find the total displacement and average velocity. Jill's home is the starting point x_0 . The following table gives Jill's time and position in the first two columns, and the displacements are calculated in the third column.

Time t_i (min)	Position x_i (km)	Displacement Δx_i (km)
$t_0 = 0$	$x_0 = 0$	$\Delta x_0 = 0$
$t_1 = 9$	$x_1 = 0.5$	$\Delta x_1 = x_1 - x_0 = 0.5$
$t_2 = 18$	$x_2 = 0$	$\Delta x_2 = x_2 - x_1 = -0.5$
$t_3 = 33$	$x_3 = 1.0$	$\Delta x_3 = x_3 - x_2 = 1.0$
$t_4 = 58$	$x_4 = -0.75$	$\Delta x_4 = x_4 - x_3 = -1.75$

Solution

- a. From the above table, the total displacement is

$$\sum \Delta x_i = 0.5 - 0.5 + 1.0 - 1.75 \text{ km} = -0.75 \text{ km}.$$

- b. The magnitude of the total displacement is $|-0.75| \text{ km} = 0.75 \text{ km}$.

- c. Average velocity = $\frac{\text{Total displacement}}{\text{Elapsed time}} = \bar{v} = \frac{-0.75 \text{ km}}{58 \text{ min}} = -0.013 \text{ km/min}$

- d. The total distance traveled (sum of magnitudes of individual displacements) is

$$x_{\text{Total}} = \sum |\Delta x_i| = 0.5 + 0.5 + 1.0 + 1.75 \text{ km} = 3.75 \text{ km}.$$

- e. We can graph Jill's position versus time as a useful aid to see the motion; the graph is shown in [Figure 3.5](#).

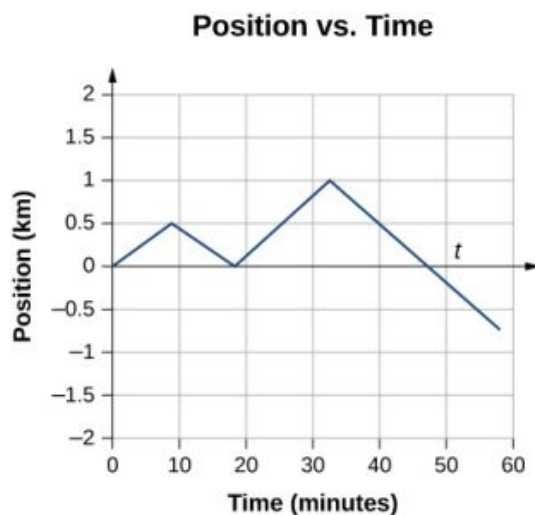


FIGURE 3.5 This graph depicts Jill's position versus time. The average velocity is the slope of a line connecting the initial and final points.

Significance

Jill's total displacement is -0.75 km, which means at the end of her trip she ends up 0.75 km due west of her home. The average velocity means if someone was to walk due west at 0.013 km/min starting at the same time Jill left her home, they both would arrive at the final stopping point at the same time. Note that if Jill were to end her trip at her house, her total displacement would be zero, as well as her average velocity. The total distance traveled during the 58 minutes of elapsed time for her trip is 3.75 km.

✓ CHECK YOUR UNDERSTANDING 3.1

A cyclist rides 3 km west and then turns around and rides 2 km east. (a) What is their displacement? (b) What is the distance traveled? (c) What is the magnitude of their displacement?



3.2 Instantaneous Velocity and Speed

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Explain the difference between average velocity and instantaneous velocity.
- Describe the difference between velocity and speed.
- Calculate the instantaneous velocity given the mathematical equation for the position.
- Calculate the speed given the instantaneous velocity.

We have now seen how to calculate the average velocity between two positions. However, since objects in the real world move continuously through space and time, we would like to find the velocity of an object at any single point. We can find the velocity of the object anywhere along its path by using some fundamental principles of calculus. This section gives us better insight into the physics of motion and will be useful in later chapters.

Instantaneous Velocity

The quantity that tells us how fast an object is moving anywhere along its path is the **instantaneous velocity**, usually called simply *velocity*. It is the average velocity between two points on the path in the limit that the time (and therefore the displacement) between the two events approaches zero. To illustrate this idea mathematically, we need to express position x as a continuous function of t denoted by $x(t)$. The expression for the average velocity between two points using this notation is $\bar{v} = \frac{x(t_2) - x(t_1)}{t_2 - t_1}$. To find the instantaneous velocity at any position, we let $t_1 = t$ and $t_2 = t + \Delta t$. After inserting these expressions into the equation for the average velocity and taking the limit as $\Delta t \rightarrow 0$, we find the expression for the instantaneous velocity:

$$v(t) = \lim_{\Delta t \rightarrow 0} \frac{x(t + \Delta t) - x(t)}{\Delta t} = \frac{dx(t)}{dt}.$$

INSTANTANEOUS VELOCITY

The instantaneous velocity of an object is the limit of the average velocity as the elapsed time approaches zero, or the derivative of x with respect to t :

$$v(t) = \frac{d}{dt}x(t). \quad 3.4$$

Like average velocity, instantaneous velocity is a vector with dimension of length per time. The instantaneous velocity at a specific time point t_0 is the rate of change of the position function, which is the slope of the position function $x(t)$ at t_0 . [Figure 3.6](#) shows how the average velocity $\bar{v} = \frac{\Delta x}{\Delta t}$ between two times approaches the instantaneous velocity at t_0 . The instantaneous velocity is shown at time t_0 , which happens to be at the maximum of the position function. The slope of the position graph is zero at this point, and thus the instantaneous velocity is zero. At other times, t_1 , t_2 , and so on, the instantaneous velocity is not zero because the slope of the position graph would be positive or negative. If the position function had a minimum, the slope of the position graph would also be zero, giving an instantaneous velocity of zero there as well. Thus, the zeros of the velocity function give the minimum and maximum of the position function.

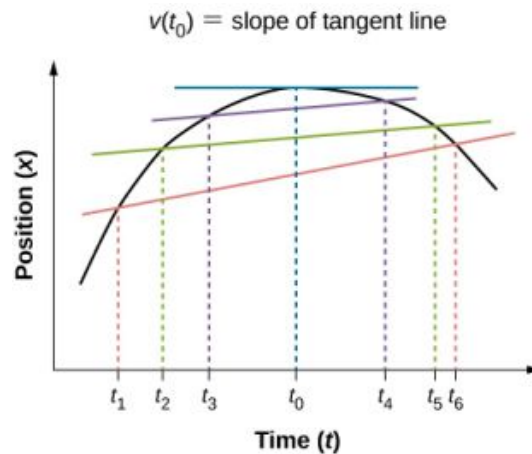


FIGURE 3.6 In a graph of position versus time, the instantaneous velocity is the slope of the tangent line at a given point. The average velocities $\bar{v} = \frac{\Delta x}{\Delta t} = \frac{x_f - x_i}{t_f - t_i}$ between times $\Delta t = t_6 - t_1$, $\Delta t = t_5 - t_2$, and $\Delta t = t_4 - t_3$ are shown. When $\Delta t \rightarrow 0$, the average velocity approaches the instantaneous velocity at $t = t_0$.



EXAMPLE 3.2

Finding Velocity from a Position-Versus-Time Graph

Given the position-versus-time graph of [Figure 3.7](#), find the velocity-versus-time graph.

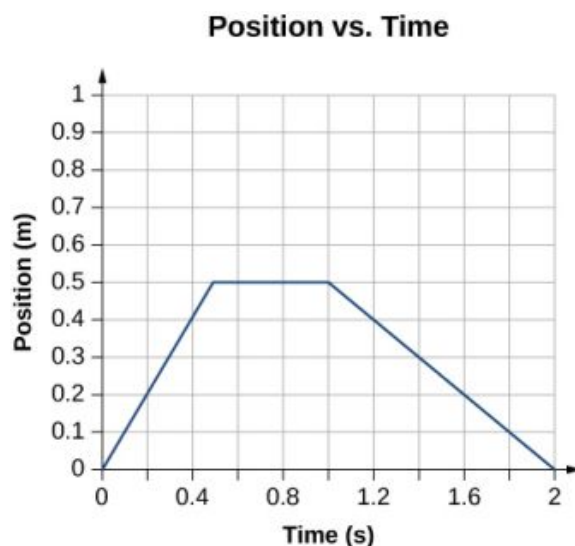


FIGURE 3.7 The object starts out in the positive direction, stops for a short time, and then reverses direction, heading back toward the origin. Notice that the object comes to rest instantaneously, which would require an infinite force. Thus, the graph is an approximation of motion in the real world. (The concept of force is discussed in [Newton's Laws of Motion](#).)

Strategy

The graph contains three straight lines during three time intervals. We find the velocity during each time interval by taking the slope of the line using the grid.

Solution

$$\text{Time interval } 0 \text{ s to } 0.5 \text{ s: } \bar{v} = \frac{\Delta x}{\Delta t} = \frac{0.5 \text{ m} - 0.0 \text{ m}}{0.5 \text{ s} - 0.0 \text{ s}} = 1.0 \text{ m/s}$$

$$\text{Time interval } 0.5 \text{ s to } 1.0 \text{ s: } \bar{v} = \frac{\Delta x}{\Delta t} = \frac{0.5 \text{ m} - 0.5 \text{ m}}{1.0 \text{ s} - 0.5 \text{ s}} = 0.0 \text{ m/s}$$

$$\text{Time interval } 1.0 \text{ s to } 2.0 \text{ s: } \bar{v} = \frac{\Delta x}{\Delta t} = \frac{0.0 \text{ m} - 0.5 \text{ m}}{2.0 \text{ s} - 1.0 \text{ s}} = -0.5 \text{ m/s}$$

The graph of these values of velocity versus time is shown in [Figure 3.8](#).

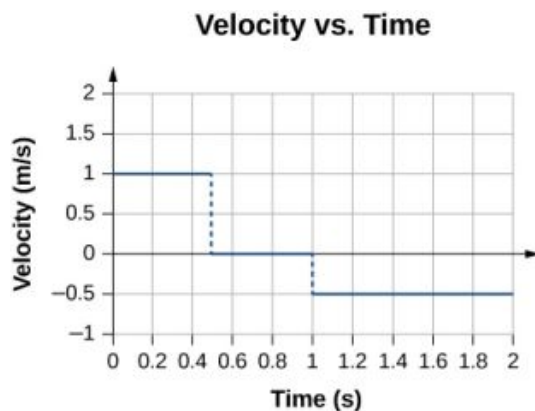


FIGURE 3.8 The velocity is positive for the first part of the trip, zero when the object is stopped, and negative when the object reverses direction.

Significance

During the time interval between 0 s and 0.5 s, the object's position is moving away from the origin and the position-versus-time curve has a positive slope. At any point along the curve during this time interval, we can find the instantaneous velocity by taking its slope, which is +1 m/s, as shown in [Figure 3.8](#). In the subsequent time interval, between 0.5 s and 1.0 s, the position doesn't change and we see the slope is zero. From 1.0 s to 2.0 s, the object is moving back toward the origin and the slope is -0.5 m/s. The object has reversed direction and has a negative velocity.

Speed

In everyday language, most people use the terms *speed* and *velocity* interchangeably. In physics, however, they do not have the same meaning and are distinct concepts. One major difference is that speed has no direction; that is, speed is a scalar.

We can calculate the **average speed** by finding the total distance traveled divided by the elapsed time:

$$\text{Average speed} = \bar{s} = \frac{\text{Total distance}}{\text{Elapsed time}}. \quad 3.5$$

Average speed is not necessarily the same as the magnitude of the average velocity, which is found by dividing the magnitude of the total displacement by the elapsed time. For example, if a trip starts and ends at the same location, the total displacement is zero, and therefore the average velocity is zero. The average speed, however, is not zero, because the total distance traveled is greater than zero. If we take a road trip of 300 km and need to be at our destination at a certain time, then we would be interested in our average speed.

However, we can calculate the **instantaneous speed** from the magnitude of the instantaneous velocity:

$$\text{Instantaneous speed} = |v(t)|. \quad 3.6$$

If a particle is moving along the x -axis at $+7.0$ m/s and another particle is moving along the same axis at -7.0 m/s, they have different velocities, but both have the same speed of 7.0 m/s. Some typical speeds are shown in the following table.

Speed	m/s	mi/h
Continental drift	10^{-7}	2×10^{-7}
Brisk walk	1.7	3.9
Cyclist	4.4	10
Sprint runner	12.2	27
Rural speed limit	24.6	56
Official land speed record	341.1	763
Speed of sound at sea level	343	768
Space shuttle on reentry	7800	17,500
Escape velocity of Earth*	11,200	25,000
Orbital speed of Earth around the Sun	29,783	66,623
Speed of light in a vacuum	299,792,458	670,616,629

TABLE 3.1 Speeds of Various Objects *Escape velocity is the velocity at which an object must be launched so that it overcomes Earth's gravity and is not pulled back toward Earth.

Calculating Instantaneous Velocity

When calculating instantaneous velocity, we need to specify the explicit form of the position function $x(t)$. If each term in the $x(t)$ equation has the form of At^n where A is a constant and n is an integer, this can be differentiated

using the power rule to be:

$$\frac{d(At^n)}{dt} = An t^{n-1}. \quad 3.7$$

Note that if there are additional terms added together, this power rule of differentiation can be done multiple times and the solution is the sum of those terms. The following example illustrates the use of [Equation 3.7](#).



EXAMPLE 3.3

Instantaneous Velocity Versus Average Velocity

The position of a particle is given by $x(t) = (3.0 \text{ m/s})t + (0.5 \text{ m/s}^3)t^3$.

- Using [Equation 3.4](#) and [Equation 3.7](#), find the instantaneous velocity at $t = 2.0 \text{ s}$.
- Calculate the average velocity between 1.0 s and 3.0 s .

Strategy

[Equation 3.4](#) gives the instantaneous velocity of the particle as the derivative of the position function. Looking at the form of the position function given, we see that it is a polynomial in t . Therefore, we can use [Equation 3.7](#), the power rule from calculus, to find the solution. We use [Equation 3.6](#) to calculate the average velocity of the particle.

Solution

$$\text{a. } v(t) = \frac{dx(t)}{dt} = (3.0 \text{ m/s}) + (1.5 \text{ m/s}^3)t^2.$$

Substituting $t = 2.0 \text{ s}$ into this equation gives $v(2.0 \text{ s}) = (3.0 \text{ m/s}) + (1.5 \text{ m/s}^3)(2.0 \text{ s})^2$.

- To determine the average velocity of the particle between 1.0 s and 3.0 s , we calculate the values of $x(1.0 \text{ s})$ and $x(3.0 \text{ s})$:

$$x(1.0 \text{ s}) = [(3.0 \text{ m/s})(1.0 \text{ s}) + (0.5 \text{ m/s}^3)(1.0 \text{ s})^3] = 3.5 \text{ m}$$

$$x(3.0 \text{ s}) = [(3.0 \text{ m/s})(3.0 \text{ s}) + (0.5 \text{ m/s}^3)(3.0 \text{ s})^3] = 22.5 \text{ m}$$

Then the average velocity is

$$\bar{v} = \frac{x(3.0 \text{ s}) - x(1.0 \text{ s})}{t(3.0 \text{ s}) - t(1.0 \text{ s})} = \frac{(22.5 - 3.5) \text{ m}}{(3.0 - 1.0) \text{ s}} = 9.5 \text{ m/s}.$$

Significance

In the limit that the time interval used to calculate $\bar{v}$ goes to zero, the value obtained for $\bar{v}$ converges to the value of v .



EXAMPLE 3.4

Instantaneous Velocity Versus Speed

Consider the motion of a particle in which the position is $x(t) = (3.0 \text{ m/s})t - (3 \text{ m/s}^2)t^2$.

- What is the instantaneous velocity at $t = 0.25 \text{ s}$, $t = 0.50 \text{ s}$, and $t = 1.0 \text{ s}$?
- What is the speed of the particle at these times?

Strategy

The instantaneous velocity is the derivative of the position function and the speed is the magnitude of the instantaneous velocity. We use [Equation 3.4](#) and [Equation 3.7](#) to solve for instantaneous velocity.

Solution

- a. $v(t) = \frac{dx(t)}{dt} = (3.0 \text{ m/s} - 6.0 \text{ m/s}^2 t)$, $v(0.25 \text{ s}) = 1.50 \text{ m/s}$,
 $v(0.5 \text{ s}) = 0 \text{ m/s}$, $v(1.0 \text{ s}) = -3.0 \text{ m/s}$
 b. Speed = $|v(t)| = 1.50 \text{ m/s}$, 0.0 m/s , and 3.0 m/s

Significance

The velocity of the particle gives us direction information, indicating the particle is moving to the left (west) or right (east). The speed gives the magnitude of the velocity. By graphing the position, velocity, and speed as functions of time, we can understand these concepts visually [Figure 3.9](#). In (a), the graph shows the particle moving in the positive direction until $t = 0.5 \text{ s}$, when it reverses direction. The reversal of direction can also be seen in (b) at 0.5 s where the velocity is zero and then turns negative. At 1.0 s it is back at the origin where it started. The particle's velocity at 1.0 s in (b) is negative, because it is traveling in the negative direction. But in (c), however, its speed is positive and remains positive throughout the travel time. We can also interpret velocity as the slope of the position-versus-time graph. The slope of $x(t)$ is decreasing toward zero, becoming zero at 0.5 s and increasingly negative thereafter. This analysis of comparing the graphs of position, velocity, and speed helps catch errors in calculations. The graphs must be consistent with each other and help interpret the calculations.

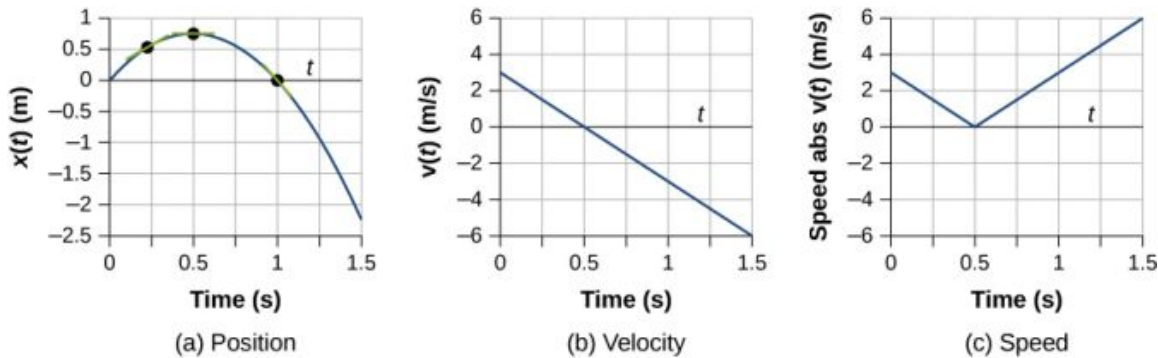


FIGURE 3.9 (a) Position: $x(t)$ versus time. (b) Velocity: $v(t)$ versus time. The slope of the position graph is the velocity. A rough comparison of the slopes of the tangent lines in (a) at 0.25 s , 0.5 s , and 1.0 s with the values for velocity at the corresponding times indicates they are the same values. (c) Speed: $|v(t)|$ versus time. Speed is always a positive number.

✓ CHECK YOUR UNDERSTANDING 3.2

The position of an object as a function of time is $x(t) = (-3 \text{ m/s}^2)t^2$. (a) What is the velocity of the object as a function of time? (b) Is the velocity ever positive? (c) What are the velocity and speed at $t = 1.0 \text{ s}$?

3.3 Average and Instantaneous Acceleration

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Calculate the average acceleration between two points in time.
- Calculate the instantaneous acceleration given the functional form of velocity.
- Explain the vector nature of instantaneous acceleration and velocity.
- Explain the difference between average acceleration and instantaneous acceleration.
- Find instantaneous acceleration at a specified time on a graph of velocity versus time.

The importance of understanding acceleration spans our day-to-day experience, as well as the vast reaches of outer space and the tiny world of subatomic physics. In everyday conversation, to *accelerate* means to speed up; applying the brake pedal causes a vehicle to slow down. We are familiar with the acceleration of our car, for example. The greater the acceleration, the greater the change in velocity over a given time. Acceleration is widely seen in experimental physics. In linear particle accelerator experiments, for example, subatomic particles are accelerated to very high velocities in collision experiments, which tell us information about the structure of the subatomic world.

as well as the origin of the universe. In space, cosmic rays are subatomic particles that have been accelerated to very high energies in supernovas (exploding massive stars) and active galactic nuclei. It is important to understand the processes that accelerate cosmic rays because these rays contain highly penetrating radiation that can damage electronics flown on spacecraft, for example.

Average Acceleration

The formal definition of acceleration is consistent with these notions just described, but is more inclusive.

AVERAGE ACCELERATION

Average acceleration is the rate at which velocity changes:

$$\bar{a} = \frac{\Delta v}{\Delta t} = \frac{v_f - v_0}{t_f - t_0}, \quad 3.8$$

where $\bar{a}$ is **average acceleration**, v is velocity, and t is time. (The bar over the a means *average* acceleration.)

Because acceleration is velocity in meters per second divided by time in seconds, the SI units for acceleration are often abbreviated m/s^2 —that is, meters per second squared or meters per second per second. This literally means by how many meters per second the velocity changes every second. Recall that velocity is a vector—it has both magnitude and direction—which means that a change in velocity can be a change in magnitude (or speed), but it can also be a change in direction. For example, if a runner traveling at 10 km/h due east slows to a stop, reverses direction, and continues her run at 10 km/h due west, her velocity has changed as a result of the change in direction, although the *magnitude* of the velocity is the same in both directions. Thus, acceleration occurs when velocity changes in magnitude (an increase or decrease in speed) or in direction, or both.

ACCELERATION AS A VECTOR

Acceleration is a vector in the same direction as the *change* in velocity, Δv . Since velocity is a vector, it can change in magnitude or in direction, or both. Acceleration is, therefore, a change in speed or direction, or both.

Keep in mind that although acceleration is in the direction of the change in velocity, it is not always in the direction of motion. When an object slows down, its acceleration is opposite to the direction of its motion. Although this is commonly referred to as *deceleration* [Figure 3.10](#), we say the train is accelerating in a direction opposite to its direction of motion.



FIGURE 3.10 A subway train in Sao Paulo, Brazil, accelerates opposite to the motion as it comes into a station. It is accelerating in a direction opposite to its direction of motion. (credit: modification of work by Yusuke Kawasaki)

The term *deceleration* can cause confusion in our analysis because it is not a vector and it does not point to a specific direction with respect to a coordinate system, so we do not use it. Acceleration is a vector, so we must choose the appropriate sign for it in our chosen coordinate system. In the case of the train in [Figure 3.10](#),

acceleration is *in the negative direction in the chosen coordinate system*, so we say the train is undergoing negative acceleration.

If an object in motion has a velocity in the positive direction with respect to a chosen origin and it acquires a constant negative acceleration, the object eventually comes to a rest and reverses direction. If we wait long enough, the object passes through the origin going in the opposite direction. This is illustrated in [Figure 3.11](#).

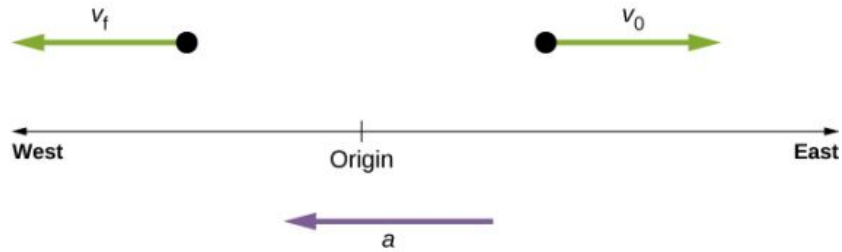


FIGURE 3.11 An object in motion with a velocity vector toward the east under negative acceleration comes to a rest and reverses direction. It passes the origin going in the opposite direction after a long enough time.

EXAMPLE 3.5

Calculating Average Acceleration: A Racehorse Leaves the Gate

A racehorse coming out of the gate accelerates from rest to a velocity of 15.0 m/s due west in 1.80 s. What is its average acceleration?



FIGURE 3.12 Racehorses accelerating out of the gate. (credit: modification of work by Jon Sullivan)

Strategy

First we draw a sketch and assign a coordinate system to the problem [Figure 3.13](#). This is a simple problem, but it always helps to visualize it. Notice that we assign east as positive and west as negative. Thus, in this case, we have negative velocity.

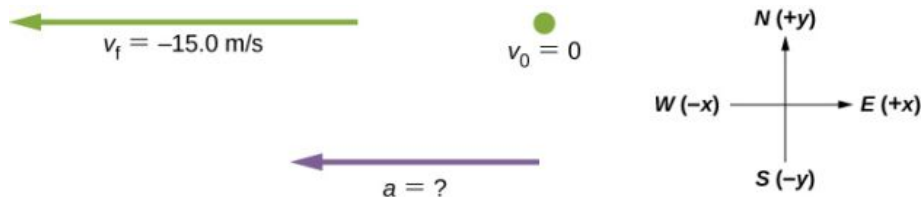


FIGURE 3.13 Identify the coordinate system, the given information, and what you want to determine.

We can solve this problem by identifying Δv and Δt from the given information, and then calculating the average acceleration directly from the equation $\bar{a} = \frac{\Delta v}{\Delta t} = \frac{v_f - v_0}{t_f - t_0}$.

Solution

First, identify the knowns: $v_0 = 0$, $v_f = -15.0 \text{ m/s}$ (the negative sign indicates direction toward the west), $\Delta t = 1.80 \text{ s}$.

Second, find the change in velocity. Since the horse is going from zero to -15.0 m/s , its change in velocity equals its final velocity:

$$\Delta v = v_f - v_0 = v_f = -15.0 \text{ m/s}.$$

Last, substitute the known values (Δv and Δt) and solve for the unknown $\bar{a}$:

$$\bar{a} = \frac{\Delta v}{\Delta t} = \frac{-15.0 \text{ m/s}}{1.80 \text{ s}} = -8.33 \text{ m/s}^2.$$

Significance

The negative sign for acceleration indicates that acceleration is toward the west. An acceleration of 8.33 m/s^2 due west means the horse increases its velocity by 8.33 m/s due west each second; that is, $8.33 \text{ meters per second per second}$, which we write as 8.33 m/s^2 . This is truly an average acceleration, because the ride is not smooth. We see later that an acceleration of this magnitude would require the rider to hang on with a force nearly equal to his weight.

 CHECK YOUR UNDERSTANDING 3.3

Protons in a linear accelerator are accelerated from rest to $2.0 \times 10^7 \text{ m/s}$ in 10^{-4} s . What is the average acceleration of the protons?

Instantaneous Acceleration

Instantaneous acceleration a , or *acceleration at a specific instant in time*, is obtained using the same process discussed for instantaneous velocity. That is, we calculate the average acceleration between two points in time separated by Δt and let Δt approach zero. The result is the derivative of the velocity function $v(t)$, which is **instantaneous acceleration** and is expressed mathematically as

$$a(t) = \frac{d}{dt}v(t). \quad 3.9$$

Thus, similar to velocity being the derivative of the position function, instantaneous acceleration is the derivative of the velocity function. We can show this graphically in the same way as instantaneous velocity. In [Figure 3.14](#), instantaneous acceleration at time t_0 is the slope of the tangent line to the velocity-versus-time graph at time t_0 . We see that average acceleration $\bar{a} = \frac{\Delta v}{\Delta t}$ approaches instantaneous acceleration as Δt approaches zero. Also in part (a) of the figure, we see that velocity has a maximum when its slope is zero. This time corresponds to the zero of the acceleration function. In part (b), instantaneous acceleration at the minimum velocity is shown, which is also zero, since the slope of the curve is zero there, too. Thus, for a given velocity function, the zeros of the acceleration function give either the minimum or the maximum velocity.

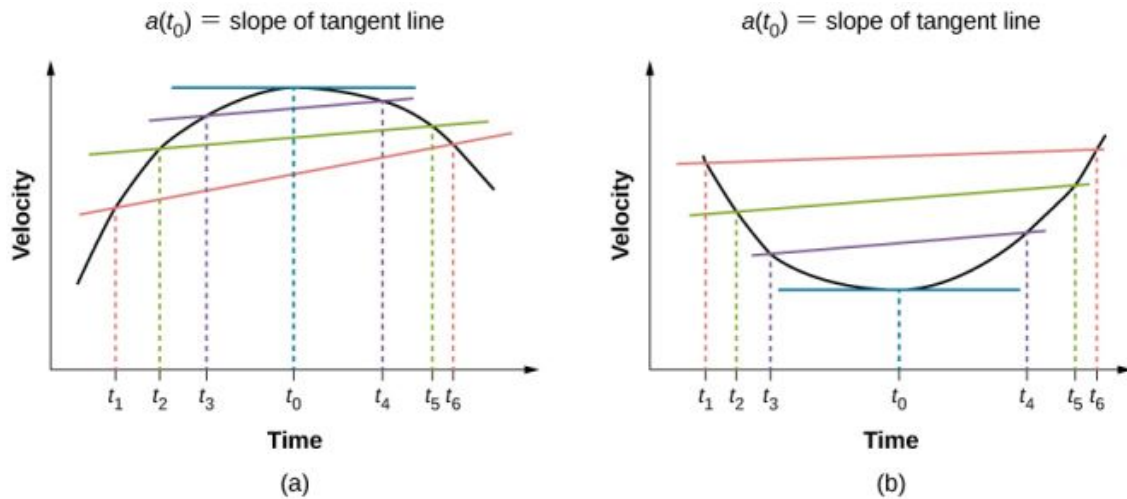


FIGURE 3.14 In a graph of velocity versus time, instantaneous acceleration is the slope of the tangent line. (a) Shown is average acceleration $\bar{a} = \frac{\Delta v}{\Delta t} = \frac{v_f - v_i}{t_f - t_i}$ between times $\Delta t = t_6 - t_1$, $\Delta t = t_5 - t_2$, and $\Delta t = t_4 - t_3$. When $\Delta t \rightarrow 0$, the average acceleration approaches instantaneous acceleration at time t_0 . In view (a), instantaneous acceleration is shown for the point on the velocity curve at maximum velocity. At this point, instantaneous acceleration is the slope of the tangent line, which is zero. At any other time, the slope of the tangent line—and thus instantaneous acceleration—would not be zero. (b) Same as (a) but shown for instantaneous acceleration at minimum velocity.

To illustrate this concept, let's look at two examples. First, a simple example is shown using [Figure 3.9\(b\)](#), the velocity-versus-time graph of [Example 3.4](#), to find acceleration graphically. This graph is depicted in [Figure 3.15\(a\)](#), which is a straight line. The corresponding graph of acceleration versus time is found from the slope of velocity and is shown in [Figure 3.15\(b\)](#). In this example, the velocity function is a straight line with a constant slope, thus acceleration is a constant. In the next example, the velocity function has a more complicated functional dependence on time.

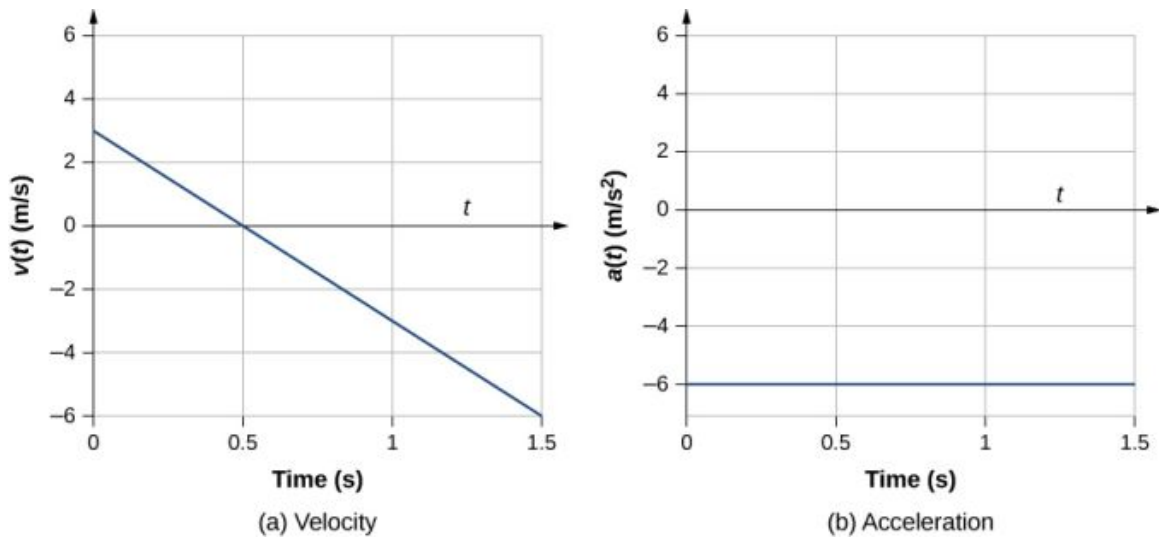


FIGURE 3.15 (a, b) The velocity-versus-time graph is linear and has a negative constant slope (a) that is equal to acceleration, shown in (b).

If we know the functional form of velocity, $v(t)$, we can calculate instantaneous acceleration $a(t)$ at any time point in the motion using [Equation 3.9](#).



EXAMPLE 3.6

Calculating Instantaneous Acceleration

A particle is in motion and is accelerating. The functional form of the velocity is $v(t) = (20 \text{ m/s}^2)t - (5 \text{ m/s}^3)t^2$.

- Find the functional form of the acceleration.

- b. Find the instantaneous velocity at $t = 1, 2, 3$, and 5 s.
- c. Find the instantaneous acceleration at $t = 1, 2, 3$, and 5 s.
- d. Interpret the results of (c) in terms of the directions of the acceleration and velocity vectors.

Strategy

We find the functional form of acceleration by taking the derivative of the velocity function. Then, we calculate the values of instantaneous velocity and acceleration from the given functions for each. For part (d), we need to compare the directions of velocity and acceleration at each time.

Solution

- a. $a(t) = \frac{dv(t)}{dt} = (20 \text{ m/s}^2) - (10 \text{ m/s}^3)t$
- b. $v(1 \text{ s}) = 15 \text{ m/s}$, $v(2 \text{ s}) = 20 \text{ m/s}$, $v(3 \text{ s}) = 15 \text{ m/s}$, $v(5 \text{ s}) = -25 \text{ m/s}$
- c. $a(1 \text{ s}) = 10 \text{ m/s}^2$, $a(2 \text{ s}) = 0 \text{ m/s}^2$, $a(3 \text{ s}) = -10 \text{ m/s}^2$, $a(5 \text{ s}) = -30 \text{ m/s}^2$
- d. At $t = 1$ s, velocity $v(1 \text{ s}) = 15 \text{ m/s}$ is positive and acceleration is positive, so both velocity and acceleration are in the same direction. The particle is moving faster.

At $t = 2$ s, velocity has increased to $v(2 \text{ s}) = 20 \text{ m/s}$, where it is maximum, which corresponds to the time when the acceleration is zero. We see that the maximum velocity occurs when the slope of the velocity function is zero, which is just the zero of the acceleration function.

At $t = 3$ s, velocity is $v(3 \text{ s}) = 15 \text{ m/s}$ and acceleration is negative. The particle has reduced its velocity and the acceleration vector is negative. The particle is slowing down.

At $t = 5$ s, velocity is $v(5 \text{ s}) = -25 \text{ m/s}$ and acceleration is increasingly negative. Between the times $t = 3$ s and $t = 5$ s the particle has decreased its velocity to zero and then become negative, thus reversing its direction. The particle is now speeding up again, but in the opposite direction.

We can see these results graphically in [Figure 3.16](#).

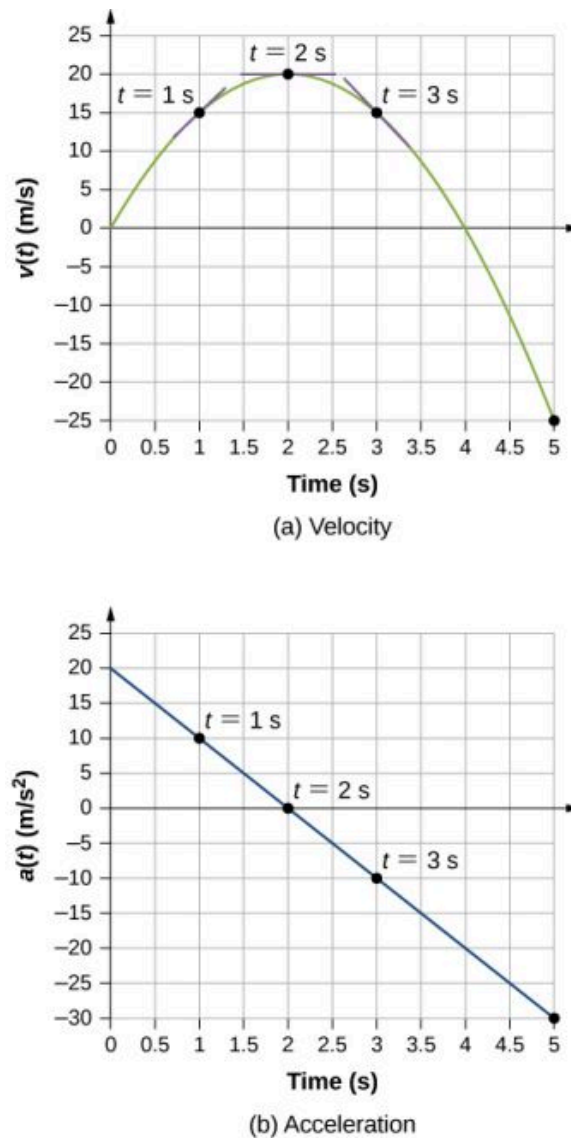


FIGURE 3.16 (a) Velocity versus time. Tangent lines are indicated at times 1, 2, and 3 s. The slopes of the tangent lines are the accelerations. At $t = 3$ s, velocity is positive. At $t = 5$ s, velocity is negative, indicating the particle has reversed direction. (b) Acceleration versus time. Comparing the values of accelerations given by the black dots with the corresponding slopes of the tangent lines (slopes of lines through black dots) in (a), we see they are identical.

Significance

By doing both a numerical and graphical analysis of velocity and acceleration of the particle, we can learn much about its motion. The numerical analysis complements the graphical analysis in giving a total view of the motion. The zero of the acceleration function corresponds to the maximum of the velocity in this example. Also in this example, when acceleration is positive and in the same direction as velocity, velocity increases. As acceleration tends toward zero, eventually becoming negative, the velocity reaches a maximum, after which it starts decreasing. If we wait long enough, velocity also becomes negative, indicating a reversal of direction. A real-world example of this type of motion is a car with a velocity that is increasing to a maximum, after which it starts slowing down, comes to a stop, then reverses direction.

✓ CHECK YOUR UNDERSTANDING 3.4

An airplane lands on a runway traveling east. Describe its acceleration.

Getting a Feel for Acceleration

You are probably used to experiencing acceleration when you step into an elevator, or step on the gas pedal in your car. However, acceleration is happening to many other objects in our universe with which we don't have direct contact. [Table 3.2](#) presents the acceleration of various objects. We can see the magnitudes of the accelerations extend over many orders of magnitude.

Acceleration	Value (m/s ²)
High-speed train	0.25
Elevator	2
Cheetah	5
Object in a free fall without air resistance near the surface of Earth	9.8
Space shuttle maximum during launch	29
Parachutist peak during normal opening of parachute	59
F16 aircraft pulling out of a dive	79
Explosive seat ejection from aircraft	147
Sprint missile	982
Fastest rocket sled peak acceleration	1540
Jumping flea	3200
Baseball struck by a bat	30,000
Closing jaws of a trap-jaw ant	1,000,000
Proton in the large Hadron collider	1.9×10^9

TABLE 3.2 Typical Values of Acceleration (credit: Wikipedia: Orders of Magnitude (acceleration))

In this table, we see that typical accelerations vary widely with different objects and have nothing to do with object size or how massive it is. Acceleration can also vary widely with time during the motion of an object. A drag racer has a large acceleration just after its start, but then it tapers off as the vehicle reaches a constant velocity. Its average acceleration can be quite different from its instantaneous acceleration at a particular time during its motion. [Figure 3.17](#) compares graphically average acceleration with instantaneous acceleration for two very different motions.

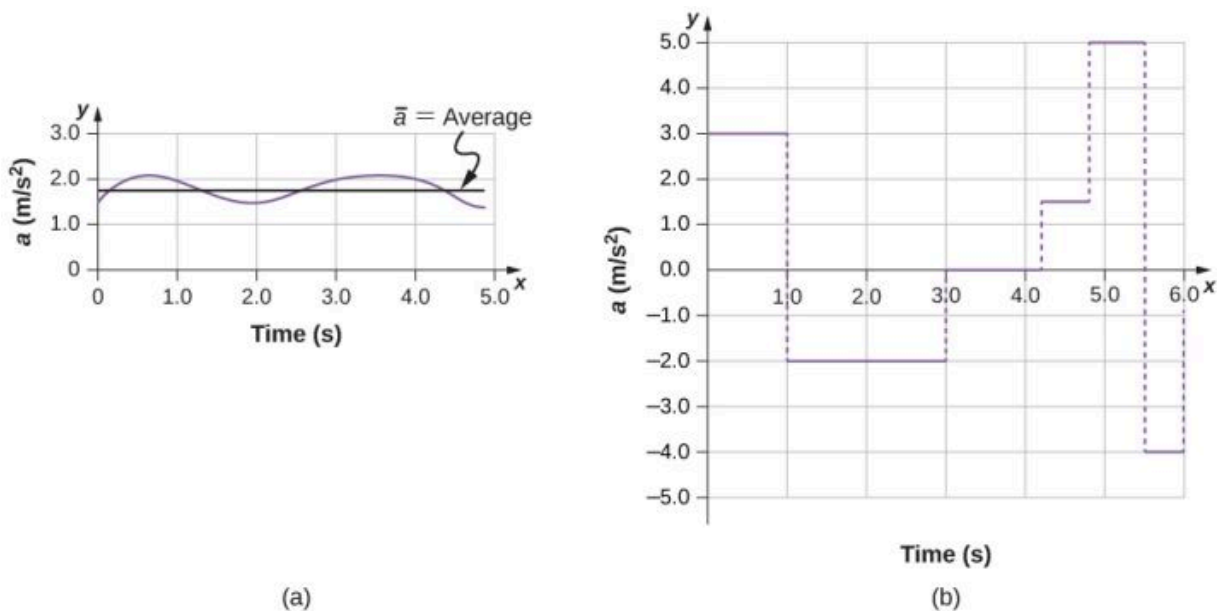


FIGURE 3.17 Graphs of instantaneous acceleration versus time for two different one-dimensional motions. (a) Acceleration varies only slightly and is always in the same direction, since it is positive. The average over the interval is nearly the same as the acceleration at any given time. (b) Acceleration varies greatly, perhaps representing a package on a post office conveyor belt that is accelerated forward and backward as it bumps along. It is necessary to consider small time intervals (such as from 0–1.0 s) with constant or nearly constant acceleration in such a situation.

INTERACTIVE

Learn about position, velocity, and acceleration graphs. Move the little man back and forth with a mouse and plot his motion. Set the position, velocity, or acceleration and let the simulation move the man for you. Visit [this link](https://openstax.org/l/21movmansimul) (<https://openstax.org/l/21movmansimul>) to use the moving man simulation.

3.4 Motion with Constant Acceleration

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Identify which equations of motion are to be used to solve for unknowns.
- Use appropriate equations of motion to solve a two-body pursuit problem.

You might guess that the greater the acceleration of, say, a car moving away from a stop sign, the greater the car's displacement in a given time. But, we have not developed a specific equation that relates acceleration and displacement. In this section, we look at some convenient equations for kinematic relationships, starting from the definitions of displacement, velocity, and acceleration. We first investigate a single object in motion, called single-body motion. Then we investigate the motion of two objects, called **two-body pursuit problems**.

Notation

First, let us make some simplifications in notation. Taking the initial time to be zero, as if time is measured with a stopwatch, is a great simplification. Since elapsed time is $\Delta t = t_f - t_0$, taking $t_0 = 0$ means that $\Delta t = t_f$, the final time on the stopwatch. When initial time is taken to be zero, we use the subscript 0 to denote initial values of position and velocity. That is, x_0 is the initial position and v_0 is the initial velocity. We put no subscripts on the final values. That is, t is the final time, x is the final position, and v is the final velocity. This gives a simpler expression for elapsed time, $\Delta t = t$. It also simplifies the expression for x displacement, which is now $\Delta x = x - x_0$. Also, it simplifies the expression for change in velocity, which is now $\Delta v = v - v_0$. To summarize, using the simplified notation, with the initial time taken to be zero,

$$\begin{aligned}\Delta t &= t \\ \Delta x &= x - x_0 \\ \Delta v &= v - v_0,\end{aligned}$$

where the subscript 0 denotes an initial value and the absence of a subscript denotes a final value in whatever motion is under consideration.

We now make the important assumption that *acceleration is constant*. This assumption allows us to avoid using calculus to find instantaneous acceleration. Since acceleration is constant, the average and instantaneous accelerations are equal—that is,

$$\bar{a} = a = \text{constant}.$$

Thus, we can use the symbol a for acceleration at all times. Assuming acceleration to be constant does not seriously limit the situations we can study nor does it degrade the accuracy of our treatment. For one thing, acceleration *is* constant in a great number of situations. Furthermore, in many other situations we can describe motion accurately by assuming a constant acceleration equal to the average acceleration for that motion. Lastly, for motion during which acceleration changes drastically, such as a car accelerating to top speed and then braking to a stop, motion can be considered in separate parts, each of which has its own constant acceleration.

Displacement and Position from Velocity

To get our first two equations, we start with the definition of average velocity:

$$\bar{v} = \frac{\Delta x}{\Delta t}.$$

Substituting the simplified notation for Δx and Δt yields

$$\bar{v} = \frac{x - x_0}{t}.$$

Solving for x gives us

$$x = x_0 + \bar{v}t, \quad 3.10$$

where the average velocity is

$$\bar{v} = \frac{v_0 + v}{2}. \quad 3.11$$

The equation $\bar{v} = \frac{v_0 + v}{2}$ reflects the fact that when acceleration is constant, $\bar{v}$ is just the simple average of the initial and final velocities. [Figure 3.18](#) illustrates this concept graphically. In part (a) of the figure, acceleration is constant, with velocity increasing at a constant rate. The average velocity during the 1-h interval from 40 km/h to 80 km/h is 60 km/h:

$$\bar{v} = \frac{v_0 + v}{2} = \frac{40 \text{ km/h} + 80 \text{ km/h}}{2} = 60 \text{ km/h}.$$

In part (b), acceleration is not constant. During the 1-h interval, velocity is closer to 80 km/h than 40 km/h. Thus, the average velocity is greater than in part (a).

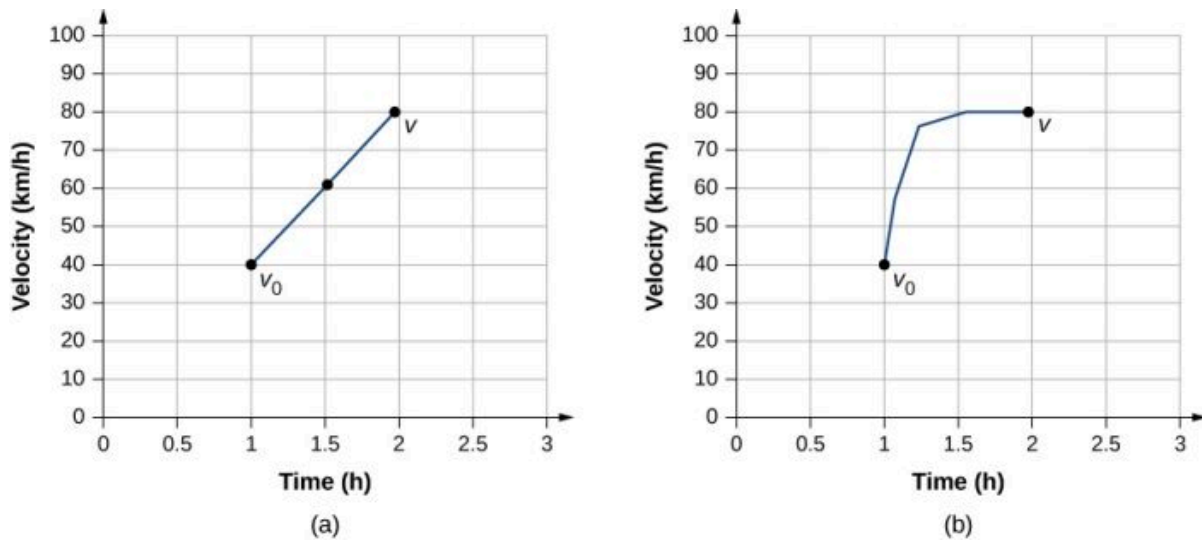


FIGURE 3.18 (a) Velocity-versus-time graph with constant acceleration showing the initial and final velocities v_0 and v . The average velocity is $\frac{1}{2}(v_0 + v) = 60$ km/h. (b) Velocity-versus-time graph with an acceleration that changes with time. The average velocity is not given by $\frac{1}{2}(v_0 + v)$, but is greater than 60 km/h.

Solving for Final Velocity from Acceleration and Time

We can derive another useful equation by manipulating the definition of acceleration:

$$a = \frac{\Delta v}{\Delta t}.$$

Substituting the simplified notation for Δv and Δt gives us

$$a = \frac{v - v_0}{t} \quad (\text{constant } a).$$

Solving for v yields

$$v = v_0 + at \quad (\text{constant } a).$$

3.12



EXAMPLE 3.7

Calculating Final Velocity

An airplane lands with an initial velocity of 70.0 m/s and then accelerates opposite to the motion at 1.50 m/s^2 for 40.0 s. What is its final velocity?

Strategy

First, we identify the knowns: $v_0 = 70 \text{ m/s}$, $a = -1.50 \text{ m/s}^2$, $t = 40 \text{ s}$.

Second, we identify the unknown; in this case, it is final velocity v_f .

Last, we determine which equation to use. To do this we figure out which kinematic equation gives the unknown in terms of the knowns. We calculate the final velocity using [Equation 3.12](#), $v = v_0 + at$.

Solution

Substitute the known values and solve:

$$v = v_0 + at = 70.0 \text{ m/s} + (-1.50 \text{ m/s}^2)(40.0 \text{ s}) = 10.0 \text{ m/s}.$$

[Figure 3.19](#) is a sketch that shows the acceleration and velocity vectors.



FIGURE 3.19 The airplane lands with an initial velocity of 70.0 m/s and slows to a final velocity of 10.0 m/s before heading for the terminal. Note the acceleration is negative because its direction is opposite to its velocity, which is positive.

Significance

The final velocity is much less than the initial velocity, as desired when slowing down, but is still positive (see figure). With jet engines, reverse thrust can be maintained long enough to stop the plane and start moving it backward, which is indicated by a negative final velocity, but is not the case here.

In addition to being useful in problem solving, the equation $v = v_0 + at$ gives us insight into the relationships among velocity, acceleration, and time. We can see, for example, that

- Final velocity depends on how large the acceleration is and how long it lasts
- If the acceleration is zero, then the final velocity equals the initial velocity ($v = v_0$), as expected (in other words, velocity is constant)
- If a is negative, then the final velocity is less than the initial velocity

All these observations fit our intuition. Note that it is always useful to examine basic equations in light of our intuition and experience to check that they do indeed describe nature accurately.

Solving for Final Position with Constant Acceleration

We can combine the previous equations to find a third equation that allows us to calculate the final position of an object experiencing constant acceleration. We start with

$$v = v_0 + at.$$

Adding v_0 to each side of this equation and dividing by 2 gives

$$\frac{v_0 + v}{2} = v_0 + \frac{1}{2}at.$$

Since $\frac{v_0 + v}{2} = \bar{v}$ for constant acceleration, we have

$$\bar{v} = v_0 + \frac{1}{2}at.$$

Now we substitute this expression for $\bar{v}$ into the equation for displacement, $x = x_0 + \bar{v}t$, yielding

$$x = x_0 + v_0t + \frac{1}{2}at^2 \quad (\text{constant } a). \quad 3.13$$



EXAMPLE 3.8

Calculating Displacement of an Accelerating Object

Dragsters can achieve an average acceleration of 26.0 m/s^2 . Suppose a dragster accelerates from rest at this rate for 5.56 s (Figure 3.20). How far does it travel in this time?



FIGURE 3.20 U.S. Army Top Fuel pilot Tony “The Sarge” Schumacher begins a race with a controlled burnout. (credit: Lt. Col. William Thurmond. Photo Courtesy of U.S. Army.)

Strategy

First, let’s draw a sketch (Figure 3.21). We are asked to find displacement, which is x if we take x_0 to be zero. (Think about x_0 as the starting line of a race. It can be anywhere, but we call it zero and measure all other positions relative to it.) We can use the equation $x = x_0 + v_0t + \frac{1}{2}at^2$ when we identify v_0 , a , and t from the statement of the problem.

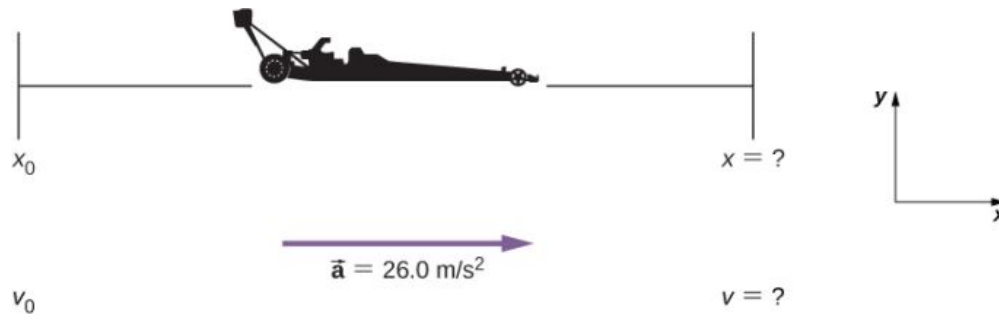


FIGURE 3.21 Sketch of an accelerating dragster.

Solution

First, we need to identify the knowns. Starting from rest means that $v_0 = 0$, a is given as 26.0 m/s^2 and t is given as 5.56 s .

Second, we substitute the known values into the equation to solve for the unknown:

$$x = x_0 + v_0t + \frac{1}{2}at^2.$$

Since the initial position and velocity are both zero, this equation simplifies to

$$x = \frac{1}{2}at^2.$$

Substituting the identified values of a and t gives

$$x = \frac{1}{2}(26.0 \text{ m/s}^2)(5.56 \text{ s})^2 = 402 \text{ m}.$$

Significance

If we convert 402 m to miles, we find that the distance covered is very close to one-quarter of a mile, the standard distance for drag racing. So, our answer is reasonable. This is an impressive displacement to cover in only 5.56 s ,

but top-notch dragsters can do a quarter mile in even less time than this. If the dragster were given an initial velocity, this would add another term to the distance equation. If the same acceleration and time are used in the equation, the distance covered would be much greater.

What else can we learn by examining the equation $x = x_0 + v_0 t + \frac{1}{2} a t^2$? We can see the following relationships:

- Displacement depends on the square of the elapsed time when acceleration is not zero. In [Example 3.8](#), the dragster covers only one-fourth of the total distance in the first half of the elapsed time.
- If acceleration is zero, then initial velocity equals average velocity ($v_0 = \bar{v}$), and $x = x_0 + v_0 t + \frac{1}{2} a t^2$ becomes $x = x_0 + v_0 t$.

Solving for Final Velocity from Distance and Acceleration

A fourth useful equation can be obtained from another algebraic manipulation of previous equations. If we solve $v = v_0 + at$ for t , we get

$$t = \frac{v - v_0}{a}.$$

Substituting this and $\bar{v} = \frac{v_0 + v}{2}$ into $x = x_0 + \bar{v}t$, we get

$$v^2 = v_0^2 + 2a(x - x_0) \quad (\text{constant } a). \quad 3.14$$



EXAMPLE 3.9

Calculating Final Velocity

Calculate the final velocity of the dragster in [Example 3.8](#) without using information about time.

Strategy

The equation $v^2 = v_0^2 + 2a(x - x_0)$ is ideally suited to this task because it relates velocities, acceleration, and displacement, and no time information is required.

Solution

First, we identify the known values. We know that $v_0 = 0$, since the dragster starts from rest. We also know that $x - x_0 = 402 \text{ m}$ (this was the answer in [Example 3.8](#)). The average acceleration was given by $a = 26.0 \text{ m/s}^2$.

Second, we substitute the knowns into the equation $v^2 = v_0^2 + 2a(x - x_0)$ and solve for v :

$$v^2 = 0 + 2 (26.0 \text{ m/s}^2) (402 \text{ m}).$$

Thus,

$$v^2 = 2.09 \times 10^4 \text{ m}^2/\text{s}^2$$

$$v = \sqrt{2.09 \times 10^4 \text{ m}^2/\text{s}^2} = 145 \text{ m/s}.$$

Significance

A velocity of 145 m/s is about 522 km/h, or about 324 mi/h, but even this breakneck speed is short of the record for the quarter mile. Also, note that a square root has two values; we took the positive value to indicate a velocity in the same direction as the acceleration.

An examination of the equation $v^2 = v_0^2 + 2a(x - x_0)$ can produce additional insights into the general relationships among physical quantities:

- The final velocity depends on how large the acceleration is and the distance over which it acts.

- For a fixed acceleration, a car that is going twice as fast doesn't simply stop in twice the distance. It takes much farther to stop. (This is why we have reduced speed zones near schools.)

Putting Equations Together

In the following examples, we continue to explore one-dimensional motion, but in situations requiring slightly more algebraic manipulation. The examples also give insight into problem-solving techniques. The note that follows is provided for easy reference to the equations needed. Be aware that these equations are not independent. In many situations we have two unknowns and need two equations from the set to solve for the unknowns. We need as many equations as there are unknowns to solve a given situation.

SUMMARY OF KINEMATIC EQUATIONS (CONSTANT A)

$$\begin{aligned}x &= x_0 + \bar{v}t \\ \bar{v} &= \frac{v_0 + v}{2} \\ v &= v_0 + at \\ x &= x_0 + v_0t + \frac{1}{2}at^2 \\ v^2 &= v_0^2 + 2a(x - x_0)\end{aligned}$$

Before we get into the examples, let's look at some of the equations more closely to see the behavior of acceleration at extreme values. Rearranging [Equation 3.12](#), we have

$$a = \frac{v - v_0}{t}.$$

From this we see that, for a finite time, if the difference between the initial and final velocities is small, the acceleration is small, approaching zero in the limit that the initial and final velocities are equal. On the contrary, in the limit $t \rightarrow 0$ for a finite difference between the initial and final velocities, acceleration becomes infinite.

Similarly, rearranging [Equation 3.14](#), we can express acceleration in terms of velocities and displacement:

$$a = \frac{v^2 - v_0^2}{2(x - x_0)}.$$

Thus, for a finite difference between the initial and final velocities acceleration becomes infinite in the limit the displacement approaches zero. Acceleration approaches zero in the limit the difference in initial and final velocities approaches zero for a finite displacement.



EXAMPLE 3.10

How Far Does a Car Go?

On dry concrete, a car can accelerate opposite to the motion at a rate of 7.00 m/s^2 , whereas on wet concrete it can accelerate opposite to the motion at only 5.00 m/s^2 . Find the distances necessary to stop a car moving at 30.0 m/s (about 110 km/h) on (a) dry concrete and (b) wet concrete. (c) Repeat both calculations and find the displacement from the point where the driver sees a traffic light turn red, taking into account their reaction time of 0.500 s to get their foot on the brake.

Strategy

First, we need to draw a sketch [Figure 3.22](#). To determine which equations are best to use, we need to list all the known values and identify exactly what we need to solve for.

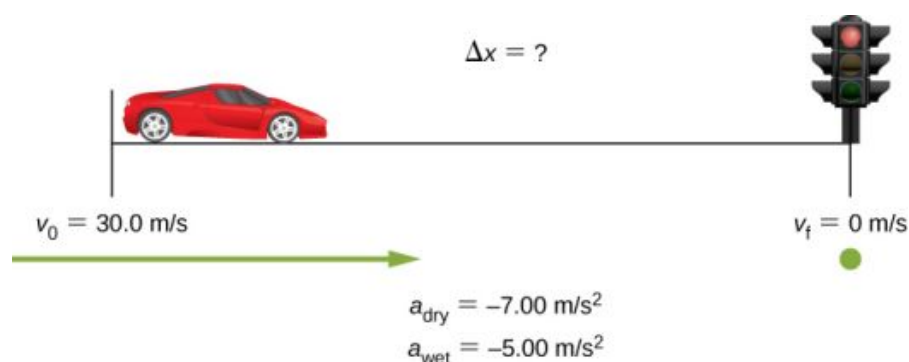


FIGURE 3.22 Sample sketch to visualize acceleration opposite to the motion and stopping distance of a car.

Solution

- a. First, we need to identify the knowns and what we want to solve for. We know that $v_0 = 30.0 \text{ m/s}$, $v = 0$, and $a = -7.00 \text{ m/s}^2$ (a is negative because it is in a direction opposite to velocity). We take x_0 to be zero. We are looking for displacement Δx , or $x - x_0$.

Second, we identify the equation that will help us solve the problem. The best equation to use is

$$v^2 = v_0^2 + 2a(x - x_0).$$

This equation is best because it includes only one unknown, x . We know the values of all the other variables in this equation. (Other equations would allow us to solve for x , but they require us to know the stopping time, t , which we do not know. We could use them, but it would entail additional calculations.)

Third, we rearrange the equation to solve for x :

$$x - x_0 = \frac{v^2 - v_0^2}{2a}$$

and substitute the known values:

$$x - 0 = \frac{0^2 - (30.0 \text{ m/s})^2}{2(-7.00 \text{ m/s}^2)}.$$

Thus,

$$x = 64.3 \text{ m on dry concrete.}$$

- b. This part can be solved in exactly the same manner as (a). The only difference is that the acceleration is -5.00 m/s^2 . The result is

$$x_{\text{wet}} = 90.0 \text{ m on wet concrete.}$$

- c. When the driver reacts, the stopping distance is the same as it is in (a) and (b) for dry and wet concrete. So, to answer this question, we need to calculate how far the car travels during the reaction time, and then add that to the stopping time. It is reasonable to assume the velocity remains constant during the driver's reaction time.

To do this, we, again, identify the knowns and what we want to solve for. We know that $\bar{v} = 30.0 \text{ m/s}$,

$t_{\text{reaction}} = 0.500 \text{ s}$, and $a_{\text{reaction}} = 0$. We take $x_{0-\text{reaction}}$ to be zero. We are looking for x_{reaction} .

Second, as before, we identify the best equation to use. In this case, $x = x_0 + \bar{v}t$ works well because the only unknown value is x , which is what we want to solve for.

Third, we substitute the knowns to solve the equation:

$$x = 0 + (30.0 \text{ m/s})(0.500 \text{ s}) = 15.0 \text{ m.}$$

This means the car travels 15.0 m while the driver reacts, making the total displacements in the two cases of dry and wet concrete 15.0 m greater than if he reacted instantly.

Last, we then add the displacement during the reaction time to the displacement when braking (Figure 3.23),

$$x_{\text{braking}} + x_{\text{reaction}} = x_{\text{total}},$$

and find (a) to be $64.3 \text{ m} + 15.0 \text{ m} = 79.3 \text{ m}$ when dry and (b) to be $90.0 \text{ m} + 15.0 \text{ m} = 105 \text{ m}$ when wet.

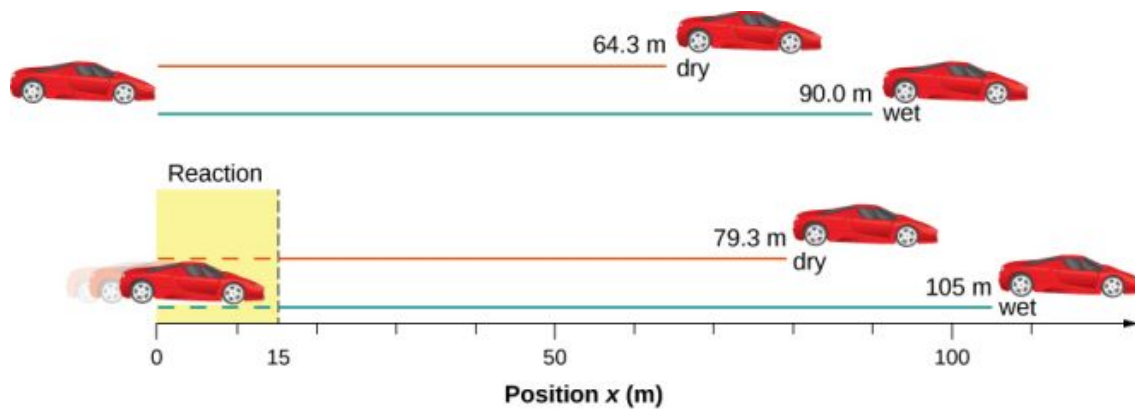


FIGURE 3.23 The distance necessary to stop a car varies greatly, depending on road conditions and driver reaction time. Shown here are the braking distances for dry and wet pavement, as calculated in this example, for a car traveling initially at 30.0 m/s. Also shown are the total distances traveled from the point when the driver first sees a light turn red, assuming a 0.500-s reaction time.

Significance

The displacements found in this example seem reasonable for stopping a fast-moving car. It should take longer to stop a car on wet pavement than dry. It is interesting that reaction time adds significantly to the displacements, but more important is the general approach to solving problems. We identify the knowns and the quantities to be determined, then find an appropriate equation. If there is more than one unknown, we need as many independent equations as there are unknowns to solve. There is often more than one way to solve a problem. The various parts of this example can, in fact, be solved by other methods, but the solutions presented here are the shortest.



EXAMPLE 3.11

Calculating Time

Suppose a car merges into freeway traffic on a 200-m-long ramp. If its initial velocity is 10.0 m/s and it accelerates at 2.00 m/s^2 , how long does it take the car to travel the 200 m up the ramp? (Such information might be useful to a traffic engineer.)

Strategy

First, we draw a sketch [Figure 3.24](#). We are asked to solve for time t . As before, we identify the known quantities to choose a convenient physical relationship (that is, an equation with one unknown, t .)

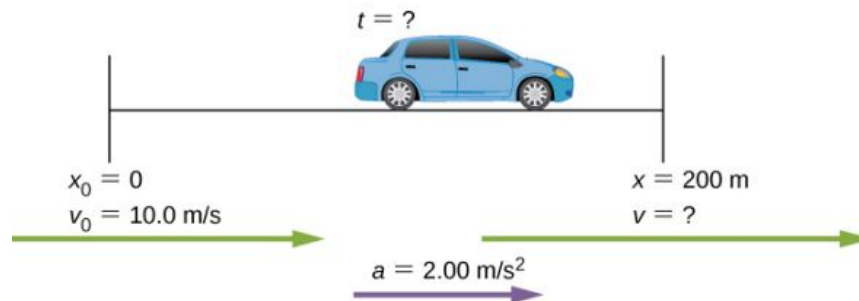


FIGURE 3.24 Sketch of a car accelerating on a freeway ramp.

Solution

Again, we identify the knowns and what we want to solve for. We know that $x_0 = 0$, $v_0 = 10 \text{ m/s}$, $a = 2.00 \text{ m/s}^2$, and $x = 200 \text{ m}$.

We need to solve for t . The equation $x = x_0 + v_0 t + \frac{1}{2} a t^2$ works best because the only unknown in the equation is the variable t , for which we need to solve. From this insight we see that when we input the knowns into the equation, we end up with a quadratic equation.

We need to rearrange the equation to solve for t , then substituting the knowns into the equation:

$$200 \text{ m} = 0 \text{ m} + (10.0 \text{ m/s})t + \frac{1}{2}(2.00 \text{ m/s}^2)t^2.$$

We then simplify the equation. The units of meters cancel because they are in each term. We can get the units of seconds to cancel by taking $t = t \text{ s}$, where t is the magnitude of time and s is the unit. Doing so leaves

$$200 = 10t + t^2.$$

We then use the quadratic formula to solve for t ,

$$t^2 + 10t - 200 = 0$$

$$t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a},$$

which yields two solutions: $t = 10.0$ and $t = -20.0$. A negative value for time is unreasonable, since it would mean the event happened 20 s before the motion began. We can discard that solution. Thus,

$$t = 10.0 \text{ s}.$$

Significance

Whenever an equation contains an unknown squared, there are two solutions. In some problems both solutions are meaningful; in others, only one solution is reasonable. The 10.0-s answer seems reasonable for a typical freeway on-ramp.

CHECK YOUR UNDERSTANDING 3.5

A rocket accelerates at a rate of 20 m/s^2 during launch. How long does it take the rocket to reach a velocity of 400 m/s ?

EXAMPLE 3.12

Acceleration of a Spaceship

A spaceship has left Earth's orbit and is on its way to the Moon. It accelerates at 20 m/s^2 for 2 min and covers a distance of 1000 km. What are the initial and final velocities of the spaceship?

Strategy

We are asked to find the initial and final velocities of the spaceship. Looking at the kinematic equations, we see that one equation will not give the answer. We must use one kinematic equation to solve for one of the velocities and substitute it into another kinematic equation to get the second velocity. Thus, we solve two of the kinematic equations simultaneously.

Solution

First we solve for v_0 using $x = x_0 + v_0t + \frac{1}{2}at^2$:

$$\begin{aligned} x - x_0 &= v_0t + \frac{1}{2}at^2 \\ 1.0 \times 10^6 \text{ m} &= v_0(120.0 \text{ s}) + \frac{1}{2}(20.0 \text{ m/s}^2)(120.0 \text{ s})^2 \\ v_0 &= 7133.3 \text{ m/s}. \end{aligned}$$

Then we substitute v_0 into $v = v_0 + at$ to solve for the final velocity:

$$v = v_0 + at = 7133.3 \text{ m/s} + (20.0 \text{ m/s}^2)(120.0 \text{ s}) = 9533.3 \text{ m/s}.$$

Significance

There are six variables in displacement, time, velocity, and acceleration that describe motion in one dimension. The initial conditions of a given problem can be many combinations of these variables. Because of this diversity, solutions may not be as easy as simple substitutions into one of the equations. This example illustrates that solutions to kinematics may require solving two simultaneous kinematic equations.

With the basics of kinematics established, we can go on to many other interesting examples and applications. In the process of developing kinematics, we have also glimpsed a general approach to problem solving that produces both correct answers and insights into physical relationships. The next level of complexity in our kinematics problems involves the motion of two interrelated bodies, called *two-body pursuit problems*.

Two-Body Pursuit Problems

Up until this point we have looked at examples of motion involving a single body. Even for the problem with two cars and the stopping distances on wet and dry roads, we divided this problem into two separate problems to find the answers. In a **two-body pursuit problem**, the motions of the objects are coupled—meaning, the unknown we seek depends on the motion of both objects. To solve these problems we write the equations of motion for each object and then solve them simultaneously to find the unknown. This is illustrated in [Figure 3.25](#).

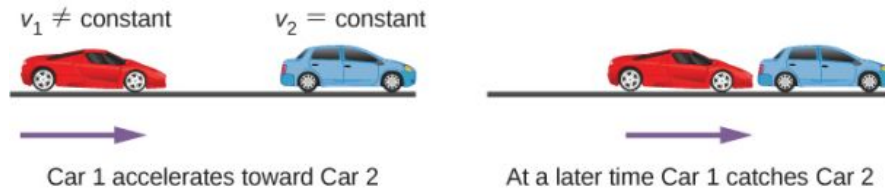


FIGURE 3.25 A two-body pursuit scenario where car 2 has a constant velocity and car 1 is behind with a constant acceleration. Car 1 catches up with car 2 at a later time.

The time and distance required for car 1 to catch car 2 depends on the initial distance car 1 is from car 2 as well as the velocities of both cars and the acceleration of car 1. The kinematic equations describing the motion of both cars must be solved to find these unknowns.

Consider the following example.

EXAMPLE 3.13

Cheetah Catching a Gazelle

A cheetah waits in hiding behind a bush. The cheetah spots a gazelle running past at 10 m/s. At the instant the gazelle passes the cheetah, the cheetah accelerates from rest at 4 m/s² to catch the gazelle. (a) How long does it take the cheetah to catch the gazelle? (b) What is the displacement of the gazelle and cheetah?

Strategy

We use the set of equations for constant acceleration to solve this problem. Since there are two objects in motion, we have separate equations of motion describing each animal. But what links the equations is a common parameter that has the same value for each animal. If we look at the problem closely, it is clear the common parameter to each animal is their position x at a later time t . Since they both start at $x_0 = 0$, their displacements are the same at a later time t , when the cheetah catches up with the gazelle. If we pick the equation of motion that solves for the displacement for each animal, we can then set the equations equal to each other and solve for the unknown, which is time.

Solution

- Equation for the gazelle: The gazelle has a constant velocity, which is its average velocity, since it is not accelerating. Therefore, we use [Equation 3.10](#) with $x_0 = 0$:

$$x = x_0 + \bar{v}t = \bar{v}t.$$

Equation for the cheetah: The cheetah is accelerating from rest, so we use [Equation 3.13](#) with $x_0 = 0$ and $v_0 = 0$:

$$x = x_0 + v_0t + \frac{1}{2}at^2 = \frac{1}{2}at^2.$$

Now we have an equation of motion for each animal with a common parameter, which can be eliminated to find the solution. In this case, we solve for t :

$$\begin{aligned}x &= \bar{v}t = \frac{1}{2}at^2 \\t &= \frac{2\bar{v}}{a}.\end{aligned}$$

The gazelle has a constant velocity of 10 m/s, which is its average velocity. The acceleration of the cheetah is 4 m/s^2 . Evaluating t , the time for the cheetah to reach the gazelle, we have

$$t = \frac{2\bar{v}}{a} = \frac{2(10 \text{ m/s})}{4 \text{ m/s}^2} = 5 \text{ s}.$$

- b. To get the displacement, we use either the equation of motion for the cheetah or the gazelle, since they should both give the same answer.

Displacement of the cheetah:

$$x = \frac{1}{2}at^2 = \frac{1}{2}(4 \text{ m/s}^2)(5)^2 = 50 \text{ m}.$$

Displacement of the gazelle:

$$x = \bar{v}t = 10 \text{ m/s}(5) = 50 \text{ m}.$$

We see that both displacements are equal, as expected.

Significance

It is important to analyze the motion of each object and to use the appropriate kinematic equations to describe the individual motion. It is also important to have a good visual perspective of the two-body pursuit problem to see the common parameter that links the motion of both objects.

CHECK YOUR UNDERSTANDING 3.6

A bicycle has a constant velocity of 10 m/s. A person starts from rest and begins to run to catch up to the bicycle in 30 s when the bicycle is at the same position as the person. What is the acceleration of the person?

3.5 Free Fall

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Use the kinematic equations with the variables y and g to analyze free-fall motion.
- Describe how the values of the position, velocity, and acceleration change during a free fall.
- Solve for the position, velocity, and acceleration as functions of time when an object is in a free fall.

An interesting application of [Equation 3.4](#) through [Equation 3.14](#) is called *free fall*, which describes the motion of an object falling in a gravitational field, such as near the surface of Earth or other celestial objects of planetary size.

Let's assume the body is falling in a straight line perpendicular to the surface, so its motion is one-dimensional. For example, we can estimate the depth of a vertical mine shaft by dropping a rock into it and listening for the rock to hit the bottom. But "falling," in the context of free fall, does not necessarily imply the body is moving from a greater height to a lesser height. If a ball is thrown upward, the equations of free fall apply equally to its ascent as well as its descent.

Gravity

The most remarkable and unexpected fact about falling objects is that if air resistance and friction are negligible, then in a given location all objects fall toward the center of Earth with the *same constant acceleration, independent of their mass*. This experimentally determined fact is unexpected because we are so accustomed to the effects of air resistance and friction that we expect light objects to fall slower than heavy ones. Until Galileo Galilei (1564–1642) proved otherwise, people believed that a heavier object has a greater acceleration in a free fall. We now know this is not the case. In the absence of air resistance, heavy objects arrive at the ground at the same time as lighter objects when dropped from the same height [Figure 3.26](#).

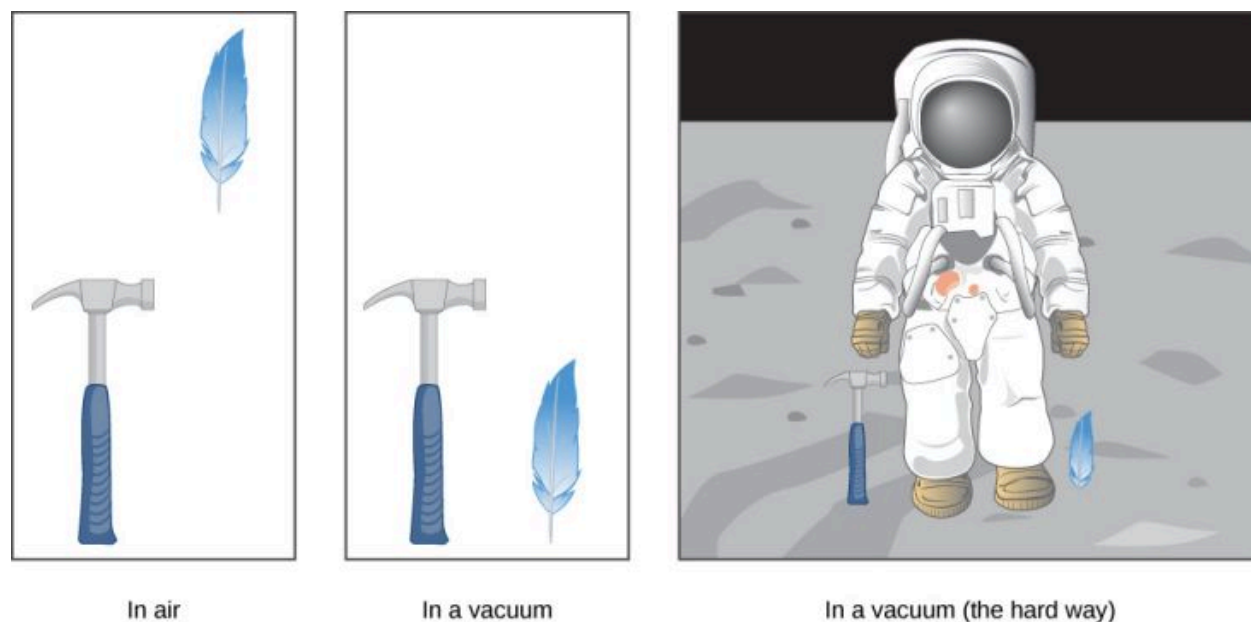


FIGURE 3.26 A hammer and a feather fall with the same constant acceleration if air resistance is negligible. This is a general characteristic of gravity not unique to Earth, as astronaut David R. Scott demonstrated in 1971 on the Moon, where the acceleration from gravity is only 1.67 m/s^2 and there is no atmosphere.

In the real world, air resistance can cause a lighter object to fall slower than a heavier object of the same size. A tennis ball reaches the ground after a baseball dropped at the same time. (It might be difficult to observe the difference if the height is not large.) Air resistance opposes the motion of an object through the air, and friction between objects—such as between clothes and a laundry chute or between a stone and a pool into which it is dropped—also opposes motion between them.

For the ideal situations of these first few chapters, an object *falling without air resistance or friction* is defined to be in **free fall**. The force of gravity causes objects to fall toward the center of Earth. The acceleration of free-falling objects is therefore called **acceleration due to gravity**. Acceleration due to gravity is constant, which means we can apply the kinematic equations to any falling object where air resistance and friction are negligible. This opens to us a broad class of interesting situations.

Acceleration due to gravity is so important that its magnitude is given its own symbol, g . It is constant at any given location on Earth and has the average value

$$g = 9.81 \text{ m/s}^2 \text{ (or } 32.2 \text{ ft/s}^2\text{)}.$$

Although g varies from 9.78 m/s^2 to 9.83 m/s^2 , depending on latitude, altitude, underlying geological formations, and local topography, let's use an average value of 9.8 m/s^2 rounded to two significant figures in this text unless specified otherwise. Neglecting these effects on the value of g as a result of position on Earth's surface, as well as effects resulting from Earth's rotation, we take the direction of acceleration due to gravity to be downward (toward the center of Earth). In fact, its direction *defines* what we call vertical. Note that whether acceleration a in the kinematic equations has the value $+g$ or $-g$ depends on how we define our coordinate system. If we define the upward direction as positive, then $a = -g = -9.8 \text{ m/s}^2$, and if we define the downward direction as positive, then $a = g = 9.8 \text{ m/s}^2$.

One-Dimensional Motion Involving Gravity

The best way to see the basic features of motion involving gravity is to start with the simplest situations and then progress toward more complex ones. So, we start by considering straight up-and-down motion with no air resistance or friction. These assumptions mean the velocity (if there is any) is vertical. If an object is dropped, we know the initial velocity is zero when in free fall. When the object has left contact with whatever held or threw it, the object is in free fall. When the object is thrown, it has the same initial speed in free fall as it did before it was released. When the object comes in contact with the ground or any other object, it is no longer in free fall and its acceleration of g is no longer valid. Under these circumstances, the motion is one-dimensional and has constant acceleration of magnitude g . We represent vertical displacement with the symbol y .

KINEMATIC EQUATIONS FOR OBJECTS IN FREE FALL

We assume here that acceleration equals $-g$ (with the positive direction upward).

$$v = v_0 - gt \quad 3.15$$

$$y = y_0 + v_0 t - \frac{1}{2}gt^2 \quad 3.16$$

$$v^2 = v_0^2 - 2g(y - y_0) \quad 3.17$$



PROBLEM-SOLVING STRATEGY

Free Fall

1. Decide on the sign of the acceleration of gravity. In [Equation 3.15](#) through [Equation 3.17](#), acceleration g is negative, which says the positive direction is upward and the negative direction is downward. In some problems, it may be useful to have acceleration g as positive, indicating the positive direction is downward.
2. Draw a sketch of the problem. This helps visualize the physics involved.
3. Record the knowns and unknowns from the problem description. This helps devise a strategy for selecting the appropriate equations to solve the problem.
4. Decide which of [Equation 3.15](#) through [Equation 3.17](#) are to be used to solve for the unknowns.



EXAMPLE 3.14

Free Fall of a Ball

[Figure 3.27](#) shows the positions of a ball, at 1-s intervals, with an initial velocity of 4.9 m/s downward, that is thrown from the top of a 98-m-high building. (a) How much time elapses before the ball reaches the ground? (b) What is the velocity when it arrives at the ground?

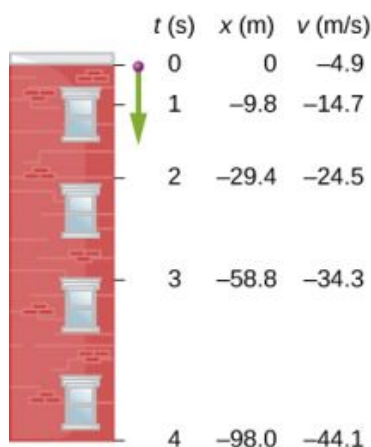


FIGURE 3.27 The positions and velocities at 1-s intervals of a ball thrown downward from a tall building at 4.9 m/s.

Strategy

Choose the origin at the top of the building with the positive direction upward and the negative direction downward. To find the time when the position is -98 m, we use [Equation 3.16](#), with $y_0 = 0$, $v_0 = -4.9$ m/s, and $g = 9.8$ m/s².

Solution

- a. Substitute the given values into the equation:

$$y = y_0 + v_0 t - \frac{1}{2} g t^2$$

$$-98.0 \text{ m} = 0 - (4.9 \text{ m/s})t - \frac{1}{2}(9.8 \text{ m/s}^2)t^2.$$

This simplifies to

$$t^2 + t - 20 = 0.$$

This is a quadratic equation with roots $t = -5.0$ s and $t = 4.0$ s. The positive root is the one we are interested in, since time $t = 0$ is the time when the ball is released at the top of the building. (The time $t = -5.0$ s represents the fact that a ball thrown upward from the ground would have been in the air for 5.0 s when it passed by the top of the building moving downward at 4.9 m/s.)

- b. Using [Equation 3.15](#), we have

$$v = v_0 - g t = -4.9 \text{ m/s} - (9.8 \text{ m/s}^2)(4.0 \text{ s}) = -44.1 \text{ m/s}.$$

Significance

For situations when two roots are obtained from a quadratic equation in the time variable, we must look at the physical significance of both roots to determine which is correct. Since $t = 0$ corresponds to the time when the ball was released, the negative root would correspond to a time before the ball was released, which is not physically meaningful. When the ball hits the ground, its velocity is not immediately zero, but as soon as the ball interacts with the ground, its acceleration is not g and it accelerates with a different value over a short time to zero velocity. This problem shows how important it is to establish the correct coordinate system and to keep the signs of g in the kinematic equations consistent.

EXAMPLE 3.15

Vertical Motion of a Baseball

A batter hits a baseball straight upward at home plate and the ball is caught 5.0 s after it is struck ([Figure 3.28](#)). (a) What is the initial velocity of the ball? (b) What is the maximum height the ball reaches? (c) How long does it take to reach the maximum height? (d) What is the acceleration at the top of its path? (e) What is the velocity of the ball

when it is caught? Assume the ball is hit and caught at the same location.

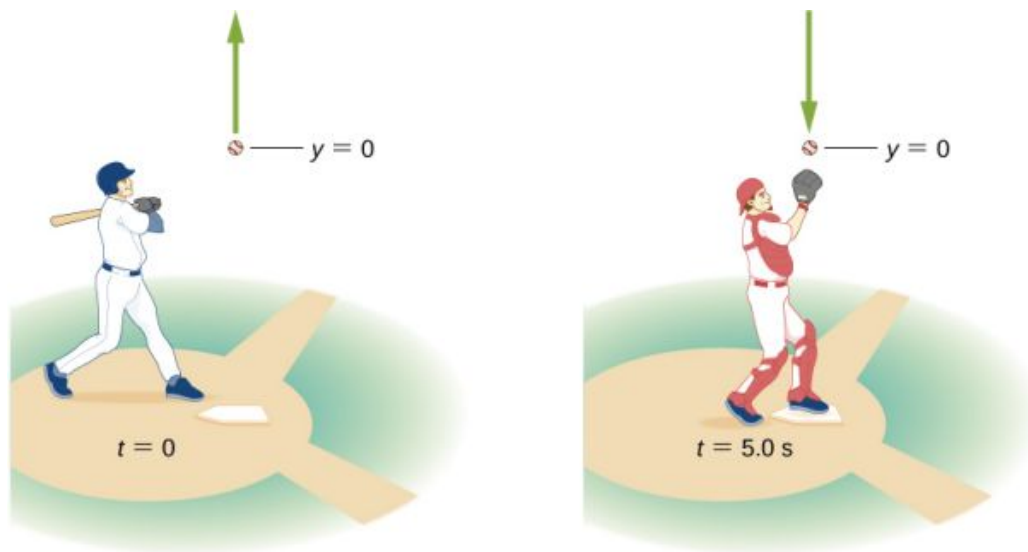


FIGURE 3.28 A baseball hit straight up is caught by the catcher 5.0 s later.

Strategy

Choose a coordinate system with a positive y -axis that is straight up and with an origin that is at the spot where the ball is hit and caught.

Solution

- a. [Equation 3.16](#) gives

$$y = y_0 + v_0 t - \frac{1}{2} g t^2$$

$$0 = 0 + v_0(5.0 \text{ s}) - \frac{1}{2} (9.8 \text{ m/s}^2) (5.0 \text{ s})^2,$$

which gives $v_0 = 24.5 \text{ m/s}$.

- b. At the maximum height, $v = 0$. With $v_0 = 24.5 \text{ m/s}$, [Equation 3.17](#) gives

$$v^2 = v_0^2 - 2g(y - y_0)$$

$$0 = (24.5 \text{ m/s})^2 - 2(9.8 \text{ m/s}^2)(y - 0)$$

or

$$y = 30.6 \text{ m}.$$

- c. To find the time when $v = 0$, we use [Equation 3.15](#):

$$v = v_0 - gt$$

$$0 = 24.5 \text{ m/s} - (9.8 \text{ m/s}^2)t.$$

This gives $t = 2.5 \text{ s}$. Since the ball rises for 2.5 s, the time to fall is 2.5 s.

- d. The acceleration is 9.8 m/s^2 everywhere, even when the velocity is zero at the top of the path. Although the velocity is zero at the top, it is changing at the rate of 9.8 m/s^2 downward.
- e. The velocity at $t = 5.0 \text{ s}$ can be determined with [Equation 3.15](#):

$$v = v_0 - gt$$

$$= 24.5 \text{ m/s} - 9.8 \text{ m/s}^2(5.0 \text{ s})$$

$$= -24.5 \text{ m/s}.$$

Significance

The ball returns with the speed it had when it left. This is a general property of free fall for any initial velocity. We used a single equation to go from throw to catch, and did not have to break the motion into two segments, upward

and downward. We are used to thinking that the effect of gravity is to create free fall downward toward Earth. It is important to understand, as illustrated in this example, that objects moving upward away from Earth are also in a state of free fall.

✓ CHECK YOUR UNDERSTANDING 3.7

A chunk of ice breaks off a glacier and falls 30.0 m before it hits the water. Assuming it falls freely (there is no air resistance), how long does it take to hit the water? Which quantity increases faster, the speed of the ice chunk or its distance traveled?

✱ EXAMPLE 3.16

Rocket Booster

A small rocket with a booster blasts off and heads straight upward. When at a height of 5.0 km and velocity of 200.0 m/s, it releases its booster. (a) What is the maximum height the booster attains? (b) What is the velocity of the booster at a height of 6.0 km? Neglect air resistance.

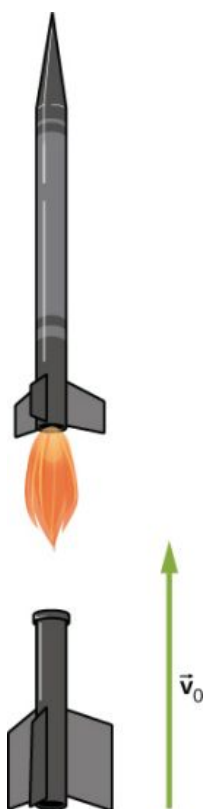


FIGURE 3.29 A rocket releases its booster at a given height and velocity. How high and how fast does the booster go?

Strategy

We need to select the coordinate system for the acceleration of gravity, which we take as negative downward. We are given the initial velocity of the booster and its height. We consider the point of release as the origin. We know the velocity is zero at the maximum position within the acceleration interval; thus, the velocity of the booster is zero at its maximum height, so we can use this information as well. From these observations, we use [Equation 3.17](#), which gives us the maximum height of the booster. We also use [Equation 3.17](#) to give the velocity at 6.0 km. The initial velocity of the booster is 200.0 m/s.

Solution

- a. From [Equation 3.17](#), $v^2 = v_0^2 - 2g(y - y_0)$. With $v = 0$ and $y_0 = 0$, we can solve for y :

$$y = \frac{v_0^2}{2g} = \frac{(2.0 \times 10^2 \text{ m/s})^2}{2(9.8 \text{ m/s}^2)} = 2040.8 \text{ m}.$$

This solution gives the maximum height of the booster in our coordinate system, which has its origin at the point of release, so the maximum height of the booster is roughly 7.0 km.

- b. An altitude of 6.0 km corresponds to $y = 1.0 \times 10^3 \text{ m}$ in the coordinate system we are using. The other initial conditions are $y_0 = 0$, and $v_0 = 200.0 \text{ m/s}$.

We have, from [Equation 3.17](#),

$$v^2 = (200.0 \text{ m/s})^2 - 2(9.8 \text{ m/s}^2)(1.0 \times 10^3 \text{ m}) \Rightarrow v = \pm 142.8 \text{ m/s}.$$

Significance

We have both a positive and negative solution in (b). Since our coordinate system has the positive direction upward, the $+142.8 \text{ m/s}$ corresponds to a positive upward velocity at 6000 m during the upward leg of the trajectory of the booster. The value $v = -142.8 \text{ m/s}$ corresponds to the velocity at 6000 m on the downward leg. This example is also important in that an object is given an initial velocity at the origin of our coordinate system, but the origin is at an altitude above the surface of Earth, which must be taken into account when forming the solution.

INTERACTIVE

Engage the Phet simulation below to learn about graphing polynomials. The shape of the curve changes as the constants are adjusted. View the curves for the individual terms (for example, $y = bx$) to see how they add to generate the polynomial curve.

Access multimedia content (<https://openstax.org/books/university-physics-volume-1/pages/3-5-free-fall>)

3.6 Finding Velocity and Displacement from Acceleration

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Derive the kinematic equations for constant acceleration using integral calculus.
- Use the integral formulation of the kinematic equations in analyzing motion.
- Find the functional form of velocity versus time given the acceleration function.
- Find the functional form of position versus time given the velocity function.

This section assumes you have enough background in calculus to be familiar with integration. In [Instantaneous Velocity and Speed](#) and [Average and Instantaneous Acceleration](#) we introduced the kinematic functions of velocity and acceleration using the derivative. By taking the derivative of the position function we found the velocity function, and likewise by taking the derivative of the velocity function we found the acceleration function. Using integral calculus, we can work backward and calculate the velocity function from the acceleration function, and the position function from the velocity function.

Kinematic Equations from Integral Calculus

Let's begin with a particle with an acceleration $a(t)$ which is a known function of time. Since the time derivative of the velocity function is acceleration,

$$\frac{d}{dt}v(t) = a(t),$$

we can take the indefinite integral of both sides, finding

$$\int \frac{d}{dt}v(t)dt = \int a(t)dt + C_1,$$

where C_1 is a constant of integration. Since $\int \frac{d}{dt}v(t)dt = v(t)$, the velocity is given by

$$v(t) = \int a(t)dt + C_1. \quad 3.18$$

Similarly, the time derivative of the position function is the velocity function,

$$\frac{d}{dt}x(t) = v(t).$$

Thus, we can use the same mathematical manipulations we just used and find

$$x(t) = \int v(t)dt + C_2, \quad 3.19$$

where C_2 is a second constant of integration.

We can derive the kinematic equations for a constant acceleration using these integrals. With $a(t) = a$ a constant, and doing the integration in [Equation 3.18](#), we find

$$v(t) = \int a dt + C_1 = at + C_1.$$

If the initial velocity is $v(0) = v_0$, then

$$v_0 = 0 + C_1.$$

Then, $C_1 = v_0$ and

$$v(t) = v_0 + at,$$

which is [Equation 3.12](#). Substituting this expression into [Equation 3.19](#) gives

$$x(t) = \int (v_0 + at)dt + C_2.$$

Doing the integration, we find

$$x(t) = v_0t + \frac{1}{2}at^2 + C_2.$$

If $x(0) = x_0$, we have

$$x_0 = 0 + 0 + C_2;$$

so, $C_2 = x_0$. Substituting back into the equation for $x(t)$, we finally have

$$x(t) = x_0 + v_0t + \frac{1}{2}at^2,$$

which is [Equation 3.13](#).



EXAMPLE 3.17

Motion of a Motorboat

A motorboat is traveling at a constant velocity of 5.0 m/s when it starts to accelerate opposite to the motion to arrive at the dock. Its acceleration is $a(t) = -\frac{1}{4}t$ m/s³. (a) What is the velocity function of the motorboat? (b) At what time does the velocity reach zero? (c) What is the position function of the motorboat? (d) What is the displacement of the motorboat from the time it begins to accelerate opposite to the motion to when the velocity is zero? (e) Graph the velocity and position functions.

Strategy

(a) To get the velocity function we must integrate and use initial conditions to find the constant of integration. (b) We set the velocity function equal to zero and solve for t . (c) Similarly, we must integrate to find the position function and use initial conditions to find the constant of integration. (d) Since the initial position is taken to be zero, we only have to evaluate the position function at the time when the velocity is zero.

Solution

We take $t = 0$ to be the time when the boat starts to accelerate opposite to the motion.

- a. From the functional form of the acceleration we can solve [Equation 3.18](#) to get $v(t)$:

$$v(t) = \int a(t)dt + C_1 = \int -\frac{1}{4}t \text{ m/s}^3 dt + C_1 = -\frac{1}{8} \text{ m/s}^3 t^2 + C_1.$$

At $t = 0$ we have $v(0) = 5.0 \text{ m/s} = 0 + C_1$, so $C_1 = 5.0 \text{ m/s}$ or $v(t) = 5.0 \text{ m/s} - \frac{1}{8} \text{ m/s}^3 t^2$.

- b. $v(t) = 0 = 5.0 \text{ m/s} - \frac{1}{8}t^2 \text{ m/s}^3 \Rightarrow t = 6.3 \text{ s}$

- c. Solve [Equation 3.19](#):

$$x(t) = \int v(t)dt + C_2 = \int (5.0 - \frac{1}{8}t^2) dt + C_2 = 5.0t \text{ m/s} - \frac{1}{24}t^3 \text{ m/s}^3 + C_2.$$

At $t = 0$, we set $x(0) = 0 = x_0$, since we are only interested in the displacement from when the boat starts to accelerate opposite to the motion. We have

$$x(0) = 0 = C_2.$$

Therefore, the equation for the position is

$$x(t) = 5.0t \text{ m/s} - \frac{1}{24}t^3 \text{ m/s}^3.$$

- d. Since the initial position is taken to be zero, we only have to evaluate the position function at the time when the velocity is zero. This occurs at $t = 6.3 \text{ s}$. Therefore, the displacement is

$$x(6.3) = 5.0t \text{ m/s} (6.3 \text{ s}) - \frac{1}{24}t^3 \text{ m/s}^3 (6.3 \text{ s})^3 = 21.1 \text{ m}.$$

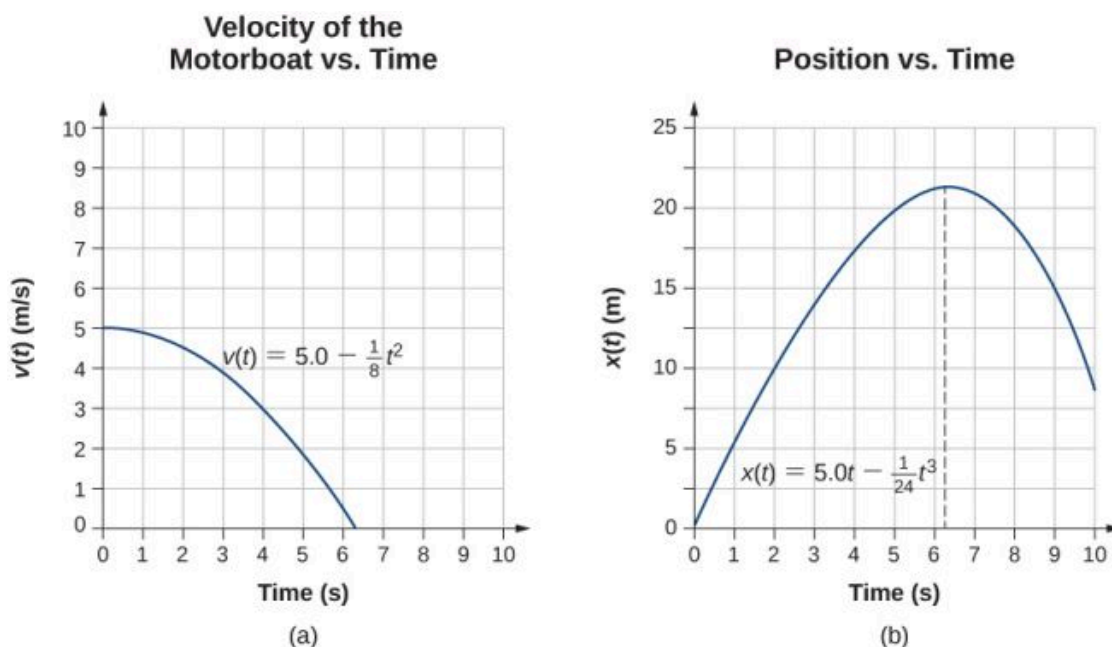


FIGURE 3.30 (a) Velocity of the motorboat as a function of time. The motorboat decreases its velocity to zero in 6.3 s. At times greater than this, velocity becomes negative—meaning, the boat is reversing direction. (b) Position of the motorboat as a function of time. At $t = 6.3 \text{ s}$, the velocity is zero and the boat has stopped. At times greater than this, the velocity becomes negative—meaning, if the boat continues to move with the same acceleration, it reverses direction and heads back toward where it originated.

Significance

The acceleration function is linear in time so the integration involves simple polynomials. In [Figure 3.30](#), we see that if we extend the solution beyond the point when the velocity is zero, the velocity becomes negative and the boat reverses direction. This tells us that solutions can give us information outside our immediate interest and we should be careful when interpreting them.

 CHECK YOUR UNDERSTANDING 3.8

A particle starts from rest and has an acceleration function $a(t) = \left(5 - \left(10\frac{1}{s}\right)t\right)\frac{m}{s^2}$. (a) What is the velocity function? (b) What is the position function? (c) When is the velocity zero?

Chapter Review

Key Terms

acceleration due to gravity acceleration of an object as a result of gravity

average acceleration the rate of change in velocity; the change in velocity over time

average speed the total distance traveled divided by elapsed time

average velocity the displacement divided by the time over which displacement occurs under constant acceleration

displacement the change in position of an object

distance traveled the total length of the path traveled between two positions

elapsed time the difference between the ending time and the beginning time

free fall the state of movement that results from gravitational force only

instantaneous acceleration acceleration at a specific point in time

instantaneous speed the absolute value of the instantaneous velocity

instantaneous velocity the velocity at a specific instant or time point

kinematics the description of motion through properties such as position, time, velocity, and acceleration

position the location of an object at a particular time

total displacement the sum of individual displacements over a given time period

two-body pursuit problem a kinematics problem in which the unknowns are calculated by solving the kinematic equations simultaneously for two moving objects

Key Equations

Displacement

$$\Delta x = x_f - x_i$$

Total displacement

$$\Delta x_{\text{Total}} = \sum \Delta x_i$$

Average velocity

$$\bar{v} = \frac{\Delta x}{\Delta t} = \frac{x_2 - x_1}{t_2 - t_1}$$

Instantaneous velocity

$$v(t) = \frac{dx(t)}{dt}$$

Average speed

$$\text{Average speed} = \bar{s} = \frac{\text{Total distance}}{\text{Elapsed time}}$$

Instantaneous speed

$$\text{Instantaneous speed} = |v(t)|$$

Average acceleration

$$\bar{a} = \frac{\Delta v}{\Delta t} = \frac{v_f - v_0}{t_f - t_0}$$

Instantaneous acceleration

$$a(t) = \frac{dv(t)}{dt}$$

Position from average velocity

$$x = x_0 + \bar{v}t$$

Average velocity (for constant acceleration)

$$\bar{v} = \frac{v_0 + v}{2}$$

Velocity from acceleration

$$v = v_0 + at \quad (\text{constant } a)$$

Position from velocity and acceleration

$$x = x_0 + v_0t + \frac{1}{2}at^2 \quad (\text{constant } a)$$

Velocity from distance

$$v^2 = v_0^2 + 2a(x - x_0) \quad (\text{constant } a)$$

Velocity of free fall

$$v = v_0 - gt \quad (\text{positive upward})$$

Height of free fall

$$y = y_0 + v_0t - \frac{1}{2}gt^2$$

Velocity of free fall from height

$$v^2 = v_0^2 - 2g(y - y_0)$$

Velocity from acceleration

$$v(t) = \int a(t)dt + C_1$$

Position from velocity

$$x(t) = \int v(t)dt + C_2$$

Summary

3.1 Position, Displacement, and Average Velocity

- Kinematics is the description of motion without considering its causes. In this chapter, it is

limited to motion along a straight line, called one-dimensional motion.

- Displacement is the change in position of an object. The SI unit for displacement is the meter.

Displacement has direction as well as magnitude.

- Distance traveled is the total length of the path traveled between two positions.
- Time is measured in terms of change. The time between two position points x_1 and x_2 is $\Delta t = t_2 - t_1$. Elapsed time for an event is $\Delta t = t_f - t_0$, where t_f is the final time and t_0 is the initial time. The initial time is often taken to be zero.
- Average velocity $\bar{v}$ is defined as displacement divided by elapsed time. If x_1, t_1 and x_2, t_2 are two position time points, the average velocity between these points is

$$\bar{v} = \frac{\Delta x}{\Delta t} = \frac{x_2 - x_1}{t_2 - t_1}.$$

3.2 Instantaneous Velocity and Speed

- Instantaneous velocity is a continuous function of time and gives the velocity at any point in time during a particle's motion. We can calculate the instantaneous velocity at a specific time by taking the derivative of the position function, which gives us the functional form of instantaneous velocity $v(t)$.
- Instantaneous velocity is a vector and can be negative.
- Instantaneous speed is found by taking the absolute value of instantaneous velocity, and it is always positive.
- Average speed is total distance traveled divided by elapsed time.
- The slope of a position-versus-time graph at a specific time gives instantaneous velocity at that time.

3.3 Average and Instantaneous Acceleration

- Acceleration is the rate at which velocity changes. The SI unit for acceleration is meters per second squared.
- Acceleration is a vector. It has both magnitude and direction and so can be the rate of change of the magnitude or direction of the velocity, or both.

Conceptual Questions

3.1 Position, Displacement, and Average Velocity

- Give an example in which there are clear distinctions among distance traveled, displacement, and magnitude of displacement. Identify each quantity in your example specifically.
- Under what circumstances does distance

- Instantaneous acceleration $a(t)$ is a continuous function of time and gives the acceleration at any specific time during the motion. It is calculated from the derivative of the velocity function. Instantaneous acceleration is the slope of the velocity-versus-time graph.
- Negative acceleration (sometimes called deceleration) is acceleration in the negative direction in the chosen coordinate system.

3.4 Motion with Constant Acceleration

- When analyzing one-dimensional motion with constant acceleration, identify the known quantities and choose the appropriate equations to solve for the unknowns. Either one or two of the kinematic equations are needed to solve for the unknowns, depending on the known and unknown quantities.
- Two-body pursuit problems always require two equations to be solved simultaneously for the unknowns.

3.5 Free Fall

- An object in free fall experiences constant acceleration if air resistance is negligible.
- On Earth, all free-falling objects have an acceleration g due to gravity, which averages $g = 9.81 \text{ m/s}^2$.
- For objects in free fall, the upward direction is normally taken as positive for displacement, velocity, and acceleration.

3.6 Finding Velocity and Displacement from Acceleration

- Integral calculus gives us a more complete formulation of kinematics.
- If acceleration $a(t)$ is known, we can use integral calculus to derive expressions for velocity $v(t)$ and position $x(t)$.
- If acceleration is constant, the integral equations reduce to [Equation 3.12](#) and [Equation 3.13](#) for motion with constant acceleration.

traveled equal magnitude of displacement? What is the only case in which magnitude of displacement and distance are exactly the same?

- Bacteria move back and forth using their flagella (structures that look like little tails). Speeds of up to $50 \text{ } \mu\text{m/s}$ ($50 \times 10^{-6} \text{ m/s}$) have been observed. The total distance traveled by a

bacterium is large for its size, whereas its displacement is small. Why is this?

4. Give an example of a device used to measure time and identify what change in that device indicates a change in time.
5. Does a car's odometer measure distance traveled or displacement?
6. During a given time interval the average velocity of an object is zero. What can you conclude about its displacement over the time interval?

3.2 Instantaneous Velocity and Speed

7. There is a distinction between average speed and the magnitude of average velocity. Give an example that illustrates the difference between these two quantities.
8. Does the speedometer of a car measure speed or velocity?
9. If you divide the total distance traveled on a car trip (as determined by the odometer) by the elapsed time of the trip, are you calculating average speed or magnitude of average velocity? Under what circumstances are these two quantities the same?
10. How are instantaneous velocity and instantaneous speed related to one another? How do they differ?

3.3 Average and Instantaneous Acceleration

11. Is it possible for speed to be constant while acceleration is not zero?
12. Is it possible for velocity to be constant while acceleration is not zero? Explain.
13. Give an example in which velocity is zero yet acceleration is not.
14. If a subway train is moving to the left (has a negative velocity) and then comes to a stop, what is the direction of its acceleration? Is the acceleration positive or negative?
15. Plus and minus signs are used in one-dimensional motion to indicate direction. What is the sign of an acceleration that reduces the magnitude of a negative velocity? Of a positive velocity?

3.4 Motion with Constant Acceleration

16. When analyzing the motion of a single object,

what is the required number of known physical variables that are needed to solve for the unknown quantities using the kinematic equations?

17. State two scenarios of the kinematics of single object where three known quantities require two kinematic equations to solve for the unknowns.

3.5 Free Fall

18. What is the acceleration of a rock thrown straight upward on the way up? At the top of its flight? On the way down? Assume there is no air resistance.
19. An object that is thrown straight up falls back to Earth. This is one-dimensional motion. (a) When is its velocity zero? (b) Does its velocity change direction? (c) Does the acceleration have the same sign on the way up as on the way down?
20. Suppose you throw a rock nearly straight up at a coconut in a palm tree and the rock just misses the coconut on the way up but hits the coconut on the way down. Neglecting air resistance and the slight horizontal variation in motion to account for the hit and miss of the coconut, how does the speed of the rock when it hits the coconut on the way down compare with what it would have been if it had hit the coconut on the way up? Is it more likely to dislodge the coconut on the way up or down? Explain.
21. The severity of a fall depends on your speed when you strike the ground. All factors but the acceleration from gravity being the same, how many times higher could a safe fall occur on the Moon than on Earth (gravitational acceleration on the Moon is about one-sixth that of the Earth)?
22. How many times higher could an astronaut jump on the Moon than on Earth if her takeoff speed is the same in both locations (gravitational acceleration on the Moon is about one-sixth of that on Earth)?

3.6 Finding Velocity and Displacement from Acceleration

23. When given the acceleration function, what additional information is needed to find the velocity function and position function?

Problems

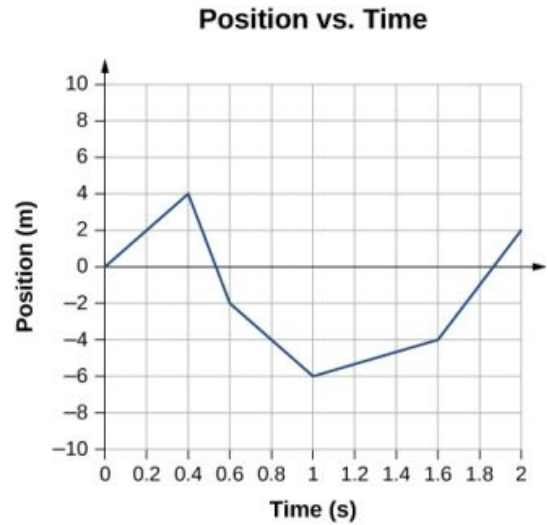
3.1 Position, Displacement, and Average Velocity

24. Consider a coordinate system in which the positive x axis is directed upward vertically. What are the positions of a particle (a) 5.0 m directly above the origin and (b) 2.0 m below the origin?
25. A car is 2.0 km west of a traffic light at $t = 0$ and 5.0 km east of the light at $t = 6.0$ min. Assume the origin of the coordinate system is the light and the positive x direction is eastward. (a) What are the car's position vectors at these two times? (b) What is the car's displacement between 0 min and 6.0 min?
26. The Shanghai maglev train connects Longyang Road to Pudong International Airport, a distance of 30 km. The journey takes 8 minutes on average. What is the maglev train's average velocity?
27. The position of a particle moving along the x -axis is given by $x(t) = 4.0 - 2.0t$ m. (a) At what time does the particle cross the origin? (b) What is the displacement of the particle between $t = 3.0$ s and $t = 6.0$ s?
28. A cyclist rides 8.0 km east for 20 minutes, then they turn and head west for 8 minutes and 3.2 km. Finally, they ride east for 16 km, which takes 40 minutes. (a) What is the final displacement of the cyclist? (b) What is their average velocity?
29. On February 15, 2013, a superbolide meteor (brighter than the Sun) entered Earth's atmosphere over Chelyabinsk, Russia, and exploded at an altitude of 23.5 km. Eyewitnesses could feel the intense heat from the fireball, and the blast wave from the explosion blew out windows in buildings. The blast wave took approximately 2 minutes 30 seconds to reach ground level. The blast wave traveled at 10° above the horizon. (a) What was the average velocity of the blast wave? (b) Compare this with the speed of sound, which is 343 m/s at sea level.

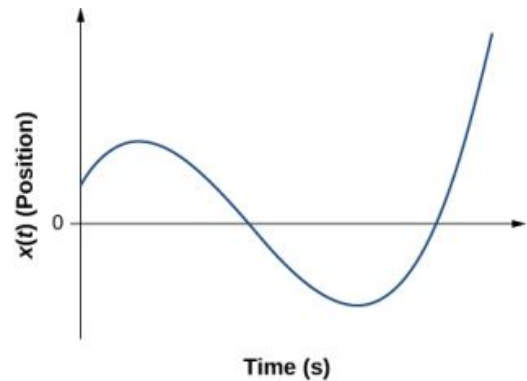
3.2 Instantaneous Velocity and Speed

30. A woodchuck runs 20 m to the right in 5 s, then turns and runs 10 m to the left in 3 s. (a) What is the average velocity of the woodchuck? (b) What is its average speed?
31. Sketch the velocity-versus-time graph from the

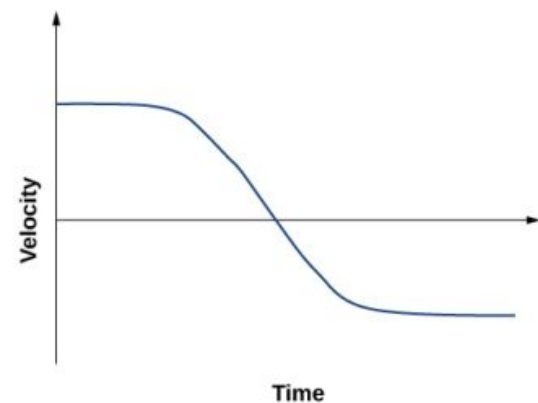
following position-versus-time graph.



32. Sketch the velocity-versus-time graph from the following position-versus-time graph.



33. Given the following velocity-versus-time graph, sketch the position-versus-time graph.

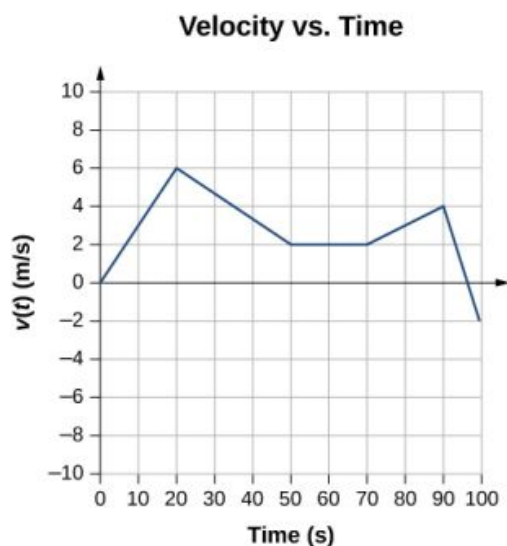


34. An object has a position function $x(t) = 5t$ m. (a) What is the velocity as a function of time? (b) Graph the position function and the velocity function.

35. A particle moves along the x -axis according to $x(t) = 10t - 2t^2$ m. (a) What is the instantaneous velocity at $t = 2$ s and $t = 3$ s? (b) What is the instantaneous speed at these times? (c) What is the average velocity between $t = 2$ s and $t = 3$ s?
36. **Unreasonable results.** A particle moves along the x -axis according to $x(t) = 3t^3 + 5t$. At what time is the velocity of the particle equal to zero? Is this reasonable?

3.3 Average and Instantaneous Acceleration

37. A cheetah can accelerate from rest to a speed of 30.0 m/s in 7.00 s. What is its acceleration?
38. Dr. John Paul Stapp was a U.S. Air Force officer who studied the effects of extreme acceleration on the human body. On December 10, 1954, Stapp rode a rocket sled, accelerating from rest to a top speed of 282 m/s (1015 km/h) in 5.00 s and was brought jarringly back to rest in only 1.40 s. Calculate his (a) acceleration in his direction of motion and (b) acceleration opposite to his direction of motion. Express each in multiples of g (9.80 m/s^2) by taking its ratio to the acceleration of gravity.
39. Sketch the acceleration-versus-time graph from the following velocity-versus-time graph.



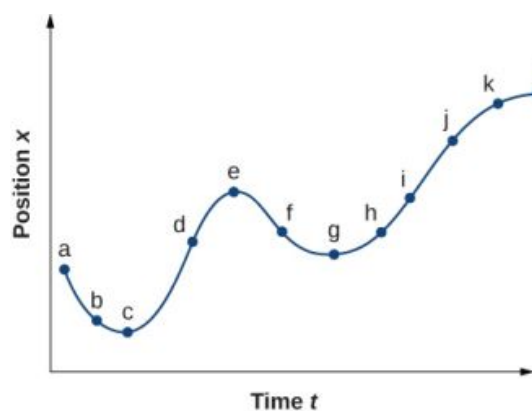
40. A commuter backs her car out of her garage with an acceleration of 1.40 m/s^2 . (a) How long does it take her to reach a speed of 2.00 m/s? (b) If she then brakes to a stop in 0.800 s, what is her acceleration?
41. Assume an intercontinental ballistic missile goes

from rest to a suborbital speed of 6.50 km/s in 60.0 s (the actual speed and time are classified). What is its average acceleration in meters per second squared and in multiples of g (9.80 m/s^2)?

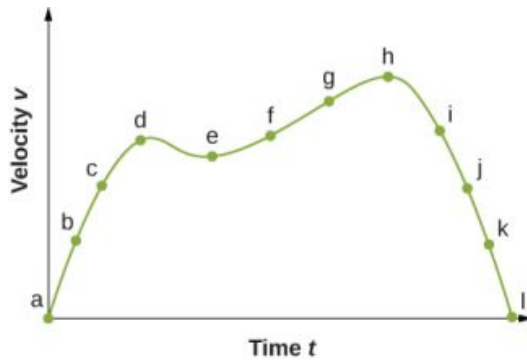
42. An airplane, starting from rest, moves down the runway at constant acceleration for 18 s and then takes off at a speed of 60 m/s. What is the average acceleration of the plane?

3.4 Motion with Constant Acceleration

43. A particle moves in a straight line at a constant velocity of 30 m/s. What is its displacement between $t = 0$ and $t = 5.0$ s?
44. A particle moves in a straight line with an initial velocity of 0 m/s and a constant acceleration of 30 m/s^2 . If $x = 0$ at $t = 0$, what is the particle's position at $t = 5$ s?
45. A particle moves in a straight line with an initial velocity of 30 m/s and constant acceleration 30 m/s^2 . (a) What is its displacement at $t = 5$ s? (b) What is its velocity at this same time?
46. (a) Sketch a graph of velocity versus time corresponding to the graph of displacement versus time given in the following figure. (b) Identify the time or times (t_a , t_b , t_c , etc.) at which the instantaneous velocity has the greatest positive value. (c) At which times is it zero? (d) At which times is it negative?



47. (a) Sketch a graph of acceleration versus time corresponding to the graph of velocity versus time given in the following figure. (b) Identify the time or times (t_a , t_b , t_c , etc.) at which the acceleration has the greatest positive value. (c) At which times is it zero? (d) At which times is it negative?



48. A particle has a constant acceleration of 6.0 m/s^2 . (a) If its initial velocity is 2.0 m/s , at what time is its displacement 5.0 m ? (b) What is its velocity at that time?
49. At $t = 10 \text{ s}$, a particle is moving from left to right with a speed of 5.0 m/s . At $t = 20 \text{ s}$, the particle is moving right to left with a speed of 8.0 m/s . Assuming the particle's acceleration is constant, determine (a) its acceleration, (b) its initial velocity, and (c) the instant when its velocity is zero.
50. A well-thrown ball is caught in a well-padded mitt. If the acceleration of the ball is $2.10 \times 10^4 \text{ m/s}^2$, and 1.85 ms ($1 \text{ ms} = 10^{-3} \text{ s}$) elapses from the time the ball first touches the mitt until it stops, what is the initial velocity of the ball?
51. A bullet in a gun is accelerated from the firing chamber to the end of the barrel at an average rate of $6.20 \times 10^5 \text{ m/s}^2$ for $8.10 \times 10^{-4} \text{ s}$. What is its muzzle velocity (that is, its final velocity)?
52. (a) A light-rail commuter train accelerates at a rate of 1.35 m/s^2 . How long does it take to reach its top speed of 80.0 km/h , starting from rest? (b) The same train ordinarily accelerates opposite to the motion at a rate of 1.65 m/s^2 . How long does it take to come to a stop from its top speed? (c) In emergencies, the train can accelerate opposite to the motion more rapidly, coming to rest from 80.0 km/h in 8.30 s . What is its emergency acceleration in meters per second squared?
53. While entering a freeway, a car accelerates from rest at a rate of 2.40 m/s^2 for 12.0 s . (a) Draw a sketch of the situation. (b) List the knowns in this problem. (c) How far does the car travel in those 12.0 s ? To solve this part, first identify the unknown, then indicate how you chose the appropriate equation to solve for it. After choosing the equation, show your steps in solving for the unknown, check your units, and discuss whether the answer is reasonable. (d) What is the car's final velocity? Solve for this unknown in the same manner as in (c), showing all steps explicitly.
54. **Unreasonable results** At the end of a race, a runner accelerates opposite to the motion from a velocity of 9.00 m/s at a rate of 2.00 m/s^2 . (a) How far does she travel in the next 5.00 s ? (b) What is her final velocity? (c) Evaluate the result. Does it make sense?
55. Blood is accelerated from rest to 30.0 cm/s in a distance of 1.80 cm by the left ventricle of the heart. (a) Make a sketch of the situation. (b) List the knowns in this problem. (c) How long does the acceleration take? To solve this part, first identify the unknown, then discuss how you chose the appropriate equation to solve for it. After choosing the equation, show your steps in solving for the unknown, checking your units. (d) Is the answer reasonable when compared with the time for a heartbeat?
56. During a slap shot, a hockey player accelerates the puck from a velocity of 8.00 m/s to 40.0 m/s in the same direction. If this shot takes $3.33 \times 10^{-2} \text{ s}$, what is the distance over which the puck accelerates?
57. A powerful motorcycle can accelerate from rest to 26.8 m/s (100 km/h) in only 3.90 s . (a) What is its average acceleration? (b) Assuming constant acceleration, how far does it travel in that time?
58. Freight trains can produce only relatively small accelerations. (a) What is the final velocity of a freight train that accelerates at a rate of 0.0500 m/s^2 for 8.00 min , starting with an initial velocity of 4.00 m/s ? (b) If the train can slow down at a rate of 0.550 m/s^2 , how long will it take to come to a stop from this velocity? (c) How far will it travel in each case?
59. A fireworks shell is accelerated from rest to a velocity of 65.0 m/s over a distance of 0.250 m . (a) Calculate the acceleration. (b) How long did the acceleration last?
60. A swan on a lake gets airborne by flapping its wings and running on top of the water. (a) If the swan must reach a velocity of 6.00 m/s to take off and it accelerates from rest at an average

rate of 0.35 m/s^2 , how far will it travel before becoming airborne? (b) How long does this take?

61. A woodpecker's brain is specially protected from large accelerations by tendon-like attachments inside the skull. While pecking on a tree, the woodpecker's head comes to a stop from an initial velocity of 0.600 m/s in a distance of only 2.00 mm . (a) Find the acceleration in meters per second squared and in multiples of g , where $g = 9.80 \text{ m/s}^2$. (b) Calculate the stopping time. (c) The tendons cradling the brain stretch, making its stopping distance 4.50 mm (greater than the head and, hence, less acceleration of the brain). What is the brain's acceleration, expressed in multiples of g ?
62. An unwary football player collides with a padded goalpost while running at a velocity of 7.50 m/s and comes to a full stop after compressing the padding and his body 0.350 m . (a) What is his acceleration? (b) How long does the collision last?
63. A care package is dropped out of a cargo plane and lands in the forest. If we assume the care package speed on impact is 54 m/s (123 mph), then what is its acceleration? Assume the trees and snow stops it over a distance of 3.0 m .
64. An express train passes through a station. It enters with an initial velocity of 22.0 m/s and accelerates opposite to the motion at a rate of 0.150 m/s^2 as it goes through. The station is 210.0 m long. (a) How fast is it going when the nose leaves the station? (b) How long is the nose of the train in the station? (c) If the train is 130 m long, what is the velocity of the end of the train as it leaves? (d) When does the end of the train leave the station?
65. **Unreasonable results** Dragsters can actually reach a top speed of 145.0 m/s in only 4.45 s . (a) Calculate the average acceleration for such a dragster. (b) Find the final velocity of this dragster starting from rest and accelerating at the rate found in (a) for 402.0 m (a quarter mile) without using any information on time. (c) Why is the final velocity greater than that used to find the average acceleration? (*Hint:* Consider whether the assumption of constant acceleration is valid for a dragster. If not, discuss whether the acceleration would be greater at the beginning or end of the run and what effect that would have on the final velocity.)

3.5 Free Fall

66. Calculate the displacement and velocity at times of (a) 0.500 s , (b) 1.00 s , (c) 1.50 s , and (d) 2.00 s for a ball thrown straight up with an initial velocity of 15.0 m/s . Take the point of release to be $y_0 = 0$.
67. Calculate the displacement and velocity at times of (a) 0.500 s , (b) 1.00 s , (c) 1.50 s , (d) 2.00 s , and (e) 2.50 s for a rock thrown straight down with an initial velocity of 14.0 m/s from the Verrazano Narrows Bridge in New York City. The roadway of this bridge is 70.0 m above the water.
68. A basketball referee tosses the ball straight up for the starting tip-off. At what velocity must a basketball player leave the ground to rise 1.25 m above the floor in an attempt to get the ball?
69. A rescue helicopter is hovering over a person whose boat has sunk. One of the rescuers throws a life preserver straight down to the victim with an initial velocity of 1.40 m/s and observes that it takes 1.8 s to reach the water. (a) List the knowns in this problem. (b) How high above the water was the preserver released? Note that the downdraft of the helicopter reduces the effects of air resistance on the falling life preserver, so that an acceleration equal to that of gravity is reasonable.
70. **Unreasonable results** A dolphin in an aquatic show jumps straight up out of the water at a velocity of 15.0 m/s . (a) List the knowns in this problem. (b) How high does his body rise above the water? To solve this part, first note that the final velocity is now a known, and identify its value. Then, identify the unknown and discuss how you chose the appropriate equation to solve for it. After choosing the equation, show your steps in solving for the unknown, checking units, and discuss whether the answer is reasonable. (c) How long a time is the dolphin in the air? Neglect any effects resulting from his size or orientation.
71. A diver bounces straight up from a diving board, avoiding the diving board on the way down, and falls feet first into a pool. She starts with a velocity of 4.00 m/s and her takeoff point is 1.80 m above the pool. (a) What is her highest point above the board? (b) How long a time are her feet in the air? (c) What is her velocity when her feet hit the water?

- 72.** (a) Calculate the height of a cliff if it takes 2.35 s for a rock to hit the ground when it is thrown straight up from the cliff with an initial velocity of 8.00 m/s. (b) How long a time would it take to reach the ground if it is thrown straight down with the same speed?
- 73.** A very strong, but inept, shot putter puts the shot straight up vertically with an initial velocity of 11.0 m/s. How long a time does he have to get out of the way if the shot was released at a height of 2.20 m and he is 1.80 m tall?
- 74.** You throw a ball straight up with an initial velocity of 15.0 m/s. It passes a tree branch on the way up at a height of 7.0 m. How much additional time elapses before the ball passes the tree branch on the way back down?
- 75.** A kangaroo can jump over an object 2.50 m high. (a) Considering just its vertical motion, calculate its vertical speed when it leaves the ground. (b) How long a time is it in the air?
- 76.** Standing at the base of one of the cliffs of Mt. Arapiles in Victoria, Australia, a hiker hears a rock break loose from a height of 105.0 m. He can't see the rock right away, but then does, 1.50 s later. (a) How far above the hiker is the rock when he can see it? (b) How much time does he have to move before the rock hits his head?
- 77.** There is a 250-m-high cliff at Half Dome in Yosemite National Park in California. Suppose a boulder breaks loose from the top of this cliff. (a) How fast will it be going when it strikes the ground? (b) Assuming a reaction time of 0.300 s, how long a time will a tourist at the bottom have to get out of the way after hearing the sound of the rock breaking loose (neglecting the height of

the tourist, which would become negligible anyway if hit)? The speed of sound is 335.0 m/s on this day.

3.6 Finding Velocity and Displacement from Acceleration

- 78.** The acceleration of a particle varies with time according to the equation $a(t) = pt^2 - qt^3$. Initially, the velocity and position are zero. (a) What is the velocity as a function of time? (b) What is the position as a function of time?
- 79.** Between $t = 0$ and $t = t_0$, a rocket moves straight upward with an acceleration given by $a(t) = A - Bt^{1/2}$, where A and B are constants. (a) If x is in meters and t is in seconds, what are the units of A and B ? (b) If the rocket starts from rest, how does the velocity vary between $t = 0$ and $t = t_0$? (c) If its initial position is zero, what is the rocket's position as a function of time during this same time interval?
- 80.** The velocity of a particle moving along the x -axis varies with time according to $v(t) = A + Bt^{-1}$, where $A = 2$ m/s, $B = 0.25$ m, and $1.0 \text{ s} \leq t \leq 8.0 \text{ s}$. Determine the acceleration and position of the particle at $t = 2.0$ s and $t = 5.0$ s. Assume that $x(t = 1 \text{ s}) = 0$.
- 81.** A particle at rest leaves the origin with its velocity increasing with time according to $v(t) = 3.2t$ m/s. At 5.0 s, the particle's velocity starts decreasing according to $[16.0 - 1.5(t - 5.0)]$ m/s. This decrease continues until $t = 11.0$ s, after which the particle's velocity remains constant at 7.0 m/s. (a) What is the acceleration of the particle as a function of time? (b) What is the position of the particle at $t = 2.0$ s, $t = 7.0$ s, and $t = 12.0$ s?

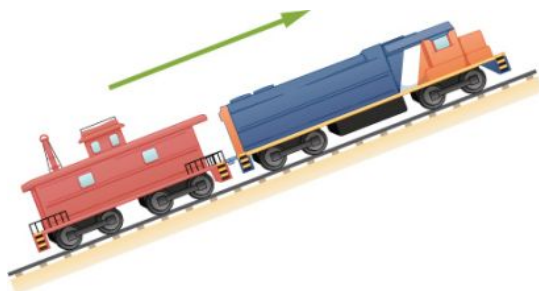
Additional Problems

- 82.** Professional baseball player Nolan Ryan could pitch a baseball at approximately 160.0 km/h. At that average velocity, how long did it take a ball thrown by Ryan to reach home plate, which is 18.4 m from the pitcher's mound? Compare this with the average reaction time of a human to a visual stimulus, which is 0.25 s.
- 83.** An airplane leaves Chicago and makes the 3000-km trip to Los Angeles in 5.0 h. A second plane leaves Chicago one-half hour later and arrives in Los Angeles at the same time. Compare the average velocities of the two

planes. Ignore the curvature of Earth and the difference in altitude between the two cities.

- 84. Unreasonable Results** A cyclist rides 16.0 km east, then 8.0 km west, then 8.0 km east, then 32.0 km west, and finally 11.2 km east. If his average velocity is 24 km/h, how long did it take him to complete the trip? Is this a reasonable time?
- 85.** An object has an acceleration of $+1.2 \text{ cm/s}^2$. At $t = 4.0$ s, its velocity is -3.4 cm/s . Determine the object's velocities at $t = 1.0$ s and $t = 6.0$ s.

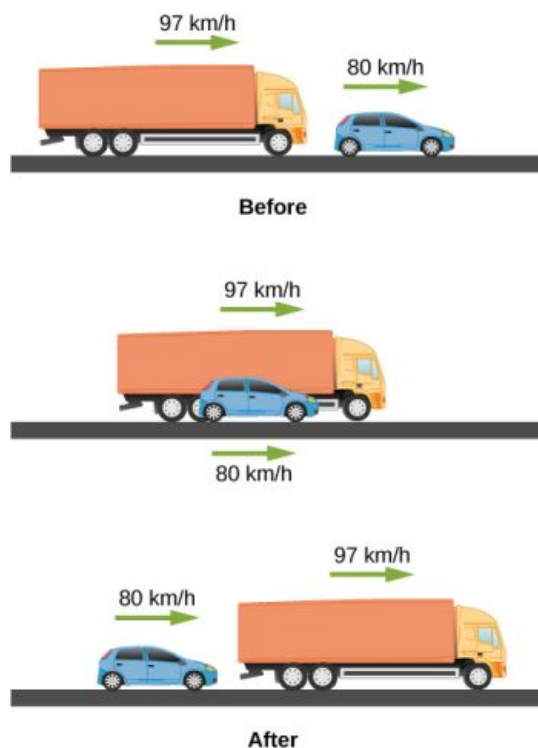
86. A particle moves along the x -axis according to the equation $x(t) = 2.0 - 4.0t^2$ m. What are the velocity and acceleration at $t = 2.0$ s and $t = 5.0$ s?
87. A particle moving at constant acceleration has velocities of 2.0 m/s at $t = 2.0$ s and -7.6 m/s at $t = 5.2$ s. What is the acceleration of the particle?
88. A train is moving up a steep grade at constant velocity (see following figure) when its caboose breaks loose and starts rolling freely along the track. After 5.0 s, the caboose is 30 m behind the train. What is the acceleration of the caboose?



89. An electron is moving in a straight line with a velocity of 4.0×10^5 m/s. It enters a region 5.0 cm long where it undergoes an acceleration of 6.0×10^{12} m/s² along the same straight line. (a) What is the electron's velocity when it emerges from this region? b) How long does the electron take to cross the region?
90. An ambulance driver is rushing a patient to the hospital. While traveling at 72 km/h, she notices the traffic light at the upcoming intersections has turned amber. To reach the intersection before the light turns red, she must travel 50 m in 2.0 s. (a) What minimum acceleration must the ambulance have to reach the intersection before the light turns red? (b) What is the speed of the ambulance when it reaches the intersection?
91. A motorcycle that is slowing down uniformly covers 2.0 successive km in 80 s and 120 s, respectively. Calculate (a) the acceleration of the motorcycle and (b) its velocity at the beginning and end of the 2 -km trip.
92. A cyclist travels from point A to point B in 10 min. During the first 2.0 min of her trip, she maintains a uniform acceleration of 0.090 m/s². She then travels at constant velocity for the next 5.0 min. Next, she accelerates opposite to the motion at a constant rate so that she comes to a

rest at point B 3.0 min later. (a) Sketch the velocity-versus-time graph for the trip. (b) What is the acceleration during the last 3 min? (c) How far does the cyclist travel?

93. Two trains are moving at 30 m/s in opposite directions on the same track. The engineers see simultaneously that they are on a collision course and apply the brakes when they are 1000 m apart. Assuming both trains have the same acceleration, what must this acceleration be if the trains are to stop just short of colliding?
94. A 10.0 -m-long truck moving with a constant velocity of 97.0 km/h passes a 3.0 -m-long car moving with a constant velocity of 80.0 km/h. How much time elapses between the moment the front of the truck is even with the back of the car and the moment the back of the truck is even with the front of the car?



95. A police car waits in hiding slightly off the highway. A speeding car is spotted by the police car doing 40 m/s. At the instant the speeding car passes the police car, the police car accelerates from rest at 4 m/s² to catch the speeding car. How long does it take the police car to catch the speeding car?
96. Pablo is running in a half marathon at a velocity of 3 m/s. Another runner, Jacob, is 50 meters behind Pablo with the same velocity. Jacob begins to accelerate at 0.05 m/s². (a) How long

- does it take Jacob to catch Pablo? (b) What is the distance covered by Jacob? (c) What is the final velocity of Jacob?
- 97. Unreasonable results** A runner approaches the finish line and is 75 m away; her speed at this position is 8 m/s. She accelerates opposite to the motion at this point at 0.5 m/s^2 . How long does it take her to cross the finish line from 75 m away? Is this reasonable?
- 98.** An airplane accelerates at 5.00 m/s^2 for 30.0 s. During this time, it covers a distance of 10.0 km. What are the initial and final velocities of the airplane?
- 99.** Compare the distance traveled of an object that undergoes a change in velocity that is twice its initial velocity with an object that changes its velocity by four times its initial velocity over the same time period. The accelerations of both objects are constant.
- 100.** An object is moving east with a constant velocity and is at position x_0 at time $t_0 = 0$. (a) With what acceleration must the object have for its total displacement to be zero at a later time t ? (b) What is the physical interpretation of the solution in the case for $t \rightarrow \infty$?
- 101.** A ball is thrown straight up. It passes a 2.00-m-high window 7.50 m off the ground on its path up and takes 1.30 s to go past the window. What was the ball's initial velocity?
- 102.** A coin is dropped from a hot-air balloon that is 300 m above the ground and rising at 10.0 m/s upward. For the coin, find (a) the maximum height reached, (b) its position and velocity 4.00 s after being released, and (c) the time before it hits the ground.
- 103.** A soft tennis ball is dropped onto a hard floor from a height of 1.50 m and rebounds to a height of 1.10 m. (a) Calculate its velocity just before it strikes the floor. (b) Calculate its velocity just after it leaves the floor on its way back up. (c) Calculate its acceleration during contact with the floor if that contact lasts 3.50 ms ($3.50 \times 10^{-3} \text{ s}$) (d) How much did the ball compress during its collision with the floor, assuming the floor is absolutely rigid?
- 104. Unreasonable results.** A raindrop falls from a cloud 100 m above the ground. Neglect air resistance. What is the speed of the raindrop when it hits the ground? Is this a reasonable number?
- 105.** Compare the time in the air of a basketball player who jumps 1.0 m vertically off the floor with that of a player who jumps 0.3 m vertically.
- 106.** Suppose that a person takes 0.5 s to react and move his hand to catch an object he has dropped. (a) How far does the object fall on Earth, where $g = 9.8 \text{ m/s}^2$? (b) How far does the object fall on the Moon, where the acceleration due to gravity is $1/6$ of that on Earth?
- 107.** A hot-air balloon rises from ground level at a constant velocity of 3.0 m/s. One minute after liftoff, a sandbag is dropped accidentally from the balloon. Calculate (a) the time it takes for the sandbag to reach the ground and (b) the velocity of the sandbag when it hits the ground.
- 108.** (a) A world record was set for the men's 100-m dash in the 2008 Olympic Games in Beijing by Usain Bolt of Jamaica. Bolt "coasted" across the finish line with a time of 9.69 s. If we assume that Bolt accelerated for 3.00 s to reach his maximum speed, and maintained that speed for the rest of the race, calculate his maximum speed and his acceleration. (b) During the same Olympics, Bolt also set the world record in the 200-m dash with a time of 19.30 s. Using the same assumptions as for the 100-m dash, what was his maximum speed for this race?
- 109.** An object is dropped from a height of 75.0 m above ground level. (a) Determine the distance traveled during the first second. (b) Determine the final velocity at which the object hits the ground. (c) Determine the distance traveled during the last second of motion before hitting the ground.
- 110.** A steel ball is dropped onto a hard floor from a height of 1.50 m and rebounds to a height of 1.45 m. (a) Calculate its velocity just before it strikes the floor. (b) Calculate its velocity just after it leaves the floor on its way back up. (c) Calculate its acceleration during contact with the floor if that contact lasts 0.0800 ms ($8.00 \times 10^{-5} \text{ s}$) (d) How much did the ball compress during its collision with the floor, assuming the floor is absolutely rigid?
- 111.** An object is dropped from a roof of a building of height h . During the last second of its descent, it drops a distance $h/3$. Calculate the height of the building.

Challenge Problems

- 112.** In a 100-m race, the winner is timed at 11.2 s. The second-place finisher's time is 11.6 s. How far is the second-place finisher behind the winner when she crosses the finish line? Assume the velocity of each runner is constant throughout the race.
- 113.** The position of a particle moving along the x -axis varies with time according to $x(t) = 5.0t^2 - 4.0t^3$ m. Find (a) the velocity and acceleration of the particle as functions of time, (b) the velocity and acceleration at $t = 2.0$ s, (c) the time at which the position is a maximum, (d) the time at which the velocity is zero, and (e) the maximum position.
- 114.** A cyclist sprints at the end of a race to clinch a victory. She has an initial velocity of 11.5 m/s and accelerates at a rate of 0.500 m/s^2 for 7.00 s. (a) What is her final velocity? (b) The cyclist continues at this velocity to the finish line. If she is 300 m from the finish line when she starts to accelerate, how much time did she save? (c) The second-place winner was 5.00 m ahead when the winner started to accelerate, but he was unable to accelerate, and traveled at 11.8 m/s until the finish line. What was the difference in finish time in seconds between the winner and runner-up? How far back was the runner-up when the winner crossed the finish line?
- 115.** In 1967, New Zealander Burt Munro set the world record for an Indian motorcycle, on the Bonneville Salt Flats in Utah, of 295.38 km/h. The one-way course was 8.00 km long. Acceleration rates are often described by the time it takes to reach 96.0 km/h from rest. If this time was 4.00 s and Burt accelerated at this rate until he reached his maximum speed, how long did it take Burt to complete the course?

CHAPTER 4

Motion in Two and Three Dimensions



FIGURE 4.1 The Red Arrows is the aerobatics display team of Britain's Royal Air Force. Based in Lincolnshire, England, they perform precision flying shows at high speeds, which requires accurate measurement of position, velocity, and acceleration in three dimensions. (credit: modification of work by Phil Long)

CHAPTER OUTLINE

4.1 Displacement and Velocity Vectors

4.2 Acceleration Vector

4.3 Projectile Motion

4.4 Uniform and Nonuniform Circular Motion

4.5 Relative Motion in One and Two Dimensions

INTRODUCTION To give a complete description of kinematics, we must explore motion in two and three dimensions. After all, most objects in our universe do not move in straight lines; rather, they follow curved paths. From kicked footballs to the flight paths of birds to the orbital motions of celestial bodies and down to the flow of blood plasma in your veins, most motion follows curved trajectories.

Fortunately, the treatment of motion in one dimension in the previous chapter has given us a foundation on which to build, as the concepts of position, displacement, velocity, and acceleration defined in one dimension can be expanded to two and three dimensions. Consider the Red Arrows, also known as the Royal Air Force Aerobatic team of the United Kingdom. Each jet follows a unique curved trajectory in three-dimensional airspace, as well as has a unique velocity and acceleration. Thus, to describe the motion of any of the jets accurately, we must assign to each jet a unique position vector in three dimensions as well as a unique velocity and acceleration vector. We can apply the same basic equations for displacement, velocity, and acceleration we derived in [Motion Along a Straight Line](#) to describe the motion of the jets in two and three dimensions, but with some modifications—in particular, the inclusion of vectors.

In this chapter we also explore two special types of motion in two dimensions: projectile motion and circular motion. Last, we conclude with a discussion of relative motion. In the chapter-opening picture, each jet has a relative motion with respect to any other jet in the group or to the people observing the air show on the ground.

4.1 Displacement and Velocity Vectors

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Calculate position vectors in a multidimensional displacement problem.
- Solve for the displacement in two or three dimensions.
- Calculate the velocity vector given the position vector as a function of time.
- Calculate the average velocity in multiple dimensions.

Displacement and velocity in two or three dimensions are straightforward extensions of the one-dimensional definitions. However, now they are vector quantities, so calculations with them have to follow the rules of vector algebra, not scalar algebra.

Displacement Vector

To describe motion in two and three dimensions, we must first establish a coordinate system and a convention for the axes. We generally use the coordinates x , y , and z to locate a particle at point $P(x, y, z)$ in three dimensions. If the particle is moving, the variables x , y , and z are functions of time (t):

$$x = x(t) \quad y = y(t) \quad z = z(t). \quad 4.1$$

The **position vector** from the origin of the coordinate system to point P is $\vec{r}(t)$. In unit vector notation, introduced in [Coordinate Systems and Components of a Vector](#), $\vec{r}(t)$ is

$$\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}. \quad 4.2$$

[Figure 4.2](#) shows the coordinate system and the vector to point P , where a particle could be located at a particular time t . Note the orientation of the x , y , and z axes. This orientation is called a right-handed coordinate system ([Coordinate Systems and Components of a Vector](#)) and it is used throughout the chapter.

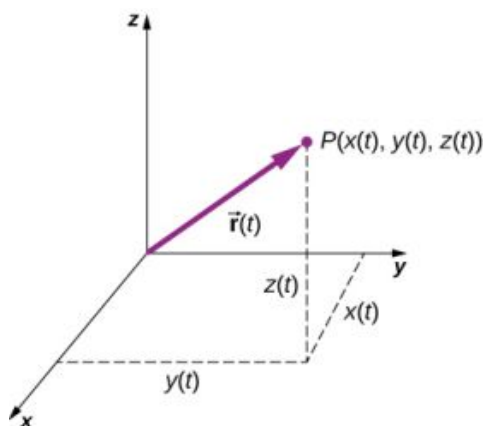


FIGURE 4.2 A three-dimensional coordinate system with a particle at position $P(x(t), y(t), z(t))$.

With our definition of the position of a particle in three-dimensional space, we can formulate the three-dimensional displacement. [Figure 4.3](#) shows a particle at time t_1 located at P_1 with position vector $\vec{r}(t_1)$. At a later time t_2 , the particle is located at P_2 with position vector $\vec{r}(t_2)$. The **displacement vector** $\Delta\vec{r}$ is found by subtracting $\vec{r}(t_1)$ from $\vec{r}(t_2)$:

$$\Delta\vec{r} = \vec{r}(t_2) - \vec{r}(t_1). \quad 4.3$$

Vector addition is discussed in [Vectors](#). Note that this is the same operation we did in one dimension, but now the vectors are in three-dimensional space.

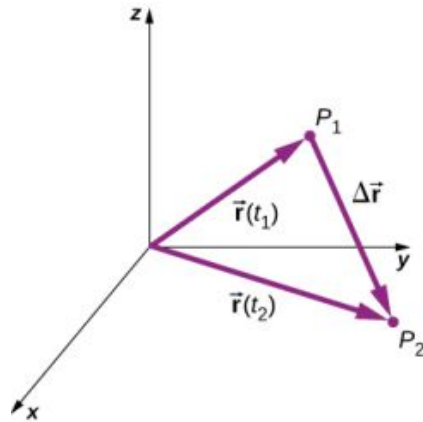


FIGURE 4.3 The displacement $\Delta \vec{r} = \vec{r}(t_2) - \vec{r}(t_1)$ is the vector from P_1 to P_2 .

The following examples illustrate the concept of displacement in multiple dimensions.



EXAMPLE 4.1

Polar Orbiting Satellite

A satellite is in a circular polar orbit around Earth at an altitude of 400 km—meaning, it passes directly overhead at the North and South Poles. What is the magnitude and direction of the displacement vector from when it is directly over the North Pole to when it is at -45° latitude?

Strategy

We make a picture of the problem to visualize the solution graphically. This will aid in our understanding of the displacement. We then use unit vectors to solve for the displacement.

Solution

[Figure 4.4](#) shows the surface of Earth and a circle that represents the orbit of the satellite. Although satellites are moving in three-dimensional space, they follow trajectories of ellipses, which can be graphed in two dimensions. The position vectors are drawn from the center of Earth, which we take to be the origin of the coordinate system, with the y -axis as north and the x -axis as east. The vector between them is the displacement of the satellite. We take the radius of Earth as 6370 km, so the length of each position vector is 6770 km.

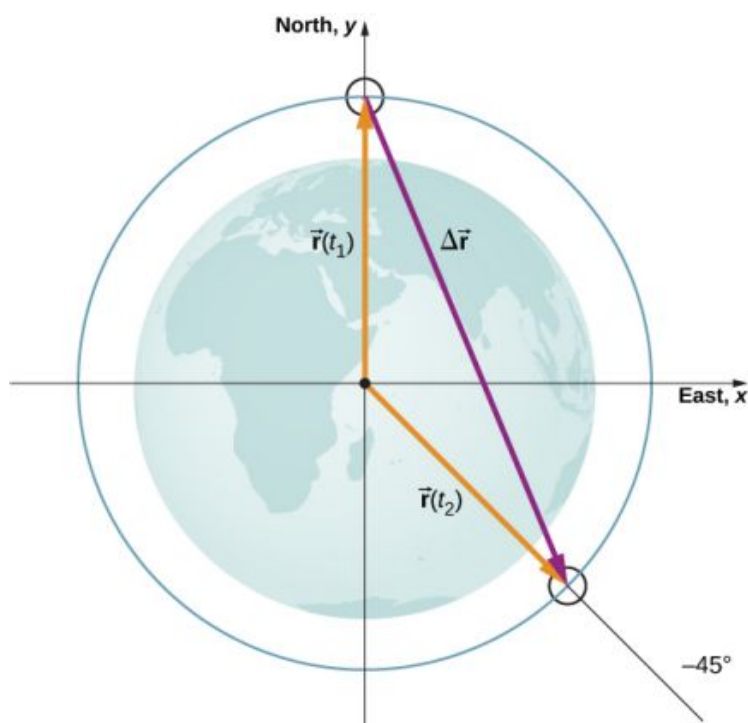


FIGURE 4.4 Two position vectors are drawn from the center of Earth, which is the origin of the coordinate system, with the y -axis as north and the x -axis as east. The vector between them is the displacement of the satellite.

In unit vector notation, the position vectors are

$$\begin{aligned}\vec{r}(t_1) &= 6770. \text{ km} \hat{j} \\ \vec{r}(t_2) &= 6770. \text{ km} (\cos(-45^\circ))\hat{i} + 6770. \text{ km} (\sin(-45^\circ))\hat{j}.\end{aligned}$$

Evaluating the sine and cosine, we have

$$\begin{aligned}\vec{r}(t_1) &= 6770. \hat{j} \\ \vec{r}(t_2) &= 4787 \hat{i} - 4787 \hat{j}.\end{aligned}$$

Now we can find $\Delta \vec{r}$, the displacement of the satellite:

$$\Delta \vec{r} = \vec{r}(t_2) - \vec{r}(t_1) = 4787 \hat{i} - 11,557 \hat{j}.$$

The magnitude of the displacement is $|\Delta \vec{r}| = \sqrt{(4787)^2 + (-11,557)^2} = 12,509 \text{ km}$. The angle the displacement makes with the x -axis is $\theta = \tan^{-1} \left(\frac{-11,557}{4787} \right) = -67.5^\circ$.

Significance

Plotting the displacement gives information and meaning to the unit vector solution to the problem. When plotting the displacement, we need to include its components as well as its magnitude and the angle it makes with a chosen axis—in this case, the x -axis ([Figure 4.5](#)).

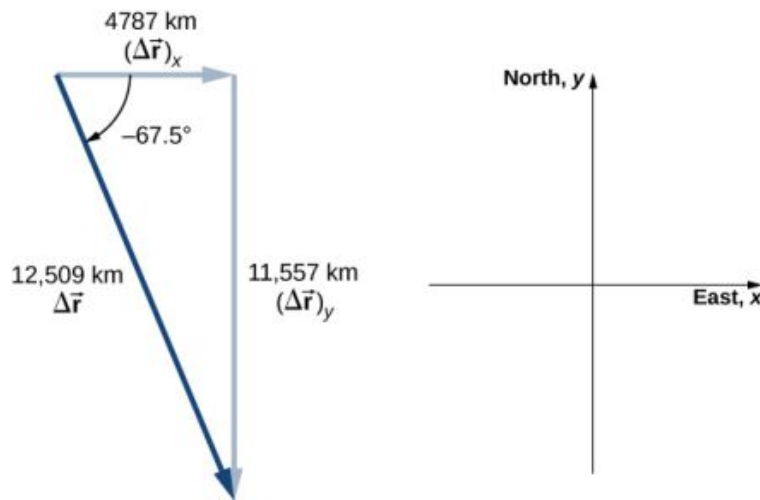


FIGURE 4.5 Displacement vector with components, angle, and magnitude.

Note that the satellite took a curved path along its circular orbit to get from its initial position to its final position in this example. It also could have traveled 4787 km east, then 11,557 km south to arrive at the same location. Both of these paths are longer than the length of the displacement vector. In fact, the displacement vector gives the shortest path between two points in one, two, or three dimensions.

Many applications in physics can have a series of displacements, as discussed in the previous chapter. The total displacement is the sum of the individual displacements, only this time, we need to be careful, because we are adding vectors. We illustrate this concept with an example of Brownian motion.

EXAMPLE 4.2

Brownian Motion

Brownian motion is a chaotic random motion of particles suspended in a fluid, resulting from collisions with the molecules of the fluid. This motion is three-dimensional. The displacements in numerical order of a particle undergoing Brownian motion could look like the following, in micrometers ([Figure 4.6](#)):

$$\begin{aligned}\Delta\vec{r}_1 &= 2.0\hat{i} + \hat{j} + 3.0\hat{k} \\ \Delta\vec{r}_2 &= -\hat{i} + 3.0\hat{k} \\ \Delta\vec{r}_3 &= 4.0\hat{i} - 2.0\hat{j} + \hat{k} \\ \Delta\vec{r}_4 &= -3.0\hat{i} + \hat{j} + 2.0\hat{k}.\end{aligned}$$

What is the total displacement of the particle from the origin?

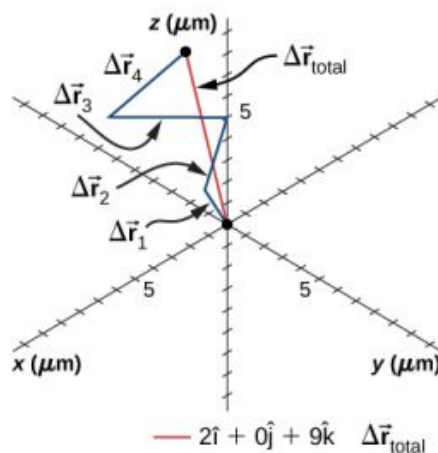


FIGURE 4.6 Trajectory of a particle undergoing random displacements of Brownian motion. The total displacement is shown in red.

Solution

We form the sum of the displacements and add them as vectors:

$$\begin{aligned}\Delta \vec{r}_{\text{Total}} &= \sum \Delta \vec{r}_i = \Delta \vec{r}_1 + \Delta \vec{r}_2 + \Delta \vec{r}_3 + \Delta \vec{r}_4 \\ &= (2.0 - 1.0 + 4.0 - 3.0)\hat{i} + (1.0 + 0 - 2.0 + 1.0)\hat{j} + (3.0 + 3.0 + 1.0 + 2.0)\hat{k} \\ &= 2.0\hat{i} + 0\hat{j} + 9.0\hat{k} \mu\text{m}.\end{aligned}$$

To complete the solution, we express the displacement as a magnitude and direction,

$$|\Delta \vec{r}_{\text{Total}}| = \sqrt{2.0^2 + 0^2 + 9.0^2} = 9.2 \mu\text{m}, \quad \theta = \tan^{-1} \left(\frac{9}{2} \right) = 77^\circ,$$

with respect to the x-axis in the xz-plane.

Significance

From the figure we can see the magnitude of the total displacement is less than the sum of the magnitudes of the individual displacements.

Velocity Vector

In the previous chapter we found the instantaneous velocity by calculating the derivative of the position function with respect to time. We can do the same operation in two and three dimensions, but we use vectors. The instantaneous **velocity vector** is now

$$\vec{v}(t) = \lim_{\Delta t \rightarrow 0} \frac{\vec{r}(t + \Delta t) - \vec{r}(t)}{\Delta t} = \frac{d\vec{r}}{dt}. \quad 4.4$$

Let's look at the relative orientation of the position vector and velocity vector graphically. In [Figure 4.7](#) we show the vectors $\vec{r}(t)$ and $\vec{r}(t + \Delta t)$, which give the position of a particle moving along a path represented by the gray line. As Δt goes to zero, the velocity vector, given by [Equation 4.4](#), becomes tangent to the path of the particle at time t .

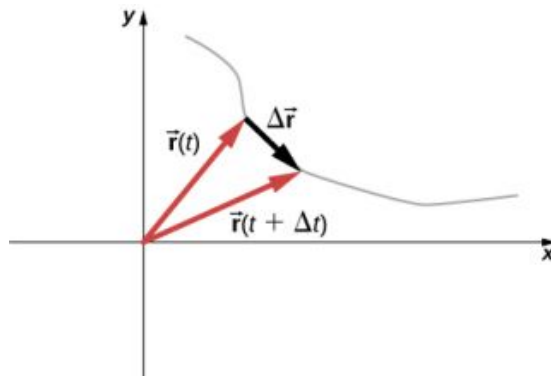


FIGURE 4.7 A particle moves along a path given by the gray line. In the limit as Δt approaches zero, the velocity vector becomes tangent to the path of the particle.

[Equation 4.4](#) can also be written in terms of the components of $\vec{v}(t)$. Since

$$\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k},$$

we can write

$$\vec{v}(t) = v_x(t)\hat{i} + v_y(t)\hat{j} + v_z(t)\hat{k} \quad 4.5$$

where

$$v_x(t) = \frac{dx(t)}{dt}, \quad v_y(t) = \frac{dy(t)}{dt}, \quad v_z(t) = \frac{dz(t)}{dt}. \quad 4.6$$

If only the average velocity is of concern, we have the vector equivalent of the one-dimensional average velocity for two and three dimensions:

$$\vec{v}_{\text{avg}} = \frac{\vec{r}(t_2) - \vec{r}(t_1)}{t_2 - t_1}. \quad 4.7$$

EXAMPLE 4.3

Calculating the Velocity Vector

The position function of a particle is $\vec{r}(t) = 2.0t^2\hat{i} + (2.0 + 3.0t)\hat{j} + 5.0t\hat{k}$ m. (a) What is the instantaneous velocity and speed at $t = 2.0$ s? (b) What is the average velocity between 1.0 s and 3.0 s?

Solution

Using [Equation 4.5](#) and [Equation 4.6](#), and taking the derivative of the position function with respect to time, we find

$$(a) \quad v(t) = \frac{d\vec{r}(t)}{dt} = 4.0t\hat{i} + 3.0\hat{j} + 5.0\hat{k} \text{ m/s}$$

$$\vec{v}(2.0 \text{ s}) = 8.0\hat{i} + 3.0\hat{j} + 5.0\hat{k} \text{ m/s}$$

$$\text{Speed } |\vec{v}(2.0 \text{ s})| = \sqrt{8^2 + 3^2 + 5^2} = 9.9 \text{ m/s}.$$

(b) From [Equation 4.7](#),

$$\begin{aligned} \vec{v}_{\text{avg}} &= \frac{\vec{r}(t_2) - \vec{r}(t_1)}{t_2 - t_1} = \frac{\vec{r}(3.0 \text{ s}) - \vec{r}(1.0 \text{ s})}{3.0 \text{ s} - 1.0 \text{ s}} = \frac{(18\hat{i} + 11\hat{j} + 15\hat{k}) \text{ m} - (2\hat{i} + 5\hat{j} + 5\hat{k}) \text{ m}}{2.0 \text{ s}} \\ &= \frac{(16\hat{i} + 6\hat{j} + 10\hat{k}) \text{ m}}{2.0 \text{ s}} = 8.0\hat{i} + 3.0\hat{j} + 5.0\hat{k} \text{ m/s}. \end{aligned}$$

Significance

We see the average velocity is the same as the instantaneous velocity at $t = 2.0$ s, as a result of the velocity function

being linear. This need not be the case in general. In fact, most of the time, instantaneous and average velocities are not the same.

✓ CHECK YOUR UNDERSTANDING 4.1

The position function of a particle is $\vec{r}(t) = 3.0t^3\hat{i} + 4.0\hat{j}$. (a) What is the instantaneous velocity at $t = 3$ s? (b) Is the average velocity between 2 s and 4 s equal to the instantaneous velocity at $t = 3$ s?

The Independence of Perpendicular Motions

When we look at the three-dimensional equations for position and velocity written in unit vector notation, [Equation 4.2](#) and [Equation 4.5](#), we see the components of these equations are separate and unique functions of time that do not depend on one another. Motion along the x direction has no part of its motion along the y and z directions, and similarly for the other two coordinate axes. Thus, the motion of an object in two or three dimensions can be divided into separate, independent motions along the perpendicular axes of the coordinate system in which the motion takes place.

To illustrate this concept with respect to displacement, consider a woman walking from point A to point B in a city with square blocks. The woman taking the path from A to B may walk east for so many blocks and then north (two perpendicular directions) for another set of blocks to arrive at B . How far she walks east is affected only by her motion eastward. Similarly, how far she walks north is affected only by her motion northward.

INDEPENDENCE OF MOTION

In the kinematic description of motion, we are able to treat the horizontal and vertical components of motion separately. In many cases, motion in the horizontal direction does not affect motion in the vertical direction, and vice versa.

An example illustrating the independence of vertical and horizontal motions is given by two baseballs. One baseball is dropped from rest. At the same instant, another is thrown horizontally from the same height and it follows a curved path. A stroboscope captures the positions of the balls at fixed time intervals as they fall ([Figure 4.8](#)).

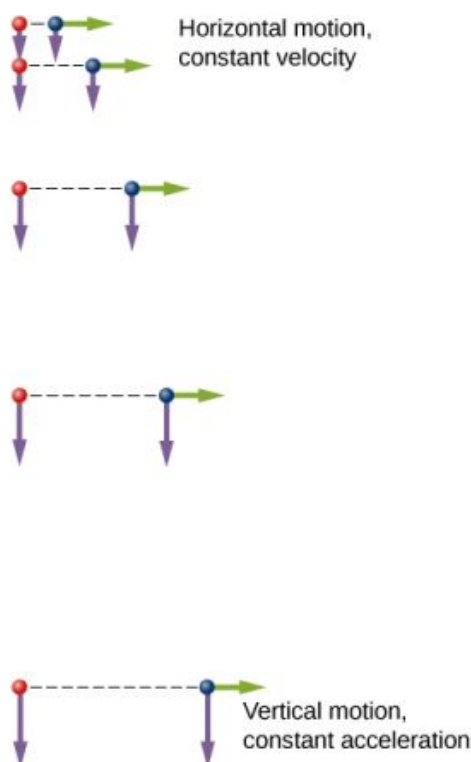


FIGURE 4.8 A diagram of the motions of two identical balls: one falls from rest and the other has an initial horizontal velocity. Each subsequent position is an equal time interval. Arrows represent the horizontal and vertical velocities at each position. The ball on the right has an initial horizontal velocity whereas the ball on the left has no horizontal velocity. Despite the difference in horizontal velocities, the vertical velocities and positions are identical for both balls, which shows the vertical and horizontal motions are independent.

It is remarkable that for each flash of the strobe, the vertical positions of the two balls are the same. This similarity implies vertical motion is independent of whether the ball is moving horizontally. (Assuming no air resistance, the vertical motion of a falling object is influenced by gravity only, not by any horizontal forces.) Careful examination of the ball thrown horizontally shows it travels the same horizontal distance between flashes. This is because there are no additional forces on the ball in the horizontal direction after it is thrown. This result means horizontal velocity is constant and is affected neither by vertical motion nor by gravity (which is vertical). Note this case is true for ideal conditions only. In the real world, air resistance affects the speed of the balls in both directions.

The two-dimensional curved path of the horizontally thrown ball is composed of two independent one-dimensional motions (horizontal and vertical). The key to analyzing such motion, called *projectile motion*, is to resolve it into motions along perpendicular directions. Resolving two-dimensional motion into perpendicular components is possible because the components are independent.

4.2 Acceleration Vector

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Calculate the acceleration vector given the velocity function in unit vector notation.
- Describe the motion of a particle with a constant acceleration in three dimensions.
- Use the one-dimensional motion equations along perpendicular axes to solve a problem in two or three dimensions with a constant acceleration.
- Express the acceleration in unit vector notation.

Instantaneous Acceleration

In addition to obtaining the displacement and velocity vectors of an object in motion, we often want to know its **acceleration vector** at any point in time along its trajectory. This acceleration vector is the instantaneous acceleration and it can be obtained from the derivative with respect to time of the velocity function, as we have seen in a previous chapter. The only difference in two or three dimensions is that these are now vector quantities. Taking the derivative with respect to time $\vec{v}(t)$, we find

$$\vec{a}(t) = \lim_{\Delta t \rightarrow 0} \frac{\vec{v}(t + \Delta t) - \vec{v}(t)}{\Delta t} = \frac{d\vec{v}(t)}{dt}. \quad 4.8$$

The acceleration in terms of components is

$$\vec{a}(t) = \frac{dv_x(t)}{dt} \hat{i} + \frac{dv_y(t)}{dt} \hat{j} + \frac{dv_z(t)}{dt} \hat{k}. \quad 4.9$$

Also, since the velocity is the derivative of the position function, we can write the acceleration in terms of the second derivative of the position function:

$$\vec{a}(t) = \frac{d^2x(t)}{dt^2} \hat{i} + \frac{d^2y(t)}{dt^2} \hat{j} + \frac{d^2z(t)}{dt^2} \hat{k}. \quad 4.10$$



EXAMPLE 4.4

Finding an Acceleration Vector

A particle has a velocity of $\vec{v}(t) = 5.0t\hat{i} + t^2\hat{j} - 2.0t^3\hat{k}$ m/s. (a) What is the acceleration function? (b) What is the acceleration vector at $t = 2.0$ s? Find its magnitude and direction.

Solution

(a) We take the first derivative with respect to time of the velocity function to find the acceleration. The derivative is taken component by component:

$$\vec{a}(t) = 5.0\hat{i} + 2.0t\hat{j} - 6.0t^2\hat{k} \text{ m/s}^2.$$

(b) Evaluating $\vec{a}(2.0 \text{ s}) = 5.0\hat{i} + 4.0\hat{j} - 24.0\hat{k}$ m/s² gives us the direction in unit vector notation. The magnitude of the acceleration is $|\vec{a}(2.0 \text{ s})| = \sqrt{5.0^2 + 4.0^2 + (-24.0)^2} = 24.8 \text{ m/s}^2$.

Significance

In this example we find that acceleration has a time dependence and is changing throughout the motion. Let's consider a different velocity function for the particle.



EXAMPLE 4.5

Finding a Particle Acceleration

A particle has a position function $\vec{r}(t) = (10t - t^2)\hat{i} + 5t\hat{j} + 5t\hat{k}$ km. (a) What is the velocity? (b) What is the acceleration? (c) Describe the motion from $t = 0$ s.

Strategy

We can gain some insight into the problem by looking at the position function. It is linear in y and z , so we know the acceleration in these directions is zero when we take the second derivative. Also, note that the position in the x direction is zero for $t = 0$ s and $t = 10$ s.

Solution

(a) Taking the derivative with respect to time of the position function, we find

$$\vec{v}(t) = (10 - 2t)\hat{i} + 5\hat{j} + 5\hat{k} \text{ m/s}.$$

The velocity function is linear in time in the x direction and is constant in the y and z directions.

(b) Taking the derivative of the velocity function, we find

$$\vec{a}(t) = -2\hat{i} \text{ m/s}^2.$$

The acceleration vector is a constant in the negative x -direction.

(c) The trajectory of the particle can be seen in [Figure 4.9](#). Let's look in the y and z directions first. The particle's position increases steadily as a function of time with a constant velocity in these directions. In the x direction, however, the particle follows a path in positive x until $t = 5$ s, when it reverses direction. We know this from looking at the velocity function, which becomes zero at this time and negative thereafter. We also know this because the acceleration is negative and constant—meaning, the particle is accelerating in the opposite direction. The particle's position reaches 25 m, where it then reverses direction and begins to accelerate in the negative x direction. The position reaches zero at $t = 10$ s.

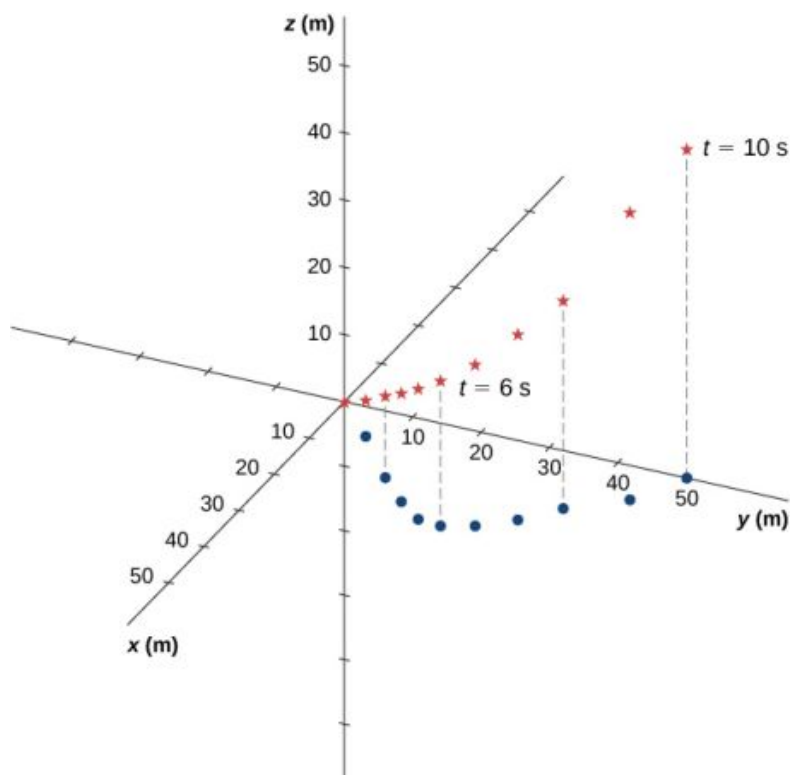


FIGURE 4.9 The particle starts at point $(x, y, z) = (0, 0, 0)$ with position vector $\vec{r} = 0$ as shown with red stars. The projection of the trajectory onto the xy -plane is shown with blue dots. The values of y and z increase linearly as a function of time, whereas x has a turning point at $t = 5$ s and 25 m, when it reverses direction. At this point, the x component of the velocity becomes negative. At $t = 10$ s, the particle is back to 0 m in the x direction.

Significance

By graphing the trajectory of the particle, we can better understand its motion, given by the numerical results of the kinematic equations.

✓ CHECK YOUR UNDERSTANDING 4.2

Suppose the acceleration function has the form $\vec{a}(t) = a\hat{i} + b\hat{j} + c\hat{k} \text{ m/s}^2$, where a , b , and c are constants. What can be said about the functional form of the velocity function?

Constant Acceleration

Multidimensional motion with constant acceleration can be treated the same way as shown in the previous chapter for one-dimensional motion. Earlier we showed that three-dimensional motion is equivalent to three one-dimensional motions, each along an axis perpendicular to the others. To develop the relevant equations in each direction, let's consider the two-dimensional problem of a particle moving in the xy plane with constant

acceleration, ignoring the z-component for the moment. The acceleration vector is

$$\vec{a} = a_{0x}\hat{i} + a_{0y}\hat{j}.$$

Each component of the motion has a separate set of equations similar to [Equation 3.10](#)–[Equation 3.14](#) of the previous chapter on one-dimensional motion. We show only the equations for position and velocity in the x- and y-directions. A similar set of kinematic equations could be written for motion in the z-direction:

$$x(t) = x_0 + (v_x)_{\text{avg}}t \quad 4.11$$

$$v_x(t) = v_{0x} + a_x t \quad 4.12$$

$$x(t) = x_0 + v_{0x}t + \frac{1}{2}a_x t^2 \quad 4.13$$

$$v_x^2(t) = v_{0x}^2 + 2a_x(x - x_0) \quad 4.14$$

$$y(t) = y_0 + (v_y)_{\text{avg}}t \quad 4.15$$

$$v_y(t) = v_{0y} + a_y t \quad 4.16$$

$$y(t) = y_0 + v_{0y}t + \frac{1}{2}a_y t^2 \quad 4.17$$

$$v_y^2(t) = v_{0y}^2 + 2a_y(y - y_0). \quad 4.18$$

Here the subscript 0 denotes the initial position or velocity. [Equation 4.11](#) to [Equation 4.18](#) can be substituted into [Equation 4.2](#) and [Equation 4.5](#) without the z-component to obtain the position vector and velocity vector as a function of time in two dimensions:

$$\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j} \text{ and } \vec{v}(t) = v_x(t)\hat{i} + v_y(t)\hat{j}.$$

The following example illustrates a practical use of the kinematic equations in two dimensions.



EXAMPLE 4.6

A Skier

[Figure 4.10](#) shows a skier moving with an acceleration of 2.1 m/s^2 down a slope of 15° at $t = 0$. With the origin of the coordinate system at the front of the lodge, her initial position and velocity are

$$\vec{r}(0) = (75.0\hat{i} - 50.0\hat{j}) \text{ m}$$

and

$$\vec{v}(0) = (4.1\hat{i} - 1.1\hat{j}) \text{ m/s}.$$

(a) What are the x- and y-components of the skier's position and velocity as functions of time? (b) What are her position and velocity at $t = 10.0 \text{ s}$?

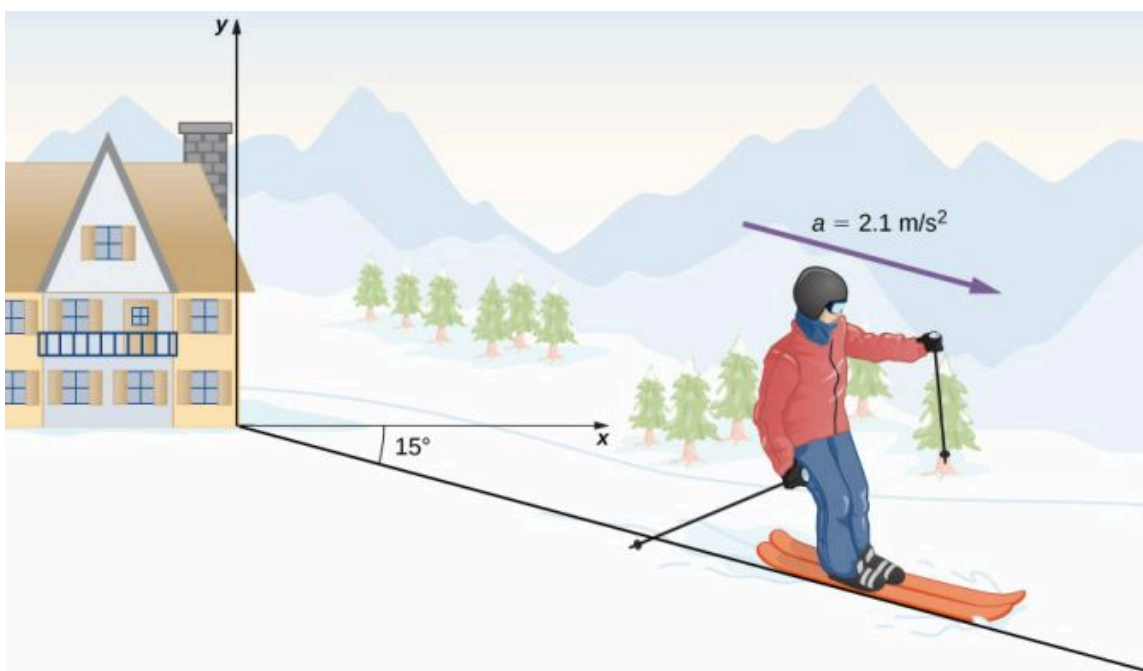


FIGURE 4.10 A skier has an acceleration of 2.1 m/s^2 down a slope of 15° . The origin of the coordinate system is at the ski lodge.

Strategy

Since we are evaluating the components of the motion equations in the x and y directions, we need to find the components of the acceleration and put them into the kinematic equations. The components of the acceleration are found by referring to the coordinate system in [Figure 4.10](#). Then, by inserting the components of the initial position and velocity into the motion equations, we can solve for her position and velocity at a later time t .

Solution

(a) The origin of the coordinate system is at the top of the hill with y -axis vertically upward and the x -axis horizontal. By looking at the trajectory of the skier, the x -component of the acceleration is positive and the y -component is negative. Since the angle is 15° down the slope, we find

$$a_x = (2.1 \text{ m/s}^2) \cos(15^\circ) = 2.0 \text{ m/s}^2$$

$$a_y = (-2.1 \text{ m/s}^2) \sin 15^\circ = -0.54 \text{ m/s}^2.$$

Inserting the initial position and velocity into [Equation 4.12](#) and [Equation 4.13](#) for x , we have

$$x(t) = 75.0 \text{ m} + (4.1 \text{ m/s})t + \frac{1}{2}(2.0 \text{ m/s}^2)t^2$$

$$v_x(t) = 4.1 \text{ m/s} + (2.0 \text{ m/s}^2)t.$$

For y , we have

$$y(t) = -50.0 \text{ m} + (-1.1 \text{ m/s})t + \frac{1}{2}(-0.54 \text{ m/s}^2)t^2$$

$$v_y(t) = -1.1 \text{ m/s} + (-0.54 \text{ m/s}^2)t.$$

(b) Now that we have the equations of motion for x and y as functions of time, we can evaluate them at $t = 10.0 \text{ s}$:

$$x(10.0 \text{ s}) = 75.0 \text{ m} + (4.1 \text{ m/s})(10.0 \text{ s}) + \frac{1}{2}(2.0 \text{ m/s}^2)(10.0 \text{ s})^2 = 216.0 \text{ m}$$

$$v_x(10.0 \text{ s}) = 4.1 \text{ m/s} + (2.0 \text{ m/s}^2)(10.0 \text{ s}) = 24.1 \text{ m/s}$$

$$y(10.0 \text{ s}) = -50.0 \text{ m} + (-1.1 \text{ m/s})(10.0 \text{ s}) + \frac{1}{2}(-0.54 \text{ m/s}^2)(10.0 \text{ s})^2 = -88.0 \text{ m}$$

$$v_y(10.0 \text{ s}) = -1.1 \text{ m/s} + (-0.54 \text{ m/s}^2)(10.0 \text{ s}) = -6.5 \text{ m/s}.$$

The position and velocity at $t = 10.0 \text{ s}$ are, finally,

$$\vec{r}(10.0 \text{ s}) = (216.0\hat{i} - 88.0\hat{j}) \text{ m}$$

$$\vec{v}(10.0 \text{ s}) = (24.1\hat{i} - 6.5\hat{j}) \text{ m/s}.$$

The magnitude of the velocity of the skier at 10.0 s is 25 m/s , which is 60 mi/h .

Significance

It is useful to know that, given the initial conditions of position, velocity, and acceleration of an object, we can find the position, velocity, and acceleration at any later time.

With [Equation 4.8](#) through [Equation 4.10](#) we have completed the set of expressions for the position, velocity, and acceleration of an object moving in two or three dimensions. If the trajectories of the objects look something like the “Red Arrows” in the opening picture for the chapter, then the expressions for the position, velocity, and acceleration can be quite complicated. In the sections to follow we examine two special cases of motion in two and three dimensions by looking at projectile motion and circular motion.

INTERACTIVE

At this [University of Colorado Boulder website \(https://openstax.org/l/21phetmotladyb\)](https://openstax.org/l/21phetmotladyb), you can explore the position velocity and acceleration of a ladybug with an interactive simulation that allows you to change these parameters.

4.3 Projectile Motion

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Use one-dimensional motion in perpendicular directions to analyze projectile motion.
- Calculate the range, time of flight, and maximum height of a projectile that is launched and impacts a flat, horizontal surface.
- Find the time of flight and impact velocity of a projectile that lands at a different height from that of launch.
- Calculate the trajectory of a projectile.

Projectile motion is the motion of an object thrown or projected into the air, subject only to acceleration as a result of gravity. The applications of projectile motion in physics and engineering are numerous. Some examples include meteors as they enter Earth’s atmosphere, fireworks, and the motion of any ball in sports. Such objects are called *projectiles* and their path is called a **trajectory**. The motion of falling objects as discussed in [Motion Along a Straight Line](#) is a simple one-dimensional type of projectile motion in which there is no horizontal movement. In this section, we consider two-dimensional projectile motion, and our treatment neglects the effects of air resistance.

The most important fact to remember here is that *motions along perpendicular axes are independent* and thus can be analyzed separately. We discussed this fact in [Displacement and Velocity Vectors](#), where we saw that vertical and horizontal motions are independent. The key to analyzing two-dimensional projectile motion is to break it into two motions: one along the horizontal axis and the other along the vertical. (This choice of axes is the most sensible because acceleration resulting from gravity is vertical; thus, there is no acceleration along the horizontal axis when air resistance is negligible.) As is customary, we call the horizontal axis the x -axis and the vertical axis the y -axis. It is not required that we use this choice of axes; it is simply convenient in the case of gravitational acceleration. In other cases we may choose a different set of axes. [Figure 4.11](#) illustrates the notation for displacement, where we define $\vec{s}$ to be the total displacement, and $\vec{x}$ and $\vec{y}$ are its component vectors along the horizontal and vertical axes, respectively. The magnitudes of these vectors are s , x , and y .

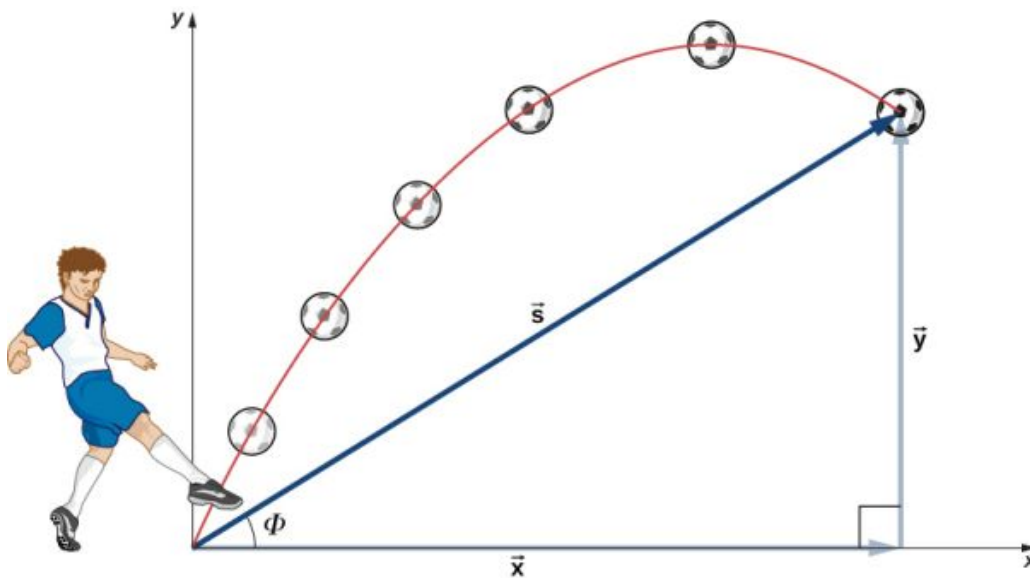


FIGURE 4.11 The total displacement s of a soccer ball at a point along its path. The vector $\vec{s}$ has components $\vec{x}$ and $\vec{y}$ along the horizontal and vertical axes. Its magnitude is s and it makes an angle ϕ with the horizontal.

To describe projectile motion completely, we must include velocity and acceleration, as well as displacement. We must find their components along the x - and y -axes. Let's assume all forces except gravity (such as air resistance and friction, for example) are negligible. Defining the positive direction to be upward, the components of acceleration are then very simple:

$$a_y = -g = -9.8 \text{ m/s}^2 \quad (-32 \text{ ft/s}^2).$$

Because gravity is vertical, $a_x = 0$. If $a_x = 0$, this means the initial velocity in the x direction is equal to the final velocity in the x direction, or $v_x = v_{0x}$. With these conditions on acceleration and velocity, we can write the kinematic [Equation 4.11](#) through [Equation 4.18](#) for motion in a uniform gravitational field, including the rest of the kinematic equations for a constant acceleration from [Motion with Constant Acceleration](#). The kinematic equations for motion in a uniform gravitational field become kinematic equations with $a_y = -g$, $a_x = 0$:

Horizontal Motion

$$v_{0x} = v_x, \quad x = x_0 + v_x t \quad 4.19$$

Vertical Motion

$$y = y_0 + \frac{1}{2}(v_{0y} + v_y)t \quad 4.20$$

$$v_y = v_{0y} - gt \quad 4.21$$

$$y = y_0 + v_{0y}t - \frac{1}{2}gt^2 \quad 4.22$$

$$v_y^2 = v_{0y}^2 - 2g(y - y_0) \quad 4.23$$

Using this set of equations, we can analyze projectile motion, keeping in mind some important points.



PROBLEM-SOLVING STRATEGY

Projectile Motion

1. Resolve the motion into horizontal and vertical components along the x - and y -axes. The magnitudes of the components of displacement $\vec{s}$ along these axes are x and y . The magnitudes of the components of velocity $\vec{v}$ are $v_x = v\cos\theta$ and $v_y = v\sin\theta$, where v is the magnitude of the velocity and θ is its direction relative to the horizontal, as shown in [Figure 4.12](#).

2. Treat the motion as two independent one-dimensional motions: one horizontal and the other vertical. Use the kinematic equations for horizontal and vertical motion presented earlier.
3. Solve for the unknowns in the two separate motions: one horizontal and one vertical. Note that the only common variable between the motions is time t . The problem-solving procedures here are the same as those for one-dimensional kinematics and are illustrated in the following solved examples.
4. Recombine quantities in the horizontal and vertical directions to find the total displacement $\vec{s}$ and velocity $\vec{v}$. Solve for the magnitude and direction of the displacement and velocity using

$$s = \sqrt{x^2 + y^2}, \quad \Phi = \tan^{-1}(y/x), \quad v = \sqrt{v_x^2 + v_y^2},$$

where Φ is the direction of the displacement $\vec{s}$.

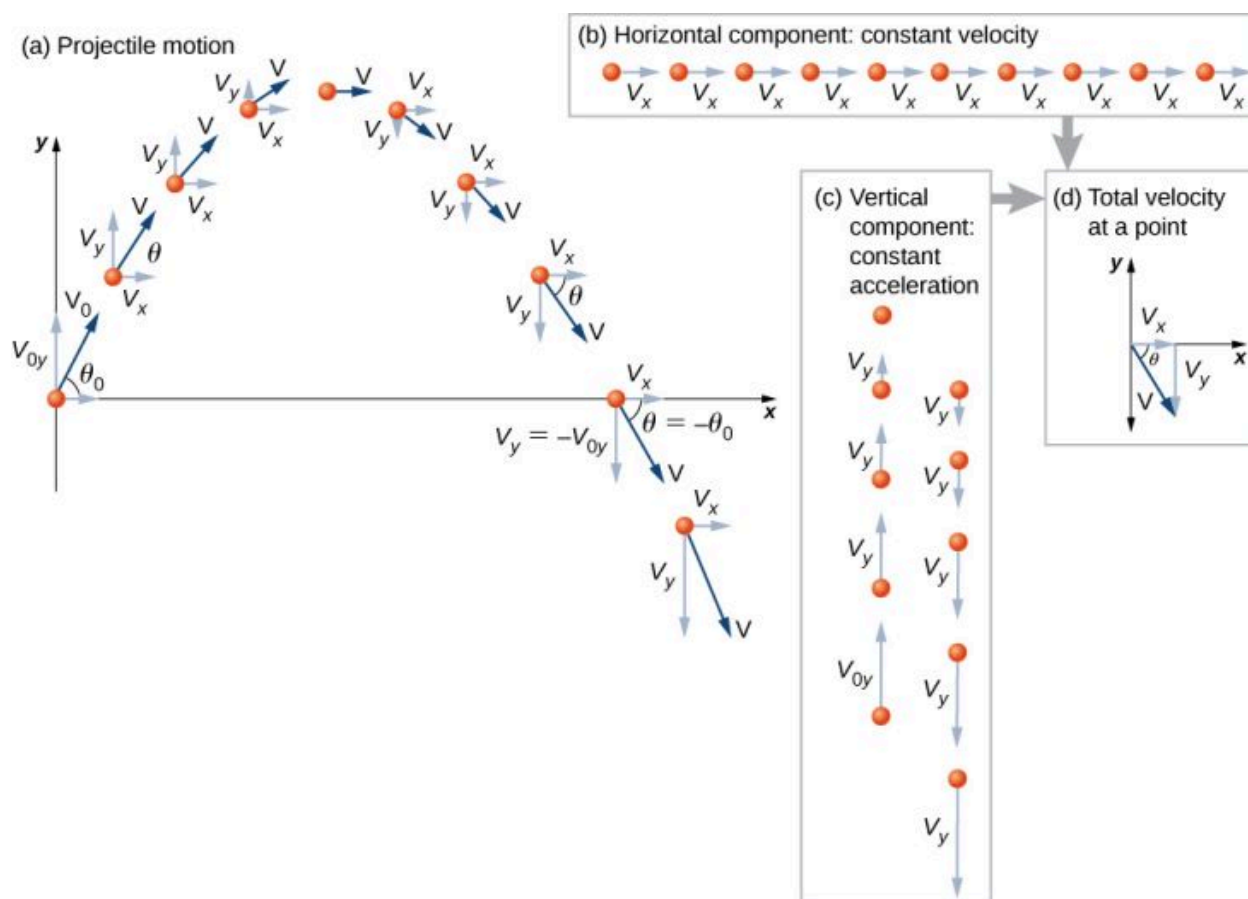


FIGURE 4.12 (a) We analyze two-dimensional projectile motion by breaking it into two independent one-dimensional motions along the vertical and horizontal axes. (b) The horizontal motion is simple, because $a_x = 0$ and v_x is a constant. (c) The velocity in the vertical direction begins to decrease as the object rises. At its highest point, the vertical velocity is zero. As the object falls toward Earth again, the vertical velocity increases again in magnitude but points in the opposite direction to the initial vertical velocity. (d) The x and y motions are recombined to give the total velocity at any given point on the trajectory.



EXAMPLE 4.7

A Fireworks Projectile Explodes High and Away

During a fireworks display, a shell is shot into the air with an initial speed of 70.0 m/s at an angle of 75.0° above the horizontal, as illustrated in [Figure 4.13](#). The fuse is timed to ignite the shell just as it reaches its highest point above the ground. (a) Calculate the height at which the shell explodes. (b) How much time passes between the launch of the shell and the explosion? (c) What is the horizontal displacement of the shell when it explodes? (d) What is the total displacement from the point of launch to the highest point?

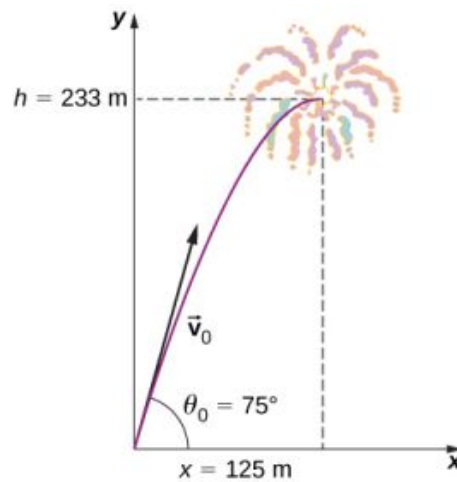


FIGURE 4.13 The trajectory of a fireworks shell. The fuse is set to explode the shell at the highest point in its trajectory, which is found to be at a height of 233 m and 125 m away horizontally.

Strategy

The motion can be broken into horizontal and vertical motions in which $a_x = 0$ and $a_y = -g$. We can then define x_0 and y_0 to be zero and solve for the desired quantities.

Solution

(a) By “height” we mean the altitude or vertical position y above the starting point. The highest point in any trajectory, called the *apex*, is reached when $v_y = 0$. Since we know the initial and final velocities, as well as the initial position, we use the following equation to find y :

$$v_y^2 = v_{0y}^2 - 2g(y - y_0).$$

Because y_0 and v_y are both zero, the equation simplifies to

$$0 = v_{0y}^2 - 2gy.$$

Solving for y gives

$$y = \frac{v_{0y}^2}{2g}.$$

Now we must find v_{0y} , the component of the initial velocity in the y direction. It is given by $v_{0y} = v_0 \sin \theta_0$, where v_0 is the initial velocity of 70.0 m/s and $\theta_0 = 75^\circ$ is the initial angle. Thus,

$$v_{0y} = v_0 \sin \theta = (70.0 \text{ m/s}) \sin 75^\circ = 67.6 \text{ m/s}$$

and y is

$$y = \frac{(67.6 \text{ m/s})^2}{2(9.80 \text{ m/s}^2)}.$$

Thus, we have

$$y = 233 \text{ m}.$$

Note that because up is positive, the initial vertical velocity is positive, as is the maximum height, but the acceleration resulting from gravity is negative. Note also that the maximum height depends only on the vertical component of the initial velocity, so that any projectile with a 67.6-m/s initial vertical component of velocity reaches a maximum height of 233 m (neglecting air resistance). The numbers in this example are reasonable for large fireworks displays, the shells of which do reach such heights before exploding. In practice, air resistance is not completely negligible, so the initial velocity would have to be somewhat larger than that given to reach the same height.

(b) As in many physics problems, there is more than one way to solve for the time the projectile reaches its highest point. In this case, the easiest method is to use $v_y = v_{0y} - gt$. Because $v_y = 0$ at the apex, this equation reduces to simply

$$0 = v_{0y} - gt$$

or

$$t = \frac{v_{0y}}{g} = \frac{67.6 \text{ m/s}}{9.80 \text{ m/s}^2} = 6.90 \text{ s}.$$

This time is also reasonable for large fireworks. If you are able to see the launch of fireworks, notice that several seconds pass before the shell explodes. Another way of finding the time is by using $y = y_0 + \frac{1}{2}(v_{0y} + v_y)t$. This is left for you as an exercise to complete.

(c) Because air resistance is negligible, $a_x = 0$ and the horizontal velocity is constant, as discussed earlier. The horizontal displacement is the horizontal velocity multiplied by time as given by $x = x_0 + v_x t$, where x_0 is equal to zero. Thus,

$$x = v_x t,$$

where v_x is the x -component of the velocity, which is given by

$$v_x = v_0 \cos \theta = (70.0 \text{ m/s}) \cos 75^\circ = 18.1 \text{ m/s}.$$

Time t for both motions is the same, so x is

$$x = (18.1 \text{ m/s}) (6.90 \text{ s}) = 125 \text{ m}.$$

Horizontal motion is a constant velocity in the absence of air resistance. The horizontal displacement found here could be useful in keeping the fireworks fragments from falling on spectators. When the shell explodes, air resistance has a major effect, and many fragments land directly below.

(d) The horizontal and vertical components of the displacement were just calculated, so all that is needed here is to find the magnitude and direction of the displacement at the highest point:

$$\begin{aligned}\vec{s} &= 125\hat{i} + 233\hat{j} \\ |\vec{s}| &= \sqrt{125^2 + 233^2} = 264 \text{ m} \\ \Phi &= \tan^{-1} \left(\frac{233}{125} \right) = 61.8^\circ.\end{aligned}$$

Note that the angle for the displacement vector is less than the initial angle of launch. To see why this is, review [Figure 4.11](#), which shows the curvature of the trajectory toward the ground level.

When solving [Example 4.7\(a\)](#), the expression we found for y is valid for any projectile motion when air resistance is negligible. Call the maximum height $y = h$. Then,

$$h = \frac{v_{0y}^2}{2g}.$$

This equation defines the *maximum height of a projectile above its launch position* and it depends only on the vertical component of the initial velocity.

CHECK YOUR UNDERSTANDING 4.3

A rock is thrown horizontally off a cliff 100.0 m high with a velocity of 15.0 m/s. (a) Define the origin of the coordinate system. (b) Which equation describes the horizontal motion? (c) Which equations describe the vertical motion? (d) What is the rock's velocity at the point of impact?

EXAMPLE 4.8

Calculating Projectile Motion: Tennis Player

A tennis player wins a match at Arthur Ashe stadium and hits a ball into the stands at 30 m/s and at an angle 45° above the horizontal (Figure 4.14). On its way down, the ball is caught by a spectator 10 m above the point where the ball was hit. (a) Calculate the time it takes the tennis ball to reach the spectator. (b) What are the magnitude and direction of the ball's velocity at impact?

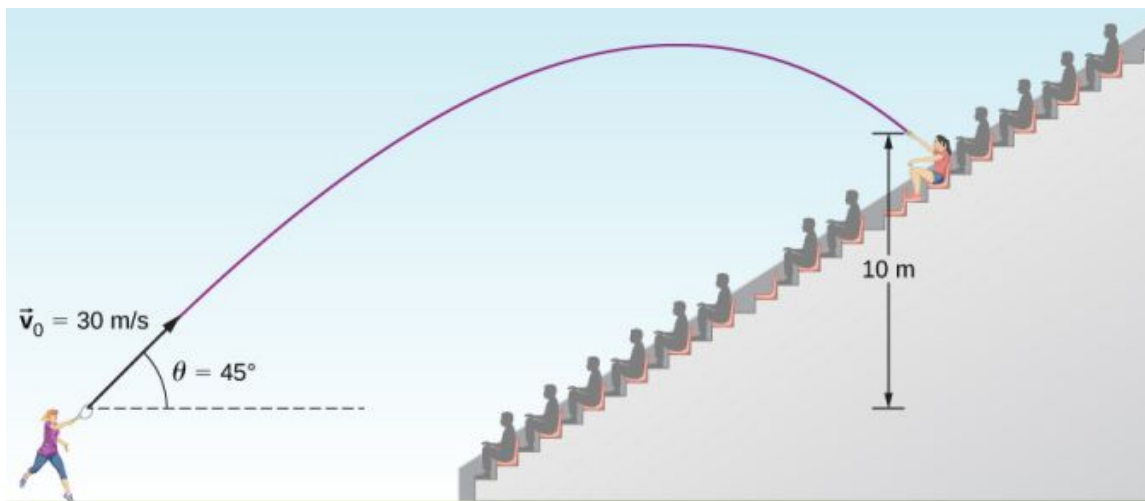


FIGURE 4.14 The trajectory of a tennis ball hit into the stands.

Strategy

Again, resolving this two-dimensional motion into two independent one-dimensional motions allows us to solve for the desired quantities. The time a projectile is in the air is governed by its vertical motion alone. Thus, we solve for t first. While the ball is rising and falling vertically, the horizontal motion continues at a constant velocity. This example asks for the final velocity. Thus, we recombine the vertical and horizontal results to obtain $\vec{v}$ at final time t , determined in the first part of the example.

Solution

(a) While the ball is in the air, it rises and then falls to a final position 10.0 m higher than its starting altitude. We can find the time for this by using Equation 4.22:

$$y = y_0 + v_{0y}t - \frac{1}{2}gt^2.$$

If we take the initial position y_0 to be zero, then the final position is $y = 10$ m. The initial vertical velocity is the vertical component of the initial velocity:

$$v_{0y} = v_0 \sin \theta_0 = (30.0 \text{ m/s}) \sin 45^\circ = 21.2 \text{ m/s}.$$

Substituting into Equation 4.22 for y gives us

$$10.0 \text{ m} = (21.2 \text{ m/s})t - (4.90 \text{ m/s}^2)t^2.$$

Rearranging terms gives a quadratic equation in t :

$$(4.90 \text{ m/s}^2)t^2 - (21.2 \text{ m/s})t + 10.0 \text{ m} = 0.$$

Use of the quadratic formula yields $t = 3.79$ s and $t = 0.54$ s. Since the ball is at a height of 10 m at two times during its trajectory—once on the way up and once on the way down—we take the longer solution for the time it takes the ball to reach the spectator:

$$t = 3.79 \text{ s}.$$

The time for projectile motion is determined completely by the vertical motion. Thus, any projectile that has an initial vertical velocity of 21.2 m/s and lands 10.0 m above its starting altitude spends 3.79 s in the air.

(b) We can find the final horizontal and vertical velocities v_x and v_y with the use of the result from (a). Then, we can combine them to find the magnitude of the total velocity vector $\vec{v}$ and the angle θ it makes with the horizontal. Since v_x is constant, we can solve for it at any horizontal location. We choose the starting point because we know both the initial velocity and the initial angle. Therefore,

$$v_x = v_0 \cos \theta_0 = (30 \text{ m/s}) \cos 45^\circ = 21.2 \text{ m/s}.$$

The final vertical velocity is given by [Equation 4.21](#):

$$v_y = v_{0y} - gt.$$

Since v_{0y} was found in part (a) to be 21.2 m/s, we have

$$v_y = 21.2 \text{ m/s} - 9.8 \text{ m/s}^2 (3.79 \text{ s}) = -15.9 \text{ m/s}.$$

The magnitude of the final velocity $\vec{v}$ is

$$v = \sqrt{v_x^2 + v_y^2} = \sqrt{(21.2 \text{ m/s})^2 + (-15.9 \text{ m/s})^2} = 26.5 \text{ m/s}.$$

The direction θ_v is found using the inverse tangent:

$$\theta_v = \tan^{-1} \left(\frac{v_y}{v_x} \right) = \tan^{-1} \left(\frac{-15.9}{21.2} \right) = 36.9^\circ \text{ below the horizon}.$$

Significance

(a) As mentioned earlier, the time for projectile motion is determined completely by the vertical motion. Thus, any projectile that has an initial vertical velocity of 21.2 m/s and lands 10.0 m above its starting altitude spends 3.79 s in the air. (b) The negative angle means the velocity is 36.9° below the horizontal at the point of impact. This result is consistent with the fact that the ball is impacting at a point on the other side of the apex of the trajectory and therefore has a negative y component of the velocity. The magnitude of the velocity is less than the magnitude of the initial velocity we expect since it is impacting 10.0 m above the launch elevation.

Time of Flight, Trajectory, and Range

Of interest are the time of flight, trajectory, and range for a projectile launched on a flat horizontal surface and impacting on the same surface. In this case, kinematic equations give useful expressions for these quantities, which are derived in the following sections.

Time of flight

We can solve for the time of flight of a projectile that is both launched and impacts on a flat horizontal surface by performing some manipulations of the kinematic equations. We note the position and displacement in y must be zero at launch and at impact on an even surface. Thus, we set the displacement in y equal to zero and find

$$y - y_0 = v_{0y}t - \frac{1}{2}gt^2 = (v_0 \sin \theta_0)t - \frac{1}{2}gt^2 = 0.$$

Factoring, we have

$$t \left(v_0 \sin \theta_0 - \frac{gt}{2} \right) = 0.$$

Solving for t gives us

$$T_{\text{tof}} = \frac{2(v_0 \sin \theta_0)}{g}. \quad 4.24$$

This is the **time of flight** for a projectile both launched and impacting on a flat horizontal surface. [Equation 4.24](#)

does not apply when the projectile lands at a different elevation than it was launched, as we saw in [Example 4.8](#) of the tennis player hitting the ball into the stands. The other solution, $t = 0$, corresponds to the time at launch. The time of flight is linearly proportional to the initial velocity in the y direction and inversely proportional to g . Thus, on the Moon, where gravity is one-sixth that of Earth, a projectile launched with the same velocity as on Earth would be airborne six times as long.

Trajectory

The trajectory of a projectile can be found by eliminating the time variable t from the kinematic equations for arbitrary t and solving for $y(x)$. We take $x_0 = y_0 = 0$ so the projectile is launched from the origin. The kinematic equation for x gives

$$x = v_{0x}t \Rightarrow t = \frac{x}{v_{0x}} = \frac{x}{v_0 \cos \theta_0}.$$

Substituting the expression for t into the equation for the position $y = (v_0 \sin \theta_0)t - \frac{1}{2}gt^2$ gives

$$y = (v_0 \sin \theta_0) \left(\frac{x}{v_0 \cos \theta_0} \right) - \frac{1}{2}g \left(\frac{x}{v_0 \cos \theta_0} \right)^2.$$

Rearranging terms, we have

$$y = (\tan \theta_0)x - \left[\frac{g}{2(v_0 \cos \theta_0)^2} \right] x^2. \quad 4.25$$

This trajectory equation is of the form $y = ax + bx^2$, which is an equation of a parabola with coefficients

$$a = \tan \theta_0, \quad b = -\frac{g}{2(v_0 \cos \theta_0)^2}.$$

Range

From the trajectory equation we can also find the **range**, or the horizontal distance traveled by the projectile. Factoring [Equation 4.25](#), we have

$$y = x \left[\tan \theta_0 - \frac{g}{2(v_0 \cos \theta_0)^2} x \right].$$

The position y is zero for both the launch point and the impact point, since we are again considering only a flat horizontal surface. Setting $y = 0$ in this equation gives solutions $x = 0$, corresponding to the launch point, and

$$x = \frac{2v_0^2 \sin \theta_0 \cos \theta_0}{g},$$

corresponding to the impact point. Using the trigonometric identity $2\sin \theta \cos \theta = \sin 2\theta$ and setting $x = R$ for range, we find

$$R = \frac{v_0^2 \sin 2\theta_0}{g}. \quad 4.26$$

Note particularly that [Equation 4.26](#) is valid only for launch and impact on a horizontal surface. We see the range is directly proportional to the square of the initial speed v_0 and $\sin 2\theta_0$, and it is inversely proportional to the acceleration of gravity. Thus, on the Moon, the range would be six times greater than on Earth for the same initial velocity. Furthermore, we see from the factor $\sin 2\theta_0$ that the range is maximum at 45° . These results are shown in [Figure 4.15](#). In (a) we see that the greater the initial velocity, the greater the range. In (b), we see that the range is maximum at 45° . This is true only for conditions neglecting air resistance. If air resistance is considered, the maximum angle is somewhat smaller. It is interesting that the same range is found for two initial launch angles that sum to 90° . The projectile launched with the smaller angle has a lower apex than the higher angle, but they both have the same range.

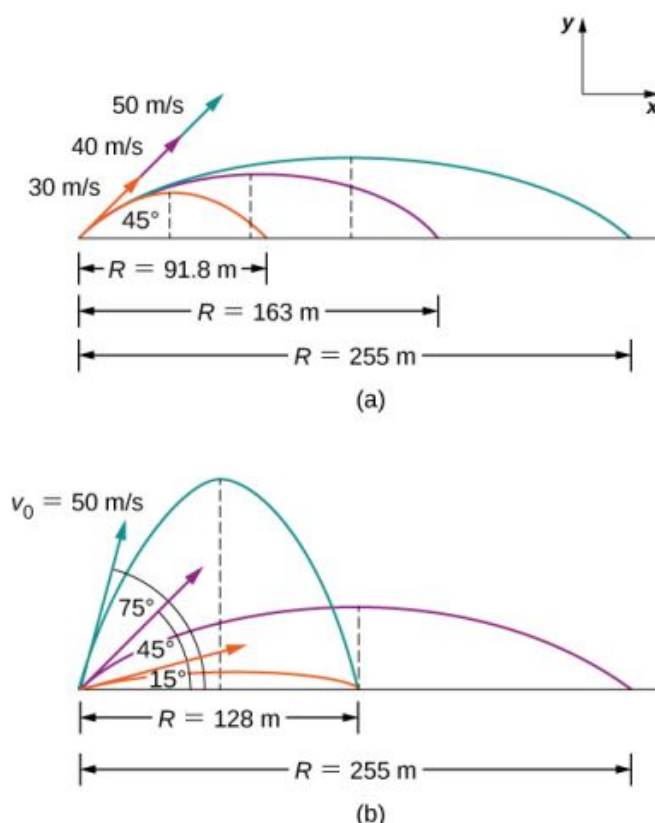


FIGURE 4.15 Trajectories of projectiles on level ground. (a) The greater the initial speed v_0 , the greater the range for a given initial angle. (b) The effect of initial angle θ_0 on the range of a projectile with a given initial speed. Note that the range is the same for initial angles of 15° and 75° , although the maximum heights of those paths are different.

EXAMPLE 4.9

Comparing Golf Shots

A golfer encounters two different situations on different holes. On the second hole they are 120 m from the green and want to hit the ball 90 m and let it run onto the green. They angle the shot low to the ground at 30° to the horizontal to let the ball roll after impact. On the fourth hole they are 90 m from the green and want to let the ball drop with a minimum amount of rolling after impact. Here, they angle the shot at 70° to the horizontal to minimize rolling after impact. Both shots are hit and impacted on a level surface.

- What is the initial speed of the ball at the second hole?
- What is the initial speed of the ball at the fourth hole?
- Write the trajectory equation for both cases.
- Graph the trajectories.

Strategy

We see that the range equation has the initial speed and angle, so we can solve for the initial speed for both (a) and (b). When we have the initial speed, we can use this value to write the trajectory equation.

Solution

$$(a) \ R = \frac{v_0^2 \sin 2\theta_0}{g} \Rightarrow v_0 = \sqrt{\frac{Rg}{\sin 2\theta_0}} = \sqrt{\frac{90.0 \text{ m}(9.8 \text{ m/s}^2)}{\sin(2(30^\circ))}} = 31.9 \text{ m/s}$$

$$(b) \ R = \frac{v_0^2 \sin 2\theta_0}{g} \Rightarrow v_0 = \sqrt{\frac{Rg}{\sin 2\theta_0}} = \sqrt{\frac{90.0 \text{ m}(9.8 \text{ m/s}^2)}{\sin(2(70^\circ))}} = 37.0 \text{ m/s}$$

(c)

$$y = x \left[\tan \theta_0 - \frac{g}{2(v_0 \cos \theta_0)^2} x \right]$$

$$\text{Second hole: } y = x \left[\tan 30^\circ - \frac{9.8 \text{ m/s}^2}{2[(31.9 \text{ m/s})(\cos 30^\circ)]^2} x \right] = 0.58x - 0.0064x^2$$

$$\text{Fourth hole: } y = x \left[\tan 70^\circ - \frac{9.8 \text{ m/s}^2}{2[(37.0 \text{ m/s})(\cos 70^\circ)]^2} x \right] = 2.75x - 0.0306x^2$$

(d) Using a graphing utility, we can compare the two trajectories, which are shown in [Figure 4.16](#).

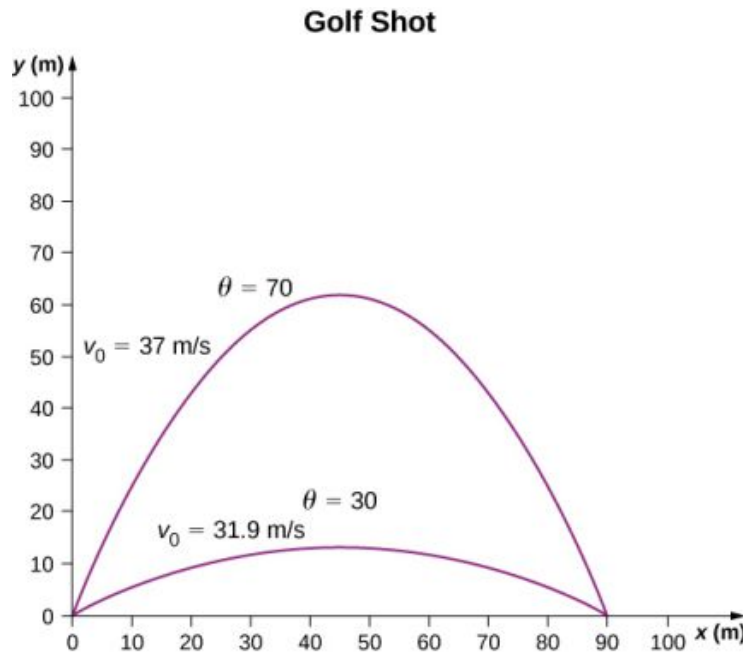


FIGURE 4.16 Two trajectories of a golf ball with a range of 90 m. The impact points of both are at the same level as the launch point.

Significance

The initial speed for the shot at 70° is greater than the initial speed of the shot at 30° . Note from [Figure 4.16](#) that if the two projectiles were launched at the same speed but at different angles, the projectiles would have the same range as long as the angles were less than 90° . The launch angles in this example add to give a number greater than 90° . Thus, the shot at 70° has to have a greater launch speed to reach 90 m, otherwise it would land at a shorter distance.

CHECK YOUR UNDERSTANDING 4.4

If the two golf shots in [Example 4.9](#) were launched at the same speed, which shot would have the greatest range?

When we speak of the range of a projectile on level ground, we assume R is very small compared with the circumference of Earth. If, however, the range is large, Earth curves away below the projectile and the acceleration resulting from gravity changes direction along the path. The range is larger than predicted by the range equation given earlier because the projectile has farther to fall than it would on level ground, as shown in [Figure 4.17](#), which is based on a drawing in Newton's *Principia*. If the initial speed is great enough, the projectile goes into orbit. Earth's surface drops 5 m every 8000 m. In 1 s an object falls 5 m without air resistance. Thus, if an object is given a

horizontal velocity of 8000 m/s (or 18,000 mi/hr) near Earth's surface, it will go into orbit around the planet because the surface continuously falls away from the object. This is roughly the speed of the Space Shuttle in a low Earth orbit when it was operational, or any satellite in a low Earth orbit. These and other aspects of orbital motion, such as Earth's rotation, are covered in greater depth in [Gravitation](#).

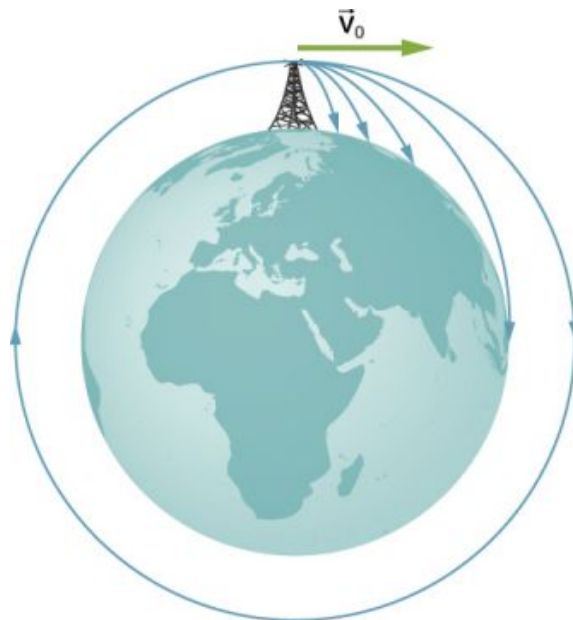


FIGURE 4.17 Projectile to satellite. In each case shown here, a projectile is launched from a very high tower to avoid air resistance. With increasing initial speed, the range increases and becomes longer than it would be on level ground because Earth curves away beneath its path. With a speed of 8000 m/s, orbit is achieved.

INTERACTIVE

At [PhET Explorations: Projectile Motion \(https://openstax.org/l/21phetpromot\)](https://openstax.org/l/21phetpromot), learn about projectile motion in terms of the launch angle and initial velocity.

4.4 Uniform and Nonuniform Circular Motion

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Solve for the centripetal acceleration of an object moving on a circular path.
- Use the equations of circular motion to find the position, velocity, and acceleration of a particle executing circular motion.
- Explain the differences between centripetal acceleration and tangential acceleration resulting from nonuniform circular motion.
- Evaluate centripetal and tangential acceleration in nonuniform circular motion, and find the total acceleration vector.

Uniform circular motion is a specific type of motion in which an object travels in a circle with a constant speed. For example, any point on a propeller spinning at a constant rate is executing uniform circular motion. Other examples are the second, minute, and hour hands of a watch. It is remarkable that points on these rotating objects are actually accelerating, although the rotation rate is a constant. To see this, we must analyze the motion in terms of vectors.

Centripetal Acceleration

In one-dimensional kinematics, objects with a constant speed have zero acceleration. However, in two- and three-dimensional kinematics, even if the speed is a constant, a particle can have acceleration if it moves along a curved trajectory such as a circle. In this case the velocity vector is changing, or $d\vec{v}/dt \neq 0$. This is shown in [Figure 4.18](#). As the particle moves counterclockwise in time Δt on the circular path, its position vector moves from $\vec{r}(t)$ to $\vec{r}(t + \Delta t)$.

The velocity vector has constant magnitude and is tangent to the path as it changes from $\vec{v}(t)$ to $\vec{v}(t + \Delta t)$, changing its direction only. Since the velocity vector $\vec{v}(t)$ is perpendicular to the position vector $\vec{r}(t)$, the triangles formed by the position vectors and $\Delta\vec{r}$, and the velocity vectors and $\Delta\vec{v}$ are similar. Furthermore, since $|\vec{r}(t)| = |\vec{r}(t + \Delta t)|$ and $|\vec{v}(t)| = |\vec{v}(t + \Delta t)|$, the two triangles are isosceles. From these facts we can make the assertion

$$\frac{\Delta v}{v} = \frac{\Delta r}{r} \text{ or } \Delta v = \frac{v}{r} \Delta r.$$

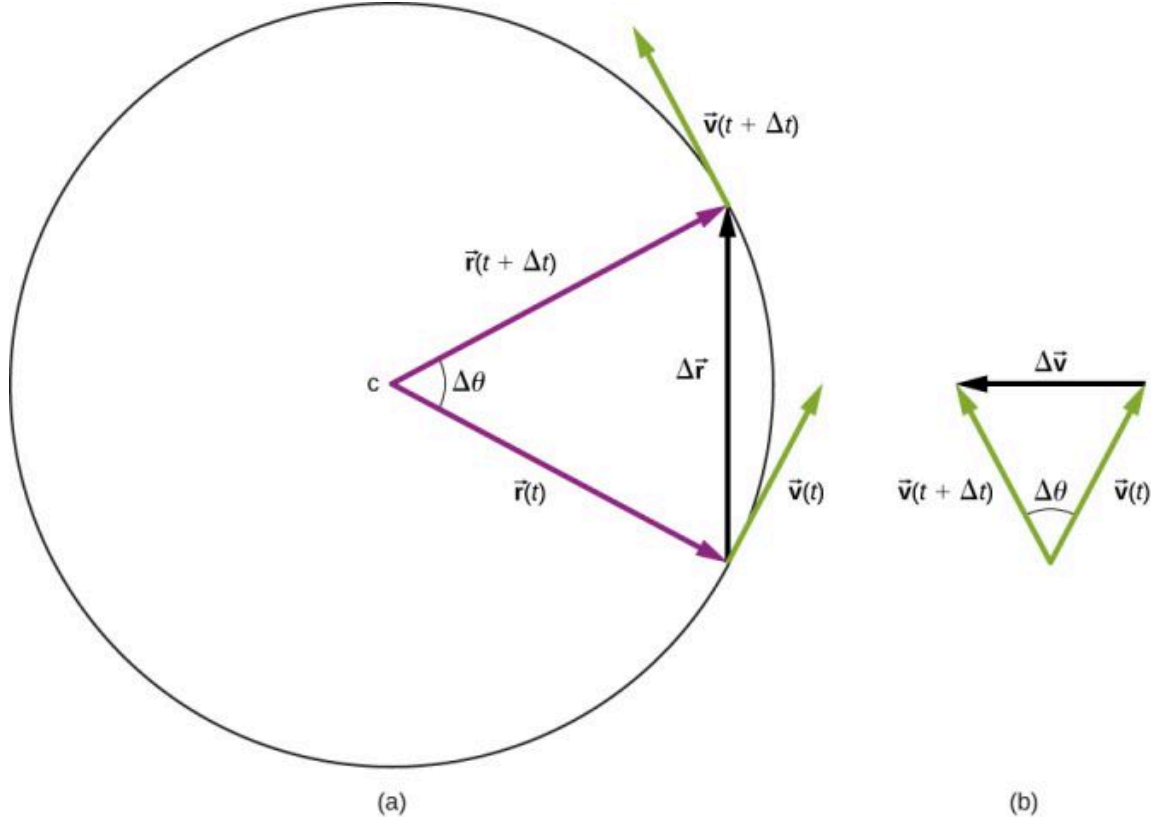


FIGURE 4.18 (a) A particle is moving in a circle at a constant speed, with position and velocity vectors at times t and $t + \Delta t$. (b) Velocity vectors forming a triangle. The two triangles in the figure are similar. The vector $\Delta\vec{v}$ points toward the center of the circle in the limit $\Delta t \rightarrow 0$.

We can find the magnitude of the acceleration from

$$a = \lim_{\Delta t \rightarrow 0} \left(\frac{\Delta v}{\Delta t} \right) = \frac{v}{r} \left(\lim_{\Delta t \rightarrow 0} \frac{\Delta r}{\Delta t} \right) = \frac{v^2}{r}.$$

The direction of the acceleration can also be found by noting that as Δt and therefore $\Delta\theta$ approach zero, the vector $\Delta\vec{v}$ approaches a direction perpendicular to $\vec{v}$. In the limit $\Delta t \rightarrow 0$, $\Delta\vec{v}$ is perpendicular to $\vec{v}$. Since $\vec{v}$ is tangent to the circle, the acceleration $d\vec{v}/dt$ points toward the center of the circle. Summarizing, a particle moving in a circle at a constant speed has an acceleration with magnitude

$$a_c = \frac{v^2}{r}.$$

4.27

The direction of the acceleration vector is toward the center of the circle (Figure 4.19). This is a radial acceleration and is called the **centripetal acceleration**, which is why we give it the subscript c . The word *centripetal* comes from the Latin words *centrum* (meaning “center”) and *petere* (meaning “to seek”), and thus takes the meaning “center seeking.”

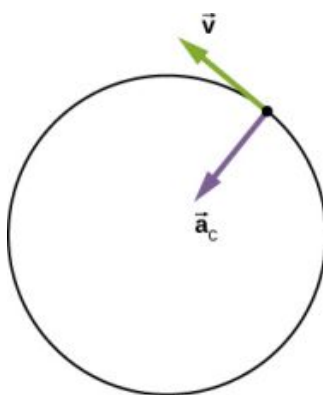


FIGURE 4.19 The centripetal acceleration vector points toward the center of the circular path of motion and is an acceleration in the radial direction. The velocity vector is also shown and is tangent to the circle.

Let's investigate some examples that illustrate the relative magnitudes of the velocity, radius, and centripetal acceleration.



EXAMPLE 4.10

Creating an Acceleration of 1 g

A jet is flying at 134.1 m/s along a straight line and makes a turn along a circular path level with the ground. What does the radius of the circle have to be to produce a centripetal acceleration of 1 g on the pilot and jet toward the center of the circular trajectory?

Strategy

Given the speed of the jet, we can solve for the radius of the circle in the expression for the centripetal acceleration.

Solution

Set the centripetal acceleration equal to the acceleration of gravity: $9.8 \text{ m/s}^2 = v^2/r$.

Solving for the radius, we find

$$r = \frac{(134.1 \text{ m/s})^2}{9.8 \text{ m/s}^2} = 1835 \text{ m} = 1.835 \text{ km}.$$

Significance

To create a greater acceleration than g on the pilot, the jet would either have to decrease the radius of its circular trajectory or increase its speed on its existing trajectory or both.



CHECK YOUR UNDERSTANDING 4.5

A flywheel has a radius of 20.0 cm. What is the speed of a point on the edge of the flywheel if it experiences a centripetal acceleration of 900.0 cm/s^2 ?

Centripetal acceleration can have a wide range of values, depending on the speed and radius of curvature of the circular path. Typical centripetal accelerations are given in the following table.

Object	Centripetal Acceleration (m/s^2 or factors of g)
Earth around the Sun	5.93×10^{-3}
Moon around the Earth	2.73×10^{-3}

TABLE 4.1 Typical Centripetal Accelerations

Object	Centripetal Acceleration (m/s ² or factors of <i>g</i>)
Satellite in geosynchronous orbit	0.233
Outer edge of a CD when playing	5.78
Jet in a barrel roll	(2–3 <i>g</i>)
Roller coaster	(5 <i>g</i>)
Electron orbiting a proton in a simple Bohr model of the atom	9.0×10^{22}

TABLE 4.1 Typical Centripetal Accelerations

Equations of Motion for Uniform Circular Motion

A particle executing circular motion can be described by its position vector $\vec{r}(t)$. Figure 4.20 shows a particle executing circular motion in a counterclockwise direction. As the particle moves on the circle, its position vector sweeps out the angle θ with the *x*-axis. Vector $\vec{r}(t)$ making an angle θ with the *x*-axis is shown with its components along the *x*- and *y*-axes. The magnitude of the position vector is $A = |\vec{r}(t)|$ and is also the radius of the circle, so that in terms of its components,

$$\vec{r}(t) = A \cos \omega t \hat{i} + A \sin \omega t \hat{j}. \quad 4.28$$

Here, ω is a constant called the **angular frequency** of the particle. The angular frequency has units of radians (rad) per second and is simply the number of radians of angular measure through which the particle passes per second. The angle θ that the position vector has at any particular time is ωt .

If T is the period of motion, or the time to complete one revolution (2π rad), then

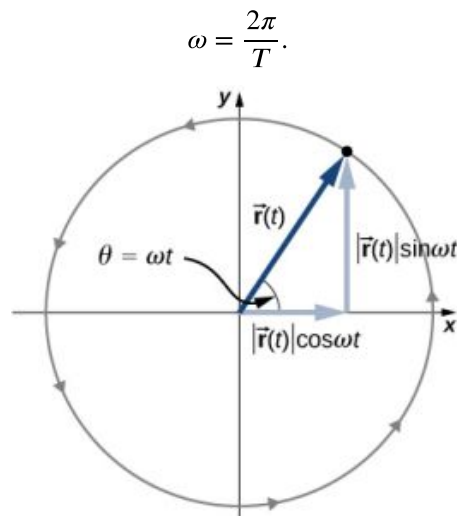


FIGURE 4.20 The position vector for a particle in circular motion with its components along the *x*- and *y*-axes. The particle moves counterclockwise. Angle θ is the angular frequency ω in radians per second multiplied by t .

Velocity and acceleration can be obtained from the position function by differentiation:

$$\vec{v}(t) = \frac{d\vec{r}(t)}{dt} = -A\omega \sin \omega t \hat{i} + A\omega \cos \omega t \hat{j}. \quad 4.29$$

It can be shown from Figure 4.20 that the velocity vector is tangential to the circle at the location of the particle, with magnitude $A\omega$. Similarly, the acceleration vector is found by differentiating the velocity:

$$\vec{\mathbf{a}}(t) = \frac{d\vec{\mathbf{v}}(t)}{dt} = -A\omega^2 \cos \omega t \hat{\mathbf{i}} - A\omega^2 \sin \omega t \hat{\mathbf{j}}. \quad 4.30$$

From this equation we see that the acceleration vector has magnitude $A\omega^2$ and is directed opposite the position vector, toward the origin, because $\vec{\mathbf{a}}(t) = -\omega^2 \vec{\mathbf{r}}(t)$.



EXAMPLE 4.11

Circular Motion of a Proton

A proton has speed 5×10^6 m/s and is moving in a circle in the xy plane of radius $r = 0.175$ m. What is its position in the xy plane at time $t = 2.0 \times 10^{-7}$ s = 200 ns? At $t = 0$, the position of the proton is 0.175 m $\hat{\mathbf{i}}$ and it circles counterclockwise. Sketch the trajectory.

Solution

According to [Equation 3.5](#),

$$\text{Average speed} = \frac{\text{Total distance}}{\text{Elapsed time}}.$$

Since the period T is the time it takes an object to go once around a circle, and the distance around a circle is $2\pi r$, we have:

$$v = \frac{2\pi r}{T}.$$

From the given data, the proton has period and angular frequency:

$$T = \frac{2\pi r}{v} = \frac{2\pi(0.175 \text{ m})}{5.0 \times 10^6 \text{ m/s}} = 2.20 \times 10^{-7} \text{ s}$$

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{2.20 \times 10^{-7} \text{ s}} = 2.856 \times 10^7 \text{ rad/s}.$$

The position of the particle at $t = 2.0 \times 10^{-7}$ s with $A = 0.175$ m is

$$\begin{aligned} \vec{\mathbf{r}}(2.0 \times 10^{-7} \text{ s}) &= A \cos \omega(2.0 \times 10^{-7} \text{ s}) \hat{\mathbf{i}} + A \sin \omega(2.0 \times 10^{-7} \text{ s}) \hat{\mathbf{j}} \text{ m} \\ &= 0.175 \cos[(2.856 \times 10^7 \text{ rad/s})(2.0 \times 10^{-7} \text{ s})] \hat{\mathbf{i}} \\ &\quad + 0.175 \sin[(2.856 \times 10^7 \text{ rad/s})(2.0 \times 10^{-7} \text{ s})] \hat{\mathbf{j}} \text{ m} \\ &= 0.175 \cos(5.712 \text{ rad}) \hat{\mathbf{i}} + 0.175 \sin(5.712 \text{ rad}) \hat{\mathbf{j}} = 0.147 \hat{\mathbf{i}} - 0.095 \hat{\mathbf{j}} \text{ m}. \end{aligned}$$

From this result we see that the proton is located slightly below the x -axis. This is shown in [Figure 4.21](#).

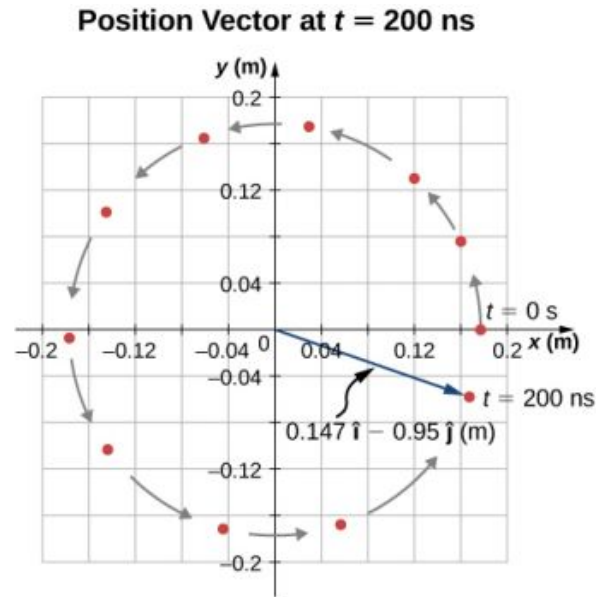


FIGURE 4.21 Position vector of the proton at $t = 2.0 \times 10^{-7} \text{ s} = 200 \text{ ns}$. The trajectory of the proton is shown. The angle through which the proton travels along the circle is 5.712 rad, which is a little less than one complete revolution.

Significance

We picked the initial position of the particle to be on the x-axis. This was completely arbitrary. If a different starting position were given, we would have a different final position at $t = 200 \text{ ns}$.

Nonuniform Circular Motion

Circular motion does not have to be at a constant speed. A particle can travel in a circle and speed up or slow down, showing an acceleration in the direction of the motion.

In uniform circular motion, the particle executing circular motion has a constant speed and the circle is at a fixed radius. If the speed of the particle is changing as well, then we introduce an additional acceleration in the direction tangential to the circle. Such accelerations occur at a point on a top that is changing its spin rate, or any accelerating rotor. In [Displacement and Velocity Vectors](#) we showed that centripetal acceleration is the time rate of change of the direction of the velocity vector. If the speed of the particle is changing, then it has a **tangential acceleration** that is the time rate of change of the magnitude of the velocity:

$$a_T = \frac{d|\vec{v}|}{dt}. \quad 4.31$$

The direction of tangential acceleration is tangent to the circle whereas the direction of centripetal acceleration is radially inward toward the center of the circle. Thus, a particle in circular motion with a tangential acceleration has a **total acceleration** that is the vector sum of the centripetal and tangential accelerations:

$$\vec{a} = \vec{a}_c + \vec{a}_T. \quad 4.32$$

The acceleration vectors are shown in [Figure 4.22](#). Note that the two acceleration vectors $\vec{a}_c$ and $\vec{a}_T$ are perpendicular to each other, with $\vec{a}_c$ in the radial direction and $\vec{a}_T$ in the tangential direction. The total acceleration $\vec{a}$ points at an angle between $\vec{a}_c$ and $\vec{a}_T$.

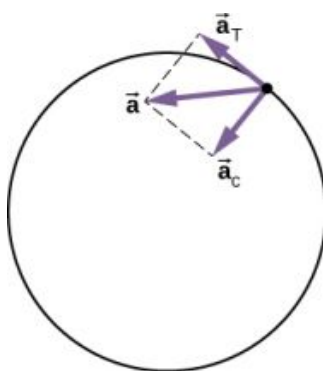


FIGURE 4.22 The centripetal acceleration points toward the center of the circle. The tangential acceleration is tangential to the circle at the particle's position. The total acceleration is the vector sum of the tangential and centripetal accelerations, which are perpendicular.



EXAMPLE 4.12

Total Acceleration during Circular Motion

A particle moves in a circle of radius $r = 2.0$ m. During the time interval from $t = 1.5$ s to $t = 4.0$ s its speed varies with time according to

$$v(t) = c_1 - \frac{c_2}{t^2}, \quad c_1 = 4.0 \text{ m/s}, \quad c_2 = 6.0 \text{ m} \cdot \text{s}.$$

What is the total acceleration of the particle at $t = 2.0$ s?

Strategy

We are given the speed of the particle and the radius of the circle, so we can calculate centripetal acceleration easily. The direction of the centripetal acceleration is toward the center of the circle. We find the magnitude of the tangential acceleration by taking the derivative with respect to time of $|v(t)|$ using [Equation 4.31](#) and evaluating it at $t = 2.0$ s. We use this and the magnitude of the centripetal acceleration to find the total acceleration.

Solution

Centripetal acceleration is

$$v(2.0\text{s}) = \left(4.0 - \frac{6.0}{(2.0)^2} \right) \text{ m/s} = 2.5 \text{ m/s}$$

$$a_c = \frac{v^2}{r} = \frac{(2.5 \text{ m/s})^2}{2.0 \text{ m}} = 3.1 \text{ m/s}^2$$

directed toward the center of the circle. Tangential acceleration is

$$a_T = \left| \frac{d\vec{v}}{dt} \right| = \frac{2c_2}{t^3} = \frac{12.0}{(2.0)^3} \text{ m/s}^2 = 1.5 \text{ m/s}^2.$$

Total acceleration is

$$|\vec{a}| = \sqrt{3.1^2 + 1.5^2} \text{ m/s}^2 = 3.44 \text{ m/s}^2$$

and $\theta = \tan^{-1} \frac{3.1}{1.5} = 64^\circ$ from the tangent to the circle. See [Figure 4.23](#).

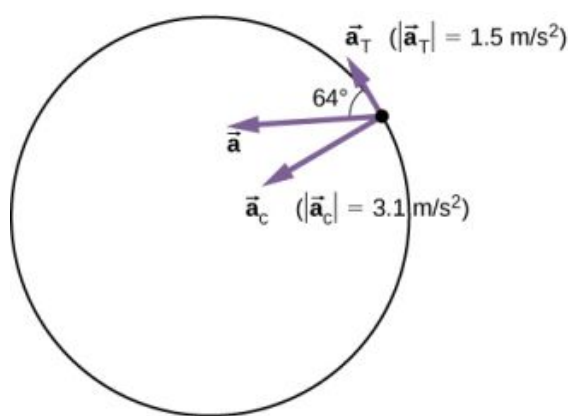


FIGURE 4.23 The tangential and centripetal acceleration vectors. The net acceleration $\vec{a}$ is the vector sum of the two accelerations.

Significance

The directions of centripetal and tangential accelerations can be described more conveniently in terms of a polar coordinate system, with unit vectors in the radial and tangential directions. This coordinate system, which is used for motion along curved paths, is discussed in detail later in the book.

4.5 Relative Motion in One and Two Dimensions

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Explain the concept of reference frames.
- Write the position and velocity vector equations for relative motion.
- Draw the position and velocity vectors for relative motion.
- Analyze one-dimensional and two-dimensional relative motion problems using the position and velocity vector equations.

Motion does not happen in isolation. If you're riding in a train moving at 10 m/s east, this velocity is measured relative to the ground on which you're traveling. However, if another train passes you at 15 m/s east, your velocity relative to this other train is different from your velocity relative to the ground. Your velocity relative to the other train is 5 m/s west. To explore this idea further, we first need to establish some terminology.

Reference Frames

To discuss relative motion in one or more dimensions, we first introduce the concept of **reference frames**. When we say an object has a certain velocity, we must state it has a velocity with respect to a given reference frame. In most examples we have examined so far, this reference frame has been Earth. If you say a person is sitting in a train moving at 10 m/s east, then you imply the person on the train is moving relative to the surface of Earth at this velocity, and Earth is the reference frame. We can expand our view of the motion of the person on the train and say Earth is spinning in its orbit around the Sun, in which case the motion becomes more complicated. In this case, the solar system is the reference frame. In summary, all discussion of relative motion must define the reference frames involved. We now develop a method to refer to reference frames in relative motion. This method makes an approximation that breaks down near the speed of light, where the more accurate methods of special relativity are needed, but is extremely accurate for everyday speeds ([Relativity \(http://openstax.org/books/university-physics-volume-3/pages/5-introduction\)](http://openstax.org/books/university-physics-volume-3/pages/5-introduction)).

Relative Motion in One Dimension

We introduce relative motion in one dimension first, because the velocity vectors simplify to having only two possible directions. Take the example of the person sitting in a train moving east. If we choose east as the positive direction and Earth as the reference frame, then we can write the velocity of the train with respect to the Earth as $\vec{v}_{TE} = 10 \text{ m/s } \hat{i}$ east, where the subscripts TE refer to train and Earth. Let's now say the person gets up out of her seat and walks toward the back of the train at 2 m/s. This tells us she has a velocity relative to the reference frame

of the train. Since the person is walking west, in the negative direction, we write her velocity with respect to the train as $\vec{v}_{PT} = -2 \text{ m/s } \hat{i}$. We can add the two velocity vectors to find the velocity of the person with respect to Earth. This **relative velocity** is written as

$$\vec{v}_{PE} = \vec{v}_{PT} + \vec{v}_{TE}. \quad 4.33$$

Note the ordering of the subscripts for the various reference frames in [Equation 4.33](#). The subscripts for the coupling reference frame, which is the train, appear consecutively in the right-hand side of the equation. [Figure 4.24](#) shows the correct order of subscripts when forming the vector equation.

$$\vec{v}_{PE} = \vec{v}_{PT} + \vec{v}_{TE}$$


FIGURE 4.24 When constructing the vector equation, the subscripts for the coupling reference frame appear consecutively on the inside. The subscripts on the left-hand side of the equation are the same as the two outside subscripts on the right-hand side of the equation.

Adding the vectors, we find $\vec{v}_{PE} = 8 \text{ m/s } \hat{i}$, so the person is moving 8 m/s east with respect to Earth. Graphically, this is shown in [Figure 4.25](#).



FIGURE 4.25 Velocity vectors of the train with respect to Earth, person with respect to the train, and person with respect to Earth.

Relative Velocity in Two Dimensions

We can now apply these concepts to describing motion in two dimensions. Consider a particle P and reference frames S and S' , as shown in [Figure 4.26](#). The position of the origin of S' as measured in S is $\vec{r}_{S'S}$, the position of P as measured in S' is $\vec{r}_{PS'}$, and the position of P as measured in S is $\vec{r}_{PS}$.

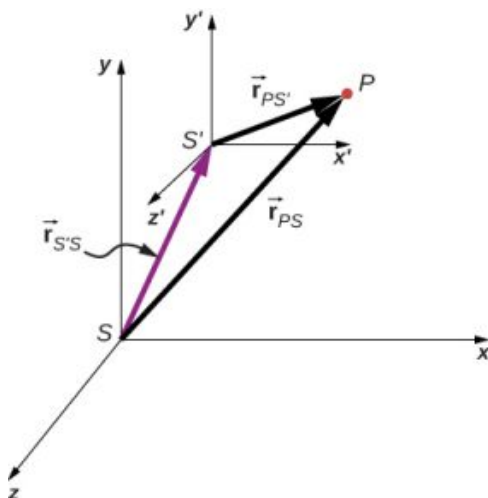


FIGURE 4.26 The positions of particle P relative to frames S and S' are $\vec{r}_{PS}$ and $\vec{r}_{PS'}$, respectively.

From [Figure 4.26](#) we see that

$$\vec{r}_{PS} = \vec{r}_{PS'} + \vec{r}_{S'S}. \quad 4.34$$

The relative velocities are the time derivatives of the position vectors. Therefore,

$$\vec{v}_{PS} = \vec{v}_{PS'} + \vec{v}_{S'S}. \quad 4.35$$

The velocity of a particle relative to S is equal to its velocity relative to S' plus the velocity of S' relative to S .

We can extend [Equation 4.35](#) to any number of reference frames. For particle P with velocities $\vec{v}_{PA}$, $\vec{v}_{PB}$, and $\vec{v}_{PC}$ in frames A , B , and C ,

$$\vec{v}_{PC} = \vec{v}_{PA} + \vec{v}_{AB} + \vec{v}_{BC}. \quad 4.36$$

We can also see how the accelerations are related as observed in two reference frames by differentiating [Equation 4.35](#):

$$\vec{a}_{PS} = \vec{a}_{PS'} + \vec{a}_{S'S}. \quad 4.37$$

We see that if the velocity of S' relative to S is a constant, then $\vec{a}_{S'S} = 0$ and

$$\vec{a}_{PS} = \vec{a}_{PS'}. \quad 4.38$$

This says the acceleration of a particle is the same as measured by two observers moving at a constant velocity relative to each other.

EXAMPLE 4.13

Motion of a Car Relative to a Truck

A truck is traveling south at a speed of 70 km/h toward an intersection. A car is traveling east toward the intersection at a speed of 80 km/h ([Figure 4.27](#)). What is the velocity of the car relative to the truck?

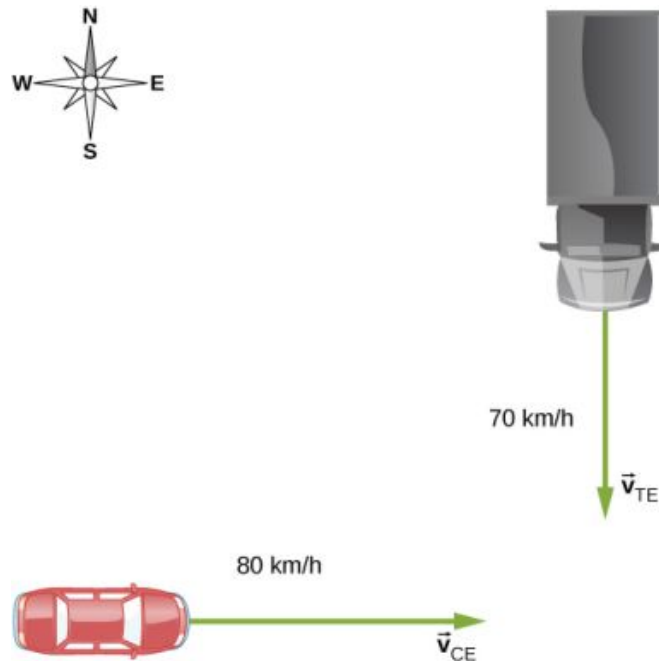


FIGURE 4.27 A car travels east toward an intersection while a truck travels south toward the same intersection.

Strategy

First, we must establish the reference frame common to both vehicles, which is Earth. Then, we write the velocities of each with respect to the reference frame of Earth, which enables us to form a vector equation that links the car, the truck, and Earth to solve for the velocity of the car with respect to the truck.

Solution

The velocity of the car with respect to Earth is $\vec{v}_{CE} = 80 \text{ km/h } \hat{i}$. The velocity of the truck with respect to Earth is $\vec{v}_{TE} = -70 \text{ km/h } \hat{j}$. Using the velocity addition rule, the relative motion equation we are seeking is

$$\vec{v}_{CT} = \vec{v}_{CE} + \vec{v}_{ET}.$$

Here, $\vec{v}_{CT}$ is the velocity of the car with respect to the truck, and Earth is the connecting reference frame. Since we have the velocity of the truck with respect to Earth, the negative of this vector is the velocity of Earth with respect to the truck: $\vec{v}_{ET} = -\vec{v}_{TE}$. The vector diagram of this equation is shown in [Figure 4.28](#).

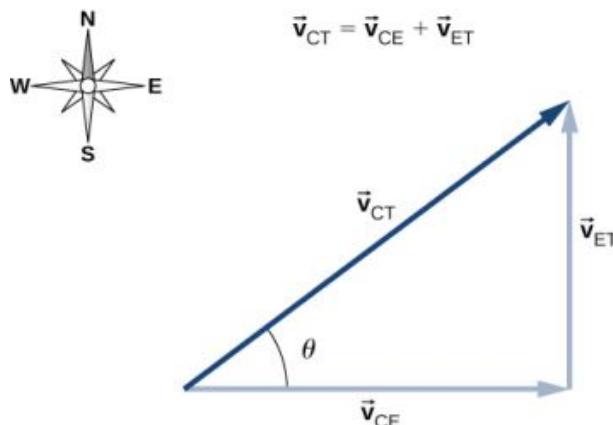


FIGURE 4.28 Vector diagram of the vector equation $\vec{v}_{CT} = \vec{v}_{CE} + \vec{v}_{ET}$.

We can now solve for the velocity of the car with respect to the truck:

$$|\vec{v}_{CT}| = \sqrt{(80.0 \text{ km/h})^2 + (70.0 \text{ km/h})^2} = 106. \text{ km/h}$$

and

$$\theta = \tan^{-1} \left(\frac{70.0}{80.0} \right) = 41.2^\circ \text{ north of east.}$$

Significance

Drawing a vector diagram showing the velocity vectors can help in understanding the relative velocity of the two objects.

✓ CHECK YOUR UNDERSTANDING 4.6

A boat heads due north, moving at 4.5 m/s relative to the water of a river that is running east at 3.0 m/s. What is the velocity of the boat with respect to Earth?

EXAMPLE 4.14

Flying a Plane in a Wind

A pilot must fly a plane due north to reach their destination. The plane can fly at 300 km/h in still air. A wind is blowing out of the northeast at 90 km/h. (a) In what direction must the pilot head the plane to fly due north? (b) What is the speed of the plane relative to the ground?

Strategy

The pilot must point the plane somewhat east of north to compensate for the wind velocity. We need to construct a vector equation that contains the velocity of the plane with respect to the ground, the velocity of the plane with respect to the air, and the velocity of the air with respect to the ground. Since these last two quantities are known, we can solve for the velocity of the plane with respect to the ground. We can graph the vectors and use this diagram to evaluate the magnitude of the plane's velocity with respect to the ground. The diagram will also tell us the angle the plane's velocity makes with north with respect to the air, which is the direction the pilot must head the plane.

Solution

The vector equation is $\vec{v}_{PG} = \vec{v}_{PA} + \vec{v}_{AG}$, where P = plane, A = air, and G = ground. From the geometry in [Figure 4.29](#), we can solve easily for the magnitude of the velocity of the plane with respect to the ground and the angle of the plane's heading, θ .

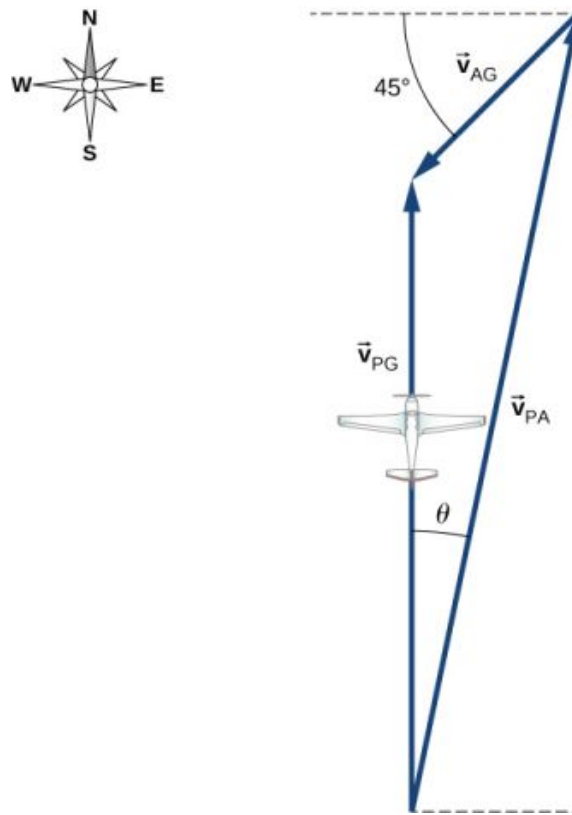


FIGURE 4.29 Vector diagram for [Equation 4.34](#) showing the vectors $\vec{v}_{PA}$, $\vec{v}_{AG}$, and $\vec{v}_{PG}$.

Known quantities:

$$|\vec{v}_{PA}| = 300 \text{ km/h}$$

$$|\vec{v}_{AG}| = 90 \text{ km/h}$$

(a) In the x direction we have the following:

- $|\vec{v}_{PA}| \sin \theta - |\vec{v}_{AG}| \sin 45^\circ = 0$
- $\theta = \sin^{-1} \left(90 \text{ km/h} \right) \left(\frac{\sin 45^\circ}{300 \text{ km/h}} \right)$
- $\theta = 12^\circ$ east of north

(b) In the y direction we have the following:

- $|\vec{v}_{PG}| = |\vec{v}_{PA}| \cos \theta - |\vec{v}_{AG}| \cos 45^\circ$
- $|\vec{v}_{PG}| = (300 \text{ km/h}) (\cos 12^\circ) - (90 \text{ km/h}) (\cos 45^\circ)$
- $|\vec{v}_{PG}| = 230 \text{ km/h}$

Chapter Review

Key Terms

acceleration vector instantaneous acceleration found by taking the derivative of the velocity function with respect to time in unit vector notation

angular frequency ω , rate of change of an angle with which an object that is moving on a circular path

centripetal acceleration component of acceleration of an object moving in a circle that is directed radially inward toward the center of the circle

displacement vector vector from the initial position to a final position on a trajectory of a particle

position vector vector from the origin of a chosen coordinate system to the position of a particle in two- or three-dimensional space

projectile motion motion of an object subject only to the acceleration of gravity

range maximum horizontal distance a projectile travels

reference frame coordinate system in which the position, velocity, and acceleration of an object at rest or moving is measured

relative velocity velocity of an object as observed from a particular reference frame, or the velocity of one reference frame with respect to another reference frame

tangential acceleration magnitude of which is the time rate of change of speed. Its direction is tangent to the circle.

time of flight elapsed time a projectile is in the air

total acceleration vector sum of centripetal and tangential accelerations

trajectory path of a projectile through the air

velocity vector vector that gives the instantaneous speed and direction of a particle; tangent to the trajectory

Key Equations

Position vector

$$\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$$

Displacement vector

$$\Delta\vec{r} = \vec{r}(t_2) - \vec{r}(t_1)$$

Velocity vector

$$\vec{v}(t) = \lim_{\Delta t \rightarrow 0} \frac{\vec{r}(t+\Delta t) - \vec{r}(t)}{\Delta t} = \frac{d\vec{r}}{dt}$$

Velocity in terms of components

$$\vec{v}(t) = v_x(t)\hat{i} + v_y(t)\hat{j} + v_z(t)\hat{k}$$

Velocity components

$$v_x(t) = \frac{dx(t)}{dt} \quad v_y(t) = \frac{dy(t)}{dt} \quad v_z(t) = \frac{dz(t)}{dt}$$

Average velocity

$$\vec{v}_{\text{avg}} = \frac{\vec{r}(t_2) - \vec{r}(t_1)}{t_2 - t_1}$$

Instantaneous acceleration

$$\vec{a}(t) = \lim_{\Delta t \rightarrow 0} \frac{\vec{v}(t+\Delta t) - \vec{v}(t)}{\Delta t} = \frac{d\vec{v}(t)}{dt}$$

Instantaneous acceleration, component form

$$\vec{a}(t) = \frac{dv_x(t)}{dt}\hat{i} + \frac{dv_y(t)}{dt}\hat{j} + \frac{dv_z(t)}{dt}\hat{k}$$

Instantaneous acceleration as second derivatives of position

$$\vec{a}(t) = \frac{d^2x(t)}{dt^2}\hat{i} + \frac{d^2y(t)}{dt^2}\hat{j} + \frac{d^2z(t)}{dt^2}\hat{k}$$

Time of flight

$$T_{\text{tof}} = \frac{2(v_0 \sin \theta_0)}{g}$$

Trajectory

$$y = (\tan \theta_0)x - \left[\frac{g}{2(v_0 \cos \theta_0)^2} \right] x^2$$

Range

$$R = \frac{v_0^2 \sin 2\theta_0}{g}$$

Centripetal acceleration

$$a_C = \frac{v^2}{r}$$

Position vector, uniform circular motion

$$\vec{r}(t) = A \cos \omega t \hat{i} + A \sin \omega t \hat{j}$$

Velocity vector, uniform circular motion

$$\vec{v}(t) = \frac{d\vec{r}(t)}{dt} = -A\omega \sin \omega t \hat{i} + A\omega \cos \omega t \hat{j}$$

Acceleration vector, uniform circular motion

$$\vec{a}(t) = \frac{d\vec{v}(t)}{dt} = -A\omega^2 \cos \omega t \hat{i} - A\omega^2 \sin \omega t \hat{j}$$

Tangential acceleration

$$a_T = \frac{d|\vec{v}|}{dt}$$

Total acceleration

$$\vec{a} = \vec{a}_C + \vec{a}_T$$

Position vector in frame

S is the position vector in frame S' plus the vector from the origin of S to the origin of S'

$$\vec{r}_{PS} = \vec{r}_{PS'} + \vec{r}_{S'S}$$

Relative velocity equation connecting two reference frames

$$\vec{v}_{PS} = \vec{v}_{PS'} + \vec{v}_{S'S}$$

Relative velocity equation connecting more than two reference frames

$$\vec{v}_{PC} = \vec{v}_{PA} + \vec{v}_{AB} + \vec{v}_{BC}$$

Relative acceleration equation

$$\vec{a}_{PS} = \vec{a}_{PS'} + \vec{a}_{S'S}$$

Summary

4.1 Displacement and Velocity Vectors

- The position function $\vec{r}(t)$ gives the position as a function of time of a particle moving in two or three dimensions. Graphically, it is a vector from the origin of a chosen coordinate system to the point where the particle is located at a specific time.
- The displacement vector $\Delta\vec{r}$ gives the shortest distance between any two points on the trajectory of a particle in two or three dimensions.
- Instantaneous velocity gives the speed and direction of a particle at a specific time on its trajectory in two or three dimensions, and is a vector in two and three dimensions.
- The velocity vector is tangent to the trajectory of the particle.
- Displacement $\vec{r}(t)$ can be written as a vector sum of the one-dimensional displacements $\vec{x}(t)$, $\vec{y}(t)$, $\vec{z}(t)$ along the x , y , and z directions.
- Velocity $\vec{v}(t)$ can be written as a vector sum of the one-dimensional velocities $v_x(t)$, $v_y(t)$, $v_z(t)$ along the x , y , and z directions.
- Motion in any given direction is independent of motion in a perpendicular direction.

4.2 Acceleration Vector

- In two and three dimensions, the acceleration vector can have an arbitrary direction and does not necessarily point along a given component of the velocity.
- The instantaneous acceleration is produced by a change in velocity taken over a very short (infinitesimal) time period. Instantaneous acceleration is a vector in two or three dimensions. It is found by taking the derivative of the velocity function with respect to time.
- In three dimensions, acceleration $\vec{a}(t)$ can be written as a vector sum of the one-dimensional accelerations $a_x(t)$, $a_y(t)$, and $a_z(t)$ along the x -, y -, and z -axes.

- The kinematic equations for constant acceleration can be written as the vector sum of the constant acceleration equations in the x , y , and z directions.

4.3 Projectile Motion

- Projectile motion is the motion of an object subject only to the acceleration of gravity, where the acceleration is constant, as near the surface of Earth.
- To solve projectile motion problems, we analyze the motion of the projectile in the horizontal and vertical directions using the one-dimensional kinematic equations for x and y .
- The time of flight of a projectile launched with initial vertical velocity v_{0y} on an even surface is given by

$$T_{tof} = \frac{2(v_0 \sin \theta)}{g}.$$

This equation is valid only when the projectile lands at the same elevation from which it was launched.

- The maximum horizontal distance traveled by a projectile is called the range. Again, the equation for range is valid only when the projectile lands at the same elevation from which it was launched.

4.4 Uniform and Nonuniform Circular Motion

- Uniform circular motion is motion in a circle at constant speed.
- Centripetal acceleration $\vec{a}_C$ is the acceleration a particle must have to follow a circular path. Centripetal acceleration always points toward the center of rotation and has magnitude $a_C = v^2/r$.
- Nonuniform circular motion occurs when there is tangential acceleration of an object executing circular motion such that the speed of the object is changing. This acceleration is called tangential acceleration $\vec{a}_T$. The magnitude of tangential acceleration is the time rate of change of the

magnitude of the velocity. The tangential acceleration vector is tangential to the circle, whereas the centripetal acceleration vector points radially inward toward the center of the circle. The total acceleration is the vector sum of tangential and centripetal accelerations.

- An object executing uniform circular motion can be described with equations of motion. The position vector of the object is $\vec{r}(t) = A \cos \omega t \hat{i} + A \sin \omega t \hat{j}$, where A is the magnitude $|\vec{r}(t)|$, which is also the radius of the circle, and ω is the angular frequency.

4.5 Relative Motion in One and Two Dimensions

- When analyzing motion of an object, the

Conceptual Questions

4.1 Displacement and Velocity Vectors

1. What form does the trajectory of a particle have if the distance from any point A to point B is equal to the magnitude of the displacement from A to B ?
2. Give an example of a trajectory in two or three dimensions caused by independent perpendicular motions.
3. If the instantaneous velocity is zero, what can be said about the slope of the position function?

4.2 Acceleration Vector

4. If the position function of a particle is a linear function of time, what can be said about its acceleration?
5. If an object has a constant x -component of the velocity and suddenly experiences an acceleration in the y direction, does the x -component of its velocity change?
6. If an object has a constant x -component of velocity and suddenly experiences an acceleration at an angle of 70° in the x direction, does the x -component of velocity change?

4.3 Projectile Motion

7. Answer the following questions for projectile motion on level ground assuming negligible air resistance, with the initial angle being neither 0° nor 90° : (a) Is the velocity ever zero? (b) When is the velocity a minimum? A maximum? (c) Can the velocity ever be the same as the initial velocity at a time other than at $t = 0$? (d) Can the speed ever be the same as the initial speed at a

reference frame in terms of position, velocity, and acceleration needs to be specified.

- Relative velocity is the velocity of an object as observed from a particular reference frame, and it varies with the choice of reference frame.
- If S and S' are two reference frames moving relative to each other at a constant velocity, then the velocity of an object relative to S is equal to its velocity relative to S' plus the velocity of S' relative to S .
- If two reference frames are moving relative to each other at a constant velocity, then the accelerations of an object as observed in both reference frames are equal.

time other than at $t = 0$?

8. Answer the following questions for projectile motion on level ground assuming negligible air resistance, with the initial angle being neither 0° nor 90° : (a) Is the acceleration ever zero? (b) Is the vector $\vec{v}$ ever parallel or antiparallel to the vector $\vec{a}$? (c) Is the vector $\vec{v}$ ever perpendicular to the vector $\vec{a}$? If so, where is this located?
9. A dime is placed at the edge of a table so it hangs over slightly. A quarter is slid horizontally on the table surface perpendicular to the edge and hits the dime head on. Which coin hits the ground first?

4.4 Uniform and Nonuniform Circular Motion

10. Can centripetal acceleration change the speed of a particle undergoing circular motion?
11. Can tangential acceleration change the speed of a particle undergoing circular motion?

4.5 Relative Motion in One and Two Dimensions

12. What frame or frames of reference do you use instinctively when driving a car? When flying in a commercial jet?
13. A basketball player dribbling down the court usually keeps his eyes fixed on the players around him. He is moving fast. Why doesn't he need to keep his eyes on the ball?
14. If someone is riding in the back of a pickup truck and throws a softball straight backward, is it possible for the ball to fall straight down as viewed by a person standing at the side of the road? Under what condition would this occur?

How would the motion of the ball appear to the person who threw it?

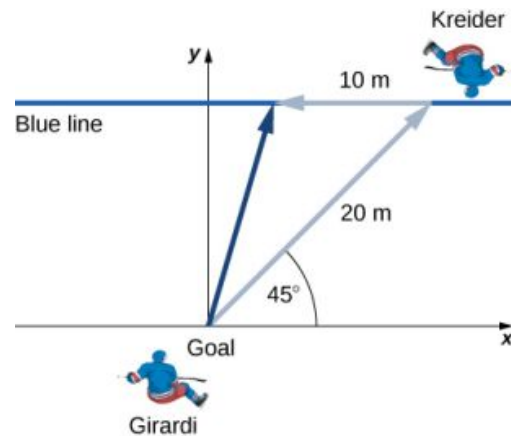
15. The hat of a jogger running at constant velocity falls off the back of his head. Draw a sketch showing the path of the hat in the jogger's frame of reference. Draw its path as viewed by a stationary observer. Neglect air resistance.

Problems

4.1 Displacement and Velocity Vectors

17. The coordinates of a particle in a rectangular coordinate system are $(1.0, -4.0, 6.0)$. What is the position vector of the particle?
18. The position of a particle changes from $\vec{r}_1 = (2.0\hat{i} + 3.0\hat{j})\text{ cm}$ to $\vec{r}_2 = (-4.0\hat{i} + 3.0\hat{j})\text{ cm}$. What is the particle's displacement?
19. The 18th hole at Pebble Beach Golf Course is a dogleg to the left of length 496.0 m. The fairway off the tee is taken to be the x direction. A golfer hits his tee shot a distance of 300.0 m, corresponding to a displacement $\Delta\vec{r}_1 = 300.0\text{ m}\hat{i}$, and hits his second shot 189.0 m with a displacement $\Delta\vec{r}_2 = 172.0\text{ m}\hat{i} + 80.3\text{ m}\hat{j}$. What is the final displacement of the golf ball from where it started?
20. A bird flies straight northeast a distance of 95.0 km for 3.0 h. With the x -axis due east and the y -axis due north, what is the displacement in unit vector notation for the bird? What is the average velocity for the trip?
21. A cyclist rides 5.0 km due east, then 10.0 km 20° west of north. From this point she rides 8.0 km due west. What is the final displacement from where the cyclist started?
22. New York Rangers defenseman Daniel Girardi stands at the goal and passes a hockey puck 20 m and 45° from straight down the ice to left wing Chris Kreider waiting at the blue line. Kreider waits for Girardi to reach the blue line and passes the puck directly across the ice to him 10 m away. What is the final displacement of the puck? See the following figure.

16. A clod of dirt falls from the bed of a moving truck. It strikes the ground directly below the end of the truck. (a) What is the direction of its velocity relative to the truck just before it hits? (b) Is this the same as the direction of its velocity relative to ground just before it hits? Explain your answers.

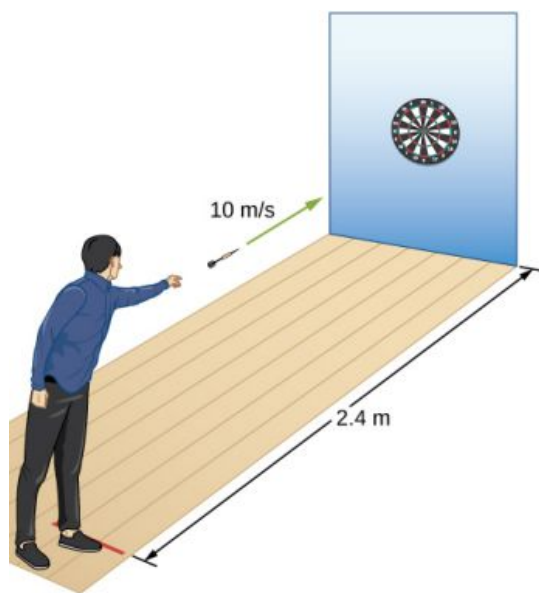


23. The position of a particle is $\vec{r}(t) = 4.0t^2\hat{i} - 3.0\hat{j} + 2.0t^3\hat{k}\text{ m}$. (a) What is the velocity of the particle at 0 s and at 1.0 s? (b) What is the average velocity between 0 s and 1.0 s?
24. Clay Matthews, a linebacker for the Green Bay Packers, can reach a speed of 10.0 m/s. At the start of a play, Matthews runs downfield at 45° with respect to the 50-yard line and covers 8.0 m in 1 s. He then runs straight down the field at 90° with respect to the 50-yard line for 12 m, with an elapsed time of 1.2 s. (a) What is Matthews' final displacement from the start of the play? (b) What is his average velocity?
25. The F-35B Lighting II is a short-takeoff and vertical landing fighter jet. If it does a vertical takeoff to 20.00-m height above the ground and then follows a flight path angled at 30° with respect to the ground for 20.00 km, what is the final displacement?

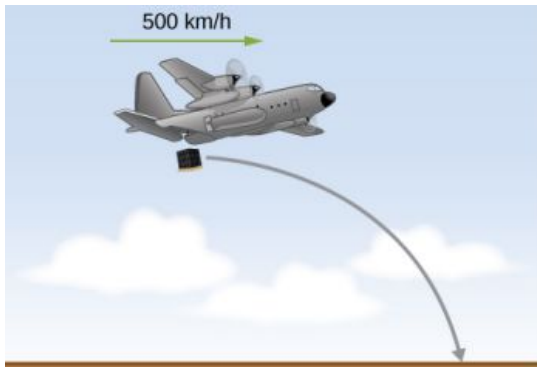
4.2 Acceleration Vector

26. The position of a particle is $\vec{r}(t) = (3.0t^2\hat{i} + 5.0\hat{j} - 6.0t\hat{k})\text{ m}$. (a) Determine its velocity and acceleration as functions of time. (b) What are its velocity and acceleration at time $t = 0$?
27. A particle's acceleration is $(4.0\hat{i} + 3.0\hat{j})\text{ m/s}^2$. At

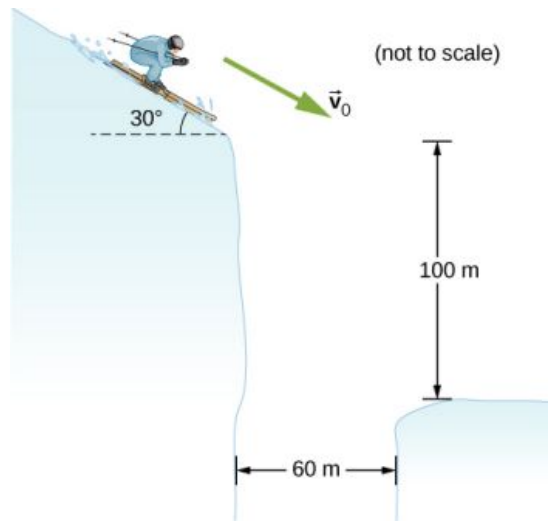
- $t = 0$, its position and velocity are zero. (a) What are the particle's position and velocity as functions of time? (b) Find the equation of the path of the particle. Draw the x - and y -axes and sketch the trajectory of the particle.
28. A boat leaves the dock at $t = 0$ and heads out into a lake with an acceleration of $2.0 \text{ m/s}^2 \hat{i}$. A strong wind is pushing the boat, giving it an additional velocity of $2.0 \text{ m/s} \hat{i} + 1.0 \text{ m/s} \hat{j}$. (a) What is the velocity of the boat at $t = 10 \text{ s}$? (b) What is the position of the boat at $t = 10 \text{ s}$? Draw a sketch of the boat's trajectory and position at $t = 10 \text{ s}$, showing the x - and y -axes.
29. The position of a particle for $t > 0$ is given by $\vec{r}(t) = (3.0t^2 \hat{i} - 7.0t^3 \hat{j} - 5.0t^{-2} \hat{k}) \text{ m}$. (a) What is the velocity as a function of time? (b) What is the acceleration as a function of time? (c) What is the particle's velocity at $t = 2.0 \text{ s}$? (d) What is its speed at $t = 1.0 \text{ s}$ and $t = 3.0 \text{ s}$? (e) What is the average velocity between $t = 1.0 \text{ s}$ and $t = 2.0 \text{ s}$?
30. The acceleration of a particle is a constant. At $t = 0$ the velocity of the particle is $(10 \hat{i} + 20 \hat{j}) \text{ m/s}$. At $t = 4 \text{ s}$ the velocity is $10 \hat{j} \text{ m/s}$. (a) What is the particle's acceleration? (b) How do the position and velocity vary with time? Assume the particle is initially at the origin.
31. A particle has a position function $\vec{r}(t) = \cos(1.0t) \hat{i} + \sin(1.0t) \hat{j} + t \hat{k}$, where the arguments of the cosine and sine functions are in radians. (a) What is the velocity vector? (b) What is the acceleration vector?
32. A Lockheed Martin F-35 II Lighting jet takes off from an aircraft carrier with a runway length of 90 m and a takeoff speed 70 m/s at the end of the runway. Jets are catapulted into airspace from the deck of an aircraft carrier with two sources of propulsion: the jet propulsion and the catapult. At the point of leaving the deck of the aircraft carrier, the F-35's acceleration decreases to a constant acceleration of 5.0 m/s^2 at 30° with respect to the horizontal. (a) What is the initial acceleration of the F-35 on the deck of the aircraft carrier to make it airborne? (b) Write the position and velocity of the F-35 in unit vector notation from the point it leaves the deck of the aircraft carrier. (c) At what altitude is the fighter 5.0 s after it leaves the deck of the aircraft carrier? (d) What is its velocity and speed at this time? (e) How far has it traveled horizontally?
- 4.3 Projectile Motion
33. A bullet is shot horizontally from shoulder height (1.5 m) with an initial speed 200 m/s . (a) How much time elapses before the bullet hits the ground? (b) How far does the bullet travel horizontally?
34. A marble rolls off a tabletop 1.0 m high and hits the floor at a point 3.0 m away from the table's edge in the horizontal direction. (a) How long is the marble in the air? (b) What is the speed of the marble when it leaves the table's edge? (c) What is its speed when it hits the floor?
35. A dart is thrown horizontally at a speed of 10 m/s at the bull's-eye of a dartboard 2.4 m away, as in the following figure. (a) How far below the intended target does the dart hit? (b) What does your answer tell you about how proficient dart players throw their darts?



36. An airplane flying horizontally with a speed of 500 km/h at a height of 800 m drops a crate of supplies (see the following figure). If the parachute fails to open, how far in front of the release point does the crate hit the ground?

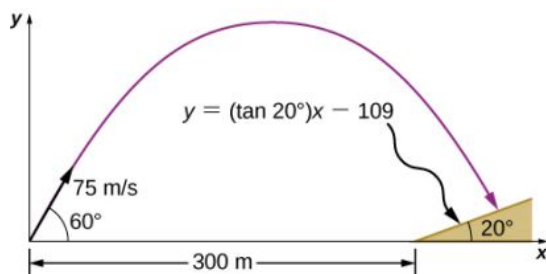


37. Suppose the airplane in the preceding problem fires a projectile horizontally in its direction of motion at a speed of 300 m/s relative to the plane. (a) How far in front of the release point does the projectile hit the ground? (b) What is its speed when it hits the ground?
38. A fastball pitcher can throw a baseball at a speed of 40 m/s (90 mi/h). (a) Assuming the pitcher can release the ball 16.7 m from home plate so the ball is moving horizontally, how long does it take the ball to reach home plate? (b) How far does the ball drop between the pitcher's hand and home plate?
39. A projectile is launched at an angle of 30° and lands 20 s later at the same height as it was launched. (a) What is the initial speed of the projectile? (b) What is the maximum altitude? (c) What is the range? (d) Calculate the displacement from the point of launch to the position on its trajectory at 15 s.
40. A basketball player shoots toward a basket 6.1 m away and 3.0 m above the floor. If the ball is released 1.8 m above the floor at an angle of 60° above the horizontal, what must the initial speed be if it were to go through the basket?
41. At a particular instant, a hot air balloon is 100 m in the air and descending at a constant speed of 2.0 m/s. At this exact instant, a girl throws a ball horizontally, relative to herself, with an initial speed of 20 m/s. When she lands, where will she find the ball? Ignore air resistance.
42. A man on a motorcycle traveling at a uniform speed of 10 m/s throws an empty can straight upward relative to himself with an initial speed of 3.0 m/s. Find the equation of the trajectory as seen by a police officer on the side of the road. Assume the initial position of the can is the point where it is thrown. Ignore air resistance.
43. An athlete can jump a distance of 8.0 m in the broad jump. What is the maximum distance the athlete can jump on the Moon, where the gravitational acceleration is one-sixth that of Earth?
44. The maximum horizontal distance a boy can throw a ball is 50 m. Assume he can throw with the same initial speed at all angles. How high does he throw the ball when he throws it straight upward?
45. A rock is thrown off a cliff at an angle of 53° with respect to the horizontal. The cliff is 100 m high. The initial speed of the rock is 30 m/s. (a) How high above the edge of the cliff does the rock rise? (b) How far has it moved horizontally when it is at maximum altitude? (c) How long after the release does it hit the ground? (d) What is the range of the rock? (e) What are the horizontal and vertical positions of the rock relative to the edge of the cliff at $t = 2.0$ s, $t = 4.0$ s, and $t = 6.0$ s?
46. Trying to escape his pursuers, a secret agent skis off a slope inclined at 30° below the horizontal at 60 km/h. To survive and land on the snow 100 m below, he must clear a gorge 60 m wide. Does he make it? Ignore air resistance.



47. A golfer on a fairway is 70 m away from the green, which sits below the level of the fairway by 20 m. If the golfer hits the ball at an angle of 40° with an initial speed of 20 m/s, how close to the green does she come?
48. A projectile is shot at a hill, the base of which is 300 m away. The projectile is shot at 60° above the horizontal with an initial speed of 75 m/s. The hill can be approximated by a plane sloped at 20° to the

horizontal. Relative to the coordinate system shown in the following figure, the equation of this straight line is $y = (\tan 20^\circ)x - 109$. Where on the hill does the projectile land?



49. An astronaut on Mars kicks a soccer ball at an angle of 45° with an initial velocity of 15 m/s. If the acceleration of gravity on Mars is 3.7 m/s^2 , (a) what is the range of the soccer kick on a flat surface? (b) What would be the range of the same kick on the Moon, where gravity is one-sixth that of Earth?
50. Mike Powell holds the record for the long jump of 8.95 m, established in 1991. If he left the ground at an angle of 15° , what was his initial speed?
51. MIT's robot cheetah can jump over obstacles 46 cm high and has speed of 12.0 km/h. (a) If the robot launches itself at an angle of 60° at this speed, what is its maximum height? (b) What would the launch angle have to be to reach a height of 46 cm?
52. Mt. Asama, Japan, is an active volcano. In 2009, an eruption threw solid volcanic rocks that landed 1 km horizontally from the crater. If the volcanic rocks were launched at an angle of 40° with respect to the horizontal and landed 900 m below the crater, (a) what would be their initial velocity and (b) what is their time of flight?
53. Drew Brees of the New Orleans Saints can throw a football 23.0 m/s (50 mph). If he angles the throw at 10° from the horizontal, what distance does it go if it is to be caught at the same elevation as it was thrown?
54. The Lunar Roving Vehicle used in NASA's late *Apollo* missions reached an unofficial lunar land speed of 5.0 m/s by astronaut Eugene Cernan. If the rover was moving at this speed on a flat lunar surface and hit a small bump that projected it off the surface at an angle of 20° , how long would it be "airborne" on the Moon?
55. A soccer goal is 2.44 m high. A player kicks the

ball at a distance 10 m from the goal at an angle of 25° . The ball hits the crossbar at the top of the goal. What is the initial speed of the soccer ball?

56. Olympus Mons on Mars is the largest volcano in the solar system, at a height of 25 km and with a radius of 312 km. If you are standing on the summit, with what initial velocity would you have to fire a projectile from a cannon horizontally to clear the volcano and land on the surface of Mars? Note that Mars has an acceleration of gravity of 3.7 m/s^2 .
57. In 1999, Robbie Knievel was the first to jump the Grand Canyon on a motorcycle. At a narrow part of the canyon (69.0 m wide) and traveling 35.8 m/s off the takeoff ramp, he reached the other side. What was his launch angle?
58. You throw a baseball at an initial speed of 15.0 m/s at an angle of 30° with respect to the horizontal. What would the ball's initial speed have to be at 30° on a planet that has twice the acceleration of gravity as Earth to achieve the same range? Consider launch and impact on a horizontal surface.
59. Aaron Rodgers throws a football at 20.0 m/s to his wide receiver, who is running straight down the field at 9.4 m/s. If Aaron throws the football when the wide receiver is 10.0 m in front of him, (a) at what angle does Aaron have to launch the ball so the ball will be at the same height as the receiver when the receiver makes it to 20.0 m in front of Aaron? (b) Will the receiver be able to catch the ball?

4.4 Uniform and Nonuniform Circular Motion

60. A flywheel is rotating at 30 rev/s. What is the total angle, in radians, through which a point on the flywheel rotates in 40 s?
61. A particle travels in a circle of radius 10 m at a constant speed of 20 m/s. What is the magnitude of the acceleration?
62. Cam Newton of the Carolina Panthers throws a perfect football spiral at 8.0 rev/s. The radius of a pro football is 8.5 cm at the middle of the short side. What is the centripetal acceleration of the laces on the football?
63. A fairground ride spins its occupants inside a flying saucer-shaped container. If the horizontal circular path the riders follow has an 8.00-m radius, at how many revolutions per minute are

the riders subjected to a centripetal acceleration equal to that of gravity?

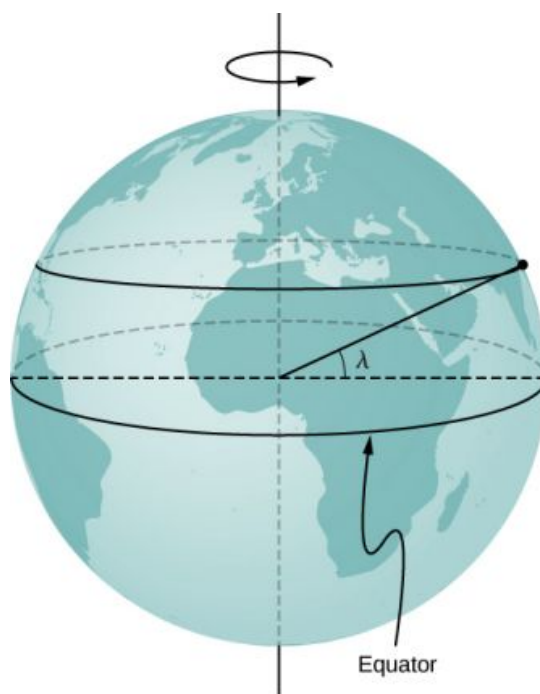
64. A runner taking part in the 200-m dash must run around the end of a track that has a circular arc with a radius of curvature of 30.0 m. The runner starts the race at a constant speed. If she completes the 200-m dash in 23.2 s and runs at constant speed throughout the race, what is her centripetal acceleration as she runs the curved portion of the track?
65. What is the acceleration of Venus toward the Sun, assuming a circular orbit?
66. An experimental jet rocket travels around Earth along its equator just above its surface. At what speed must the jet travel if the magnitude of its acceleration is g ?
67. A fan is rotating at a constant 360.0 rev/min. What is the magnitude of the acceleration of a point on one of its blades 10.0 cm from the axis of rotation?
68. A point located on the second hand of a large clock has a radial acceleration of 0.1 cm/s^2 . How far is the point from the axis of rotation of the second hand?

4.5 Relative Motion in One and Two Dimensions

69. The coordinate axes of the reference frame S' remain parallel to those of S , as S' moves away from S at a constant velocity $\vec{v}_{S'S} = (4.0\hat{i} + 3.0\hat{j} + 5.0\hat{k}) \text{ m/s}$. (a) If at time $t = 0$ the origins coincide, what is the position of the origin O' in the S frame as a function of time? (b) How is particle position for $\vec{r}(t)$ and $\vec{r}'(t)$, as measured in S and S' , respectively, related? (c) What is the relationship between particle velocities $\vec{v}(t)$ and $\vec{v}'(t)$? (d) How are accelerations $\vec{a}(t)$ and $\vec{a}'(t)$ related?
70. The coordinate axes of the reference frame S' remain parallel to those of S , as S' moves away from S at a constant velocity $\vec{v}_{S'S} = (1.0\hat{i} + 2.0\hat{j} + 3.0\hat{k})t \text{ m/s}$. (a) If at time $t = 0$ the origins coincide, what is the position of origin O' in the S frame as a function of time? (b) How is particle position for $\vec{r}(t)$ and $\vec{r}'(t)$, as measured in S and S' , respectively, related? (c) What is the relationship between particle velocities $\vec{v}(t)$ and $\vec{v}'(t)$? (d) How are accelerations $\vec{a}(t)$ and $\vec{a}'(t)$ related?
71. The velocity of a particle in reference frame A is $(2.0\hat{i} + 3.0\hat{j}) \text{ m/s}$. The velocity of reference frame A with respect to reference frame B is $4.0\hat{k} \text{ m/s}$, and the velocity of reference frame B with respect to C is $2.0\hat{j} \text{ m/s}$. What is the velocity of the particle in reference frame C ?
72. Raindrops fall vertically at 4.5 m/s relative to the earth. What does an observer in a car moving at 22.0 m/s in a straight line measure as the velocity of the raindrops?
73. A seagull can fly at a velocity of 9.00 m/s in still air. (a) If it takes the bird 20.0 min to travel 6.00 km straight into an oncoming wind, what is the velocity of the wind? (b) If the bird turns around and flies with the wind, how long will it take the bird to return 6.00 km?
74. A ship sets sail from Rotterdam, heading due north at 7.00 m/s relative to the water. The local ocean current is 1.50 m/s in a direction 40.0° north of east. What is the velocity of the ship relative to Earth?
75. A boat can be rowed at 8.0 km/h in still water. (a) How much time is required to row 1.5 km downstream in a river moving 3.0 km/h relative to the shore? (b) How much time is required for the return trip? (c) In what direction must the boat be aimed to row straight across the river? (d) Suppose the river is 0.8 km wide. What is the velocity of the boat with respect to Earth and how much time is required to get to the opposite shore? (e) Suppose, instead, the boat is aimed straight across the river. How much time is required to get across and how far downstream is the boat when it reaches the opposite shore?
76. A small plane flies at 200 km/h in still air. If the wind blows directly out of the west at 50 km/h, (a) in what direction must the pilot head her plane to move directly north across land and (b) how long does it take her to reach a point 300 km directly north of her starting point?
77. A cyclist traveling southeast along a road at 15 km/h feels a wind blowing from the southwest at 25 km/h. To a stationary observer, what are the speed and direction of the wind?
78. A river is moving east at 4.0 m/s. A boat starts from the dock heading 30° north of west at 7.0 m/s. If the river is 1800 m wide, (a) what is the velocity of the boat with respect to Earth and (b) how long does it take the boat to cross the river?

Additional Problems

- 79.** A Formula One race car is traveling at 89.0 m/s along a straight track enters a turn on the race track with radius of curvature of 200.0 m. What centripetal acceleration must the car have to stay on the track?
- 80.** A particle travels in a circular orbit of radius 10 m. Its speed is changing at a rate of 15.0 m/s^2 at an instant when its speed is 40.0 m/s. What is the magnitude of the acceleration of the particle?
- 81.** The driver of a car moving at 90.0 km/h presses down on the brake as the car enters a circular curve of radius 150.0 m. If the speed of the car is decreasing at a rate of 9.0 km/h each second, what is the magnitude of the acceleration of the car at the instant its speed is 60.0 km/h?
- 82.** A race car entering the curved part of the track at the Daytona 500 drops its speed from 85.0 m/s to 80.0 m/s in 2.0 s. If the radius of the curved part of the track is 316.0 m, calculate the total acceleration of the race car at the beginning and ending of reduction of speed.
- 83.** An elephant is located on Earth's surface at a latitude λ . Calculate the centripetal acceleration of the elephant resulting from the rotation of Earth around its polar axis. Express your answer in terms of λ , the radius R_E of Earth, and time T for one rotation of Earth. Compare your answer with g for $\lambda = 40^\circ$.



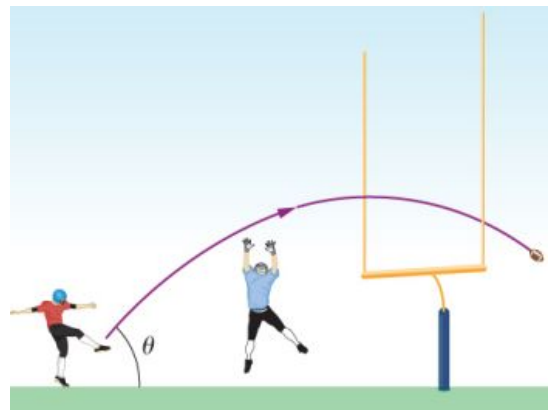
- 84.** A proton in a synchrotron is moving in a circle of radius 1 km and increasing its speed by $v(t) = c_1 + c_2 t^2$, where $c_1 = 2.0 \times 10^5 \text{ m/s}$, $c_2 = 10^5 \text{ m/s}^3$. (a) What is the proton's total acceleration at $t = 5.0 \text{ s}$? (b) At what time does the expression for the velocity become unphysical?
- 85.** A propeller blade at rest starts to rotate from $t = 0 \text{ s}$ to $t = 5.0 \text{ s}$ with a tangential acceleration of the tip of the blade at 3.00 m/s^2 . The tip of the blade is 1.5 m from the axis of rotation. At $t = 5.0 \text{ s}$, what is the total acceleration of the tip of the blade?
- 86.** A particle is executing circular motion with a constant angular frequency of $\omega = 4.00 \text{ rad/s}$. If time $t = 0$ corresponds to the position of the particle being located at $y = 0 \text{ m}$ and $x = 5 \text{ m}$, (a) what is the position of the particle at $t = 10 \text{ s}$? (b) What is its velocity at this time? (c) What is its acceleration?
- 87.** A particle's centripetal acceleration is $a_C = 4.0 \text{ m/s}^2$ at $t = 0 \text{ s}$ where it is on the x -axis and moving counterclockwise in the xy plane. It is executing uniform circular motion about an axis at a distance of 5.0 m. What is its velocity at $t = 10 \text{ s}$?
- 88.** A rod 3.0 m in length is rotating at 2.0 rev/s about an axis at one end. Compare the

- centripetal accelerations at radii of (a) 1.0 m, (b) 2.0 m, and (c) 3.0 m.
89. A particle located initially at $(1.5\hat{j} + 4.0\hat{k})\text{m}$ undergoes a displacement of $(2.5\hat{i} + 3.2\hat{j} - 1.2\hat{k})\text{m}$. What is the final position of the particle?
90. The position of a particle is given by $\vec{r}(t) = (50\text{ m/s})t\hat{i} - (4.9\text{ m/s}^2)t^2\hat{j}$. (a) What are the particle's velocity and acceleration as functions of time? (b) What are the initial conditions to produce the motion?
91. A spaceship is traveling at a constant velocity of $\vec{v}(t) = 250.0\hat{i}\text{m/s}$ when its rockets fire, giving it an acceleration of $\vec{a}(t) = (3.0\hat{i} + 4.0\hat{k})\text{m/s}^2$. What is its velocity 5 s after the rockets fire?
92. A crossbow is aimed horizontally at a target 40 m away. The arrow hits 30 cm below the spot at which it was aimed. What is the initial velocity of the arrow?
93. A long jumper can jump a distance of 8.0 m when he takes off at an angle of 45° with respect to the horizontal. Assuming he can jump with the same initial speed at all angles, how much distance does he lose by taking off at 30° ?
94. On planet Arcon, the maximum horizontal range of a projectile launched at 10 m/s is 20 m. What is the acceleration of gravity on this planet?
95. A mountain biker encounters a jump on a race course that sends him into the air at 60° to the horizontal. If he lands at a horizontal distance of 45.0 m and 20 m below his launch point, what is his initial speed?
96. Which has the greater centripetal acceleration, a car with a speed of 15.0 m/s along a circular track of radius 100.0 m or a car with a speed of 12.0 m/s along a circular track of radius 75.0 m?
97. A geosynchronous satellite orbits Earth at a distance of 42,250.0 km and has a period of 1 day. What is the centripetal acceleration of the satellite?
98. Two speedboats are traveling at the same speed relative to the water in opposite directions in a moving river. An observer on the riverbank sees the boats moving at 4.0 m/s and 5.0 m/s. (a) What is the speed of the boats relative to the river? (b) How fast is the river moving relative to the shore?

Challenge Problems

99. World's Longest Par 3. The tee of the world's longest par 3 sits atop South Africa's Hanglip Mountain at 400.0 m above the green and can only be reached by helicopter. The horizontal distance to the green is 359.0 m. Neglect air resistance and answer the following questions. (a) If a golfer launches a shot that is 40° with respect to the horizontal, what initial velocity must she give the ball? (b) What is the time to reach the green?
100. When a field goal kicker kicks a football as hard as he can at 45° to the horizontal, the ball just clears the 3-m-high crossbar of the goalposts 45.7 m away. (a) What is the maximum speed the kicker can impart to the football? (b) In addition to clearing the crossbar, the football must be high enough in the air early during its flight to clear the reach of the onrushing defensive lineman. If the lineman is 4.6 m away and has a vertical reach of 2.5 m, can he block the 45.7-m field goal attempt? (c) What if the

lineman is 1.0 m away?



101. A truck is traveling east at 80 km/h. At an intersection 32 km ahead, a car is traveling north at 50 km/h. (a) How long after this moment will the vehicles be closest to each other? (b) How far apart will they be at that point?

CHAPTER 5

Newton's Laws of Motion



FIGURE 5.1 The Golden Gate Bridge, one of the greatest works of modern engineering, was the longest suspension bridge in the world in the year it opened, 1937. It is still among the longest suspension bridges. In designing and building a bridge, what physics must we consider? What forces act on the bridge? What forces keep the bridge from falling? How do the towers, cables, and ground interact to maintain stability?

CHAPTER OUTLINE

5.1 Forces

5.2 Newton's First Law

5.3 Newton's Second Law

5.4 Mass and Weight

5.5 Newton's Third Law

5.6 Common Forces

5.7 Drawing Free-Body Diagrams

INTRODUCTION When you drive across a bridge, you expect it to remain stable. You also expect to speed up or slow your car in response to traffic changes. In both cases, you deal with forces. The forces on the bridge are in equilibrium, so it stays in place. In contrast, the force produced by your car engine causes a change in motion. Isaac Newton discovered the laws of motion that describe these situations.

Forces affect every moment of your life. Your body is held to Earth by force and held together by the forces of charged particles. When you open a door, walk down a street, lift your fork, or touch a baby's face, you are applying forces. Zooming in deeper, your body's atoms are held together by electrical forces, and the core of the atom, called the nucleus, is held together by the strongest force we know—strong nuclear force.

5.1 Forces

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Distinguish between kinematics and dynamics
- Understand the definition of force
- Identify simple free-body diagrams
- Define the SI unit of force, the newton
- Describe force as a vector

The study of motion is called *kinematics*, but kinematics only describes the way objects move—their velocity and their acceleration. **Dynamics** is the study of how forces affect the motion of objects and systems. It considers the causes of motion of objects and systems of interest, where a system is anything being analyzed. The foundation of dynamics are the laws of motion stated by Isaac Newton (1642–1727). These laws provide an example of the breadth and simplicity of principles under which nature functions. They are also universal laws in that they apply to situations on Earth and in space.

Newton's laws of motion were just one part of the monumental work that has made him legendary (Figure 5.2). The development of Newton's laws marks the transition from the Renaissance to the modern era. Not until the advent of modern physics was it discovered that Newton's laws produce a good description of motion only when the objects are moving at speeds much less than the speed of light and when those objects are larger than the size of most molecules (about 10^{-9} m in diameter). These constraints define the realm of Newtonian mechanics. At the beginning of the twentieth century, Albert Einstein (1879–1955) developed the theory of relativity and, along with many other scientists, quantum mechanics. Quantum mechanics does not have the constraints present in Newtonian physics. All of the situations we consider in this chapter, and all those preceding the introduction of relativity in [Relativity \(http://openstax.org/books/university-physics-volume-3/pages/5-introduction\)](http://openstax.org/books/university-physics-volume-3/pages/5-introduction), are in the realm of Newtonian physics.

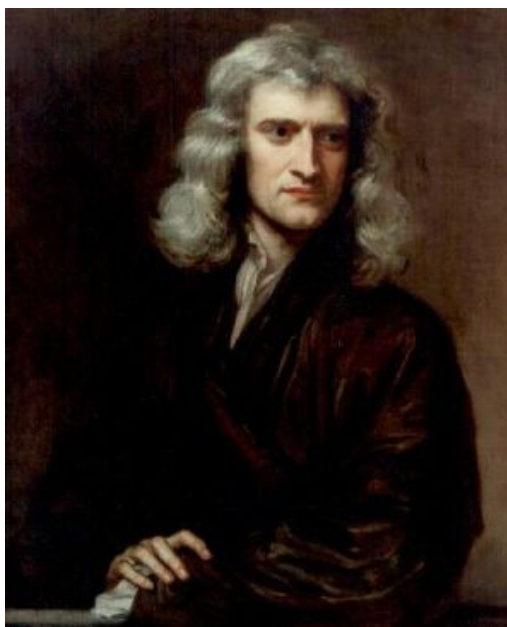


FIGURE 5.2 Isaac Newton (1642–1727) published his amazing work, *Philosophiae Naturalis Principia Mathematica*, in 1687. It proposed scientific laws that still apply today to describe the motion of objects (the laws of motion). Newton also discovered the law of gravity, invented calculus, and made great contributions to the theories of light and color.

Working Definition of Force

Dynamics is the study of the forces that cause objects and systems to move. To understand this, we need a working definition of force. An intuitive definition of **force**—that is, a push or a pull—is a good place to start. We know that a push or a pull has both magnitude and direction (therefore, it is a vector quantity), so we can define force as the push or pull on an object with a specific magnitude and direction. Force can be represented by vectors or expressed

as a multiple of a standard force.

The push or pull on an object can vary considerably in either magnitude or direction. For example, a cannon exerts a strong force on a cannonball that is launched into the air. In contrast, Earth exerts only a tiny downward pull on a flea. Our everyday experiences also give us a good idea of how multiple forces add. If two people push in different directions on a third person, as illustrated in [Figure 5.3](#), we might expect the total force to be in the direction shown. Since force is a vector, it adds just like other vectors. Forces, like other vectors, are represented by arrows and can be added using the familiar head-to-tail method or trigonometric methods. These ideas were developed in [Vectors](#).

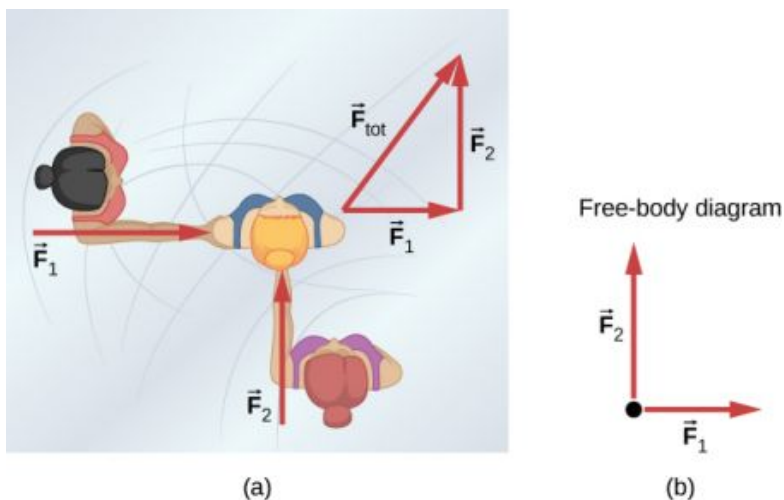


FIGURE 5.3 (a) An overhead view of two ice skaters pushing on a third skater. Forces are vectors and add like other vectors, so the total force on the third skater is in the direction shown. (b) A free-body diagram representing the forces acting on the third skater.

[Figure 5.3](#)(b) is our first example of a **free-body diagram**, which is a sketch showing all external forces acting on an object or system. The object or system is represented by a single isolated point (or free body), and only those forces acting *on* it that originate outside of the object or system—that is, **external forces**—are shown. (These forces are the only ones shown because only external forces acting on the free body affect its motion. We can ignore any internal forces within the body.) The forces are represented by vectors extending outward from the free body.

Free-body diagrams are useful in analyzing forces acting on an object or system, and are employed extensively in the study and application of Newton's laws of motion. You will see them throughout this text and in all your studies of physics. The following steps briefly explain how a free-body diagram is created; we examine this strategy in more detail in [Drawing Free-Body Diagrams](#).



PROBLEM-SOLVING STRATEGY

Drawing Free-Body Diagrams

1. Draw the object under consideration. If you are treating the object as a particle, represent the object as a point. Place this point at the origin of an xy -coordinate system.
2. Include all forces that act on the object, representing these forces as vectors. However, do not include the net force on the object or the forces that the object exerts on its environment.
3. Resolve all force vectors into x - and y -components.
4. Draw a separate free-body diagram for each object in the problem.

We illustrate this strategy with two examples of free-body diagrams ([Figure 5.4](#)). The terms used in this figure are explained in more detail later in the chapter.

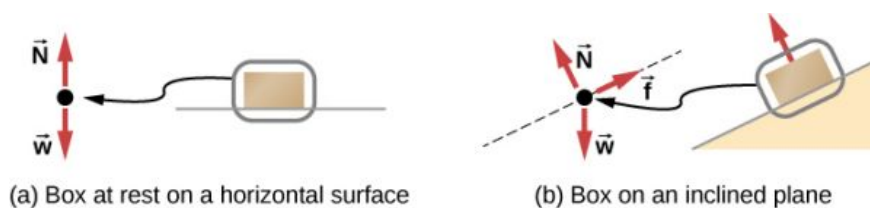


FIGURE 5.4 In these free-body diagrams, $\vec{N}$ is the normal force, $\vec{w}$ is the weight of the object, and $\vec{f}$ is the friction.

The steps given here are sufficient to guide you in this important problem-solving strategy. The final section of this chapter explains in more detail how to draw free-body diagrams when working with the ideas presented in this chapter.

Development of the Force Concept

A quantitative definition of force can be based on some standard force, just as distance is measured in units relative to a standard length. One possibility is to stretch a spring a certain fixed distance (Figure 5.5) and use the force it exerts to pull itself back to its relaxed shape—called a *restoring force*—as a standard. The magnitude of all other forces can be considered as multiples of this standard unit of force. Many other possibilities exist for standard forces. Some alternative definitions of force will be given later in this chapter.

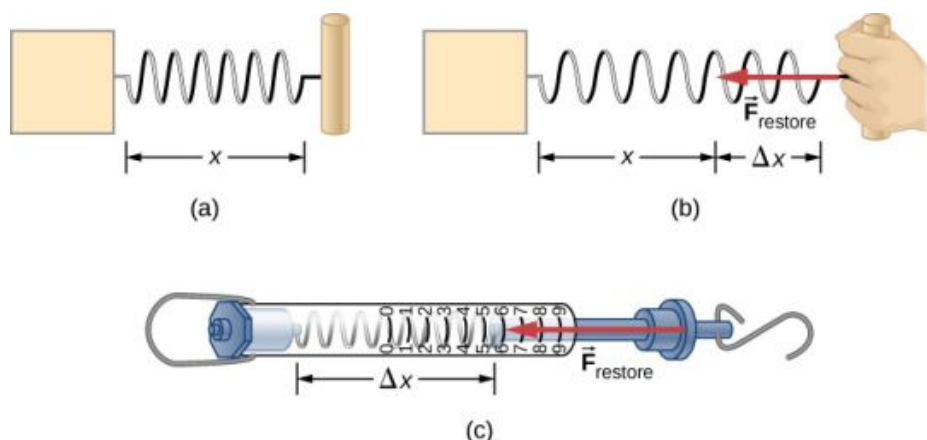


FIGURE 5.5 The force exerted by a stretched spring can be used as a standard unit of force. (a) This spring has a length x when undistorted. (b) When stretched a distance Δx , the spring exerts a restoring force $\vec{F}_{\text{restore}}$, which is reproducible. (c) A spring scale is one device that uses a spring to measure force. The force $\vec{F}_{\text{restore}}$ is exerted on whatever is attached to the hook. Here, this force has a magnitude of six units of the force standard being employed.

Let's analyze force more deeply. Suppose a physics student sits at a table, working diligently on his homework (Figure 5.6). What external forces act on him? Can we determine the origin of these forces?

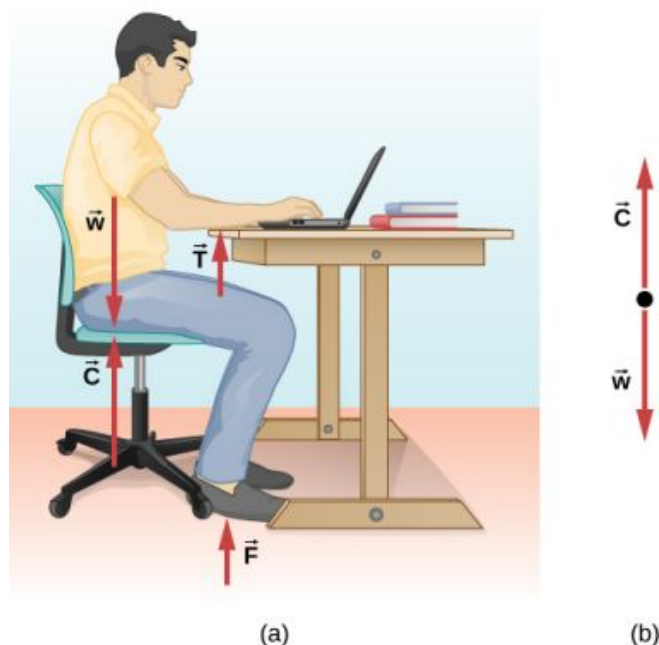


FIGURE 5.6 (a) The forces acting on the student are due to the chair, the table, the floor, and Earth's gravitational attraction. (b) In solving a problem involving the student, we may want to consider only the forces acting along the line running through his torso. A free-body diagram for this situation is shown.

In most situations, forces are grouped into two categories: *contact forces* and *field forces*. As you might guess, contact forces are due to direct physical contact between objects. For example, the student in [Figure 5.6](#) experiences the contact forces $\vec{C}$, $\vec{F}$, and $\vec{T}$, which are exerted by the chair on his posterior, the floor on his feet, and the table on his forearms, respectively. Field forces, however, act without the necessity of physical contact between objects. They depend on the presence of a “field” in the region of space surrounding the body under consideration. Since the student is in Earth's gravitational field, he feels a gravitational force $\vec{w}$; in other words, he has weight.

You can think of a field as a property of space that is detectable by the forces it exerts. Scientists think there are only four fundamental force fields in nature. These are the gravitational, electromagnetic, strong nuclear, and weak fields (we consider these four forces in nature later in this text). As noted for $\vec{w}$ in [Figure 5.6](#), the gravitational field is responsible for the weight of a body. The forces of the electromagnetic field include those of static electricity and magnetism; they are also responsible for the attraction among atoms in bulk matter. Both the strong nuclear and the weak force fields are effective only over distances roughly equal to a length of scale no larger than an atomic nucleus (10^{-15} m). Their range is so small that neither field has influence in the macroscopic world of Newtonian mechanics.

Contact forces are fundamentally electromagnetic. While the elbow of the student in [Figure 5.6](#) is in contact with the tabletop, the atomic charges in his skin interact electromagnetically with the charges in the surface of the table. The net (total) result is the force $\vec{T}$. Similarly, when adhesive tape sticks to a piece of paper, the atoms of the tape are intermingled with those of the paper to cause a net electromagnetic force between the two objects. However, in the context of Newtonian mechanics, the electromagnetic origin of contact forces is not an important concern.

Vector Notation for Force

As previously discussed, force is a vector; it has both magnitude and direction. The SI unit of force is called the **newton** (abbreviated N), and 1 N is the force needed to accelerate an object with a mass of 1 kg at a rate of 1 m/s^2 : $1 \text{ N} = 1 \text{ kg} \cdot \text{m/s}^2$. An easy way to remember the size of a newton is to imagine holding a small apple; it has a weight of about 1 N.

We can thus describe a two-dimensional force in the form $\vec{F} = a\hat{i} + b\hat{j}$ (the unit vectors $\hat{i}$ and $\hat{j}$ indicate the direction of these forces along the x-axis and the y-axis, respectively) and a three-dimensional force in the form $\vec{F} = a\hat{i} + b\hat{j} + c\hat{k}$. In [Figure 5.3](#), let's suppose that ice skater 1, on the left side of the figure, pushes horizontally

with a force of 30.0 N to the right; we represent this as $\vec{F}_1 = 30.0\hat{i}$ N. Similarly, if ice skater 2 pushes with a force of 40.0 N in the positive vertical direction shown, we would write $\vec{F}_2 = 40.0\hat{j}$ N. The resultant of the two forces causes a mass to accelerate—in this case, the third ice skater. This resultant is called the **net external force** $\vec{F}_{\text{net}}$ and is found by taking the vector sum of all external forces acting on an object or system (thus, we can also represent net external force as $\sum \vec{F}$):

$$\vec{F}_{\text{net}} = \sum \vec{F} = \vec{F}_1 + \vec{F}_2 + \dots \quad 5.1$$

This equation can be extended to any number of forces.

In this example, we have $\vec{F}_{\text{net}} = \sum \vec{F} = \vec{F}_1 + \vec{F}_2 = 30.0\hat{i} + 40.0\hat{j}$ N. The hypotenuse of the triangle shown in [Figure 5.3](#) is the resultant force, or net force. It is a vector. To find its magnitude (the size of the vector, without regard to direction), we use the rule given in [Vectors](#), taking the square root of the sum of the squares of the components:

$$F_{\text{net}} = \sqrt{(30.0 \text{ N})^2 + (40.0 \text{ N})^2} = 50.0 \text{ N}.$$

The direction is given by

$$\theta = \tan^{-1} \left(\frac{F_2}{F_1} \right) = \tan^{-1} \left(\frac{40.0}{30.0} \right) = 53.1^\circ,$$

measured from the positive x-axis, as shown in the free-body diagram in [Figure 5.3\(b\)](#).

Let's suppose the ice skaters now push the third ice skater with $\vec{F}_1 = 3.0\hat{i} + 8.0\hat{j}$ N and $\vec{F}_2 = 5.0\hat{i} + 4.0\hat{j}$ N. What is the resultant of these two forces? We must recognize that force is a vector; therefore, we must add using the rules for vector addition:

$$\vec{F}_{\text{net}} = \vec{F}_1 + \vec{F}_2 = (3.0\hat{i} + 8.0\hat{j}) + (5.0\hat{i} + 4.0\hat{j}) = 8.0\hat{i} + 12\hat{j} \text{ N}$$

CHECK YOUR UNDERSTANDING 5.1

Find the magnitude and direction of the net force in the ice skater example just given.

INTERACTIVE

Engage the Phet simulation below to learn how to add vectors. Drag vectors onto a graph, change their length and angle, and sum them together. The magnitude, angle, and components of each vector can be displayed in several formats.

[Access multimedia content \(https://openstax.org/books/university-physics-volume-1/pages/5-1-forces\)](https://openstax.org/books/university-physics-volume-1/pages/5-1-forces)

5.2 Newton's First Law

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Describe Newton's first law of motion
- Recognize friction as an external force
- Define inertia
- Identify inertial reference frames
- Calculate equilibrium for a system

Experience suggests that an object at rest remains at rest if left alone and that an object in motion tends to slow down and stop unless some effort is made to keep it moving. However, **Newton's first law** gives a deeper explanation of this observation.

NEWTON'S FIRST LAW OF MOTION

A body at rest remains at rest or, if in motion, remains in motion at constant velocity unless acted on by a net external force.

Note the repeated use of the verb “remains.” We can think of this law as preserving the status quo of motion. Also note the expression “constant velocity;” this means that the object maintains a path along a straight line, since neither the magnitude nor the direction of the velocity vector changes. We can use [Figure 5.7](#) to consider the two parts of Newton's first law.



FIGURE 5.7 (a) A hockey puck is shown at rest; it remains at rest until an outside force such as a hockey stick changes its state of rest; (b) a hockey puck is shown in motion; it continues in motion in a straight line until an outside force causes it to change its state of motion. Although it is slick, an ice surface provides some friction that slows the puck.

Rather than contradicting our experience, Newton's first law says that there must be a cause for any change in velocity (a change in either magnitude or direction) to occur. This cause is a net external force, which we defined earlier in the chapter. An object sliding across a table or floor slows down due to the net force of friction acting on the object. If friction disappears, will the object still slow down?

The idea of cause and effect is crucial in accurately describing what happens in various situations. For example, consider what happens to an object sliding along a rough horizontal surface. The object quickly grinds to a halt. If we spray the surface with talcum powder to make the surface smoother, the object slides farther. If we make the surface even smoother by rubbing lubricating oil on it, the object slides farther yet. Extrapolating to a frictionless surface and ignoring air resistance, we can imagine the object sliding in a straight line indefinitely. Friction is thus the cause of slowing (consistent with Newton's first law). The object would not slow down if friction were eliminated.

Consider an air hockey table ([Figure 5.8](#)). When the air is turned off, the puck slides only a short distance before friction slows it to a stop. However, when the air is turned on, it creates a nearly frictionless surface, and the puck glides long distances without slowing down. Additionally, if we know enough about the friction, we can accurately predict how quickly the object slows down.

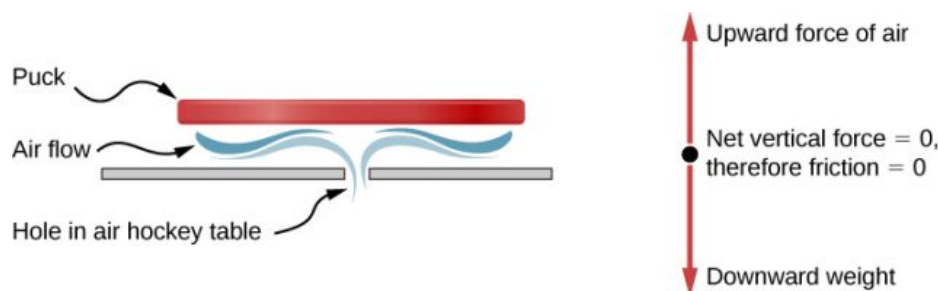


FIGURE 5.8 An air hockey table is useful in illustrating Newton's laws. When the air is off, friction quickly slows the puck; but when the air is on, it minimizes contact between the puck and the hockey table, and the puck glides far down the table.

Newton's first law is general and can be applied to anything from an object sliding on a table to a satellite in orbit to blood pumped from the heart. Experiments have verified that any change in velocity (speed or direction) must be caused by an external force. The idea of *generally applicable or universal laws* is important—it is a basic feature of all laws of physics. Identifying these laws is like recognizing patterns in nature from which further patterns can be discovered. The genius of Galileo, who first developed the idea for the first law of motion, and Newton, who clarified it, was to ask the fundamental question: “What is the cause?” Thinking in terms of cause and effect is fundamentally different from the typical ancient Greek approach, when questions such as “Why does a tiger have stripes?” would have been answered in Aristotelian fashion, such as “That is the nature of the beast.” The ability to think in terms of cause and effect is the ability to make a connection between an observed behavior and the surrounding world.

Gravitation and Inertia

Regardless of the scale of an object, whether a molecule or a subatomic particle, two properties remain valid and thus of interest to physics: gravitation and inertia. Both are connected to mass. Roughly speaking, *mass* is a measure of the amount of matter in something. *Gravitation* is the attraction of one mass to another, such as the attraction between yourself and Earth that holds your feet to the floor. The magnitude of this attraction is your weight, and it is a force.

Mass is also related to **inertia**, the ability of an object to resist changes in its motion—in other words, to resist acceleration. Newton's first law is often called the **law of inertia**. As we know from experience, some objects have more inertia than others. It is more difficult to change the motion of a large boulder than that of a basketball, for example, because the boulder has more mass than the basketball. In other words, the inertia of an object is measured by its mass. The relationship between mass and weight is explored later in this chapter.

Inertial Reference Frames

Earlier, we stated Newton's first law as “A body at rest remains at rest or, if in motion, remains in motion at constant velocity unless acted on by a net external force.” It can also be stated as “Every body remains in its state of uniform motion in a straight line unless it is compelled to change that state by forces acting on it.” To Newton, “uniform motion in a straight line” meant constant velocity, which includes the case of zero velocity, or rest. Therefore, the first law says that the velocity of an object remains constant if the net force on it is zero.

Newton's first law is usually considered to be a statement about reference frames. It provides a method for identifying a special type of reference frame: the **inertial reference frame**. In principle, we can make the net force on a body zero. If its velocity relative to a given frame is constant, then that frame is said to be inertial. So by definition, an inertial reference frame is a reference frame in which Newton's first law is valid. Newton's first law applies to objects with constant velocity. From this fact, we can infer the following statement.

INERTIAL REFERENCE FRAME

A reference frame moving at constant velocity relative to an inertial frame is also inertial. A reference frame accelerating relative to an inertial frame is not inertial.

Are inertial frames common in nature? It turns out that well within experimental error, a reference frame at rest relative to the most distant, or “fixed,” stars is inertial. All frames moving uniformly with respect to this fixed-star frame are also inertial. For example, a nonrotating reference frame attached to the Sun is, for all practical purposes, inertial, because its velocity relative to the fixed stars does not vary by more than one part in 10^{10} . Earth accelerates relative to the fixed stars because it rotates on its axis and revolves around the Sun; hence, a reference frame attached to its surface is not inertial. For most problems, however, such a frame serves as a sufficiently accurate approximation to an inertial frame, because the acceleration of a point on Earth's surface relative to the fixed stars is rather small ($< 3.4 \times 10^{-2} \text{ m/s}^2$). Thus, unless indicated otherwise, we consider reference frames fixed on Earth to be inertial.

Finally, no particular inertial frame is more special than any other. As far as the laws of nature are concerned, all inertial frames are equivalent. In analyzing a problem, we choose one inertial frame over another simply on the basis of convenience.

Newton's First Law and Equilibrium

Newton's first law tells us about the equilibrium of a system, which is the state in which the forces on the system are balanced. Returning to [Forces](#) and the ice skaters in [Figure 5.3](#), we know that the forces $\vec{F}_1$ and $\vec{F}_2$ combine to form a resultant force, or the net external force: $\vec{F}_R = \vec{F}_{\text{net}} = \vec{F}_1 + \vec{F}_2$. To create equilibrium, we require a balancing force that will produce a net force of zero. This force must be equal in magnitude but opposite in direction to $\vec{F}_R$, which means the vector must be $-\vec{F}_R$. Referring to the ice skaters, for which we found $\vec{F}_R$ to be $30.0\hat{i} + 40.0\hat{j}$ N, we can determine the balancing force by simply finding $-\vec{F}_R = -30.0\hat{i} - 40.0\hat{j}$ N. See the free-body diagram in [Figure 5.3\(b\)](#).

We can give Newton's first law in vector form:

$$\vec{v} = \text{constant when } \vec{F}_{\text{net}} = \vec{0} \text{ N.}$$

5.2

This equation says that a net force of zero implies that the velocity $\vec{v}$ of the object is constant. (The word “constant” can indicate zero velocity.)

Newton's first law is deceptively simple. If a car is at rest, the only forces acting on the car are weight and the contact force of the pavement pushing up on the car ([Figure 5.9](#)). It is easy to understand that a nonzero net force is required to change the state of motion of the car. As a car moves with constant velocity, the friction force propels the car forward and opposes the drag force against it.

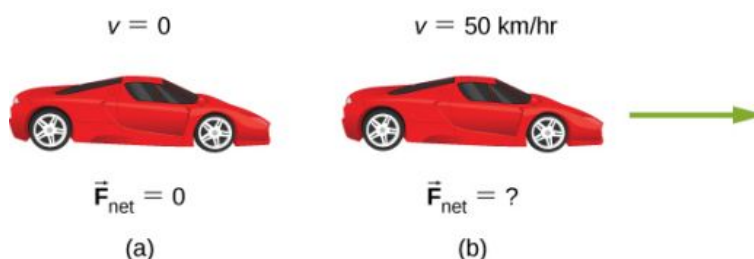


FIGURE 5.9 A car is shown (a) parked and (b) moving at constant velocity. How do Newton's laws apply to the parked car? What does the knowledge that the car is moving at constant velocity tell us about the net horizontal force on the car?



EXAMPLE 5.1

When Does Newton's First Law Apply to Your Car?

Newton's laws can be applied to all physical processes involving force and motion, including something as mundane as driving a car.

- Your car is parked outside your house. Does Newton's first law apply in this situation? Why or why not?
- Your car moves at constant velocity down the street. Does Newton's first law apply in this situation? Why or why not?

Strategy

In (a), we are considering the first part of Newton's first law, dealing with a body at rest; in (b), we look at the second part of Newton's first law for a body in motion.

Solution

- When your car is parked, all forces on the car must be balanced; the vector sum is 0 N. Thus, the net force is zero, and Newton's first law applies. The acceleration of the car is zero, and in this case, the velocity is also zero.
- When your car is moving at constant velocity down the street, the net force must also be zero according to Newton's first law. The car's frictional force between the road and tires opposes the drag force on the car with the same magnitude, producing a net force of zero. The body continues in its state of constant velocity until

the net force becomes nonzero. Realize that *a net force of zero means that an object is either at rest or moving with constant velocity, that is, it is not accelerating*. What do you suppose happens when the car accelerates? We explore this idea in the next section.

Significance

As this example shows, there are two kinds of equilibrium. In (a), the car is at rest; we say it is in *static equilibrium*. In (b), the forces on the car are balanced, but the car is moving; we say that it is in *dynamic equilibrium*. (We examine this idea in more detail in [Static Equilibrium and Elasticity](#).) Again, it is possible for two (or more) forces to act on an object yet for the object to move. In addition, a net force of zero cannot produce acceleration.

✓ CHECK YOUR UNDERSTANDING 5.2

A skydiver opens their parachute, and shortly thereafter, they are moving at constant velocity. (a) What forces are acting on them? (b) Which force is bigger?

🔗 INTERACTIVE

Engage the simulation below to predict, qualitatively, how an external force will affect the speed and direction of an object's motion. Explain the effects with the help of a free-body diagram. Use free-body diagrams to draw position, velocity, acceleration, and force graphs, and vice versa. Explain how the graphs relate to one another. Given a scenario or a graph, sketch all four graphs.

[Access multimedia content \(https://openstax.org/books/university-physics-volume-1/pages/5-2-newtons-first-law\)](https://openstax.org/books/university-physics-volume-1/pages/5-2-newtons-first-law)

5.3 Newton's Second Law

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Distinguish between external and internal forces
- Describe Newton's second law of motion
- Explain the dependence of acceleration on net force and mass

Newton's second law is closely related to his first law. It mathematically gives the cause-and-effect relationship between force and changes in motion. Newton's second law is quantitative and is used extensively to calculate what happens in situations involving a force. Before we can write down Newton's second law as a simple equation that gives the exact relationship of force, mass, and acceleration, we need to sharpen some ideas we mentioned earlier.

Force and Acceleration

First, what do we mean by a change in motion? The answer is that a change in motion is equivalent to a change in velocity. A change in velocity means, by definition, that there is acceleration. Newton's first law says that a net external force causes a change in motion; thus, we see that a *net external force causes nonzero acceleration*.

We defined external force in [Forces](#) as force acting on an object or system that originates outside of the object or system. Let's consider this concept further. An intuitive notion of *external* is correct—it is outside the system of interest. For example, in [Figure 5.10\(a\)](#), the system of interest is the car plus the person within it. The two forces exerted by the two students are external forces. In contrast, an internal force acts between elements of the system. Thus, the force the person in the car exerts to hang on to the steering wheel is an internal force between elements of the system of interest. Only external forces affect the motion of a system, according to Newton's first law. (The internal forces cancel each other out, as explained in the next section.) Therefore, we must define the boundaries of the system before we can determine which forces are external. Sometimes, the system is obvious, whereas at other times, identifying the boundaries of a system is more subtle. The concept of a system is fundamental to many areas of physics, as is the correct application of Newton's laws. This concept is revisited many times in the study of physics.

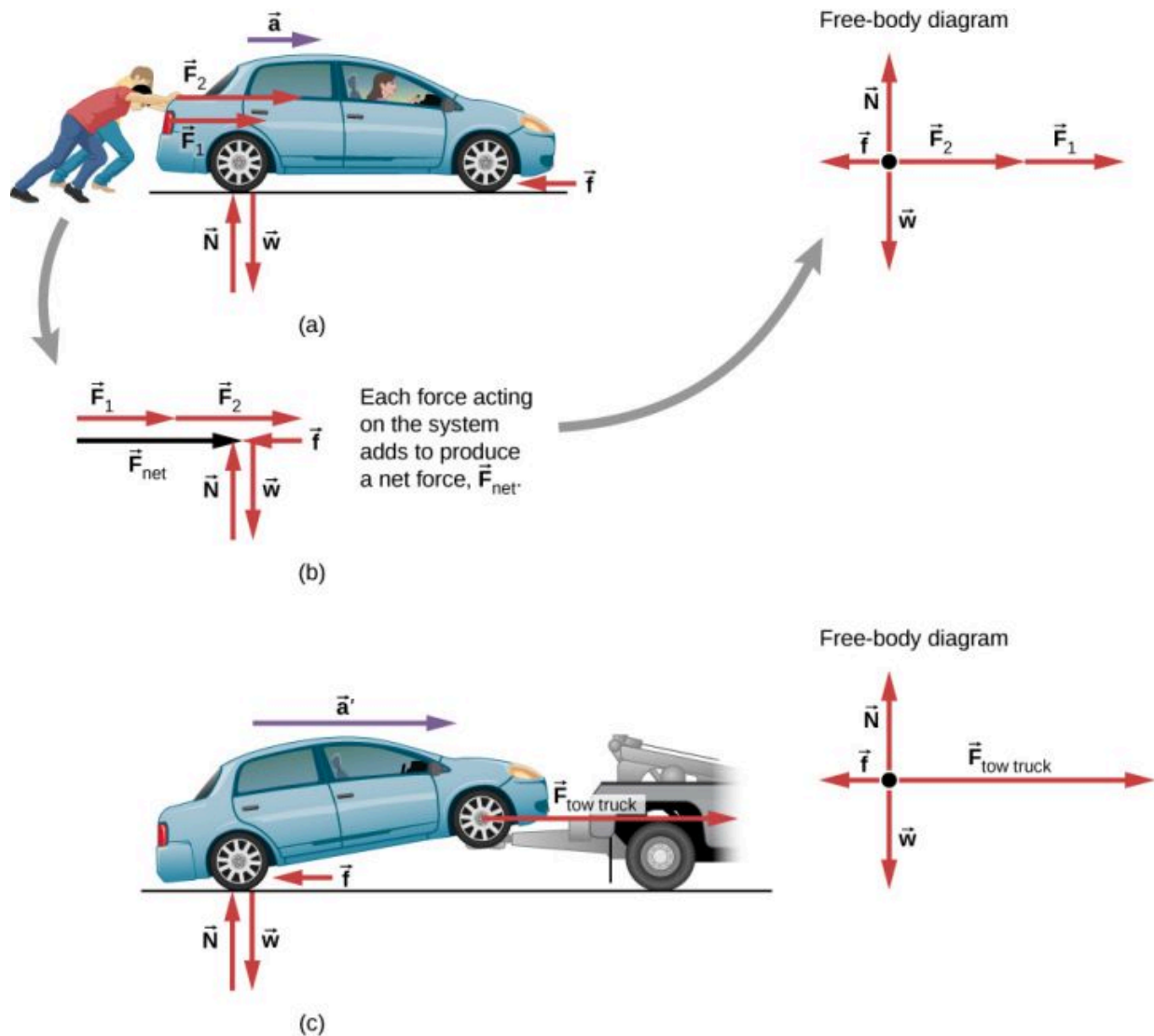


FIGURE 5.10 Different forces exerted on the same mass produce different accelerations. (a) Two students push a stalled car. All external forces acting on the car are shown. (b) The forces acting on the car are transferred to a coordinate plane (free-body diagram) for simpler analysis. (c) The tow truck can produce greater external force on the same mass, and thus greater acceleration.

From this example, you can see that different forces exerted on the same mass produce different accelerations. In [Figure 5.10\(a\)](#), the two students push a car with a driver in it. Arrows representing all external forces are shown. The system of interest is the car and its driver. The weight $\vec{w}$ of the system and the support of the ground $\vec{N}$ are also shown for completeness and are assumed to cancel (because there was no vertical motion and no imbalance of forces in the vertical direction to create a change in motion). The vector $\vec{f}$ represents the friction acting on the car, and it acts to the left, opposing the motion of the car. (We discuss friction in more detail in the next chapter.) In [Figure 5.10\(b\)](#), all external forces acting on the system add together to produce the net force $\vec{F}_{net}$. The free-body diagram shows all of the forces acting on the system of interest. The dot represents the center of mass of the system. Each force vector extends from this dot. Because there are two forces acting to the right, the vectors are shown collinearly. Finally, in [Figure 5.10\(c\)](#), a larger net external force produces a larger acceleration ($\vec{a}' > \vec{a}$) when the tow truck pulls the car.

It seems reasonable that acceleration would be directly proportional to and in the same direction as the net external force acting on a system. This assumption has been verified experimentally and is illustrated in [Figure 5.10](#). To obtain an equation for Newton's second law, we first write the relationship of acceleration $\vec{a}$ and net external force $\vec{F}_{net}$ as the proportionality

$$\vec{a} \propto \vec{F}_{\text{net}}$$

where the symbol $\propto$ means “proportional to.” (Recall from [Forces](#) that the net external force is the vector sum of all external forces and is sometimes indicated as $\sum \vec{F}$.) This proportionality shows what we have said in words—acceleration is directly proportional to net external force. Once the system of interest is chosen, identify the external forces and ignore the internal ones. It is a tremendous simplification to disregard the numerous internal forces acting between objects within the system, such as muscular forces within the students’ bodies, let alone the myriad forces between the atoms in the objects. Still, this simplification helps us solve some complex problems.

It also seems reasonable that acceleration should be inversely proportional to the mass of the system. In other words, the larger the mass (the inertia), the smaller the acceleration produced by a given force. As illustrated in [Figure 5.11](#), the same net external force applied to a basketball produces a much smaller acceleration when it is applied to an SUV. The proportionality is written as

$$a \propto \frac{1}{m},$$

where m is the mass of the system and a is the magnitude of the acceleration. Experiments have shown that acceleration is exactly inversely proportional to mass, just as it is directly proportional to net external force.

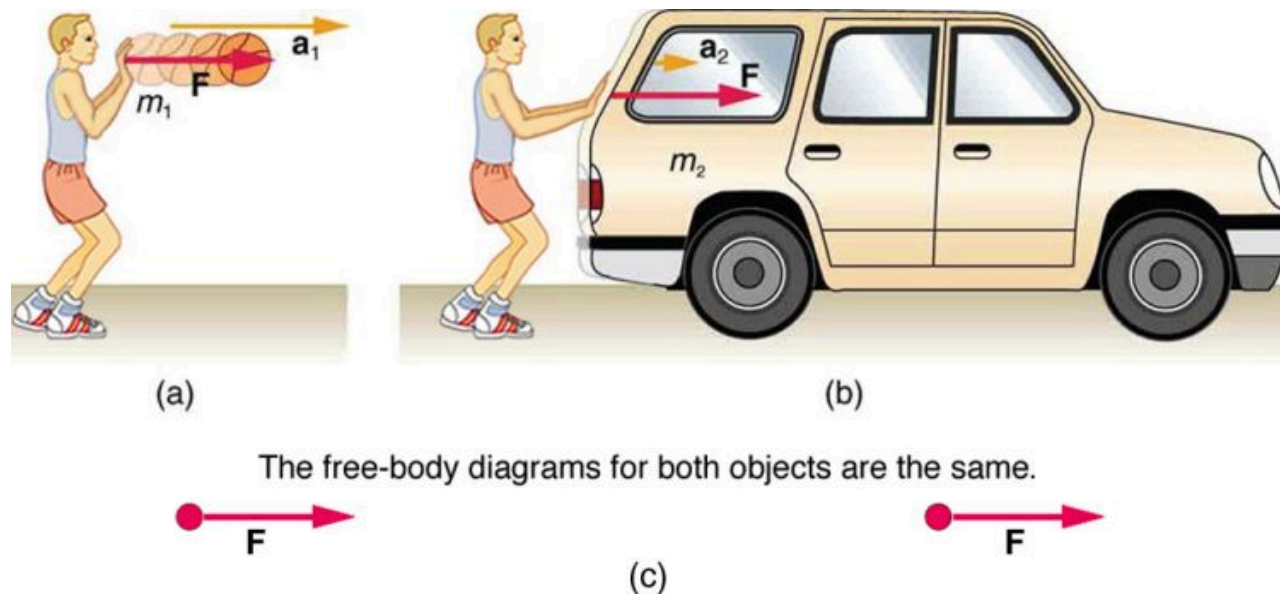


FIGURE 5.11 The same force exerted on systems of different masses produces different accelerations. (a) A basketball player pushes on a basketball to make a pass. (Ignore the effect of gravity on the ball.) (b) The same player exerts an identical force on a stalled SUV and produces far less acceleration. (c) The free-body diagrams are identical, permitting direct comparison of the two situations. A series of patterns for free-body diagrams will emerge as you do more problems and learn how to draw them in [Drawing Free-Body Diagrams](#).

It has been found that the acceleration of an object depends only on the net external force and the mass of the object. Combining the two proportionalities just given yields **Newton's second law**.

NEWTON'S SECOND LAW OF MOTION

The acceleration of a system is directly proportional to and in the same direction as the net external force acting on the system and is inversely proportional to its mass. In equation form, Newton's second law is

$$\vec{a} = \frac{\vec{F}_{\text{net}}}{m},$$

where $\vec{a}$ is the acceleration, $\vec{F}_{\text{net}}$ is the net force, and m is the mass. This is often written in the more familiar form

$$\vec{F}_{\text{net}} = \sum \vec{F} = m\vec{a}, \quad 5.3$$

but the first equation gives more insight into what Newton's second law means. When only the magnitude of force and acceleration are considered, this equation can be written in the simpler scalar form:

$$F_{\text{net}} = ma. \quad 5.4$$

The law is a cause-and-effect relationship among three quantities that is not simply based on their definitions. The validity of the second law is based on experimental verification. The free-body diagram, which you will learn to draw in [Drawing Free-Body Diagrams](#), is the basis for writing Newton's second law.

EXAMPLE 5.2

What Acceleration Can a Person Produce When Pushing a Lawn Mower?

Suppose that the net external force (push minus friction) exerted on a lawn mower is 51 N (about 11 lb.) parallel to the ground ([Figure 5.12](#)). The mass of the mower is 24 kg. What is its acceleration?

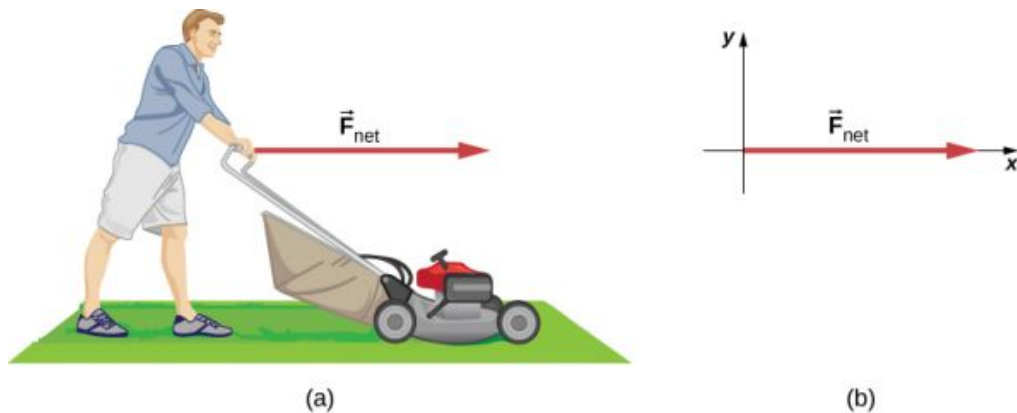


FIGURE 5.12 (a) The net force on a lawn mower is 51 N to the right. At what rate does the lawn mower accelerate to the right? (b) The free-body diagram for this problem is shown.

Strategy

This problem involves only motion in the horizontal direction; we are also given the net force, indicated by the single vector, but we can suppress the vector nature and concentrate on applying Newton's second law. Since F_{net} and m are given, the acceleration can be calculated directly from Newton's second law as $F_{\text{net}} = ma$.

Solution

The magnitude of the acceleration a is $a = F_{\text{net}}/m$. Entering known values gives

$$a = \frac{51 \text{ N}}{24 \text{ kg}}.$$

Substituting the unit of kilograms times meters per square second for newtons yields

$$a = \frac{51 \text{ kg} \cdot \text{m/s}^2}{24 \text{ kg}} = 2.1 \text{ m/s}^2.$$

Significance

The direction of the acceleration is the same direction as that of the net force, which is parallel to the ground. This is a result of the vector relationship expressed in Newton's second law, that is, the vector representing net force is the scalar multiple of the acceleration vector. There is no information given in this example about the individual external forces acting on the system, but we can say something about their relative magnitudes. For example, the force exerted by the person pushing the mower must be greater than the friction opposing the motion (since we know the mower moved forward), and the vertical forces must cancel because no acceleration occurs in the vertical direction.

(the mower is moving only horizontally). The acceleration found is small enough to be reasonable for a person pushing a mower. Such an effort would not last too long, because the person's top speed would soon be reached.

✓ CHECK YOUR UNDERSTANDING 5.3

At the time of its launch, the HMS *Titanic* was the most massive mobile object ever built, with a mass of 6.0×10^7 kg. If a force of 6 MN (6×10^6 N) was applied to the ship, what acceleration would it experience?

In the preceding example, we dealt with net force only for simplicity. However, several forces act on the lawn mower. The weight $\vec{w}$ (discussed in detail in [Mass and Weight](#)) pulls down on the mower, toward the center of Earth; this produces a contact force on the ground. The ground must exert an upward force on the lawn mower, known as the normal force $\vec{N}$, which we define in [Common Forces](#). These forces are balanced and therefore do not produce vertical acceleration. In the next example, we show both of these forces. As you continue to solve problems using Newton's second law, be sure to show multiple forces.

✳ EXAMPLE 5.3

Which Force Is Bigger?

- (a) The car shown in [Figure 5.13](#) is moving at a constant speed. Which force is bigger, $\vec{F}_{\text{friction}}$ or $\vec{F}_{\text{drag}}$? Explain.
- (b) The same car is now accelerating to the right. Which force is bigger, $\vec{F}_{\text{friction}}$ or $\vec{F}_{\text{drag}}$? Explain.

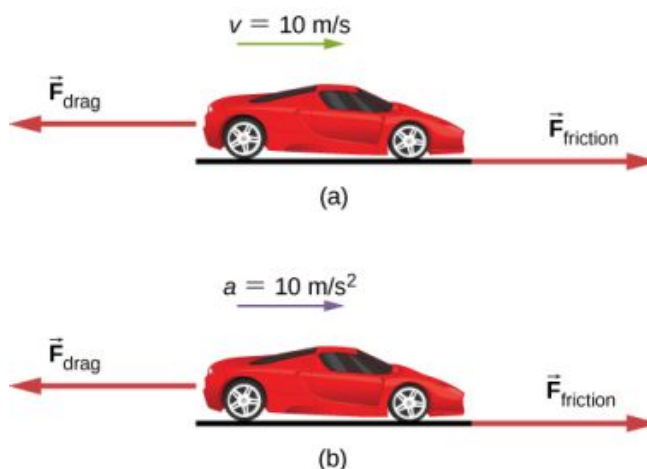


FIGURE 5.13 A car is shown (a) moving at constant speed and (b) accelerating. How do the forces acting on the car compare in each case? (a) What does the knowledge that the car is moving at constant velocity tell us about the net horizontal force on the car compared to the friction force? (b) What does the knowledge that the car is accelerating tell us about the horizontal force on the car compared to the friction force?

Strategy

We must consider Newton's first and second laws to analyze the situation. We need to decide which law applies; this, in turn, will tell us about the relationship between the forces.

Solution

- The forces are equal. According to Newton's first law, if the net force is zero, the velocity is constant.
- In this case, $\vec{F}_{\text{friction}}$ must be larger than $\vec{F}_{\text{drag}}$. According to Newton's second law, a net force is required to cause acceleration.

Significance

These questions may seem trivial, but they are commonly answered incorrectly. For a car or any other object to

move, it must be accelerated from rest to the desired speed; this requires that the friction force be greater than the drag force. Once the car is moving at constant velocity, the net force must be zero; otherwise, the car will accelerate (gain speed). To solve problems involving Newton's laws, we must understand whether to apply Newton's first law (where $\sum \vec{F} = \vec{0}$) or Newton's second law (where $\sum \vec{F}$ is not zero). This will be apparent as you see more examples and attempt to solve problems on your own.

EXAMPLE 5.4

What Rocket Thrust Accelerates This Sled?

Before space flights carrying astronauts, rocket sleds were used to test aircraft, missile equipment, and physiological effects on human subjects at high speeds. They consisted of a platform that was mounted on one or two rails and propelled by several rockets.

Calculate the magnitude of force exerted by each rocket, called its thrust T , for the four-rocket propulsion system shown in [Figure 5.14](#). The sled's initial acceleration is 49 m/s^2 , the mass of the system is 2100 kg , and the force of friction opposing the motion is 650 N .

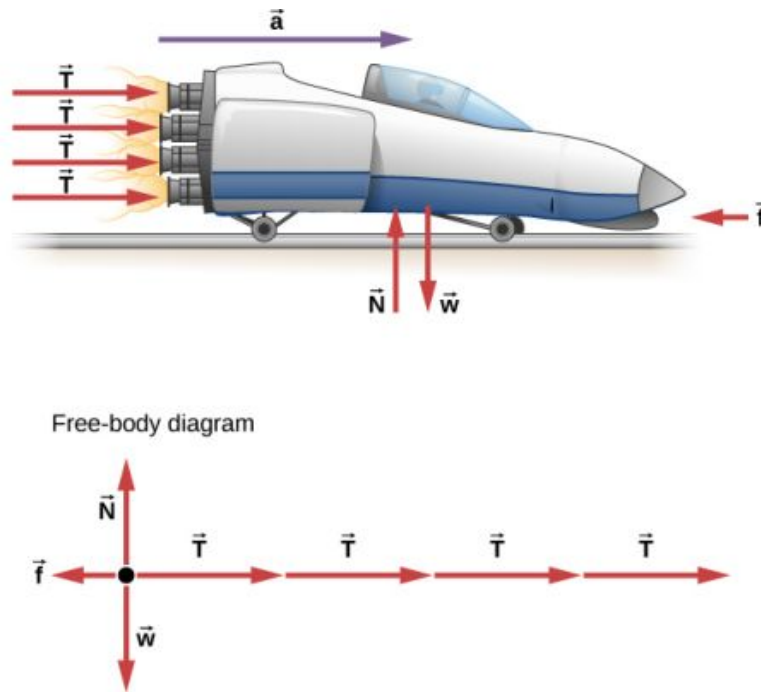


FIGURE 5.14 A sled experiences a rocket thrust that accelerates it to the right. Each rocket creates an identical thrust T . The system here is the sled, its rockets, and its rider, so none of the forces between these objects are considered. The arrow representing friction ($\vec{f}$) is drawn larger than scale.

Strategy

Although forces are acting both vertically and horizontally, we assume the vertical forces cancel because there is no vertical acceleration. This leaves us with only horizontal forces and a simpler one-dimensional problem. Directions are indicated with plus or minus signs, with right taken as the positive direction. See the free-body diagram in [Figure 5.14](#).

Solution

Since acceleration, mass, and the force of friction are given, we start with Newton's second law and look for ways to find the thrust of the engines. We have defined the direction of the force and acceleration as acting "to the right," so we need to consider only the magnitudes of these quantities in the calculations. Hence we begin with

$$F_{\text{net}} = ma$$

where F_{net} is the net force along the horizontal direction. We can see from the figure that the engine thrusts add, whereas friction opposes the thrust. In equation form, the net external force is

$$F_{\text{net}} = 4T - f.$$

Substituting this into Newton's second law gives us

$$F_{\text{net}} = ma = 4T - f.$$

Using a little algebra, we solve for the total thrust $4T$:

$$4T = ma + f.$$

Substituting known values yields

$$4T = ma + f = (2100 \text{ kg})(49 \text{ m/s}^2) + 650 \text{ N}.$$

Therefore, the total thrust is

$$4T = 1.0 \times 10^5 \text{ N},$$

and the individual thrusts are

$$T = \frac{1.0 \times 10^5 \text{ N}}{4} = 2.5 \times 10^4 \text{ N}.$$

Significance

The numbers are quite large, so the result might surprise you. Experiments such as this were performed in the early 1960s to test the limits of human endurance, and the setup was designed to protect human subjects in jet fighter emergency ejections. Speeds of 1000 km/h were obtained, with accelerations of $45 g$'s. (Recall that g , acceleration due to gravity, is 9.80 m/s^2 . When we say that acceleration is $45 g$'s, it is $45 \times 9.8 \text{ m/s}^2$, which is approximately 440 m/s^2 .) Although living subjects are not used anymore, land speeds of 10,000 km/h have been obtained with a rocket sled.

In this example, as in the preceding one, the system of interest is obvious. We see in later examples that choosing the system of interest is crucial—and the choice is not always obvious.

Newton's second law is more than a definition; it is a relationship among acceleration, force, and mass. It can help us make predictions. Each of those physical quantities can be defined independently, so the second law tells us something basic and universal about nature.

CHECK YOUR UNDERSTANDING 5.4

A 550-kg sports car collides with a 2200-kg truck, and during the collision, the net force on each vehicle is the force exerted by the other. If the magnitude of the truck's acceleration is 10 m/s^2 , what is the magnitude of the sports car's acceleration?

Component Form of Newton's Second Law

We have developed Newton's second law and presented it as a vector equation in [Equation 5.3](#). This vector equation can be written as three component equations:

$$\sum \mathbf{F}_x = m\mathbf{a}_x, \sum \mathbf{F}_y = m\mathbf{a}_y, \text{ and } \sum \mathbf{F}_z = m\mathbf{a}_z. \quad 5.5$$

The second law is a description of how a body responds mechanically to its environment. The influence of the environment is the net force $\vec{\mathbf{F}}_{\text{net}}$, the body's response is the acceleration $\vec{\mathbf{a}}$, and the strength of the response is inversely proportional to the mass m . The larger the mass of an object, the smaller its response (its acceleration) to the influence of the environment (a given net force). Therefore, a body's mass is a measure of its inertia, as we explained in [Newton's First Law](#).



EXAMPLE 5.5

Force on a Soccer Ball

A 0.400-kg soccer ball is kicked across the field by a player; it undergoes acceleration given by $\vec{a} = 3.00\hat{i} + 7.00\hat{j} \text{ m/s}^2$. Find (a) the resultant force acting on the ball and (b) the magnitude and direction of the resultant force.

Strategy

The vectors in $\hat{i}$ and $\hat{j}$ format, which indicate force direction along the x-axis and the y-axis, respectively, are involved, so we apply Newton's second law in vector form.

Solution

- a. We apply Newton's second law:

$$\vec{F}_{\text{net}} = m\vec{a} = (0.400 \text{ kg}) (3.00\hat{i} + 7.00\hat{j} \text{ m/s}^2) = 1.20\hat{i} + 2.80\hat{j} \text{ N}.$$

- b. Magnitude and direction are found using the components of $\vec{F}_{\text{net}}$:

$$F_{\text{net}} = \sqrt{(1.20 \text{ N})^2 + (2.80 \text{ N})^2} = 3.05 \text{ N} \text{ and } \theta = \tan^{-1} \left(\frac{2.80}{1.20} \right) = 66.8^\circ.$$

Significance

We must remember that Newton's second law is a vector equation. In (a), we are multiplying a vector by a scalar to determine the net force in vector form. While the vector form gives a compact representation of the force vector, it does not tell us how “big” it is, or where it goes, in intuitive terms. In (b), we are determining the actual size (magnitude) of this force and the direction in which it travels.



EXAMPLE 5.6

Mass of a Car

Find the mass of a car if a net force of $-600.0\hat{j} \text{ N}$ produces an acceleration of $-0.2\hat{j} \text{ m/s}^2$.

Strategy

Vector division is not defined, so $m = \vec{F}_{\text{net}}/\vec{a}$ cannot be performed. However, mass m is a scalar, so we can use the scalar form of Newton's second law, $m = F_{\text{net}}/a$.

Solution

We use $m = F_{\text{net}}/a$ and substitute the magnitudes of the two vectors: $F_{\text{net}} = 600.0 \text{ N}$ and $a = 0.2 \text{ m/s}^2$. Therefore,

$$m = \frac{F_{\text{net}}}{a} = \frac{600.0 \text{ N}}{0.2 \text{ m/s}^2} = 3000 \text{ kg}.$$

Significance

Force and acceleration were given in the $\hat{i}$ and $\hat{j}$ format, but the answer, mass m , is a scalar and thus is not given in $\hat{i}$ and $\hat{j}$ form.



EXAMPLE 5.7

Several Forces on a Particle

A particle of mass $m = 4.0 \text{ kg}$ is acted upon by four forces of magnitudes.

$F_1 = 10.0 \text{ N}$, $F_2 = 40.0 \text{ N}$, $F_3 = 5.0 \text{ N}$, and $F_4 = 2.0 \text{ N}$, with the directions as shown in the free-body diagram in [Figure 5.15](#). What is the acceleration of the particle?

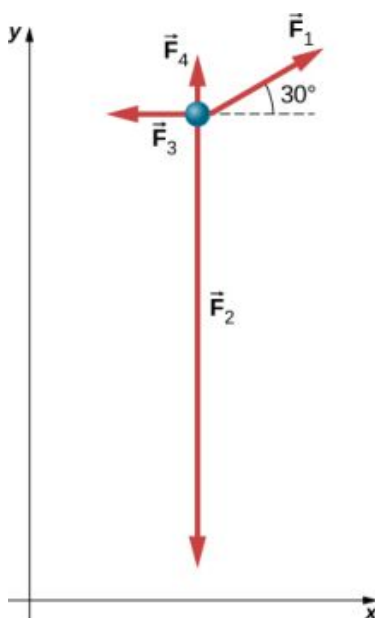


FIGURE 5.15 Four forces in the xy -plane are applied to a 4.0-kg particle.

Strategy

Because this is a two-dimensional problem, we must use a free-body diagram. First, $\vec{F}_1$ must be resolved into x - and y -components. We can then apply the second law in each direction.

Solution

We draw a free-body diagram as shown in [Figure 5.15](#). Now we apply Newton's second law. We consider all vectors resolved into x - and y -components:

$$\sum F_x = ma_x$$

$$F_{1x} - F_{3x} = ma_x$$

$$F_1 \cos 30^\circ - F_{3x} = ma_x$$

$$(10.0 \text{ N}) (\cos 30^\circ) - 5.0 \text{ N} = (4.0 \text{ kg}) a_x$$

$$a_x = 0.92 \text{ m/s}^2.$$

$$\sum F_y = ma_y$$

$$F_{1y} + F_{4y} - F_{2y} = ma_y$$

$$F_1 \sin 30^\circ + F_{4y} - F_{2y} = ma_y$$

$$(10.0 \text{ N}) (\sin 30^\circ) + 2.0 \text{ N} - 40.0 \text{ N} = (4.0 \text{ kg}) a_y$$

$$a_y = -8.3 \text{ m/s}^2.$$

Thus, the net acceleration is

$$\vec{a} = (0.92\hat{i} - 8.3\hat{j}) \text{ m/s}^2,$$

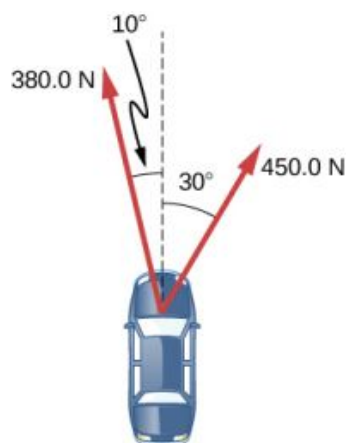
which is a vector of magnitude 8.4 m/s^2 directed at 276° to the positive x -axis.

Significance

Numerous examples in everyday life can be found that involve three or more forces acting on a single object, such as cables running from the Golden Gate Bridge or a football player being tackled by three defenders. We can see that the solution of this example is just an extension of what we have already done.

✓ CHECK YOUR UNDERSTANDING 5.5

A car has forces acting on it, as shown below. The mass of the car is 1000.0 kg. The road is slick, so friction can be ignored. (a) What is the net force on the car? (b) What is the acceleration of the car?



Newton's Second Law and Momentum

Newton actually stated his second law in terms of momentum: “The instantaneous rate at which a body’s momentum changes is equal to the net force acting on the body.” (“Instantaneous rate” implies that the derivative is involved.) This can be given by the vector equation

$$\vec{F}_{\text{net}} = \frac{d\vec{p}}{dt}. \quad 5.6$$

This means that Newton’s second law addresses the central question of motion: What causes a change in motion of an object? Momentum was described by Newton as “quantity of motion,” a way of combining both the velocity of an object and its mass. We devote [Linear Momentum and Collisions](#) to the study of momentum.

For now, it is sufficient to define *momentum* $\vec{p}$ as the product of the mass of the object m and its velocity $\vec{v}$:

$$\vec{p} = m\vec{v}. \quad 5.7$$

Since velocity is a vector, so is momentum.

It is easy to visualize momentum. A train moving at 10 m/s has more momentum than one that moves at 2 m/s. In everyday life, we speak of one sports team as “having momentum” when they score points against the opposing team.

If we substitute [Equation 5.7](#) into [Equation 5.6](#), we obtain

$$\vec{F}_{\text{net}} = \frac{d\vec{p}}{dt} = \frac{d(m\vec{v})}{dt}.$$

When m is constant, we have

$$\vec{F}_{\text{net}} = m \frac{d(\vec{v})}{dt} = m\vec{a}.$$

Thus, we see that the momentum form of Newton’s second law reduces to the form given earlier in this section.

INTERACTIVE

Explore the [forces at work \(https://openstax.org/l/21forcesatwork\)](https://openstax.org/l/21forcesatwork) when [pulling a cart \(https://openstax.org/l/21pullacart\)](https://openstax.org/l/21pullacart) or pushing a refrigerator, crate, or person. Put [an object on a ramp \(https://openstax.org/l/21ramp\)](https://openstax.org/l/21ramp) and see how it affects its motion. Engage the simulation below and see how applied forces make objects move.

[Access multimedia content \(https://openstax.org/books/university-physics-volume-1/pages/5-3-newtons-second-law\)](https://openstax.org/books/university-physics-volume-1/pages/5-3-newtons-second-law)

5.4 Mass and Weight

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Explain the difference between mass and weight
- Explain why objects falling through the air are never truly in free fall
- Describe the concept of weightlessness

Mass and weight are often used interchangeably in everyday conversation. For example, our medical records often show our weight in kilograms but never in the correct units of newtons. In physics, however, there is an important distinction. Weight is the pull of Earth on an object. It depends on the distance from the center of Earth. Unlike weight, mass does not vary with location. The mass of an object is the same on Earth, in orbit, or on the surface of the Moon.

Units of Force

The equation $F_{\text{net}} = ma$ is used to define net force in terms of mass, length, and time. As explained earlier, the SI unit of force is the newton. Since $F_{\text{net}} = ma$,

$$1 \text{ N} = 1 \text{ kg} \cdot \text{m/s}^2.$$

Although almost the entire world uses the newton for the unit of force, in the United States, the most familiar unit of force is the pound (lb), where $1 \text{ N} = 0.225 \text{ lb}$. Thus, a 225-lb person weighs 1000 N.

Weight and Gravitational Force

When an object is dropped, it accelerates toward the center of Earth. Newton's second law says that a net force on an object is responsible for its acceleration. If air resistance is negligible, the net force on a falling object is the gravitational force, commonly called its **weight** $\vec{w}$, or its force due to gravity acting on an object of mass m . Weight can be denoted as a vector because it has a direction; *down* is, by definition, the direction of gravity, and hence, weight is a downward force. The magnitude of weight is denoted as w . Galileo was instrumental in showing that, in the absence of air resistance, all objects fall with the same acceleration g . Using Galileo's result and Newton's second law, we can derive an equation for weight.

Consider an object with mass m falling toward Earth. It experiences only the downward force of gravity, which is the weight $\vec{w}$. Newton's second law says that the magnitude of the net external force on an object is $\vec{F}_{\text{net}} = m\vec{a}$. We know that the acceleration of an object due to gravity is $\vec{g}$, or $\vec{a} = \vec{g}$. Substituting these into Newton's second law gives us the following equations.

WEIGHT

The gravitational force on a mass is its weight. We can write this in vector form, where $\vec{w}$ is weight and m is mass, as

$$\vec{w} = m\vec{g}. \quad 5.8$$

In scalar form, we can write

$$w = mg. \quad 5.9$$

Since $g = 9.80 \text{ m/s}^2$ on Earth, the weight of a 1.00-kg object on Earth is 9.80 N:

$$w = mg = (1.00 \text{ kg})(9.80 \text{ m/s}^2) = 9.80 \text{ N}.$$

When the net external force on an object is its weight, we say that it is in **free fall**, that is, the only force acting on the object is gravity. However, when objects on Earth fall downward through the air, they are never truly in free fall because there is always some upward resistance force from the air acting on the object.

Acceleration due to gravity g varies slightly over the surface of Earth, so the weight of an object depends on its location and is not an intrinsic property of the object. Weight varies dramatically if we leave Earth's surface. On the

Moon, for example, acceleration due to gravity is only 1.62 m/s^2 . A 1.0-kg mass thus has a weight of 9.8 N on Earth and only about 1.6 N on the Moon.

The broadest definition of weight in this sense is that the weight of an object is the gravitational force on it from the nearest large body, such as Earth, the Moon, or the Sun. This is the most common and useful definition of weight in physics. It differs dramatically, however, from the definition of weight used by NASA and the popular media in relation to space travel and exploration. When they speak of “weightlessness” and “microgravity,” they are referring to the phenomenon we call “free fall” in physics. We use the preceding definition of weight, force $\vec{w}$ due to gravity acting on an object of mass m , and we make careful distinctions between free fall and actual weightlessness.

Be aware that weight and mass are different physical quantities, although they are closely related. Mass is an intrinsic property of an object: It is a quantity of matter. The quantity or amount of matter of an object is determined by the numbers of atoms and molecules of various types it contains. Because these numbers do not vary, in Newtonian physics, mass does not vary; therefore, its response to an applied force does not vary. In contrast, weight is the gravitational force acting on an object, so it does vary depending on gravity. For example, a person closer to the center of Earth, at a low elevation such as New Orleans, weighs slightly more than a person who is located in the higher elevation of Denver, even though they may have the same mass.

It is tempting to equate mass to weight, because most of our examples take place on Earth, where the weight of an object varies only a little with the location of the object. In addition, it is difficult to count and identify all of the atoms and molecules in an object, so mass is rarely determined in this manner. If we consider situations in which $\vec{g}$ is a constant on Earth, we see that weight $\vec{w}$ is directly proportional to mass m , since $\vec{w} = m\vec{g}$, that is, the more massive an object is, the more it weighs. Operationally, the masses of objects are determined by comparison with the standard kilogram, as we discussed in [Units and Measurement](#). But by comparing an object on Earth with one on the Moon, we can easily see a variation in weight but not in mass. For instance, on Earth, a 5.0-kg object weighs 49 N ; on the Moon, where g is 1.67 m/s^2 , the object weighs 8.4 N . However, the mass of the object is still 5.0 kg on the Moon.



EXAMPLE 5.8

Clearing a Field

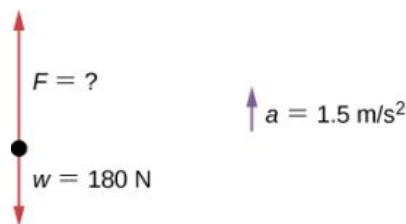
A farmer is lifting some moderately heavy rocks from a field to plant crops. He lifts a stone that weighs 40.0 lb . (about 180 N). What force does he apply if the stone accelerates at a rate of 1.5 m/s^2 ?

Strategy

We were given the weight of the stone, which we use in finding the net force on the stone. However, we also need to know its mass to apply Newton’s second law, so we must apply the equation for weight, $w = mg$, to determine the mass.

Solution

No forces act in the horizontal direction, so we can concentrate on vertical forces, as shown in the following free-body diagram. We label the acceleration to the side; technically, it is not part of the free-body diagram, but it helps to remind us that the object accelerates upward (so the net force is upward).



$$\begin{aligned}
 w &= mg \\
 m &= \frac{w}{g} = \frac{180 \text{ N}}{9.8 \text{ m/s}^2} = 18 \text{ kg} \\
 \sum F &= ma \\
 F - w &= ma \\
 F - 180 \text{ N} &= (18 \text{ kg})(1.5 \text{ m/s}^2) \\
 F - 180 \text{ N} &= 27 \text{ N} \\
 F &= 207 \text{ N} = 210 \text{ N to two significant figures}
 \end{aligned}$$

Significance

To apply Newton's second law as the primary equation in solving a problem, we sometimes have to rely on other equations, such as the one for weight or one of the kinematic equations, to complete the solution.

✓ CHECK YOUR UNDERSTANDING 5.6

For [Example 5.8](#), find the acceleration when the farmer's applied force is 230.0 N.

5.5 Newton's Third Law

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- State Newton's third law of motion
- Identify the action and reaction forces in different situations
- Apply Newton's third law to define systems and solve problems of motion

We have thus far considered force as a push or a pull; however, if you think about it, you realize that no push or pull ever occurs by itself. When you push on a wall, the wall pushes back on you. This brings us to **Newton's third law**.

NEWTON'S THIRD LAW OF MOTION

Whenever one body exerts a force on a second body, the first body experiences a force that is equal in magnitude and opposite in direction to the force that it exerts. Mathematically, if a body *A* exerts a force $\vec{F}$ on body *B*, then *B* simultaneously exerts a force $-\vec{F}$ on *A*, or in vector equation form,

$$\vec{F}_{AB} = -\vec{F}_{BA}. \quad 5.10$$

Newton's third law represents a certain symmetry in nature: Forces always occur in pairs, and one body cannot exert a force on another without experiencing a force itself. We sometimes refer to this law loosely as "action-reaction," where the force exerted is the action and the force experienced as a consequence is the reaction. Newton's third law has practical uses in analyzing the origin of forces and understanding which forces are external to a system.

We can readily see Newton's third law at work by taking a look at how people move about. Consider a swimmer pushing off the side of a pool ([Figure 5.16](#)). She pushes against the wall of the pool with her feet and accelerates in the direction opposite that of her push. The wall has exerted an equal and opposite force on the swimmer. You might think that two equal and opposite forces would cancel, but they do not *because they act on different systems*. In this case, there are two systems that we could investigate: the swimmer and the wall. If we select the swimmer to be the system of interest, as in the figure, then $F_{\text{wall on feet}}$ is an external force on this system and affects its motion. The swimmer moves in the direction of this force. In contrast, the force $F_{\text{feet on wall}}$ acts on the wall, not on our system of interest. Thus, $F_{\text{feet on wall}}$ does not directly affect the motion of the system and does not cancel $F_{\text{wall on feet}}$. The swimmer pushes in the direction opposite that in which she wishes to move. The reaction to her push is thus in the desired direction. In a free-body diagram, such as the one shown in [Figure 5.16](#), we never include both forces of an action-reaction pair; in this case, we only use $F_{\text{wall on feet}}$, not $F_{\text{feet on wall}}$.

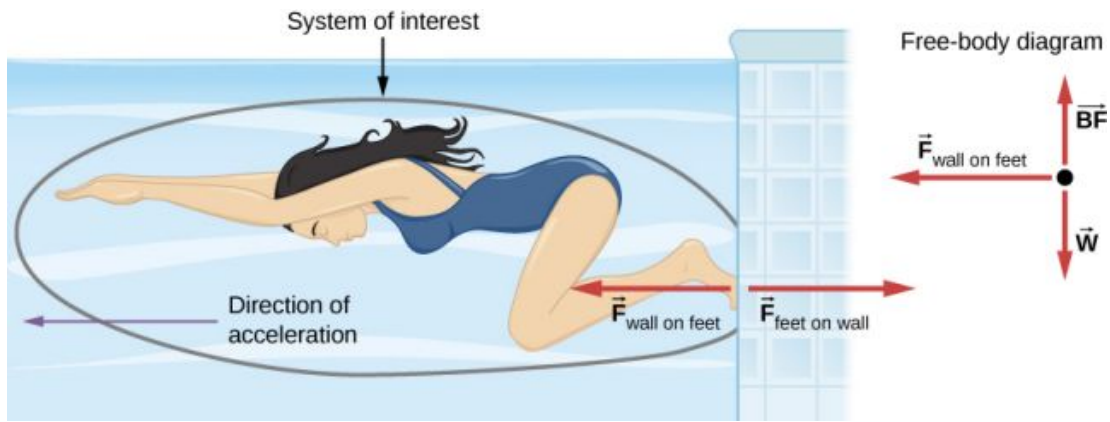


FIGURE 5.16 When the swimmer exerts a force on the wall, she accelerates in the opposite direction; in other words, the net external force on her is in the direction opposite of $F_{\text{feet on wall}}$. This opposition occurs because, in accordance with Newton's third law, the wall exerts a force $F_{\text{wall on feet}}$ on the swimmer that is equal in magnitude but in the direction opposite to the one she exerts on it. The line around the swimmer indicates the system of interest. Thus, the free-body diagram shows only $F_{\text{wall on feet}}$, w (the gravitational force), and BF , which is the buoyant force of the water supporting the swimmer's weight. The vertical forces w and BF cancel because there is no vertical acceleration.

Other examples of Newton's third law are easy to find:

- As a professor paces in front of a whiteboard, he exerts a force backward on the floor. The floor exerts a reaction force forward on the professor that causes him to accelerate forward.
- A car accelerates forward because the ground pushes forward on the drive wheels, in reaction to the drive wheels pushing backward on the ground. You can see evidence of the wheels pushing backward when tires spin on a gravel road and throw the rocks backward.
- Rockets move forward by expelling gas backward at high velocity. This means the rocket exerts a large backward force on the gas in the rocket combustion chamber; therefore, the gas exerts a large reaction force forward on the rocket. This reaction force, which pushes a body forward in response to a backward force, is called **thrust**. It is a common misconception that rockets propel themselves by pushing on the ground or on the air behind them. They actually work better in a vacuum, where they can more readily expel the exhaust gases.
- Helicopters create lift by pushing air down, thereby experiencing an upward reaction force.
- Birds and airplanes also fly by exerting force on the air in a direction opposite that of whatever force they need. For example, the wings of a bird force air downward and backward to get lift and move forward.
- An octopus propels itself in the water by ejecting water through a funnel from its body, similar to a jet ski.
- When a person pulls down on a vertical rope, the rope pulls up on the person ([Figure 5.17](#)).

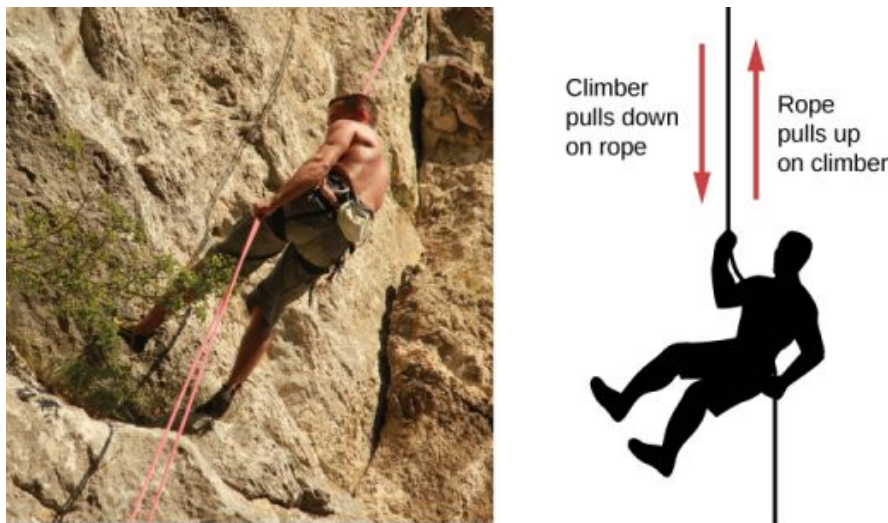


FIGURE 5.17 When the mountain climber pulls down on the rope, the rope pulls up on the mountain climber. (credit left: modification of work by Cristian Bortes)

There are two important features of Newton's third law. First, the forces exerted (the action and reaction) are always equal in magnitude but opposite in direction. Second, these forces are acting on different bodies or systems: A 's force acts on B and B 's force acts on A . In other words, the two forces are distinct forces that do not act on the same body. Thus, they do not cancel each other.

For the situation shown in [Figure 5.6](#), the third law indicates that because the chair is pushing upward on the boy with force $\vec{C}$, he is pushing downward on the chair with force $-\vec{C}$. Similarly, he is pushing downward with forces $-\vec{F}$ and $-\vec{T}$ on the floor and table, respectively. Finally, since Earth pulls downward on the boy with force $\vec{w}$, he pulls upward on Earth with force $-\vec{w}$. If that student were to angrily pound the table in frustration, he would quickly learn the painful lesson (avoidable by studying Newton's laws) that the table hits back just as hard.

A person who is walking or running applies Newton's third law instinctively. For example, the runner in [Figure 5.18](#) pushes backward on the ground so that it pushes him forward.

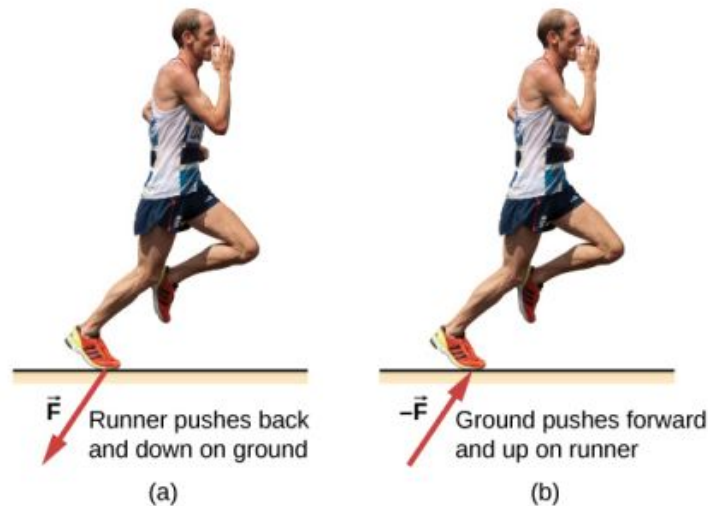


FIGURE 5.18 The runner experiences Newton's third law. (a) A force is exerted by the runner on the ground. (b) The reaction force of the ground on the runner pushes him forward. (credit "runner": modification of work by "Greenwich Photography"/Flickr)

EXAMPLE 5.9

Forces on a Stationary Object

The package in [Figure 5.19](#) is sitting on a scale. For the scale and package at rest (i.e., not accelerating), show that the scale indicates the weight of the package.

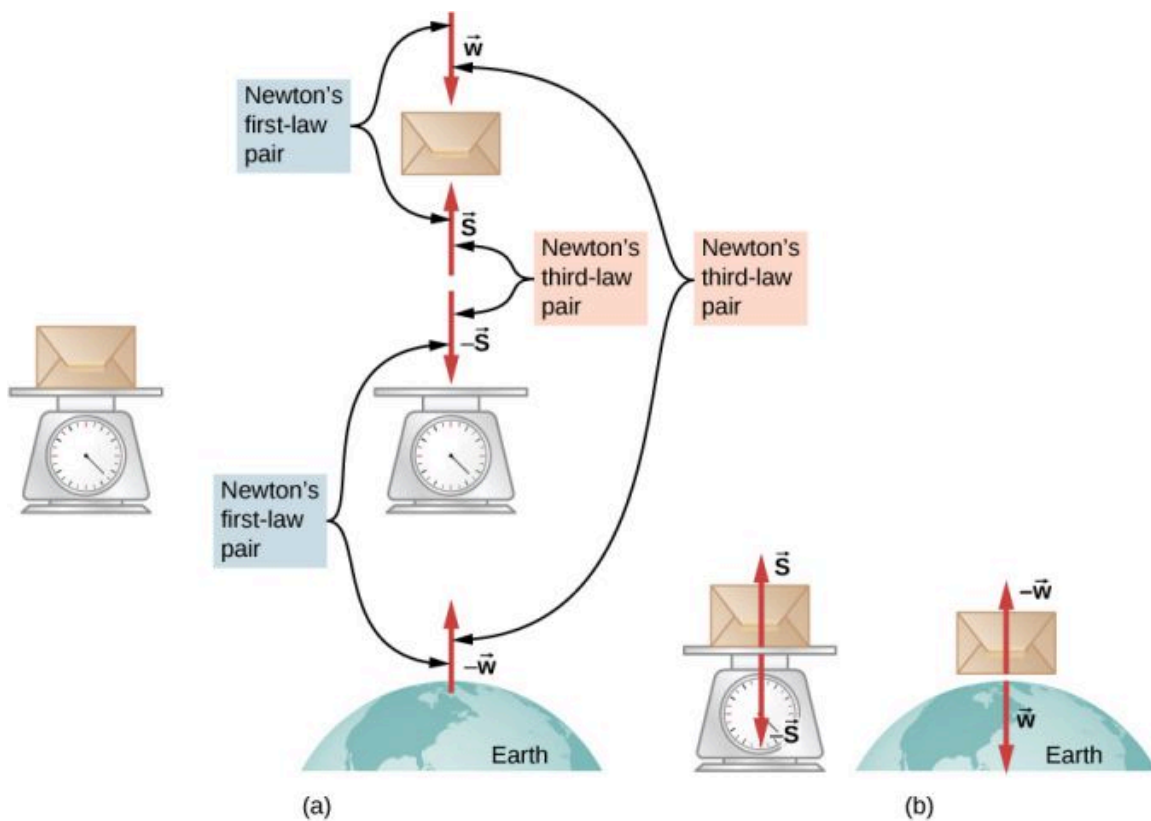


FIGURE 5.19 (a) The forces on a package sitting on a scale, along with their reaction forces. The force $\vec{w}$ is the weight of the package (the force due to Earth's gravity) and $\vec{S}$ is the force of the scale on the package. (b) Isolation of the package-scale system and the package-Earth system makes the action and reaction pairs clear.

Strategy

The forces on the package are $\vec{S}$, which is due to the scale, and $-\vec{w}$, which is due to Earth's gravitational field. The reaction forces that the package exerts are $-\vec{S}$ on the scale and $\vec{w}$ on Earth. Because the package is not accelerating, application of the second law yields

$$\vec{S} - \vec{w} = m\vec{a} = \vec{0},$$

so

$$\vec{S} = \vec{w}.$$

The scale reading indicates the magnitude of the force $\vec{S}$, which is equal to the magnitude of the package's weight.

Significance

The scale actually does not measure the weight of the package; it measures the force $-\vec{S}$ on its surface. If the system is accelerating, $\vec{S}$ and $-\vec{w}$ would not be equal, as explained in [Applications of Newton's Laws](#).



EXAMPLE 5.10

Getting Up to Speed: Choosing the Correct System

A physics professor pushes a cart of demonstration equipment to a lecture hall ([Figure 5.20](#)). Her mass is 65.0 kg, the cart's mass is 12.0 kg, and the equipment's mass is 7.0 kg. Calculate the acceleration produced when the professor exerts a backward force of 150 N on the floor. All forces opposing the motion, such as friction on the cart's wheels and air resistance, total 24.0 N.

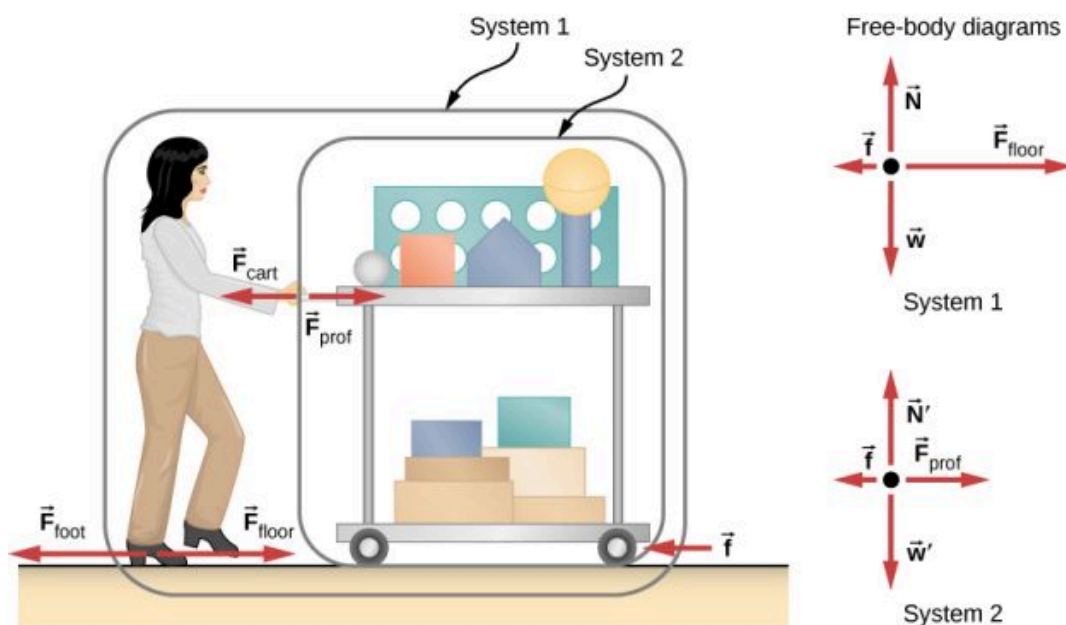


FIGURE 5.20 A professor pushes the cart with her demonstration equipment. The lengths of the arrows are proportional to the magnitudes of the forces (except for $\vec{f}$, because it is too small to be drawn to scale). System 1 is appropriate for this example, because it asks for the acceleration of the entire group of objects. Only $\vec{F}_{\text{floor}}$ and $\vec{f}$ are external forces acting on System 1 along the line of motion. All other forces either cancel or act on the outside world. System 2 is chosen for the next example so that $\vec{F}_{\text{prof}}$ is an external force and enters into Newton's second law. The free-body diagrams, which serve as the basis for Newton's second law, vary with the system chosen.

Strategy

Since they accelerate as a unit, we define the system to be the professor, cart, and equipment. This is System 1 in [Figure 5.20](#). The professor pushes backward with a force F_{foot} of 150 N. According to Newton's third law, the floor exerts a forward reaction force F_{floor} of 150 N on System 1. Because all motion is horizontal, we can assume there is no net force in the vertical direction. Therefore, the problem is one-dimensional along the horizontal direction. As noted, friction f opposes the motion and is thus in the opposite direction of F_{floor} . We do not include the forces F_{prof} or F_{cart} because these are internal forces, and we do not include F_{foot} because it acts on the floor, not on the system. There are no other significant forces acting on System 1. If the net external force can be found from all this information, we can use Newton's second law to find the acceleration as requested. See the free-body diagram in the figure.

Solution

Newton's second law is given by

$$a = \frac{F_{\text{net}}}{m}.$$

The net external force on System 1 is deduced from [Figure 5.20](#) and the preceding discussion to be

$$F_{\text{net}} = F_{\text{floor}} - f = 150 \text{ N} - 24.0 \text{ N} = 126 \text{ N}.$$

The mass of System 1 is

$$m = (65.0 + 12.0 + 7.0) \text{ kg} = 84 \text{ kg}.$$

These values of F_{net} and m produce an acceleration of

$$a = \frac{F_{\text{net}}}{m} = \frac{126 \text{ N}}{84 \text{ kg}} = 1.5 \text{ m/s}^2.$$

Significance

None of the forces between components of System 1, such as between the professor's hands and the cart, contribute to the net external force because they are internal to System 1. Another way to look at this is that forces

between components of a system cancel because they are equal in magnitude and opposite in direction. For example, the force exerted by the professor on the cart results in an equal and opposite force back on the professor. In this case, both forces act on the same system and therefore cancel. Thus, internal forces (between components of a system) cancel. Choosing System 1 was crucial to solving this problem.



EXAMPLE 5.11

Force on the Cart: Choosing a New System

Calculate the force the professor exerts on the cart in [Figure 5.20](#), using data from the previous example if needed.

Strategy

If we define the system of interest as the cart plus the equipment (System 2 in [Figure 5.20](#)), then the net external force on System 2 is the force the professor exerts on the cart minus friction. The force she exerts on the cart, F_{prof} , is an external force acting on System 2. F_{prof} was internal to System 1, but it is external to System 2 and thus enters Newton's second law for this system.

Solution

Newton's second law can be used to find F_{prof} . We start with

$$a = \frac{F_{\text{net}}}{m}.$$

The magnitude of the net external force on System 2 is

$$F_{\text{net}} = F_{\text{prof}} - f.$$

We solve for F_{prof} , the desired quantity:

$$F_{\text{prof}} = F_{\text{net}} + f.$$

The value of f is given, so we must calculate net F_{net} . That can be done because both the acceleration and the mass of System 2 are known. Using Newton's second law, we see that

$$F_{\text{net}} = ma,$$

where the mass of System 2 is 19.0 kg ($m = 12.0 \text{ kg} + 7.0 \text{ kg}$) and its acceleration was found to be $a = 1.5 \text{ m/s}^2$ in the previous example. Thus,

$$F_{\text{net}} = ma = (19.0 \text{ kg})(1.5 \text{ m/s}^2) = 29 \text{ N}.$$

Now we can find the desired force:

$$F_{\text{prof}} = F_{\text{net}} + f = 29 \text{ N} + 24.0 \text{ N} = 53 \text{ N}.$$

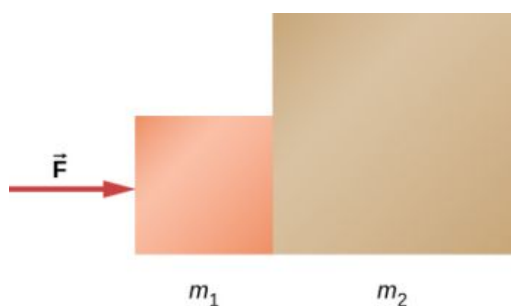
Significance

This force is significantly less than the 150-N force the professor exerted backward on the floor. Not all of that 150-N force is transmitted to the cart; some of it accelerates the professor. The choice of a system is an important analytical step both in solving problems and in thoroughly understanding the physics of the situation (which are not necessarily the same things).



CHECK YOUR UNDERSTANDING 5.7

Two blocks are at rest and in contact on a frictionless surface as shown below, with $m_1 = 2.0 \text{ kg}$, $m_2 = 6.0 \text{ kg}$, and applied force 24 N. (a) Find the acceleration of the system of blocks. (b) Suppose that the blocks are later separated. What force will give the second block, with the mass of 6.0 kg, the same acceleration as the system of blocks?



INTERACTIVE

View this [video \(https://openstax.org/l/21actionreact\)](https://openstax.org/l/21actionreact) to watch examples of action and reaction.

INTERACTIVE

View this [video \(https://openstax.org/l/21NewtonsLaws\)](https://openstax.org/l/21NewtonsLaws) to watch examples of Newton's laws and internal and external forces.

5.6 Common Forces

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Define normal and tension forces
- Distinguish between real and fictitious forces
- Apply Newton's laws of motion to solve problems involving a variety of forces

Forces are given many names, such as push, pull, thrust, and weight. Traditionally, forces have been grouped into several categories and given names relating to their source, how they are transmitted, or their effects. Several of these categories are discussed in this section, together with some interesting applications. Further examples of forces are discussed later in this text.

A Catalog of Forces: Normal, Tension, and Other Examples of Forces

A catalog of forces will be useful for reference as we solve various problems involving force and motion. These forces include normal force, tension, friction, and spring force.

Normal force

Weight (also called the force of gravity) is a pervasive force that acts at all times and must be counteracted to keep an object from falling. You must support the weight of a heavy object by pushing up on it when you hold it stationary, as illustrated in [Figure 5.21\(a\)](#). But how do inanimate objects like a table support the weight of a mass placed on them, such as shown in [Figure 5.21\(b\)](#)? When the bag of dog food is placed on the table, the table sags slightly under the load. This would be noticeable if the load were placed on a card table, but even a sturdy oak table deforms when a force is applied to it. Unless an object is deformed beyond its limit, it will exert a restoring force much like a deformed spring (or a trampoline or diving board). The greater the deformation, the greater the restoring force. Thus, when the load is placed on the table, the table sags until the restoring force becomes as large as the weight of the load. At this point, the net external force on the load is zero. That is the situation when the load is stationary on the table. The table sags quickly and the sag is slight, so we do not notice it. But it is similar to the sagging of a trampoline when you climb onto it.

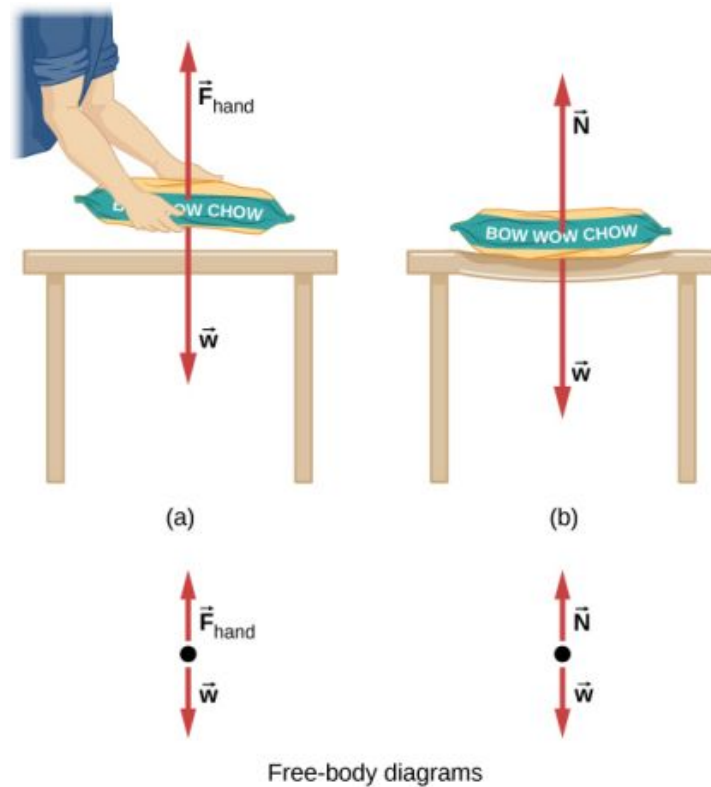


FIGURE 5.21 (a) The person holding the bag of dog food must supply an upward force $\vec{F}_{\text{hand}}$ equal in magnitude and opposite in direction to the weight of the food $\vec{w}$ so that it doesn't drop to the ground. (b) The card table sags when the dog food is placed on it, much like a stiff trampoline. Elastic restoring forces in the table grow as it sags until they supply a force $\vec{N}$ equal in magnitude and opposite in direction to the weight of the load.

We must conclude that whatever supports a load, be it animate or not, must supply an upward force equal to the weight of the load, as we assumed in a few of the previous examples. If the force supporting the weight of an object, or a load, is perpendicular to the surface of contact between the load and its support, this force is defined as a **normal force** and here is given by the symbol $\vec{N}$. (This is not the newton unit for force, or N.) The word *normal* means perpendicular to a surface. This means that the normal force experienced by an object resting on a horizontal surface can be expressed in vector form as follows:

$$\vec{N} = -m\vec{g}.$$

In scalar form, this becomes

$$N = mg.$$

The normal force can be less than the object's weight if the object is on an incline.



EXAMPLE 5.12

Weight on an Incline

Consider the skier on the slope in [Figure 5.22](#). Her mass including equipment is 60.0 kg. (a) What is her acceleration if friction is negligible? (b) What is her acceleration if friction is 45.0 N?

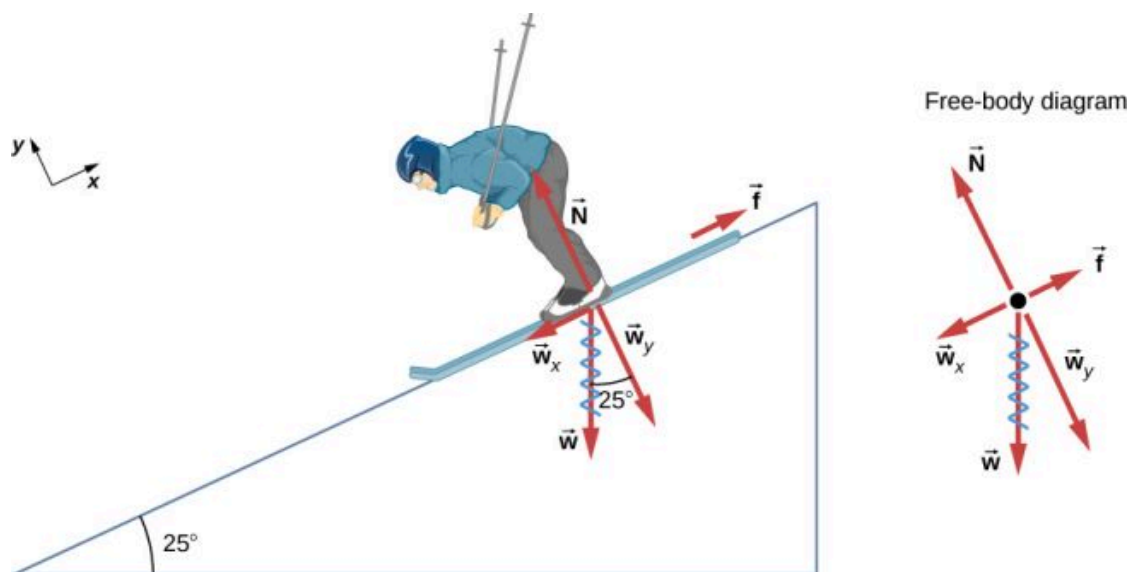


FIGURE 5.22 Since the acceleration is parallel to the slope and acting down the slope, it is most convenient to project all forces onto a coordinate system where one axis is parallel to the slope and the other is perpendicular to it (axes shown to the left of the skier). $\vec{N}$ is perpendicular to the slope and $\vec{f}$ is parallel to the slope, but $\vec{w}$ has components along both axes, namely, w_y and w_x . Here, $\vec{w}$ has a squiggly line to show that it has been replaced by these components. The force $\vec{N}$ is equal in magnitude to w_y , so there is no acceleration perpendicular to the slope, but f is less than w_x , so there is a downslope acceleration (along the axis parallel to the slope).

Strategy

This is a two-dimensional problem, since not all forces on the skier (the system of interest) are parallel. The approach we have used in two-dimensional kinematics also works well here. Choose a convenient coordinate system and project the vectors onto its axes, creating two one-dimensional problems to solve. The most convenient coordinate system for motion on an incline is one that has one coordinate parallel to the slope and one perpendicular to the slope. (Motions along mutually perpendicular axes are independent.) We use x and y for the parallel and perpendicular directions, respectively. This choice of axes simplifies this type of problem, because there is no motion perpendicular to the slope and the acceleration is downslope. Regarding the forces, friction is drawn in opposition to motion (friction always opposes forward motion) and is always parallel to the slope, w_x is drawn parallel to the slope and downslope (it causes the motion of the skier down the slope), and w_y is drawn as the component of weight perpendicular to the slope. Then, we can consider the separate problems of forces parallel to the slope and forces perpendicular to the slope.

Solution

The magnitude of the component of weight parallel to the slope is

$$w_x = w \sin 25^\circ = mg \sin 25^\circ,$$

and the magnitude of the component of the weight perpendicular to the slope is

$$w_y = w \cos 25^\circ = mg \cos 25^\circ.$$

a. Neglect friction. Since the acceleration is parallel to the slope, we need only consider forces parallel to the slope. (Forces perpendicular to the slope add to zero, since there is no acceleration in that direction.) The forces parallel to the slope are the component of the skier's weight parallel to slope w_x and friction f . Using Newton's second law, with subscripts to denote quantities parallel to the slope,

$$a_x = \frac{F_{\text{net } x}}{m}$$

where $F_{\text{net } x} = w_x = -mg \sin 25^\circ$, assuming no friction for this part. Therefore,

$$a_x = \frac{F_{\text{net } x}}{m} = \frac{mg \sin 25^\circ}{m} = g \sin 25^\circ$$

$$(9.80 \text{ m/s}^2)(0.4226) = 4.14 \text{ m/s}^2$$

is the acceleration.

b. Include friction. We have a given value for friction, and we know its direction is parallel to the slope and it opposes motion between surfaces in contact. So the net external force is

$$F_{\text{net } x} = w_x - f.$$

Substituting this into Newton's second law, $a_x = F_{\text{net } x}/m$, gives

$$a_x = \frac{F_{\text{net } x}}{m} = \frac{w_x - f}{m} = \frac{mg \sin 25^\circ - f}{m}.$$

We substitute known values to obtain

$$a_x = \frac{(60.0 \text{ kg})(9.80 \text{ m/s}^2)(0.4226) - 45.0 \text{ N}}{60.0 \text{ kg}}.$$

This gives us

$$a_x = 3.39 \text{ m/s}^2,$$

which is the acceleration parallel to the incline when there is 45.0 N of opposing friction.

Significance

Since friction always opposes motion between surfaces, the acceleration is smaller when there is friction than when there is none. It is a general result that if friction on an incline is negligible, then the acceleration down the incline is $a = g \sin \theta$, regardless of mass. As discussed previously, all objects fall with the same acceleration in the absence of air resistance. Similarly, all objects, regardless of mass, slide down a frictionless incline with the same acceleration (if the angle is the same).

When an object rests on an incline that makes an angle θ with the horizontal, the force of gravity acting on the object is divided into two components: a force acting perpendicular to the plane, w_y , and a force acting parallel to the plane, w_x (Figure 5.23). The normal force $\vec{N}$ is typically equal in magnitude and opposite in direction to the perpendicular component of the weight w_y . The force acting parallel to the plane, w_x , causes the object to accelerate down the incline.

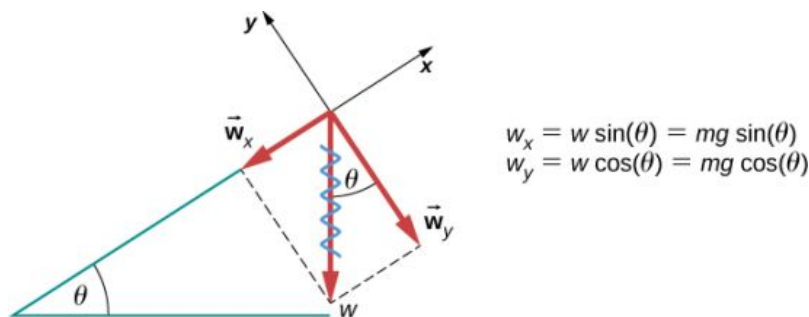


FIGURE 5.23 An object rests on an incline that makes an angle θ with the horizontal.

Be careful when resolving the weight of the object into components. If the incline is at an angle θ to the horizontal, then the magnitudes of the weight components are

$$w_x = w \sin \theta = mg \sin \theta$$

and

$$w_y = w \cos \theta = mg \cos \theta.$$

We use the second equation to write the normal force experienced by an object resting on an inclined plane:

$$N = mg \cos \theta,$$

object at rest or sliding on an inclined plane. Instead of memorizing these equations, it is helpful to be able to determine them from reason. To do this, we draw the right angle formed by the three weight vectors. The angle θ of the incline is the same as the angle formed between w and w_y . Knowing this property, we can use trigonometry to determine the magnitude of the weight components:

$$\begin{aligned}\cos \theta &= \frac{w_y}{w}, & w_y &= w \cos \theta = mg \cos \theta \\ \sin \theta &= \frac{w_x}{w}, & w_x &= w \sin \theta = mg \sin \theta.\end{aligned}$$

CHECK YOUR UNDERSTANDING 5.8

A force of 1150 N acts parallel to a ramp to push a 250-kg gun safe into a moving van. The ramp is frictionless and inclined at 17° . (a) What is the acceleration of the safe up the ramp? (b) If we consider friction in this problem, with a friction force of 120 N, what is the acceleration of the safe?

Tension

A **tension** is a force along the length of a medium; in particular, it is a pulling force that acts along a stretched flexible connector, such as a rope or cable. The word “tension” comes from a Latin word meaning “to stretch.” Not coincidentally, the flexible cords that carry muscle forces to other parts of the body are called *tendons*.

Any flexible connector, such as a string, rope, chain, wire, or cable, can only exert a pull parallel to its length; thus, a force carried by a flexible connector is a tension with a direction parallel to the connector. Tension is a pull in a connector. Consider the phrase: “You can’t push a rope.” Instead, tension force pulls outward along the two ends of a rope.

Consider a person holding a mass on a rope, as shown in [Figure 5.24](#). If the 5.00-kg mass in the figure is stationary, then its acceleration is zero and the net force is zero. The only external forces acting on the mass are its weight and the tension supplied by the rope. Thus,

$$F_{\text{net}} = T - w = 0,$$

where T and w are the magnitudes of the tension and weight, respectively, and their signs indicate direction, with up being positive. As we proved using Newton’s second law, the tension equals the weight of the supported mass:

$$T = w = mg,$$

object hanging at rest. Thus, for a 5.00-kg mass (neglecting the mass of the rope), we see that

$$T = mg = (5.00 \text{ kg}) (9.80 \text{ m/s}^2) = 49.0 \text{ N}.$$

If we cut the rope and insert a spring, the spring would extend a length corresponding to a force of 49.0 N, providing a direct observation and measure of the tension force in the rope.

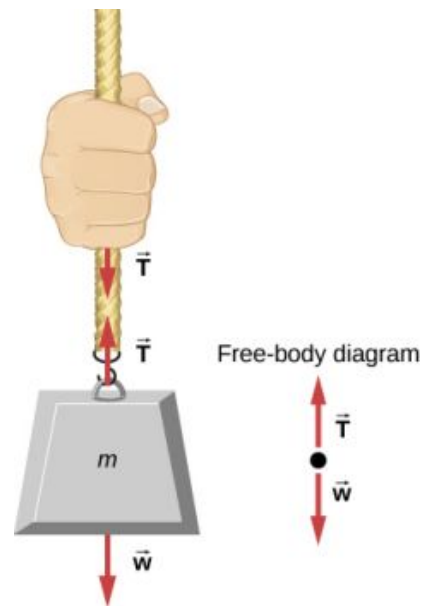


FIGURE 5.24 When a perfectly flexible connector (one requiring no force to bend it) such as this rope transmits a force $\vec{T}$, that force must be parallel to the length of the rope, as shown. By Newton's third law, the rope pulls with equal force but in opposite directions on the hand and the supported mass (neglecting the weight of the rope). The rope is the medium that carries the equal and opposite forces between the two objects. The tension anywhere in the rope between the hand and the mass is equal. Once you have determined the tension in one location, you have determined the tension at all locations along the rope.

Flexible connectors are often used to transmit forces around corners, such as in a hospital traction system, a tendon, or a bicycle brake cable. If there is no friction, the tension transmission is undiminished; only its direction changes, and it is always parallel to the flexible connector, as shown in [Figure 5.25](#).

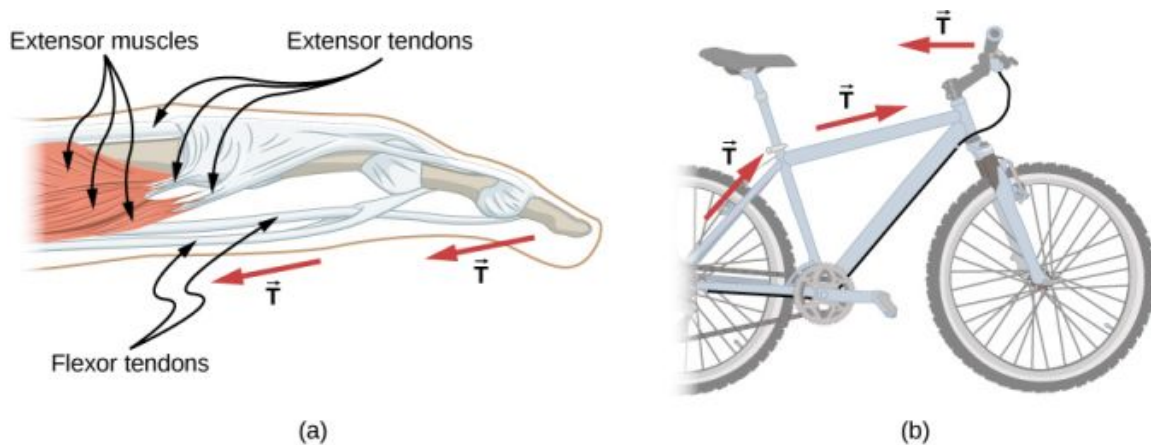


FIGURE 5.25 (a) Tendons in the finger carry force T from the muscles to other parts of the finger, usually changing the force's direction but not its magnitude (the tendons are relatively friction free). (b) The brake cable on a bicycle carries the tension T from the brake lever on the handlebars to the brake mechanism. Again, the direction but not the magnitude of T is changed.



EXAMPLE 5.13

What Is the Tension in a Tightrope?

Calculate the tension in the wire supporting the 70.0-kg tightrope walker shown in [Figure 5.26](#).

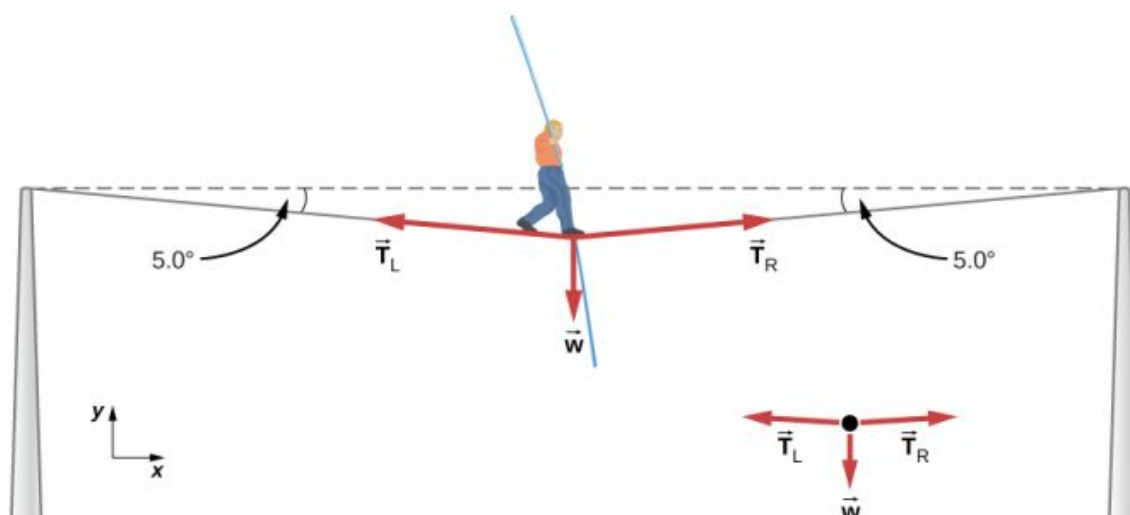


FIGURE 5.26 The weight of a tightrope walker causes a wire to sag by 5.0° . The system of interest is the point in the wire at which the tightrope walker is standing.

Strategy

As you can see in [Figure 5.26](#), the wire is bent under the person's weight. Thus, the tension on either side of the person has an upward component that can support his weight. As usual, forces are vectors represented pictorially by arrows that have the same direction as the forces and lengths proportional to their magnitudes. The system is the tightrope walker, and the only external forces acting on him are his weight $\vec{W}$ and the two tensions $\vec{T}_L$ (left tension) and $\vec{T}_R$ (right tension). It is reasonable to neglect the weight of the wire. The net external force is zero, because the system is static. We can use trigonometry to find the tensions. One conclusion is possible at the outset—we can see from [Figure 5.26\(b\)](#) that the magnitudes of the tensions T_L and T_R must be equal. We know this because there is no horizontal acceleration in the rope and the only forces acting to the left and right are T_L and T_R . Thus, the magnitude of those horizontal components of the forces must be equal so that they cancel each other out.

Whenever we have two-dimensional vector problems in which no two vectors are parallel, the easiest method of solution is to pick a convenient coordinate system and project the vectors onto its axes. In this case, the best coordinate system has one horizontal axis (x) and one vertical axis (y).

Solution

First, we need to resolve the tension vectors into their horizontal and vertical components. It helps to look at a new free-body diagram showing all horizontal and vertical components of each force acting on the system ([Figure 5.27](#)).

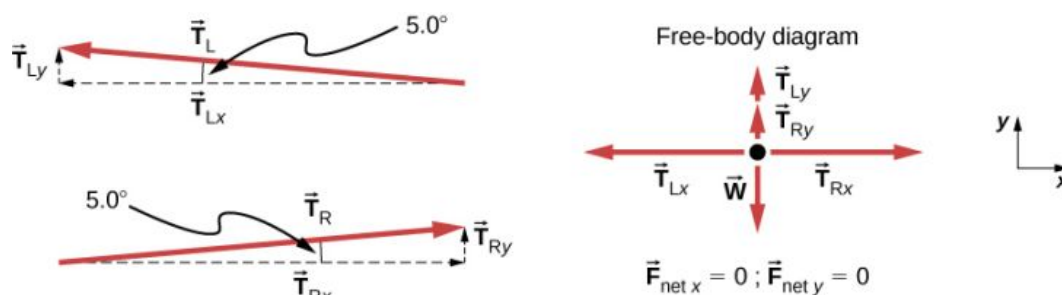


FIGURE 5.27 When the vectors are projected onto vertical and horizontal axes, their components along these axes must add to zero, since the tightrope walker is stationary. The small angle results in T being much greater than w .

Consider the horizontal components of the forces (denoted with a subscript x):

$$F_{\text{net } x} = T_{Rx} - T_{Lx}.$$

The net external horizontal force $F_{\text{net } x} = 0$, since the person is stationary. Thus,

$$F_{\text{net } x} = 0 = T_{Rx} - T_{Lx}$$

$$T_{Lx} = T_{Rx}.$$

Now observe [Figure 5.27](#). You can use trigonometry to determine the magnitude of T_L and T_R :

$$\cos 5.0^\circ = \frac{T_{Lx}}{T_L}, \quad T_{Lx} = T_L \cos 5.0^\circ$$

$$\cos 5.0^\circ = \frac{T_{Rx}}{T_R}, \quad T_{Rx} = T_R \cos 5.0^\circ.$$

Equating T_{Lx} and T_{Rx} :

$$T_L \cos 5.0^\circ = T_R \cos 5.0^\circ.$$

Thus,

$$T_L = T_R = T,$$

as predicted. Now, considering the vertical components (denoted by a subscript y), we can solve for T . Again, since the person is stationary, Newton's second law implies that $F_{\text{net } y} = 0$. Thus, as illustrated in the free-body diagram,

$$F_{\text{net } y} = T_{Ly} + T_{Ry} - w = 0.$$

We can use trigonometry to determine the relationships among T_{Ly} , T_{Ry} , and T . As we determined from the analysis in the horizontal direction, $T_L = T_R = T$:

$$\sin 5.0^\circ = \frac{T_{Ly}}{T_L}, \quad T_{Ly} = T_L \sin 5.0^\circ = T \sin 5.0^\circ$$

$$\sin 5.0^\circ = \frac{T_{Ry}}{T_R}, \quad T_{Ry} = T_R \sin 5.0^\circ = T \sin 5.0^\circ.$$

Now we can substitute the values for T_{Ly} and T_{Ry} , into the net force equation in the vertical direction:

$$F_{\text{net } y} = T_{Ly} + T_{Ry} - w = 0$$

$$F_{\text{net } y} = T \sin 5.0^\circ + T \sin 5.0^\circ - w = 0$$

$$2T \sin 5.0^\circ - w = 0$$

$$2T \sin 5.0^\circ = w$$

and

$$T = \frac{w}{2 \sin 5.0^\circ} = \frac{mg}{2 \sin 5.0^\circ},$$

so

$$T = \frac{(70.0 \text{ kg})(9.80 \text{ m/s}^2)}{2(0.0872)},$$

and the tension is

$$T = 3930 \text{ N}.$$

Significance

The vertical tension in the wire acts as a force that supports the weight of the tightrope walker. The tension is almost six times the 686-N weight of the tightrope walker. Since the wire is nearly horizontal, the vertical component of its tension is only a fraction of the tension in the wire. The large horizontal components are in opposite directions and cancel, so most of the tension in the wire is not used to support the weight of the tightrope walker.

If we wish to create a large tension, all we have to do is exert a force perpendicular to a taut flexible connector, as

illustrated in [Figure 5.26](#). As we saw in [Example 5.13](#), the weight of the tightrope walker acts as a force perpendicular to the rope. We saw that the tension in the rope is related to the weight of the tightrope walker in the following way:

$$T = \frac{w}{2 \sin \theta}.$$

We can extend this expression to describe the tension T created when a perpendicular force ($F_{\perp}$) is exerted at the middle of a flexible connector:

$$T = \frac{F_{\perp}}{2 \sin \theta}.$$

The angle between the horizontal and the bent connector is represented by θ . In this case, T becomes large as θ approaches zero. Even the relatively small weight of any flexible connector will cause it to sag, since an infinite tension would result if it were horizontal (i.e., $\theta = 0$ and $\sin \theta = 0$). For example, [Figure 5.28](#) shows a situation where we wish to pull a car out of the mud when no tow truck is available. Each time the car moves forward, the chain is tightened to keep it as straight as possible. The tension in the chain is given by $T = \frac{F_{\perp}}{2 \sin \theta}$, and since θ is small, T is large. This situation is analogous to the tightrope walker, except that the tensions shown here are those transmitted to the car and the tree rather than those acting at the point where $F_{\perp}$ is applied.

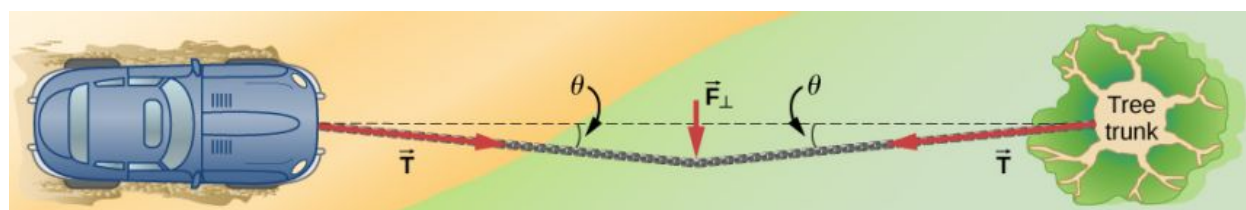


FIGURE 5.28 We can create a large tension in the chain—and potentially a big mess—by pushing on it perpendicular to its length, as shown.

✓ CHECK YOUR UNDERSTANDING 5.9

One end of a 3.0-m rope is tied to a tree; the other end is tied to a car stuck in the mud. The motorist pulls sideways on the midpoint of the rope, displacing it a distance of 0.25 m. If he exerts a force of 200.0 N under these conditions, determine the force exerted on the car.

In [Applications of Newton's Laws](#), we extend the discussion on tension in a cable to include cases in which the angles shown are not equal.

Friction

Friction is a resistive force opposing motion or its tendency. Imagine an object at rest on a horizontal surface. The net force acting on the object must be zero, leading to equality of the weight and the normal force, which act in opposite directions. If the surface is tilted, the normal force balances the component of the weight perpendicular to the surface. If the object does not slide downward, the component of the weight parallel to the inclined plane is balanced by friction. Friction is discussed in greater detail in the next chapter.

Spring force

A spring is a special medium with a specific atomic structure that has the ability to restore its shape, if deformed. To restore its shape, a spring exerts a restoring force that is proportional to and in the opposite direction in which it is stretched or compressed. This is the statement of a law known as Hooke's law, which has the mathematical form

$$\vec{F} = -k\vec{x}. \quad 5.11$$

The constant of proportionality k is a measure of the spring's stiffness. The line of action of this force is parallel to the spring axis, and the sense of the force is in the opposite direction of the displacement vector ([Figure 5.29](#)). The displacement must be measured from the relaxed position; $x = 0$ when the spring is relaxed.

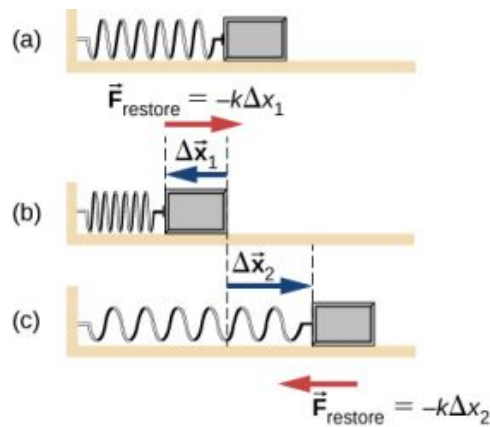


FIGURE 5.29 A spring exerts its force proportional to a displacement, whether it is compressed or stretched. (a) The spring is in a relaxed position and exerts no force on the block. (b) The spring is compressed by displacement $\Delta\vec{x}_1$ of the object and exerts restoring force $-k\Delta\vec{x}_1$. (c) The spring is stretched by displacement $\Delta\vec{x}_2$ of the object and exerts restoring force $-k\Delta\vec{x}_2$.

Real Forces and Inertial Frames

There is another distinction among forces: Some forces are real, whereas others are not. *Real forces* have some physical origin, such as a gravitational pull. In contrast, *fictitious forces* arise simply because an observer is in an accelerating or noninertial frame of reference, such as one that rotates (like a merry-go-round) or undergoes linear acceleration (like a car slowing down). For example, if a satellite is heading due north above Earth's Northern Hemisphere, then to an observer on Earth, it will appear to experience a force to the west that has no physical origin. Instead, Earth is rotating toward the east and moves east under the satellite. In Earth's frame, this looks like a westward force on the satellite, or it can be interpreted as a violation of Newton's first law (the law of inertia). We can identify a fictitious force by asking the question, "What is the reaction force?" If we cannot name the reaction force, then the force we are considering is fictitious. In the example of the satellite, the reaction force would have to be an eastward force on Earth. Recall that an inertial frame of reference is one in which all forces are real and, equivalently, one in which Newton's laws have the simple forms given in this chapter.

Earth's rotation is slow enough that Earth is nearly an inertial frame. You ordinarily must perform precise experiments to observe fictitious forces and the slight departures from Newton's laws, such as the effect just described. On a large scale, such as for the rotation of weather systems and ocean currents, the effects can be easily observed ([Figure 5.30](#)).



FIGURE 5.30 Hurricane Fran is shown heading toward the southeastern coast of the United States in September 1996. Notice the characteristic "eye" shape of the hurricane. This is a result of the Coriolis effect, which is the deflection of objects (in this case, air) when considered in a rotating frame of reference, like the spin of Earth. This hurricane shows a counter-clockwise rotation because it is a low pressure storm. (credit "runner": modification of work by "Greenwich Photography"/Flickr)

The crucial factor in determining whether a frame of reference is inertial is whether it accelerates or rotates relative to a known inertial frame. Unless stated otherwise, all phenomena discussed in this text are in inertial frames.

The forces discussed in this section are real forces, but they are not the only real forces. Lift and thrust, for example, are more specialized real forces. In the long list of forces, are some more basic than others? Are some different manifestations of the same underlying force? The answer to both questions is yes, as you will see in the treatment of modern physics later in the text.

INTERACTIVE

Explore forces and motion in this [interactive simulation \(https://openstax.org/l/21ramp\)](https://openstax.org/l/21ramp) as you push household objects up and down a ramp. Lower and raise the ramp to see how the angle of inclination affects the parallel forces. Graphs show forces, energy, and work.

INTERACTIVE

Stretch and compress springs in the simulation below to explore the relationships among force, spring constant, and displacement. Investigate what happens when two springs are connected in series and in parallel.

[Access multimedia content \(https://openstax.org/books/university-physics-volume-1/pages/5-6-common-forces\)](https://openstax.org/books/university-physics-volume-1/pages/5-6-common-forces)

5.7 Drawing Free-Body Diagrams

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Explain the rules for drawing a free-body diagram
- Construct free-body diagrams for different situations

The first step in describing and analyzing most phenomena in physics involves the careful drawing of a free-body diagram. Free-body diagrams have been used in examples throughout this chapter. Remember that a free-body diagram must only include the external forces acting on the body of interest. Once we have drawn an accurate free-body diagram, we can apply Newton's first law if the body is in equilibrium (balanced forces; that is, $F_{\text{net}} = 0$) or Newton's second law if the body is accelerating (unbalanced force; that is, $F_{\text{net}} \neq 0$).

In [Forces](#), we gave a brief problem-solving strategy to help you understand free-body diagrams. Here, we add some details to the strategy that will help you in constructing these diagrams.



PROBLEM-SOLVING STRATEGY

Constructing Free-Body Diagrams

Observe the following rules when constructing a free-body diagram:

1. Draw the object under consideration; it does not have to be artistic. At first, you may want to draw a circle around the object of interest to be sure you focus on labeling the forces acting on the object. If you are treating the object as a particle (no size or shape and no rotation), represent the object as a point. We often place this point at the origin of an xy -coordinate system.
2. Include all forces that act on the object, representing these forces as vectors. Consider the types of forces described in [Common Forces](#)—normal force, friction, tension, and spring force—as well as weight and applied force. Do not include the net force on the object. With the exception of gravity, all of the forces we have discussed require direct contact with the object. However, forces that the object exerts on its environment must not be included. We never include both forces of an action-reaction pair.
3. Convert the free-body diagram into a more detailed diagram showing the x - and y -components of a given force (this is often helpful when solving a problem using Newton's first or second law). In this case, place a squiggly line through the original vector to show that it is no longer in play—it has been replaced by its x - and y -components.
4. If there are two or more objects, or bodies, in the problem, draw a separate free-body diagram for each object.

Note: If there is acceleration, we do not directly include it in the free-body diagram; however, it may help to indicate acceleration outside the free-body diagram. You can label it in a different color to indicate that it is separate from the free-body diagram.

Let's apply the problem-solving strategy in drawing a free-body diagram for a sled. In [Figure 5.31\(a\)](#), a sled is pulled by force $\vec{P}$ at an angle of 30° . In part (b), we show a free-body diagram for this situation, as described by steps 1 and 2 of the problem-solving strategy. In part (c), we show all forces in terms of their x - and y -components, in keeping with step 3.

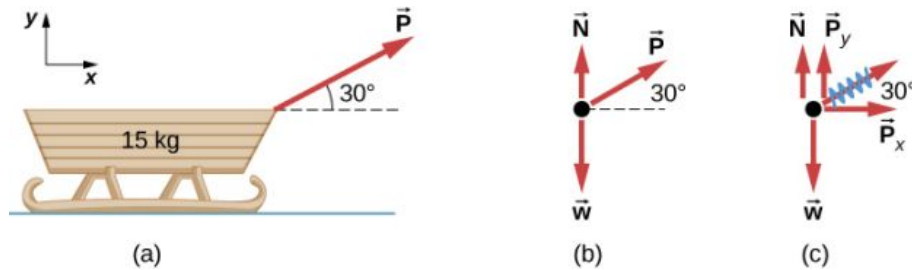


FIGURE 5.31 (a) A moving sled is shown as (b) a free-body diagram and (c) a free-body diagram with force components.



EXAMPLE 5.14

Two Blocks on an Inclined Plane

Construct the free-body diagram for object A and object B in [Figure 5.32](#).

Strategy

We follow the four steps listed in the problem-solving strategy.

Solution

We start by creating a diagram for the first object of interest. In [Figure 5.32\(a\)](#), object A is isolated (circled) and represented by a dot.

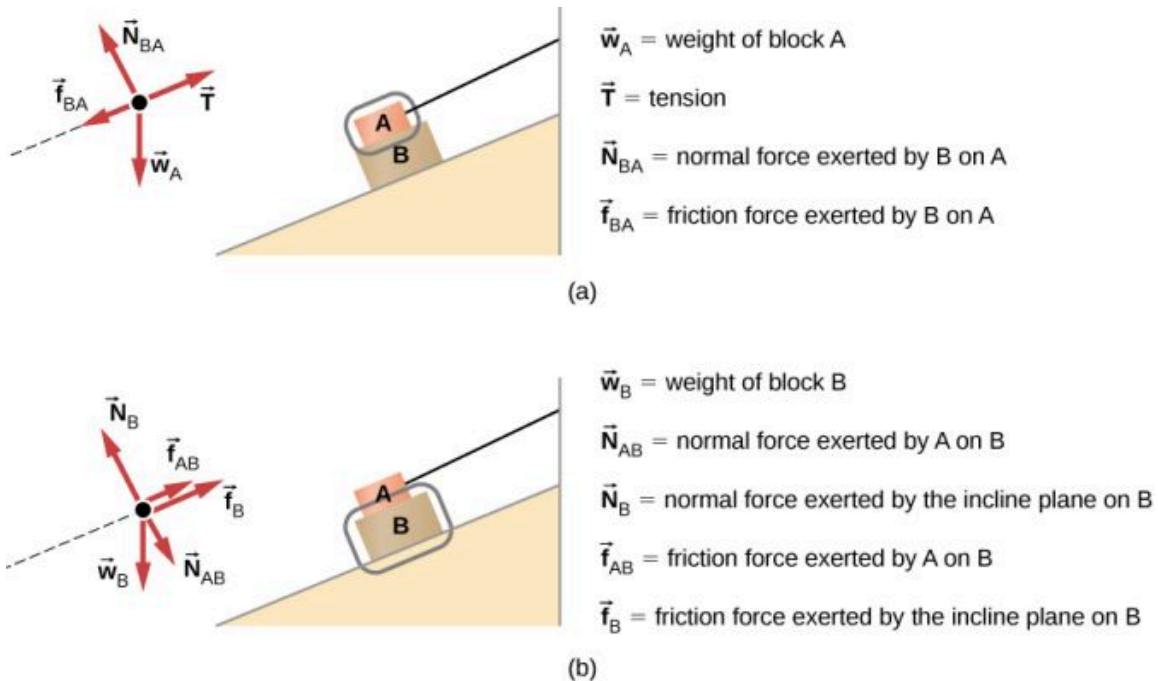


FIGURE 5.32 (a) The free-body diagram for isolated object A. (b) The free-body diagram for isolated object B. Comparing the two drawings, we see that friction acts in the opposite direction in the two figures. Because object A experiences a force that tends to pull it to the right, friction must act to the left. Because object B experiences a component of its weight that pulls it to the left, down the incline, the friction force must oppose it and act up the ramp. Friction always acts opposite the intended direction of motion.

We now include any force that acts on the body. Here, no applied force is present. The weight of the object acts as a force pointing vertically downward, and the presence of the cord indicates a force of tension pointing away from the object. Object A has one interface and hence experiences a normal force, directed away from the interface. The source of this force is object B, and this normal force is labeled accordingly. Since object B has a tendency to slide down, object A has a tendency to slide up with respect to the interface, so the friction f_{BA} is directed downward parallel to the inclined plane.

As noted in step 4 of the problem-solving strategy, we then construct the free-body diagram in [Figure 5.32\(b\)](#) using the same approach. Object B experiences two normal forces and two friction forces due to the presence of two contact surfaces. The interface with the inclined plane exerts external forces of N_B and f_B , and the interface with object A exerts the normal force N_{AB} and friction f_{AB} ; N_{AB} is directed away from object B, and f_{AB} is opposing the tendency of the relative motion of object B with respect to object A.

Significance

The object under consideration in each part of this problem was circled in gray. When you are first learning how to draw free-body diagrams, you will find it helpful to circle the object before deciding what forces are acting on that particular object. This focuses your attention, preventing you from considering forces that are not acting on the body.

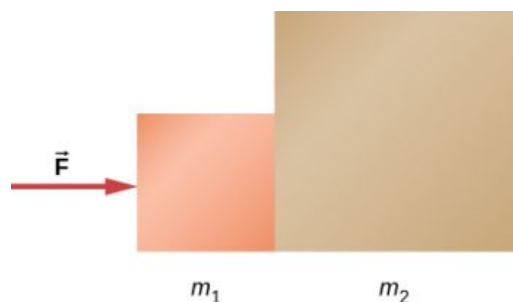
EXAMPLE 5.15

Two Blocks in Contact

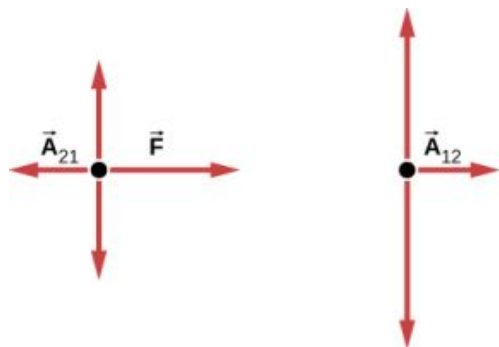
A force is applied to two blocks in contact, as shown. Draw the reaction forces between the two blocks.

Strategy

Draw a free-body diagram for each block. Be sure to consider Newton's third law at the interface where the two blocks touch.



Solution



Significance

$\vec{A}_{21}$ is the action force of block 2 on block 1. $\vec{A}_{12}$ is the reaction force of block 1 on block 2. We use these free-body diagrams in [Applications of Newton's Laws](#).



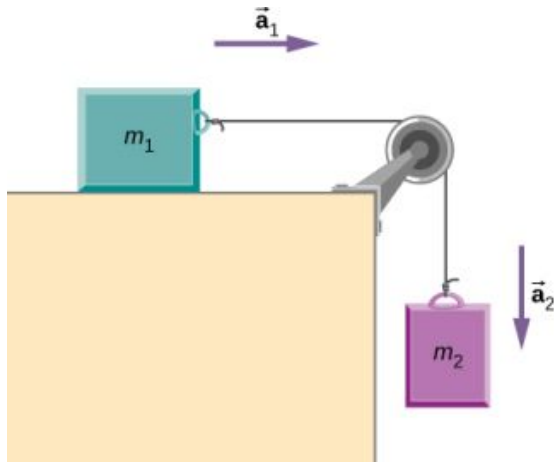
EXAMPLE 5.16

Block on the Table (Coupled Blocks)

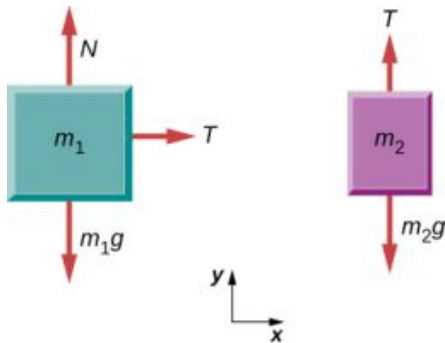
A block rests on the table, as shown. A light rope is attached to it and runs over a pulley. The other end of the rope is attached to a second block. The two blocks are said to be coupled. Block m_2 exerts a force due to its weight, which causes the system (two blocks and a string) to accelerate.

Strategy

We assume that the string has no mass so that we do not have to consider it as a separate object. Draw a free-body diagram for each block.



Solution



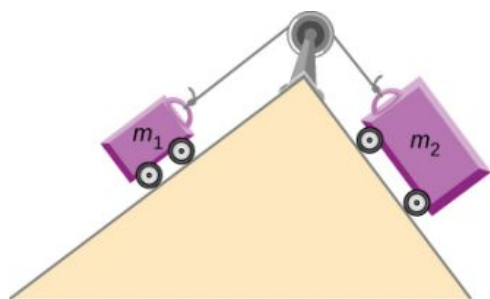
Significance

Each block accelerates (notice the labels shown for $\vec{a}_1$ and $\vec{a}_2$); however, assuming the string remains taut, the magnitudes of acceleration are equal. Thus, we have $|\vec{a}_1| = |\vec{a}_2|$. If we were to continue solving the problem, we could simply call the acceleration $\vec{a}$. Also, we use two free-body diagrams because we are usually finding tension T , which may require us to use a system of two equations in this type of problem. The tension is the same on both m_1 and m_2 .



CHECK YOUR UNDERSTANDING 5.10

(a) Draw the free-body diagram for the situation shown. (b) Redraw it showing components; use x -axes parallel to the two ramps.



INTERACTIVE

Engage the simulation below to predict, qualitatively, how an external force will affect the speed and direction of an object's motion. Explain the effects with the help of a free-body diagram. Use free-body diagrams to draw position, velocity, acceleration, and force graphs, and vice versa. Explain how the graphs relate to one another. Given a scenario or a graph, sketch all four graphs.

Access multimedia content (<https://openstax.org/books/university-physics-volume-1/pages/5-7-drawing-free-body-diagrams>)

Chapter Review

Key Terms

dynamics study of how forces affect the motion of objects and systems

external force force acting on an object or system that originates outside of the object or system

force push or pull on an object with a specific magnitude and direction; can be represented by vectors or expressed as a multiple of a standard force

free fall situation in which the only force acting on an object is gravity

free-body diagram sketch showing all external forces acting on an object or system; the system is represented by a single isolated point, and the forces are represented by vectors extending outward from that point

Hooke's law in a spring, a restoring force proportional to and in the opposite direction of the imposed displacement

inertia ability of an object to resist changes in its motion

inertial reference frame reference frame moving at constant velocity relative to an inertial frame is also inertial; a reference frame accelerating relative to an inertial frame is not inertial

law of inertia see Newton's first law of motion

net external force vector sum of all external forces acting on an object or system; causes a mass to

accelerate

newton SI unit of force; 1 N is the force needed to accelerate an object with a mass of 1 kg at a rate of 1 m/s^2

Newton's first law of motion body at rest remains at rest or, if in motion, remains in motion at constant velocity unless acted on by a net external force; also known as the law of inertia

Newton's second law of motion acceleration of a system is directly proportional to and in the same direction as the net external force acting on the system and is inversely proportional to its mass

Newton's third law of motion whenever one body exerts a force on a second body, the first body experiences a force that is equal in magnitude and opposite in direction to the force that it exerts

normal force force supporting the weight of an object, or a load, that is perpendicular to the surface of contact between the load and its support; the surface applies this force to an object to support the weight of the object

tension pulling force that acts along a stretched flexible connector, such as a rope or cable

thrust reaction force that pushes a body forward in response to a backward force

weight force $\vec{w}$ due to gravity acting on an object of mass m

Key Equations

Net external force

$$\vec{F}_{\text{net}} = \sum \vec{F} = \vec{F}_1 + \vec{F}_2 + \dots$$

Newton's first law

$$\vec{v} = \text{constant when } \vec{F}_{\text{net}} = \vec{0} \text{ N}$$

Newton's second law, vector form

$$\vec{F}_{\text{net}} = \sum \vec{F} = m\vec{a}$$

Newton's second law, scalar form

$$F_{\text{net}} = ma$$

Newton's second law, component form

$$\sum \vec{F}_x = m\vec{a}_x, \sum \vec{F}_y = m\vec{a}_y, \text{ and } \sum \vec{F}_z = m\vec{a}_z.$$

Newton's second law, momentum form

$$\vec{F}_{\text{net}} = \frac{d\vec{p}}{dt}$$

Definition of weight, vector form

$$\vec{w} = m\vec{g}$$

Definition of weight, scalar form

$$w = mg$$

Newton's third law

$$\vec{F}_{AB} = -\vec{F}_{BA}$$

Normal force on an object resting on a horizontal surface, vector form

$$\vec{N} = -m\vec{g}$$

Normal force on an object resting on a horizontal surface, scalar form

$$N = mg$$

Normal force on an object resting on an inclined plane, scalar form

$$N = mg \cos \theta$$

Restoring force by a spring displaced by x from the relaxed position $\vec{F} = -k\vec{x}$.

Tension in a cable supporting an object of mass m at rest, scalar form $T = w = mg$

Summary

5.1 Forces

- Dynamics is the study of how forces affect the motion of objects, whereas kinematics simply describes the way objects move.
- Force is a push or pull that can be defined in terms of various standards, and it is a vector that has both magnitude and direction.
- External forces are any outside forces that act on a body. A free-body diagram is a drawing of all external forces acting on a body.
- The SI unit of force is the newton (N).

5.2 Newton's First Law

- According to Newton's first law, there must be a cause for any change in velocity (a change in either magnitude or direction) to occur. This law is also known as the law of inertia.
- Friction is an external force that causes an object to slow down.
- Inertia is the tendency of an object to remain at rest or remain in motion. Inertia is related to an object's mass.
- If an object's velocity relative to a given frame is constant, then the frame is inertial. This means that for an inertial reference frame, Newton's first law is valid.
- Equilibrium is achieved when the forces on a system are balanced.
- A net force of zero means that an object is either at rest or moving with constant velocity; that is, it is not accelerating.

5.3 Newton's Second Law

- An external force acts on a system from outside the system, as opposed to internal forces, which act between components within the system.
- Newton's second law of motion says that the net external force on an object with a certain mass is directly proportional to and in the same direction as the acceleration of the object.
- Newton's second law can also describe net force as the instantaneous rate of change of momentum. Thus, a net external force causes nonzero acceleration.

5.4 Mass and Weight

- Mass is the quantity of matter in a substance.
- The weight of an object is the net force on a falling object, or its gravitational force. The object experiences acceleration due to gravity.
- Some upward resistance force from the air acts on all falling objects on Earth, so they can never truly be in free fall.
- Careful distinctions must be made between free fall and weightlessness using the definition of weight as force due to gravity acting on an object of a certain mass.

5.5 Newton's Third Law

- Newton's third law of motion represents a basic symmetry in nature, with an experienced force equal in magnitude and opposite in direction to an exerted force.
- Two equal and opposite forces do not cancel because they act on different systems.
- Action-reaction pairs include a swimmer pushing off a wall, helicopters creating lift by pushing air down, and an octopus propelling itself forward by ejecting water from its body. Rockets, airplanes, and cars are pushed forward by a thrust reaction force.
- Choosing a system is an important analytical step in understanding the physics of a problem and solving it.

5.6 Common Forces

- When an object rests on a surface, the surface applies a force to the object that supports the weight of the object. This supporting force acts perpendicular to and away from the surface. It is called a normal force.
- When an object rests on a nonaccelerating horizontal surface, the magnitude of the normal force is equal to the weight of the object.
- When an object rests on an inclined plane that makes an angle θ with the horizontal surface, the weight of the object can be resolved into components that act perpendicular and parallel to the surface of the plane.
- The pulling force that acts along a stretched flexible connector, such as a rope or cable, is

called tension. When a rope supports the weight of an object at rest, the tension in the rope is equal to the weight of the object. If the object is accelerating, tension is greater than weight, and if it is accelerating opposite to the motion, tension is less than weight.

- The force of friction is a force experienced by a moving object (or an object that has a tendency to move) parallel to the interface opposing the motion (or its tendency).
- The force developed in a spring obeys Hooke's law, according to which its magnitude is proportional to the displacement and has a sense in the opposite direction of the displacement.
- Real forces have a physical origin, whereas

fictitious forces occur because the observer is in an accelerating or noninertial frame of reference.

5.7 Drawing Free-Body Diagrams

- To draw a free-body diagram, we draw the object of interest, draw all forces acting on that object, and resolve all force vectors into x - and y -components. We must draw a separate free-body diagram for each object in the problem.
- A free-body diagram is a useful means of describing and analyzing all the forces that act on a body to determine equilibrium according to Newton's first law or acceleration according to Newton's second law.

Conceptual Questions

5.1 Forces

1. What properties do forces have that allow us to classify them as vectors?

5.2 Newton's First Law

2. Taking a frame attached to Earth as inertial, which of the following objects cannot have inertial frames attached to them, and which are inertial reference frames?
 - (a) A car moving at constant velocity
 - (b) A car that is accelerating
 - (c) An elevator in free fall
 - (d) A space capsule orbiting Earth
 - (e) An elevator descending uniformly
3. A woman was transporting an open box of cupcakes to a school party. The car in front of her stopped suddenly; she applied her brakes immediately. She was wearing her seat belt and suffered no physical harm (just a great deal of embarrassment), but the cupcakes flew into the dashboard and became "smushcakes." Explain what happened.

5.3 Newton's Second Law

4. Why can we neglect forces such as those holding a body together when we apply Newton's second law?
5. A rock is thrown straight up. At the top of the trajectory, the velocity is momentarily zero. Does this imply that the force acting on the object is zero? Explain your answer.

5.4 Mass and Weight

6. What is the relationship between weight and

mass? Which is an intrinsic, unchanging property of a body?

7. How much does a 70-kg astronaut weight in space, far from any celestial body? What is her mass at this location?
8. Which of the following statements is accurate?
 - (a) Mass and weight are the same thing expressed in different units.
 - (b) If an object has no weight, it must have no mass.
 - (c) If the weight of an object varies, so must the mass.
 - (d) Mass and inertia are different concepts.
 - (e) Weight is always proportional to mass.
9. When you stand on Earth, your feet push against it with a force equal to your weight. Why doesn't Earth accelerate away from you?
10. How would you give the value of $\vec{g}$ in vector form?

5.5 Newton's Third Law

11. Identify the action and reaction forces in the following situations: (a) Earth attracts the Moon, (b) a boy kicks a football, (c) a rocket accelerates upward, (d) a car accelerates forward, (e) a high jumper leaps, and (f) a bullet is shot from a gun.
12. Suppose that you are holding a cup of coffee in your hand. Identify all forces on the cup and the reaction to each force.
13. (a) Why does an ordinary rifle recoil (kick backward) when fired? (b) The barrel of a recoilless rifle is open at both ends. Describe how Newton's third law applies when one is

fired. (c) Can you safely stand close behind one when it is fired?

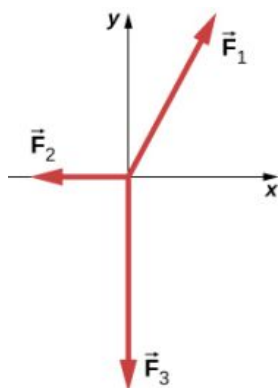
5.6 Common Forces

14. A table is placed on a rug. Then a book is placed on the table. What does the floor exert a normal force on?
15. A particle is moving to the right. (a) Can the force on it be acting to the left? If yes, what would happen? (b) Can that force be acting downward? If yes, why?

Problems

5.1 Forces

19. Two ropes are attached to a tree, and forces of $\vec{F}_1 = 2.0\hat{i} + 4.0\hat{j}$ N and $\vec{F}_2 = 3.0\hat{i} + 6.0\hat{j}$ N are applied. The forces are coplanar (in the same plane). (a) What is the resultant (net force) of these two force vectors? (b) Find the magnitude and direction of this net force.
20. A telephone pole has three cables pulling as shown from above, with $\vec{F}_1 = (300.0\hat{i} + 500.0\hat{j})$, $\vec{F}_2 = -200.0\hat{i}$, and $\vec{F}_3 = -800.0\hat{j}$. (a) Find the net force on the telephone pole in component form. (b) Find the magnitude and direction of this net force.



21. Two teenagers are pulling on ropes attached to a tree. The angle between the ropes is 30.0° . David pulls with a force of 400.0 N and Stephanie pulls with a force of 300.0 N. (a) Find the component form of the net force. (b) Find the magnitude of the resultant (net) force on the tree and the angle it makes with David's rope.

5.2 Newton's First Law

22. Two forces of $\vec{F}_1 = \frac{75.0}{\sqrt{2}}(\hat{i} - \hat{j})$ N and

5.7 Drawing Free-Body Diagrams

16. In completing the solution for a problem involving forces, what do we do after constructing the free-body diagram? That is, what do we apply?
17. If a book is located on a table, how many forces should be shown in a free-body diagram of the book? Describe them.
18. If the book in the previous question is in free fall, how many forces should be shown in a free-body diagram of the book? Describe them.

$\vec{F}_2 = \frac{150.0}{\sqrt{2}}(\hat{i} - \hat{j})$ N act on an object. Find the third force $\vec{F}_3$ that is needed to balance the first two forces.

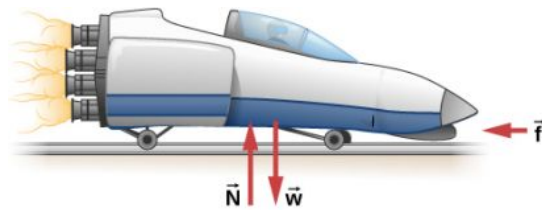
23. While sliding a couch across a floor, Andrea and Jennifer exert forces $\vec{F}_A$ and $\vec{F}_J$ on the couch. Andrea's force is due north with a magnitude of 130.0 N and Jennifer's force is 32° east of north with a magnitude of 180.0 N. (a) Find the net force in component form. (b) Find the magnitude and direction of the net force. (c) If Andrea and Jennifer's housemates, David and Stephanie, disagree with the move and want to prevent its relocation, with what combined force $\vec{F}_{DS}$ should they push so that the couch does not move?

5.3 Newton's Second Law

24. Andrea, a 63.0-kg sprinter, starts a race with an acceleration of 4.200 m/s^2 . What is the net external force on her?
25. If the sprinter from the previous problem accelerates at that rate for 20.00 m and then maintains that velocity for the remainder of a 100.00-m dash, what will her time be for the race?
26. A cleaner pushes a 4.50-kg laundry cart in such a way that the net external force on it is 60.0 N. Calculate the magnitude of his cart's acceleration.
27. Astronauts in orbit are apparently weightless. This means that a clever method of measuring the mass of astronauts is needed to monitor their mass gains or losses, and adjust their diet. One way to do this is to exert a known force on an astronaut and measure the acceleration

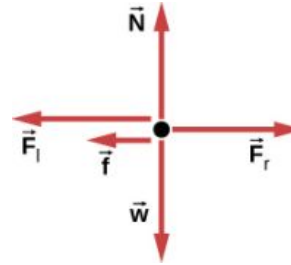
produced. Suppose a net external force of 50.0 N is exerted, and an astronaut's acceleration is measured to be 0.893 m/s^2 . (a) Calculate her mass. (b) By exerting a force on the astronaut, the vehicle in which she orbits experiences an equal and opposite force. Use this knowledge to find an equation for the acceleration of the system (astronaut and spaceship) that would be measured by a nearby observer. (c) Discuss how this would affect the measurement of the astronaut's acceleration. Propose a method by which recoil of the vehicle is avoided.

28. In [Figure 5.12](#), the net external force on the 24-kg mower is given as 51 N. If the force of friction opposing the motion is 24 N, what force F (in newtons) is the person exerting on the mower? Suppose the mower is moving at 1.5 m/s when the force F is removed. How far will the mower go before stopping?
29. The rocket sled shown below accelerates opposite to the motion at a rate of 196 m/s^2 . What force is necessary to produce this acceleration opposite to the motion? Assume that the rockets are off. The mass of the system is $2.10 \times 10^3 \text{ kg}$.

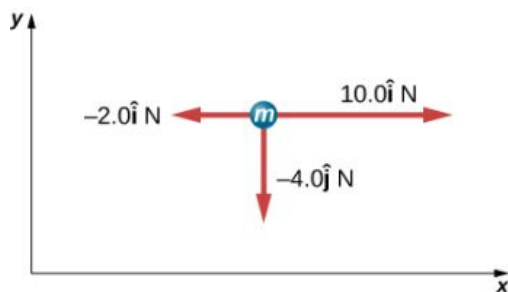


30. If the rocket sled shown in the previous problem starts with only one rocket burning, what is the magnitude of this acceleration? Assume that the mass of the system is $2.10 \times 10^3 \text{ kg}$, the thrust T is $2.40 \times 10^4 \text{ N}$, and the force of friction opposing the motion is 650.0 N. (b) Why is the acceleration not one-fourth of what it is with all rockets burning?
31. What is the acceleration opposite to the motion of the rocket sled if it comes to rest in 1.10 s from a speed of 1000.0 km/h? (Such acceleration opposite to the motion caused one test subject to black out and have temporary blindness.)
32. Suppose two children push horizontally, but in exactly opposite directions, on a third child in a wagon. The first child exerts a force of 75.0 N, the second exerts a force of 90.0 N, friction is 12.0 N, and the mass of the third child plus

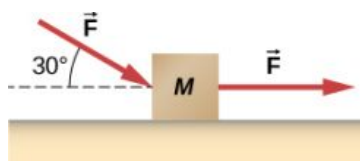
wagon is 23.0 kg. (a) What is the system of interest if the acceleration of the child in the wagon is to be calculated? (See the free-body diagram.) (b) Calculate the acceleration. (c) What would the acceleration be if friction were 15.0 N?



33. A powerful motorcycle can produce an acceleration of 3.50 m/s^2 while traveling at 90.0 km/h. At that speed, the forces resisting motion, including friction and air resistance, total 400.0 N. (Air resistance is analogous to air friction. It always opposes the motion of an object.) What is the magnitude of the force that motorcycle exerts backward on the ground to produce its acceleration if the mass of the motorcycle with rider is 245 kg?
34. A car with a mass of 1000.0 kg accelerates from 0 to 90.0 km/h in 10.0 s. (a) What is its acceleration? (b) What is the net force on the car?
35. The driver in the previous problem applies the brakes when the car is moving at 90.0 km/h, and the car comes to rest after traveling 40.0 m. What is the net force on the car during its acceleration opposite to the motion?
36. An 80.0-kg passenger in an SUV traveling at $1.00 \times 10^2 \text{ km/h}$ is wearing a seat belt. The driver slams on the brakes and the SUV stops in 45.0 m. Find the force of the seat belt on the passenger.
37. A particle of mass 2.0 kg is acted on by a single force $\vec{F}_1 = 18\hat{i} \text{ N}$. (a) What is the particle's acceleration? (b) If the particle starts at rest, how far does it travel in the first 5.0 s?
38. Suppose that the particle of the previous problem also experiences forces $\vec{F}_2 = -15\hat{i} \text{ N}$ and $\vec{F}_3 = 6.0\hat{j} \text{ N}$. What is its acceleration in this case?
39. Find the acceleration of the body of mass 5.0 kg shown below.



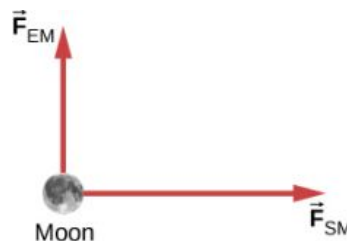
40. In the following figure, the horizontal surface on which this block slides is frictionless. If the two forces acting on it each have magnitude $F = 30.0\text{ N}$ and $M = 10.0\text{ kg}$, what is the magnitude of the resulting acceleration of the block?



5.4 Mass and Weight

41. The weight of an astronaut plus their space suit on the Moon is only 250 N. (a) How much does the suited astronaut weigh on Earth? (b) What is the mass on the Moon? On Earth?
42. Suppose the mass of a fully loaded module in which astronauts take off from the Moon is $1.00 \times 10^4\text{ kg}$. The thrust of its engines is $3.00 \times 10^4\text{ N}$. (a) Calculate the module's magnitude of acceleration in a vertical takeoff from the Moon. (b) Could it lift off from Earth? If not, why not? If it could, calculate the magnitude of its acceleration.
43. A rocket sled accelerates at a rate of 49.0 m/s^2 . Its passenger has a mass of 75.0 kg. (a) Calculate the horizontal component of the force the seat exerts against his body. Compare this with his weight using a ratio. (b) Calculate the direction and magnitude of the total force the seat exerts against his body.
44. Repeat the previous problem for a situation in which the rocket sled accelerates opposite to the motion at a rate of 201 m/s^2 . In this problem, the forces are exerted by the seat and the seat belt.
45. A body of mass 2.00 kg is pushed straight upward by a 25.0 N vertical force. What is its acceleration?

46. A car weighing 12,500 N starts from rest and accelerates to 83.0 km/h in 5.00 s. The friction force is 1350 N. Find the applied force produced by the engine.
47. A body with a mass of 10.0 kg is assumed to be in Earth's gravitational field with $g = 9.80\text{ m/s}^2$. What is the net force on the body if there are no other external forces acting on the object?
48. A firefighter has mass m ; they hear the fire alarm and slides down the pole with acceleration a (which is less than g in magnitude). (a) Write an equation giving the vertical force they must apply to the pole. (b) If their mass is 90.0 kg and they accelerate at 5.00 m/s^2 , what is the magnitude of their applied force?
49. A baseball catcher is performing a stunt for a television commercial. He will catch a baseball (mass 145 g) dropped from a height of 60.0 m above his glove. His glove stops the ball in 0.0100 s. What is the force exerted by his glove on the ball?
50. When the Moon is directly overhead at sunset, the force by Earth on the Moon, F_{EM} , is essentially at 90° to the force by the Sun on the Moon, F_{SM} , as shown below. Given that $F_{EM} = 1.98 \times 10^{20}\text{ N}$ and $F_{SM} = 4.36 \times 10^{20}\text{ N}$, all other forces on the Moon are negligible, and the mass of the Moon is $7.35 \times 10^{22}\text{ kg}$, determine the magnitude of the Moon's acceleration.



5.5 Newton's Third Law

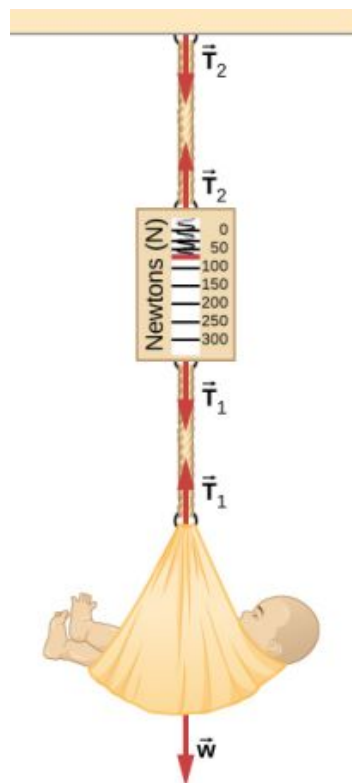
51. (a) What net external force is exerted on a 1100.0-kg artillery shell fired from a battleship if the shell is accelerated at $2.40 \times 10^4\text{ m/s}^2$? (b) What is the magnitude of the force exerted on the ship by the artillery shell, and why?
52. A rugby player is being pushed backward by an opposing player who is exerting a force of 800.0 N on him. The mass of the losing player plus equipment is 90.0 kg, and he is accelerating backward at 1.20 m/s^2 . (a) What is the force of

vertical strand (find their ratio).

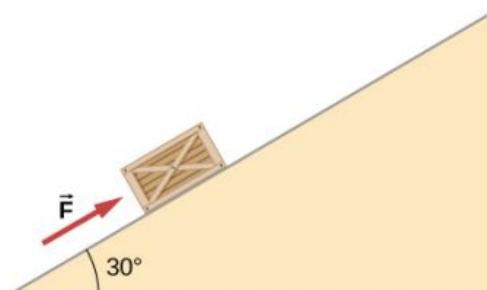
60. Suppose Kevin, a 60.0-kg gymnast, climbs a rope. (a) What is the tension in the rope if he climbs at a constant speed? (b) What is the tension in the rope if he accelerates upward at a rate of 1.50 m/s^2 ?
61. Show that, as explained in the text, a force $F_{\perp}$ exerted on a flexible medium at its center and perpendicular to its length (such as on the tightrope wire in Figure 5.26) gives rise to a tension of magnitude $T = F_{\perp}/2 \sin(\theta)$.
62. Consider Figure 5.28. The driver attempts to get the car out of the mud by exerting a perpendicular force of 610.0 N, and the distance she pushes in the middle of the rope is 1.00 m while she stands 6.00 m away from the car on the left and 6.00 m away from the tree on the right. What is the tension T in the rope, and how do you find the answer?
63. A bird has a mass of 26 g and perches in the middle of a stretched telephone line. (a) Show that the tension in the line can be calculated using the equation $T = \frac{mg}{2 \sin \theta}$. Determine the tension when (b) $\theta = 5^\circ$ and (c) $\theta = 0.5^\circ$. Assume that each half of the line is straight.



64. One end of a 30-m rope is tied to a tree; the other end is tied to a car stuck in the mud. The motorist pulls sideways on the midpoint of the rope, displacing it a distance of 2 m. If he exerts a force of 80 N under these conditions, determine the force exerted on the car.
65. Consider the baby being weighed in the following figure. (a) What is the mass of the infant and basket if a scale reading of 55 N is observed? (b) What is tension T_1 in the cord attaching the baby to the scale? (c) What is tension T_2 in the cord attaching the scale to the ceiling, if the scale has a mass of 0.500 kg? (d) Sketch the situation, indicating the system of interest used to solve each part. The masses of the cords are negligible.



66. What force must be applied to a 100.0-kg crate on a frictionless plane inclined at 30° to cause an acceleration of 2.0 m/s^2 up the plane?



67. A 2.0-kg block is on a perfectly smooth ramp that makes an angle of 30° with the horizontal. (a) What is the block's acceleration down the ramp and the force of the ramp on the block? (b) What force applied upward along and parallel to the ramp would allow the block to move with constant velocity?

5.7 Drawing Free-Body Diagrams

68. A ball of mass m hangs at rest, suspended by a string. (a) Sketch all forces. (b) Draw the free-body diagram for the ball.
69. A car moves along a horizontal road. Draw a free-body diagram; be sure to include the friction of the road that opposes the forward

motion of the car.

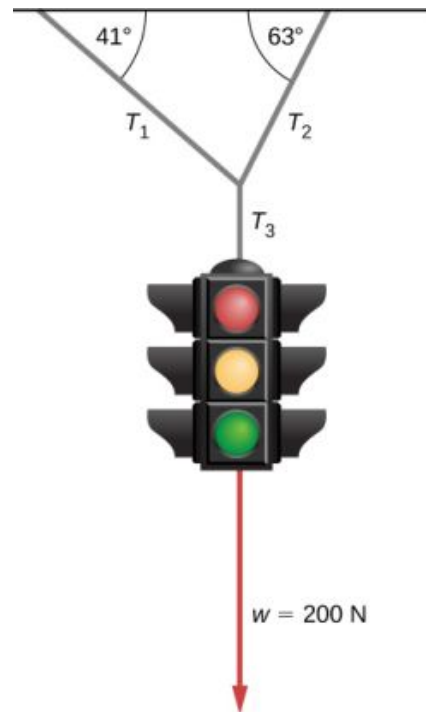
70. A runner pushes against the track, as shown. (a) Provide a free-body diagram showing all the forces on the runner. (*Hint:* Place all forces at the center of his body, and include his weight.) (b) Give a revised diagram showing the xy -component form.



(credit: modification of work by "Greenwich Photography"/Flickr)

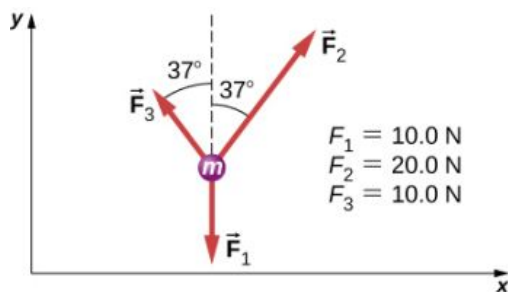
71. The traffic light hangs from the cables as shown.

Draw a free-body diagram on a coordinate plane for this situation.

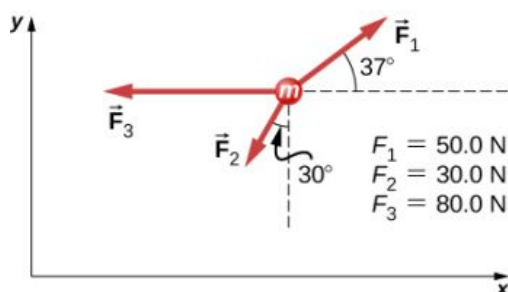


Additional Problems

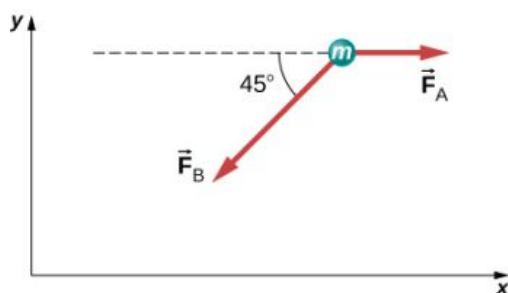
72. Two small forces, $\vec{F}_1 = -2.40\hat{i} - 6.10\hat{j}$ N and $\vec{F}_2 = 8.50\hat{i} - 9.70\hat{j}$ N, are exerted on a rogue asteroid by a pair of space tractors. (a) Find the net force. (b) What are the magnitude and direction of the net force? (c) If the mass of the asteroid is 125 kg, what acceleration does it experience (in vector form)? (d) What are the magnitude and direction of the acceleration?
73. Two forces of 25 and 45 N act on an object. Their directions differ by 70° . The resulting acceleration has magnitude of 10.0 m/s^2 . What is the mass of the body?
74. A force of 1600 N acts parallel to a ramp to push a 300-kg piano into a moving van. The ramp is inclined at 20° . (a) What is the acceleration of the piano up the ramp? (b) What is the velocity of the piano when it reaches the top if the ramp is 4.0 m long and the piano starts from rest?
75. Draw a free-body diagram of a diver who has entered the water, moved downward, and is acted on by an upward force due to the water which balances the weight (that is, the diver is suspended).
76. For a swimmer who has just jumped off a diving board, assume air resistance is negligible. The swimmer has a mass of 80.0 kg and jumps off a board 10.0 m above the water. Three seconds after entering the water, her downward motion is stopped. What average upward force did the water exert on her?
77. (a) Find an equation to determine the magnitude of the net force required to stop a car of mass m , given that the initial speed of the car is v_0 and the stopping distance is x . (b) Find the magnitude of the net force if the mass of the car is 1050 kg, the initial speed is 40.0 km/h, and the stopping distance is 25.0 m.
78. A sailboat has a mass of 1.50×10^3 kg and is acted on by a force of 2.00×10^3 N toward the east, while the wind acts behind the sails with a force of 3.00×10^3 N in a direction 45° north of east. Find the magnitude and direction of the resulting acceleration.
79. Find the acceleration of the body of mass 10.0 kg shown below.



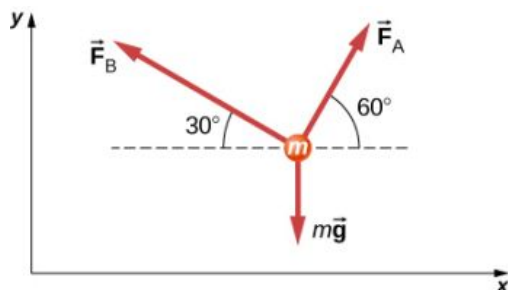
80. A body of mass 2.0 kg is moving along the x -axis with a speed of 3.0 m/s at the instant represented below. (a) What is the acceleration of the body? (b) What is the body's velocity 10.0 s later? (c) What is its displacement after 10.0 s?



81. Force $\vec{F}_B$ has twice the magnitude of force $\vec{F}_A$. Find the direction in which the particle accelerates in this figure.



82. Shown below is a body of mass 1.0 kg under the influence of the forces $\vec{F}_A$, $\vec{F}_B$, and $m\vec{g}$. If the body accelerates to the left at 20 m/s^2 , what are $\vec{F}_A$ and $\vec{F}_B$?



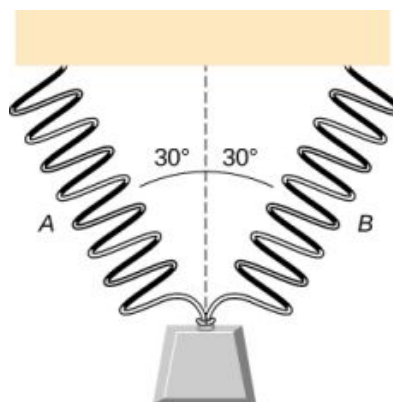
83. A force acts on a car of mass m so that the speed v of the car increases with position x as $v = kx^2$, where k is constant and all quantities are in SI units. Find the force acting on the car as a function of position.

84. A 7.0-N force parallel to an incline is applied to a 1.0-kg crate. The ramp is tilted at 20° and is frictionless. (a) What is the acceleration of the crate? (b) If all other conditions are the same but the ramp has a friction force of 1.9 N, what is the acceleration?

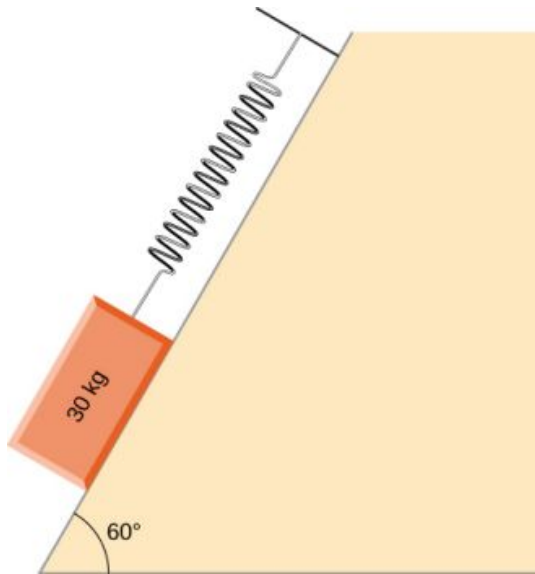
85. Two boxes, A and B, are at rest. Box A is on level ground, while box B rests on an inclined plane tilted at angle θ with the horizontal. (a) Write expressions for the normal force acting on each block. (b) Compare the two forces; that is, tell which one is larger or whether they are equal in magnitude. (c) If the angle of incline is 10° , which force is greater?

86. A mass of 250.0 g is suspended from a spring hanging vertically. The spring stretches 6.00 cm. How much will the spring stretch if the suspended mass is 530.0 g?

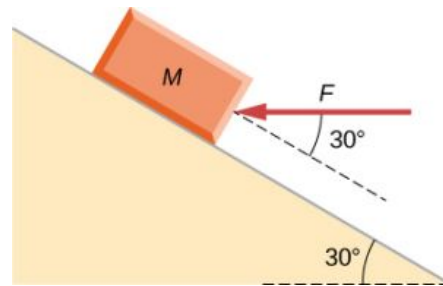
87. As shown below, two identical springs, each with the spring constant 20 N/m, support a 15.0-N weight. (a) What is the tension in spring A? (b) What is the amount of stretch of spring A from the rest position?



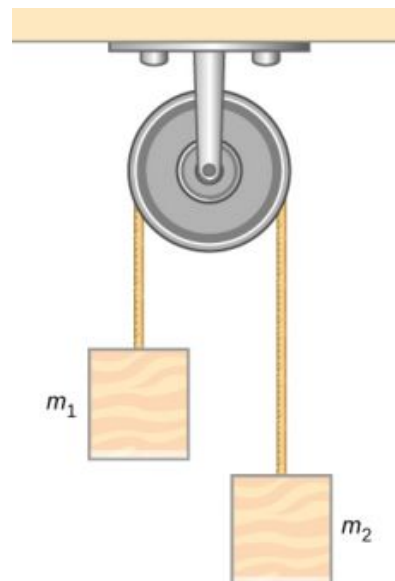
88. Shown below is a 30.0-kg block resting on a frictionless ramp inclined at 60° to the horizontal. The block is held by a spring that is stretched 5.0 cm. What is the force constant of the spring?



89. In building a house, carpenters use nails from a large box. The box is suspended from a spring twice during the day to measure the usage of nails. At the beginning of the day, the spring stretches 50 cm. At the end of the day, the spring stretches 30 cm. What fraction or percentage of the nails have been used?
90. A force is applied to a block to move it up a 30° incline. The incline is frictionless. If $F = 65.0$ N and $M = 5.00$ kg, what is the magnitude of the acceleration of the block?

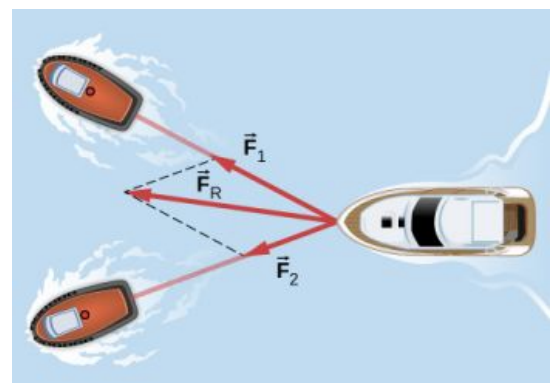


91. Two forces are applied to a 5.0-kg object, and it accelerates at a rate of 2.0 m/s^2 in the positive y-direction. If one of the forces acts in the positive x-direction with magnitude 12.0 N, find the magnitude of the other force.
92. The block on the right shown below has more mass than the block on the left ($m_2 > m_1$). Draw free-body diagrams for each block.



Challenge Problems

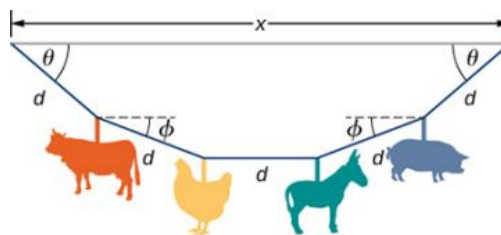
93. If two tugboats pull on a disabled vessel, as shown here in an overhead view, the disabled vessel will be pulled along the direction indicated by the result of the exerted forces. (a) Draw a free-body diagram for the vessel. Assume no friction or drag forces affect the vessel. (b) Did you include all forces in the overhead view in your free-body diagram? Why or why not?



94. A 10.0-kg object is initially moving east at 15.0 m/s. Then a force acts on it for 2.00 s, after

which it moves northwest, also at 15.0 m/s. What are the magnitude and direction of the average force that acted on the object over the 2.00-s interval?

95. On June 25, 1983, shot-putter Udo Beyer of East Germany threw the 7.26-kg shot 22.22 m, which at that time was a world record. (a) If the shot was released at a height of 2.20 m with a projection angle of 45.0° , what was its initial velocity? (b) If while in Beyer's hand the shot was accelerated uniformly over a distance of 1.20 m, what was the net force on it?
96. A body of mass m moves in a horizontal direction such that at time t its position is given by $x(t) = at^4 + bt^3 + ct$, where a , b , and c are constants. (a) What is the acceleration of the body? (b) What is the time-dependent force acting on the body?
97. A body of mass m has initial velocity v_0 in the positive x -direction. It is acted on by a constant force F for time t until the velocity becomes zero; the force continues to act on the body until its velocity becomes $-v_0$ in the same amount of time. Write an expression for the total distance the body travels in terms of the variables indicated.
98. The velocities of a 3.0-kg object at $t = 6.0$ s and $t = 8.0$ s are $(3.0\hat{i} - 6.0\hat{j} + 4.0\hat{k})$ m/s and $(-2.0\hat{i} + 4.0\hat{k})$ m/s, respectively. If the object is moving at constant acceleration, what is the force acting on it?
99. A 120-kg astronaut is riding in a rocket sled that is sliding along an inclined plane. The sled has a horizontal component of acceleration of 5.0 m/s^2 and a downward component of 3.8 m/s^2 . Calculate the magnitude of the force on the rider by the sled. (*Hint:* Remember that gravitational acceleration must be considered.)
100. Two forces are acting on a 5.0-kg object that moves with acceleration 2.0 m/s^2 in the positive y -direction. If one of the forces acts in the positive x -direction and has magnitude of 12 N, what is the magnitude of the other force?
101. Suppose that you are viewing a soccer game from a helicopter above the playing field. Two soccer players simultaneously kick a stationary soccer ball on the flat field; the soccer ball has mass 0.420 kg. The first player kicks with force 162 N at 9.0° north of west. At the same instant,
- the second player kicks with force 215 N at 15° east of south. Find the acceleration of the ball in $\hat{i}$ and $\hat{j}$ form.
102. A 10.0-kg mass hangs from a spring that has the spring constant 535 N/m. Find the position of the end of the spring away from its rest position. (Use $g = 9.80 \text{ m/s}^2$.)
103. A 0.0502-kg pair of fuzzy dice is attached to the rearview mirror of a car by a short string. The car accelerates at constant rate, and the dice hang at an angle of 3.20° from the vertical because of the car's acceleration. What is the magnitude of the acceleration of the car?
104. At a circus, a donkey pulls on a sled carrying a small clown with a force given by $2.48\hat{i} + 4.33\hat{j}$ N. A horse pulls on the same sled, aiding the hapless donkey, with a force of $6.56\hat{i} + 5.33\hat{j}$ N. The mass of the sled is 575 kg. Using $\hat{i}$ and $\hat{j}$ form for the answer to each problem, find (a) the net force on the sled when the two animals act together, (b) the acceleration of the sled, and (c) the velocity after 6.50 s.
105. Hanging from the ceiling over a baby bed, well out of baby's reach, is a string with plastic shapes, as shown here. The string is taut (there is no slack), as shown by the straight segments. Each plastic shape has the same mass m , and they are equally spaced by a distance d , as shown. The angles labeled θ describe the angle formed by the end of the string and the ceiling at each end. The center length of string is horizontal. The remaining two segments each form an angle with the horizontal, labeled ϕ . Let T_1 be the tension in the leftmost section of the string, T_2 be the tension in the section adjacent to it, and T_3 be the tension in the horizontal segment. (a) Find an equation for the tension in each section of the string in terms of the variables m , g , and θ . (b) Find the angle ϕ in terms of the angle θ . (c) If $\theta = 5.10^\circ$, what is the value of ϕ ? (d) Find the distance x between the endpoints in terms of d and θ .



- 106.** A bullet shot from a rifle has mass of 10.0 g and travels to the right at 350 m/s. It strikes a target, a large bag of sand, penetrating it a distance of 34.0 cm. Find the magnitude and direction of the retarding force that slows and stops the bullet.
- 107.** An object is acted on by three simultaneous forces: $\vec{F}_1 = (-3.00\hat{i} + 2.00\hat{j})$ N, $\vec{F}_2 = (6.00\hat{i} - 4.00\hat{j})$ N, and $\vec{F}_3 = (2.00\hat{i} + 5.00\hat{j})$ N. The object experiences acceleration of 4.23 m/s^2 . (a) Find the acceleration vector in terms of m . (b) Find the mass of the object. (c) If the object begins from rest, find its speed after 5.00 s. (d) Find the components of the velocity of the object after 5.00 s.
- 108.** In a particle accelerator, a proton has mass $1.67 \times 10^{-27} \text{ kg}$ and an initial speed of $2.00 \times 10^5 \text{ m/s}$. It moves in a straight line, and its speed increases to $9.00 \times 10^5 \text{ m/s}$ in a distance of 10.0 cm. Assume that the acceleration is constant. Find the magnitude of the force exerted on the proton.
- 109.** A drone is being directed across a frictionless ice-covered lake. The mass of the drone is 1.50 kg, and its velocity is $3.00\hat{i} \text{ m/s}$. After 10.0 s, the velocity is $9.00\hat{i} + 4.00\hat{j} \text{ m/s}$. If a constant force in the horizontal direction is causing this change in motion, find (a) the components of the force and (b) the magnitude of the force.

CHAPTER 6

Applications of Newton's Laws



FIGURE 6.1 Stock cars racing in the Grand National Divisional race at Iowa Speedway in May, 2015. Cars often reach speeds of 200 mph (320 km/h). (credit: modification of work by Erik Schneider/U.S. Navy)

CHAPTER OUTLINE

6.1 Solving Problems with Newton's Laws

6.2 Friction

6.3 Centripetal Force

6.4 Drag Force and Terminal Speed

INTRODUCTION As a racecar moves in a curved path around the turn, its wheels also spin rapidly. The wheels complete many revolutions while the car makes only part of one (a circular arc). How can we describe the velocities, accelerations, and forces involved? What force keeps a racecar from spinning out, hitting the wall bordering the track? What provides this force? Why is the track banked? We answer all of these questions in this chapter as we expand our consideration of Newton's laws of motion.

6.1 Solving Problems with Newton's Laws

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Apply problem-solving techniques to solve for quantities in more complex systems of forces
- Use concepts from kinematics to solve problems using Newton's laws of motion
- Solve more complex equilibrium problems
- Solve more complex acceleration problems
- Apply calculus to more advanced dynamics problems

Success in problem solving is necessary to understand and apply physical principles. We developed a pattern of analyzing and setting up the solutions to problems involving Newton's laws in [Newton's Laws of Motion](#); in this chapter, we continue to discuss these strategies and apply a step-by-step process.

Problem-Solving Strategies

We follow here the basics of problem solving presented earlier in this text, but we emphasize specific strategies that are useful in applying Newton's laws of motion. Once you identify the physical principles involved in the problem and determine that they include Newton's laws of motion, you can apply these steps to find a solution. These techniques also reinforce concepts that are useful in many other areas of physics. Many problem-solving strategies are stated outright in the worked examples, so the following techniques should reinforce skills you have already begun to develop.



PROBLEM-SOLVING STRATEGY

Applying Newton's Laws of Motion

1. Identify the physical principles involved by listing the givens and the quantities to be calculated.
2. Sketch the situation, using arrows to represent all forces.
3. Determine the system of interest. The result is a free-body diagram that is essential to solving the problem.
4. Apply Newton's second law to solve the problem. If necessary, apply appropriate kinematic equations from the chapter on motion along a straight line.
5. Check the solution to see whether it is reasonable.

Let's apply this problem-solving strategy to the challenge of lifting a grand piano into a second-story apartment. Once we have determined that Newton's laws of motion are involved (if the problem involves forces), it is particularly important to draw a careful sketch of the situation. Such a sketch is shown in [Figure 6.2\(a\)](#). Then, as in [Figure 6.2\(b\)](#), we can represent all forces with arrows. Whenever sufficient information exists, it is best to label these arrows carefully and make the length and direction of each correspond to the represented force.

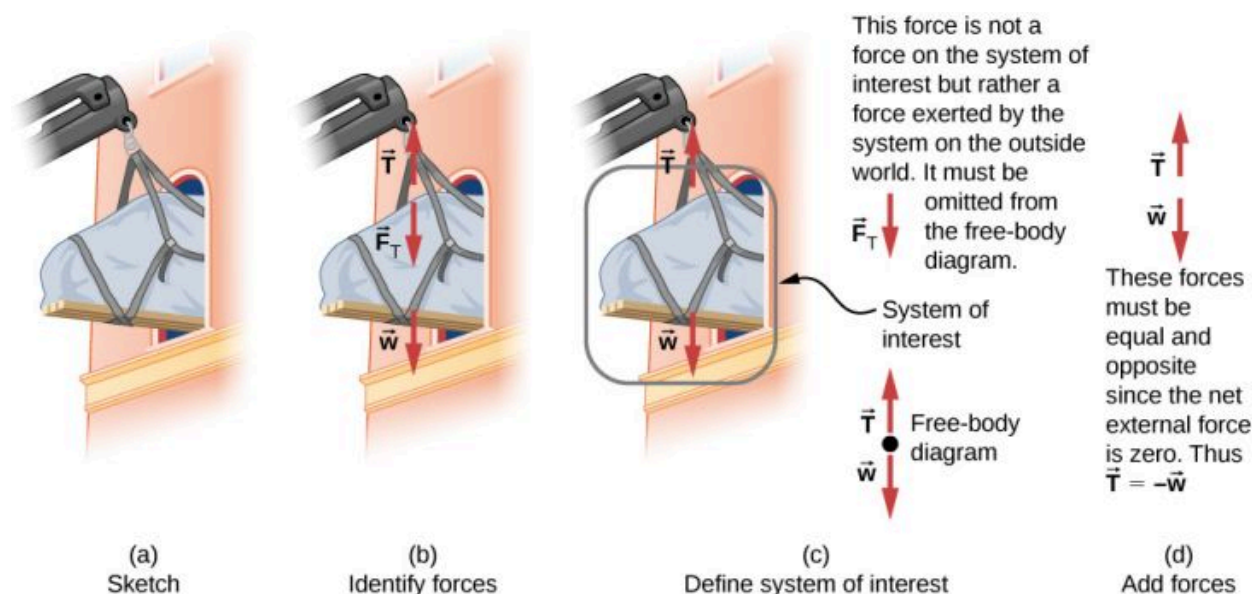


FIGURE 6.2 (a) A grand piano is being lifted to a second-story apartment. (b) Arrows are used to represent all forces: $\vec{T}$ is the tension in the rope above the piano, $\vec{F}_T$ is the force that the piano exerts on the rope, and $\vec{w}$ is the weight of the piano. All other forces, such as the nudge of a breeze, are assumed to be negligible. (c) Suppose we are given the piano's mass and asked to find the tension in the rope. We then define the system of interest as shown and draw a free-body diagram. Now $\vec{F}_T$ is no longer shown, because it is not a force acting on the system of interest; rather, $\vec{F}_T$ acts on the outside world. (d) Showing only the arrows, the head-to-tail method of addition is used. It is apparent that if the piano is stationary, $\vec{T} = -\vec{w}$.

As with most problems, we next need to identify what needs to be determined and what is known or can be inferred from the problem as stated, that is, make a list of knowns and unknowns. It is particularly crucial to identify the system of interest, since Newton's second law involves only external forces. We can then determine which forces are external and which are internal, a necessary step to employ Newton's second law. (See [Figure 6.2\(c\)](#).) Newton's third law may be used to identify whether forces are exerted between components of a system (internal) or between

the system and something outside (external). As illustrated in [Newton's Laws of Motion](#), the system of interest depends on the question we need to answer. Only forces are shown in free-body diagrams, not acceleration or velocity. We have drawn several free-body diagrams in previous worked examples. [Figure 6.2\(c\)](#) shows a free-body diagram for the system of interest. Note that no internal forces are shown in a free-body diagram.

Once a free-body diagram is drawn, we apply Newton's second law. This is done in [Figure 6.2\(d\)](#) for a particular situation. In general, once external forces are clearly identified in free-body diagrams, it should be a straightforward task to put them into equation form and solve for the unknown, as done in all previous examples. If the problem is one-dimensional—that is, if all forces are parallel—then the forces can be handled algebraically. If the problem is two-dimensional, then it must be broken down into a pair of one-dimensional problems. We do this by projecting the force vectors onto a set of axes chosen for convenience. As seen in previous examples, the choice of axes can simplify the problem. For example, when an incline is involved, a set of axes with one axis parallel to the incline and one perpendicular to it is most convenient. It is almost always convenient to make one axis parallel to the direction of motion, if this is known. Generally, just write Newton's second law in components along the different directions. Then, you have the following equations:

$$\sum F_x = ma_x, \quad \sum F_y = ma_y.$$

(If, for example, the system is accelerating horizontally, then you can then set $a_y = 0$.) We need this information to determine unknown forces acting on a system.

As always, we must check the solution. In some cases, it is easy to tell whether the solution is reasonable. For example, it is reasonable to find that friction causes an object to slide down an incline more slowly than when no friction exists. In practice, intuition develops gradually through problem solving; with experience, it becomes progressively easier to judge whether an answer is reasonable. Another way to check a solution is to check the units. If we are solving for force and end up with units of millimeters per second, then we have made a mistake.

There are many interesting applications of Newton's laws of motion, a few more of which are presented in this section. These serve also to illustrate some further subtleties of physics and to help build problem-solving skills. We look first at problems involving particle equilibrium, which make use of Newton's first law, and then consider particle acceleration, which involves Newton's second law.

Particle Equilibrium

Recall that a particle in equilibrium is one for which the external forces are balanced. Static equilibrium involves objects at rest, and dynamic equilibrium involves objects in motion without acceleration, but it is important to remember that these conditions are relative. For example, an object may be at rest when viewed from our frame of reference, but the same object would appear to be in motion when viewed by someone moving at a constant velocity. We now make use of the knowledge attained in [Newton's Laws of Motion](#), regarding the different types of forces and the use of free-body diagrams, to solve additional problems in particle equilibrium.



EXAMPLE 6.1

Different Tensions at Different Angles

Consider the traffic light (mass of 15.0 kg) suspended from two wires as shown in [Figure 6.3](#). Find the tension in each wire, neglecting the masses of the wires.

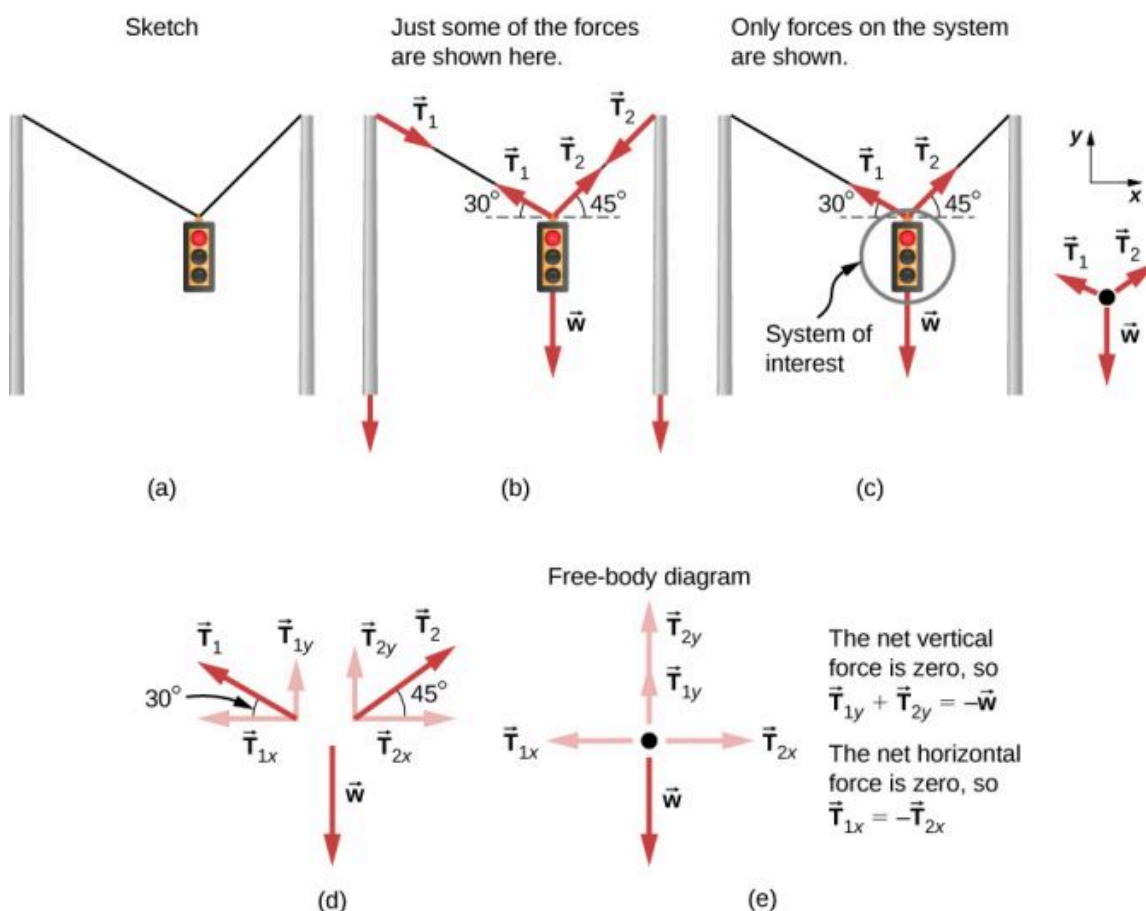


FIGURE 6.3 A traffic light is suspended from two wires. (b) Some of the forces involved. (c) Only forces acting on the system are shown here. The free-body diagram for the traffic light is also shown. (d) The forces projected onto vertical (y) and horizontal (x) axes. The horizontal components of the tensions must cancel, and the sum of the vertical components of the tensions must equal the weight of the traffic light. (e) The free-body diagram shows the vertical and horizontal forces acting on the traffic light.

Strategy

The system of interest is the traffic light, and its free-body diagram is shown in [Figure 6.3\(c\)](#). The three forces involved are not parallel, and so they must be projected onto a coordinate system. The most convenient coordinate system has one axis vertical and one horizontal, and the vector projections on it are shown in [Figure 6.3\(d\)](#). There are two unknowns in this problem (T_1 and T_2), so two equations are needed to find them. These two equations come from applying Newton's second law along the vertical and horizontal axes, noting that the net external force is zero along each axis because acceleration is zero.

Solution

First consider the horizontal or x-axis:

$$F_{\text{net } x} = T_{2x} + T_{1x} = 0.$$

Thus, as you might expect,

$$|T_{1x}| = |T_{2x}|.$$

This gives us the following relationship:

$$T_1 \cos 30^\circ = T_2 \cos 45^\circ.$$

Thus,

$$T_2 = 1.225T_1.$$

Note that T_1 and T_2 are not equal in this case because the angles on either side are not equal. It is reasonable that T_2 ends up being greater than T_1 because it is exerted more vertically than T_1 .

Now consider the force components along the vertical or y -axis:

$$F_{\text{net } y} = T_{1y} + T_{2y} - w = 0.$$

This implies

$$T_{1y} + T_{2y} = w.$$

Substituting the expressions for the vertical components gives

$$T_1 \sin 30^\circ + T_2 \sin 45^\circ = w.$$

There are two unknowns in this equation, but substituting the expression for T_2 in terms of T_1 reduces this to one equation with one unknown:

$$T_1(0.500) + (1.225T_1)(0.707) = w = mg,$$

which yields

$$1.366T_1 = (15.0 \text{ kg})(9.80 \text{ m/s}^2).$$

Solving this last equation gives the magnitude of T_1 to be

$$T_1 = 108 \text{ N}.$$

Finally, we find the magnitude of T_2 by using the relationship between them, $T_2 = 1.225T_1$, found above. Thus we obtain

$$T_2 = 132 \text{ N}.$$

Significance

Both tensions would be larger if both wires were more horizontal, and they will be equal if and only if the angles on either side are the same (as they were in the earlier example of a tightrope walker in [Newton's Laws of Motion](#)).

Particle Acceleration

We have given a variety of examples of particles in equilibrium. We now turn our attention to particle acceleration problems, which are the result of a nonzero net force. Refer again to the steps given at the beginning of this section, and notice how they are applied to the following examples.



EXAMPLE 6.2

Drag Force on a Barge

Two tugboats push on a barge at different angles ([Figure 6.4](#)). The first tugboat exerts a force of $2.7 \times 10^5 \text{ N}$ in the x -direction, and the second tugboat exerts a force of $3.6 \times 10^5 \text{ N}$ in the y -direction. The mass of the barge is $5.0 \times 10^6 \text{ kg}$ and its acceleration is observed to be $7.5 \times 10^{-2} \text{ m/s}^2$ in the direction shown. What is the drag force of the water on the barge resisting the motion? (*Note:* Drag force is a frictional force exerted by fluids, such as air or water. The drag force opposes the motion of the object. Since the barge is flat bottomed, we can assume that the drag force is in the direction opposite of motion of the barge.)

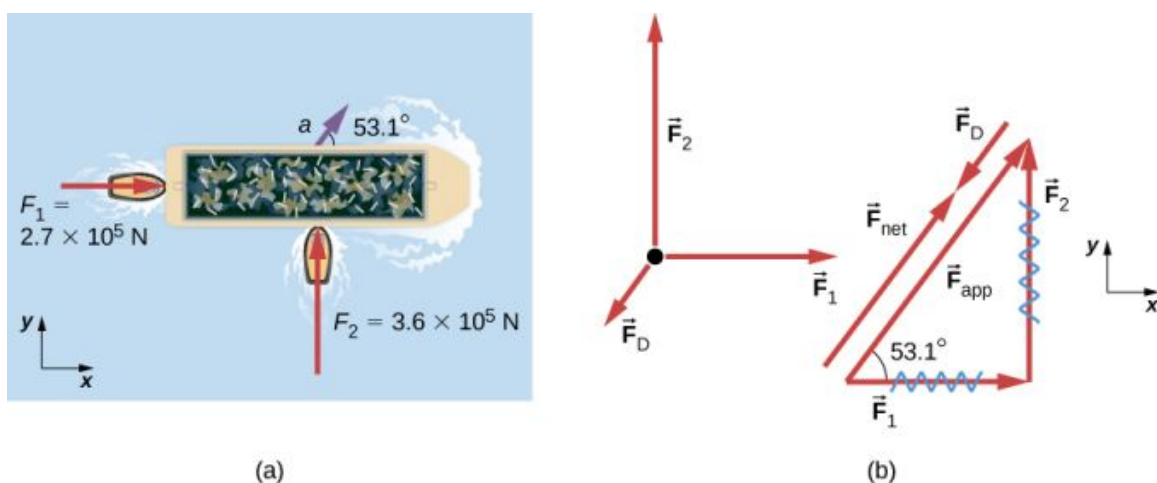


FIGURE 6.4 (a) A view from above of two tugboats pushing on a barge. (b) The free-body diagram for the ship contains only forces acting in the plane of the water. It omits the two vertical forces—the weight of the barge and the buoyant force of the water supporting it cancel and are not shown. Note that $\vec{F}_{\text{app}}$ is the total applied force of the tugboats.

Strategy

The directions and magnitudes of acceleration and the applied forces are given in [Figure 6.4\(a\)](#). We define the total force of the tugboats on the barge as $\vec{F}_{\text{app}}$ so that

$$\vec{F}_{\text{app}} = \vec{F}_1 + \vec{F}_2.$$

The drag of the water $\vec{F}_D$ is in the direction opposite to the direction of motion of the boat; this force thus works against $\vec{F}_{\text{app}}$, as shown in the free-body diagram in [Figure 6.4\(b\)](#). The system of interest here is the barge, since the forces on it are given as well as its acceleration. Because the applied forces are perpendicular, the x - and y -axes are in the same direction as $\vec{F}_1$ and $\vec{F}_2$. The problem quickly becomes a one-dimensional problem along the direction of $\vec{F}_{\text{app}}$, since friction is in the direction opposite to $\vec{F}_{\text{app}}$. Our strategy is to find the magnitude and direction of the net applied force $\vec{F}_{\text{app}}$ and then apply Newton's second law to solve for the drag force $\vec{F}_D$.

Solution

Since F_x and F_y are perpendicular, we can find the magnitude and direction of $\vec{F}_{\text{app}}$ directly. First, the resultant magnitude is given by the Pythagorean theorem:

$$F_{\text{app}} = \sqrt{F_1^2 + F_2^2} = \sqrt{(2.7 \times 10^5 \text{ N})^2 + (3.6 \times 10^5 \text{ N})^2} = 4.5 \times 10^5 \text{ N}.$$

The angle is given by

$$\theta = \tan^{-1} \left(\frac{F_2}{F_1} \right) = \tan^{-1} \left(\frac{3.6 \times 10^5 \text{ N}}{2.7 \times 10^5 \text{ N}} \right) = 53.1^\circ.$$

From Newton's first law, we know this is the same direction as the acceleration. We also know that $\vec{F}_D$ is in the opposite direction of $\vec{F}_{\text{app}}$, since it acts to slow down the acceleration. Therefore, the net external force is in the same direction as $\vec{F}_{\text{app}}$, but its magnitude is slightly less than $\vec{F}_{\text{app}}$. The problem is now one-dimensional. From the free-body diagram, we can see that

$$F_{\text{net}} = F_{\text{app}} - F_D.$$

However, Newton's second law states that

$$F_{\text{net}} = ma.$$

Thus,

$$F_{\text{app}} - F_D = ma.$$

This can be solved for the magnitude of the drag force of the water F_D in terms of known quantities:

$$F_D = F_{\text{app}} - ma.$$

Substituting known values gives

$$F_D = (4.5 \times 10^5 \text{ N}) - (5.0 \times 10^6 \text{ kg}) (7.5 \times 10^{-2} \text{ m/s}^2) = 7.5 \times 10^4 \text{ N}.$$

The direction of $\vec{F}_D$ has already been determined to be in the direction opposite to $\vec{F}_{\text{app}}$, or at an angle of 53° south of west.

Significance

The numbers used in this example are reasonable for a moderately large barge. It is certainly difficult to obtain larger accelerations with tugboats, and small speeds are desirable to avoid running the barge into the docks. Drag is relatively small for a well-designed hull at low speeds, consistent with the answer to this example, where F_D is less than 1/600th of the weight of the ship.

In [Newton's Laws of Motion](#), we discussed the normal force, which is a contact force that acts normal to the surface so that an object does not have an acceleration perpendicular to the surface. The bathroom scale is an excellent example of a normal force acting on a body. It provides a quantitative reading of how much it must push upward to support the weight of an object. But can you predict what you would see on the dial of a bathroom scale if you stood on it during an elevator ride? Will you see a value greater than your weight when the elevator starts up? What about when the elevator moves upward at a constant speed? Take a guess before reading the next example.



EXAMPLE 6.3

What Does the Bathroom Scale Read in an Elevator?

[Figure 6.5](#) shows a 75.0-kg man (weight of about 165 lb.) standing on a bathroom scale in an elevator. Calculate the scale reading: (a) if the elevator accelerates upward at a rate of 1.20 m/s^2 , and (b) if the elevator moves upward at a constant speed of 1 m/s .

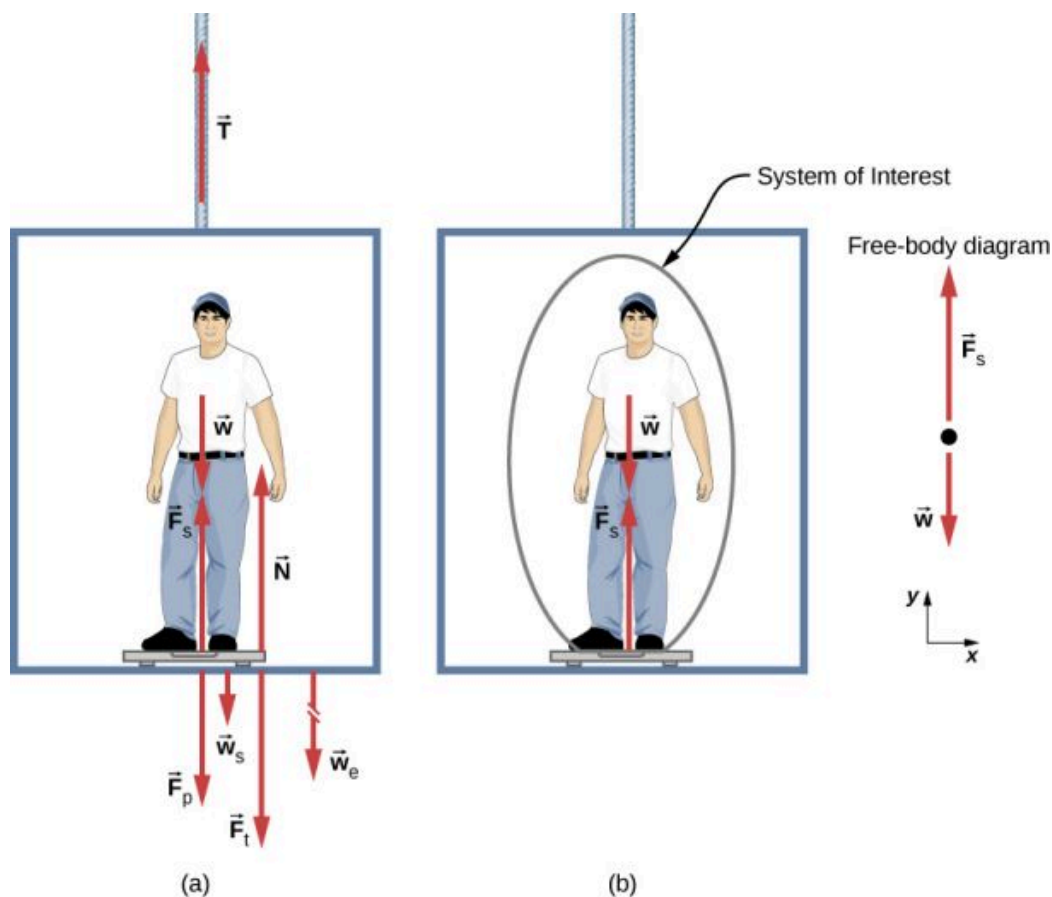


FIGURE 6.5 (a) The various forces acting when a person stands on a bathroom scale in an elevator. The arrows are approximately correct for when the elevator is accelerating upward—broken arrows represent forces too large to be drawn to scale. $\vec{T}$ is the tension in the supporting cable, $\vec{w}$ is the weight of the person, $\vec{w}_s$ is the weight of the scale, $\vec{w}_e$ is the weight of the elevator, $\vec{F}_s$ is the force of the scale on the person, $\vec{F}_p$ is the force of the person on the scale, $\vec{F}_t$ is the force of the scale on the floor of the elevator, and $\vec{N}$ is the force of the floor upward on the scale. (b) The free-body diagram shows only the external forces acting on the designated system of interest—the person—and is the diagram we use for the solution of the problem.

Strategy

If the scale at rest is accurate, its reading equals $\vec{F}_p$, the magnitude of the force the person exerts downward on it. [Figure 6.5\(a\)](#) shows the numerous forces acting on the elevator, scale, and person. It makes this one-dimensional problem look much more formidable than if the person is chosen to be the system of interest and a free-body diagram is drawn, as in [Figure 6.5\(b\)](#). Analysis of the free-body diagram using Newton's laws can produce answers to both [Figure 6.5\(a\)](#) and (b) of this example, as well as some other questions that might arise. The only forces acting on the person are his weight $\vec{w}$ and the upward force of the scale $\vec{F}_s$. According to Newton's third law, $\vec{F}_p$ and $\vec{F}_s$ are equal in magnitude and opposite in direction, so that we need to find F_s in order to find what the scale reads. We can do this, as usual, by applying Newton's second law,

$$\vec{F}_{\text{net}} = m\vec{a}.$$

From the free-body diagram, we see that $\vec{F}_{\text{net}} = \vec{F}_s - \vec{w}$, so we have

$$F_s - w = ma.$$

Solving for F_s gives us an equation with only one unknown:

$$F_s = ma + w,$$

or, because $w = mg$, simply

$$F_s = ma + mg.$$

No assumptions were made about the acceleration, so this solution should be valid for a variety of accelerations in addition to those in this situation. (*Note:* We are considering the case when the elevator is accelerating upward. If the elevator is accelerating downward, Newton's second law becomes $F_s - w = -ma$.)

Solution

- a. We have $a = 1.20 \text{ m/s}^2$, so that

$$F_s = (75.0 \text{ kg})(9.80 \text{ m/s}^2) + (75.0 \text{ kg})(1.20 \text{ m/s}^2)$$

yielding

$$F_s = 825 \text{ N}.$$

- b. Now, what happens when the elevator reaches a constant upward velocity? Will the scale still read more than his weight? For any constant velocity—up, down, or stationary—acceleration is zero because $a = \frac{\Delta v}{\Delta t}$ and $\Delta v = 0$. Thus,

$$F_s = ma + mg = 0 + mg$$

or

$$F_s = (75.0 \text{ kg})(9.80 \text{ m/s}^2),$$

which gives

$$F_s = 735 \text{ N}.$$

Significance

The scale reading in [Figure 6.5\(a\)](#) is about 185 lb. What would the scale have read if he were stationary? Since his acceleration would be zero, the force of the scale would be equal to his weight:

$$F_{\text{net}} = ma = 0 = F_s - w$$

$$F_s = w = mg$$

$$F_s = (75.0 \text{ kg})(9.80 \text{ m/s}^2) = 735 \text{ N}.$$

Thus, the scale reading in the elevator is greater than his 735-N (165-lb.) weight. This means that the scale is pushing up on the person with a force greater than his weight, as it must in order to accelerate him upward. Clearly, the greater the acceleration of the elevator, the greater the scale reading, consistent with what you feel in rapidly accelerating versus slowly accelerating elevators. In [Figure 6.5\(b\)](#), the scale reading is 735 N, which equals the person's weight. This is the case whenever the elevator has a constant velocity—moving up, moving down, or stationary.

CHECK YOUR UNDERSTANDING 6.1

Now calculate the scale reading when the elevator accelerates downward at a rate of 1.20 m/s^2 .

The solution to the previous example also applies to an elevator accelerating downward, as mentioned. When an elevator accelerates downward, a is negative, and the scale reading is *less* than the weight of the person. If a constant downward velocity is reached, the scale reading again becomes equal to the person's weight. If the elevator is in free fall and accelerating downward at g , then the scale reading is zero and the person appears to be weightless.



EXAMPLE 6.4

Two Attached Blocks

[Figure 6.6](#) shows a block of mass m_1 on a frictionless, horizontal surface. It is pulled by a light string that passes over a frictionless and massless pulley. The other end of the string is connected to a block of mass m_2 . Find the acceleration of the blocks and the tension in the string in terms of m_1 , m_2 , and g .

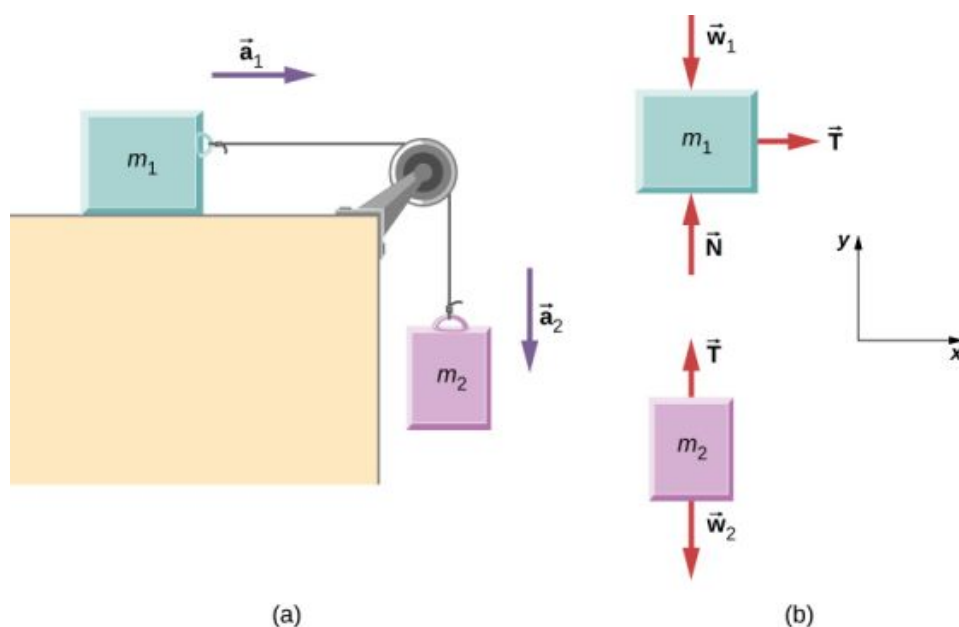


FIGURE 6.6 (a) Block 1 is connected by a light string to block 2. (b) The free-body diagrams of the blocks.

Strategy

We draw a free-body diagram for each mass separately, as shown in Figure 6.6. Then we analyze each one to find the required unknowns. The forces on block 1 are the gravitational force, the contact force of the surface, and the tension in the string. Block 2 is subjected to the gravitational force and the string tension. Newton's second law applies to each, so we write two vector equations:

$$\text{For block 1: } \vec{T} + \vec{w}_1 + \vec{N} = m_1 \vec{a}_1$$

$$\text{For block 2: } \vec{T} + \vec{w}_2 = m_2 \vec{a}_2.$$

Notice that $\vec{T}$ is the same for both blocks. Since the string and the pulley have negligible mass, and since there is no friction in the pulley, the tension is the same throughout the string. We can now write component equations for each block. All forces are either horizontal or vertical, so we can use the same horizontal/vertical coordinate system for both objects

Solution

The component equations follow from the vector equations above. We see that block 1 has the vertical forces balanced, so we ignore them and write an equation relating the x-components. There are no horizontal forces on block 2, so only the y-equation is written. We obtain these results:

Block 1	Block 2
$\sum F_x = ma_x$	$\sum F_y = ma_y$
$T_x = m_1 a_{1x}$	$T_y - m_2 g = m_2 a_{2y}.$

When block 1 moves to the right, block 2 travels an equal distance downward; thus, $a_{1x} = -a_{2y}$. Writing the common acceleration of the blocks as $a = a_{1x} = -a_{2y}$, we now have

$$T = m_1 a$$

and

$$T - m_2 g = -m_2 a.$$

From these two equations, we can express a and T in terms of the masses m_1 and m_2 , and g :

$$a = \frac{m_2}{m_1 + m_2} g$$

and

$$T = \frac{m_1 m_2}{m_1 + m_2} g.$$

Significance

Notice that the tension in the string is *less* than the weight of the block hanging from the end of it. A common error in problems like this is to set $T = m_2 g$. You can see from the free-body diagram of block 2 that cannot be correct if the block is accelerating.

✓ CHECK YOUR UNDERSTANDING 6.2

Calculate the acceleration of the system, and the tension in the string, when the masses are $m_1 = 5.00$ kg and $m_2 = 3.00$ kg.

✳ EXAMPLE 6.5

Atwood Machine

A classic problem in physics, similar to the one we just solved, is that of the Atwood machine, which consists of a rope running over a pulley, with two objects of different mass attached. It is particularly useful in understanding the connection between force and motion. In [Figure 6.7](#), $m_1 = 2.00$ kg and $m_2 = 4.00$ kg. Consider the pulley to be frictionless. (a) If m_2 is released, what will its acceleration be? (b) What is the tension in the string?

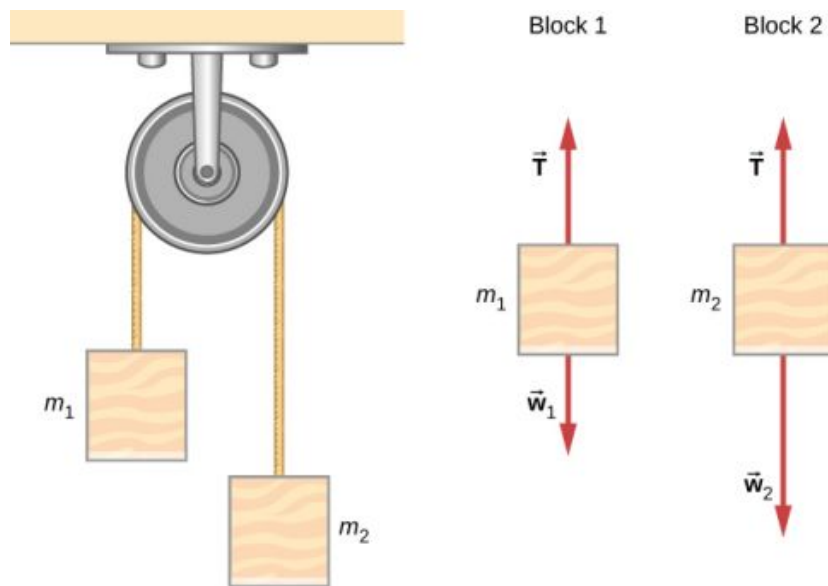


FIGURE 6.7 An Atwood machine and free-body diagrams for each of the two blocks.

Strategy

We draw a free-body diagram for each mass separately, as shown in the figure. Then we analyze each diagram to find the required unknowns. This may involve the solution of simultaneous equations. It is also important to note the similarity with the previous example. As block 2 accelerates with acceleration a_2 in the downward direction, block 1 accelerates upward with acceleration a_1 . Thus, $a = a_1 = -a_2$.

Solution

a. We have

$$\text{For } m_1, \sum F_y = T - m_1 g = m_1 a. \quad \text{For } m_2, \sum F_y = T - m_2 g = -m_2 a.$$

(The negative sign in front of $m_2 a$ indicates that m_2 accelerates downward; both blocks accelerate at the same rate, but in opposite directions.) Solve the two equations simultaneously (subtract them) and the result is

$$(m_2 - m_1)g = (m_1 + m_2)a.$$

Solving for a :

$$a = \frac{m_2 - m_1}{m_1 + m_2}g = \frac{4 \text{ kg} - 2 \text{ kg}}{4 \text{ kg} + 2 \text{ kg}}(9.8 \text{ m/s}^2) = 3.27 \text{ m/s}^2.$$

- b. Observing the first block, we see that

$$T - m_1 g = m_1 a$$

$$T = m_1(g + a) = (2 \text{ kg})(9.8 \text{ m/s}^2 + 3.27 \text{ m/s}^2) = 26.1 \text{ N}.$$

Significance

The result for the acceleration given in the solution can be interpreted as the ratio of the unbalanced force on the system, $(m_2 - m_1)g$, to the total mass of the system, $m_1 + m_2$. We can also use the Atwood machine to measure local gravitational field strength.

CHECK YOUR UNDERSTANDING 6.3

Determine a general formula in terms of m_1 , m_2 and g for calculating the tension in the string for the Atwood machine shown above.

Newton's Laws of Motion and Kinematics

Physics is most interesting and most powerful when applied to general situations that involve more than a narrow set of physical principles. Newton's laws of motion can also be integrated with other concepts that have been discussed previously in this text to solve problems of motion. For example, forces produce accelerations, a topic of kinematics, and hence the relevance of earlier chapters.

When approaching problems that involve various types of forces, acceleration, velocity, and/or position, listing the givens and the quantities to be calculated will allow you to identify the principles involved. Then, you can refer to the chapters that deal with a particular topic and solve the problem using strategies outlined in the text. The following worked example illustrates how the problem-solving strategy given earlier in this chapter, as well as strategies presented in other chapters, is applied to an integrated concept problem.

EXAMPLE 6.6

What Force Must a Soccer Player Exert to Reach Top Speed?

A soccer player starts at rest and accelerates forward, reaching a velocity of 8.00 m/s in 2.50 s. (a) What is her average acceleration? (b) What average force does the ground exert forward on the runner so that she achieves this acceleration? The player's mass is 70.0 kg, and air resistance is negligible.

Strategy

To find the answers to this problem, we use the problem-solving strategy given earlier in this chapter. The solutions to each part of the example illustrate how to apply specific problem-solving steps. In this case, we do not need to use all of the steps. We simply identify the physical principles, and thus the knowns and unknowns; apply Newton's second law; and check to see whether the answer is reasonable.

Solution

- a. We are given the initial and final velocities (zero and 8.00 m/s forward); thus, the change in velocity is $\Delta v = 8.00 \text{ m/s}$. We are given the elapsed time, so $\Delta t = 2.50 \text{ s}$. The unknown is acceleration, which can be found from its definition:

$$a = \frac{\Delta v}{\Delta t}.$$

Substituting the known values yields

$$a = \frac{8.00 \text{ m/s}}{2.50 \text{ s}} = 3.20 \text{ m/s}^2.$$

- b. Here we are asked to find the average force the ground exerts on the runner to produce this acceleration. (Remember that we are dealing with the force or forces acting on the object of interest.) This is the reaction force to that exerted by the player backward against the ground, by Newton's third law. Neglecting air resistance, this would be equal in magnitude to the net external force on the player, since this force causes her acceleration. Since we now know the player's acceleration and are given her mass, we can use Newton's second law to find the force exerted. That is,

$$F_{\text{net}} = ma.$$

Substituting the known values of m and a gives

$$F_{\text{net}} = (70.0 \text{ kg})(3.20 \text{ m/s}^2) = 224 \text{ N}.$$

This is a reasonable result: The acceleration is attainable for an athlete in good condition. The force is about 50 pounds, a reasonable average force.

Significance

This example illustrates how to apply problem-solving strategies to situations that include topics from different chapters. The first step is to identify the physical principles, the knowns, and the unknowns involved in the problem. The second step is to solve for the unknown, in this case using Newton's second law. Finally, we check our answer to ensure it is reasonable. These techniques for integrated concept problems will be useful in applications of physics outside of a physics course, such as in your profession, in other science disciplines, and in everyday life.

CHECK YOUR UNDERSTANDING 6.4

The soccer player stops after completing the play described above, but now notices that the ball is in position to be stolen. If she now experiences a force of 126 N to attempt to steal the ball, which is 2.00 m away from her, how long will it take her to get to the ball?

EXAMPLE 6.7

What Force Acts on a Model Helicopter?

A 1.50-kg model helicopter has a velocity of $5.00\hat{\mathbf{j}}$ m/s at $t = 0$. It is accelerated at a constant rate for two seconds (2.00 s) after which it has a velocity of $(6.00\hat{\mathbf{i}} + 12.00\hat{\mathbf{j}})$ m/s. What is the magnitude of the resultant force acting on the helicopter during this time interval?

Strategy

We can easily set up a coordinate system in which the x -axis ($\hat{\mathbf{i}}$ direction) is horizontal, and the y -axis ($\hat{\mathbf{j}}$ direction) is vertical. We know that $\Delta t = 2.00 \text{ s}$ and $\Delta \mathbf{v} = (6.00\hat{\mathbf{i}} + 12.00\hat{\mathbf{j}} \text{ m/s}) - (5.00\hat{\mathbf{j}} \text{ m/s})$. From this, we can calculate the acceleration by the definition; we can then apply Newton's second law.

Solution

We have

$$a = \frac{\Delta v}{\Delta t} = \frac{(6.00\hat{\mathbf{i}} + 12.00\hat{\mathbf{j}} \text{ m/s}) - (5.00\hat{\mathbf{j}} \text{ m/s})}{2.00 \text{ s}} = 3.00\hat{\mathbf{i}} + 3.50\hat{\mathbf{j}} \text{ m/s}^2$$

$$\sum \vec{\mathbf{F}} = m\vec{\mathbf{a}} = (1.50 \text{ kg})(3.00\hat{\mathbf{i}} + 3.50\hat{\mathbf{j}} \text{ m/s}^2) = 4.50\hat{\mathbf{i}} + 5.25\hat{\mathbf{j}} \text{ N}.$$

The magnitude of the force is now easily found:

$$F = \sqrt{(4.50 \text{ N})^2 + (5.25 \text{ N})^2} = 6.91 \text{ N}.$$

Significance

The original problem was stated in terms of $\hat{i} - \hat{j}$ vector components, so we used vector methods. Compare this example with the previous example.

✓ CHECK YOUR UNDERSTANDING 6.5

Find the direction of the resultant for the 1.50-kg model helicopter.

✿ EXAMPLE 6.8

Baggage Tractor

Figure 6.8(a) shows a baggage tractor pulling luggage carts from an airplane. The tractor has mass 650.0 kg, while cart A has mass 250.0 kg and cart B has mass 150.0 kg. The driving force acting for a brief period of time accelerates the system from rest and acts for 3.00 s. (a) If this driving force is given by $F = (820.0t) \text{ N}$, find the speed after 3.00 seconds. (b) What is the horizontal force acting on the connecting cable between the tractor and cart A at this instant?

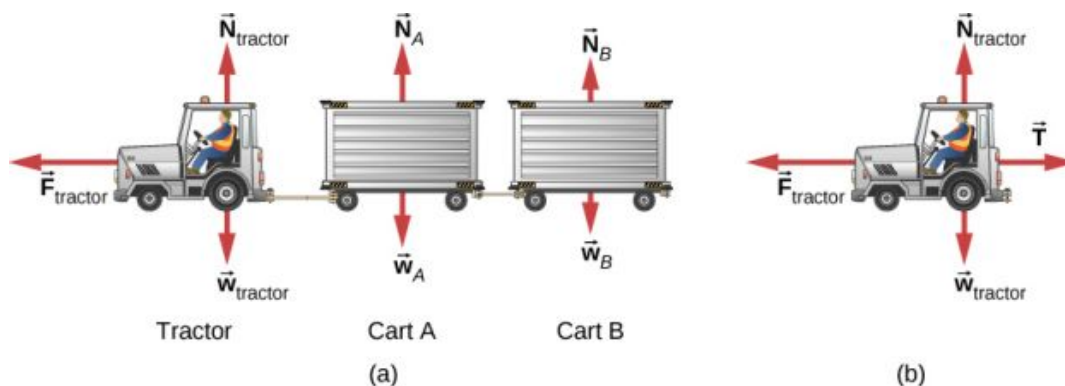


FIGURE 6.8 (a) A free-body diagram is shown, which indicates all the external forces on the system consisting of the tractor and baggage carts for carrying airline luggage. (b) A free-body diagram of the tractor only is shown isolated in order to calculate the tension in the cable to the carts.

Strategy

A free-body diagram shows the driving force of the tractor, which gives the system its acceleration. We only need to consider motion in the horizontal direction. The vertical forces balance each other and it is not necessary to consider them. For part b, we make use of a free-body diagram of the tractor alone to determine the force between it and cart A. This exposes the coupling force $\vec{T}$, which is our objective.

Solution

$$\begin{aligned} \text{a. } \sum F_x &= m_{\text{system}} a_x \text{ and } \sum F_x = 820.0t, \text{ so} \\ 820.0t &= (650.0 + 250.0 + 150.0)a \\ a &= 0.7809t. \end{aligned}$$

Since acceleration is a function of time, we can determine the velocity of the tractor by using $a = \frac{dv}{dt}$ with the initial condition that $v_0 = 0$ at $t = 0$. We integrate from $t = 0$ to $t = 3$:

$$dv = a dt, \quad \int_0^3 dv = \int_0^{3.00} a dt = \int_0^{3.00} 0.7809t dt, \quad v = 0.3905t^2 \Big|_0^{3.00} = 3.51 \text{ m/s}.$$

- b. Refer to the free-body diagram in [Figure 6.8\(b\)](#).

$$\begin{aligned}\sum F_x &= m_{\text{tractor}} a_x \\ 820.0t - T &= m_{\text{tractor}}(0.7805)t \\ (820.0)(3.00) - T &= (650.0)(0.7805)(3.00) \\ T &= 938 \text{ N.}\end{aligned}$$

Significance

Since the force varies with time, we must use calculus to solve this problem. Notice how the total mass of the system was important in solving [Figure 6.8\(a\)](#), whereas only the mass of the truck (since it supplied the force) was of use in [Figure 6.8\(b\)](#).

Recall that $v = \frac{ds}{dt}$ and $a = \frac{dv}{dt}$. If acceleration is a function of time, we can use the calculus forms developed in [Motion Along a Straight Line](#), as shown in this example. However, sometimes acceleration is a function of displacement. In this case, we can derive an important result from these calculus relations. Solving for dt in each, we have $dt = \frac{ds}{v}$ and $dt = \frac{dv}{a}$. Now, equating these expressions, we have $\frac{ds}{v} = \frac{dv}{a}$. We can rearrange this to obtain $ads = vdv$.

EXAMPLE 6.9

Motion of a Projectile Fired Vertically

A 10.0-kg mortar shell is fired vertically upward from the ground, with an initial velocity of 50.0 m/s (see [Figure 6.9](#)). Determine the maximum height it will travel if atmospheric resistance is measured as $F_D = (0.0100v^2) \text{ N}$, where v is the speed at any instant.

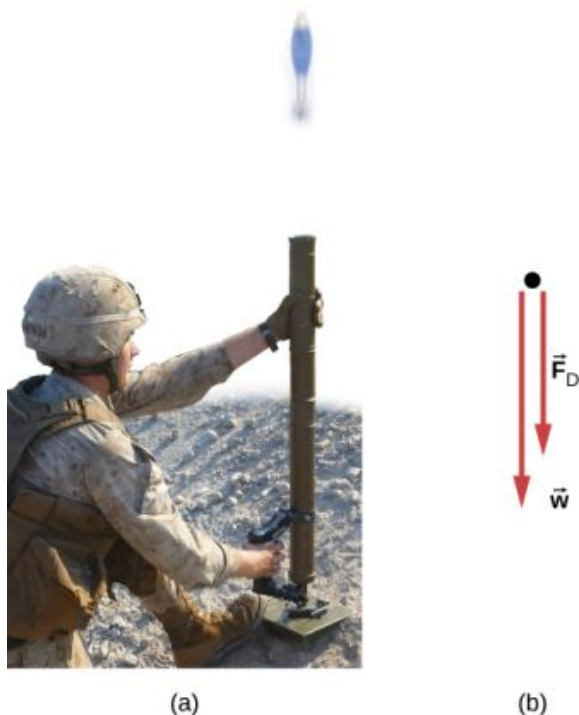


FIGURE 6.9 (a) The mortar fires a shell straight up; we consider the friction force provided by the air. (b) A free-body diagram is shown which indicates all the forces on the mortar shell. (credit a: modification of work by OS541/DoD; The appearance of U.S. Department of Defense (DoD) visual information does not imply or constitute DoD endorsement.)

Strategy

The known force on the mortar shell can be related to its acceleration using the equations of motion. Kinematics can then be used to relate the mortar shell's acceleration to its position.

Solution

Initially, $y_0 = 0$ and $v_0 = 50.0$ m/s. At the maximum height $y = h$, $v = 0$. The free-body diagram shows F_D to act downward, because it slows the upward motion of the mortar shell. Thus, we can write

$$\sum F_y = ma_y$$

$$\begin{aligned} -F_D - w &= ma_y \\ -0.0100v^2 - 98.0 &= 10.0a \\ a &= -0.00100v^2 - 9.80. \end{aligned}$$

The acceleration depends on v and is therefore variable. Since $a = f(v)$, we can relate a to v using the rearrangement described above,

$$ads = vdv.$$

We replace ds with dy because we are dealing with the vertical direction,

$$ady = vdv, \quad (-0.00100v^2 - 9.80)dy = vdv.$$

We now separate the variables (v 's and dv 's on one side; dy on the other):

$$\begin{aligned} \int_0^h dy &= \int_{50.0}^0 \frac{v dv}{(-0.00100v^2 - 9.80)} \\ \int_0^h dy &= - \int_{50.0}^0 \frac{v dv}{(0.00100v^2 + 9.80)} = (-5 \times 10^2) \ln(0.00100v^2 + 9.80) \Big|_{50.0}^0. \end{aligned}$$

Thus, $h = 114$ m.

Significance

Notice the need to apply calculus since the force is not constant, which also means that acceleration is not constant. To make matters worse, the force depends on v (not t), and so we must use the trick explained prior to the example. The answer for the height indicates a lower elevation if there were air resistance. We will deal with the effects of air resistance and other drag forces in greater detail in [Drag Force and Terminal Speed](#).

** CHECK YOUR UNDERSTANDING 6.6**

If atmospheric resistance is neglected, find the maximum height for the mortar shell. Is calculus required for this solution?

** INTERACTIVE**

Explore the forces at work in this [simulation \(https://openstax.org/l/21forcesatwork\)](https://openstax.org/l/21forcesatwork) when you try to push a filing cabinet. Create an applied force and see the resulting frictional force and total force acting on the cabinet. Charts show the forces, position, velocity, and acceleration vs. time. View a free-body diagram of all the forces (including gravitational and normal forces).

6.2 Friction

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Describe the general characteristics of friction
- List the various types of friction
- Calculate the magnitude of static and kinetic friction, and use these in problems involving Newton's laws of motion

When a body is in motion, it has resistance because the body interacts with its surroundings. This resistance is a force of friction. Friction opposes relative motion between systems in contact but also allows us to move, a concept that becomes obvious if you try to walk on ice. Friction is a common yet complex force, and its behavior still not completely understood. Still, it is possible to understand the circumstances in which it behaves.

Static and Kinetic Friction

The basic definition of **friction** is relatively simple to state.

FRICITION

Friction is a force that opposes relative motion between systems in contact.

There are several forms of friction. One of the simpler characteristics of sliding friction is that it is parallel to the contact surfaces between systems and is always in a direction that opposes motion or attempted motion of the systems relative to each other. If two systems are in contact and moving relative to one another, then the friction between them is called kinetic friction. For example, friction slows a hockey puck sliding on ice. When objects are stationary, static friction can act between them; the static friction is usually greater than the kinetic friction between two objects.

STATIC AND KINETIC FRICTION

If two systems are in contact and stationary relative to one another, then the friction between them is called **static friction**. If two systems are in contact and moving relative to one another, then the friction between them is called **kinetic friction**.

Imagine, for example, trying to slide a heavy crate across a concrete floor—you might push very hard on the crate and not move it at all. This means that the static friction responds to what you do—it increases to be equal to and in the opposite direction of your push. If you finally push hard enough, the crate seems to slip suddenly and starts to move. Now static friction gives way to kinetic friction. Once in motion, it is easier to keep it in motion than it was to get it started, indicating that the kinetic frictional force is less than the static frictional force. If you add mass to the crate, say by placing a box on top of it, you need to push even harder to get it started and also to keep it moving. Furthermore, if you oiled the concrete you would find it easier to get the crate started and keep it going (as you might expect).

[Figure 6.10](#) is a crude pictorial representation of how friction occurs at the interface between two objects. Close-up inspection of these surfaces shows them to be rough. Thus, when you push to get an object moving (in this case, a crate), you must raise the object until it can skip along with just the tips of the surface hitting, breaking off the points, or both. A considerable force can be resisted by friction with no apparent motion. The harder the surfaces are pushed together (such as if another box is placed on the crate), the more force is needed to move them. Part of the friction is due to adhesive forces between the surface molecules of the two objects, which explains the dependence of friction on the nature of the substances. For example, rubber-soled shoes slip less than those with leather soles. Adhesion varies with substances in contact and is a complicated aspect of surface physics. Once an object is moving, there are fewer points of contact (fewer molecules adhering), so less force is required to keep the object moving. At small but nonzero speeds, friction is nearly independent of speed.

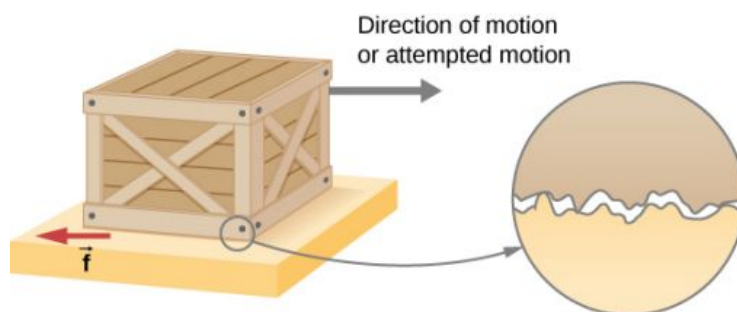


FIGURE 6.10 Frictional forces, such as $\vec{f}$, always oppose motion or attempted motion between objects in contact. Friction arises in part because of the roughness of the surfaces in contact, as seen in the expanded view. For the object to move, it must rise to where the peaks of the top surface can skip along the bottom surface. Thus, a force is required just to set the object in motion. Some of the peaks will be broken off, also requiring a force to maintain motion. Much of the friction is actually due to attractive forces between molecules making up the two objects, so that even perfectly smooth surfaces are not friction-free. (In fact, perfectly smooth, clean surfaces of similar materials would adhere, forming a bond called a “cold weld.”)

The magnitude of the frictional force has two forms: one for static situations (static friction), the other for situations involving motion (kinetic friction). What follows is an approximate empirical (experimentally determined) model only. These equations for static and kinetic friction are not vector equations.

MAGNITUDE OF STATIC FRICTION

The magnitude of static friction f_s is

$$f_s \leq \mu_s N, \quad 6.1$$

where μ_s is the coefficient of static friction and N is the magnitude of the normal force.

The symbol $\leq$ means *less than or equal to*, implying that static friction can have a maximum value of $\mu_s N$. Static friction is a responsive force that increases to be equal and opposite to whatever force is exerted, up to its maximum limit. Once the applied force exceeds

$f_s(\text{max})$, the object moves. Thus,

$$f_s(\text{max}) = \mu_s N.$$

MAGNITUDE OF KINETIC FRICTION

The magnitude of kinetic friction f_k is given by

$$f_k = \mu_k N, \quad 6.2$$

where μ_k is the coefficient of kinetic friction.

A system in which $f_k = \mu_k N$ is described as a system in which *friction behaves simply*. The transition from static friction to kinetic friction is illustrated in [Figure 6.11](#).

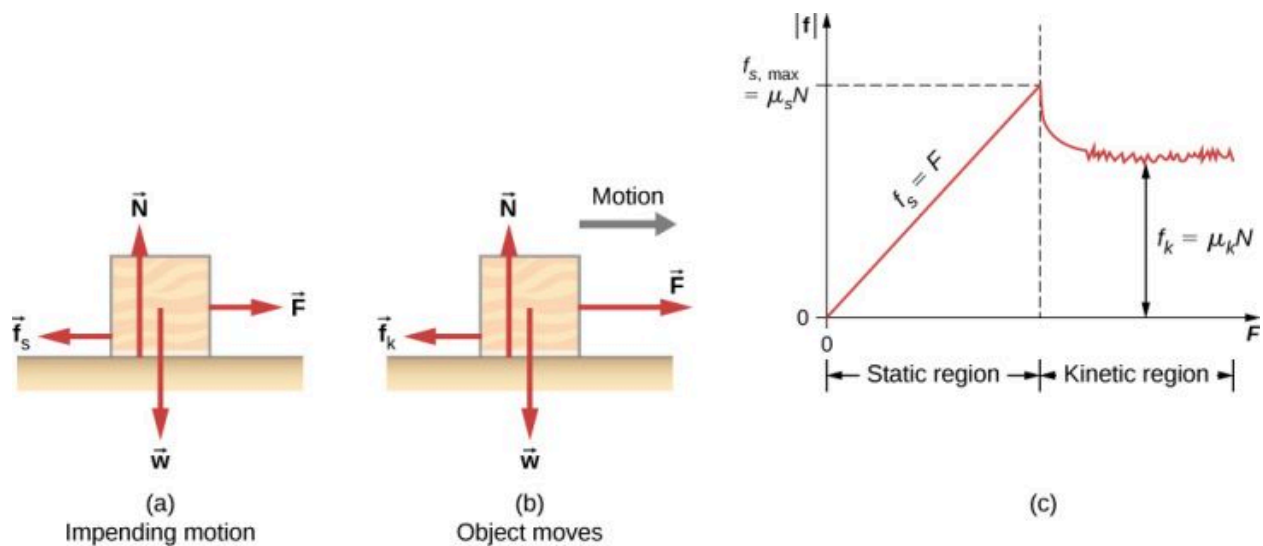


FIGURE 6.11 (a) The force of friction $\vec{f}$ between the block and the rough surface opposes the direction of the applied force $\vec{F}$. The magnitude of the static friction balances that of the applied force. This is shown in the left side of the graph in (c). (b) At some point, the magnitude of the applied force is greater than the force of kinetic friction, and the block moves to the right. This is shown in the right side of the graph. (c) The graph of the frictional force versus the applied force; note that $f_{s, \max} > f_k$. This means that $\mu_s > \mu_k$.

As you can see in [Table 6.1](#), the coefficients of kinetic friction are less than their static counterparts. The approximate values of μ are stated to only one or two digits to indicate the approximate description of friction given by the preceding two equations.

System	Static Friction μ_s	Kinetic Friction μ_k
Rubber on dry concrete	1.0	0.7
Rubber on wet concrete	0.5-0.7	0.3-0.5
Wood on wood	0.5	0.3
Waxed wood on wet snow	0.14	0.1
Metal on wood	0.5	0.3
Steel on steel (dry)	0.6	0.3
Steel on steel (oiled)	0.05	0.03
Teflon on steel	0.04	0.04
Bone lubricated by synovial fluid	0.016	0.015
Shoes on wood	0.9	0.7
Shoes on ice	0.1	0.05
Ice on ice	0.1	0.03
Steel on ice	0.04	0.02

TABLE 6.1 Approximate Coefficients of Static and Kinetic Friction

[Equation 6.1](#) and [Equation 6.2](#) include the dependence of friction on materials and the normal force. The direction of friction is always opposite that of motion, parallel to the surface between objects, and perpendicular to the normal force. For example, if the crate you try to push (with a force parallel to the floor) has a mass of 100 kg, then the normal force is equal to its weight,

$$w = mg = (100 \text{ kg})(9.80 \text{ m/s}^2) = 980 \text{ N},$$

perpendicular to the floor. If the coefficient of static friction is 0.45, you would have to exert a force parallel to the floor greater than

$$f_s(\text{max}) = \mu_s N = (0.45)(980 \text{ N}) = 440 \text{ N}$$

to move the crate. Once there is motion, friction is less and the coefficient of kinetic friction might be 0.30, so that a force of only

$$f_k = \mu_k N = (0.30)(980 \text{ N}) = 290 \text{ N}$$

keeps it moving at a constant speed. If the floor is lubricated, both coefficients are considerably less than they would be without lubrication. Coefficient of friction is a unitless quantity with a magnitude usually between 0 and 1.0. The actual value depends on the two surfaces that are in contact.

Many people have experienced the slipperiness of walking on ice. However, many parts of the body, especially the joints, have much smaller coefficients of friction—often three or four times less than ice. A joint is formed by the ends of two bones, which are connected by thick tissues. The knee joint is formed by the lower leg bone (the tibia) and the thighbone (the femur). The hip is a ball (at the end of the femur) and socket (part of the pelvis) joint. The ends of the bones in the joint are covered by cartilage, which provides a smooth, almost-glassy surface. The joints also produce a fluid (synovial fluid) that reduces friction and wear. A damaged or arthritic joint can be replaced by an artificial joint ([Figure 6.12](#)). These replacements can be made of metals (stainless steel or titanium) or plastic (polyethylene), also with very small coefficients of friction.

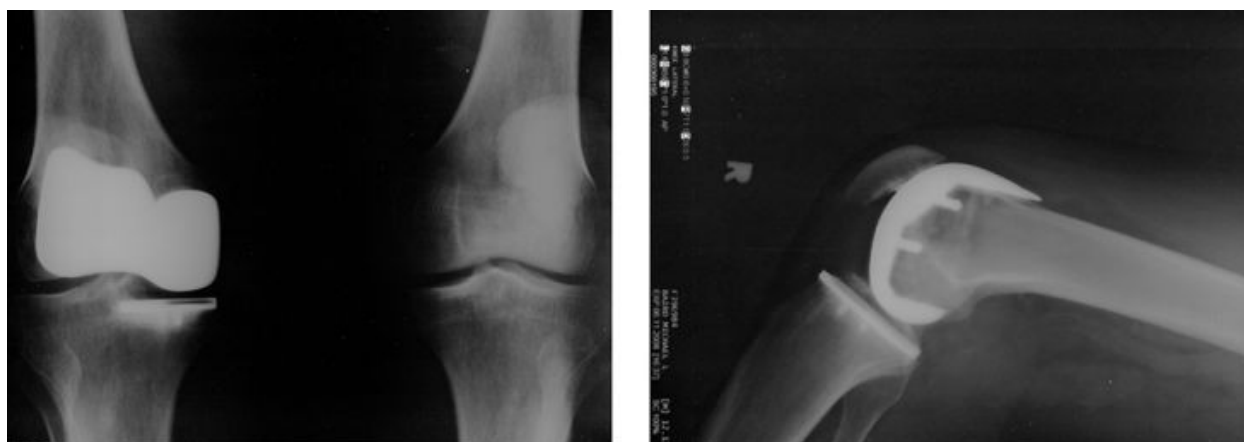


FIGURE 6.12 Artificial knee replacement is a procedure that has been performed for more than 20 years. These post-operative X-rays show a right knee joint replacement. (credit: modification of work by Mike Baird)

Natural lubricants include saliva produced in our mouths to aid in the swallowing process, and the slippery mucus found between organs in the body, allowing them to move freely past each other during heartbeats, during breathing, and when a person moves. Hospitals and doctor's clinics commonly use artificial lubricants, such as gels, to reduce friction.

The equations given for static and kinetic friction are empirical laws that describe the behavior of the forces of friction. While these formulas are very useful for practical purposes, they do not have the status of mathematical statements that represent general principles (e.g., Newton's second law). In fact, there are cases for which these equations are not even good approximations. For instance, neither formula is accurate for lubricated surfaces or for two surfaces sliding across each other at high speeds. Unless specified, we will not be concerned with these exceptions.

EXAMPLE 6.10

Static and Kinetic Friction

A 20.0-kg crate is at rest on a floor as shown in Figure 6.13. The coefficient of static friction between the crate and floor is 0.700 and the coefficient of kinetic friction is 0.600. A horizontal force $\vec{P}$ is applied to the crate. Find the force of friction if (a) $\vec{P} = 20.0 \text{ N}\hat{i}$, (b) $\vec{P} = 30.0 \text{ N}\hat{i}$, (c) $\vec{P} = 120.0 \text{ N}\hat{i}$, and (d) $\vec{P} = 180.0 \text{ N}\hat{i}$.

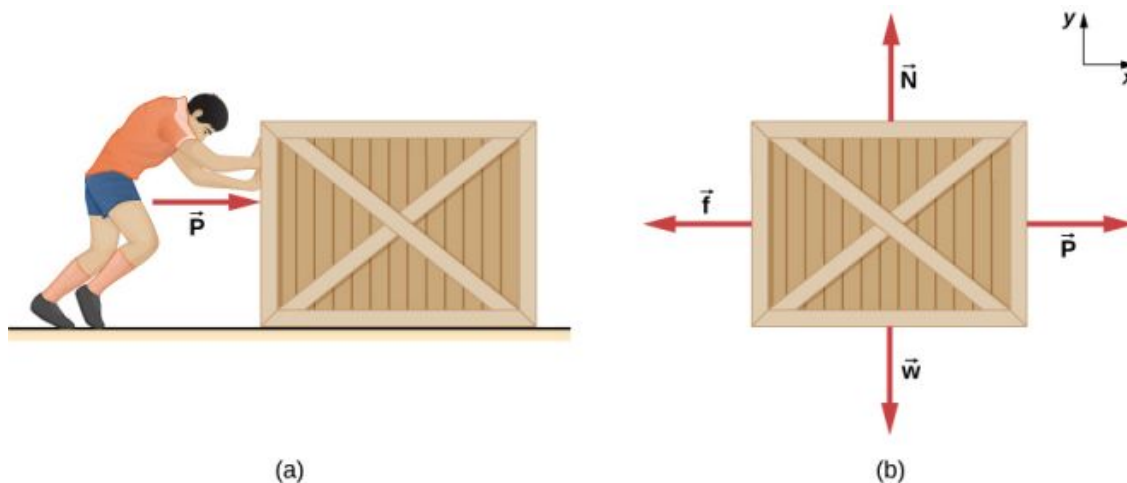


FIGURE 6.13 (a) A crate on a horizontal surface is pushed with a force $\vec{P}$. (b) The forces on the crate. Here, $\vec{f}$ may represent either the static or the kinetic frictional force.

Strategy

The free-body diagram of the crate is shown in Figure 6.13(b). We apply Newton's second law in the horizontal and vertical directions, including the friction force in opposition to the direction of motion of the box.

Solution

Newton's second law GIVES

$$\begin{aligned} \sum F_x &= ma_x & \sum F_y &= ma_y \\ P - f &= ma_x & N - w &= 0. \end{aligned}$$

Here we are using the symbol f to represent the frictional force since we have not yet determined whether the crate is subject to static friction or kinetic friction. We do this whenever we are unsure what type of friction is acting. Now the weight of the crate is

$$w = (20.0 \text{ kg})(9.80 \text{ m/s}^2) = 196 \text{ N},$$

which is also equal to N . The maximum force of static friction is therefore $(0.700)(196 \text{ N}) = 137 \text{ N}$. As long as $\vec{P}$ is less than 137 N, the force of static friction keeps the crate stationary and $f_s = \vec{P}$. Thus, (a) $f_s = 20.0 \text{ N}$, (b) $f_s = 30.0 \text{ N}$, and (c) $f_s = 120.0 \text{ N}$.

(d) If $\vec{P} = 180.0 \text{ N}$, the applied force is greater than the maximum force of static friction (137 N), so the crate can no longer remain at rest. Once the crate is in motion, kinetic friction acts. Then

$$f_k = \mu_k N = (0.600)(196 \text{ N}) = 118 \text{ N},$$

and the acceleration is

$$a_x = \frac{P - f_k}{m} = \frac{180.0 \text{ N} - 118 \text{ N}}{20.0 \text{ kg}} = 3.10 \text{ m/s}^2.$$

Significance

This example illustrates how we consider friction in a dynamics problem. Notice that static friction has a value that

matches the applied force, until we reach the maximum value of static friction. Also, no motion can occur until the applied force equals the force of static friction, but the force of kinetic friction will then become smaller.

✓ CHECK YOUR UNDERSTANDING 6.7

A block of mass 1.0 kg rests on a horizontal surface. The frictional coefficients for the block and surface are $\mu_s = 0.50$ and $\mu_k = 0.40$. (a) What is the minimum horizontal force required to move the block? (b) What is the block's acceleration when this force is applied?

Friction and the Inclined Plane

One situation where friction plays an obvious role is that of an object on a slope. It might be a crate being pushed up a ramp to a loading dock or a skateboarder coasting down a mountain, but the basic physics is the same. We usually generalize the sloping surface and call it an inclined plane but then pretend that the surface is flat. Let's look at an example of analyzing motion on an inclined plane with friction.

✿ EXAMPLE 6.11

Downhill Skier

A skier with a mass of 62 kg is sliding down a snowy slope at a constant acceleration. Find the coefficient of kinetic friction for the skier if friction is known to be 45.0 N.

Strategy

The magnitude of kinetic friction is given as 45.0 N. Kinetic friction is related to the normal force N by $f_k = \mu_k N$; thus, we can find the coefficient of kinetic friction if we can find the normal force on the skier. The normal force is always perpendicular to the surface, and since there is no motion perpendicular to the surface, the normal force should equal the component of the skier's weight perpendicular to the slope. (See [Figure 6.14](#), which repeats a figure from the chapter on Newton's laws of motion.)

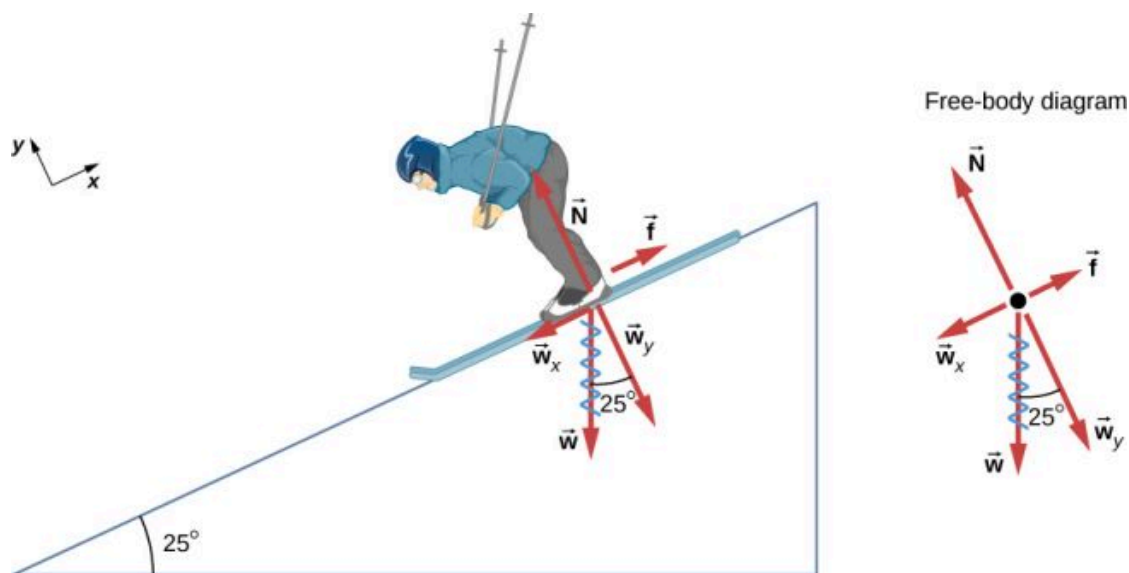


FIGURE 6.14 The motion of the skier and friction are parallel to the slope, so it is most convenient to project all forces onto a coordinate system where one axis is parallel to the slope and the other is perpendicular (axes shown to left of skier). The normal force $\vec{N}$ is perpendicular to the slope, and friction $\vec{f}$ is parallel to the slope, but the skier's weight $\vec{W}$ has components along both axes, namely $\vec{W}_y$ and $\vec{W}_x$. The normal force $\vec{N}$ is equal in magnitude to $\vec{W}_y$, so there is no motion perpendicular to the slope.

We have

$$N = w_y = w \cos 25^\circ = mg \cos 25^\circ.$$

Substituting this into our expression for kinetic friction, we obtain

$$f_k = \mu_k mg \cos 25^\circ,$$

which can now be solved for the coefficient of kinetic friction μ_k .

Solution

Solving for μ_k gives

$$\mu_k = \frac{f_k}{N} = \frac{f_k}{w \cos 25^\circ} = \frac{f_k}{mg \cos 25^\circ}.$$

Substituting known values on the right-hand side of the equation,

$$\mu_k = \frac{45.0 \text{ N}}{(62 \text{ kg})(9.80 \text{ m/s}^2)(0.906)} = 0.082.$$

Significance

This result is a little smaller than the coefficient listed in [Table 6.1](#) for waxed wood on snow, but it is still reasonable since values of the coefficients of friction can vary greatly. In situations like this, where an object of mass m slides down a slope that makes an angle θ with the horizontal, friction is given by $f_k = \mu_k mg \cos \theta$. All objects slide down a slope with constant acceleration under these circumstances.

We have discussed that when an object rests on a horizontal surface, the normal force supporting it is equal in magnitude to its weight. Furthermore, simple friction is always proportional to the normal force. When an object is not on a horizontal surface, as with the inclined plane, we must find the force acting on the object that is directed perpendicular to the surface; it is a component of the weight.

We now derive a useful relationship for calculating coefficient of friction on an inclined plane. Notice that the result applies only for situations in which the object slides at constant speed down the ramp.

An object slides down an inclined plane at a constant velocity if the net force on the object is zero. We can use this fact to measure the coefficient of kinetic friction between two objects. As shown in [Example 6.11](#), the kinetic friction on a slope is $f_k = \mu_k mg \cos \theta$. The component of the weight down the slope is equal to $mg \sin \theta$ (see the free-body diagram in [Figure 6.14](#)). These forces act in opposite directions, so when they have equal magnitude, the acceleration is zero. Writing these out,

$$\mu_k mg \cos \theta = mg \sin \theta.$$

Solving for μ_k , we find that

$$\mu_k = \frac{mg \sin \theta}{mg \cos \theta} = \tan \theta.$$

Put a coin on a book and tilt it until the coin slides at a constant velocity down the book. You might need to tap the book lightly to get the coin to move. Measure the angle of tilt relative to the horizontal and find μ_k . Note that the coin does not start to slide at all until an angle greater than θ is attained, since the coefficient of static friction is larger than the coefficient of kinetic friction. Think about how this may affect the value for μ_k and its uncertainty.

Atomic-Scale Explanations of Friction

The simpler aspects of friction dealt with so far are its macroscopic (large-scale) characteristics. Great strides have been made in the atomic-scale explanation of friction during the past several decades. Researchers are finding that the atomic nature of friction seems to have several fundamental characteristics. These characteristics not only explain some of the simpler aspects of friction—they also hold the potential for the development of nearly friction-free environments that could save hundreds of billions of dollars in energy which is currently being converted (unnecessarily) into heat.

[Figure 6.15](#) illustrates one macroscopic characteristic of friction that is explained by microscopic (small-scale)

research. We have noted that friction is proportional to the normal force, but not to the amount of area in contact, a somewhat counterintuitive notion. When two rough surfaces are in contact, the actual contact area is a tiny fraction of the total area because only high spots touch. When a greater normal force is exerted, the actual contact area increases, and we find that the friction is proportional to this area.

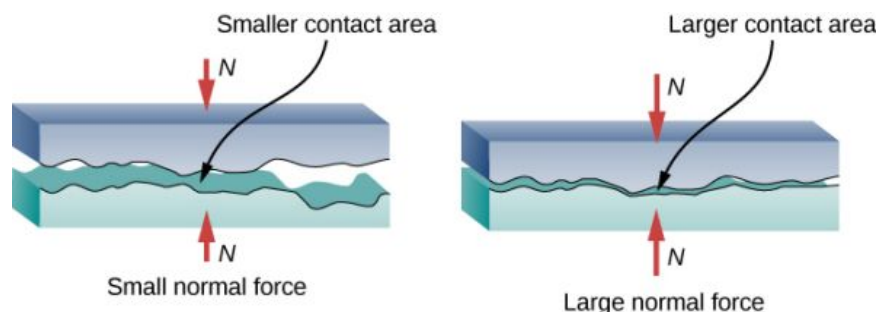


FIGURE 6.15 Two rough surfaces in contact have a much smaller area of actual contact than their total area. When the normal force is larger as a result of a larger applied force, the area of actual contact increases, as does friction.

However, the atomic-scale view promises to explain far more than the simpler features of friction. The mechanism for how heat is generated is now being determined. In other words, why do surfaces get warmer when rubbed? Essentially, atoms are linked with one another to form lattices. When surfaces rub, the surface atoms adhere and cause atomic lattices to vibrate—essentially creating sound waves that penetrate the material. The sound waves diminish with distance, and their energy is converted into heat. Chemical reactions that are related to frictional wear can also occur between atoms and molecules on the surfaces. Figure 6.16 shows how the tip of a probe drawn across another material is deformed by atomic-scale friction. The force needed to drag the tip can be measured and is found to be related to shear stress, which is discussed in [Static Equilibrium and Elasticity](#). The variation in shear stress is remarkable (more than a factor of 10^{12}) and difficult to predict theoretically, but shear stress is yielding a fundamental understanding of a large-scale phenomenon known since ancient times—friction.

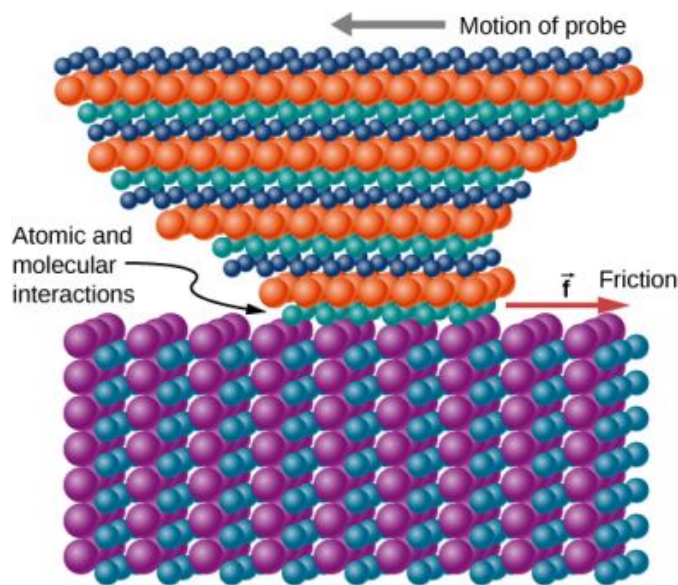


FIGURE 6.16 The tip of a probe is deformed sideways by frictional force as the probe is dragged across a surface. Measurements of how the force varies for different materials are yielding fundamental insights into the atomic nature of friction.

INTERACTIVE

Engaging the simulation below, describe a model for friction on a molecular level. Describe matter in terms of molecular motion. The description should include diagrams to support the description; how the temperature affects the image; what are the differences and similarities between solid, liquid, and gas particle motion; and how the size and speed of gas molecules relate to everyday objects.

[Access multimedia content \(https://openstax.org/books/university-physics-volume-1/pages/6-2-friction\)](https://openstax.org/books/university-physics-volume-1/pages/6-2-friction)

EXAMPLE 6.12

Sliding Blocks

The two blocks of [Figure 6.17](#) are attached to each other by a massless string that is wrapped around a frictionless pulley. When the bottom 4.00-kg block is pulled to the left by the constant force $\vec{P}$, the top 2.00-kg block slides across it to the right. Find the magnitude of the force necessary to move the blocks at constant speed. Assume that the coefficient of kinetic friction between all surfaces is 0.400.

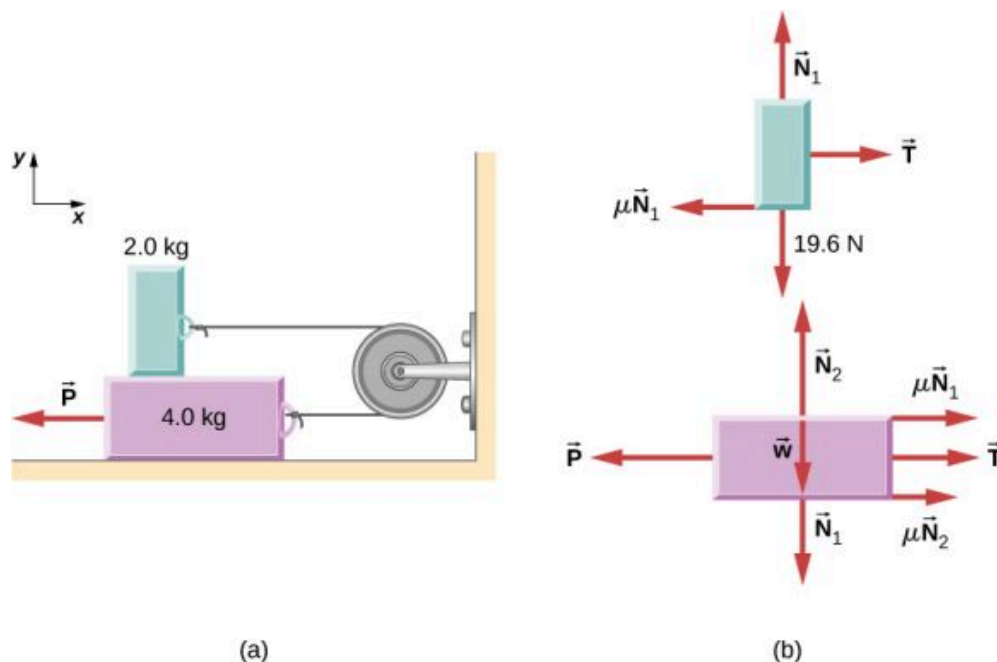


FIGURE 6.17 (a) Each block moves at constant velocity. (b) Free-body diagrams for the blocks.

Strategy

We analyze the motions of the two blocks separately. The top block is subjected to a contact force exerted by the bottom block. The components of this force are the normal force N_1 and the frictional force $-0.400N_1$. Other forces on the top block are the tension T in the string and the weight of the top block itself, 19.6 N. The bottom block is subjected to contact forces due to the top block and due to the floor. The first contact force has components $-N_1$ and $0.400N_1$, which are simply reaction forces to the contact forces that the bottom block exerts on the top block. The components of the contact force of the floor are N_2 and $0.400N_2$. Other forces on this block are $-P$, the tension T , and the weight -39.2 N.

Solution

Since the top block is moving horizontally to the right at constant velocity, its acceleration is zero in both the horizontal and the vertical directions. From Newton's second law,

$$\begin{aligned} \sum F_x &= m_1 a_x & \sum F_y &= m_1 a_y \\ T - 0.400N_1 &= 0 & N_1 - 19.6 \text{ N} &= 0. \end{aligned}$$

Solving for the two unknowns, we obtain $N_1 = 19.6$ N and $T = 0.40N_1 = 7.84$ N. The bottom block is also not accelerating, so the application of Newton's second law to this block gives

$$\begin{aligned} \sum F_x &= m_2 a_x & \sum F_y &= m_2 a_y \\ T - P + 0.400 N_1 + 0.400 N_2 &= 0 & N_2 - 39.2 \text{ N} - N_1 &= 0. \end{aligned}$$

The values of N_1 and T were found with the first set of equations. When these values are substituted into the second set of equations, we can determine N_2 and P . They are

$$N_2 = 58.8 \text{ N and } P = 39.2 \text{ N.}$$

Significance

Understanding what direction in which to draw the friction force is often troublesome. Notice that each friction force labeled in [Figure 6.17](#) acts in the direction opposite the motion of its corresponding block.



EXAMPLE 6.13

A Crate on an Accelerating Truck

A 50.0-kg crate rests on the bed of a truck as shown in [Figure 6.18](#). The coefficients of friction between the surfaces are $\mu_k = 0.300$ and $\mu_s = 0.400$. Find the frictional force on the crate when the truck is accelerating forward relative to the ground at (a) 2.00 m/s^2 , and (b) 5.00 m/s^2 .

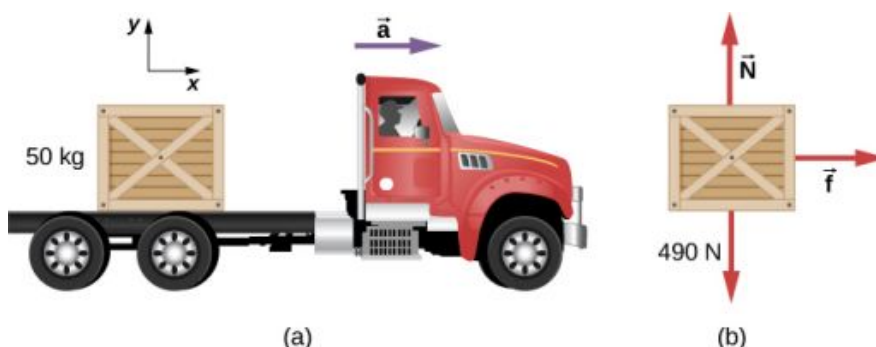


FIGURE 6.18 (a) A crate rests on the bed of the truck that is accelerating forward. (b) The free-body diagram of the crate.

Strategy

The forces on the crate are its weight and the normal and frictional forces due to contact with the truck bed. We start by *assuming* that the crate is not slipping. In this case, the static frictional force f_s acts on the crate. Furthermore, the accelerations of the crate and the truck are equal.

Solution

- a. Application of Newton's second law to the crate, using the reference frame attached to the ground, yields

$$\begin{aligned} \sum F_x &= ma_x & \sum F_y &= ma_y \\ f_s &= (50.0 \text{ kg})(2.00 \text{ m/s}^2) & N - 4.90 \times 10^2 \text{ N} &= (50.0 \text{ kg})(0) \\ &= 1.00 \times 10^2 \text{ N} & N &= 4.90 \times 10^2 \text{ N.} \end{aligned}$$

We can now check the validity of our no-slip assumption. The maximum value of the force of static friction is

$$\mu_s N = (0.400)(4.90 \times 10^2 \text{ N}) = 196 \text{ N,}$$

whereas the *actual* force of static friction that acts when the truck accelerates forward at 2.00 m/s^2 is only $1.00 \times 10^2 \text{ N}$. Thus, the assumption of no slipping is valid.

- b. If the crate is to move with the truck when it accelerates at 5.0 m/s^2 , the force of static friction must be

$$f_s = ma_x = (50.0 \text{ kg})(5.00 \text{ m/s}^2) = 250 \text{ N.}$$

Since this exceeds the maximum of 196 N, the crate must slip. The frictional force is therefore kinetic and is

$$f_k = \mu_k N = (0.300)(4.90 \times 10^2 \text{ N}) = 147 \text{ N.}$$

The horizontal acceleration of the crate relative to the ground is now found from

$$\begin{aligned} \sum F_x &= ma_x \\ 147 \text{ N} &= (50.0 \text{ kg})a_x, \\ \text{so } a_x &= 2.94 \text{ m/s}^2. \end{aligned}$$

Significance

Relative to the ground, the truck is accelerating forward at 5.0 m/s^2 and the crate is accelerating forward at 2.94 m/s^2 . Hence the crate is sliding backward relative to the bed of the truck with an acceleration $2.94 \text{ m/s}^2 - 5.00 \text{ m/s}^2 = -2.06 \text{ m/s}^2$.

EXAMPLE 6.14

Snowboarding

Earlier, we analyzed the situation of a downhill skier moving at constant velocity to determine the coefficient of kinetic friction. Now let's do a similar analysis to determine acceleration. The snowboarder of [Figure 6.19](#) glides down a slope that is inclined at 13° to the horizontal. The coefficient of kinetic friction between the board and the snow is $\mu_k = 0.20$. What is the acceleration of the snowboarder?

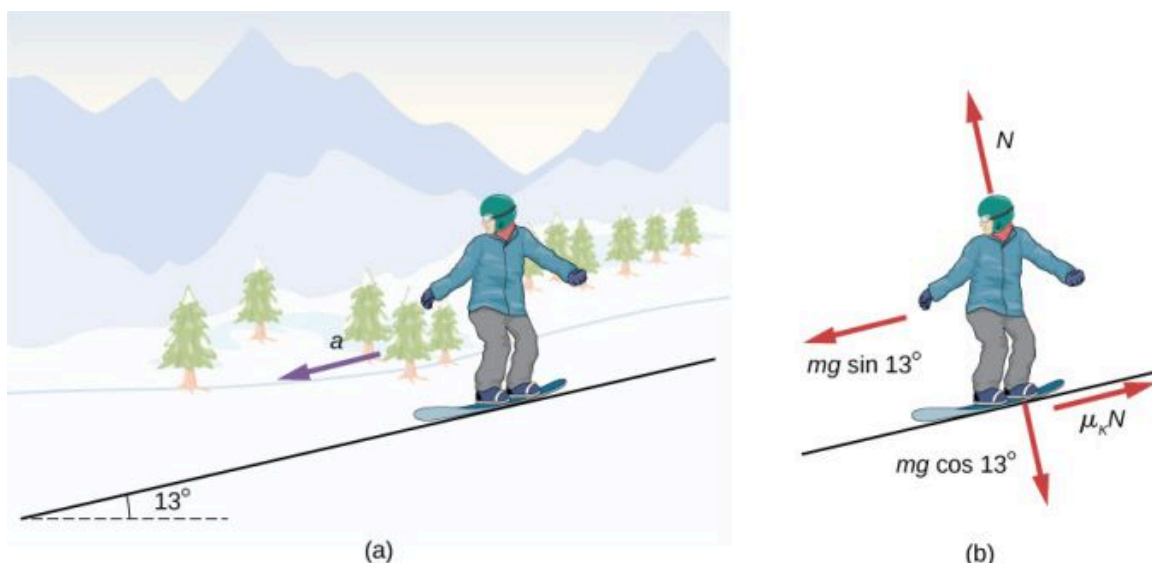


FIGURE 6.19 (a) A snowboarder glides down a slope inclined at 13° to the horizontal. (b) The free-body diagram of the snowboarder.

Strategy

The forces acting on the snowboarder are her weight and the contact force of the slope, which has a component normal to the incline and a component along the incline (force of kinetic friction). Because she moves along the slope, the most convenient reference frame for analyzing her motion is one with the x -axis along and the y -axis perpendicular to the incline. In this frame, both the normal and the frictional forces lie along coordinate axes, the components of the weight are $mg \sin \theta$ along the slope and $mg \cos \theta$ at right angles into the slope, and the only acceleration is along the x -axis ($a_y = 0$).

Solution

We can now apply Newton's second law to the snowboarder:

$$\begin{aligned} \sum F_x &= ma_x & \sum F_y &= ma_y \\ mg \sin \theta - \mu_k N &= ma_x & N - mg \cos \theta &= m(0). \end{aligned}$$

From the second equation, $N = mg \cos \theta$. Upon substituting this into the first equation, we find

$$\begin{aligned} a_x &= g(\sin \theta - \mu_k \cos \theta) \\ &= g(\sin 13^\circ - 0.20 \cos 13^\circ) = 0.29 \text{ m/s}^2. \end{aligned}$$

Significance

Notice from this equation that if θ is small enough or μ_k is large enough, a_x is negative, that is, the snowboarder

slows down.

✓ CHECK YOUR UNDERSTANDING 6.8

The snowboarder is now moving down a hill with incline 10.0° . What is the snowboarder's acceleration?

6.3 Centripetal Force

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Explain the equation for centripetal acceleration
- Apply Newton's second law to develop the equation for centripetal force
- Use circular motion concepts in solving problems involving Newton's laws of motion

In [Motion in Two and Three Dimensions](#), we examined the basic concepts of circular motion. An object undergoing circular motion, like one of the race cars shown at the beginning of this chapter, must be accelerating because it is changing the direction of its velocity. We proved that this centrally directed acceleration, called centripetal acceleration, is given by the formula

$$a_c = \frac{v^2}{r}$$

where v is the velocity of the object, directed along a tangent line to the curve at any instant. If we know the angular velocity ω , then we can use

$$a_c = r\omega^2.$$

Angular velocity gives the rate at which the object is turning through the curve, in units of rad/s. This acceleration acts along the radius of the curved path and is thus also referred to as a radial acceleration.

An acceleration must be produced by a force. Any force or combination of forces can cause a centripetal or radial acceleration. Just a few examples are the tension in the rope on a tether ball, the force of Earth's gravity on the Moon, friction between roller skates and a rink floor, a banked roadway's force on a car, and forces on the tube of a spinning centrifuge. Any net force causing uniform circular motion is called a **centripetal force**. The direction of a centripetal force is toward the center of curvature, the same as the direction of centripetal acceleration. According to Newton's second law of motion, net force is mass times acceleration: $F_{\text{net}} = ma$. For uniform circular motion, the acceleration is the centripetal acceleration: $a = a_c$. Thus, the magnitude of centripetal force F_c is

$$F_c = ma_c.$$

By substituting the expressions for centripetal acceleration a_c ($a_c = \frac{v^2}{r}$; $a_c = r\omega^2$), we get two expressions for the centripetal force F_c in terms of mass, velocity, angular velocity, and radius of curvature:

$$F_c = m\frac{v^2}{r}; \quad F_c = mr\omega^2. \quad 6.3$$

You may use whichever expression for centripetal force is more convenient. Centripetal force $\vec{F}_c$ is always perpendicular to the path and points to the center of curvature, because $\vec{a}_c$ is perpendicular to the velocity and points to the center of curvature. Note that if you solve the first expression for r , you get

$$r = \frac{mv^2}{F_c}.$$

This implies that for a given mass and velocity, a large centripetal force causes a small radius of curvature—that is, a tight curve, as in [Figure 6.20](#).

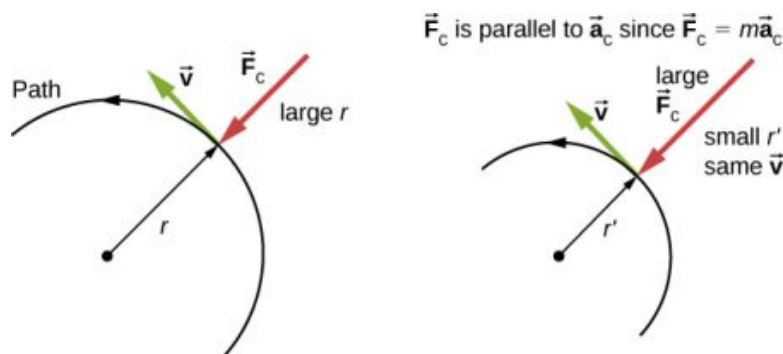


FIGURE 6.20 The frictional force supplies the centripetal force and is numerically equal to it. Centripetal force is perpendicular to velocity and causes uniform circular motion. The larger the F_c , the smaller the radius of curvature r and the sharper the curve. The second curve has the same v , but a larger F_c produces a smaller r .

EXAMPLE 6.15

What Coefficient of Friction Do Cars Need on a Flat Curve?

- (a) Calculate the centripetal force exerted on a 900.0-kg car that negotiates a 500.0-m radius curve at 25.00 m/s.
 (b) Assuming an unbanked curve, find the minimum static coefficient of friction between the tires and the road, static friction being the reason that keeps the car from slipping ([Figure 6.21](#)).

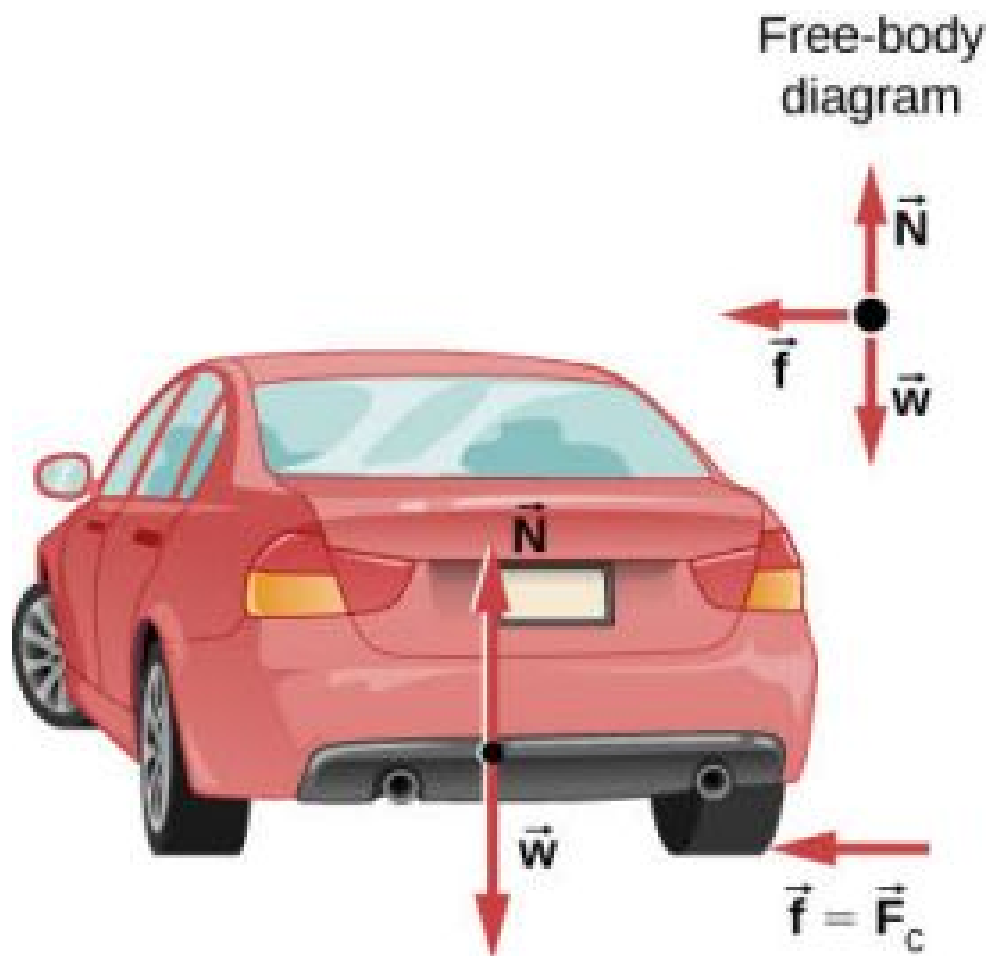


FIGURE 6.21 This car on level ground is moving away and turning to the left. The centripetal force causing the car to turn in a circular path is due to friction between the tires and the road. A minimum coefficient of friction is needed, or the car will move in a larger-radius curve and leave the roadway.

Strategy

- a. We know that $F_c = \frac{mv^2}{r}$. Thus,

$$F_c = \frac{mv^2}{r} = \frac{(900.0 \text{ kg})(25.00 \text{ m/s})^2}{(500.0 \text{ m})} = 1125 \text{ N}.$$

- b. [Figure 6.21](#) shows the forces acting on the car on an unbanked (level ground) curve. Friction is to the left, keeping the car from slipping, and because it is the only horizontal force acting on the car, the friction is the centripetal force in this case. We know that the maximum static friction (at which the tires roll but do not slip) is $\mu_s N$, where μ_s is the static coefficient of friction and N is the normal force. The normal force equals the car's weight on level ground, so $N = mg$. Thus the centripetal force in this situation is
- $$F_c \equiv f = \mu_s N = \mu_s mg.$$

Now we have a relationship between centripetal force and the coefficient of friction. Using the equation

$$F_c = m \frac{v^2}{r},$$

we obtain

$$m \frac{v^2}{r} = \mu_s mg.$$

We solve this for μ_s , noting that mass cancels, and obtain

$$\mu_s = \frac{v^2}{rg}.$$

Substituting the knowns,

$$\mu_s = \frac{(25.00 \text{ m/s})^2}{(500.0 \text{ m})(9.80 \text{ m/s}^2)} = 0.13.$$

(Because coefficients of friction are approximate, the answer is given to only two digits.)

Significance

The coefficient of friction found in [Figure 6.21](#)(b) is much smaller than is typically found between tires and roads. The car still negotiates the curve if the coefficient is greater than 0.13, because static friction is a responsive force, able to assume a value less than but no more than $\mu_s N$. A higher coefficient would also allow the car to negotiate the curve at a higher speed, but if the coefficient of friction is less, the safe speed would be less than 25 m/s. Note that mass cancels, implying that, in this example, it does not matter how heavily loaded the car is to negotiate the turn. Mass cancels because friction is assumed proportional to the normal force, which in turn is proportional to mass. If the surface of the road were banked, the normal force would be greater, as discussed next.

** CHECK YOUR UNDERSTANDING 6.9**

A car moving at 96.8 km/h travels around a circular curve of radius 182.9 m on a flat country road. What must be the minimum coefficient of static friction to keep the car from slipping?

Banked Curves

Let us now consider **banked curves**, where the slope of the road helps you negotiate the curve ([Figure 6.22](#)). The greater the angle θ , the faster you can take the curve. Race tracks for bikes as well as cars, for example, often have steeply banked curves. In an “ideally banked curve,” the angle θ is such that you can negotiate the curve at a certain speed without the aid of friction between the tires and the road. We will derive an expression for θ for an ideally banked curve and consider an example related to it.

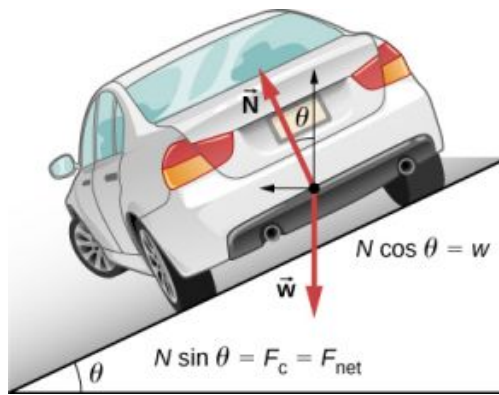


FIGURE 6.22 The car on this banked curve is moving away and turning to the left.

For **ideal banking**, the net external force equals the horizontal centripetal force in the absence of friction. The components of the normal force N in the horizontal and vertical directions must equal the centripetal force and the weight of the car, respectively. In cases in which forces are not parallel, it is most convenient to consider components along perpendicular axes—in this case, the vertical and horizontal directions.

[Figure 6.22](#) shows a free-body diagram for a car on a frictionless banked curve. If the angle θ is ideal for the speed and radius, then the net external force equals the necessary centripetal force. The only two external forces acting on the car are its weight $\vec{w}$ and the normal force of the road $\vec{N}$. (A frictionless surface can only exert a force perpendicular to the surface—that is, a normal force.) These two forces must add to give a net external force that is horizontal toward the center of curvature and has magnitude mv^2/r . Because this is the crucial force and it is horizontal, we use a coordinate system with vertical and horizontal axes. Only the normal force has a horizontal component, so this must equal the centripetal force, that is,

$$N \sin \theta = \frac{mv^2}{r}.$$

Because the car does not leave the surface of the road, the net vertical force must be zero, meaning that the vertical components of the two external forces must be equal in magnitude and opposite in direction. From [Figure 6.22](#), we see that the vertical component of the normal force is $N \cos \theta$, and the only other vertical force is the car's weight. These must be equal in magnitude; thus,

$$N \cos \theta = mg.$$

Now we can combine these two equations to eliminate N and get an expression for θ , as desired. Solving the second equation for $N = mg/(\cos \theta)$ and substituting this into the first yields

$$\begin{aligned} mg \frac{\sin \theta}{\cos \theta} &= \frac{mv^2}{r} \\ mg \tan \theta &= \frac{mv^2}{r} \\ \tan \theta &= \frac{v^2}{rg}. \end{aligned}$$

Taking the inverse tangent gives

$$\theta = \tan^{-1} \left(\frac{v^2}{rg} \right). \quad 6.4$$

This expression can be understood by considering how θ depends on v and r . A large θ is obtained for a large v and a small r . That is, roads must be steeply banked for high speeds and sharp curves. Friction helps, because it allows you to take the curve at greater or lower speed than if the curve were frictionless. Note that θ does not depend on the mass of the vehicle.

EXAMPLE 6.16

What Is the Ideal Speed to Take a Steeply Banked Tight Curve?

Curves on some test tracks and race courses, such as Daytona International Speedway in Florida, are very steeply banked. This banking, with the aid of tire friction and very stable car configurations, allows the curves to be taken at very high speed. To illustrate, calculate the speed at which a 100.0-m radius curve banked at 31.0° should be driven if the road were frictionless.

Strategy

We first note that all terms in the expression for the ideal angle of a banked curve except for speed are known; thus, we need only rearrange it so that speed appears on the left-hand side and then substitute known quantities.

Solution

Starting with

$$\tan \theta = \frac{v^2}{rg},$$

we get

$$v = \sqrt{rg \tan \theta}.$$

Noting that $\tan 31.0^\circ = 0.6009$, we obtain

$$v = \sqrt{(100.0 \text{ m})(9.80 \text{ m/s}^2)(0.609)} = 24.3 \text{ m/s}.$$

Significance

This is just about 87.5 km/h, consistent with a very steeply banked and rather sharp curve. Tire friction enables a vehicle to take the curve at significantly higher speeds.

Airplanes also make turns by banking. The lift force, due to the force of the air on the wing, acts at right angles to the wing. When the airplane banks, the pilot is obtaining greater lift than necessary for level flight. The vertical component of lift balances the airplane's weight, and the horizontal component accelerates the plane. The banking angle shown in [Figure 6.23](#) is given by θ . We analyze the forces in the same way we treat the case of the car rounding a banked curve.

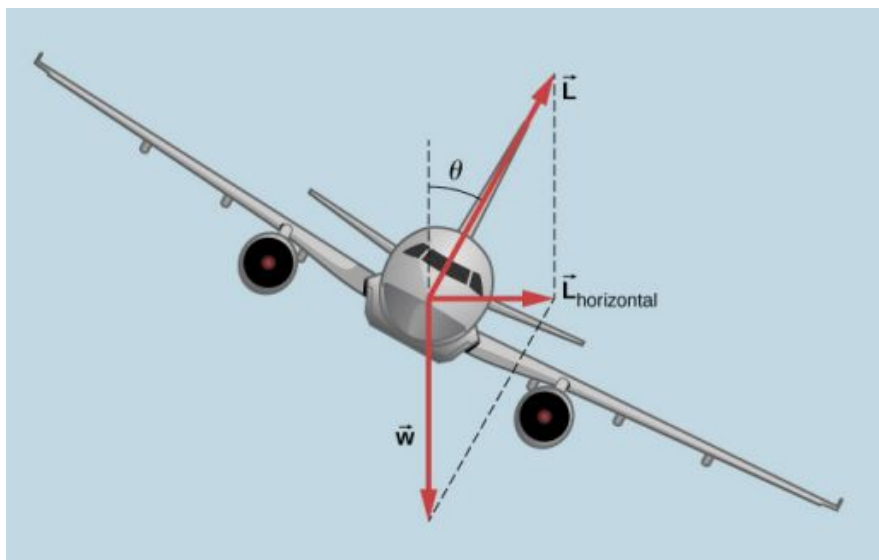


FIGURE 6.23 In a banked turn, the horizontal component of lift is unbalanced and accelerates the plane. The normal component of lift balances the plane's weight. The banking angle is given by θ . Compare the vector diagram with that shown in [Figure 6.22](#).

INTERACTIVE

Join the [ladybug](https://openstax.org/l/21ladybug_es) (https://openstax.org/l/21ladybug_es) in an exploration of rotational motion. Rotate the merry-go-round to change its angle or choose a constant angular velocity or angular acceleration. Explore how circular motion relates to the bug's xy -position, velocity, and acceleration using vectors or graphs.

INTERACTIVE

A circular motion requires a force, the so-called centripetal force, which is directed to the axis of rotation. This simplified [model of a carousel](https://openstax.org/l/21carousel) (<https://openstax.org/l/21carousel>) demonstrates this force.

Inertial Forces and Noninertial (Accelerated) Frames: The Coriolis Force

What do taking off in a jet airplane, turning a corner in a car, riding a merry-go-round, and the circular motion of a tropical cyclone have in common? Each exhibits inertial forces—fictitious forces that merely seem to arise from motion, because the observer's frame of reference is accelerating or rotating. When taking off in a jet, most people would agree it feels as if you are being pushed back into the seat as the airplane accelerates down the runway. Yet a physicist would say that *you* tend to remain stationary while the *seat* pushes forward on you. An even more common experience occurs when you make a tight curve in your car—say, to the right ([Figure 6.24](#)). You feel as if you are thrown (that is, *forced*) toward the left relative to the car. Again, a physicist would say that *you* are going in a straight line (recall Newton's first law) but the *car* moves to the right, not that you are experiencing a force from the left.

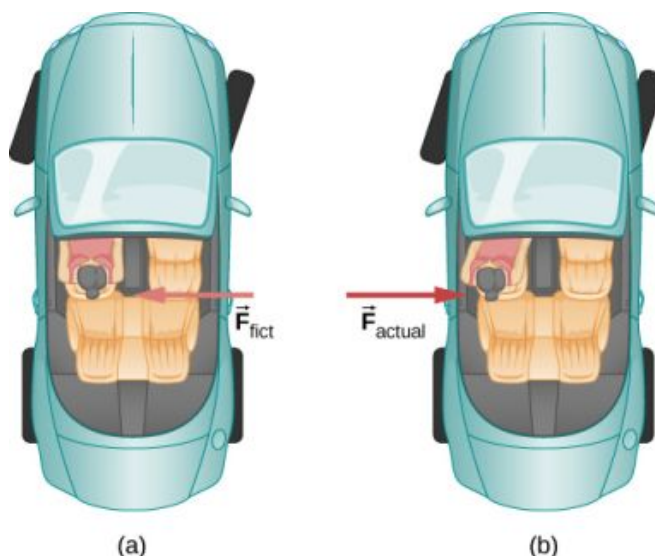


FIGURE 6.24 (a) The car driver feels herself forced to the left relative to the car when she makes a right turn. This is a fictitious inertial force arising from the use of the car as a frame of reference. (b) In Earth's frame of reference, the driver moves in a straight line, obeying Newton's first law, and the car moves to the right. There is no force to the left on the driver relative to Earth. Instead, there is a force to the right on the car to make it turn.

We can reconcile these points of view by examining the frames of reference used. Let us concentrate on people in a car. Passengers instinctively use the car as a frame of reference, whereas a physicist might use Earth. The physicist might make this choice because Earth is nearly an inertial frame of reference, in which all forces have an identifiable physical origin. In such a frame of reference, Newton's laws of motion take the form given in [Newton's Laws of Motion](#). The car is a **noninertial frame of reference** because it is accelerated to the side. The force to the left sensed by car passengers is an **inertial force** having no physical origin (it is due purely to the inertia of the passenger, not to some physical cause such as tension, friction, or gravitation). The car, as well as the driver, is actually accelerating to the right.

A physicist will choose whatever reference frame is most convenient for the situation being analyzed. There is no problem to a physicist in including inertial forces and Newton's second law, as usual, if that is more convenient, for

example, on a merry-go-round or on a rotating planet. Noninertial (accelerated) frames of reference are used when it is useful to do so. Different frames of reference must be considered in discussing the motion of an astronaut in a spacecraft traveling at speeds near the speed of light, as you will appreciate in the study of the special theory of relativity.

Let us now take a mental ride on a merry-go-round—specifically, a rapidly rotating playground merry-go-round (Figure 6.25). You take the merry-go-round to be your frame of reference because you rotate together. When rotating in that noninertial frame of reference, you feel an inertial force that tends to throw you off; this is often referred to as a *centrifugal force* (not to be confused with centripetal force). Centrifugal force is a commonly used term, but it does not actually exist. You must hang on tightly to counteract your inertia (which people often refer to as centrifugal force). In Earth's frame of reference, there is no force trying to throw you off; we emphasize that centrifugal force is a fiction. You must hang on to make yourself go in a circle because otherwise you would go in a straight line, right off the merry-go-round, in keeping with Newton's first law. But the force you exert acts toward the center of the circle.

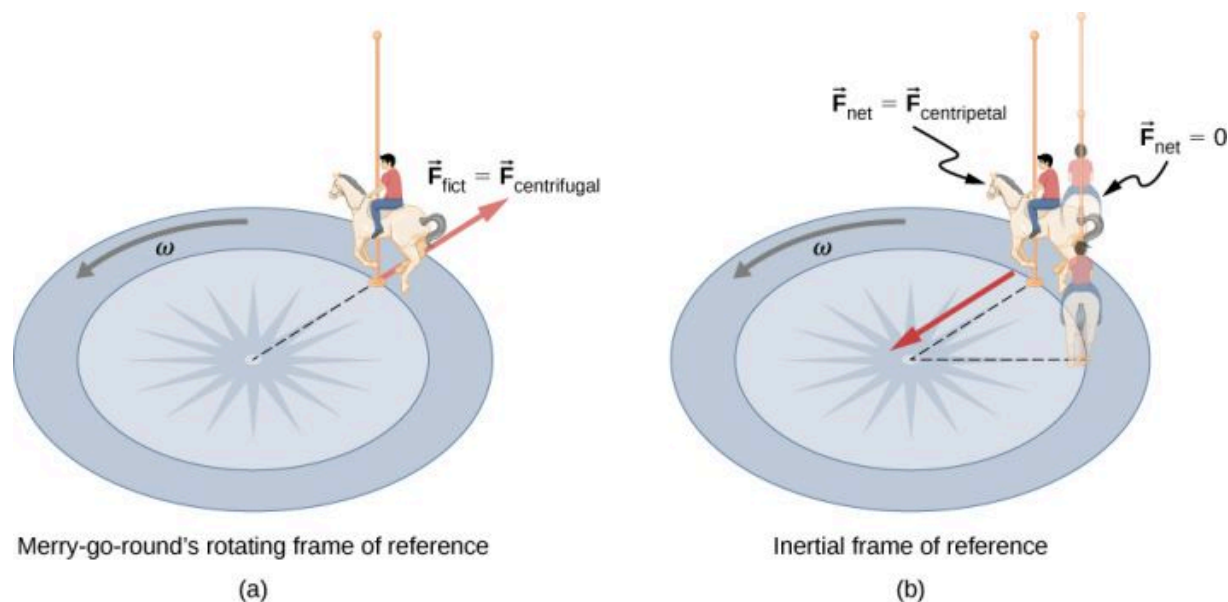


FIGURE 6.25 (a) A rider on a merry-go-round feels as if he is being thrown off. This inertial force is sometimes mistakenly called the centrifugal force in an effort to explain the rider's motion in the rotating frame of reference. (b) In an inertial frame of reference and according to Newton's laws, it is his inertia that carries him off (the unshaded rider has $F_{\text{net}} = 0$ and heads in a straight line). A force, $F_{\text{centripetal}}$, is needed to cause a circular path.

This inertial effect, carrying you away from the center of rotation if there is no centripetal force to cause circular motion, is put to good use in centrifuges (Figure 6.26). A centrifuge spins a sample very rapidly, as mentioned earlier in this chapter. Viewed from the rotating frame of reference, the inertial force throws particles outward, hastening their sedimentation. The greater the angular velocity, the greater the centrifugal force. But what really happens is that the inertia of the particles carries them along a line tangent to the circle while the test tube is forced in a circular path by a centripetal force.

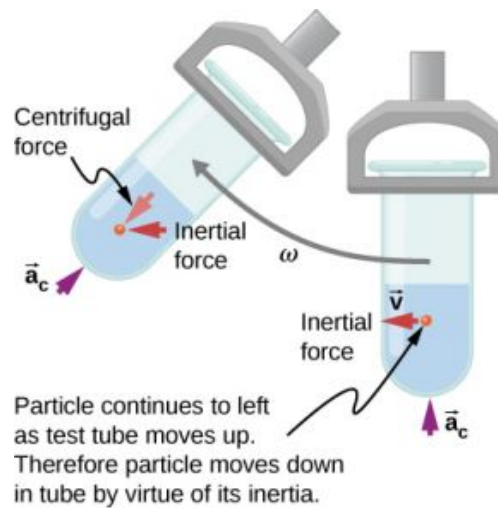


FIGURE 6.26 Centrifuges use inertia to perform their task. Particles in the fluid sediment settle out because their inertia carries them away from the center of rotation. The large angular velocity of the centrifuge quickens the sedimentation. Ultimately, the particles come into contact with the test tube walls, which then supply the centripetal force needed to make them move in a circle of constant radius.

Let us now consider what happens if something moves in a rotating frame of reference. For example, what if you slide a ball directly away from the center of the merry-go-round, as shown in [Figure 6.27](#)? The ball follows a straight path relative to Earth (assuming negligible friction) and a path curved to the right on the merry-go-round's surface. A person standing next to the merry-go-round sees the ball moving straight and the merry-go-round rotating underneath it. In the merry-go-round's frame of reference, we explain the apparent curve to the right by using an inertial force, called the **Coriolis force**, which causes the ball to curve to the right. The Coriolis force can be used by anyone in that frame of reference to explain why objects follow curved paths and allows us to apply Newton's laws in noninertial frames of reference.

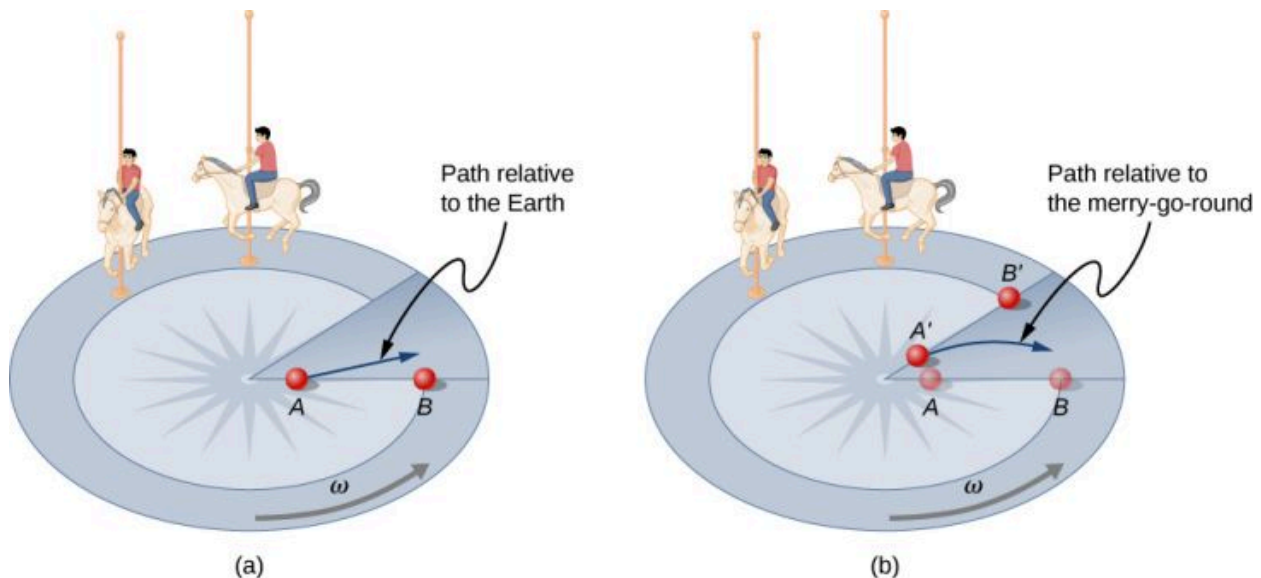


FIGURE 6.27 Looking down on the counterclockwise rotation of a merry-go-round, we see that a ball slid straight toward the edge follows a path curved to the right. The person slides the ball toward point B, starting at point A. Both points rotate to the shaded positions (A' and B') shown in the time that the ball follows the curved path in the rotating frame and a straight path in Earth's frame.

Up until now, we have considered Earth to be an inertial frame of reference with little or no worry about effects due to its rotation. Yet such effects *do* exist—in the rotation of weather systems, for example. Most consequences of Earth's rotation can be qualitatively understood by analogy with the merry-go-round. Viewed from above the North Pole, Earth rotates counterclockwise, as does the merry-go-round in [Figure 6.27](#). As on the merry-go-round, any motion in Earth's Northern Hemisphere experiences a Coriolis force to the right. Just the opposite occurs in the Southern Hemisphere; there, the force is to the left. Because Earth's angular velocity is small, the Coriolis force is usually negligible, but for large-scale motions, such as wind patterns, it has substantial effects.

The Coriolis force causes hurricanes in the Northern Hemisphere to rotate in the counterclockwise direction, whereas tropical cyclones in the Southern Hemisphere rotate in the clockwise direction. (The terms hurricane, typhoon, and tropical storm are regionally specific names for cyclones, which are storm systems characterized by low pressure centers, strong winds, and heavy rains.) [Figure 6.28](#) helps show how these rotations take place. Air flows toward any region of low pressure, and tropical cyclones contain particularly low pressures. Thus winds flow toward the center of a tropical cyclone or a low-pressure weather system at the surface. In the Northern Hemisphere, these inward winds are deflected to the right, as shown in the figure, producing a counterclockwise circulation at the surface for low-pressure zones of any type. Low pressure at the surface is associated with rising air, which also produces cooling and cloud formation, making low-pressure patterns quite visible from space. Conversely, wind circulation around high-pressure zones is clockwise in the Southern Hemisphere but is less visible because high pressure is associated with sinking air, producing clear skies.

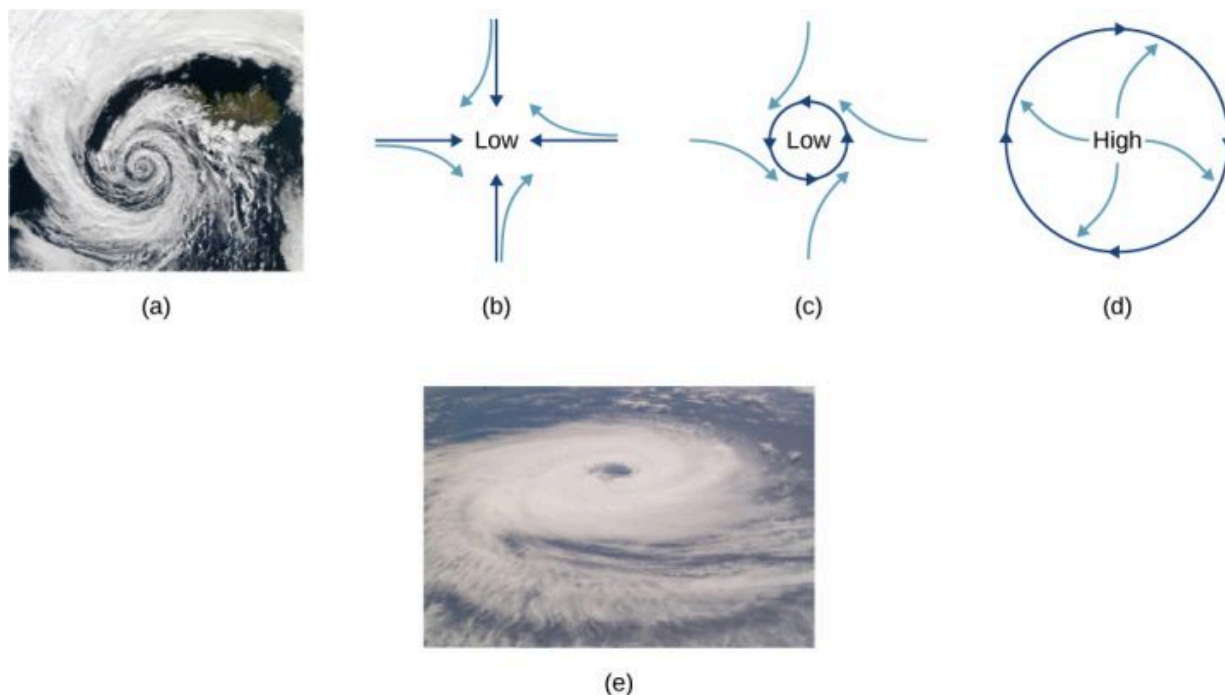


FIGURE 6.28 (a) The counterclockwise rotation of this Northern Hemisphere hurricane is a major consequence of the Coriolis force. (b) Without the Coriolis force, air would flow straight into a low-pressure zone, such as that found in tropical cyclones. (c) The Coriolis force deflects the winds to the right, producing a counterclockwise rotation. (d) Wind flowing away from a high-pressure zone is also deflected to the right, producing a clockwise rotation. (e) The opposite direction of rotation is produced by the Coriolis force in the Southern Hemisphere, leading to tropical cyclones. (credit a and credit e: modifications of work by NASA)

The rotation of tropical cyclones and the path of a ball on a merry-go-round can just as well be explained by inertia and the rotation of the system underneath. When noninertial frames are used, inertial forces, such as the Coriolis force, must be invented to explain the curved path. There is no identifiable physical source for these inertial forces. In an inertial frame, inertia explains the path, and no force is found to be without an identifiable source. Either view allows us to describe nature, but a view in an inertial frame is the simplest in the sense that all forces have origins and explanations.

6.4 Drag Force and Terminal Speed

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Express the drag force mathematically
- Describe applications of the drag force
- Define terminal velocity
- Determine an object's terminal velocity given its mass

Another interesting force in everyday life is the force of drag on an object when it is moving in a fluid (either a gas or a liquid). You feel the drag force when you move your hand through water. You might also feel it if you move your

hand during a strong wind. The faster you move your hand, the harder it is to move. You feel a smaller drag force when you tilt your hand so only the side goes through the air—you have decreased the area of your hand that faces the direction of motion.

Drag Forces

Like friction, the **drag force** always opposes the motion of an object. Unlike simple friction, the drag force is proportional to some function of the velocity of the object in that fluid. This functionality is complicated and depends upon the shape of the object, its size, its velocity, and the fluid it is in. For most large objects such as cyclists, cars, and baseballs not moving too slowly, the magnitude of the drag force F_D is proportional to the square of the speed of the object. We can write this relationship mathematically as $F_D \propto v^2$. When taking into account other factors, this relationship becomes

$$F_D = \frac{1}{2} C \rho A v^2, \quad 6.5$$

where C is the drag coefficient, A is the area of the object facing the fluid, and ρ is the density of the fluid. (Recall that density is mass per unit volume.) This equation can also be written in a more generalized fashion as $F_D = b v^n$, where b is a constant equivalent to $0.5 C \rho A$. We have set the exponent n for these equations as 2 because when an object is moving at high velocity through air, the magnitude of the drag force is proportional to the square of the speed. As we shall see in [Fluid Mechanics](#), for small particles moving at low speeds in a fluid, the exponent n is equal to 1.

DRAG FORCE

Drag force F_D is proportional to the square of the speed of the object. Mathematically,

$$F_D = \frac{1}{2} C \rho A v^2,$$

where C is the drag coefficient, A is the area of the object facing the fluid, and ρ is the density of the fluid.

Athletes as well as car designers seek to reduce the drag force to lower their race times ([Figure 6.29](#)). Aerodynamic shaping of an automobile can reduce the drag force and thus increase a car's gas mileage.



FIGURE 6.29 From racing cars to bobsled racers, aerodynamic shaping is crucial to achieving top speeds. Bobsleds are designed for speed and are shaped like a bullet with tapered fins. (credit: "U.S. Army"/Wikimedia Commons)

The value of the drag coefficient C is determined empirically, usually with the use of a wind tunnel ([Figure 6.30](#)).



FIGURE 6.30 NASA researchers test a model plane in a wind tunnel. (credit: NASA/Ames)

The drag coefficient can depend upon velocity, but we assume that it is a constant here. [Table 6.2](#) lists some typical drag coefficients for a variety of objects. Notice that the drag coefficient is a dimensionless quantity. At highway speeds, over 50% of the power of a car is used to overcome air drag. The most fuel-efficient cruising speed is about 70–80 km/h (about 45–50 mi/h). For this reason, during the 1970s oil crisis in the United States, maximum speeds on highways were set at about 90 km/h (55 mi/h).

Object	C
Airfoil	0.05
Toyota Camry	0.28
Ford Focus	0.32
Honda Civic	0.36
Ferrari Testarossa	0.37
Dodge Ram Pickup	0.43
Sphere	0.45
Hummer H2 SUV	0.64
Skydiver (feet first)	0.70

TABLE 6.2 Typical Values of Drag Coefficient C

Object	C
Bicycle	0.90
Skydiver (horizontal)	1.0
Circular flat plate	1.12

TABLE 6.2 Typical Values of Drag Coefficient C

Substantial research is under way in the sporting world to minimize drag. The dimples on golf balls are being redesigned, as are the clothes that athletes wear. Bicycle racers and some swimmers and runners wear full bodysuits. Australian Cathy Freeman wore a full body suit in the 2000 Sydney Olympics and won a gold medal in the 400-m race. Many swimmers in the 2008 Beijing Olympics wore (Speedo) body suits; it might have made a difference in breaking many world records ([Figure 6.31](#)). Most elite swimmers (and cyclists) shave their body hair. Such innovations can have the effect of slicing away milliseconds in a race, sometimes making the difference between a gold and a silver medal. One consequence is that careful and precise guidelines must be continuously developed to maintain the integrity of the sport.



FIGURE 6.31 Body suits, such as this LZR Racer Suit, have been credited with aiding in many world records after their release in 2008. Smoother “skin” and more compression forces on a swimmer’s body provide at least 10% less drag. (credit: NASA/Kathy Barnstorff)

Terminal Velocity

Some interesting situations connected to Newton’s second law occur when considering the effects of drag forces upon a moving object. For instance, consider a skydiver falling through air under the influence of gravity. The two forces acting on him are the force of gravity and the drag force (ignoring the small buoyant force). The downward force of gravity remains constant regardless of the velocity at which the person is moving. However, as the person’s velocity increases, the magnitude of the drag force increases until the magnitude of the drag force is equal to the gravitational force, thus producing a net force of zero. A zero net force means that there is no acceleration, as shown by Newton’s second law. At this point, the person’s velocity remains constant and we say that the person has reached his **terminal velocity** (v_T). Since F_D is proportional to the speed squared, a heavier skydiver must go faster for F_D to equal his weight. Let’s see how this works out more quantitatively.

At the terminal velocity,

$$F_{\text{net}} = mg - F_D = ma = 0.$$

Thus,

$$mg = F_D.$$

Using the equation for drag force, we have

$$mg = \frac{1}{2}C\rho Av_T^2.$$

Solving for the velocity, we obtain

$$v_T = \sqrt{\frac{2mg}{\rho CA}}.$$

Assume the density of air is $\rho = 1.21 \text{ kg/m}^3$. A 75-kg skydiver descending head first has a cross-sectional area of approximately $A = 0.18 \text{ m}^2$ and a drag coefficient of approximately $C = 0.70$. We find that

$$v_T = \sqrt{\frac{2(75 \text{ kg})(9.80 \text{ m/s}^2)}{(1.21 \text{ kg/m}^3)(0.70)(0.18 \text{ m}^2)}} = 98 \text{ m/s} = 350 \text{ km/h}.$$

This means a skydiver with a mass of 75 kg achieves a terminal velocity of about 350 km/h while traveling in a headfirst position, minimizing the area and his drag. In a spread-eagle position, that terminal velocity may decrease to about 200 km/h as the area increases. This terminal velocity becomes much smaller after the parachute opens.



EXAMPLE 6.17

Terminal Velocity of a Skydiver

Find the terminal velocity of an 85-kg skydiver falling in a spread-eagle position. Assume that in the spread-eagle position, the diver has a cross-sectional area of 0.70 m^2 .

Strategy

At terminal velocity, $F_{\text{net}} = 0$. Thus, the drag force on the skydiver must equal the force of gravity (the person's weight). Using the equation of drag force, we find $mg = \frac{1}{2}\rho CA v_T^2$.

Solution

The terminal velocity v_T can be written as

$$v_T = \sqrt{\frac{2mg}{\rho CA}} = \sqrt{\frac{2(85 \text{ kg})(9.80 \text{ m/s}^2)}{(1.21 \text{ kg/m}^3)(1.0)(0.70 \text{ m}^2)}} = 44 \text{ m/s}.$$

Significance

This result is consistent with the value for v_T mentioned earlier. The 75-kg skydiver going feet first had a terminal velocity of $v_T = 98 \text{ m/s}$. He weighed less but had a smaller frontal area and so a smaller drag due to the air.



CHECK YOUR UNDERSTANDING 6.10

Find the terminal velocity of a 50-kg skydiver falling in spread-eagle fashion.

The size of the object that is falling through air presents another interesting application of air drag. If you fall from a 5-m-high branch of a tree, you will likely get hurt—possibly fracturing a bone. However, a small squirrel does this all the time, without getting hurt. You do not reach a terminal velocity in such a short distance, but the squirrel does.

The following interesting quote on animal size and terminal velocity is from a 1928 essay by a British biologist, J. B. S. Haldane, titled “On Being the Right Size.”

“To the mouse and any smaller animal, [gravity] presents practically no dangers. You can drop a mouse down a thousand-yard mine shaft; and, on arriving at the bottom, it gets a slight shock and walks away, provided that the ground is fairly soft. A rat is killed, a man is broken, and a horse splashes. For the resistance presented to movement by the air is proportional to the surface of the moving object. Divide an animal's length, breadth, and height each by

ten; its weight is reduced to a thousandth, but its surface only to a hundredth. So the resistance to falling in the case of the small animal is relatively ten times greater than the driving force.”

The above quadratic dependence of air drag upon velocity does not hold if the object is very small, is going very slow, or is in a denser medium than air. Then we find that the drag force is proportional just to the velocity. This relationship is given by Stokes’ law.

STOKES’ LAW

For a spherical object falling in a medium, the drag force is

$$F_s = 6\pi r\eta v,$$

6.6

where r is the radius of the object, η is the viscosity of the fluid, and v is the object’s velocity.

Good examples of Stokes’ law are provided by microorganisms, pollen, and dust particles. Because each of these objects is so small, we find that many of these objects travel unaided only at a constant (terminal) velocity. Terminal velocities for bacteria (size about $1\ \mu\text{m}$) can be about $2\ \mu\text{m/s}$. To move at a greater speed, many bacteria swim using flagella (organelles shaped like little tails) that are powered by little motors embedded in the cell.

Sediment in a lake can move at a greater terminal velocity (about $5\ \mu\text{m/s}$), so it can take days for it to reach the bottom of the lake after being deposited on the surface.

If we compare animals living on land with those in water, you can see how drag has influenced evolution. Fish, dolphins, and even massive whales are streamlined in shape to reduce drag forces. Birds are streamlined and migratory species that fly large distances often have particular features such as long necks. Flocks of birds fly in the shape of a spearhead as the flock forms a streamlined pattern (Figure 6.32). In humans, one important example of streamlining is the shape of sperm, which need to be efficient in their use of energy.



FIGURE 6.32 Geese fly in a V formation during their long migratory travels. This shape reduces drag and energy consumption for individual birds, and also allows them a better way to communicate. (credit: modification of work by “Julo”/Wikimedia Commons)

INTERACTIVE

In lecture demonstrations, we do [measurements of the drag force \(https://openstax.org/l/21dragforce\)](https://openstax.org/l/21dragforce) on different objects. The objects are placed in a uniform airstream created by a fan. Calculate the drag coefficient.

The Calculus of Velocity-Dependent Frictional Forces

When a body slides across a surface, the frictional force on it is approximately constant and given by $\mu_k N$.

Unfortunately, the frictional force on a body moving through a liquid or a gas does not behave so simply. This drag

force is generally a complicated function of the body's velocity. However, for a body moving in a straight line at moderate speeds through a liquid such as water, the frictional force can often be approximated by

$$f_R = -bv,$$

where b is a constant whose value depends on the dimensions and shape of the body and the properties of the liquid, and v is the velocity of the body. Two situations for which the frictional force can be represented by this equation are a motorboat moving through water and a small object falling slowly through a liquid.

Let's consider the object falling through a liquid. The free-body diagram of this object with the positive direction downward is shown in [Figure 6.33](#). Newton's second law in the vertical direction gives the differential equation

$$mg - bv = m \frac{dv}{dt},$$

where we have written the acceleration as dv/dt . As v increases, the frictional force $-bv$ increases until it matches mg . At this point, there is no acceleration and the velocity remains constant at the terminal velocity v_T . From the previous equation,

$$mg - bv_T = 0,$$

so

$$v_T = \frac{mg}{b}.$$

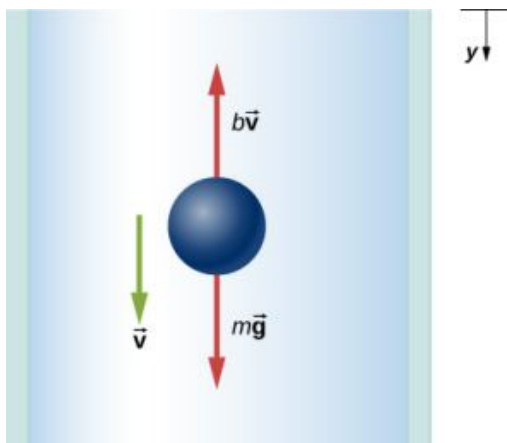


FIGURE 6.33 Free-body diagram of an object falling through a resistive medium.

We can find the object's velocity by integrating the differential equation for v . First, we rearrange terms in this equation to obtain

$$\frac{dv}{g - (b/m)v} = dt.$$

Assuming that $v = 0$ at $t = 0$, integration of this equation yields

$$\int_0^v \frac{dv'}{g - (b/m)v'} = \int_0^t dt',$$

or

$$-\frac{m}{b} \ln \left(g - \frac{b}{m} v' \right) \Big|_0^v = t' \Big|_0^t,$$

where v' and t' are dummy variables of integration. With the limits given, we find

$$-\frac{m}{b} \left[\ln \left(g - \frac{b}{m} v \right) - \ln g \right] = t.$$

Since $\ln A - \ln B = \ln(A/B)$, and $\ln(A/B) = x$ implies $e^x = A/B$, we obtain

$$\frac{g - (bv/m)}{g} = e^{-bt/m},$$

and

$$v = \frac{mg}{b}(1 - e^{-bt/m}).$$

Notice that as $t \rightarrow \infty$, $v \rightarrow mg/b = v_T$, which is the terminal velocity.

The position at any time may be found by integrating the equation for v . With $v = dy/dt$,

$$dy = \frac{mg}{b}(1 - e^{-bt/m})dt.$$

Assuming $y = 0$ when $t = 0$,

$$\int_0^y dy' = \frac{mg}{b} \int_0^t (1 - e^{-bt'/m}) dt',$$

which integrates to

$$y = \frac{mg}{b}t + \frac{m^2g}{b^2}(e^{-bt/m} - 1).$$



EXAMPLE 6.18

Effect of the Resistive Force on a Motorboat

A motorboat is moving across a lake at a speed v_0 when its motor suddenly freezes up and stops. The boat then slows down under the frictional force $f_R = -bv$. (a) What are the velocity and position of the boat as functions of time? (b) If the boat slows down from 4.0 to 1.0 m/s in 10 s, how far does it travel before stopping?

Solution

- a. With the motor stopped, the only horizontal force on the boat is $f_R = -bv$, so from Newton's second law,

$$m \frac{dv}{dt} = -bv,$$

which we can write as

$$\frac{dv}{v} = -\frac{b}{m}dt.$$

Integrating this equation between the time zero when the velocity is v_0 and the time t when the velocity is v , we have

$$\int_0^v \frac{dv'}{v'} = -\frac{b}{m} \int_0^t dt'.$$

Thus,

$$\ln \frac{v}{v_0} = -\frac{b}{m}t,$$

which, since $\ln A = x$ implies $e^x = A$, we can write this as

$$v = v_0 e^{-bt/m}.$$

Now from the definition of velocity,

$$\frac{dx}{dt} = v_0 e^{-bt/m},$$

so we have

$$dx = v_0 e^{-bt/m} dt.$$

With the initial position zero, we have

$$\int_0^x dx' = v_0 \int_0^t e^{-bt'/m} dt',$$

and

$$x = -\frac{mv_0}{b} e^{-bt'/m} \Big|_0^t = \frac{mv_0}{b} (1 - e^{-bt/m}).$$

As time increases, $e^{-bt/m} \rightarrow 0$, and the position of the boat approaches a limiting value

$$x_{\max} = \frac{mv_0}{b}.$$

Although this tells us that the boat takes an infinite amount of time to reach $x_{\max}$, the boat effectively stops after a reasonable time. For example, at $t = 10m/b$, we have

$$v = v_0 e^{-10} \simeq 4.5 \times 10^{-5} v_0,$$

whereas we also have

$$x = x_{\max} (1 - e^{-10}) \simeq 0.99995 x_{\max}.$$

Therefore, the boat's velocity and position have essentially reached their final values.

- b. With $v_0 = 4.0$ m/s and $v = 1.0$ m/s, we have $1.0 \text{ m/s} = (4.0 \text{ m/s})e^{-(b/m)(10 \text{ s})}$, so

$$\ln 0.25 = -\ln 4.0 = -\frac{b}{m}(10 \text{ s}),$$

and

$$\frac{b}{m} = \frac{1}{10} \ln 4.0 \text{ s}^{-1} = 0.14 \text{ s}^{-1}.$$

Now the boat's limiting position is

$$x_{\max} = \frac{mv_0}{b} = \frac{4.0 \text{ m/s}}{0.14 \text{ s}^{-1}} = 29 \text{ m}.$$

Significance

In the both of the previous examples, we found “limiting” values. The terminal velocity is the same as the limiting velocity, which is the velocity of the falling object after a (relatively) long time has passed. Similarly, the limiting distance of the boat is the distance the boat will travel after a long amount of time has passed. Due to the properties of exponential decay, the time involved to reach either of these values is actually not too long (certainly not an infinite amount of time!) but they are quickly found by taking the limit to infinity.

CHECK YOUR UNDERSTANDING 6.11

Suppose the resistive force of the air on a skydiver can be approximated by $f = -bv^2$. If the terminal velocity of a 100-kg skydiver is 60 m/s, what is the value of b ?

Chapter Review

Key Terms

banked curve curve in a road that is sloping in a manner that helps a vehicle negotiate the curve

centripetal force any net force causing uniform circular motion

Coriolis force inertial force causing the apparent deflection of moving objects when viewed in a rotating frame of reference

drag force force that always opposes the motion of an object in a fluid; unlike simple friction, the drag force is proportional to some function of the velocity of the object in that fluid

friction force that opposes relative motion or attempts at motion between systems in contact

ideal banking sloping of a curve in a road, where the angle of the slope allows the vehicle to negotiate the curve at a certain speed without the aid of

friction between the tires and the road; the net external force on the vehicle equals the horizontal centripetal force in the absence of friction

inertial force force that has no physical origin

kinetic friction force that opposes the motion of two systems that are in contact and moving relative to each other

noninertial frame of reference accelerated frame of reference

static friction force that opposes the motion of two systems that are in contact and are not moving relative to each other

terminal velocity constant velocity achieved by a falling object, which occurs when the weight of the object is balanced by the upward drag force

Key Equations

Magnitude of static friction	$f_s \leq \mu_s N$
Magnitude of kinetic friction	$f_k = \mu_k N$
Centripetal force	$F_c = m \frac{v^2}{r}$ or $F_c = m r \omega^2$
Ideal angle of a banked curve	$\tan \theta = \frac{v^2}{r g}$
Drag force	$F_D = \frac{1}{2} C \rho A v^2$
Stokes' law	$F_s = 6 \pi r \eta v$

Summary

6.1 Solving Problems with Newton's Laws

- Newton's laws of motion can be applied in numerous situations to solve motion problems.
- Some problems contain multiple force vectors acting in different directions on an object. Be sure to draw diagrams, resolve all force vectors into horizontal and vertical components, and draw a free-body diagram. Always analyze the direction in which an object accelerates so that you can determine whether $F_{\text{net}} = ma$ or $F_{\text{net}} = 0$.
- The normal force on an object is not always equal in magnitude to the weight of the object. If an object is accelerating vertically, the normal force is less than or greater than the weight of the object. Also, if the object is on an inclined plane, the normal force is always less than the full weight of the object.
- Some problems contain several physical quantities, such as forces, acceleration, velocity, or position. You can apply concepts from

kinematics and dynamics to solve these problems.

6.2 Friction

- Friction is a contact force that opposes the motion or attempted motion between two systems. Simple friction is proportional to the normal force N supporting the two systems.
- The magnitude of static friction force between two materials stationary relative to each other is determined using the coefficient of static friction, which depends on both materials.
- The kinetic friction force between two materials moving relative to each other is determined using the coefficient of kinetic friction, which also depends on both materials and is always less than the coefficient of static friction.

6.3 Centripetal Force

- Centripetal force $\vec{F}_c$ is a "center-seeking" force that always points toward the center of rotation. It is perpendicular to linear velocity and has the

magnitude

$$F_c = ma_c.$$

- Rotating and accelerated frames of reference are noninertial. Inertial forces, such as the Coriolis force, are needed to explain motion in such frames.

6.4 Drag Force and Terminal Speed

- Drag forces acting on an object moving in a fluid

Conceptual Questions

6.1 Solving Problems with Newton's Laws

- To simulate the apparent weightlessness of space orbit, astronauts are trained in the hold of a cargo aircraft that is accelerating downward at g . Why do they appear to be weightless, as measured by standing on a bathroom scale, in this accelerated frame of reference? Is there any difference between their apparent weightlessness in orbit and in the aircraft?

6.2 Friction

- The glue on a piece of tape can exert forces. Can these forces be a type of simple friction? Explain, considering especially that tape can stick to vertical walls and even to ceilings.
- When you learn to drive, you discover that you need to let up slightly on the brake pedal as you come to a stop or the car will stop with a jerk. Explain this in terms of the relationship between static and kinetic friction.
- When you push a piece of chalk across a chalkboard, it sometimes screeches because it rapidly alternates between slipping and sticking to the board. Describe this process in more detail, in particular, explaining how it is related to the fact that kinetic friction is less than static friction. (The same slip-grab process occurs when tires screech on pavement.)
- A physics major is cooking breakfast when she notices that the frictional force between her steel spatula and Teflon frying pan is only 0.200 N. Knowing the coefficient of kinetic friction between the two materials, she quickly calculates the normal force. What is it?

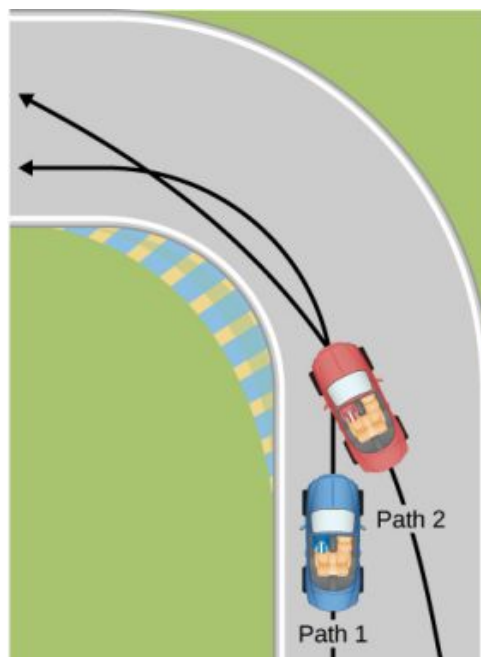
6.3 Centripetal Force

- If you wish to reduce the stress (which is related to centripetal force) on high-speed tires, would you use large- or small-diameter tires? Explain.

oppose the motion. For larger objects (such as a baseball) moving at a velocity in air, the drag force is determined using the drag coefficient (typical values are given in [Table 6.2](#)), the area of the object facing the fluid, and the fluid density.

- For small objects (such as a bacterium) moving in a denser medium (such as water), the drag force is given by Stokes' law.

- Define centripetal force. Can any type of force (for example, tension, gravitational force, friction, and so on) be a centripetal force? Can any combination of forces be a centripetal force?
- If centripetal force is directed toward the center, why do you feel that you are 'thrown' away from the center as a car goes around a curve? Explain.
- Race car drivers routinely cut corners, as shown below (Path 2). Explain how this allows the curve to be taken at the greatest speed.

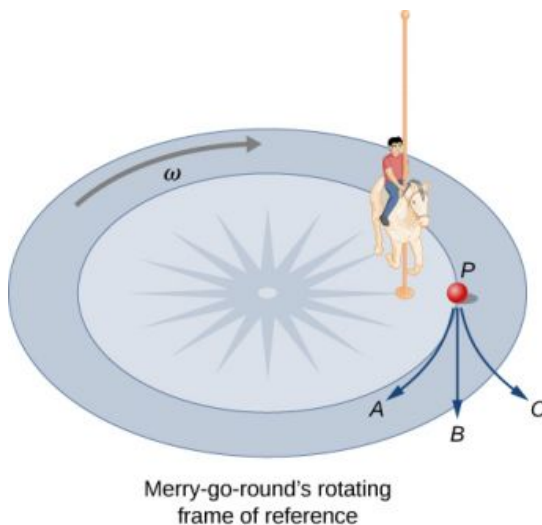


- Many amusement parks have rides that make vertical loops like the one shown below. For safety, the cars are attached to the rails in such a way that they cannot fall off. If the car goes over the top at just the right speed, gravity alone will supply the centripetal force. What other force acts and what is its direction if:
 - The car goes over the top at faster than this speed?
 - The car goes over the top at slower than this

speed?



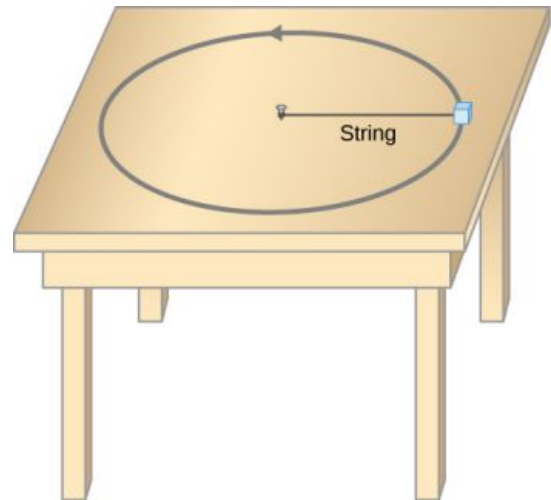
11. What causes water to be removed from clothes in a spin-dryer?
12. As a skater forms a circle, what force is responsible for making his turn? Use a free-body diagram in your answer.
13. Suppose a child is riding on a merry-go-round at a distance about halfway between its center and edge. She has a lunch box resting on wax paper, so that there is very little friction between it and the merry-go-round. (a) Which path shown below, as viewed from the ground, will the lunch box take when she lets go? (b) The lunch box leaves a trail in the dust on the merry-go-round. What is the shape of the dust trail? Explain your answers.



14. Do you feel yourself thrown to either side when you negotiate a curve that is ideally banked for

your car's speed? What is the direction of the force exerted on you by the car seat?

15. Suppose a mass is moving in a circular path on a frictionless table as shown below. In Earth's frame of reference, there is no centrifugal force pulling the mass away from the center of rotation, yet there is a force stretching the string attaching the mass to the nail. Using concepts related to centripetal force and Newton's third law, explain what force stretches the string, identifying its physical origin.



16. When a toilet is flushed or a sink is drained, the water (and other material) begins to rotate about the drain on the way down. Assuming no initial rotation and a flow initially directly straight toward the drain, explain what causes the rotation and which direction it has in the Northern Hemisphere. (Note that this is a small effect and in most toilets the rotation is caused by directional water jets.) Would the direction of rotation reverse if water were forced up the drain?
17. A car rounds a curve and encounters a patch of ice with a very low coefficient of kinetic friction. The car slides off the road. Describe the path of the car as it leaves the road.
18. In one amusement park ride, riders enter a large vertical barrel and stand against the wall on its horizontal floor. The barrel is spun up and the floor drops away. Riders feel as if they are pinned to the wall by a force something like the gravitational force. This is an inertial force sensed and used by the riders to explain events in the rotating frame of reference of the barrel. Explain in an inertial frame of reference (Earth is nearly one) what pins the riders to the wall, and identify all forces acting on them.

19. Two friends are having a conversation. Anna says a satellite in orbit is in free fall because the satellite keeps falling toward Earth. Tom says a satellite in orbit is not in free fall because the acceleration due to gravity is not 9.80 m/s^2 . Who do you agree with and why?
20. A nonrotating frame of reference placed at the center of the Sun is very nearly an inertial one. Why is it not exactly an inertial frame?

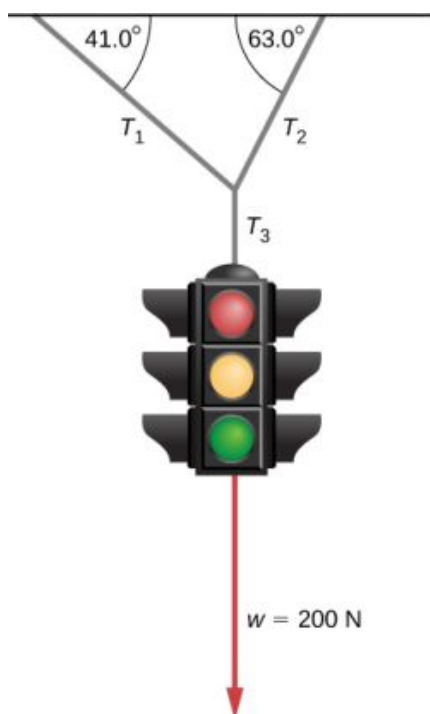
6.4 Drag Force and Terminal Speed

21. Athletes such as swimmers and bicyclists wear body suits in competition. Formulate a list of pros and cons of such suits.

Problems

6.1 Solving Problems with Newton's Laws

25. A 30.0-kg girl in a swing is pushed to one side and held at rest by a horizontal force $\vec{F}$ so that the swing ropes are 30.0° with respect to the vertical. (a) Calculate the tension in each of the two ropes supporting the swing under these conditions. (b) Calculate the magnitude of $\vec{F}$.
26. Find the tension in each of the three cables supporting the traffic light if it weighs $2.00 \times 10^2 \text{ N}$.

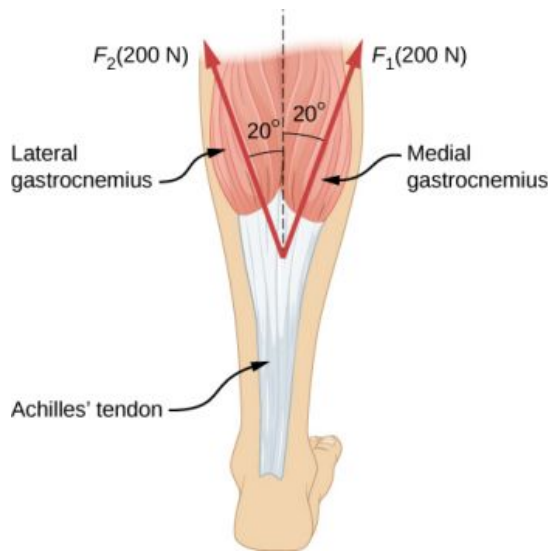


27. Three forces act on an object, considered to be a particle, which moves with constant velocity

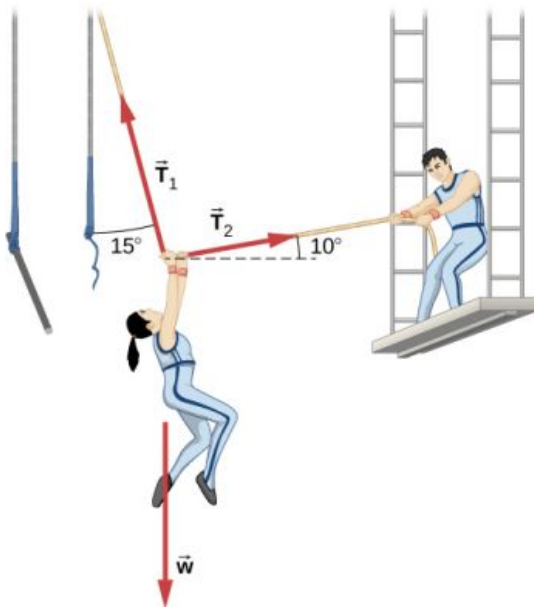
22. Two expressions were used for the drag force experienced by a moving object in a liquid. One depended upon the speed, while the other was proportional to the square of the speed. In which types of motion would each of these expressions be more applicable than the other one?
23. As cars travel, oil and gasoline leaks onto the road surface. If a light rain falls, what does this do to the control of the car? Does a heavy rain make any difference?
24. Why can a squirrel jump from a tree branch to the ground and run away undamaged, while a human could break a bone in such a fall?

$v = (3\hat{i} - 2\hat{j}) \text{ m/s}$. Two of the forces are $\vec{F}_1 = (3\hat{i} + 5\hat{j}) \text{ N}$ and $\vec{F}_2 = (4\hat{i} - 7\hat{j}) \text{ N}$. Find the third force.

28. A flea jumps by exerting a force of $1.20 \times 10^{-5} \text{ N}$ straight down on the ground. A breeze blowing on the flea parallel to the ground exerts a force of $0.500 \times 10^{-6} \text{ N}$ on the flea while the flea is still in contact with the ground. Find the direction and magnitude of the acceleration of the flea if its mass is $6.00 \times 10^{-7} \text{ kg}$. Do not neglect the gravitational force.
29. Two muscles in the back of the leg pull upward on the Achilles tendon, as shown below. (These muscles are called the medial and lateral heads of the gastrocnemius muscle.) Find the magnitude and direction of the total force on the Achilles tendon. What type of movement could be caused by this force?



30. After a mishap, a 76.0-kg circus performer clings to a trapeze, which is being pulled to the side by another circus artist, as shown here. Calculate the tension in the two ropes if the person is momentarily motionless. Include a free-body diagram in your solution.



31. A 35.0-kg dolphin accelerates opposite to the motion from 12.0 to 7.50 m/s in 2.30 s to join another dolphin in play. What average force was exerted to slow the first dolphin if it was moving horizontally? (The gravitational force is balanced by the buoyant force of the water.)
32. When starting a foot race, a 70.0-kg sprinter exerts an average force of 650 N backward on the ground for 0.800 s. (a) What is their final speed? (b) How far do they travel?

33. A large rocket has a mass of 2.00×10^6 kg at takeoff, and its engines produce a thrust of 3.50×10^7 N. (a) Find its initial acceleration if it takes off vertically. (b) How long does it take to reach a velocity of 120 km/h straight up, assuming constant mass and thrust?

34. A basketball player jumps straight up for a ball. To do this, he lowers his body 0.300 m and then accelerates through this distance by forcefully straightening his legs. This player leaves the floor with a vertical velocity sufficient to carry him 0.900 m above the floor. (a) Calculate his velocity when he leaves the floor. (b) Calculate his acceleration while he is straightening his legs. He goes from zero to the velocity found in (a) in a distance of 0.300 m. (c) Calculate the force he exerts on the floor to do this, given that his mass is 110.0 kg.

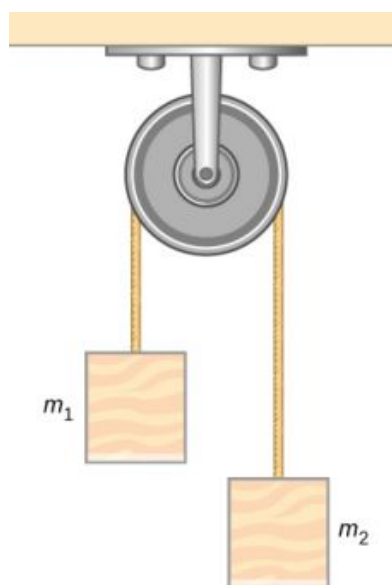
35. A 2.50-kg fireworks shell is fired straight up from a mortar and reaches a height of 110.0 m. (a) Neglecting air resistance (a poor assumption, but we will make it for this example), calculate the shell's velocity when it leaves the mortar. (b) The mortar itself is a tube 0.450 m long. Calculate the average acceleration of the shell in the tube as it goes from zero to the velocity found in (a). (c) What is the average force on the shell in the mortar? Express your answer in newtons and as a ratio to the weight of the shell.

36. A 0.500-kg potato is fired at an angle of 80.0° above the horizontal from a PVC pipe used as a "potato gun" and reaches a height of 110.0 m. (a) Neglecting air resistance, calculate the potato's velocity when it leaves the gun. (b) The gun itself is a tube 0.450 m long. Calculate the average acceleration of the potato in the tube as it goes from zero to the velocity found in (a). (c) What is the average force on the potato in the gun? Express your answer in newtons and as a ratio to the weight of the potato.

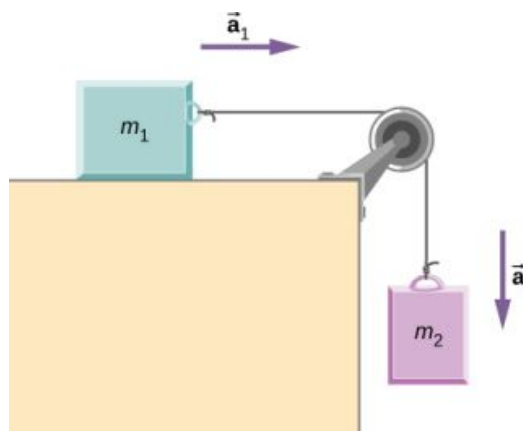
37. An elevator filled with passengers has a mass of 1.70×10^3 kg. (a) The elevator accelerates upward from rest at a rate of 1.20 m/s^2 for 1.50 s. Calculate the tension in the cable supporting the elevator. (b) The elevator continues upward at constant velocity for 8.50 s. What is the tension in the cable during this time? (c) The elevator accelerates opposite to the motion at a rate of 0.600 m/s^2 for 3.00 s. What is the tension in the cable during acceleration opposite to the motion? (d) How high has the elevator moved

above its original starting point, and what is its final velocity?

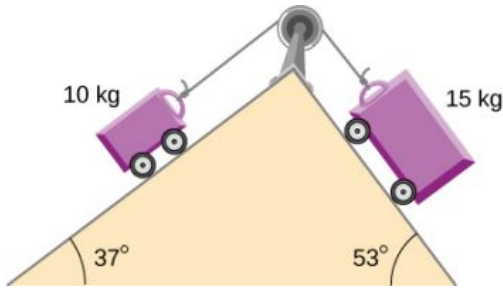
38. A 20.0-g ball hangs from the roof of a freight car by a string. When the freight car begins to move, the string makes an angle of 35.0° with the vertical. (a) What is the acceleration of the freight car? (b) What is the tension in the string?
39. A student's backpack, full of textbooks, is hung from a spring scale attached to the ceiling of an elevator. When the elevator is accelerating downward at 3.8 m/s^2 , the scale reads 60 N. (a) What is the mass of the backpack? (b) What does the scale read if the elevator moves upward while speeding up at a rate 3.8 m/s^2 ? (c) What does the scale read if the elevator moves upward at constant velocity? (d) If the elevator had no brakes and the cable supporting it were to break loose so that the elevator could fall freely, what would the spring scale read?
40. A service elevator takes a load of garbage, mass 10.0 kg, from a floor of a skyscraper under construction, down to ground level, accelerating downward at a rate of 1.2 m/s^2 . Find the magnitude of the force the garbage exerts on the floor of the service elevator?
41. A roller coaster car starts from rest at the top of a track 30.0 m long and inclined at 20.0° to the horizontal. Assume that friction can be ignored. (a) What is the acceleration of the car? (b) How much time elapses before it reaches the bottom of the track?
42. The device shown below is the Atwood's machine considered in [Example 6.5](#). Assuming that the masses of the string and the frictionless pulley are negligible, (a) find an equation for the acceleration of the two blocks; (b) find an equation for the tension in the string; and (c) find both the acceleration and tension when block 1 has mass 2.00 kg and block 2 has mass 4.00 kg.



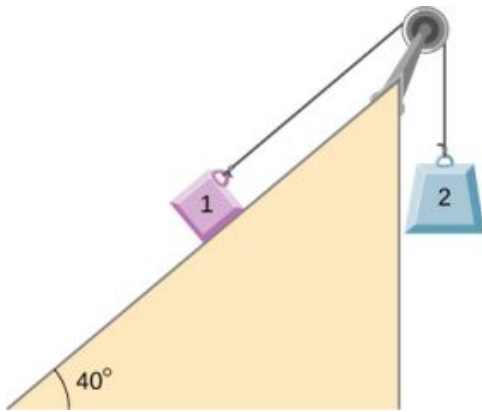
43. Two blocks are connected by a massless rope as shown below. The mass of the block on the table is 4.0 kg and the hanging mass is 1.0 kg. The table and the pulley are frictionless. (a) Find the acceleration of the system. (b) Find the tension in the rope. (c) Find the speed with which the hanging mass hits the floor if it starts from rest and is initially located 1.0 m from the floor.



44. Shown below are two carts connected by a cord that passes over a small frictionless pulley. Each cart rolls freely with negligible friction. Calculate the acceleration of the carts and the tension in the cord.

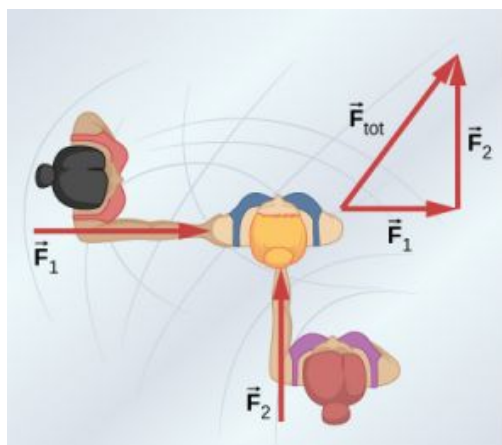


45. A 2.00 kg block (mass 1) and a 4.00 kg block (mass 2) are connected by a light string as shown; the inclination of the ramp is 40.0° . Friction is negligible. What is (a) the acceleration of each block and (b) the tension in the string?



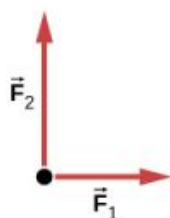
6.2 Friction

46. (a) When rebuilding an engine, a physics major must exert 3.00×10^2 N of force to insert a dry steel piston into a steel cylinder. What is the normal force between the piston and cylinder? (b) What force would they have to exert if the steel parts were oiled?
47. (a) What is the maximum frictional force in the knee joint of a person who supports 66.0 kg of her mass on that knee? (b) During strenuous exercise, it is possible to exert forces to the joints that are easily 10 times greater than the weight being supported. What is the maximum force of friction under such conditions? The frictional forces in joints are relatively small in all circumstances except when the joints deteriorate, such as from injury or arthritis. Increased frictional forces can cause further damage and pain.
48. Suppose you have a 120-kg wooden crate resting on a wood floor, with coefficient of static friction 0.500 between these wood surfaces. (a) What maximum force can you exert horizontally on the crate without moving it? (b) If you continue to exert this force once the crate starts to slip, what will its acceleration then be? The coefficient of sliding friction is known to be 0.300 for this situation.
49. (a) If half of the weight of a small 1.00×10^3 -kg utility truck is supported by its two drive wheels, what is the maximum acceleration it can achieve on dry concrete? (b) Will a metal cabinet lying on the wooden bed of the truck slip if it accelerates at this rate? (c) Solve both problems assuming the truck has four-wheel drive.
50. A team of eight dogs pulls a sled with waxed wood runners on wet snow (mush!). The dogs have average masses of 19.0 kg, and the loaded sled with its rider has a mass of 210 kg. (a) Calculate the acceleration of the dogs starting from rest if each dog exerts an average force of 185 N backward on the snow. (b) Calculate the force in the coupling between the dogs and the sled.
51. Consider the 65.0-kg ice skater being pushed by two others shown below. (a) Find the direction and magnitude of $\mathbf{F}_{\text{tot}}$, the total force exerted on her by the others, given that the magnitudes F_1 and F_2 are 26.4 N and 18.6 N, respectively. (b) What is her initial acceleration if she is initially stationary and wearing steel-bladed skates that point in the direction of $\mathbf{F}_{\text{tot}}$? (c) What is her acceleration assuming she is already moving in the direction of $\mathbf{F}_{\text{tot}}$? (Remember that friction always acts in the direction opposite that of motion or attempted motion between surfaces in contact.)



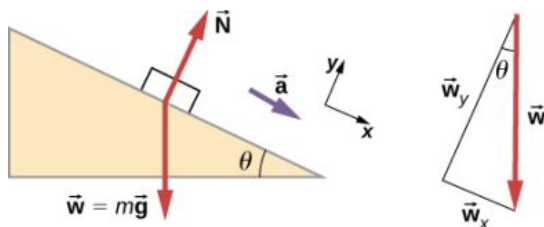
(a)

Free-body diagram

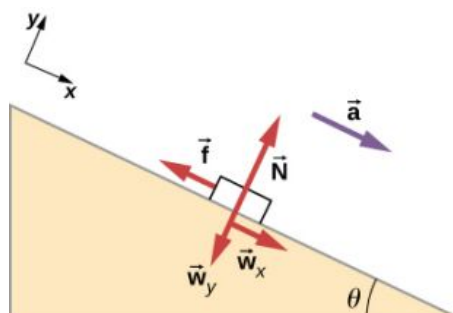


(b)

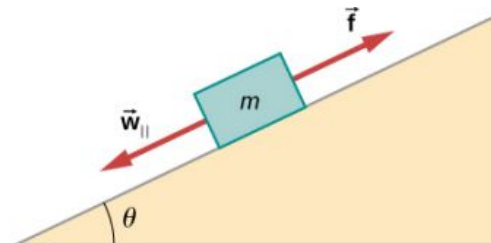
52. Show that the acceleration of any object down a frictionless incline that makes an angle θ with the horizontal is $a = g \sin \theta$. (Note that this acceleration is independent of mass.)



53. Show that the acceleration of any object down an incline where friction behaves simply (that is, where $f_k = \mu_k N$) is $a = g(\sin \theta - \mu_k \cos \theta)$. Note that the acceleration is independent of mass and reduces to the expression found in the previous problem when friction becomes negligibly small ($\mu_k = 0$).



54. Calculate the acceleration opposite to the motion of a snow boarder going up a 5.00° slope, assuming the coefficient of friction for waxed wood on wet snow. The result of the preceding problem may be useful, but be careful to consider the fact that the snow boarder is going uphill.
55. A machine at a post office sends packages out a chute and down a ramp to be loaded into delivery vehicles. (a) Calculate the acceleration of a box heading down a 10.0° slope, assuming the coefficient of friction for a parcel on waxed wood is 0.100. (b) Find the angle of the slope down which this box could move at a constant velocity. You can neglect air resistance in both parts.
56. If an object is to rest on an incline without slipping, then friction must equal the component of the weight of the object parallel to the incline. This requires greater and greater friction for steeper slopes. Show that the maximum angle of an incline above the horizontal for which an object will not slide down is $\theta = \tan^{-1} \mu_s$. You may use the result of the previous problem. Assume that $a = 0$ and that static friction has reached its maximum value.



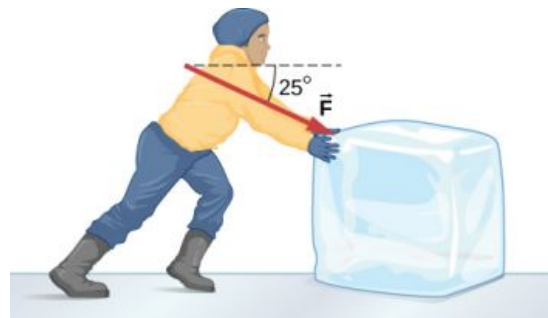
57. Calculate the maximum acceleration of a car that is heading down a 6.00° slope (one that makes an angle of 6.00° with the horizontal) under the following road conditions. You may assume that the weight of the car is evenly distributed on all four tires and that the coefficient of static friction is involved—that is,

the tires are not allowed to slip during the acceleration opposite to the motion. (Ignore rolling.) Calculate for a car: (a) On dry concrete. (b) On wet concrete. (c) On ice, assuming that $\mu_s = 0.100$, the same as for shoes on ice.

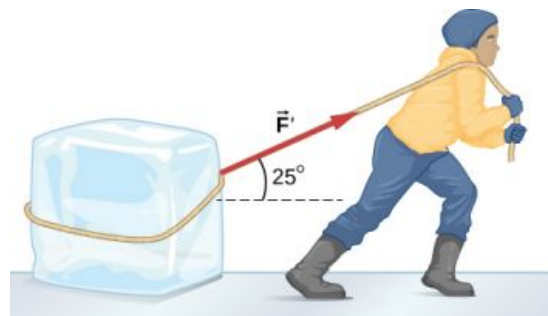
58. Calculate the maximum acceleration of a car that is heading up a 4.00° slope (one that makes an angle of 4.00° with the horizontal) under the following road conditions. Assume that only half the weight of the car is supported by the two drive wheels and that the coefficient of static friction is involved—that is, the tires are not allowed to slip during the acceleration. (Ignore rolling.) (a) On dry concrete. (b) On wet concrete. (c) On ice, assuming that $\mu_s = 0.100$, the same as for shoes on ice.
59. Repeat the preceding problem for a car with four-wheel drive.
60. A freight train consists of two 8.00×10^5 -kg engines and 45 cars with average masses of 5.50×10^5 kg. (a) What force must each engine exert backward on the track to accelerate the train at a rate of $5.00 \times 10^{-2} \text{ m/s}^2$ if the force of friction is $7.50 \times 10^5 \text{ N}$, assuming the engines exert identical forces? This is not a large frictional force for such a massive system. Rolling friction for trains is small, and consequently, trains are very energy-efficient transportation systems. (b) What is the force in the coupling between the 37th and 38th cars (this is the force each exerts on the other), assuming all cars have the same mass and that friction is evenly distributed among all of the cars and engines?
61. Consider the 52.0-kg mountain climber shown below. (a) Find the tension in the rope and the force that the mountain climber must exert with her feet on the vertical rock face to remain stationary. Assume that the force is exerted parallel to her legs. Also, assume negligible force exerted by her arms. (b) What is the minimum coefficient of friction between her shoes and the cliff?



62. A contestant in a winter sporting event pushes a 45.0-kg block of ice across a frozen lake as shown below. The coefficient of friction of ice can be found in [Table 6.1](#). (a) Calculate the minimum force F he must exert to get the block moving. (b) What is its acceleration once it starts to move, if that force is maintained?



63. The contestant now pulls the block of ice with a rope over his shoulder at the same angle above the horizontal as shown below. The coefficient of friction of ice can be found in [Table 6.1](#). Calculate the minimum force F he must exert to get the block moving. (b) What is its acceleration once it starts to move, if that force is maintained?



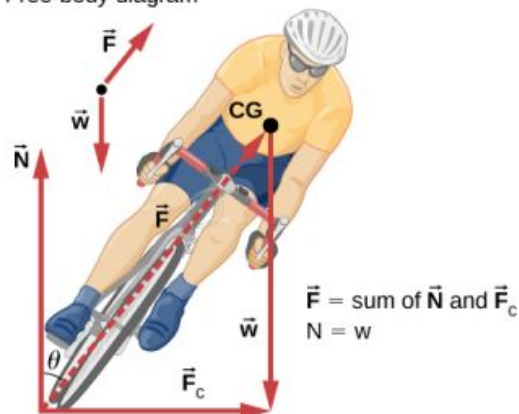
64. At a post office, a parcel that is a 20.0-kg box

slides down a ramp inclined at 30.0° with the horizontal. The coefficient of kinetic friction between the box and plane is 0.0300. (a) Find the acceleration of the box. (b) Find the velocity of the box as it reaches the end of the plane, if the length of the plane is 2 m and the box starts at rest.

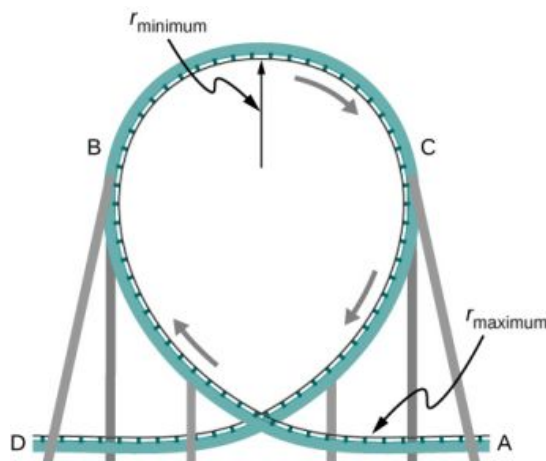
6.3 Centripetal Force

65. (a) A 22.0-kg child is riding a playground merry-go-round that is rotating at 40.0 rev/min. What centripetal force is exerted if he is 1.25 m from its center? (b) What centripetal force is exerted if the merry-go-round rotates at 3.00 rev/min and he is 8.00 m from its center? (c) Compare each force with his weight.
66. Calculate the centripetal force on the end of a 100-m (radius) wind turbine blade that is rotating at 0.5 rev/s. Assume the mass is 4 kg.
67. What is the ideal banking angle for a gentle turn of 1.20-km radius on a highway with a 105 km/h speed limit (about 65 mi/h), assuming everyone travels at the limit?
68. What is the ideal speed to take a 100.0-m-radius curve banked at a 20.0° angle?
69. (a) What is the radius of a bobsled turn banked at 75.0° and taken at 30.0 m/s, assuming it is ideally banked? (b) Calculate the centripetal acceleration. (c) Does this acceleration seem large to you?
70. Part of riding a bicycle involves leaning at the correct angle when making a turn, as seen below. To be stable, the force exerted by the ground must be on a line going through the center of gravity. The force on the bicycle wheel can be resolved into two perpendicular components—friction parallel to the road (this must supply the centripetal force) and the vertical normal force (which must equal the system's weight). (a) Show that θ (as defined as shown) is related to the speed v and radius of curvature r of the turn in the same way as for an ideally banked roadway—that is, $\theta = \tan^{-1}(v^2/rg)$. (b) Calculate θ for a 12.0-m/s turn of radius 30.0 m (as in a race).

Free-body diagram

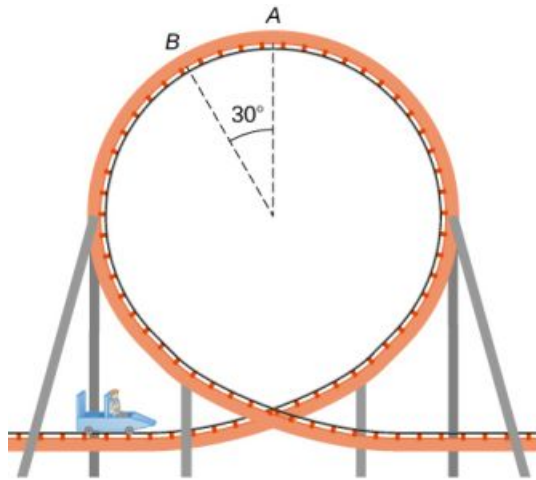


71. If a car takes a banked curve at less than the ideal speed, friction is needed to keep it from sliding toward the inside of the curve (a problem on icy mountain roads). (a) Calculate the ideal speed to take a 100.0 m radius curve banked at 15.0° . (b) What is the minimum coefficient of friction needed for a frightened driver to take the same curve at 20.0 km/h?
72. Modern roller coasters have vertical loops like the one shown here. The radius of curvature is smaller at the top than on the sides so that the downward centripetal acceleration at the top will be greater than the acceleration due to gravity, keeping the passengers pressed firmly into their seats. What is the speed of the roller coaster at the top of the loop if the radius of curvature there is 15.0 m and the downward acceleration of the car is $1.50 g$?



73. A child of mass 40.0 kg is in a roller coaster car that travels in a loop of radius 7.00 m. At point A the speed of the car is 10.0 m/s, and at point B, the speed is 10.5 m/s. Assume the child is not holding on and does not wear a seat belt. (a) What is the force of the car seat on the child at point A? (b) What

is the force of the car seat on the child at point B? (c) What minimum speed is required to keep the child in his seat at point A?



74. In the simple Bohr model of the ground state of the hydrogen atom, the electron travels in a circular orbit around a fixed proton. The radius of the orbit is 5.28×10^{-11} m, and the speed of the electron is 2.18×10^6 m/s. The mass of an electron is 9.11×10^{-31} kg. What is the force on the electron?
75. Railroad tracks follow a circular curve of radius 500.0 m and are banked at an angle of 5.0° . For trains of what speed are these tracks designed?
76. The CERN particle accelerator is circular with a circumference of 26.7 km. (a) What is the acceleration of the protons ($m = 1.67 \times 10^{-27}$ kg) that move around the accelerator at 5% of the speed of light? (The speed of light is $v = 3.00 \times 10^8$ m/s.) (b) What is the force on the protons?
77. A car rounds an unbanked curve of radius 65 m. If the coefficient of static friction between the road and car is 0.70, what is the maximum speed at which the car can traverse the curve without slipping?
78. A banked highway is designed for traffic moving at 90.0 km/h. The radius of the curve is 310 m. What is the angle of banking of the highway?
- position with a surface area of 0.140 m^2 .
80. A 60.0-kg and a 90.0-kg skydiver jump from an airplane at an altitude of 6.00×10^3 m, both falling in a headfirst position. Make some assumption on their frontal areas and calculate their terminal velocities. How long will it take for each skydiver to reach the ground (assuming the time to reach terminal velocity is small)? Assume all values are accurate to three significant digits.
81. A 560-g squirrel with a surface area of 930 cm^2 falls from a 5.0-m tree to the ground. Estimate its terminal velocity. (Use a drag coefficient for a skydiver falling feet first.) What will be the velocity of a 56-kg person hitting the ground, assuming no drag contribution in such a short distance?
82. To maintain a constant speed, the force provided by a car's engine must equal the drag force plus the force of friction of the road (the rolling resistance). (a) What are the drag forces at 70 km/h and 100 km/h for a Toyota Camry? (Drag area is 0.70 m^2) (b) What is the drag force at 70 km/h and 100 km/h for a Hummer H2? (Drag area is 2.44 m^2) Assume all values are accurate to three significant digits.
83. By what factor does the drag force on a car increase as it goes from 65 to 110 km/h?
84. Calculate the velocity a spherical rain drop would achieve falling from 5.00 km (a) in the absence of air drag (b) with air drag. Take the size across of the drop to be 4 mm, the density to be $1.00 \times 10^3 \text{ kg/m}^3$, and the surface area to be πr^2 .
85. Using Stokes' law, verify that the units for viscosity are kilograms per meter per second.
86. Find the terminal velocity of a spherical bacterium (diameter $2.00 \mu\text{m}$) falling in water. You will first need to note that the drag force is equal to the weight at terminal velocity. Take the density of the bacterium to be $1.10 \times 10^3 \text{ kg/m}^3$.
87. Stokes' law describes sedimentation of particles in liquids and can be used to measure viscosity. Particles in liquids achieve terminal velocity quickly. One can measure the time it takes for a particle to fall a certain distance and then use Stokes' law to calculate the viscosity of the liquid. Suppose a steel ball bearing (density

6.4 Drag Force and Terminal Speed

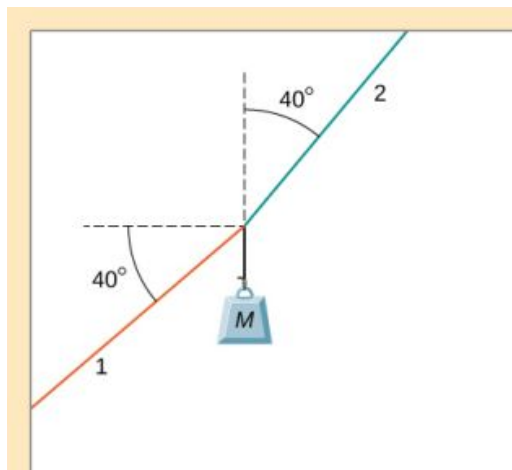
79. The terminal velocity of a person falling in air depends upon the weight and the area of the person facing the fluid. Find the terminal velocity (in meters per second and kilometers per hour) of an 80.0-kg skydiver falling in a headfirst

$7.8 \times 10^3 \text{ kg/m}^3$, diameter 3.0 mm) is dropped in a container of motor oil. It takes 12 s to fall a distance of 0.60 m. Calculate the viscosity of the oil.

88. Suppose that the resistive force of the air on a skydiver can be approximated by $f = -bv^2$. If the terminal velocity of a 50.0-kg skydiver is 60.0 m/s, what is the value of b ?

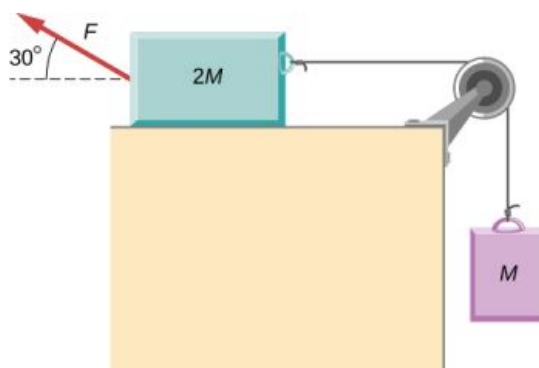
Additional Problems

90. (a) What is the final velocity of a car originally traveling at 50.0 km/h that accelerates opposite to the motion at a rate of 0.400 m/s^2 for 50.0 s? Assume a coefficient of friction of 1.0. (b) What is unreasonable about the result? (c) Which premise is unreasonable, or which premises are inconsistent?
91. A 75.0-kg woman stands on a bathroom scale in an elevator that accelerates from rest to 30.0 m/s in 2.00 s. (a) Calculate the scale reading in newtons and compare it with her weight. (The scale exerts an upward force on her equal to its reading.) (b) What is unreasonable about the result? (c) Which premise is unreasonable, or which premises are inconsistent?
92. (a) Calculate the minimum coefficient of friction needed for a car to negotiate an unbanked 50.0 m radius curve at 30.0 m/s. (b) What is unreasonable about the result? (c) Which premises are unreasonable or inconsistent?
93. As shown below, if $M = 5.50 \text{ kg}$, what is the tension in string 1?

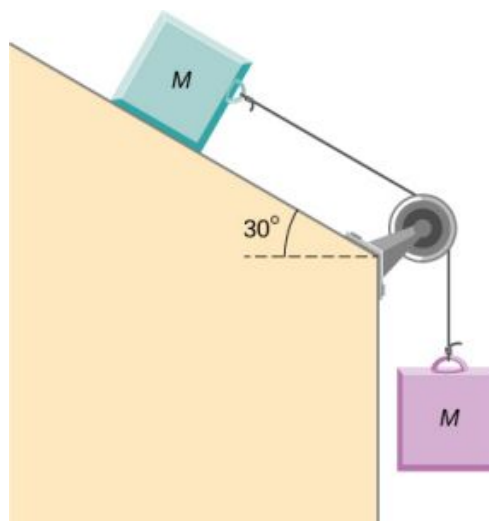


94. As shown below, if $F = 60.0 \text{ N}$ and $M = 4.00 \text{ kg}$, what is the magnitude of the acceleration of the suspended object? All surfaces are frictionless.

89. A small diamond of mass 10.0 g drops from a swimmer's earring and falls through the water, reaching a terminal velocity of 2.0 m/s. (a) Assuming the frictional force on the diamond obeys $f = -bv$, what is b ? (b) How far does the diamond fall before it reaches 90 percent of its terminal speed?



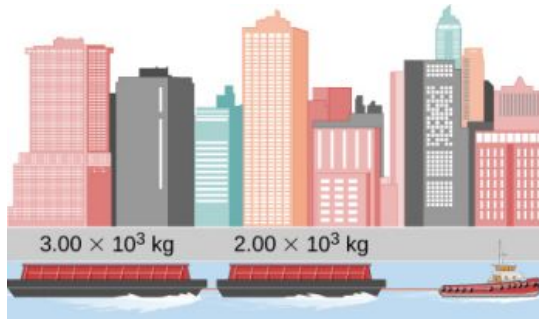
95. As shown below, if $M = 6.0 \text{ kg}$, what is the tension in the connecting string? The pulley and all surfaces are frictionless.



96. A small space probe is released from a spaceship. The space probe has mass 20.0 kg and contains 90.0 kg of fuel. It starts from rest in deep space, from the origin of a coordinate system based on the spaceship, and burns fuel at the rate of 3.00 kg/s. The engine provides a constant thrust of 120.0 N. (a) Write an expression for the mass of the space probe as a function of time, between 0 and 30 seconds, assuming that the engine ignites fuel beginning at $t = 0$. (b) What is the velocity after 15.0 s? (c)

What is the position of the space probe after 15.0 s, with initial position at the origin? (d) Write an expression for the position as a function of time, for $t > 30.0$ s.

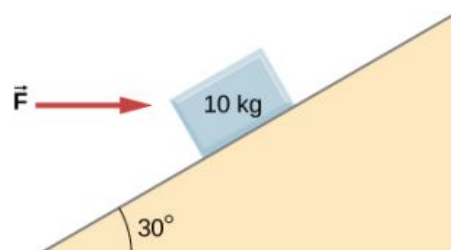
97. A half-full recycling bin has mass 3.0 kg and is pushed up a 40.0° incline with constant speed under the action of a 26-N force acting up and parallel to the incline. The incline has friction. What magnitude force must act up and parallel to the incline for the bin to move down the incline at constant velocity?
98. A child has mass 6.0 kg and slides down a 35° incline with constant speed under the action of a 13.4-N force acting up and parallel to the incline. What is the coefficient of kinetic friction between the child and the surface of the incline?
99. The two barges shown here are coupled by a cable of negligible mass. The mass of the front barge is 2.00×10^3 kg and the mass of the rear barge is 3.00×10^3 kg. A tugboat pulls the front barge with a horizontal force of magnitude 20.0×10^3 N, and the frictional forces of the water on the front and rear barges are 8.00×10^3 N and 10.0×10^3 N, respectively. Find the horizontal acceleration of the barges and the tension in the connecting cable.



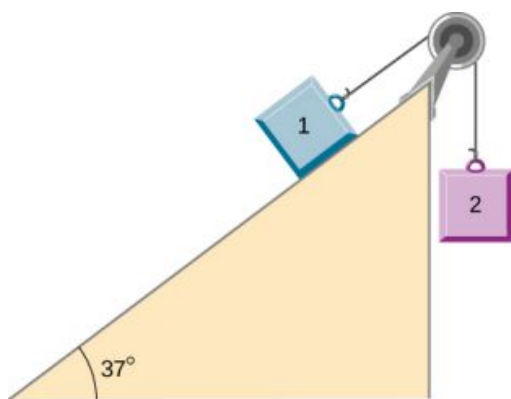
100. If the order of the barges of the preceding exercise is reversed so that the tugboat pulls the 3.00×10^3 -kg barge with a force of 20.0×10^3 N, what are the acceleration of the barges and the tension in the coupling cable?
101. An object with mass m moves along the x -axis. Its position at any time is given by $x(t) = pt^3 + qt^2$ where p and q are constants. Find the net force on this object for any time t .
102. A helicopter with mass 2.35×10^4 kg has a position given by $\vec{r}(t) = (0.020 t^3)\hat{i} + (2.2t)\hat{j} - (0.060 t^2)\hat{k}$. Find the net force on the helicopter at $t = 3.0$ s.
103. Located at the origin, an electric car of mass m is

at rest and in equilibrium. A time dependent force of $\vec{F}(t)$ is applied at time $t = 0$, and its components are $F_x(t) = p + nt$ and $F_y(t) = qt$ where p , q , and n are constants. Find the position $\vec{r}(t)$ and velocity $\vec{v}(t)$ as functions of time t .

104. A particle of mass m is located at the origin. It is at rest and in equilibrium. A time-dependent force of $\vec{F}(t)$ is applied at time $t = 0$, and its components are $F_x(t) = pt$ and $F_y(t) = n + qt$ where p , q , and n are constants. Find the position $\vec{r}(t)$ and velocity $\vec{v}(t)$ as functions of time t .
105. A 2.0-kg object has a velocity of $4.0\hat{i}$ m/s at $t = 0$. A constant resultant force of $(2.0\hat{i} + 4.0\hat{j})$ N then acts on the object for 3.0 s. What is the magnitude of the object's velocity at the end of the 3.0-s interval?
106. A 1.5-kg mass has an acceleration of $(4.0\hat{i} - 3.0\hat{j})$ m/s². Only two forces act on the mass. If one of the forces is $(2.0\hat{i} - 1.4\hat{j})$ N, what is the magnitude of the other force?
107. A box is dropped onto a conveyor belt moving at 3.4 m/s. If the coefficient of friction between the box and the belt is 0.27, how long will it take before the box moves without slipping?
108. Shown below is a 10.0-kg block being pushed by a horizontal force $\vec{F}$ of magnitude 200.0 N. The coefficient of kinetic friction between the two surfaces is 0.50. Find the acceleration of the block.

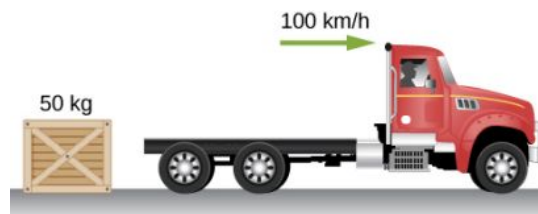


109. As shown below, the mass of block 1 is $m_1 = 4.0$ kg, while the mass of block 2 is $m_2 = 8.0$ kg. The coefficient of friction between m_1 and the inclined surface is $\mu_k = 0.40$. What is the acceleration of the system?



- 110.** A student is attempting to move a 30-kg mini-fridge into her dorm room. During a moment of inattention, the mini-fridge slides down a 35 degree incline at constant speed when she applies a force of 25 N acting up and parallel to the incline. What is the coefficient of kinetic friction between the fridge and the surface of the incline?
- 111.** A crate of mass 100.0 kg rests on a rough surface inclined at an angle of 37.0° with the horizontal. A massless rope to which a force can be applied parallel to the surface is attached to the crate and leads to the top of the incline. In its present state, the crate is just ready to slip and start to move down the plane. The coefficient of friction is 80% of that for the static case. (a) What is the coefficient of static friction? (b) What is the maximum force that can be applied upward along the plane on the rope and not move the block? (c) With a slightly greater applied force, the block will slide up the plane. Once it begins to move, what is its acceleration and what reduced force is necessary to keep it moving upward at constant speed? (d) If the block is given a slight nudge to get it started down the plane, what will be its acceleration in that direction? (e) Once the block begins to slide downward, what upward force on the rope is required to keep the block from accelerating downward?
- 112.** A car is moving at high speed along a highway when the driver makes an emergency braking. The wheels become locked (stop rolling), and the resulting skid marks are 32.0 meters long. If the coefficient of kinetic friction between tires and road is 0.550, and the acceleration was constant during braking, how fast was the car going when the wheels became locked?
- 113.** A crate having mass 50.0 kg falls horizontally off the

back of the flatbed truck, which is traveling at 100 km/h. Find the value of the coefficient of kinetic friction between the road and crate if the crate slides 50 m on the road in coming to rest. The initial speed of the crate is the same as the truck, 100 km/h.



- 114.** A 15-kg sled is pulled across a horizontal, snow-covered surface by a force applied to a rope at 30 degrees with the horizontal. The coefficient of kinetic friction between the sled and the snow is 0.20. (a) If the force is 33 N, what is the horizontal acceleration of the sled? (b) What must the force be in order to pull the sled at constant velocity?
- 115.** A 30.0-g ball at the end of a string is swung in a vertical circle with a radius of 25.0 cm. The tangential velocity is 200.0 cm/s. Find the tension in the string: (a) at the top of the circle, (b) at the bottom of the circle, and (c) at a distance of 12.5 cm from the center of the circle ($r = 12.5$ cm).
- 116.** A particle of mass 0.50 kg starts moves through a circular path in the xy -plane with a position given by $\vec{r}(t) = (4.0 \cos 3t)\hat{i} + (4.0 \sin 3t)\hat{j}$ where r is in meters and t is in seconds. (a) Find the velocity and acceleration vectors as functions of time. (b) Show that the acceleration vector always points toward the center of the circle (and thus represents centripetal acceleration). (c) Find the centripetal force vector as a function of time.
- 117.** A stunt cyclist rides on the interior of a cylinder 12 m in radius. The coefficient of static friction between the tires and the wall is 0.68. Find the value of the minimum speed for the cyclist to perform the stunt.
- 118.** When a body of mass 0.25 kg is attached to a vertical massless spring, it is extended 5.0 cm from its unstretched length of 4.0 cm. The body and spring are placed on a horizontal frictionless surface and rotated about the held end of the spring at 2.0 rev/s. How far is the spring stretched?

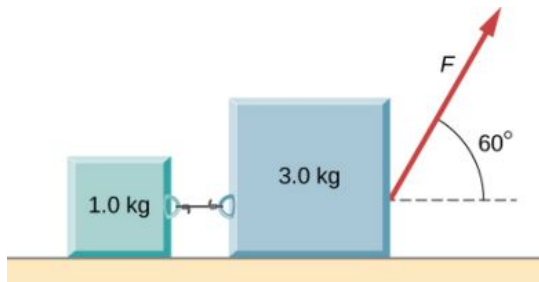
- 119.** A piece of bacon starts to slide down the pan when one side of a pan is raised up 5.0 cm. If the length of the pan from pivot to the raising point is 23.5 cm, what is the coefficient of static friction between the pan and the bacon?

- 120.** A plumb bob hangs from the roof of a railroad car. The car rounds a circular track of radius 300.0 m at a speed of 90.0 km/h. At what angle relative to the vertical does the plumb bob hang?

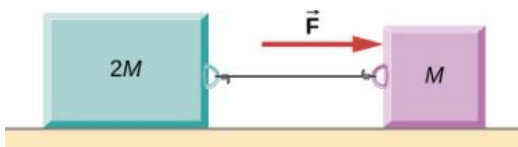
- 121.** An airplane flies at 120.0 m/s and banks at a 30° angle. If its mass is 2.50×10^3 kg, (a) what is the magnitude of the lift force? (b) what is the radius of the turn?

- 122.** The position of a particle is given by $\vec{r}(t) = A(\cos \omega t \hat{i} + \sin \omega t \hat{j})$, where ω is a constant. (a) Show that the particle moves in a circle of radius A . (b) Calculate $d\vec{r}/dt$ and then show that the speed of the particle is a constant $A\omega$. (c) Determine $d^2\vec{r}/dt^2$ and show that a is given by $a_c = r\omega^2$. (d) Calculate the centripetal force on the particle. [Hint: For (b) and (c), you will need to use $(d/dt)(\cos \omega t) = -\omega \sin \omega t$ and $(d/dt)(\sin \omega t) = \omega \cos \omega t$.]

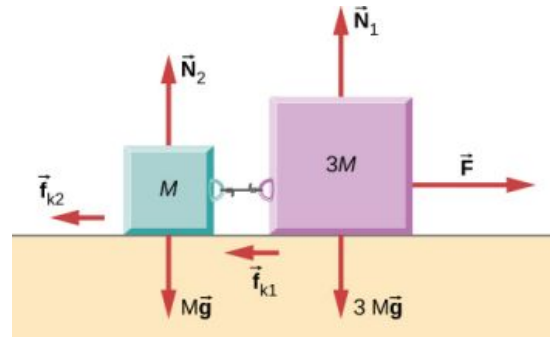
- 123.** Two blocks connected by a string are pulled across a horizontal surface by a force applied to one of the blocks, as shown below. The coefficient of kinetic friction between the blocks and the surface is 0.25. If each block has an acceleration of 2.0 m/s^2 to the right, what is the magnitude F of the applied force?



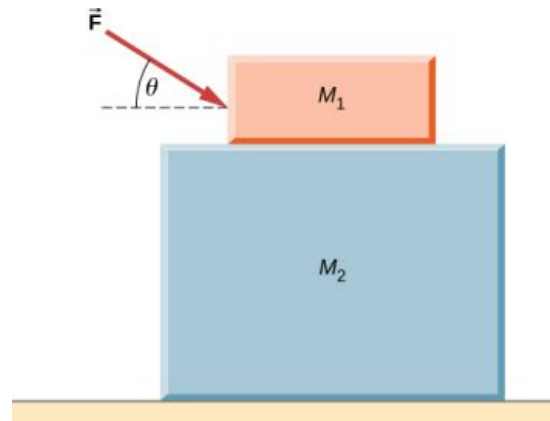
- 124.** As shown below, the coefficient of kinetic friction between the surface and the larger block is 0.20, and the coefficient of kinetic friction between the surface and the smaller block is 0.30. If $F = 10 \text{ N}$ and $M = 1.0 \text{ kg}$, what is the tension in the connecting string?



- 125.** In the figure, the coefficient of kinetic friction between the surface and the blocks is μ_k . If $M = 1.0 \text{ kg}$, find an expression for the magnitude of the acceleration of either block (in terms of F , μ_k , and g).

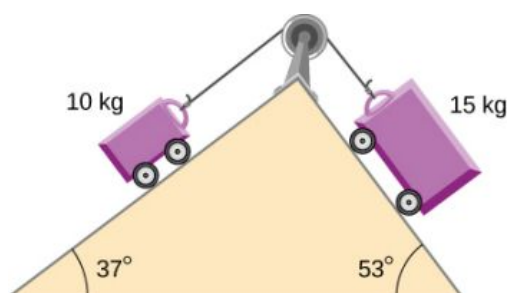


- 126.** Two blocks are stacked as shown below, and rest on a frictionless surface. There is friction between the two blocks (coefficient of friction μ). An external force is applied to the top block at an angle θ with the horizontal. What is the maximum force F that can be applied for the two blocks to move together?



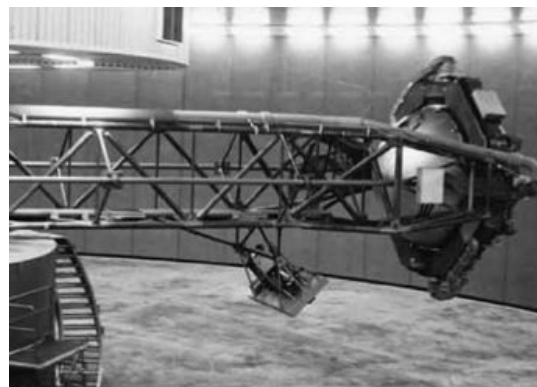
- 127.** A box rests on the (horizontal) back of a truck. The coefficient of static friction between the box and the surface on which it rests is 0.24. What maximum distance can the truck travel (starting from rest and moving horizontally with constant acceleration) in 3.0 s without having the box slide?

- 128.** A double-incline plane is shown below. The coefficient of friction on the left surface is 0.30, and on the right surface 0.16. Calculate the acceleration of the system.

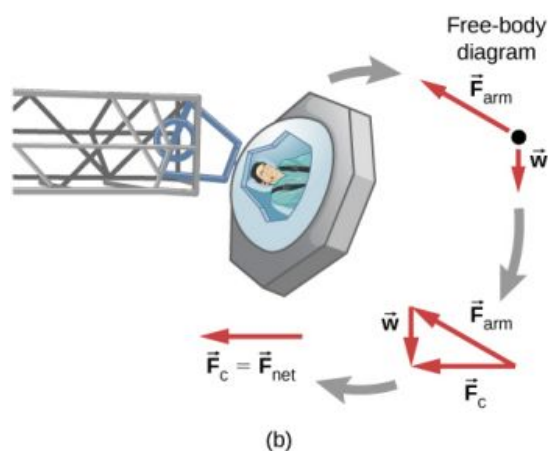


Challenge Problems

- 129.** In a later chapter, you will find that the weight of a particle varies with altitude such that $w = \frac{mgr_0^2}{r^2}$ where r_0 is the radius of Earth and r is the distance from Earth's center. If the particle is fired vertically with velocity v_0 from Earth's surface, determine its velocity as a function of position r . (*Hint: use $adr = vdv$, the rearrangement mentioned in the text.*)
- 130.** A large centrifuge, like the one shown below, is used to expose aspiring astronauts to accelerations similar to those experienced in rocket launches and atmospheric reentries. (a) At what angular velocity is the centripetal acceleration $10g$ if the rider is 15.0 m from the center of rotation? (b) The rider's cage hangs on a pivot at the end of the arm, allowing it to swing outward during rotation as shown in the bottom accompanying figure. At what angle θ below the horizontal will the cage hang when the centripetal acceleration is $10g$? (*Hint: The arm supplies centripetal force and supports the weight of the cage. Draw a free-body diagram of the forces to see what the angle θ should be.*)



(a)



- 131.** A car of mass 1000.0 kg is traveling along a level road at 100.0 km/h when its brakes are applied. Calculate the stopping distance if the coefficient of kinetic friction of the tires is 0.500 . Neglect air resistance. (*Hint: since the distance traveled is of interest rather than the time, x is the desired independent variable and not t . Use the Chain Rule to change the variable: $\frac{dv}{dt} = \frac{dv}{dx} \frac{dx}{dt} = v \frac{dv}{dx}$.)*
- 132.** An airplane flying at 200.0 m/s makes a turn that takes 4.0 min. What bank angle is required? What is the percentage increase in the perceived

weight of the passengers?

- 133.** A skydiver is at an altitude of 1520 m. After 10.0 seconds of free fall, he opens his parachute and finds that the air resistance, F_D , is given by the formula $F_D = -bv$, where b is a constant and v is the velocity. If $b = 7.50$, and the mass of the skydiver is 82.0 kg, first set up differential equations for the velocity and the position, and then find: (a) the speed of the skydiver when the parachute opens, (b) the distance fallen before the parachute opens, (c) the terminal velocity after the parachute opens (find the limiting velocity), and (d) the time the skydiver is in the air after the parachute opens.
- 134.** In a television commercial, a small, spherical bead of mass 4.00 g is released from rest at

$t = 0$ in a bottle of liquid shampoo. The terminal speed is observed to be 2.00 cm/s. Find (a) the value of the constant b in the equation $v = \frac{mg}{b}(1 - e^{-bt/m})$, and (b) the value of the resistive force when the bead reaches terminal speed.

- 135.** A boater and motor boat are at rest on a lake. Together, they have mass 200.0 kg. If the thrust of the motor is a constant force of 40.0 N in the direction of motion, and if the resistive force of the water is numerically equivalent to 2 times the speed v of the boat, set up and solve the differential equation to find: (a) the velocity of the boat at time t ; (b) the limiting velocity (the velocity after a long time has passed).

CHAPTER 7

Work and Kinetic Energy



FIGURE 7.1 A sprinter exerts her maximum power with the greatest force in the short time her foot is in contact with the ground. This adds to her kinetic energy, preventing her from slowing down during the race. Pushing back hard on the track generates a reaction force that propels the sprinter forward to win at the finish. (credit: modification of work by Marie-Lan Nguyen)

CHAPTER OUTLINE

7.1 Work

7.2 Kinetic Energy

7.3 Work-Energy Theorem

7.4 Power

INTRODUCTION In this chapter, we discuss some basic physical concepts involved in every physical motion in the universe, going beyond the concepts of force and change in motion, which we discussed in [Motion in Two and Three Dimensions](#) and [Newton's Laws of Motion](#). These concepts are work, kinetic energy, and power. We explain how these quantities are related to one another, which will lead us to a fundamental relationship called the work-energy theorem. In the next chapter, we generalize this idea to the broader principle of conservation of energy.

The application of Newton's laws usually requires solving differential equations that relate the forces acting on an object to the accelerations they produce. Often, an analytic solution is intractable or impossible, requiring lengthy numerical solutions or simulations to get approximate results. In such situations, more general relations, like the work-energy theorem (or the conservation of energy), can still provide useful answers to many questions and require a more modest amount of mathematical calculation. In particular, you will see how the work-energy theorem is useful in relating the speeds of a particle, at different points along its trajectory, to the forces acting on it, even when the trajectory is otherwise too complicated to deal with. Thus, some aspects of motion can be addressed with fewer equations and without vector decompositions.

7.1 Work

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Represent the work done by any force
- Evaluate the work done for various forces

In physics, **work** is done on an object when energy is transferred to the object. In other words, work is done when a force acts on something that undergoes a displacement from one position to another. Forces can vary as a function of position, and displacements can be along various paths between two points. We first define the increment of work dW done by a force $\vec{F}$ acting through an infinitesimal displacement $d\vec{r}$ as the dot product of these two vectors:

$$dW = \vec{F} \cdot d\vec{r} = |\vec{F}| |d\vec{r}| \cos \theta. \quad 7.1$$

Then, we can add up the contributions for infinitesimal displacements, along a path between two positions, to get the total work.

WORK DONE BY A FORCE

The work done by a force is the integral of the force with respect to displacement along the path of the displacement:

$$W_{AB} = \int_{\text{path } AB} \vec{F} \cdot d\vec{r}. \quad 7.2$$

The vectors involved in the definition of the work done by a force acting on a particle are illustrated in [Figure 7.2](#). While in general, [Equation 7.2](#) requires mathematics beyond the scope of this text, in many simple situations this integral becomes a familiar integral in one variable. We will examine several such examples and restrict our discussion to these cases.

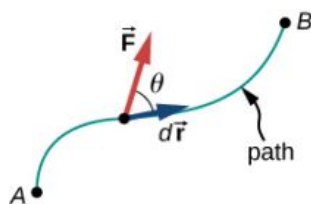


FIGURE 7.2 Vectors used to define work. The force acting on a particle and its infinitesimal displacement are shown at one point along the path between A and B. The infinitesimal work is the dot product of these two vectors; the total work is the integral of the dot product along the path.

We choose to express the dot product in terms of the magnitudes of the vectors and the cosine of the angle between them, because the meaning of the dot product for work can be put into words more directly in terms of magnitudes and angles. We could equally well have expressed the dot product in terms of the various components introduced in [Vectors](#). In two dimensions, these were the x - and y -components in Cartesian coordinates, or the r - and φ -components in polar coordinates; in three dimensions, it was just x -, y -, and z -components. Which choice is more convenient depends on the situation. In words, you can express [Equation 7.1](#) for the work done by a force acting over a displacement as a product of one component acting parallel to the other component. From the properties of vectors, it doesn't matter if you take the component of the force parallel to the displacement or the component of the displacement parallel to the force—you get the same result either way.

Recall that the magnitude of a force times the cosine of the angle the force makes with a given direction is the component of the force in the given direction. The components of a vector can be positive, negative, or zero, depending on whether the angle between the vector and the component-direction is between 0° and 90° or 90° and 180° , or is equal to 90° . As a result, the work done by a force can be positive, negative, or zero, depending on whether the force is generally in the direction of the displacement, generally opposite to the displacement, or

perpendicular to the displacement. The maximum work is done by a given force when it is along the direction of the displacement ($\cos \theta = \pm 1$), and zero work is done when the force is perpendicular to the displacement ($\cos \theta = 0$).

The units of work are units of force multiplied by units of length, which in the SI system is newtons times meters, $\text{N} \cdot \text{m}$. This combination is called a joule, for historical reasons that we will mention later, and is abbreviated as J. In the English system, still used in the United States, the unit of force is the pound (lb) and the unit of distance is the foot (ft), so the unit of work is the foot-pound ($\text{ft} \cdot \text{lb}$).

Work Done by Constant Forces and Contact Forces

The simplest work to evaluate is that done by a force that is constant in magnitude and direction. In this case, we can factor out the force; the remaining integral is just the total displacement, which only depends on the end points A and B , but not on the path between them:

$$W_{AB} = \vec{\mathbf{F}} \cdot \int_A^B d\vec{\mathbf{r}} = \vec{\mathbf{F}} \cdot (\vec{\mathbf{r}}_B - \vec{\mathbf{r}}_A) = |\vec{\mathbf{F}}| |\vec{\mathbf{r}}_B - \vec{\mathbf{r}}_A| \cos \theta \quad (\text{constant force}).$$

Figure 7.3(a) shows a person exerting a constant force $\vec{\mathbf{F}}$ along the handle of a lawn mower, which makes an angle θ with the horizontal. The horizontal displacement of the lawn mower, over which the force acts, is $\vec{\mathbf{d}}$. The work done on the lawn mower is $W = \vec{\mathbf{F}} \cdot \vec{\mathbf{d}} = Fd \cos \theta$, which the figure also illustrates as the horizontal component of the force times the magnitude of the displacement.

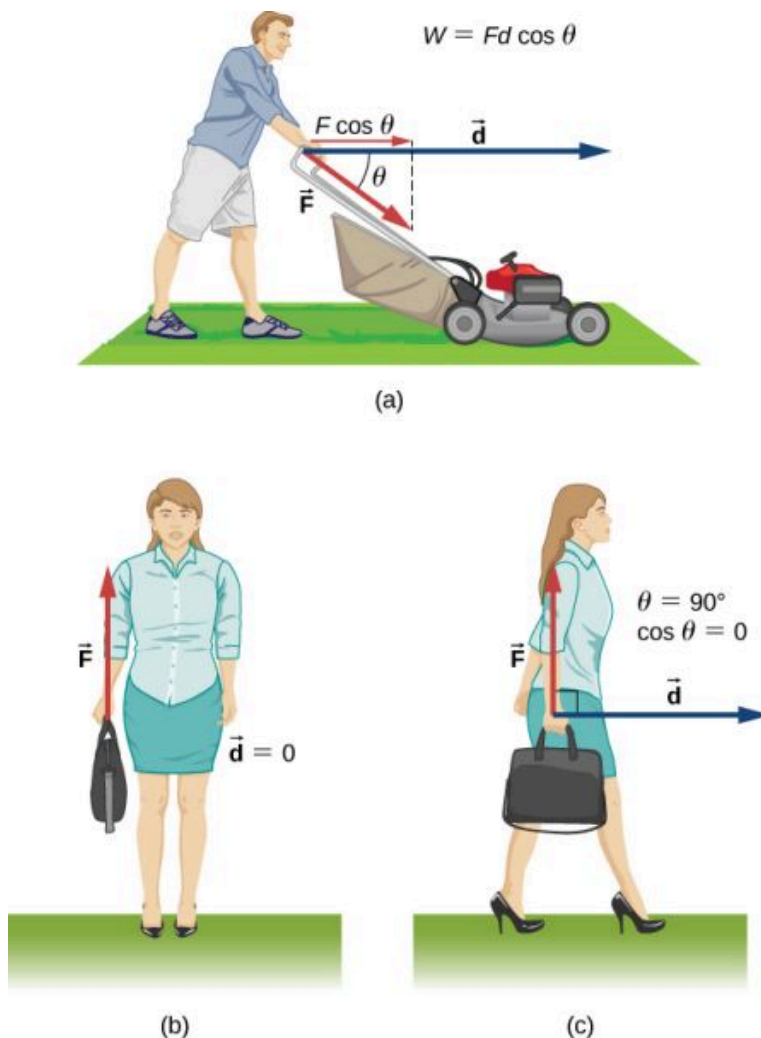


FIGURE 7.3 Work done by a constant force. (a) A person pushes a lawn mower with a constant force. The component of the force parallel to the displacement is the work done, as shown in the equation in the figure. (b) A person holds a briefcase. No work is done because the displacement is zero. (c) The person in (b) walks horizontally while holding the briefcase. No work is done because $\cos \theta$ is zero.

Figure 7.3(b) shows a person holding a briefcase. The person must exert an upward force, equal in magnitude to the weight of the briefcase, but this force does no work, because the displacement over which it acts is zero.

In Figure 7.3(c), where the person in (b) is walking horizontally with constant speed, the work done by the person on the briefcase is still zero, but now because the angle between the force exerted and the displacement is 90° ($\vec{F}$ perpendicular to $\vec{d}$) and $\cos 90^\circ = 0$.



EXAMPLE 7.1

Calculating the Work You Do to Push a Lawn Mower

How much work is done on the lawn mower by the person in Figure 7.3(a) if he exerts a constant force of 75.0 N at an angle 35° below the horizontal and pushes the mower 25.0 m on level ground?

Strategy

We can solve this problem by substituting the given values into the definition of work done on an object by a constant force, stated in the equation $W = Fd \cos \theta$. The force, angle, and displacement are given, so that only the work W is unknown.

Solution

The equation for the work is

$$W = Fd \cos \theta.$$

Substituting the known values gives

$$W = (75.0 \text{ N})(25.0 \text{ m})\cos(35.0^\circ) = 1.54 \times 10^3 \text{ J}.$$

Significance

Even though one and a half kilojoules may seem like a lot of work, we will see in [Potential Energy and Conservation of Energy](#) that it's only about as much work as you could do by burning one sixth of a gram of fat.

When you mow the grass, other forces act on the lawn mower besides the force you exert—namely, the contact force of the ground and the gravitational force of Earth. Let's consider the work done by these forces in general. For an object moving on a surface, the displacement $d\vec{r}$ is tangent to the surface. The part of the contact force on the object that is perpendicular to the surface is the normal force $\vec{N}$. Since the cosine of the angle between the normal and the tangent to a surface is zero, we have

$$dW_N = \vec{N} \cdot d\vec{r} = 0.$$

The normal force never does work under these circumstances. (Note that if the displacement $d\vec{r}$ did have a relative component perpendicular to the surface, the object would either leave the surface or break through it, and there would no longer be any normal contact force. However, if the object is more than a particle, and has an internal structure, the normal contact force can do work on it, for example, by displacing it or deforming its shape. This will be mentioned in the next chapter.)

The part of the contact force on the object that is parallel to the surface is friction, $\vec{f}$. For this object sliding along the surface, kinetic friction $\vec{f}_k$ is opposite to $d\vec{r}$, relative to the surface, so the work done by kinetic friction is negative. If the magnitude of $\vec{f}_k$ is constant (as it would be if all the other forces on the object were constant), then the work done by friction is

$$W_{\text{fr}} = \int_A^B \vec{f}_k \cdot d\vec{r} = -f_k \int_A^B |dr| = -f_k |l_{AB}|, \quad 7.3$$

where $|l_{AB}|$ is the path length on the surface. The force of static friction does no work in the reference frame between two surfaces because there is never displacement between the surfaces. As an external force, static

friction can do work. Static friction can keep someone from sliding off a sled when the sled is moving and perform positive work on the person. If you're driving your car at the speed limit on a straight, level stretch of highway, the negative work done by air resistance is balanced by the positive work done by the static friction of the road on the drive wheels. You can pull the rug out from under an object in such a way that it slides backward relative to the rug, but forward relative to the floor. In this case, kinetic friction exerted by the rug on the object could be in the same direction as the displacement of the object, relative to the floor, and do positive work. The bottom line is that you need to analyze each particular case to determine the work done by the forces, whether positive, negative or zero.

EXAMPLE 7.2

Moving a Couch

You decide to move your couch to a new position on your horizontal living room floor. The normal force on the couch is 1 kN and the coefficient of friction is 0.6. (a) You first push the couch 3 m parallel to a wall and then 1 m perpendicular to the wall (A to B in Figure 7.4). How much work is done by the frictional force? (b) You don't like the new position, so you move the couch straight back to its original position (B to A in Figure 7.4). What was the total work done against friction moving the couch away from its original position and back again?

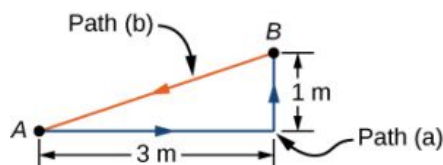


FIGURE 7.4 Top view of paths for moving a couch.

Strategy

The magnitude of the force of kinetic friction on the couch is constant, equal to the coefficient of friction times the normal force, $f_K = \mu_K N$. Therefore, the work done by it is $W_{fr} = -f_K d$, where d is the path length traversed. The segments of the paths are the sides of a right triangle, so the path lengths are easily calculated. In part (b), you can use the fact that the work done against a force is the negative of the work done by the force.

Solution

- a. The work done by friction is

$$W = -(0.6)(1 \text{ kN})(3 \text{ m} + 1 \text{ m}) = -2.4 \text{ kJ}.$$

- b. The length of the path along the hypotenuse is $\sqrt{10} \text{ m}$, so the total work done against friction is

$$W = (0.6)(1 \text{ kN})(3 \text{ m} + 1 \text{ m} + \sqrt{10} \text{ m}) = 4.3 \text{ kJ}.$$

Significance

The total path over which the work of friction was evaluated began and ended at the same point (it was a closed path), so that the total displacement of the couch was zero. However, the total work was not zero. The reason is that forces like friction are classified as nonconservative forces, or dissipative forces, as we discuss in the next chapter.

CHECK YOUR UNDERSTANDING 7.1

Can kinetic friction ever be a constant force for all paths?

The other force on the lawn mower mentioned above was Earth's gravitational force, or the weight of the mower. Near the surface of Earth, the gravitational force on an object of mass m has a constant magnitude, mg , and constant direction, vertically down. Therefore, the work done by gravity on an object is the dot product of its weight and its displacement. In many cases, it is convenient to express the dot product for gravitational work in terms of the x -, y -, and z -components of the vectors. A typical coordinate system has the x -axis horizontal and the y -axis vertically up. Then the gravitational force is $-mg\hat{j}$, so the work done by gravity, over any path from A to B, is

$$W_{\text{grav},AB} = -mg\hat{j} \cdot (\vec{r}_B - \vec{r}_A) = -mg(y_B - y_A).$$

7.4

The work done by a constant force of gravity on an object depends only on the object's weight and the difference in height through which the object is displaced. Gravity does negative work on an object that moves upward ($y_B > y_A$), or, in other words, you must do positive work against gravity to lift an object upward. Alternately, gravity does positive work on an object that moves downward ($y_B < y_A$), or you do negative work against gravity to “lift” an object downward, controlling its descent so it doesn't drop to the ground. (“Lift” is used as opposed to “drop”.)

EXAMPLE 7.3

Shelving a Book

You lift an oversized library book, weighing 20 N, 1 m vertically down from a shelf, and carry it 3 m horizontally to a table (Figure 7.5). (a) How much work does gravity do on the book? (b) When you're finished, you move the book in a straight line back to its original place on the shelf. What was the total work done against gravity, moving the book away from its original position on the shelf and back again?

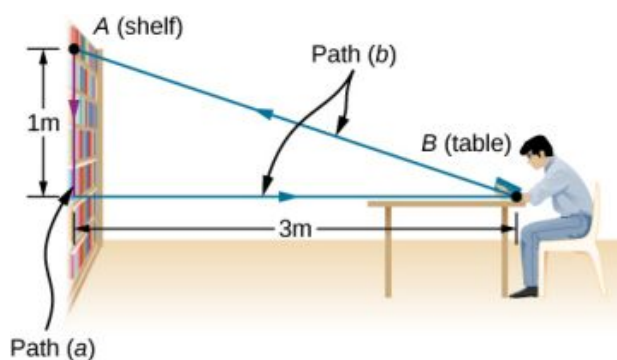


FIGURE 7.5 Side view of the paths for moving a book to and from a shelf.

Strategy

We have just seen that the work done by a constant force of gravity depends only on the weight of the object moved and the difference in height for the path taken, $W_{AB} = -mg(y_B - y_A)$. We can evaluate the difference in height to answer (a) and (b).

Solution

- Since the book starts on the shelf and is lifted down $y_B - y_A = -1$ m, we have

$$W = -(20 \text{ N})(-1 \text{ m}) = 20 \text{ J}.$$
- There is zero difference in height for any path that begins and ends at the same place on the shelf, so $W = 0$.

Significance

Gravity does positive work (20 J) when the book moves down from the shelf. The gravitational force between two objects is an attractive force, which does positive work when the objects get closer together. Gravity does zero work (0 J) when the book moves horizontally from the shelf to the table and negative work (−20 J) when the book moves from the table back to the shelf. The total work done by gravity is zero $[20 \text{ J} + 0 \text{ J} + (-20 \text{ J}) = 0]$. Unlike friction or other dissipative forces, described in Example 7.2, the total work done against gravity, over any closed path, is zero. Positive work is done against gravity on the upward parts of a closed path, but an equal amount of negative work is done against gravity on the downward parts. In other words, work done *against* gravity, lifting an object *up*, is “given back” when the object comes back down. Forces like gravity (those that do zero work over any closed path) are classified as conservative forces and play an important role in physics.

✓ CHECK YOUR UNDERSTANDING 7.2

Can Earth's gravity ever be a constant force for all paths?

Work Done by Forces that Vary

In general, forces may vary in magnitude and direction at points in space, and paths between two points may be curved. The infinitesimal work done by a variable force can be expressed in terms of the components of the force and the displacement along the path,

$$dW = F_x dx + F_y dy + F_z dz.$$

Here, the components of the force are functions of position along the path, and the displacements depend on the equations of the path. (Although we chose to illustrate dW in Cartesian coordinates, other coordinates are better suited to some situations.) Equation 7.2 defines the total work as a line integral, or the limit of a sum of infinitesimal amounts of work. The physical concept of work is straightforward: you calculate the work for tiny displacements and add them up. Sometimes the mathematics can seem complicated, but the following example demonstrates how cleanly they can operate.



EXAMPLE 7.4

Work Done by a Variable Force over a Curved Path

An object moves along a parabolic path $y = (0.5 \text{ m}^{-1})x^2$ from the origin $A = (0, 0)$ to the point $B = (2 \text{ m}, 2 \text{ m})$ under the action of a force $\vec{F} = (5 \text{ N/m})y\hat{i} + (10 \text{ N/m})x\hat{j}$ (Figure 7.6). Calculate the work done.

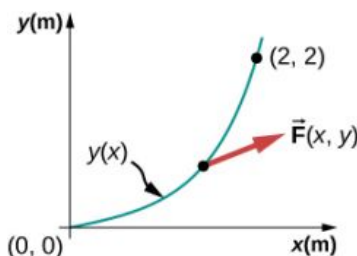


FIGURE 7.6 The parabolic path of a particle acted on by a given force.

Strategy

The components of the force are given functions of x and y . We can use the equation of the path to express y and dy in terms of x and dx ; namely,

$$y = (0.5 \text{ m}^{-1})x^2 \text{ and } dy = 2(0.5 \text{ m}^{-1})x dx.$$

Then, the integral for the work is just a definite integral of a function of x .

Solution

The infinitesimal element of work is

$$\begin{aligned} dW &= F_x dx + F_y dy = (5 \text{ N/m})y dx + (10 \text{ N/m})x dy \\ &= (5 \text{ N/m})(0.5 \text{ m}^{-1})x^2 dx + (10 \text{ N/m})2(0.5 \text{ m}^{-1})x^2 dx = (12.5 \text{ N/m}^2)x^2 dx. \end{aligned}$$

The integral of x^2 is $x^3/3$, so

$$W = \int_0^{2 \text{ m}} (12.5 \text{ N/m}^2)x^2 dx = (12.5 \text{ N/m}^2) \frac{x^3}{3} \bigg|_0^{2 \text{ m}} = (12.5 \text{ N/m}^2) \left(\frac{8}{3} \right) = 33.3 \text{ J}.$$

Significance

This integral was not hard to do. You can follow the same steps, as in this example, to calculate line integrals

representing work for more complicated forces and paths. In this example, everything was given in terms of x - and y -components, which are easiest to use in evaluating the work in this case. In other situations, magnitudes and angles might be easier.

✓ CHECK YOUR UNDERSTANDING 7.3

Find the work done by the same force in [Example 7.4](#) over a cubic path, $y = (0.25 \text{ m}^{-2})x^3$, between the same points $A = (0, 0)$ and $B = (2 \text{ m}, 2 \text{ m})$.

One very important and widely applicable variable force is the force exerted by a perfectly elastic spring, which satisfies Hooke's law $\vec{F} = -k\Delta\vec{x}$, where k is the spring constant, and $\Delta\vec{x} = \vec{x} - \vec{x}_{\text{eq}}$ is the displacement from the spring's unstretched (equilibrium) position ([Newton's Laws of Motion](#)). Note that the unstretched position is only the same as the equilibrium position if no other forces are acting (or, if they are, they cancel one another). Forces between molecules, or in any system undergoing small displacements from a stable equilibrium, behave approximately like a spring force.

To calculate the work done by a spring force, we can choose the x -axis along the length of the spring, in the direction of increasing length, as in [Figure 7.7](#), with the origin at the equilibrium position $x_{\text{eq}} = 0$. (Then positive x corresponds to a stretch and negative x to a compression.) With this choice of coordinates, the spring force has only an x -component, $F_x = -kx$, and the work done when x changes from x_A to x_B is

$$W_{\text{spring}, AB} = \int_A^B F_x dx = -k \int_A^B x dx = -k \left. \frac{x^2}{2} \right|_A^B = -\frac{1}{2}k(x_B^2 - x_A^2). \quad 7.5$$

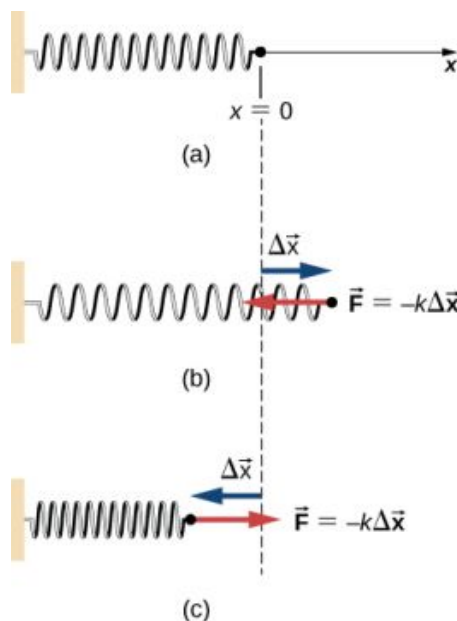


FIGURE 7.7 (a) The spring exerts no force at its equilibrium position. The spring exerts a force in the opposite direction to (b) an extension or stretch, and (c) a compression.

Notice that W_{AB} depends only on the starting and ending points, A and B , and is independent of the actual path between them, as long as it starts at A and ends at B . That is, the actual path could involve going back and forth before ending.

Another interesting thing to notice about [Equation 7.5](#) is that, for this one-dimensional case, you can readily see the correspondence between the work done by a force and the area under the curve of the force versus its displacement. Recall that, in general, a one-dimensional integral is the limit of the sum of infinitesimals, $f(x)dx$, representing the area of strips, as shown in [Figure 7.8](#). In [Equation 7.5](#), since $F = -kx$ is a straight line with slope

$-k$, when plotted versus x , the “area” under the line is just an algebraic combination of triangular “areas,” where “areas” above the x -axis are positive and those below are negative, as shown in Figure 7.9. The magnitude of one of these “areas” is just one-half the triangle’s base, along the x -axis, times the triangle’s height, along the force axis. (There are quotation marks around “area” because this base-height product has the units of work, rather than square meters.)

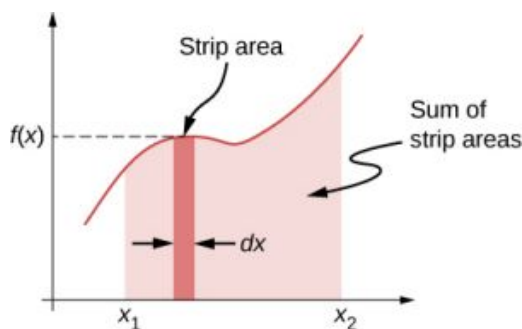


FIGURE 7.8 A curve of $f(x)$ versus x showing the area of an infinitesimal strip, $f(x)dx$, and the sum of such areas, which is the integral of $f(x)$ from x_1 to x_2 .

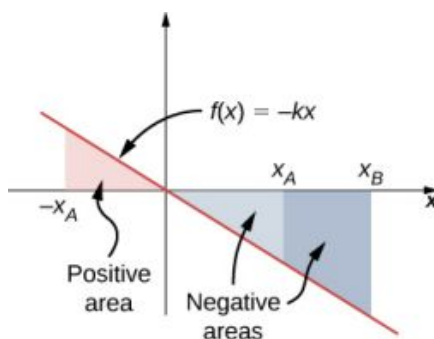


FIGURE 7.9 Curve of the spring force $f(x) = -kx$ versus x , showing areas under the line, between x_A and x_B , for both positive and negative values of x_A . When x_A is negative, the total area under the curve for the integral in Equation 7.5 is the sum of positive and negative triangular areas. When x_A is positive, the total area under the curve is the difference between two negative triangles.

EXAMPLE 7.5

Work Done by a Spring Force

A perfectly elastic spring requires 0.54 J of work to stretch 6 cm from its equilibrium position, as in Figure 7.7(b). (a) What is its spring constant k ? (b) How much work is required to stretch it an additional 6 cm?

Strategy

Work “required” means work done against the spring force, which is the negative of the work in Equation 7.5, that is

$$W = \frac{1}{2}k(x_B^2 - x_A^2).$$

For part (a), $x_A = 0$ and $x_B = 6\text{ cm}$; for part (b), $x_A = 6\text{ cm}$ and $x_B = 12\text{ cm}$. In part (a), the work is given and you can solve for the spring constant; in part (b), you can use the value of k , from part (a), to solve for the work.

Solution

- $W = 0.54\text{ J} = \frac{1}{2}k[(6\text{ cm})^2 - 0]$, so $k = 3\text{ N/cm}$.
- $W = \frac{1}{2}(3\text{ N/cm})[(12\text{ cm})^2 - (6\text{ cm})^2] = 1.62\text{ J}$.

Significance

Since the work done by a spring force is independent of the path, you only needed to calculate the difference in the quantity $\frac{1}{2}kx^2$ at the end points. Notice that the work required to stretch the spring from 0 to 12 cm is four times that required to stretch it from 0 to 6 cm, because that work depends on the square of the amount of stretch from

equilibrium, $\frac{1}{2}kx^2$. In this circumstance, the work to stretch the spring from 0 to 12 cm is also equal to the work for a composite path from 0 to 6 cm followed by an additional stretch from 6 cm to 12 cm. Therefore, $4W(0 \text{ cm to } 6 \text{ cm}) = W(0 \text{ cm to } 6 \text{ cm}) + W(6 \text{ cm to } 12 \text{ cm})$, or $W(6 \text{ cm to } 12 \text{ cm}) = 3W(0 \text{ cm to } 6 \text{ cm})$, as we found above.

✓ CHECK YOUR UNDERSTANDING 7.4

The spring in [Example 7.5](#) is compressed 6 cm from its equilibrium length. (a) Does the spring force do positive or negative work and (b) what is the magnitude?

7.2 Kinetic Energy

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Calculate the kinetic energy of a particle given its mass and its velocity or momentum
- Evaluate the kinetic energy of a body, relative to different frames of reference

It's plausible to suppose that the greater the velocity of a body, the greater effect it could have on other bodies. This does not depend on the direction of the velocity, only its magnitude. At the end of the seventeenth century, a quantity was introduced into mechanics to explain collisions between two perfectly elastic bodies, in which one body makes a head-on collision with an identical body at rest. The first body stops, and the second body moves off with the initial velocity of the first body. (If you have ever played billiards or croquet, or seen a model of Newton's Cradle, you have observed this type of collision.) The idea behind this quantity was related to the forces acting on a body and was referred to as "the energy of motion." Later on, during the eighteenth century, the name **kinetic energy** was given to energy of motion.

With this history in mind, we can now state the classical definition of kinetic energy. Note that when we say "classical," we mean non-relativistic, that is, at speeds much less than the speed of light. At speeds comparable to the speed of light, the special theory of relativity requires a different expression for the kinetic energy of a particle, as discussed in [Relativity \(http://openstax.org/books/university-physics-volume-3/pages/5-introduction\)](http://openstax.org/books/university-physics-volume-3/pages/5-introduction).

Since objects (or systems) of interest vary in complexity, we first define the kinetic energy of a particle with mass m .

KINETIC ENERGY

The kinetic energy of a particle is one-half the product of the particle's mass m and the square of its speed v :

$$K = \frac{1}{2}mv^2. \quad 7.6$$

We then extend this definition to any system of particles by adding up the kinetic energies of all the constituent particles:

$$K = \sum \frac{1}{2}mv^2. \quad 7.7$$

Note that just as we can express Newton's second law in terms of either the rate of change of momentum or mass times the rate of change of velocity, so the kinetic energy of a particle can be expressed in terms of its mass and momentum ($\vec{p} = m\vec{v}$), instead of its mass and velocity. Since $v = p/m$, we see that

$$K = \frac{1}{2}m\left(\frac{p}{m}\right)^2 = \frac{p^2}{2m}$$

also expresses the kinetic energy of a single particle. Sometimes, this expression is more convenient to use than [Equation 7.6](#).

The units of kinetic energy are mass times the square of speed, or $\text{kg} \cdot \text{m}^2/\text{s}^2$. But the units of force are mass times

acceleration, $\text{kg} \cdot \text{m/s}^2$, so the units of kinetic energy are also the units of force times distance, which are the units of work, or joules. You will see in the next section that work and kinetic energy have the same units, because they are different forms of the same, more general, physical property.



EXAMPLE 7.6

Kinetic Energy of an Object

(a) What is the kinetic energy of an 80-kg athlete, running at 10 m/s? (b) The Chicxulub crater in Yucatan, one of the largest existing impact craters on Earth, is thought to have been created by an asteroid, traveling at 22 km/s and releasing 4.2×10^{23} J of kinetic energy upon impact. What was its mass? (c) In nuclear reactors, thermal neutrons, traveling at about 2.2 km/s, play an important role. What is the kinetic energy of such a particle?

Strategy

To answer these questions, you can use the definition of kinetic energy in [Equation 7.6](#). You also have to look up the mass of a neutron.

Solution

Don't forget to convert km into m to do these calculations, although, to save space, we omitted showing these conversions.

- $K = \frac{1}{2}(80 \text{ kg})(10 \text{ m/s})^2 = 4.0 \text{ kJ}.$
- $m = 2K/v^2 = 2(4.2 \times 10^{23} \text{ J})/(22 \text{ km/s})^2 = 1.7 \times 10^{15} \text{ kg}.$
- $K = \frac{1}{2}(1.68 \times 10^{-27} \text{ kg})(2.2 \text{ km/s})^2 = 4.1 \times 10^{-21} \text{ J}.$

Significance

In this example, we used the way mass and speed are related to kinetic energy, and we encountered a very wide range of values for the kinetic energies. Different units are commonly used for such very large and very small values. The energy of the impactor in part (b) can be compared to the explosive yield of TNT and nuclear explosions, 1 megaton = 4.18×10^{15} J. The Chicxulub asteroid's kinetic energy was about a hundred million megatons. At the other extreme, the energy of subatomic particle is expressed in electron-volts, $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$. The thermal neutron in part (c) has a kinetic energy of about one fortieth of an electron-volt.

CHECK YOUR UNDERSTANDING 7.5

(a) A car and a truck are each moving with the same kinetic energy. Assume that the truck has more mass than the car. Which has the greater speed? (b) A car and a truck are each moving with the same speed. Which has the greater kinetic energy?

Because velocity is a relative quantity, you can see that the value of kinetic energy must depend on your frame of reference. You can generally choose a frame of reference that is suited to the purpose of your analysis and that simplifies your calculations. One such frame of reference is the one in which the observations of the system are made (likely an external frame). Another choice is a frame that is attached to, or moves with, the system (likely an internal frame). The equations for relative motion, discussed in [Motion in Two and Three Dimensions](#), provide a link to calculating the kinetic energy of an object with respect to different frames of reference.



EXAMPLE 7.7

Kinetic Energy Relative to Different Frames

A 75.0-kg person walks down the central aisle of a subway car at a speed of 1.50 m/s relative to the car, whereas the train is moving at 15.0 m/s relative to the tracks. (a) What is the person's kinetic energy relative to the car? (b) What is the person's kinetic energy relative to the tracks? (c) What is the person's kinetic energy relative to a frame

moving with the person?

Strategy

Since speeds are given, we can use $\frac{1}{2}mv^2$ to calculate the person's kinetic energy. However, in part (a), the person's speed is relative to the subway car (as given); in part (b), it is relative to the tracks; and in part (c), it is zero. If we denote the car frame by C, the track frame by T, and the person by P, the relative velocities in part (b) are related by $\vec{v}_{PT} = \vec{v}_{PC} + \vec{v}_{CT}$. We can assume that the central aisle and the tracks lie along the same line, but the direction the person is walking relative to the car isn't specified, so we will give an answer for each possibility, $v_{PT} = v_{CT} \pm v_{PC}$, as shown in [Figure 7.10](#).

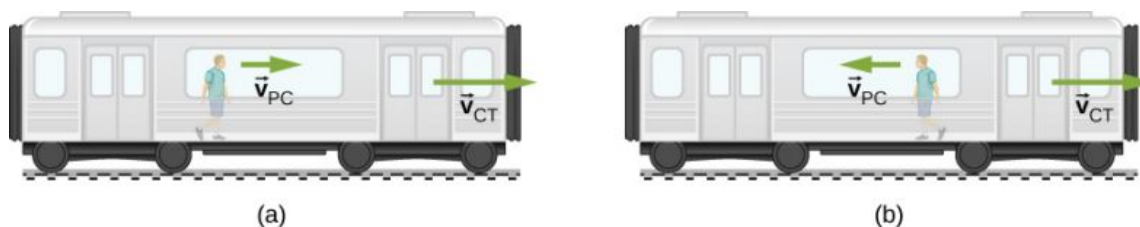


FIGURE 7.10 The possible motions of a person walking in a train are (a) toward the front of the car and (b) toward the back of the car.

Solution

a. $K = \frac{1}{2}(75.0 \text{ kg})(1.50 \text{ m/s})^2 = 84.4 \text{ J}.$

b. $v_{PT} = (15.0 \pm 1.50) \text{ m/s}.$ Therefore, the two possible values for kinetic energy relative to the track are

$$K = \frac{1}{2}(75.0 \text{ kg})(13.5 \text{ m/s})^2 = 6.83 \text{ kJ}$$

and

$$K = \frac{1}{2}(75.0 \text{ kg})(16.5 \text{ m/s})^2 = 10.2 \text{ kJ}.$$

c. In a frame where $v_P = 0$, $K = 0$ as well.

Significance

You can see that the kinetic energy of an object can have very different values, depending on the frame of reference. However, the kinetic energy of an object can never be negative, since it is the product of the mass and the square of the speed, both of which are always positive or zero.

✓ CHECK YOUR UNDERSTANDING 7.6

You are rowing a boat parallel to the banks of a river. Your kinetic energy relative to the banks is less than your kinetic energy relative to the water. Are you rowing with or against the current?

The kinetic energy of a particle is a single quantity, but the kinetic energy of a system of particles can sometimes be divided into various types, depending on the system and its motion. For example, if all the particles in a system have the same velocity, the system is undergoing translational motion and has translational kinetic energy. If an object is rotating, it could have rotational kinetic energy, or if it's vibrating, it could have vibrational kinetic energy. The kinetic energy of a system, relative to an internal frame of reference, may be called internal kinetic energy. The kinetic energy associated with random molecular motion may be called thermal energy. These names will be used in later chapters of the book, when appropriate. Regardless of the name, every kind of kinetic energy is the same physical quantity, representing energy associated with motion.



EXAMPLE 7.8

Special Names for Kinetic Energy

(a) A player lobs a mid-court pass with a 624-g basketball, which covers 15 m in 2 s. What is the basketball's horizontal translational kinetic energy while in flight? (b) An average molecule of air, in the basketball in part (a), has a mass of 29 u, and an average speed of 500 m/s, relative to the basketball. There are about 3×10^{23} molecules inside it, moving in random directions, when the ball is properly inflated. What is the average translational kinetic energy of the random motion of all the molecules inside, relative to the basketball? (c) How fast would the basketball have to travel relative to the court, as in part (a), so as to have a kinetic energy equal to the amount in part (b)?

Strategy

In part (a), first find the horizontal speed of the basketball and then use the definition of kinetic energy in terms of mass and speed, $K = \frac{1}{2}mv^2$. Then in part (b), convert unified units to kilograms and then use $K = \frac{1}{2}mv^2$ to get the average translational kinetic energy of one molecule, relative to the basketball. Then multiply by the number of molecules to get the total result. Finally, in part (c), we can substitute the amount of kinetic energy in part (b), and the mass of the basketball in part (a), into the definition $K = \frac{1}{2}mv^2$, and solve for v .

Solution

- a. The horizontal speed is (15 m)/(2 s), so the horizontal kinetic energy of the basketball is

$$\frac{1}{2}(0.624 \text{ kg})(7.5 \text{ m/s})^2 = 17.6 \text{ J}.$$

- b. The average translational kinetic energy of a molecule is

$$\frac{1}{2}(29 \text{ u})(1.66 \times 10^{-27} \text{ kg/u})(500 \text{ m/s})^2 = 6.02 \times 10^{-21} \text{ J},$$

and the total kinetic energy of all the molecules is

$$(3 \times 10^{23})(6.02 \times 10^{-21} \text{ J}) = 1.80 \text{ kJ}.$$

- c. $v = \sqrt{2(1.8 \text{ kJ})/(0.624 \text{ kg})} = 76.0 \text{ m/s}.$

Significance

In part (a), this kind of kinetic energy can be called the horizontal kinetic energy of an object (the basketball), relative to its surroundings (the court). If the basketball were spinning, all parts of it would have not just the average speed, but it would also have rotational kinetic energy. Part (b) reminds us that this kind of kinetic energy can be called internal or thermal kinetic energy. Notice that this energy is about a hundred times the energy in part (a). How to make use of thermal energy will be the subject of the chapters on thermodynamics. In part (c), since the energy in part (b) is about 100 times that in part (a), the speed should be about 10 times as big, which it is (76 compared to 7.5 m/s).

7.3 Work-Energy Theorem

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Apply the work-energy theorem to find information about the motion of a particle, given the forces acting on it
- Use the work-energy theorem to find information about the forces acting on a particle, given information about its motion

We have discussed how to find the work done on a particle by the forces that act on it, but how is that work manifested in the motion of the particle? According to Newton's second law of motion, the sum of all the forces acting on a particle, or the net force, determines the rate of change in the momentum of the particle, or its motion. Therefore, we should consider the work done by all the forces acting on a particle, or the **net work**, to see what

effect it has on the particle's motion.

Let's start by looking at the net work done on a particle as it moves over an infinitesimal displacement, which is the dot product of the net force and the displacement: $dW_{\text{net}} = \vec{F}_{\text{net}} \cdot d\vec{r}$. Newton's second law tells us that $\vec{F}_{\text{net}} = m(d\vec{v}/dt)$, so $dW_{\text{net}} = m(d\vec{v}/dt) \cdot d\vec{r}$. For the mathematical functions describing the motion of a physical particle, we can rearrange the differentials dt , etc., as algebraic quantities in this expression, that is,

$$dW_{\text{net}} = m \left(\frac{d\vec{v}}{dt} \right) \cdot d\vec{r} = m d\vec{v} \cdot \left(\frac{d\vec{r}}{dt} \right) = m \vec{v} \cdot d\vec{v},$$

where we substituted the velocity for the time derivative of the displacement and used the commutative property of the dot product [Equation 2.30]. Since derivatives and integrals of scalars are probably more familiar to you at this point, we express the dot product in terms of Cartesian coordinates before we integrate between any two points A and B on the particle's trajectory. This gives us the net work done on the particle:

$$\begin{aligned} W_{\text{net},AB} &= \int_A^B (mv_x dv_x + mv_y dv_y + mv_z dv_z) \\ &= \frac{1}{2} m |v_x^2 + v_y^2 + v_z^2|_A^B = \left| \frac{1}{2} mv^2 \right|_A^B = K_B - K_A. \end{aligned} \quad 7.8$$

In the middle step, we used the fact that the square of the velocity is the sum of the squares of its Cartesian components, and in the last step, we used the definition of the particle's kinetic energy. This important result is called the **work-energy theorem** (Figure 7.11).

WORK-ENERGY THEOREM

The net work done on a particle equals the change in the particle's kinetic energy:

$$W_{\text{net}} = K_B - K_A. \quad 7.9$$



FIGURE 7.11 Horse pulls are common events at state fairs. The work done by the horses pulling on the load results in a change in kinetic energy of the load, ultimately going faster. (credit: modification of work by “Jassen”/ Flickr)

According to this theorem, when an object slows down, its final kinetic energy is less than its initial kinetic energy, the change in its kinetic energy is negative, and so is the net work done on it. If an object speeds up, the net work done on it is positive. When calculating the net work, you must include all the forces that act on an object. If you leave out any forces that act on an object, or if you include any forces that don't act on it, you will get a wrong result.

The importance of the work-energy theorem, and the further generalizations to which it leads, is that it makes some types of calculations much simpler to accomplish than they would be by trying to solve Newton's second law. For example, in [Newton's Laws of Motion](#), we found the speed of an object sliding down a frictionless plane by solving

Newton's second law for the acceleration and using kinematic equations for constant acceleration, obtaining

$$v_f^2 = v_i^2 + 2g(s_f - s_i)\sin\theta,$$

where s is the displacement down the plane.

We can also get this result from the work-energy theorem in [Equation 7.1](#). Since only two forces are acting on the object-gravity and the normal force-and the normal force doesn't do any work, the net work is just the work done by gravity. The work dW is the dot product of the force of gravity or $\vec{F} = -mg\hat{j}$ and the displacement $\vec{dr} = dx\hat{i} + dy\hat{j}$. After taking the dot product and integrating from an initial position y_i to a final position y_f , one finds the net work as

$$W_{\text{net}} = W_{\text{grav}} = -mg(y_f - y_i),$$

where y is positive up. The work-energy theorem says that this equals the change in kinetic energy:

$$-mg(y_f - y_i) = \frac{1}{2}m(v_f^2 - v_i^2).$$

Using a right triangle, we can see that $(y_f - y_i) = (s_f - s_i)\sin\theta$, so the result for the final speed is the same.

What is gained by using the work-energy theorem? The answer is that for a frictionless plane surface, not much. However, Newton's second law is easy to solve only for this particular case, whereas the work-energy theorem gives the final speed for any shaped frictionless surface. For an arbitrary curved surface, the normal force is not constant, and Newton's second law may be difficult or impossible to solve analytically. Constant or not, for motion along a surface, the normal force never does any work, because it's perpendicular to the displacement. A calculation using the work-energy theorem avoids this difficulty and applies to more general situations.



PROBLEM-SOLVING STRATEGY

Work-Energy Theorem

1. Draw a free-body diagram for each force on the object.
2. Determine whether or not each force does work over the displacement in the diagram. Be sure to keep any positive or negative signs in the work done.
3. Add up the total amount of work done by each force.
4. Set this total work equal to the change in kinetic energy and solve for any unknown parameter.
5. Check your answers. If the object is traveling at a constant speed or zero acceleration, the total work done should be zero and match the change in kinetic energy. If the total work is positive, the object must have sped up or increased kinetic energy. If the total work is negative, the object must have slowed down or decreased kinetic energy.



EXAMPLE 7.9

Loop-the-Loop

The frictionless track for a toy car includes a loop-the-loop of radius R . How high, measured from the bottom of the loop, must the car be placed to start from rest on the approaching section of track and go all the way around the loop?

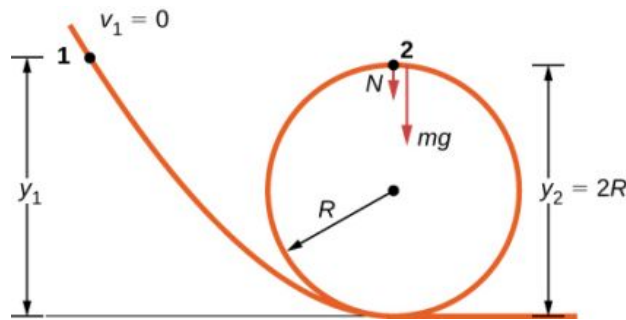


FIGURE 7.12 A frictionless track for a toy car has a loop-the-loop in it. How high must the car start so that it can go around the loop without falling off?

Strategy

The free-body diagram at the final position of the object is drawn in [Figure 7.12](#). The gravitational work is the only work done over the displacement that is not zero. Since the weight points in the same direction as the net vertical displacement, the total work done by the gravitational force is positive. From the work-energy theorem, the starting height determines the speed of the car at the top of the loop,

$$-mg(y_2 - y_1) = \frac{1}{2}mv_2^2,$$

where the notation is shown in the accompanying figure. At the top of the loop, the normal force and gravity are both down and the acceleration is centripetal, so

$$a_{\text{top}} = \frac{F}{m} = \frac{N + mg}{m} = \frac{v_2^2}{R}.$$

The condition for maintaining contact with the track is that there must be some normal force, however slight; that is, $N > 0$. Substituting for v_2^2 and N , we can find the condition for y_1 .

Solution

Implement the steps in the strategy to arrive at the desired result:

$$N = -mg + \frac{mv_2^2}{R} = \frac{-mgR + 2mg(y_1 - 2R)}{R} > 0 \quad \text{or} \quad y_1 > \frac{5R}{2}.$$

Significance

On the surface of the loop, the normal component of gravity and the normal contact force must provide the centripetal acceleration of the car going around the loop. The tangential component of gravity slows down or speeds up the car. A child would find out how high to start the car by trial and error, but now that you know the work-energy theorem, you can predict the minimum height (as well as other more useful results) from physical principles. By using the work-energy theorem, you did not have to solve a differential equation to determine the height.

✓ CHECK YOUR UNDERSTANDING 7.7

Suppose the radius of the loop-the-loop in [Example 7.9](#) is 15 cm and the toy car starts from rest at a height of 45 cm above the bottom. What is its speed at the top of the loop?

🔗 INTERACTIVE

Watch this [video \(https://openstax.org/l/21rollcoas\)](https://openstax.org/l/21rollcoas) to see a looping rollercoaster.

In situations where the motion of an object is known, but the values of one or more of the forces acting on it are not known, you may be able to use the work-energy theorem to get some information about the forces. Work depends

on the force and the distance over which it acts, so the information is provided via their product.

EXAMPLE 7.10

Determining a Stopping Force

A bullet has a mass of 40 grains (2.60 g) and a muzzle velocity of 1100 ft./s (335 m/s). It can penetrate eight 1-inch pine boards, each with thickness 0.75 inches. What is the average stopping force exerted by the wood, as shown in Figure 7.13?

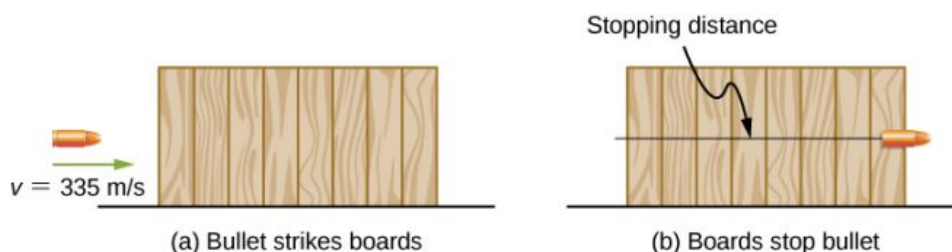


FIGURE 7.13 The boards exert a force to stop the bullet. As a result, the boards do work and the bullet loses kinetic energy.

Strategy

We can assume that under the general conditions stated, the bullet loses all its kinetic energy penetrating the boards, so the work-energy theorem says its initial kinetic energy is equal to the average stopping force times the distance penetrated. The change in the bullet's kinetic energy and the net work done stopping it are both negative, so when you write out the work-energy theorem, with the net work equal to the average force times the stopping distance, that's what you get. The total thickness of eight 1-inch pine boards that the bullet penetrates is $8 \times \frac{3}{4} \text{ in.} = 6 \text{ in.} = 15.2 \text{ cm.}$

Solution

Applying the work-energy theorem, we get

$$W_{\text{net}} = -F_{\text{ave}} \Delta s_{\text{stop}} = -K_{\text{initial}},$$

so

$$F_{\text{ave}} = \frac{\frac{1}{2}mv^2}{\Delta s_{\text{stop}}} = \frac{\frac{1}{2}(2.6 \times 10^{-3} \text{ kg})(335 \text{ m/s})^2}{0.152 \text{ m}} = 960 \text{ N.}$$

Significance

We could have used Newton's second law and kinematics in this example, but the work-energy theorem also supplies an answer to less simple situations. The penetration of a bullet, fired vertically upward into a block of wood, is discussed in one section of Asif Shakur's recent article ["Bullet-Block Science Video Puzzle." *The Physics Teacher* (January 2015) 53(1): 15-16]. If the bullet is fired dead center into the block, it loses all its kinetic energy and penetrates slightly farther than if fired off-center. The reason is that if the bullet hits off-center, it has a little kinetic energy after it stops penetrating, because the block rotates. The work-energy theorem implies that a smaller change in kinetic energy results in a smaller penetration. You will understand more of the physics in this interesting article after you finish reading [Angular Momentum](#).

INTERACTIVE

Learn more about work and energy in this [PhET simulation \(https://openstax.org/l/21ramp\)](https://openstax.org/l/21ramp) called "the ramp." Try changing the force pushing the box and the frictional force along the incline. The work and energy plots can be examined to note the total work done and change in kinetic energy of the box.

7.4 Power

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Relate the work done during a time interval to the power delivered
- Find the power expended by a force acting on a moving body

The concept of work involves force and displacement; the work-energy theorem relates the net work done on a body to the difference in its kinetic energy, calculated between two points on its trajectory. None of these quantities or relations involves time explicitly, yet we know that the time available to accomplish a particular amount of work is frequently just as important to us as the amount itself. In the chapter-opening figure, several sprinters may have achieved the same velocity at the finish, and therefore did the same amount of work, but the winner of the race did it in the least amount of time.

We express the relation between work done and the time interval involved in doing it, by introducing the concept of power. Since work can vary as a function of time, we first define **average power** as the work done during a time interval, divided by the interval,

$$P_{\text{ave}} = \frac{\Delta W}{\Delta t}. \quad 7.10$$

Then, we can define the **instantaneous power** (frequently referred to as just plain **power**).

POWER

Power is defined as the rate of doing work, or the limit of the average power for time intervals approaching zero,

$$P = \frac{dW}{dt}. \quad 7.11$$

If the power is constant over a time interval, the average power for that interval equals the instantaneous power, and the work done by the agent supplying the power is $W = P\Delta t$. If the power during an interval varies with time, then the work done is the time integral of the power,

$$W = \int P dt.$$

The work-energy theorem relates how work can be transformed into kinetic energy. Since there are other forms of energy as well, as we discuss in the next chapter, we can also define power as the rate of transfer of energy. Work and energy are measured in units of joules, so power is measured in units of joules per second, which has been given the SI name watts, abbreviation W: $1 \text{ J/s} = 1 \text{ W}$. Another common unit for expressing the power capability of everyday devices is horsepower: $1 \text{ hp} = 746 \text{ W}$.



EXAMPLE 7.11

Pull-Up Power

An 80-kg army trainee does pull-ups on a horizontal bar ([Figure 7.14](#)). It takes the trainee 0.8 seconds to raise the body from a lower position to where the chin is above the bar. How much power do the trainee's muscles supply moving his body from the lower position to where the chin is above the bar? (*Hint: Make reasonable estimates for any quantities needed.*)

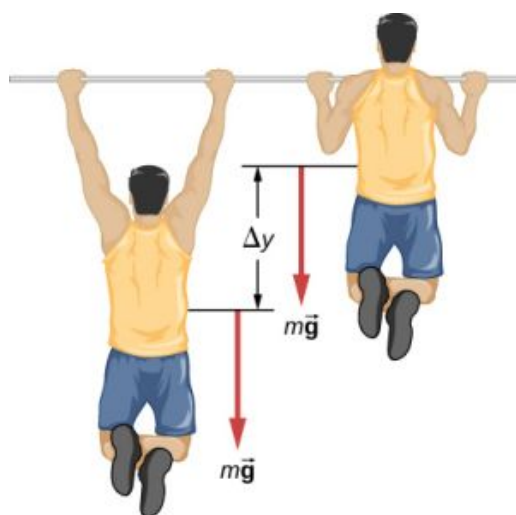


FIGURE 7.14 What is the power expended in doing ten pull-ups in ten seconds?

Strategy

The work done against gravity, going up or down a distance Δy , is $mg\Delta y$. Let's assume that $\Delta y = 2\text{ ft} \approx 60\text{ cm}$. Also, assume that the arms comprise 10% of the body mass and are not included in the moving mass. With these assumptions, we can calculate the work done.

Solution

The result we get, applying our assumptions, is

$$P = \frac{mg(\Delta y)}{t} = \frac{0.9(80\text{ kg})(9.8\text{ m/s}^2)(0.60\text{ m})}{0.8\text{ s}} = 529\text{ W}.$$

Significance

This is typical for power expenditure in strenuous exercise; in everyday units, it's somewhat more than one horsepower ($1\text{ hp} = 746\text{ W}$).

✓ CHECK YOUR UNDERSTANDING 7.8

Estimate the power expended by a weightlifter raising a 150-kg barbell 2 m in 3 s.

The power involved in moving a body can also be expressed in terms of the forces acting on it. If a force $\vec{F}$ acts on a body that is displaced $d\vec{r}$ in a time dt , the power expended by the force is

$$P = \frac{dW}{dt} = \frac{\vec{F} \cdot d\vec{r}}{dt} = \vec{F} \cdot \left(\frac{d\vec{r}}{dt} \right) = \vec{F} \cdot \vec{v}, \quad 7.12$$

where $\vec{v}$ is the velocity of the body. The fact that the limits implied by the derivatives exist, for the motion of a real body, justifies the rearrangement of the infinitesimals.

✳ EXAMPLE 7.12

Automotive Power Driving Uphill

How much power must an automobile engine expend to move a 1200-kg car up a 15% grade at 90 km/h (Figure 7.15)? Assume that 25% of this power is dissipated overcoming air resistance and friction.

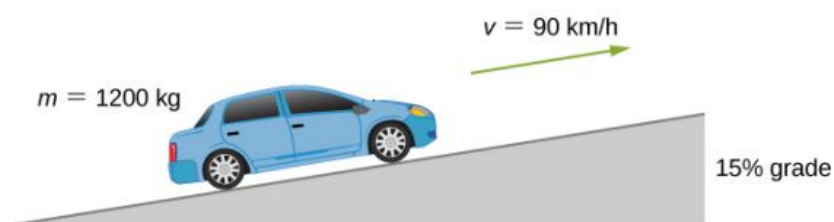


FIGURE 7.15 We want to calculate the power needed to move a car up a hill at constant speed.

Strategy

At constant velocity, there is no change in kinetic energy, so the net work done to move the car is zero. Therefore the power supplied by the engine to move the car equals the power expended against gravity and air resistance. By assumption, 75% of the power is supplied against gravity, which equals $m\vec{g} \cdot \vec{v} = mgv \sin \theta$, where θ is the angle of the incline. A 15% grade means $\tan \theta = 0.15$. This reasoning allows us to solve for the power required.

Solution

Carrying out the suggested steps, we find

$$0.75 P = mgv \sin(\tan^{-1} 0.15),$$

or

$$P = \frac{(1200 \times 9.8 \text{ N})(90 \text{ m}/3.6 \text{ s})\sin(8.53^\circ)}{0.75} = 58 \text{ kW},$$

or about 78 hp. (You should supply the steps used to convert units.)

Significance

This is a reasonable amount of power for the engine of a small to mid-size car to supply ($1 \text{ hp} = 0.746 \text{ kW}$). Note that this is only the power expended to move the car. Much of the engine's power goes elsewhere, for example, into waste heat. That's why cars need radiators. Any remaining power could be used for acceleration, or to operate the car's accessories.

Chapter Review

Key Terms

average power work done in a time interval divided by the time interval

kinetic energy energy of motion, one-half an object's mass times the square of its speed

net work work done by all the forces acting on an object

power (or instantaneous power) rate of doing work

work done when a force acts on something that undergoes a displacement from one position to another

work-energy theorem net work done on a particle is equal to the change in its kinetic energy

Key Equations

Work done by a force over an infinitesimal displacement

$$dW = \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = |\vec{\mathbf{F}}| |d\vec{\mathbf{r}}| \cos \theta$$

Work done by a force acting along a path from A to B

$$W_{AB} = \int_{\text{path } AB} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$$

Work done by a constant force of kinetic friction

$$W_{\text{fr}} = -f_k |l_{AB}|$$

Work done going from A to B by Earth's gravity, near its surface

$$W_{\text{grav}, AB} = -mg(y_B - y_A)$$

Work done going from A to B by one-dimensional spring force

$$W_{\text{spring}, AB} = -\left(\frac{1}{2}k\right)(x_B^2 - x_A^2)$$

Kinetic energy of a non-relativistic particle

$$K = \frac{1}{2}mv^2 = \frac{p^2}{2m}$$

Work-energy theorem

$$W_{\text{net}} = K_B - K_A$$

Power as rate of doing work

$$P = \frac{dW}{dt}$$

Power as the dot product of force and velocity

$$P = \vec{\mathbf{F}} \cdot \vec{\mathbf{v}}$$

Summary

7.1 Work

- The infinitesimal increment of work done by a force, acting over an infinitesimal displacement, is the dot product of the force and the displacement.
- The work done by a force, acting over a finite path, is the integral of the infinitesimal increments of work done along the path.
- The work done *against* a force is the negative of the work done *by* the force.
- The work done by a normal or frictional contact force must be determined in each particular case.
- The work done by the force of gravity, on an object near the surface of Earth, depends only on the weight of the object and the difference in height through which it moved.
- The work done by a spring force, acting from an initial position to a final position, depends only on the spring constant and the squares of those positions.

7.2 Kinetic Energy

- The kinetic energy of a particle is the product of one-half its mass and the square of its speed, for

non-relativistic speeds.

- The kinetic energy of a system is the sum of the kinetic energies of all the particles in the system.
- Kinetic energy is relative to a frame of reference, is always positive, and is sometimes given special names for different types of motion.

7.3 Work-Energy Theorem

- Because the net force on a particle is equal to its mass times the derivative of its velocity, the integral for the net work done on the particle is equal to the change in the particle's kinetic energy. This is the work-energy theorem.
- You can use the work-energy theorem to find certain properties of a system, without having to solve the differential equation for Newton's second law.

7.4 Power

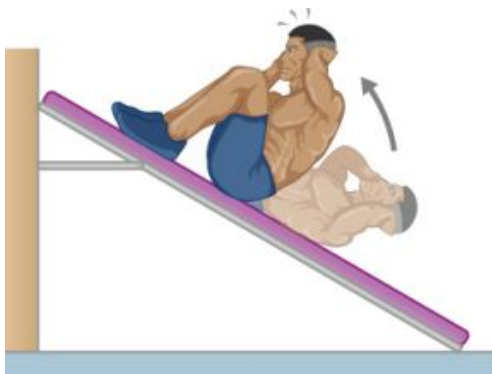
- Power is the rate of doing work; that is, the derivative of work with respect to time.
- Alternatively, the work done, during a time interval, is the integral of the power supplied over the time interval.
- The power delivered by a force, acting on a moving particle, is the dot product of the force

and the particle's velocity.

Conceptual Questions

7.1 Work

1. Give an example of something we think of as work in everyday circumstances that is not work in the scientific sense. Is energy transferred or changed in form in your example? If so, explain how this is accomplished without doing work.
2. Give an example of a situation in which there is a force and a displacement, but the force does no work. Explain why it does no work.
3. Describe a situation in which a force is exerted for a long time but does no work. Explain.
4. A body moves in a circle at constant speed. Does the centripetal force that accelerates the body do any work? Explain.
5. Suppose you throw a ball upward and catch it when it returns at the same height. How much work does the gravitational force do on the ball over its entire trip?
6. Why is it more difficult to do sit-ups while on a slant board than on a horizontal surface? (See below.)



7. As a young man, Tarzan climbed up a vine to reach his tree house. As he got older, he decided to build and use a staircase instead. Since the work of the gravitational force mg is path independent, what did the King of the Apes gain

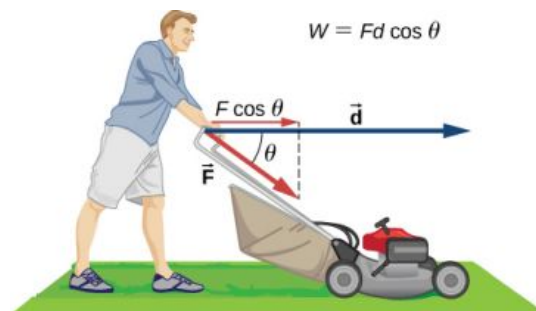
in using stairs?

7.2 Kinetic Energy

8. A particle of m has a velocity of $v_x\hat{i} + v_y\hat{j} + v_z\hat{k}$. Is its kinetic energy given by $m(v_x^2\hat{i} + v_y^2\hat{j} + v_z^2\hat{k})/2$? If not, what is the correct expression?
9. One particle has mass m and a second particle has mass $2m$. The second particle is moving with speed v and the first with speed $2v$. How do their kinetic energies compare?
10. A person drops a pebble of mass m_1 from a height h , and it hits the floor with kinetic energy K . The person drops another pebble of mass m_2 from a height of $2h$, and it hits the floor with the same kinetic energy K . How do the masses of the pebbles compare?

7.3 Work-Energy Theorem

11. The person shown below does work on the lawn mower. Under what conditions would the mower gain energy from the person pushing the mower? Under what conditions would it lose energy?



12. Work done on a system puts energy into it. Work done by a system removes energy from it. Give an example for each statement.
13. Two marbles of masses m and $2m$ are dropped from a height h . Compare their kinetic energies when they reach the ground.
14. Compare the work required to accelerate a car of mass 2000 kg from 30.0 to 40.0 km/h with that required for an acceleration from 50.0 to 60.0 km/h.
15. Suppose you are jogging at constant velocity. Are you doing any work on the environment and vice versa?

16. Two forces act to double the speed of a particle, initially moving with kinetic energy of 1 J. One of the forces does 4 J of work. How much work does the other force do?

7.4 Power

17. Most electrical appliances are rated in watts. Does this rating depend on how long the appliance is on? (When off, it is a zero-watt device.) Explain in terms of the definition of power.
18. Explain, in terms of the definition of power, why energy consumption is sometimes listed in kilowatt-hours rather than joules. What is the

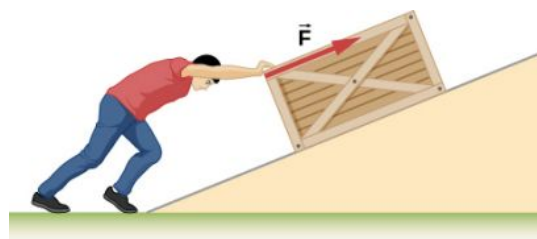
relationship between these two energy units?

19. A spark of static electricity, such as that you might receive from a doorknob on a cold dry day, may carry a few hundred watts of power. Explain why you are not injured by such a spark.
20. Does the work done in lifting an object depend on how fast it is lifted? Does the power expended depend on how fast it is lifted?
21. Can the power expended by a force be negative?
22. How can a 50-W light bulb use more energy than a 1000-W oven?

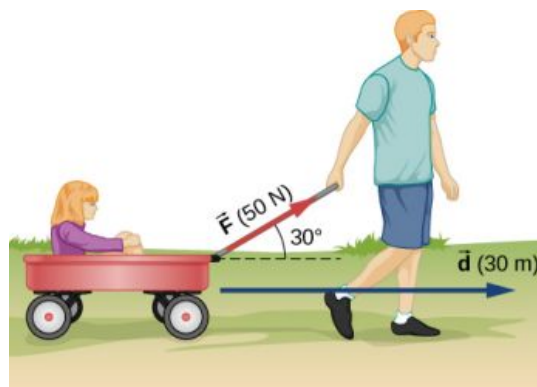
Problems

7.1 Work

23. How much work does a supermarket checkout attendant do on a can of soup he pushes 0.600 m horizontally with a force of 5.00 N?
24. A 75.0-kg person climbs stairs, gaining 2.50 m in height. Find the work done to accomplish this task.
25. (a) Calculate the work done on a 1500-kg elevator car by its cable to lift it 40.0 m at constant speed, assuming friction averages 100 N. (b) What is the work done on the elevator car by the gravitational force in this process? (c) What is the total work done on the elevator car?
26. Suppose a car travels 108 km at a speed of 30.0 m/s, and uses 2.0 gal of gasoline. Only 30% of the gasoline goes into useful work by the force that keeps the car moving at constant speed despite friction. (The energy content of gasoline is about 140 MJ/gal.) (a) What is the magnitude of the force exerted to keep the car moving at constant speed? (b) If the required force is directly proportional to speed, how many gallons will be used to drive 108 km at a speed of 28.0 m/s?
27. Calculate the work done by an 85.0-kg man who pushes a crate 4.00 m up along a ramp that makes an angle of 20.0° with the horizontal (see below). He exerts a force of 500 N on the crate parallel to the ramp and moves at a constant speed. Be certain to include the work he does on the crate and on his body to get up the ramp.

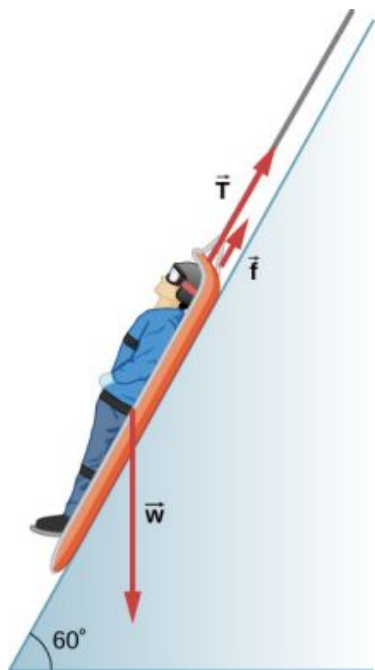


28. How much work is done by the boy pulling his sister 30.0 m in a wagon as shown below?



29. A shopper pushes a grocery cart 20.0 m at constant speed on level ground, against a 35.0 N frictional force. He pushes in a direction 25.0° below the horizontal. (a) What is the work done on the cart by friction? (b) What is the work done on the cart by the gravitational force? (c) What is the work done on the cart by the shopper? (d) Find the force the shopper exerts, using energy considerations. (e) What is the total work done on the cart?
30. Suppose the ski patrol lowers a rescue sled and victim, having a total mass of 90.0 kg, down a 60.0° slope at constant speed, as shown below.

The coefficient of friction between the sled and the snow is 0.100. (a) How much work is done by friction as the sled moves 30.0 m along the hill? (b) How much work is done by the rope on the sled in this distance? (c) What is the work done by the gravitational force on the sled? (d) What is the total work done?



31. A constant 20-N force pushes a small ball in the direction of the force over a distance of 5.0 m. What is the work done by the force?
32. A toy cart is pulled a distance of 6.0 m in a straight line across the floor. The force pulling the cart has a magnitude of 20 N and is directed at 37° above the horizontal. What is the work done by this force?
33. A 5.0-kg box rests on a horizontal surface. The coefficient of kinetic friction between the box and surface is $\mu_k = 0.50$. A horizontal force pulls the box at constant velocity for 10 cm. Find the work done by (a) the applied horizontal force, (b) the frictional force, and (c) the net force.
34. A sled plus passenger with total mass 50 kg is pulled 20 m across the snow ($\mu_k = 0.20$) at constant velocity by a force directed 25° above the horizontal. Calculate (a) the work of the applied force, (b) the work of friction, and (c) the total work.
35. Suppose that the sled plus passenger of the preceding problem is pushed 20 m across the

snow at constant velocity by a force directed 30° below the horizontal. Calculate (a) the work of the applied force, (b) the work of friction, and (c) the total work.

36. How much work does the force $F(x) = (-2.0/x)$ N do on a particle as it moves from $x = 2.0$ m to $x = 5.0$ m?
37. How much work is done against the gravitational force on a 5.0-kg briefcase when it is carried from the ground floor to the roof of the Empire State Building, a vertical climb of 380 m?
38. It takes 500 J of work to compress a spring 10 cm. What is the force constant of the spring?
39. A bungee cord is essentially a very long rubber band that can stretch up to four times its unstretched length. However, its spring constant varies over its stretch [see Menz, P.G. "The Physics of Bungee Jumping." *The Physics Teacher* (November 1993) 31: 483-487]. Take the length of the cord to be along the x -direction and define the stretch x as the length of the cord l minus its un-stretched length l_0 ; that is, $x = l - l_0$ (see below). Suppose a particular bungee cord has a spring constant, for $0 \leq x \leq 4.88$ m, of $k_1 = 204$ N/m and for $x \geq 4.88$ m, of $k_2 = 111$ N/m. (Recall that the spring constant is the slope of the force $F(x)$ versus its stretch x .) (a) What is the tension in the cord when the stretch is 16.7 m (the maximum desired for a given jump)? (b) How much work must be done against the elastic force of the bungee cord to stretch it 16.7 m?

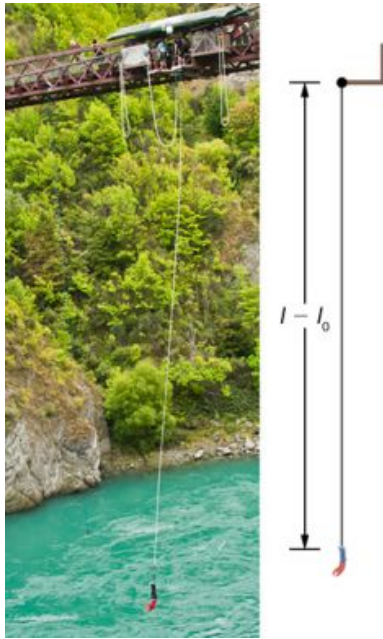


FIGURE 7.16 (credit: modification of work by Graeme Churchard)

- 40.** A bungee cord exerts a nonlinear elastic force of magnitude $F(x) = k_1 x + k_2 x^3$, where x is the distance the cord is stretched, $k_1 = 204 \text{ N/m}$ and $k_2 = -0.233 \text{ N/m}^3$. How much work must be done on the cord to stretch it 16.7 m?
- 41.** Engineers desire to model the magnitude of the elastic force of a bungee cord using the equation $F(x) = a \left[\frac{x+9 \text{ m}}{9 \text{ m}} - \left(\frac{9 \text{ m}}{x+9 \text{ m}} \right)^2 \right]$, where x is the stretch of the cord along its length and a is a constant. If it takes 22.0 kJ of work to stretch the cord by 16.7 m, determine the value of the constant a .
- 42.** A particle moving in the xy -plane is subject to a force $\vec{F}(x, y) = (50 \text{ N/m})(x\hat{i} + \frac{y^2}{3\text{m}}\hat{j})$ where x and y are in meters. Calculate the work done on the particle by this force, as it moves in a straight line from the point (3 m, 4 m) to the point (6 m, 8 m).
- 43.** A particle moves along a curved path $y(x) = (10 \text{ m})\{1 + \cos[(0.1 \text{ m}^{-1})x]\}$, from $x = 0$ to $x = 10\pi \text{ m}$, subject to a tangential force of variable magnitude $F(x) = (10 \text{ N})\sin[(0.1 \text{ m}^{-1})x]$. How much work does the force do? (*Hint:* Consult a table of integrals or use a numerical integration program.)
- 44.** Compare the kinetic energy of a 20,000-kg truck moving at 110 km/h with that of an 80.0-kg astronaut in orbit moving at 27,500 km/h.
- 45.** (a) How fast must a 3000-kg elephant move to have the same kinetic energy as a 65.0-kg sprinter running at 10.0 m/s? (b) Discuss how the larger energies needed for the movement of larger animals would relate to metabolic rates.
- 46.** Estimate the kinetic energy of a 90,000-ton aircraft carrier moving at a speed of 30 knots. You will need to look up the definition of a nautical mile to use in converting the unit for speed, where 1 knot equals 1 nautical mile per hour. Furthermore for this problem, 1 ton is equivalent to 2,000 pounds.
- 47.** Calculate the kinetic energies of (a) a 2000.0-kg automobile moving at 100.0 km/h; (b) an 80.-kg runner sprinting at 10. m/s; and (c) a 9.1×10^{-31} -kg electron moving at $2.0 \times 10^7 \text{ m/s}$.
- 48.** A 5.0-kg body has three times the kinetic energy of an 8.0-kg body. Calculate the ratio of the speeds of these bodies.
- 49.** An 8.0-g bullet has a speed of 800 m/s. (a) What is its kinetic energy? (b) What is its kinetic energy if the speed is halved?

7.2 Kinetic Energy

7.3 Work-Energy Theorem

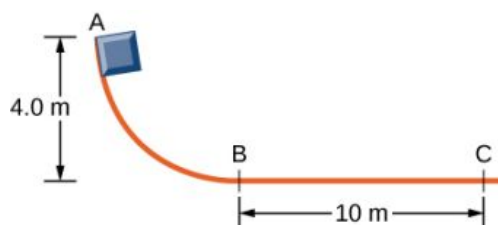
- 50.** (a) Calculate the force needed to bring a 950-kg car to rest from a speed of 90.0 km/h in a distance of 120 m (a fairly typical distance for a non-panic stop). (b) Suppose instead the car hits a concrete abutment at full speed and is brought to a stop in 2.00 m. Calculate the force exerted on the car and compare it with the force found in part (a).
- 51.** A car's bumper is designed to withstand a 4.0-km/h (1.1-m/s) collision with an immovable object without damage to the body of the car. The bumper cushions the shock by absorbing the force over a distance. Calculate the magnitude of the average force on a bumper that collapses 0.200 m while bringing a 900-kg car to rest from an initial speed of 1.1 m/s.
- 52.** Boxing gloves are padded to lessen the force of a blow. (a) Calculate the force exerted by a boxing glove on an opponent's face, if the glove and face compress 7.50 cm during a blow in

which the 7.00-kg arm and glove are brought to rest from an initial speed of 10.0 m/s. (b) Calculate the force exerted by an identical blow in the days when no gloves were used, and the knuckles and face would compress only 2.00 cm. Assume the change in mass by removing the glove is negligible. (c) Discuss the magnitude of the force with glove on. Does it seem high enough to cause damage even though it is lower than the force with no glove?

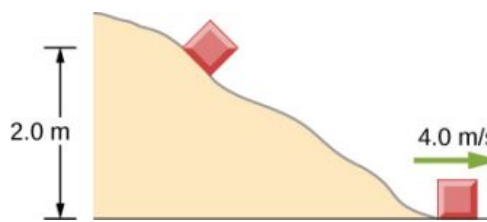
53. Using energy considerations, calculate the average force a 60.0-kg sprinter exerts backward on the track to accelerate from 2.00 to 8.00 m/s in a distance of 25.0 m, if he encounters a headwind that exerts an average force of 30.0 N against him.
54. A 5.0-kg box has an acceleration of 2.0 m/s^2 when it is pulled by a horizontal force across a surface with $\mu_K = 0.50$. Find the work done over a distance of 10 cm by (a) the horizontal force, (b) the frictional force, and (c) the net force. (d) What is the change in kinetic energy of the box?
55. A constant 10-N horizontal force is applied to a 20-kg cart at rest on a level floor. If friction is negligible, what is the speed of the cart when it has been pushed 8.0 m?
56. In the preceding problem, the 10-N force is applied at an angle of 45° below the horizontal. What is the speed of the cart when it has been pushed 8.0 m?
57. Compare the work required to stop a 100-kg crate sliding at 1.0 m/s and an 8.0-g bullet traveling at 500 m/s.
58. A wagon with its passenger sits at the top of a hill. The wagon is given a slight push and rolls 100 m down a 10° incline to the bottom of the hill. What is the wagon's speed when it reaches the end of the incline. Assume that the retarding force of friction is negligible.
59. An 8.0-g bullet with a speed of 800 m/s is shot into a wooden block and penetrates 20 cm before stopping. What is the average force of the wood on the bullet? Assume the block does not move.
60. A 2.0-kg block starts with a speed of 10 m/s at the bottom of a plane inclined at 37° to the horizontal. The coefficient of sliding friction between the block and plane is $\mu_k = 0.30$. (a)

Use the work-energy principle to determine how far the block slides along the plane before momentarily coming to rest. (b) After stopping, the block slides back down the plane. What is its speed when it reaches the bottom? (*Hint:* For the round trip, only the force of friction does work on the block.)

61. When a 3.0-kg block is pushed against a massless spring of force constant $4.5 \times 10^3 \text{ N/m}$, the spring is compressed 8.0 cm. The block is released, and it slides 2.0 m (from the point at which it is released) across a horizontal surface before friction stops it. What is the coefficient of kinetic friction between the block and the surface?
62. A small block of mass 200 g starts at rest at A, slides to B where its speed is $v_B = 8.0 \text{ m/s}$, then slides along the horizontal surface a distance 10 m before coming to rest at C. (See below.) (a) What is the work of friction along the curved surface? (b) What is the coefficient of kinetic friction along the horizontal surface?



63. A small object is placed at the top of an incline that is essentially frictionless. The object slides down the incline onto a rough horizontal surface, where it stops in 5.0 s after traveling 60 m. (a) What is the speed of the object at the bottom of the incline and its acceleration along the horizontal surface? (b) What is the height of the incline?
64. When released, a 100-g block slides down the path shown below, reaching the bottom with a speed of 4.0 m/s. How much work does the force of friction do?



65. A 0.22LR-caliber bullet like that mentioned in [Example 7.10](#) is fired into a door made of a

single thickness of 1-inch pine boards. How fast would the bullet be traveling after it penetrated through the door?

66. A sled starts from rest at the top of a snow-covered incline that makes a 22° angle with the horizontal. After sliding 75 m down the slope, its speed is 14 m/s. Use the work-energy theorem to calculate the coefficient of kinetic friction between the runners of the sled and the snowy surface.

7.4 Power

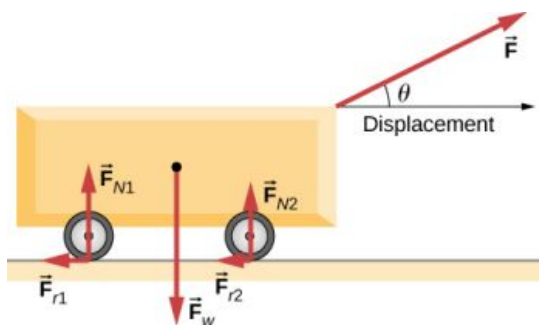
67. A person in good physical condition can put out 100 W of useful power for several hours at a stretch, perhaps by pedaling a mechanism that drives an electric generator. Neglecting any problems of generator efficiency and practical considerations such as resting time: (a) How many people would it take to run a 4.00-kW electric clothes dryer? (b) How many people would it take to replace a large electric power plant that generates 800 MW?
68. What is the cost of operating a 3.00-W electric clock for a year if the cost of electricity is \$0.0900 per kW · h?
69. A large household air conditioner may consume 15.0 kW of power. What is the cost of operating this air conditioner 3.00 h per day for 30.0 d if the cost of electricity is \$0.110 per kW · h?
70. (a) What is the average power consumption in watts of an appliance that uses 5.00 kW · h of energy per day? (b) How many joules of energy does this appliance consume in a year?
71. (a) What is the average useful power output of a person who does 6.00×10^6 J of useful work in 8.00 h? (b) Working at this rate, how long will it take this person to lift 2000 kg of bricks 1.50 m to a platform? (Work done to lift his body can be omitted because it is not considered useful output here.)
72. A 500-kg dragster accelerates from rest to a final speed of 110 m/s in 400 m (about a quarter of a mile) and encounters an average frictional force of 1200 N. What is its average power output in watts and horsepower if this takes 7.30 s?
73. (a) How long will it take an 850-kg car with a useful power output of 40.0 hp (1 hp equals 746 W) to reach a speed of 15.0 m/s, neglecting friction? (b) How long will this acceleration take if the car also climbs a 3.00-m high hill in the process?
74. (a) Find the useful power output of an elevator motor that lifts a 2500-kg load a height of 35.0 m in 12.0 s, if it also increases the speed from rest to 4.00 m/s. Note that the total mass of the counterbalanced system is 10,000 kg—so that only 2500 kg is raised in height, but the full 10,000 kg is accelerated. (b) What does it cost, if electricity is \$0.0900 per kW · h?
75. (a) How long would it take a 1.50×10^5 -kg airplane with engines that produce 100 MW of power to reach a speed of 250 m/s and an altitude of 12.0 km if air resistance were negligible? (b) If it actually takes 900 s, what is the power? (c) Given this power, what is the average force of air resistance if the airplane takes 1200 s? (*Hint:* You must find the distance the plane travels in 1200 s assuming constant acceleration.)
76. Calculate the power output needed for a 950-kg car to climb a 2.00° slope at a constant 30.0 m/s while encountering wind resistance and friction totaling 600 N.
77. A man of mass 80 kg runs up a flight of stairs 20 m high in 10 s. (a) how much power is used to lift the man? (b) If the man's body is 25% efficient, how much power does he expend?
78. The man of the preceding problem consumes approximately 1.05×10^7 J (2500 food calories) of energy per day in maintaining a constant weight. What is the average power he produces over a day? Compare this with his power production when he runs up the stairs.
79. An electron in a television tube is accelerated uniformly from rest to a speed of 8.4×10^7 m/s over a distance of 2.5 cm. What is the power delivered to the electron at the instant that its displacement is 1.0 cm?
80. Coal is lifted out of a mine a vertical distance of 50 m by an engine that supplies 500 W to a conveyor belt. How much coal per minute can be brought to the surface? Ignore the effects of friction.
81. A girl pulls her 15-kg wagon along a flat sidewalk by applying a 10-N force at 37° to the horizontal. Assume that friction is negligible and that the wagon starts from rest. (a) How much work does the girl do on the wagon in the first 2.0 s. (b)

How much instantaneous power does she exert at $t = 2.0$ s?

82. A typical automobile engine has an efficiency of 25%. Suppose that the engine of a 1000-kg automobile has a maximum power output of 140 hp. What is the maximum grade that the automobile can climb at 50 km/h if the frictional retarding force on it is 300 N?

Additional Problems

84. A cart is pulled a distance D on a flat, horizontal surface by a constant force F that acts at an angle θ with the horizontal direction. The other forces on the object during this time are gravity (F_w), normal forces (F_{N1}) and (F_{N2}), and rolling frictions F_{r1} and F_{r2} , as shown below. What is the work done by each force?

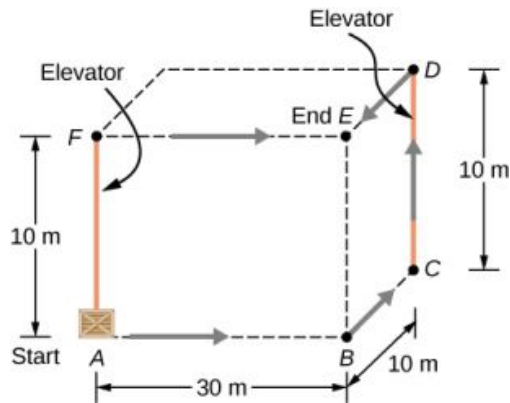


85. Consider a particle on which several forces act, one of which is known to be constant in time: $\vec{F}_1 = (3 \text{ N})\hat{i} + (4 \text{ N})\hat{j}$. As a result, the particle moves along the x -axis from $x = 0$ to $x = 5$ m in some time interval. What is the work done by $\vec{F}_1$?
86. Consider a particle on which several forces act, one of which is known to be constant in time: $\vec{F}_1 = (3 \text{ N})\hat{i} + (4 \text{ N})\hat{j}$. As a result, the particle moves first along the x -axis from $x = 0$ to $x = 5$ m and then parallel to the y -axis from $y = 0$ to $y = 6$ m. What is the work done by $\vec{F}_1$?
87. Consider a particle on which several forces act, one of which is known to be constant in time: $\vec{F}_1 = (3 \text{ N})\hat{i} + (4 \text{ N})\hat{j}$. As a result, the particle moves along a straight path from a Cartesian coordinate of $(0 \text{ m}, 0 \text{ m})$ to $(5 \text{ m}, 6 \text{ m})$. What is the work done by $\vec{F}_1$?
88. Consider a particle on which a force acts that depends on the position of the particle. This force is given by $\vec{F}_1 = (2y)\hat{i} + (3x)\hat{j}$. Find the work done by this force when the particle moves

83. When jogging at 13 km/h on a level surface, a 70-kg man uses energy at a rate of approximately 850 W. Using the facts that the “human engine” is approximately 25% efficient, determine the rate at which this man uses energy when jogging up a 5.0° slope at this same speed. Assume that the frictional retarding force is the same in both cases.

from the origin to a point 5 meters to the right on the x -axis.

89. A boy pulls a 5-kg cart with a 20-N force at an angle of 30° above the horizontal for a length of time. Over this time frame, the cart moves a distance of 12 m on the horizontal floor. (a) Find the work done on the cart by the boy. (b) What will be the work done by the boy if he pulled with the same force horizontally instead of at an angle of 30° above the horizontal over the same distance?
90. A crate of mass 200 kg is to be brought from a site on the ground floor to a third floor apartment. The workers know that they can either use the elevator first, then slide it along the third floor to the apartment, or first slide the crate to another location marked C below, and then take the elevator to the third floor and slide it on the third floor a shorter distance. The trouble is that the third floor is very rough compared to the ground floor. Given that the coefficient of kinetic friction between the crate and the ground floor is 0.100 and between the crate and the third floor surface is 0.300, find the work needed by the workers for each path shown from A to E. Assume that the force the workers need to do is just enough to slide the crate at constant velocity (zero acceleration). *Note:* The work by the elevator against the force of gravity is not done by the workers.



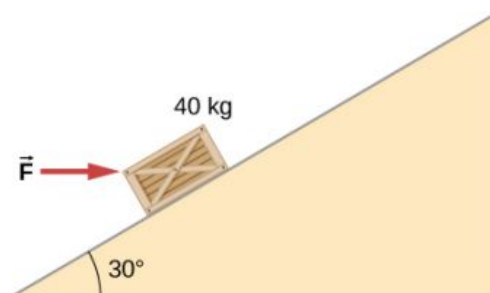
91. A hockey puck of mass 0.17 kg is shot across a rough floor with the roughness different at different places, which can be described by a position-dependent coefficient of kinetic friction. For a puck moving along the x -axis, the coefficient of kinetic friction is the following function of x , where x is in m : $\mu(x) = 0.1 + 0.05x$. Find the work done by the kinetic frictional force on the hockey puck when it has moved (a) from $x = 0$ to $x = 2 \text{ m}$, and (b) from $x = 2 \text{ m}$ to $x = 4 \text{ m}$.
92. A horizontal force of 20 N is required to keep a 5.0 kg box traveling at a constant speed up a frictionless incline for a vertical height change of 3.0 m . (a) What is the work done by gravity during this change in height? (b) What is the work done by the normal force? (c) What is the work done by the horizontal force?
93. A 7.0-kg box slides along a horizontal frictionless floor at 1.7 m/s and collides with a relatively massless spring that compresses 23 cm before the box comes to a stop. (a) How much kinetic energy does the box have before it collides with the spring? (b) Calculate the work done by the spring. (c) Determine the spring

Challenge Problems

99. Shown below is a 40-kg crate that is pushed at constant velocity a distance 8.0 m along a 30° incline by the horizontal force $\vec{F}$. The coefficient of kinetic friction between the crate and the incline is $\mu_k = 0.40$. Calculate the work done by (a) the applied force, (b) the frictional force, (c) the gravitational force, and (d) the net force.

constant of the spring.

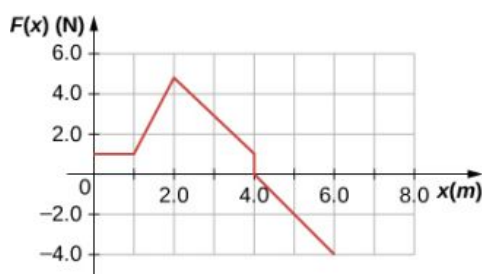
94. You are driving your car on a straight road with a coefficient of friction between the tires and the road of 0.55 . A large piece of debris falls in front of your view and you immediately slam on the brakes, leaving a skid mark of 30.5 m (100-feet) long before coming to a stop. A policeman sees your car stopped on the road, looks at the skid mark, and gives you a ticket for traveling over the 13.4 m/s (30 mph) speed limit. Should you fight the speeding ticket in court?
95. A crate is being pushed across a rough floor surface. If no force is applied on the crate, the crate will slow down and come to a stop. If the crate of mass 50 kg moving at speed 8 m/s comes to rest in 10 seconds, what is the rate at which the frictional force on the crate takes energy away from the crate?
96. Suppose a horizontal force of 20 N is required to maintain a speed of 8 m/s of a 50 kg crate. (a) What is the power of this force? (b) Note that the acceleration of the crate is zero despite the fact that 20 N force acts on the crate horizontally. What happens to the energy given to the crate as a result of the work done by this 20 N force?
97. Grains from a hopper falls at a rate of 10 kg/s vertically onto a conveyor belt that is moving horizontally at a constant speed of 2 m/s . (a) What force is needed to keep the conveyor belt moving at the constant velocity? (b) What is the minimum power of the motor driving the conveyor belt?
98. A cyclist in a race must climb a 5° hill at a speed of 8 m/s . If the mass of the bike and the biker together is 80 kg , what must be the power output of the biker to achieve the goal?



100. The surface of the preceding problem is modified so that the coefficient of kinetic friction

is decreased. The same horizontal force is applied to the crate, and after being pushed 8.0 m, its speed is 5.0 m/s. How much work is now done by the force of friction? Assume that the crate starts at rest.

- 101.** The force $F(x)$ varies with position, as shown below. Find the work done by this force on a particle as it moves from $x = 1.0$ m to $x = 5.0$ m.



- 102.** Find the work done by the same force in [Example 7.4](#), between the same points, $A = (0, 0)$ and $B = (2 \text{ m}, 2 \text{ m})$, over a circular arc of radius 2 m, centered at $(0, 2 \text{ m})$. Evaluate the path integral using Cartesian coordinates. (*Hint: You will probably need to consult a table of integrals.*)
- 103.** Answer the preceding problem using polar coordinates.
- 104.** Find the work done by the same force in [Example 7.4](#), between the same points, $A = (0, 0)$ and $B = (2 \text{ m}, 2 \text{ m})$, over a circular arc of radius 2 m, centered at $(2 \text{ m}, 0)$. Evaluate the path integral using Cartesian coordinates.
- (*Hint: You will probably need to consult a table of integrals.*)
- 105.** Answer the preceding problem using polar coordinates.
- 106.** Constant power P is delivered to a car of mass m by its engine. Show that if air resistance can be ignored, the distance covered in a time t by the car, starting from rest, is given by $s = (8P/9m)^{1/2} t^{3/2}$.
- 107.** Suppose that the air resistance a car encounters is independent of its speed. When the car travels at 15 m/s, its engine delivers 20 hp to its wheels. (a) What is the power delivered to the wheels when the car travels at 30 m/s? (b) How much energy does the car use in covering 10 km at 15 m/s? At 30 m/s? Assume that the engine is 25% efficient. (c) Answer the same questions if the force of air resistance is proportional to the speed of the automobile. (d) What do these results, plus your experience with gasoline consumption, tell you about air resistance?
- 108.** Consider a linear spring, as in [Figure 7.7\(a\)](#), with mass M uniformly distributed along its length. The left end of the spring is fixed, but the right end, at the equilibrium position $x = 0$, is moving with speed v in the x -direction. What is the total kinetic energy of the spring? (*Hint: First express the kinetic energy of an infinitesimal element of the spring dm in terms of the total mass, equilibrium length, speed of the right-hand end, and position along the spring; then integrate.*)

CHAPTER 8

Potential Energy and Conservation of Energy



FIGURE 8.1 Shown here is part of a Ball Machine sculpture by George Rhoads. A ball in this contraption is lifted, rolls, falls, bounces, and collides with various objects, but throughout its travels, its kinetic energy changes in definite, predictable amounts, which depend on its position and the objects with which it interacts. (credit: modification of work by Roland Tanglao)

CHAPTER OUTLINE

8.1 Potential Energy of a System

8.2 Conservative and Non-Conservative Forces

8.3 Conservation of Energy

8.4 Potential Energy Diagrams and Stability

8.5 Sources of Energy

INTRODUCTION In George Rhoads' rolling ball sculpture, the principle of conservation of energy governs the changes in the ball's kinetic energy and relates them to changes and transfers for other types of energy associated with the ball's interactions. In this chapter, we introduce the important concept of potential energy. This will enable us to formulate the law of conservation of mechanical energy and to apply it to simple systems, making solving problems easier. In the final section on sources of energy, we will consider energy transfers and the general law of conservation of energy. Throughout this book, the law of conservation of energy will be applied in increasingly more detail, as you encounter more complex and varied systems, and other forms of energy.

8.1 Potential Energy of a System

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Relate the difference of potential energy to work done on a particle for a system without friction or air drag
- Explain the meaning of the zero of the potential energy function for a system
- Calculate and apply the gravitational potential energy for an object near Earth's surface and the elastic potential energy of a mass-spring system

In [Work](#), we saw that the work done on an object by the constant gravitational force, near the surface of Earth, over

any displacement is a function only of the difference in the positions of the end-points of the displacement. This property allows us to define a different kind of energy for the system than its kinetic energy, which is called **potential energy**. We consider various properties and types of potential energy in the following subsections.

Potential Energy Basics

In [Motion in Two and Three Dimensions](#), we analyzed the motion of a projectile, like kicking a football in [Figure 8.2](#). For this example, let's ignore friction and air resistance and begin the analysis immediately after the kick, at the start of the projectile motion of the ball. As the football rises, the work done by the gravitational force on the football is negative, because the ball's displacement is positive vertically and the force due to gravity is negative vertically. We also noted that the ball slowed down until it reached its highest point in the motion, thereby decreasing the ball's kinetic energy. This loss in kinetic energy translates to a gain in gravitational potential energy of the football-Earth system.

As the football falls toward Earth, the work done on the football is now positive, because the displacement and the gravitational force both point vertically downward. The ball also speeds up, which indicates an increase in kinetic energy. Therefore, energy is converted from gravitational potential energy back into kinetic energy.

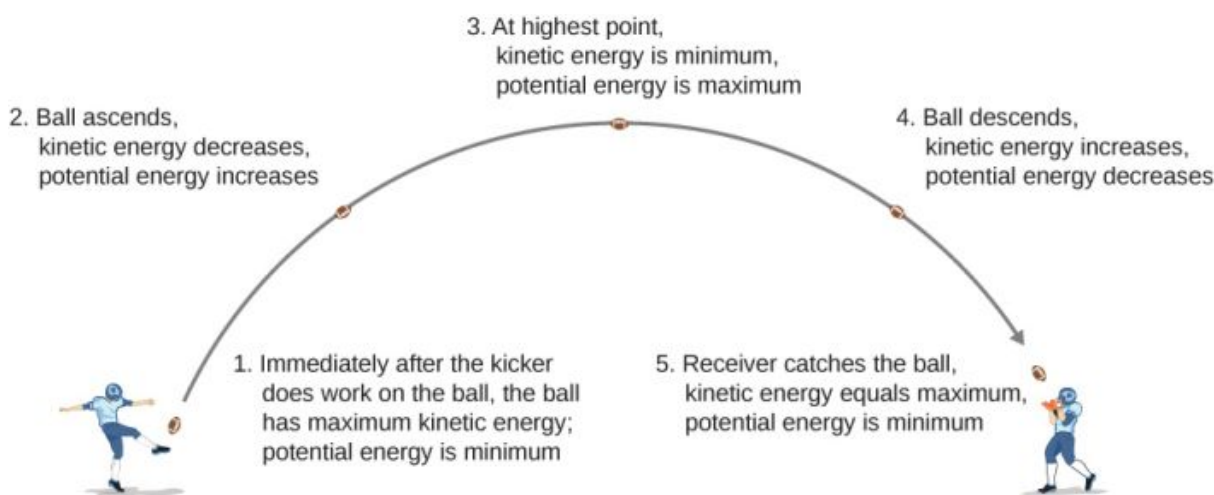


FIGURE 8.2 As a football starts its descent toward the wide receiver, gravitational potential energy is converted back into kinetic energy.

Based on this scenario, we can define the difference of potential energy from point *A* to point *B* as the negative of the work done:

$$\Delta U_{AB} = U_B - U_A = -W_{AB}. \quad 8.1$$

This formula explicitly states a **potential energy difference**, not just an absolute potential energy. Therefore, we need to define potential energy at a given position in such a way as to state standard values of potential energy on their own, rather than potential energy differences. We do this by rewriting the potential energy function in terms of an arbitrary constant,

$$\Delta U = U(\vec{r}) - U(\vec{r}_0). \quad 8.2$$

The choice of the potential energy at a starting location of $\vec{r}_0$ is made out of convenience in the given problem. Most importantly, whatever choice is made should be stated and kept consistent throughout the given problem. There are some well-accepted choices of initial potential energy. For example, the lowest height in a problem is usually defined as zero potential energy, or if an object is in space, the farthest point away from the system is often defined as zero potential energy. Then, the potential energy, with respect to zero at $\vec{r}_0$, is just $U(\vec{r})$.

As long as there is no friction or air resistance, the change in kinetic energy of the football equals negative of the change in gravitational potential energy of the football. This can be generalized to any potential energy:

$$\Delta K_{AB} = -\Delta U_{AB}. \quad 8.3$$

Let's look at a specific example, choosing zero potential energy for gravitational potential energy at convenient points.



EXAMPLE 8.1

Basic Properties of Potential Energy

A particle moves along the x -axis under the action of a force given by $F = -ax^2$, where $a = 3 \text{ N/m}^2$. (a) What is the difference in its potential energy as it moves from $x_A = 1 \text{ m}$ to $x_B = 2 \text{ m}$? (b) What is the particle's potential energy at $x = 1 \text{ m}$ with respect to a given 0.5 J of potential energy at $x = 0$?

Strategy

(a) The difference in potential energy is the negative of the work done, as defined by [Equation 8.1](#). The work is defined in the previous chapter as the dot product of the force with the distance. Since the particle is moving forward in the x -direction, the dot product simplifies to a multiplication ($\hat{\mathbf{i}} \cdot \hat{\mathbf{i}} = 1$). To find the total work done, we need to integrate the function between the given limits. After integration, we can state the work or the change in potential energy. (b) The potential energy function, with respect to zero at $x = 0$, is the indefinite integral encountered in part (a), with the constant of integration determined from [Equation 8.3](#). Then, we substitute the x -value into the function of potential energy to calculate the potential energy at $x = 1 \text{ m}$.

Solution

- a. The work done by the given force as the particle moves from coordinate x to $x + dx$ in one dimension is

$$dW = \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = Fdx = -ax^2 dx.$$

Substituting this expression into [Equation 8.1](#), we obtain

$$\Delta U = -W = \int_{x_1}^{x_2} ax^2 dx = \frac{1}{3}(3 \text{ N/m}^2)x^3 \Big|_{1 \text{ m}}^{2 \text{ m}} = 7 \text{ J}.$$

- b. The indefinite integral for the potential energy function in part (a) is

$$U(x) = \frac{1}{3}ax^3 + \text{const.},$$

and we want the constant to be determined by

$$U(0) = 0.5 \text{ J}.$$

Thus, the potential energy with respect to zero at $x = 0$ is just

$$U(x) = \frac{1}{3}ax^3 + 0.5 \text{ J}.$$

Therefore, the potential energy at $x = 1 \text{ m}$ is

$$U(1 \text{ m}) = \frac{1}{3}(3 \text{ N/m}^2)(1 \text{ m})^3 + 0.5 \text{ J} = 1.5 \text{ J}.$$

Significance

In this one-dimensional example, any function we can integrate, independent of path, is conservative. Notice how we applied the definition of potential energy difference to determine the potential energy function with respect to zero at a chosen point. Also notice that the potential energy, as determined in part (b), at $x = 1 \text{ m}$ is $U(1 \text{ m}) = 1 \text{ J}$ and at $x = 2 \text{ m}$ is $U(2 \text{ m}) = 8 \text{ J}$; their difference is the result in part (a).



CHECK YOUR UNDERSTANDING 8.1

In [Example 8.1](#), what are the potential energies of the particle at $x = 1 \text{ m}$ and $x = 2 \text{ m}$ with respect to zero at $x = 1.5 \text{ m}$? Verify that the difference of potential energy is still 7 J .

Systems of Several Particles

In general, a system of interest could consist of several particles. The difference in the potential energy of the system is the negative of the work done by gravitational or elastic forces, which, as we will see in the next section, are conservative forces. The potential energy difference depends only on the initial and final positions of the particles, and on some parameters that characterize the interaction (like mass for gravity or the spring constant for a Hooke's law force).

It is important to remember that potential energy is a property of the interactions between objects in a chosen system, and not just a property of each object. This is especially true for electric forces, although in the examples of potential energy we consider below, parts of the system are either so big (like Earth, compared to an object on its surface) or so small (like a massless spring), that the changes those parts undergo are negligible when included in the system.

Types of Potential Energy

For each type of interaction present in a system, you can label a corresponding type of potential energy. The total potential energy of the system is the sum of the potential energies of all the types. (This follows from the additive property of the dot product in the expression for the work done.) Let's look at some specific examples of types of potential energy discussed in [Work](#). First, we consider each of these forces when acting separately, and then when both act together.

Gravitational potential energy near Earth's surface

The system of interest consists of our planet, Earth, and one or more particles near its surface (or bodies small enough to be considered as particles, compared to Earth). The gravitational force on each particle (or body) is just its weight mg near the surface of Earth, acting vertically down. According to Newton's third law, each particle exerts a force on Earth of equal magnitude but in the opposite direction. Newton's second law tells us that the magnitude of the acceleration produced by each of these forces on Earth is mg divided by Earth's mass. Since the ratio of the mass of any ordinary object to the mass of Earth is vanishingly small, the motion of Earth can be completely neglected. Therefore, we consider this system to be a group of single-particle systems, subject to the uniform gravitational force of Earth.

In [Work](#), the work done on a body by Earth's uniform gravitational force, near its surface, depended on the mass of the body, the acceleration due to gravity, and the difference in height the body traversed, as given by [Equation 7.4](#). By definition, this work is the negative of the difference in the gravitational potential energy, so that difference is

$$\Delta U_{\text{grav}} = -W_{\text{grav},AB} = mg(y_B - y_A). \quad 8.4$$

You can see from this that the gravitational potential energy function, near Earth's surface, is

$$U(y) = mgy + \text{const.} \quad 8.5$$

You can choose the value of the constant, as described in the discussion of [Equation 8.2](#); however, for solving most problems, the most convenient constant to choose is zero for when $y = 0$, which is the lowest vertical position in the problem.



EXAMPLE 8.2

Gravitational Potential Energy of a Hiker

The summit of Great Blue Hill in Milton, MA, is 147 m above its base and has an elevation above sea level of 195 m ([Figure 8.3](#)). (Its Native American name, *Massachusetts*, was adopted by settlers for naming the Bay Colony and state near its location.) A 75-kg hiker ascends from the base to the summit. What is the gravitational potential energy of the hiker-Earth system with respect to zero gravitational potential energy at base height, when the hiker is (a) at the base of the hill, (b) at the summit, and (c) at sea level, afterward?



FIGURE 8.3 Sketch of the profile of Great Blue Hill, Milton, MA. The altitudes of the three levels are indicated.

Strategy

First, we need to pick an origin for the y -axis and then determine the value of the constant that makes the potential energy zero at the height of the base. Then, we can determine the potential energies from [Equation 8.5](#), based on the relationship between the zero potential energy height and the height at which the hiker is located.

Solution

- Let's choose the origin for the y -axis at base height, where we also want the zero of potential energy to be. This choice makes the constant equal to zero and

$$U(\text{base}) = U(0) = 0.$$

- At the summit, $y = 147$ m, so

$$U(\text{summit}) = U(147 \text{ m}) = mgh = (75 \times 9.8 \text{ N})(147 \text{ m}) = 108 \text{ kJ}.$$

- At sea level, $y = (147 - 195) \text{ m} = -48 \text{ m}$, so

$$U(\text{sea-level}) = (75 \times 9.8 \text{ N})(-48 \text{ m}) = -35.3 \text{ kJ}.$$

Significance

Besides illustrating the use of [Equation 8.4](#) and [Equation 8.5](#), the values of gravitational potential energy we found are reasonable. The gravitational potential energy is higher at the summit than at the base, and lower at sea level than at the base. Gravity does work on you on your way up, too! It does negative work and not quite as much (in magnitude), as your muscles do. But it certainly does work. Similarly, your muscles do work on your way down, as negative work. The numerical values of the potential energies depend on the choice of zero of potential energy, but the physically meaningful differences of potential energy do not. [Note that since [Equation 8.2](#) is a difference, the numerical values do not depend on the origin of coordinates.]

CHECK YOUR UNDERSTANDING 8.2

What are the values of the gravitational potential energy of the hiker at the base, summit, and sea level, with respect to a sea-level zero of potential energy?

Elastic potential energy

In [Work](#), we saw that the work done by a perfectly elastic spring, in one dimension, depends only on the spring constant and the squares of the displacements from the unstretched position, as given in [Equation 7.5](#). This work involves only the properties of a Hooke's law interaction and not the properties of real springs and whatever objects are attached to them. Therefore, we can define the difference of elastic potential energy for a spring force as the negative of the work done by the spring force in this equation, before we consider systems that embody this type of force. Thus,

$$\Delta U = -W_{AB} = \frac{1}{2}k(x_B^2 - x_A^2), \quad 8.6$$

where the object travels from point A to point B . The potential energy function corresponding to this difference is

$$U(x) = \frac{1}{2}kx^2 + \text{const.} \quad 8.7$$

If the spring force is the only force acting, it is simplest to take the zero of potential energy at $x = 0$, when the spring is at its unstretched length. Then, the constant in [Equation 8.7](#) is zero. (Other choices may be more convenient if other forces are acting.)



EXAMPLE 8.3

Spring Potential Energy

A system contains a perfectly elastic spring, with an unstretched length of 20 cm and a spring constant of 4 N/cm. (a) How much elastic potential energy does the spring contribute when its length is 23 cm? (b) How much more potential energy does it contribute if its length increases to 26 cm?

Strategy

When the spring is at its unstretched length, it contributes nothing to the potential energy of the system, so we can use [Equation 8.7](#) with the constant equal to zero. The value of x is the length minus the unstretched length. When the spring is expanded, the spring's displacement or difference between its relaxed length and stretched length should be used for the x -value in calculating the potential energy of the spring.

Solution

- The displacement of the spring is $x = 23 \text{ cm} - 20 \text{ cm} = 3 \text{ cm}$, so the contributed potential energy is $U = \frac{1}{2} kx^2 = \frac{1}{2} (4 \text{ N/cm}) (3 \text{ cm})^2 = 0.18 \text{ J}$.
- When the spring's displacement is $x = 26 \text{ cm} - 20 \text{ cm} = 6 \text{ cm}$, the potential energy is $U = \frac{1}{2} kx^2 = \frac{1}{2} (4 \text{ N/cm}) (6 \text{ cm})^2 = 0.72 \text{ J}$, which is a 0.54-J increase over the amount in part (a).

Significance

Calculating the elastic potential energy and potential energy differences from [Equation 8.7](#) involves solving for the potential energies based on the given lengths of the spring. Since U depends on x^2 , the potential energy for a compression (negative x) is the same as for an extension of equal magnitude.



CHECK YOUR UNDERSTANDING 8.3

When the length of the spring in [Example 8.3](#) changes from an initial value of 22.0 cm to a final value, the elastic potential energy it contributes changes by -0.0800 J . Find the final length.

Gravitational and elastic potential energy

A simple system embodying both gravitational and elastic types of potential energy is a one-dimensional, vertical mass-spring system. This consists of a massive particle (or block), hung from one end of a perfectly elastic, massless spring, the other end of which is fixed, as illustrated in [Figure 8.4](#).

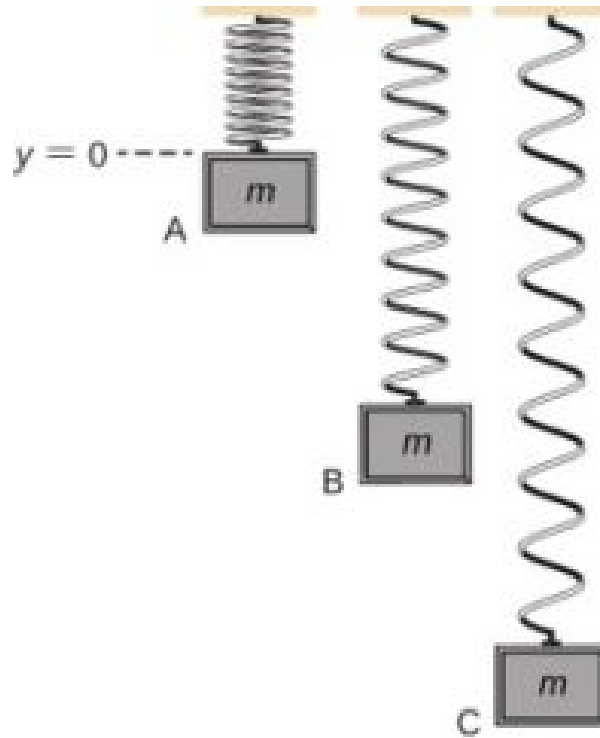


FIGURE 8.4 A vertical mass-spring system, with the positive y -axis pointing upward. The mass is initially at an unstretched spring length, point A. Then it is released, expanding past point B to point C, where it comes to a stop.

First, let's consider the potential energy of the system. We need to define the constant in the potential energy function of [Equation 8.5](#). Often, the ground is a suitable choice for when the gravitational potential energy is zero; however, in this case, the highest point or when $y = 0$ is a convenient location for zero gravitational potential energy. Note that this choice is arbitrary, and the problem can be solved correctly even if another choice is picked.

We must also define the elastic potential energy of the system and the corresponding constant, as detailed in [Equation 8.7](#). This is where the spring is unstretched, or at the $y = 0$ position.

If we consider that the total energy of the system is conserved, then the energy at point A equals point C. The block is placed just on the spring so its initial kinetic energy is zero. By the setup of the problem discussed previously, both the gravitational potential energy and elastic potential energy are equal to zero. Therefore, the initial energy of the system is zero. When the block arrives at point C, its kinetic energy is zero. However, it now has both gravitational potential energy and elastic potential energy. Therefore, we can solve for the distance y that the block travels before coming to a stop:

$$\begin{aligned}
 K_A + U_A &= K_C + U_C \\
 0 &= 0 + mgy_C + \frac{1}{2}k(y_C)^2 \\
 y_C &= \frac{-2mg}{k}
 \end{aligned}$$



FIGURE 8.5 A bungee jumper transforms gravitational potential energy at the start of the jump into elastic potential energy at the bottom of the jump.



EXAMPLE 8.4

Potential Energy of a Vertical Mass-Spring System

A block weighing 1.2 N is hung from a spring with a spring constant of 6.0 N/m, as shown in [Figure 8.4](#). (a) What is the maximum expansion of the spring, as seen at point C? (b) What is the total potential energy at point B, halfway between A and C? (c) What is the speed of the block at point B?

Strategy

In part (a) we calculate the distance y_C as discussed in the previous text. Then in part (b), we use half of the y value to calculate the potential energy at point B using equations [Equation 8.4](#) and [Equation 8.6](#). This energy must be equal to the kinetic energy, [Equation 7.6](#), at point B since the initial energy of the system is zero. By calculating the kinetic energy at point B, we can now calculate the speed of the block at point B.

Solution

- a. Since the total energy of the system is zero at point A as discussed previously, the maximum expansion of the spring is calculated to be:

$$y_C = \frac{-2mg}{k}$$

$$y_C = \frac{-2(1.2 \text{ N})}{(6.0 \text{ N/m})} = -0.40 \text{ m}$$

- b. The position of y_B is half of the position at y_C or $y_B = y_C/2 = -mg/k$. The total potential energy at point B would therefore be:

$$U_B = mg(y_C/2) + \frac{1}{2}k(y_C/2)^2$$

$$U_B = mg(-mg/k) + (\frac{1}{2})k(-mg/k)^2$$

$$U_B = -(mg)^2/(2k)$$

$$U_B = (1.2 \text{ N})(-0.20 \text{ m}) + \frac{1}{2}(6 \text{ N/m})(-0.20 \text{ m})^2$$

$$U_B = -0.12 \text{ J where we have used } y_C = -2mg/k \text{ from part (a)}$$

- c. The mass of the block is the weight divided by gravity.

$$m = \frac{F_w}{g} = \frac{1.2 \text{ N}}{9.8 \text{ m/s}^2} = 0.12 \text{ kg}$$

The kinetic energy at point B therefore is 0.12 J because the total energy is zero. Therefore, the speed of the block at point B is equal to

$$K = \frac{1}{2}mv^2$$

$$v = \sqrt{\frac{2K}{m}} = \sqrt{\frac{2(0.12 \text{ J})}{(0.12 \text{ kg})}} = 1.4 \text{ m/s}$$

Significance

Even though the potential energy due to gravity is relative to a chosen zero location, the solutions to this problem would be the same if the zero energy points were chosen at different locations.

✓ CHECK YOUR UNDERSTANDING 8.4

Suppose the mass in [Equation 8.6](#) is doubled while keeping the all other conditions the same. Would the maximum expansion of the spring increase, decrease, or remain the same? Would the speed at point B be larger, smaller, or the same compared to the original mass?

🔗 INTERACTIVE

Engage the simulation below to learn about conservation of energy with a skater! Build tracks, ramps and jumps for the skater and view the kinetic energy, potential energy and friction as he moves. You can also take the skater to different planets or even space!

[Access multimedia content \(https://openstax.org/books/university-physics-volume-1/pages/8-1-potential-energy-of-a-system\)](https://openstax.org/books/university-physics-volume-1/pages/8-1-potential-energy-of-a-system)

A sample chart of a variety of energies is shown in [Table 8.1](#) to give you an idea about typical energy values associated with certain events. Some of these are calculated using kinetic energy, whereas others are calculated by using quantities found in a form of potential energy that may not have been discussed at this point.

Object/phenomenon	Energy in joules
Big Bang	10^{68}
Annual world energy use	4.0×10^{20}
Large fusion bomb (9 megaton)	3.8×10^{16}
Hiroshima-size fission bomb (10 kiloton)	4.2×10^{13}
1 barrel crude oil	5.9×10^9
1 metric ton TNT	4.2×10^9
1 gallon of gasoline	1.2×10^8
Daily adult food intake (recommended)	1.2×10^7
1000-kg car at 90 km/h	3.1×10^5

TABLE 8.1 Energy of Various Objects and Phenomena

Object/phenomenon	Energy in joules
Tennis ball at 100 km/h	22
Mosquito (10^{-2} g at 0.5 m/s)	1.3×10^{-6}
Single electron in a TV tube beam	4.0×10^{-15}
Energy to break one DNA strand	10^{-19}
1 kg hydrogen (fusion to helium)	6.4×10^{14}
1 kg uranium (nuclear fission)	8.0×10^{13}

TABLE 8.1 Energy of Various Objects and Phenomena

8.2 Conservative and Non-Conservative Forces

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Characterize a conservative force in several different ways
- Specify mathematical conditions that must be satisfied by a conservative force and its components
- Relate the conservative force between particles of a system to the potential energy of the system
- Calculate the components of a conservative force in various cases

In the interactions we considered in [Potential Energy and Conservation of Energy](#), any transition between kinetic and potential energy did not change the total energy of the system. This was path independent, meaning that we can start and stop at any two points in the problem, and the total energy of the system—kinetic plus potential—at these points are equal to each other. This is characteristic of a **conservative force**. We dealt with conservative forces in the preceding section, such as the gravitational force and spring force. When comparing the motion of the football in [Figure 8.2](#), the total energy of the system never changes, even though the gravitational potential energy of the football increases, as the ball rises relative to ground and falls back to the initial gravitational potential energy when the football player catches the ball. **Non-conservative forces** are dissipative forces such as friction or air resistance. These forces take energy away from the system as the system progresses, energy that you can't get back. These forces are path dependent; therefore it matters where the object starts and stops.

CONSERVATIVE FORCE

The work done by a conservative force is independent of the path; in other words, the work done by a conservative force is the same for any path connecting two points:

$$W_{AB, \text{path-1}} = \int_{AB, \text{path-1}} \vec{\mathbf{F}}_{\text{cons}} \cdot d\vec{\mathbf{r}} = W_{AB, \text{path-2}} = \int_{AB, \text{path-2}} \vec{\mathbf{F}}_{\text{cons}} \cdot d\vec{\mathbf{r}}. \quad 8.8$$

The work done by a non-conservative force depends on the path taken.

Equivalently, a force is conservative if the work it does around any closed path is zero:

$$W_{\text{closed path}} = \oint \vec{\mathbf{F}}_{\text{cons}} \cdot d\vec{\mathbf{r}} = 0. \quad 8.9$$

[In [Equation 8.9](#), we use the notation of a circle in the middle of the integral sign for a line integral over a closed path, a notation found in most physics and engineering texts.] [Equation 8.8](#) and [Equation 8.9](#) are equivalent because any closed path is the sum of two paths: the first going from A to B, and the second going from B to A. The work done going along a path from B to A is the negative of the work done going along the same path from A to B, where A

and B are any two points on the closed path:

$$\begin{aligned} 0 &= \int \vec{\mathbf{F}}_{\text{cons}} \cdot d\vec{\mathbf{r}} = \int_{AB, \text{path-1}} \vec{\mathbf{F}}_{\text{cons}} \cdot d\vec{\mathbf{r}} + \int_{BA, \text{path-2}} \vec{\mathbf{F}}_{\text{cons}} \cdot d\vec{\mathbf{r}} \\ &= \int_{AB, \text{path-1}} \vec{\mathbf{F}}_{\text{cons}} \cdot d\vec{\mathbf{r}} - \int_{AB, \text{path-2}} \vec{\mathbf{F}}_{\text{cons}} \cdot d\vec{\mathbf{r}} = 0. \end{aligned}$$

You might ask how we go about proving whether or not a force is conservative, since the definitions involve any and all paths from A to B , or any and all closed paths, but to do the integral for the work, you have to choose a particular path. One answer is that the work done is independent of path if the infinitesimal work $\vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$ is an **exact differential**, the way the infinitesimal net work was equal to the exact differential of the kinetic energy, $dW_{\text{net}} = m\vec{\mathbf{v}} \cdot d\vec{\mathbf{v}} = d\left(\frac{1}{2}mv^2\right)$,

when we derived the work-energy theorem in [Work-Energy Theorem](#). There are mathematical conditions that you can use to test whether the infinitesimal work done by a force is an exact differential, and the force is conservative. These conditions only involve differentiation and are thus relatively easy to apply. In two dimensions, the condition for $\vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = F_x dx + F_y dy$ to be an exact differential is

$$\frac{dF_x}{dy} = \frac{dF_y}{dx}. \quad \mathbf{8.10}$$

You may recall that the work done by the force in [Example 7.4](#) depended on the path. For that force,

$$F_x = (5 \text{ N/m}) y \text{ and } F_y = (10 \text{ N/m}) x.$$

Therefore,

$$(dF_x/dy) = 5 \text{ N/m} \neq (dF_y/dx) = 10 \text{ N/m},$$

which indicates it is a non-conservative force. Can you see what you could change to make it a conservative force?



FIGURE 8.6 A grinding wheel applies a non-conservative force, because the work done depends on how many rotations the wheel makes, so it is path-dependent. (credit: modification of work by Grantez Stephens, U.S. Navy)



EXAMPLE 8.5

Conservative or Not?

Which of the following two-dimensional forces are conservative and which are not? Assume a and b are constants with appropriate units:

(a) $axy^3\hat{\mathbf{i}} + ayx^3\hat{\mathbf{j}}$, (b) $a \left[(y^2/x)\hat{\mathbf{i}} + 2y \ln(x/b)\hat{\mathbf{j}} \right]$, (c) $\frac{ax\hat{\mathbf{i}} + ay\hat{\mathbf{j}}}{x^2 + y^2}$

Strategy

Apply the condition stated in [Equation 8.10](#), namely, using the derivatives of the components of each force indicated. If the derivative of the y -component of the force with respect to x is equal to the derivative of the x -component of the force with respect to y , the force is a conservative force, which means the path taken for potential energy or work calculations always yields the same results.

Solution

- a. $\frac{dF_x}{dy} = \frac{d(axy^3)}{dy} = 3axy^2$ and $\frac{dF_y}{dx} = \frac{d(ayx^3)}{dx} = 3ayx^2$, so this force is non-conservative.
- b. $\frac{dF_x}{dy} = \frac{d(ay^2/x)}{dy} = \frac{2ay}{x}$ and $\frac{dF_y}{dx} = \frac{d(2ay \ln(x/b))}{dx} = \frac{2ay}{x}$, so this force is conservative.
- c. $\frac{dF_x}{dy} = \frac{d(ax/(x^2+y^2))}{dy} = -\frac{ax(2y)}{(x^2+y^2)^2} = \frac{dF_y}{dx} = \frac{d(ay/(x^2+y^2))}{dx}$, again conservative.

Significance

The conditions in [Equation 8.10](#) are derivatives as functions of a single variable; in three dimensions, similar conditions exist that involve more derivatives.

✓ CHECK YOUR UNDERSTANDING 8.5

A two-dimensional, conservative force is zero on the x - and y -axes, and satisfies the condition $(dF_x/dy) = (dF_y/dx) = (4 \text{ N/m}^3) xy$. What is the magnitude of the force at the point $x = y = 1 \text{ m}$?

Before leaving this section, we note that non-conservative forces do not have potential energy associated with them because the energy is lost to the system and can't be turned into useful work later. So there is always a conservative force associated with every potential energy. We have seen that potential energy is defined in relation to the work done by conservative forces. That relation, [Equation 8.1](#), involved an integral for the work; starting with the force and displacement, you integrated to get the work and the change in potential energy. However, integration is the inverse operation of differentiation; you could equally well have started with the potential energy and taken its derivative, with respect to displacement, to get the force. The infinitesimal increment of potential energy is the dot product of the force and the infinitesimal displacement,

$$dU = -\vec{F} \cdot d\vec{l} = -F_l dl.$$

Here, we chose to represent the displacement in an arbitrary direction by $d\vec{l}$, so as not to be restricted to any particular coordinate direction. We also expressed the dot product in terms of the magnitude of the infinitesimal displacement and the component of the force in its direction. Both these quantities are scalars, so you can divide by dl to get

$$F_l = -\frac{dU}{dl}. \quad 8.11$$

This equation gives the relation between force and the potential energy associated with it. In words, the component of a conservative force, in a particular direction, equals the negative of the derivative of the corresponding potential energy, with respect to a displacement in that direction. For one-dimensional motion, say along the x -axis, [Equation 8.11](#) give the entire vector force, $\vec{F} = F_x \hat{i} = -\frac{\partial U}{\partial x} \hat{i}$.

In two dimensions,

$$\vec{F} = F_x \hat{i} + F_y \hat{j} = -\left(\frac{\partial U}{\partial x}\right) \hat{i} - \left(\frac{\partial U}{\partial y}\right) \hat{j}.$$

From this equation, you can see why [Equation 8.11](#) is the condition for the work to be an exact differential, in terms

of the derivatives of the components of the force. In general, a partial derivative notation is used. If a function has many variables in it, the derivative is taken only of the variable the partial derivative specifies. The other variables are held constant. In three dimensions, you add another term for the z-component, and the result is that the force is the negative of the gradient of the potential energy. However, we won't be looking at three-dimensional examples just yet.



EXAMPLE 8.6

Force due to a Quartic Potential Energy

The potential energy for a particle undergoing one-dimensional motion along the x-axis is

$$U(x) = \frac{1}{4}cx^4,$$

where $c = 8 \text{ N/m}^3$. Its total energy at $x = 0$ is 2 J, and it is not subject to any non-conservative forces. Find (a) the positions where its kinetic energy is zero and (b) the forces at those positions.

Strategy

(a) We can find the positions where $K = 0$, so the potential energy equals the total energy of the given system. (b) Using [Equation 8.11](#), we can find the force evaluated at the positions found from the previous part, since the mechanical energy is conserved.

Solution

- a. The total energy of the system of 2 J equals the quartic elastic energy as given in the problem,

$$2 \text{ J} = \frac{1}{4}(8 \text{ N/m}^3) x_f^4.$$

Solving for x_f results in $x_f = \pm 1 \text{ m}$.

- b. From [Equation 8.11](#),

$$F_x = -dU/dx = -cx^3.$$

Thus, evaluating the force at $\pm 1 \text{ m}$, we get

$$\vec{F} = -(8 \text{ N/m}^3)(\pm 1 \text{ m})^3 \hat{i} = \pm 8 \text{ N} \hat{i}.$$

At both positions, the magnitude of the forces is 8 N and the directions are toward the origin, since this is the potential energy for a restoring force.

Significance

Finding the force from the potential energy is mathematically easier than finding the potential energy from the force, because differentiating a function is generally easier than integrating one.



CHECK YOUR UNDERSTANDING 8.6

Find the forces on the particle in [Example 8.6](#) when its kinetic energy is 1.0 J at $x = 0$.

8.3 Conservation of Energy

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Formulate the principle of conservation of mechanical energy, with or without the presence of non-conservative forces
- Use the conservation of mechanical energy to calculate various properties of simple systems

In this section, we elaborate and extend the result we derived in [Potential Energy of a System](#), where we re-wrote the work-energy theorem in terms of the change in the kinetic and potential energies of a particle. This will lead us to a discussion of the important principle of the conservation of mechanical energy. As you continue to examine

other topics in physics, in later chapters of this book, you will see how this conservation law is generalized to encompass other types of energy and energy transfers. The last section of this chapter provides a preview.

The terms ‘conserved quantity’ and ‘conservation law’ have specific, scientific meanings in physics, which are different from the everyday meanings associated with the use of these words. (The same comment is also true about the scientific and everyday uses of the word ‘work.’) In everyday usage, you could conserve water by not using it, or by using less of it, or by re-using it. Water is composed of molecules consisting of two atoms of hydrogen and one of oxygen. Bring these atoms together to form a molecule and you create water; dissociate the atoms in such a molecule and you destroy water. However, in scientific usage, a **conserved quantity** for a system stays constant, changes by a definite amount that is transferred to other systems, and/or is converted into other forms of that quantity. A conserved quantity, in the scientific sense, can be transformed, but not strictly created or destroyed. Thus, there is no physical law of conservation of water.

Systems with a Single Particle or Object

We first consider a system with a single particle or object. Returning to our development of [Equation 8.2](#), recall that we first separated all the forces acting on a particle into conservative and non-conservative types, and wrote the work done by each type of force as a separate term in the work-energy theorem. We then replaced the work done by the conservative forces by the change in the potential energy of the particle, combining it with the change in the particle’s kinetic energy to get [Equation 8.2](#). Now, we write this equation without the middle step and define the sum of the kinetic and potential energies, $K + U = E$; to be the **mechanical energy** of the particle.

CONSERVATION OF ENERGY

The mechanical energy E of a particle stays constant unless forces outside the system or non-conservative forces do work on it, in which case, the change in the mechanical energy is equal to the work done by the non-conservative forces:

$$W_{\text{nc},AB} = \Delta(K + U)_{AB} = \Delta E_{AB}. \quad 8.12$$

This statement expresses the concept of **energy conservation** for a classical particle as long as there is no non-conservative work. Recall that a classical particle is just a point mass, is nonrelativistic, and obeys Newton’s laws of motion. In [Relativity \(http://openstax.org/books/university-physics-volume-3/pages/5-introduction\)](http://openstax.org/books/university-physics-volume-3/pages/5-introduction), we will see that conservation of energy still applies to a non-classical particle, but for that to happen, we have to make a slight adjustment to the definition of energy.

It is sometimes convenient to separate the case where the work done by non-conservative forces is zero, either because no such forces are assumed present, or, like the normal force, they do zero work when the motion is parallel to the surface. Then

$$0 = W_{\text{nc},AB} = \Delta(K + U)_{AB} = \Delta E_{AB}. \quad 8.13$$

In this case, the conservation of mechanical energy can be expressed as follows: The mechanical energy of a particle does not change if all the non-conservative forces that may act on it do no work. Understanding the concept of energy conservation is the important thing, not the particular equation you use to express it.



PROBLEM-SOLVING STRATEGY

Conservation of Energy

1. Identify the body or bodies to be studied (the system). Often, in applications of the principle of mechanical energy conservation, we study more than one body at the same time.
2. Identify all forces acting on the body or bodies.
3. Determine whether each force that does work is conservative. If a non-conservative force (e.g., friction) is doing work, then mechanical energy is not conserved. The system must then be analyzed with non-

conservative work, [Equation 8.12](#).

4. For every conservative force that does work, choose a reference point and determine the potential energy function for the force. The reference points for the various potential energies do not have to be at the same location.
5. If no non-conservative work is done, apply the principle of mechanical energy conservation by setting the sum of the kinetic energies and potential energies equal at every point of interest.

EXAMPLE 8.7

Simple Pendulum

A particle of mass m is hung from the ceiling by a massless string of length 1.0 m, as shown in [Figure 8.7](#). The particle is released from rest, when the angle between the string and the downward vertical direction is 30° . What is its speed when it reaches the lowest point of its arc?

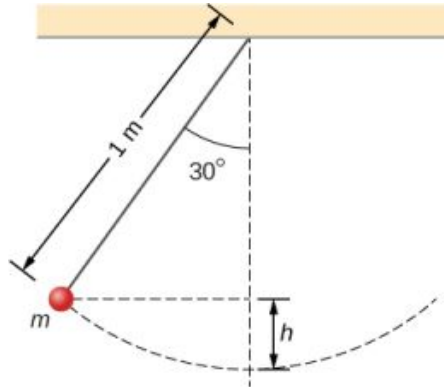


FIGURE 8.7 A particle hung from a string constitutes a simple pendulum. It is shown when released from rest, along with some distances used in analyzing the motion.

Strategy

Using our problem-solving strategy, the first step is to define that we are interested in the particle-Earth system. Second, only the gravitational force is acting on the particle, which is conservative (step 3). We neglect air resistance in the problem, and no work is done by the string tension, which is perpendicular to the arc of the motion. Therefore, the mechanical energy of the system is conserved, as represented by [Equation 8.13](#), $0 = \Delta(K + U)$. Because the particle starts from rest, the increase in the kinetic energy is just the kinetic energy at the lowest point. This increase in kinetic energy equals the decrease in the gravitational potential energy, which we can calculate from the geometry. In step 4, we choose a reference point for zero gravitational potential energy to be at the lowest vertical point the particle achieves, which is mid-swing. Lastly, in step 5, we set the sum of energies at the highest point (initial) of the swing to the lowest point (final) of the swing to ultimately solve for the final speed.

Solution

We are neglecting non-conservative forces, so we write the energy conservation formula relating the particle at the highest point (initial) and the lowest point in the swing (final) as

$$K_i + U_i = K_f + U_f.$$

Since the particle is released from rest, the initial kinetic energy is zero. At the lowest point, we define the gravitational potential energy to be zero. Therefore our conservation of energy formula reduces to

$$\begin{aligned} 0 + mgh &= \frac{1}{2}mv^2 + 0 \\ v &= \sqrt{2gh}. \end{aligned}$$

The vertical height of the particle is not given directly in the problem. This can be solved for by using trigonometry and two givens: the length of the pendulum and the angle through which the particle is vertically pulled up. Looking at the diagram, the vertical dashed line is the length of the pendulum string. The vertical height is labeled h . The

other partial length of the vertical string can be calculated with trigonometry. That piece is solved for by

$$\cos \theta = x/L, x = L \cos \theta.$$

Therefore, by looking at the two parts of the string, we can solve for the height h ,

$$\begin{aligned} x + h &= L \\ L \cos \theta + h &= L \\ h &= L - L \cos \theta = L(1 - \cos \theta). \end{aligned}$$

We substitute this height into the previous expression solved for speed to calculate our result:

$$v = \sqrt{2gL(1 - \cos \theta)} = \sqrt{2(9.8 \text{ m/s}^2)(1 \text{ m})(1 - \cos 30^\circ)} = 1.62 \text{ m/s}.$$

Significance

We found the speed directly from the conservation of mechanical energy, without having to solve the differential equation for the motion of a pendulum (see [Oscillations](#)). We can approach this problem in terms of bar graphs of total energy. Initially, the particle has all potential energy, being at the highest point, and no kinetic energy. When the particle crosses the lowest point at the bottom of the swing, the energy moves from the potential energy column to the kinetic energy column. Therefore, we can imagine a progression of this transfer as the particle moves between its highest point, lowest point of the swing, and back to the highest point ([Figure 8.8](#)). As the particle travels from the lowest point in the swing to the highest point on the far right hand side of the diagram, the energy bars go in reverse order from (c) to (b) to (a).

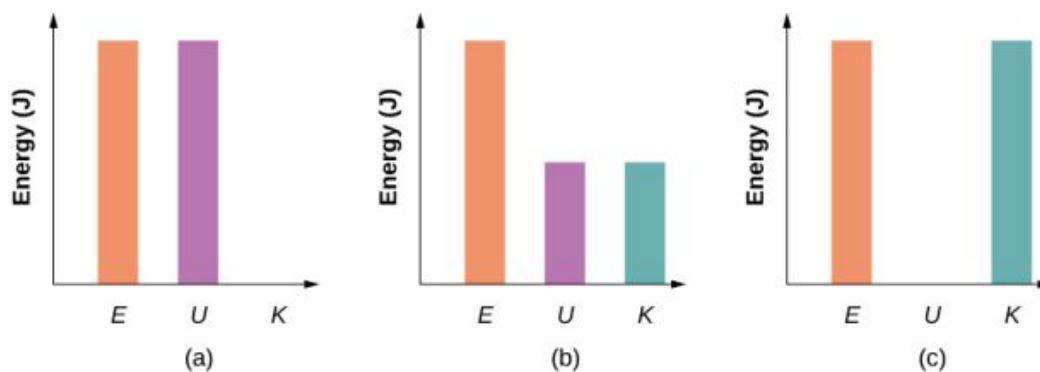


FIGURE 8.8 Bar graphs representing the total energy (E), potential energy (U), and kinetic energy (K) of the particle in different positions. (a) The total energy of the system equals the potential energy and the kinetic energy is zero, which is found at the highest point the particle reaches. (b) The particle is midway between the highest and lowest point, so the kinetic energy plus potential energy bar graphs equal the total energy. (c) The particle is at the lowest point of the swing, so the kinetic energy bar graph is the highest and equal to the total energy of the system.

✓ CHECK YOUR UNDERSTANDING 8.7

How high above the bottom of its arc is the particle in the simple pendulum above, when its speed is 0.81 m/s?

EXAMPLE 8.8

Air Resistance on a Falling Object

A helicopter is hovering at an altitude of 1 km when a panel from its underside breaks loose and plummets to the ground ([Figure 8.9](#)). The mass of the panel is 15 kg, and it hits the ground with a speed of 45 m/s. How much mechanical energy was dissipated by air resistance during the panel's descent?

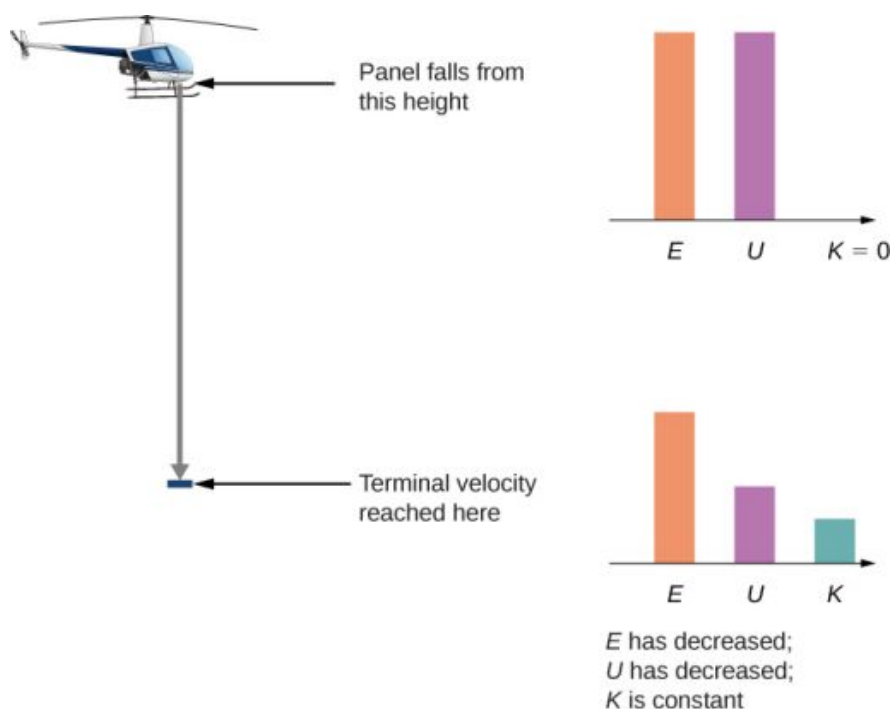


FIGURE 8.9 A helicopter loses a panel that falls until it reaches terminal velocity of 45 m/s. How much did air resistance contribute to the dissipation of energy in this problem?

Strategy

Step 1: Here only one body is being investigated.

Step 2: Gravitational force is acting on the panel, as well as air resistance, which is stated in the problem.

Step 3: Gravitational force is conservative; however, the non-conservative force of air resistance does negative work on the falling panel, so we can use the conservation of mechanical energy, in the form expressed by [Equation 8.12](#), to find the energy dissipated. This energy is the magnitude of the work:

$$\Delta E_{\text{diss}} = |W_{\text{nc,if}}| = |\Delta(K + U)_{\text{if}}|.$$

Step 4: The initial kinetic energy, at $y_i = 1 \text{ km}$, is zero. We set the gravitational potential energy to zero at ground level out of convenience.

Step 5: The non-conservative work is set equal to the energies to solve for the work dissipated by air resistance.

Solution

The mechanical energy dissipated by air resistance is the algebraic sum of the gain in the kinetic energy and loss in potential energy. Therefore the calculation of this energy is

$$\begin{aligned} \Delta E_{\text{diss}} &= |K_f - K_i + U_f - U_i| \\ &= \left| \frac{1}{2}(15 \text{ kg})(45 \text{ m/s})^2 - 0 + 0 - (15 \text{ kg})(9.8 \text{ m/s}^2)(1000 \text{ m}) \right| = 130 \text{ kJ}. \end{aligned}$$

Significance

Most of the initial mechanical energy of the panel (U_i), 147 kJ, was lost to air resistance. Notice that we were able to calculate the energy dissipated without knowing what the force of air resistance was, only that it was dissipative.

CHECK YOUR UNDERSTANDING 8.8

You probably recall that, neglecting air resistance, if you throw a projectile straight up, the time it takes to reach its maximum height equals the time it takes to fall from the maximum height back to the starting height. Suppose you

cannot neglect air resistance, as in [Example 8.8](#). Is the time the projectile takes to go up (a) greater than, (b) less than, or (c) equal to the time it takes to come back down? Explain.

In these examples, we were able to use conservation of energy to calculate the speed of a particle just at particular points in its motion. But the method of analyzing particle motion, starting from energy conservation, is more powerful than that. More advanced treatments of the theory of mechanics allow you to calculate the full time dependence of a particle's motion, for a given potential energy. In fact, it is often the case that a better model for particle motion is provided by the form of its kinetic and potential energies, rather than an equation for force acting on it. (This is especially true for the quantum mechanical description of particles like electrons or atoms.)

We can illustrate some of the simplest features of this energy-based approach by considering a particle in one-dimensional motion, with potential energy $U(x)$ and no non-conservative interactions present. [Equation 8.12](#) and the definition of velocity require

$$K = \frac{1}{2}mv^2 = E - U(x)$$

$$v = \frac{dx}{dt} = \sqrt{\frac{2(E - U(x))}{m}}.$$

Separate the variables x and t and integrate, from an initial time $t = 0$ to an arbitrary time, to get

$$t = \int_0^t dt = \int_{x_0}^x \frac{dx}{\sqrt{2[E - U(x)]/m}}. \quad 8.14$$

If you can do the integral in [Equation 8.14](#), then you can solve for x as a function of t .



EXAMPLE 8.9

Constant Acceleration

Use the potential energy $U(x) = E(x/x_0)$, for $E > 0$, in [Equation 8.14](#) to find the position x of a particle as a function of time t .

Strategy

Since we know how the potential energy changes as a function of x , we can substitute for $U(x)$ in [Equation 8.14](#), integrate, and then solve for x . This results in an expression of x as a function of time with constants of energy E , mass m , and the initial position x_0 .

Solution

Following the first two suggested steps in the above strategy,

$$t = \int_{x_0}^x \frac{dx}{\sqrt{(2E/mx_0)(x_0 - x)}} = \frac{1}{\sqrt{(2E/mx_0)}} \left| -2\sqrt{(x_0 - x)} \right|_{x_0}^x = -\frac{2\sqrt{(x_0 - x)}}{\sqrt{(2E/mx_0)}}.$$

Solving for the position, we obtain $x(t) = x_0 - \frac{1}{2}(E/mx_0)t^2$.

Significance

The position as a function of time, for this potential, represents one-dimensional motion with constant acceleration, $a = (-E/mx_0)$, starting at rest from position x_0 . This is not so surprising, since this is a potential energy for a constant force, $F = -dU/dx = -E/x_0$, and $a = F/m$.



CHECK YOUR UNDERSTANDING 8.9

What potential energy $U(x)$ can you substitute in [Equation 8.13](#) that will result in motion with constant velocity of 2

m/s for a particle of mass 1 kg and mechanical energy 1 J?

We will look at another more physically appropriate example of the use of [Equation 8.13](#) after we have explored some further implications that can be drawn from the functional form of a particle's potential energy.

Systems with Several Particles or Objects

Systems generally consist of more than one particle or object. However, the conservation of mechanical energy, in one of the forms in [Equation 8.12](#) or [Equation 8.13](#), is a fundamental law of physics and applies to any system. You just have to include the kinetic and potential energies of all the particles, and the work done by all the non-conservative forces acting on them. Until you learn more about the dynamics of systems composed of many particles, in [Linear Momentum and Collisions](#), [Fixed-Axis Rotation](#), and [Angular Momentum](#), it is better to postpone discussing the application of energy conservation to then.

8.4 Potential Energy Diagrams and Stability

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Create and interpret graphs of potential energy
- Explain the connection between stability and potential energy

Often, you can get a good deal of useful information about the dynamical behavior of a mechanical system just by interpreting a graph of its potential energy as a function of position, called a **potential energy diagram**. This is most easily accomplished for a one-dimensional system, whose potential energy can be plotted in one two-dimensional graph—for example, $U(x)$ versus x —on a piece of paper or a computer program. For systems whose motion is in more than one dimension, the motion needs to be studied in three-dimensional space. We will simplify our procedure for one-dimensional motion only.

First, let's look at an object, freely falling vertically, near the surface of Earth, in the absence of air resistance. The mechanical energy of the object is conserved, $E = K + U$, and the potential energy, with respect to zero at ground level, is $U(y) = mgy$, which is a straight line through the origin with slope mg . In the graph shown in [Figure 8.10](#), the x -axis is the height above the ground y and the y -axis is the object's energy.

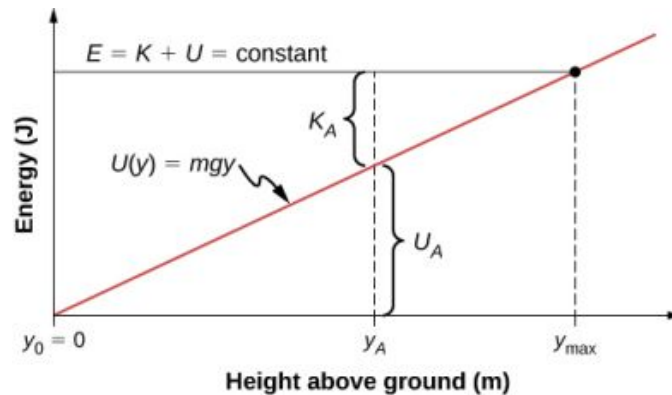


FIGURE 8.10 The potential energy graph for an object in vertical free fall, with various quantities indicated.

The line at energy E represents the constant mechanical energy of the object, whereas the kinetic and potential energies, K_A and U_A , are indicated at a particular height y_A . You can see how the total energy is divided between kinetic and potential energy as the object's height changes. Since kinetic energy can never be negative, there is a maximum potential energy and a maximum height, which an object with the given total energy cannot exceed:

$$K = E - U \geq 0,$$

$$U \leq E.$$

If we use the gravitational potential energy reference point of zero at y_0 , we can rewrite the gravitational potential energy U as mgy . Solving for y results in

$$y \leq E/mg = y_{\max}.$$

We note in this expression that the quantity of the total energy divided by the weight (mg) is located at the maximum height of the particle, or $y_{\max}$. At the maximum height, the kinetic energy and the speed are zero, so if the object were initially traveling upward, its velocity would go through zero there, and $y_{\max}$ would be a turning point in the motion. At ground level, $y_0 = 0$, the potential energy is zero, and the kinetic energy and the speed are maximum:

$$\begin{aligned} U_0 &= 0 = E - K_0, \\ E &= K_0 = \frac{1}{2}mv_0^2, \\ v_0 &= \pm\sqrt{2E/m}. \end{aligned}$$

The maximum speed $\pm v_0$ gives the initial velocity necessary to reach $y_{\max}$, the maximum height, and $-v_0$ represents the final velocity, after falling from $y_{\max}$. You can read all this information, and more, from the potential energy diagram we have shown.

Consider a mass-spring system on a frictionless, stationary, horizontal surface, so that gravity and the normal contact force do no work and can be ignored (Figure 8.11). This is like a one-dimensional system, whose mechanical energy E is a constant and whose potential energy, with respect to zero energy at zero displacement from the spring's unstretched length, $x = 0$, is $U(x) = \frac{1}{2}kx^2$.

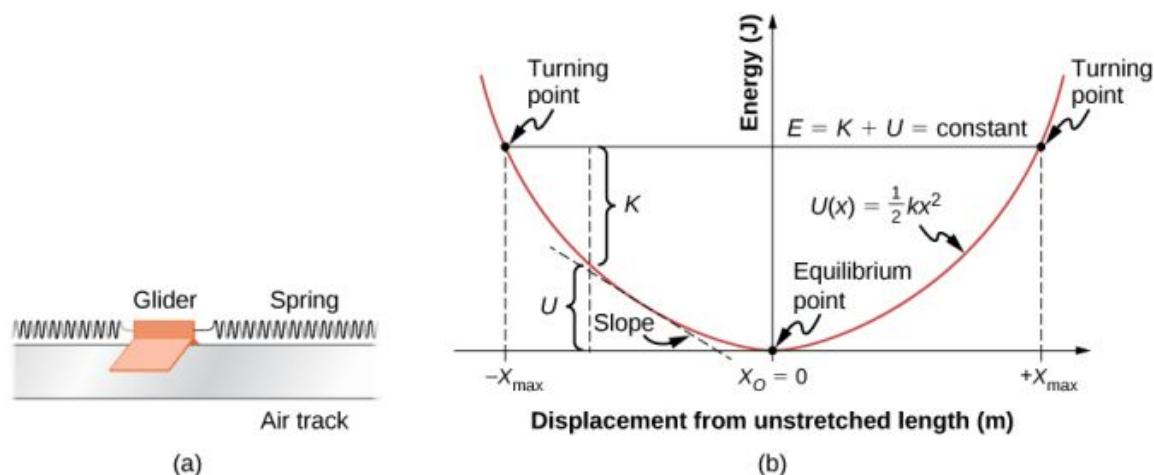


FIGURE 8.11 (a) A glider between springs on an air track is an example of a horizontal mass-spring system. (b) The potential energy diagram for this system, with various quantities indicated.

You can read off the same type of information from the potential energy diagram in this case, as in the case for the body in vertical free fall, but since the spring potential energy describes a variable force, you can learn more from this graph. As for the object in vertical free fall, you can deduce the physically allowable range of motion and the maximum values of distance and speed, from the limits on the kinetic energy, $0 \leq K \leq E$. Therefore, $K = 0$ and $U = E$ at a **turning point**, of which there are two for the elastic spring potential energy,

$$x_{\max} = \pm\sqrt{2E/k}.$$

The glider's motion is confined to the region between the turning points, $-x_{\max} \leq x \leq x_{\max}$. This is true for any (positive) value of E because the potential energy is unbounded with respect to x . For this reason, as well as the shape of the potential energy curve, $U(x)$ is called an infinite potential well. At the bottom of the potential well, $x = 0$, $U = 0$ and the kinetic energy is a maximum, $K = E$, so $v_{\max} = \pm\sqrt{2E/m}$.

However, from the slope of this potential energy curve, you can also deduce information about the force on the glider and its acceleration. We saw earlier that the negative of the slope of the potential energy is the spring force, which in this case is also the net force, and thus is proportional to the acceleration. When $x = 0$, the slope, the force, and the acceleration are all zero, so this is an **equilibrium point**. The negative of the slope, on either side of the equilibrium point, gives a force pointing back to the equilibrium point, $F = \pm kx$, so the equilibrium is termed stable and the force is called a restoring force. This implies that $U(x)$ has a relative minimum there. If the force on

either side of an equilibrium point has a direction that is the same as the direction of position change, pointing away from the equilibrium point, the equilibrium is termed unstable, and this implies that $U(x)$ has a relative maximum there.



EXAMPLE 8.10

Quartic and Quadratic Potential Energy Diagram

The potential energy for a particle undergoing one-dimensional motion along the x -axis is $U(x) = 2(x^4 - x^2)$, where U is in joules and x is in meters. The particle is not subject to any non-conservative forces and its mechanical energy is constant at $E = -0.25$ J. (a) Is the motion of the particle confined to any regions on the x -axis, and if so, what are they? (b) Are there any equilibrium points, and if so, where are they and are they stable or unstable?

Strategy

First, we need to graph the potential energy as a function of x . The function is zero at the origin, becomes negative as x increases in the positive or negative directions (x^2 is larger than x^4 for $x < 1$), and then becomes positive at sufficiently large $|x|$. Your graph should look like a double potential well, with the zeros determined by solving the equation $U(x) = 0$, and the extremes determined by examining the first and second derivatives of $U(x)$, as shown in Figure 8.12.

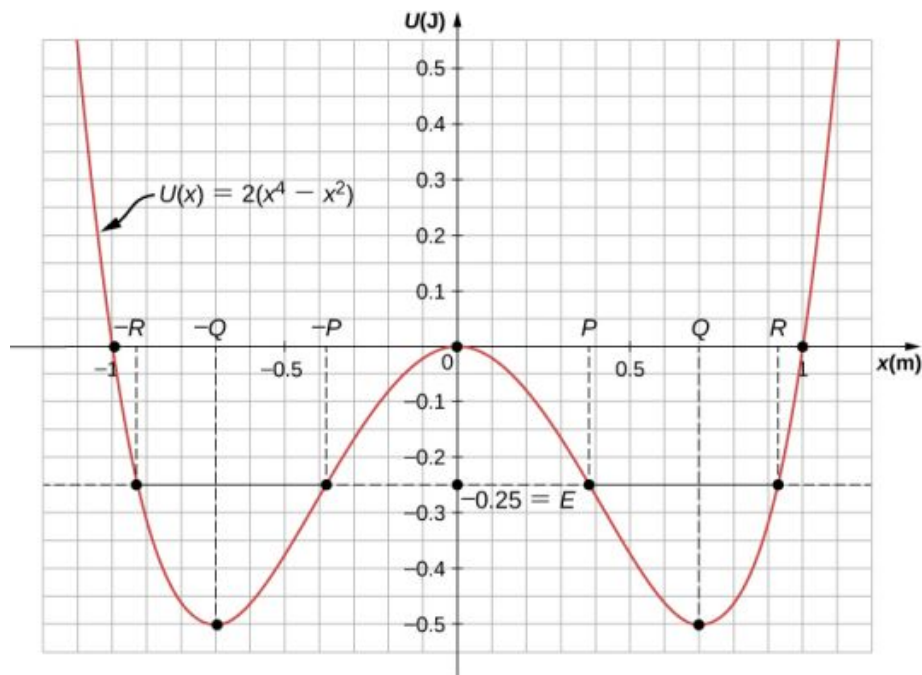


FIGURE 8.12 The potential energy graph for a one-dimensional, quartic and quadratic potential energy, with various quantities indicated.

You can find the values of (a) the allowed regions along the x -axis, for the given value of the mechanical energy, from the condition that the kinetic energy can't be negative, and (b) the equilibrium points and their stability from the properties of the force (stable for a relative minimum and unstable for a relative maximum of potential energy).

You can just eyeball the graph to reach qualitative answers to the questions in this example. That, after all, is the value of potential energy diagrams. You can see that there are two allowed regions for the motion ($E > U$) and three equilibrium points (slope $dU/dx = 0$), of which the central one is unstable ($d^2U/dx^2 < 0$), and the other two are stable ($d^2U/dx^2 > 0$).

Solution

- a. To find the allowed regions for x , we use the condition

$$K = E - U = -\frac{1}{4} - 2(x^4 - x^2) \geq 0.$$

If we complete the square in x^2 , this condition simplifies to $2\left(x^2 - \frac{1}{2}\right)^2 \leq \frac{1}{4}$, which we can solve to obtain

$$\frac{1}{2} - \sqrt{\frac{1}{8}} \leq x^2 \leq \frac{1}{2} + \sqrt{\frac{1}{8}}.$$

This represents two allowed regions, $x_p \leq x \leq x_R$ and $-x_R \leq x \leq -x_p$, where $x_p = 0.38$ and $x_R = 0.92$ (in meters).

- b. To find the equilibrium points, we solve the equation

$$dU/dx = 8x^3 - 4x = 0$$

and find $x = 0$ and $x = \pm x_Q$, where $x_Q = 1/\sqrt{2} = 0.707$ (meters). The second derivative

$$d^2U/dx^2 = 24x^2 - 4$$

is negative at $x = 0$, so that position is a relative maximum and the equilibrium there is unstable. The second derivative is positive at $x = \pm x_Q$, so these positions are relative minima and represent stable equilibria.

Significance

The particle in this example can oscillate in the allowed region about either of the two stable equilibrium points we found, but it does not have enough energy to escape from whichever potential well it happens to initially be in. The conservation of mechanical energy and the relations between kinetic energy and speed, and potential energy and force, enable you to deduce much information about the qualitative behavior of the motion of a particle, as well as some quantitative information, from a graph of its potential energy.

CHECK YOUR UNDERSTANDING 8.10

Repeat [Example 8.10](#) when the particle's mechanical energy is $+0.25$ J.

Before ending this section, let's practice applying the method based on the potential energy of a particle to find its position as a function of time, for the one-dimensional, mass-spring system considered earlier in this section.

EXAMPLE 8.11

Sinusoidal Oscillations

Find $x(t)$ for a particle moving with a constant mechanical energy $E > 0$ and a potential energy $U(x) = \frac{1}{2}kx^2$, when the particle starts from rest at time $t = 0$.

Strategy

We follow the same steps as we did in [Example 8.9](#). Substitute the potential energy U into [Equation 8.14](#) and factor out the constants, like m or k . Integrate the function and solve the resulting expression for position, which is now a function of time.

Solution

Substitute the potential energy in [Equation 8.14](#) and integrate using an integral solver found on a web search:

$$t = \int_{x_0}^x \frac{dx}{\sqrt{(k/m)[(2E/k) - x^2]}} = \sqrt{\frac{m}{k}} \left[\sin^{-1} \left(\frac{x}{\sqrt{2E/k}} \right) - \sin^{-1} \left(\frac{x_0}{\sqrt{2E/k}} \right) \right].$$

From the initial conditions at $t = 0$, the initial kinetic energy is zero and the initial potential energy is $\frac{1}{2}kx_0^2 = E$, from which you can see that $x_0/\sqrt{(2E/k)} = \pm 1$ and $\sin^{-1}(\pm 1) = \pm 90^\circ$. Now you can solve for x :

$$x(t) = \sqrt{(2E/k)} \sin \left[\left(\sqrt{k/m} \right) t \pm 90^\circ \right] = \pm \sqrt{(2E/k)} \cos \left[\left(\sqrt{k/m} \right) t \right].$$

Significance

A few paragraphs earlier, we referred to this mass-spring system as an example of a harmonic oscillator. Here, we anticipate that a harmonic oscillator executes sinusoidal oscillations with a maximum displacement of $\sqrt{2E/k}$ (called the amplitude) and a rate of oscillation of $(1/2\pi) \sqrt{k/m}$ (called the frequency). Further discussions about oscillations can be found in [Oscillations](#).

CHECK YOUR UNDERSTANDING 8.11

Find $x(t)$ for the mass-spring system in [Example 8.11](#) if the particle starts from $x_0 = 0$ at $t = 0$. What is the particle's initial velocity?

8.5 Sources of Energy

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Describe energy transformations and conversions in general terms
- Explain what it means for an energy source to be renewable or nonrenewable

In this chapter, we have studied energy. We learned that energy can take different forms and can be transferred from one form to another. You will find that energy is discussed in many everyday, as well as scientific, contexts, because it is involved in all physical processes. It will also become apparent that many situations are best understood, or most easily conceptualized, by considering energy. So far, no experimental results have contradicted the conservation of energy. In fact, whenever measurements have appeared to conflict with energy conservation, new forms of energy have been discovered or recognized in accordance with this principle.

What are some other forms of energy? Many of these are covered in later chapters (also see [Figure 8.13](#)), but let's detail a few here:

- Atoms and molecules inside all objects are in random motion. The internal kinetic energy from these random motions is called *thermal energy*, because it is related to the temperature of the object. In this case, the energy is known as *heat energy*.
- *Electrical energy* is a common form that is converted to many other forms and does work in a wide range of practical situations.
- Fuels, such as gasoline and food, have *chemical energy*, which is potential energy arising from their molecular structure. Chemical energy can be converted into thermal energy by reactions like oxidation. Chemical reactions can also produce electrical energy, such as in batteries. Electrical energy can, in turn, produce thermal energy and light, such as in an electric heater or a light bulb.
- Light is just one kind of electromagnetic radiation, or *radiant energy*, which also includes radio, infrared, ultraviolet, X-rays, and gamma rays. All bodies with thermal energy can radiate energy in electromagnetic waves.
- *Nuclear energy* comes from reactions and processes that convert measurable amounts of mass into energy. Nuclear energy is transformed into radiant energy in the Sun, into thermal energy in the boilers of nuclear power plants, and then into electrical energy in the generators of power plants. These and all other forms of energy can be transformed into one another and, to a certain degree, can be converted into mechanical work.

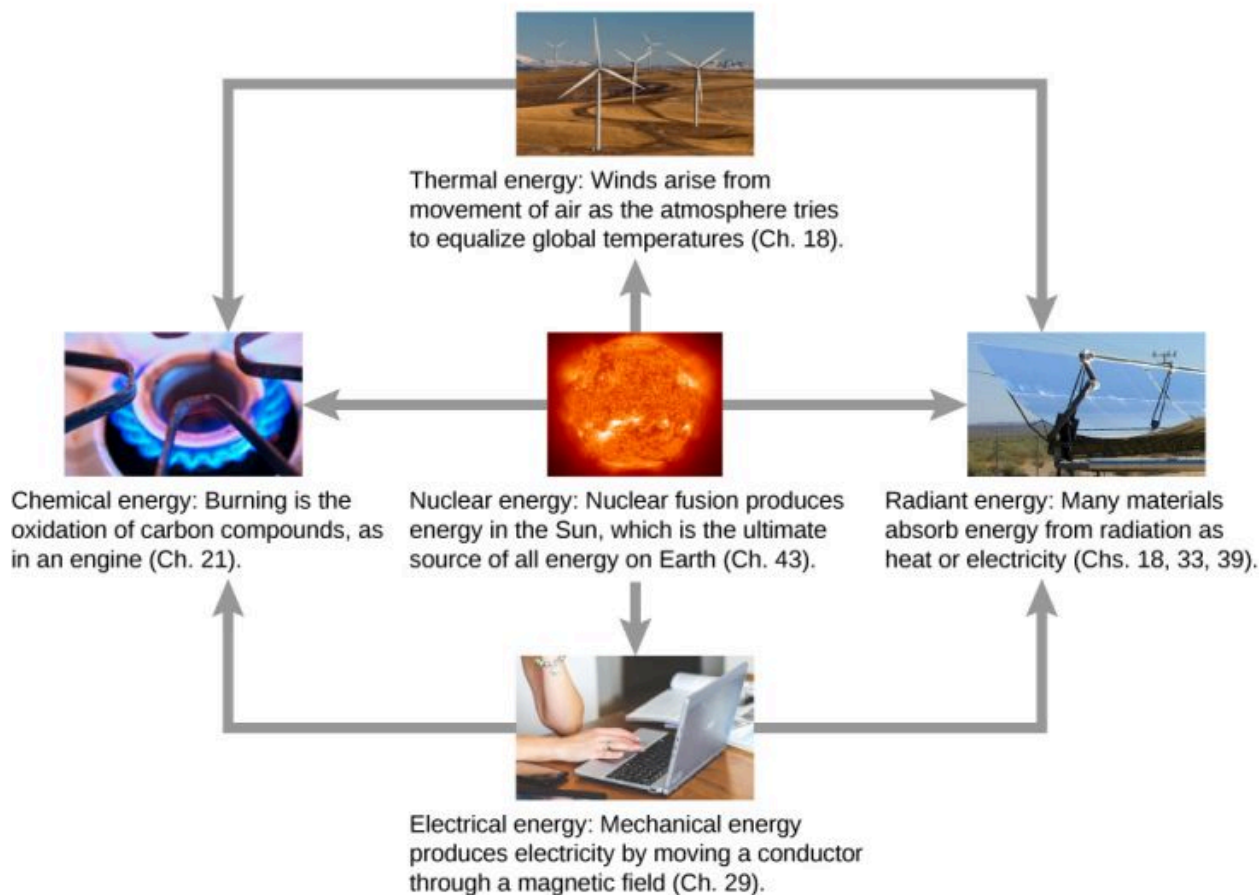


FIGURE 8.13 Energy that we use in society takes many forms, which be converted from one into another depending on the process involved. We will study many of these forms of energy in later chapters in this text. (credit "sun": modification of work by EIT - SOHO Consortium, ESA, NASA credit "solar panels": "modification of work by "kjkolb"/Wikimedia Commons; credit "gas burner": modification of work by Steven Depolo)

The transformation of energy from one form into another happens all the time. The chemical energy in food is converted into thermal energy through metabolism; light energy is converted into chemical energy through photosynthesis. Another example of energy conversion occurs in a solar cell. Sunlight impinging on a solar cell produces electricity, which can be used to run electric motors or heat water. In an example encompassing many steps, the chemical energy contained in coal is converted into thermal energy as it burns in a furnace, to transform water into steam, in a boiler. Some of the thermal energy in the steam is then converted into mechanical energy as it expands and spins a turbine, which is connected to a generator to produce electrical energy. In these examples, not all of the initial energy is converted into the forms mentioned, because some energy is always transferred to the environment.

Energy is an important element at all levels of society. We live in a very interdependent world, and access to adequate and reliable energy resources is crucial for economic growth and for maintaining the quality of our lives. The principal energy resources used in the world are shown in [Figure 8.14](#). The figure distinguishes between two major types of energy sources: **renewable** and **non-renewable**, and further divides each type into a few more specific kinds. Renewable sources are energy sources that are replenished through naturally occurring, ongoing processes, on a time scale that is much shorter than the anticipated lifetime of the civilization using the source. Non-renewable sources are depleted once some of the energy they contain is extracted and converted into other kinds of energy. The natural processes by which non-renewable sources are formed typically take place over geological time scales.

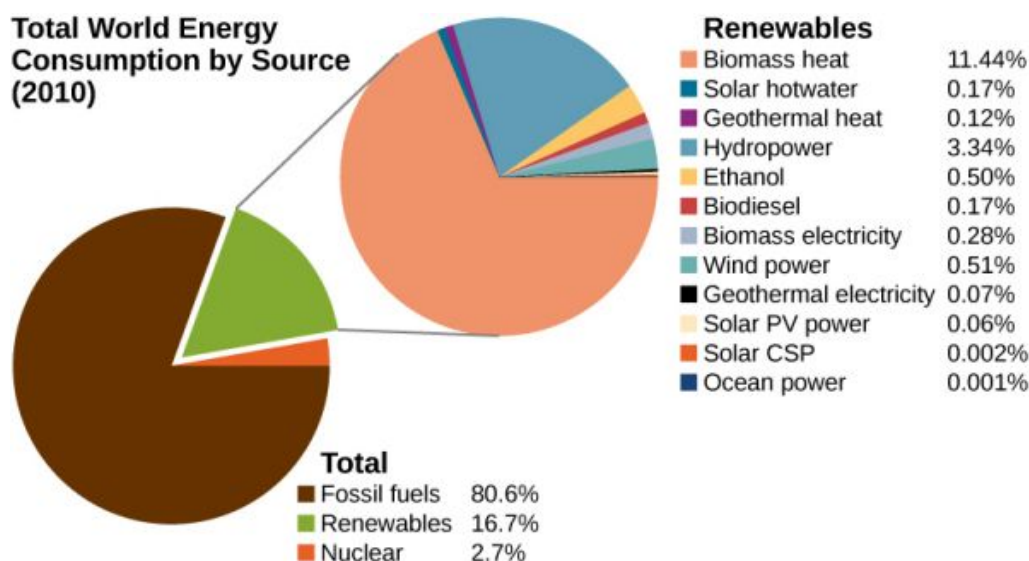


FIGURE 8.14 World energy consumption by source; the percentage of renewables is increasing, accounting for 19% in 2012.

Our most important non-renewable energy sources are fossil fuels, such as coal, petroleum, and natural gas. These account for about 81% of the world's energy consumption, as shown in the figure. Burning fossil fuels creates chemical reactions that transform potential energy, in the molecular structures of the reactants, into thermal energy and products. This thermal energy can be used to heat buildings or to operate steam-driven machinery. Internal combustion and jet engines convert some of the energy of rapidly expanding gases, released from burning gasoline, into mechanical work. Electrical power generation is mostly derived from transferring energy in expanding steam, via turbines, into mechanical work, which rotates coils of wire in magnetic fields to generate electricity. Nuclear energy is the other non-renewable source shown in [Figure 8.14](#) and supplies about 3% of the world's consumption. Nuclear reactions release energy by transforming potential energy, in the structure of nuclei, into thermal energy, analogous to energy release in chemical reactions. The thermal energy obtained from nuclear reactions can be transferred and converted into other forms in the same ways that energy from fossil fuels are used.

An unfortunate byproduct of relying on energy produced from the combustion of fossil fuels is the release of carbon dioxide into the atmosphere and its contribution to global warming. Nuclear energy poses environmental problems as well, including the safety and disposal of nuclear waste. Besides these important consequences, reserves of non-renewable sources of energy are limited and, given the rapidly growing rate of world energy consumption, may not last for more than a few hundred years. Considerable effort is going on to develop and expand the use of renewable sources of energy, involving a significant percentage of the world's physicists and engineers.

Four of the renewable energy sources listed in [Figure 8.14](#)—those using material from plants as fuel (biomass heat, ethanol, biodiesel, and biomass electricity)—involve the same types of energy transformations and conversions as just discussed for fossil and nuclear fuels. The other major types of renewable energy sources are hydropower, wind power, geothermal power, and solar power.

Hydropower is produced by converting the gravitational potential energy of falling or flowing water into kinetic energy and then into work to run electric generators or machinery. Converting the mechanical energy in ocean surface waves and tides is in development. Wind power also converts kinetic energy into work, which can be used directly to generate electricity, operate mills, and propel sailboats.

The interior of Earth has a great deal of thermal energy, part of which is left over from its original formation (gravitational potential energy converted into thermal energy) and part of which is released from radioactive minerals (a form of natural nuclear energy). It will take a very long time for this geothermal energy to escape into space, so people generally regard it as a renewable source, when actually, it's just inexhaustible on human time scales.

The source of solar power is energy carried by the electromagnetic waves radiated by the Sun. Most of this energy is carried by visible light and infrared (heat) radiation. When suitable materials absorb electromagnetic waves, radiant energy is converted into thermal energy, which can be used to heat water, or when concentrated, to make steam and

generate electricity ([Figure 8.15](#)). However, in another important physical process, known as the photoelectric effect, energetic radiation impinging on certain materials is directly converted into electricity. Materials that do this are called photovoltaics (PV in [Figure 8.14](#)). Some solar power systems use lenses or mirrors to concentrate the Sun's rays, before converting their energy through photovoltaics, and these are qualified as CSP in [Figure 8.14](#).



FIGURE 8.15 Solar cell arrays found in a sunny area converting the solar energy into stored electrical energy. (credit: modification of work by Sarah Swenty, U.S. Fish and Wildlife Service)

As we finish this chapter on energy and work, it is relevant to draw some distinctions between two sometimes misunderstood terms in the area of energy use. As we mentioned earlier, the “law of conservation of energy” is a very useful principle in analyzing physical processes. It cannot be proven from basic principles but is a very good bookkeeping device, and no exceptions have ever been found. It states that the total amount of energy in an isolated system always remains constant. Related to this principle, but remarkably different from it, is the important philosophy of energy conservation. This concept has to do with seeking to decrease the amount of energy used by an individual or group through reducing activities (e.g., turning down thermostats, driving fewer kilometers) and/or increasing conversion efficiencies in the performance of a particular task, such as developing and using more efficient room heaters, cars that have greater miles-per-gallon ratings, energy-efficient compact fluorescent lights, etc.

Since energy in an isolated system is not destroyed, created, or generated, you might wonder why we need to be concerned about our energy resources, since energy is a conserved quantity. The problem is that the final result of most energy transformations is waste heat, that is, work that has been “degraded” in the energy transformation. We will discuss this idea in more detail in the chapters on thermodynamics.

Chapter Review

Key Terms

conservative force force that does work independent of path

conserved quantity one that cannot be created or destroyed, but may be transformed between different forms of itself

energy conservation total energy of an isolated system is constant

equilibrium point position where the assumed conservative, net force on a particle, given by the slope of its potential energy curve, is zero

exact differential is the total differential of a function and requires the use of partial derivatives if the function involves more than one dimension

mechanical energy sum of the kinetic and potential energies

non-conservative force force that does work that depends on path

non-renewable energy source that is not renewable, but is depleted by human consumption

potential energy function of position, energy possessed by an object relative to the system considered

potential energy diagram graph of a particle's potential energy as a function of position

potential energy difference negative of the work done acting between two points in space

renewable energy source that is replenished by natural processes, over human time scales

turning point position where the velocity of a particle, in one-dimensional motion, changes sign

Key Equations

Difference of potential energy

Potential energy with respect to zero of potential energy at $\vec{r}_0$

Gravitational potential energy near Earth's surface

Potential energy for an ideal spring

Work done by conservative force over a closed path

Condition for conservative force in two dimensions

Conservative force is the negative derivative of potential energy

Conservation of energy with no non-conservative forces

$$\Delta U_{AB} = U_B - U_A = -W_{AB}$$

$$\Delta U = U(\vec{r}) - U(\vec{r}_0)$$

$$U(y) = mgy + \text{const.}$$

$$U(x) = \frac{1}{2}kx^2 + \text{const.}$$

$$W_{\text{closed path}} = \int \vec{F}_{\text{cons}} \cdot d\vec{r} = 0$$

$$\left(\frac{dF_x}{dy} \right) = \left(\frac{dF_y}{dx} \right)$$

$$F_l = -\frac{dU}{dl}$$

$$0 = W_{nc,AB} = \Delta(K + U)_{AB} = \Delta E_{AB}.$$

Summary

8.1 Potential Energy of a System

- For a single-particle system, the difference of potential energy is the opposite of the work done by the forces acting on the particle as it moves from one position to another.
- Since only differences of potential energy are physically meaningful, the zero of the potential energy function can be chosen at a convenient location.
- The potential energies for Earth's constant gravity, near its surface, and for a Hooke's law force are linear and quadratic functions of position, respectively.

8.2 Conservative and Non-Conservative Forces

- A conservative force is one for which the work done is independent of path. Equivalently, a force

is conservative if the work done over any closed path is zero.

- A non-conservative force is one for which the work done depends on the path.
- For a conservative force, the infinitesimal work is an exact differential. This implies conditions on the derivatives of the force's components.
- The component of a conservative force, in a particular direction, equals the negative of the derivative of the potential energy for that force, with respect to a displacement in that direction.

8.3 Conservation of Energy

- A conserved quantity is a physical property that stays constant regardless of the path taken.
- A form of the work-energy theorem says that the change in the mechanical energy of a particle equals the work done on it by non-conservative

forces.

- If non-conservative forces do no work and there are no external forces, the mechanical energy of a particle stays constant. This is a statement of the conservation of mechanical energy and there is no change in the total mechanical energy.
- For one-dimensional particle motion, in which the mechanical energy is constant and the potential energy is known, the particle's position, as a function of time, can be found by evaluating an integral that is derived from the conservation of mechanical energy.

8.4 Potential Energy Diagrams and Stability

- Interpreting a one-dimensional potential energy diagram allows you to obtain qualitative, and some quantitative, information about the motion of a particle.
- At a turning point, the potential energy equals the mechanical energy and the kinetic energy is zero, indicating that the direction of the velocity reverses there.

Conceptual Questions

8.1 Potential Energy of a System

1. The kinetic energy of a system must always be positive or zero. Explain whether this is true for the potential energy of a system.
2. The force exerted by a diving board is conservative, provided the internal friction is negligible. Assuming friction is negligible, describe changes in the potential energy of a diving board as a swimmer dives from it, starting just before the swimmer steps on the board until just after their feet leave it.
3. Describe the gravitational potential energy transfers and transformations for a javelin, starting from the point at which an athlete picks up the javelin and ending when the javelin is stuck into the ground after being thrown.
4. A couple of soccer balls of equal mass are kicked off the ground at the same speed but at different angles. Soccer ball A is kicked off at an angle slightly above the horizontal, whereas ball B is kicked slightly below the vertical. How do each of the following compare for ball A and ball B? (a) The initial kinetic energy and (b) the change in gravitational potential energy from the ground to the highest point? If the energy in part (a) differs from part (b), explain why there is a difference between the two energies.

- The negative of the slope of the potential energy curve, for a particle, equals the one-dimensional component of the conservative force on the particle. At an equilibrium point, the slope is zero and is a stable (unstable) equilibrium for a potential energy minimum (maximum).

8.5 Sources of Energy

- Energy can be transferred from one system to another and transformed or converted from one type into another. Some of the basic types of energy are kinetic, potential, thermal, and electromagnetic.
- Renewable energy sources are those that are replenished by ongoing natural processes, over human time scales. Examples are wind, water, geothermal, and solar power.
- Non-renewable energy sources are those that are depleted by consumption, over human time scales. Examples are fossil fuel and nuclear power.

5. What is the dominant factor that affects the speed of an object that started from rest down a frictionless incline if the only work done on the object is from gravitational forces?
6. Two people observe a leaf falling from a tree. One person is standing on a ladder and the other is on the ground. If each person were to compare the energy of the leaf observed, would each person find the following to be the same or different for the leaf, from the point where it falls off the tree to when it hits the ground: (a) the kinetic energy of the leaf; (b) the change in gravitational potential energy; (c) the final gravitational potential energy?

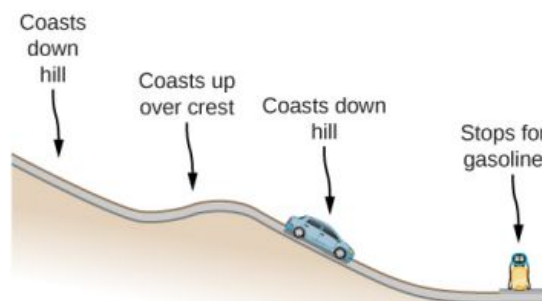
8.2 Conservative and Non-Conservative Forces

7. What is the physical meaning of a non-conservative force?
8. A bottle rocket is shot straight up in the air with a speed 30 m/s. If the air resistance is ignored, the bottle would go up to a height of approximately 46 m. However, the rocket goes up to only 35 m before returning to the ground. What happened? Explain, giving only a qualitative response.
9. An external force acts on a particle during a trip from one point to another and back to that same point. This particle is only effected by

conservative forces. Does this particle's kinetic energy and potential energy change as a result of this trip?

8.3 Conservation of Energy

10. When a body slides down an inclined plane, does the work of friction depend on the body's initial speed? Answer the same question for a body sliding down a curved surface.
11. Consider the following scenario. A car for which friction is *not* negligible accelerates from rest down a hill, running out of gasoline after a short distance (see below). The driver lets the car coast farther down the hill, then up and over a small crest. He then coasts down that hill into a gas station, where he brakes to a stop and fills the tank with gasoline. Identify the forms of energy the car has, and how they are changed and transferred in this series of events.



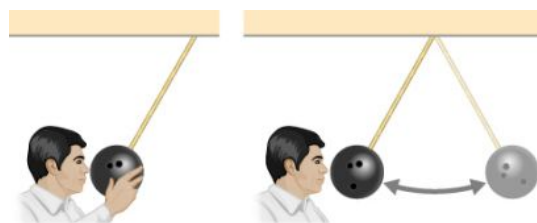
12. A dropped ball bounces to one-half its original height. Discuss the energy transformations that take place.
13. " $E = K + U$ constant is a special case of the work-energy theorem." Discuss this statement.

Problems

8.1 Potential Energy of a System

19. Using values from [Table 8.1](#), how many DNA molecules could be broken by the energy carried by a single electron in the beam of an old-fashioned TV tube? (These electrons were not dangerous in themselves, but they did create dangerous X-rays. Later-model tube TVs had shielding that absorbed X-rays before they escaped and exposed viewers.)
20. If the energy in fusion bombs were used to supply the energy needs of the world, how many of the 9-megaton variety would be needed for a year's supply of energy (using data from [link](#))?
21. A camera weighing 10 N falls from a small drone

14. In a common physics demonstration, a bowling ball is suspended from the ceiling by a rope. The professor pulls the ball away from its equilibrium position and holds it adjacent to his nose, as shown below. He releases the ball so that it swings directly away from him. Does he get struck by the ball on its return swing? What is he trying to show in this demonstration?



15. A child jumps up and down on a bed, reaching a higher height after each bounce. Explain how the child can increase his maximum gravitational potential energy with each bounce.
16. Can a non-conservative force increase the mechanical energy of the system?
17. Neglecting air resistance, how much would I have to raise the vertical height if I wanted to double the impact speed of a falling object?
18. A box is dropped onto a spring at its equilibrium position. The spring compresses with the box attached and comes to rest. Since the spring is in the vertical position, does the change in the gravitational potential energy of the box while the spring is compressing need to be considered in this problem?

hovering 20 m overhead and enters free fall. What is the gravitational potential energy change of the camera from the drone to the ground if you take a reference point of (a) the ground being zero gravitational potential energy? (b) The drone being zero gravitational potential energy? What is the gravitational potential energy of the camera (c) before it falls from the drone and (d) after the camera lands on the ground if the reference point of zero gravitational potential energy is taken to be a second person looking out of a building 30 m from the ground?

22. Someone drops a 50-g pebble off of a docked cruise ship, 70.0 m above the water level. A person on a dock 3.0 m above the water level

holds out a net to catch the pebble. (a) How much work is done on the pebble by gravity during the drop? (b) What is the change in the gravitational potential energy during the drop? If the gravitational potential energy is zero at the water line, what is the gravitational potential energy (c) when the pebble is dropped? (d) When it reaches the net? What if the gravitational potential energy was 30.0 Joules at water level? (e) Find the answers to the same questions in (c) and (d).

23. A cat's crinkle ball toy of mass 15 g is thrown straight up with an initial speed of 3 m/s. Assume in this problem that air drag is negligible. (a) What is the kinetic energy of the ball as it leaves the hand? (b) How much work is done by the gravitational force during the ball's rise to its peak? (c) What is the change in the gravitational potential energy of the ball during the rise to its peak? (d) If the gravitational potential energy is taken to be zero at the point where it leaves your hand, what is the gravitational potential energy when it reaches the maximum height? (e) What if the gravitational potential energy is taken to be zero at the maximum height the ball reaches, what would the gravitational potential energy be when it leaves the hand? (f) What is the maximum height the ball reaches?

8.2 Conservative and Non-Conservative Forces

24. A force $F(x) = (3.0/x)$ N acts on a particle as it moves along the positive x -axis. (a) How much work does the force do on the particle as it moves from $x = 2.0$ m to $x = 5.0$ m? (b) Picking a convenient reference point of the potential energy to be zero at $x = \infty$, find the potential energy for this force.
25. A force $F(x) = (-5.0x^2 + 7.0x)$ N acts on a particle. How much work does the force do on the particle as it moves from $x = 2.0$ m to $x = 5.0$ m?
26. Find the force corresponding to the potential energy $U(x) = -a/x + b/x^2$.
27. The potential energy function for either one of the two atoms in a diatomic molecule is often approximated by $U(x) = a/x^{12} - b/x^6$ where x is the distance between the atoms. (a) At what distance of separation does the potential energy have a local minimum (not at $x = \infty$)? (b) What

is the force on an atom at this separation? (c) How does the force vary with the separation distance?

28. A particle of mass 2.0 kg moves under the influence of the force $F(x) = (3/\sqrt{x})$ N. If its speed at $x = 2.0$ m is $v = 6.0$ m/s, what is its speed at $x = 7.0$ m?
29. A particle of mass 2.0 kg moves under the influence of the force $F(x) = (-5x^2 + 7x)$ N. If its speed at $x = -4.0$ m is $v = 20.0$ m/s, what is its speed at $x = 4.0$ m?
30. A crate on rollers is being pushed without frictional loss of energy across the floor of a freight car (see the following figure). The car is moving to the right with a constant speed v_0 . If the crate starts at rest relative to the freight car, then from the work-energy theorem, $Fd = mv^2/2$, where d , the distance the crate moves, and v , the speed of the crate, are both measured relative to the freight car. (a) To an observer at rest beside the tracks, what distance d' is the crate pushed when it moves the distance d in the car? (b) What are the crate's initial and final speeds v_0' and v' as measured by the observer beside the tracks? (c) Show that $Fd' = m(v')^2/2 - m(v_0')^2/2$ and, consequently, that work is equal to the change in kinetic energy in both reference systems.

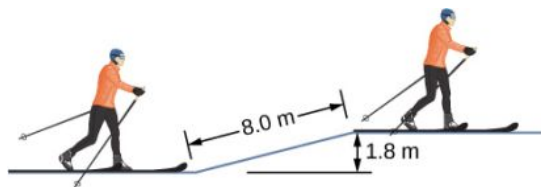


8.3 Conservation of Energy

31. A boy throws a ball of mass 0.25 kg straight upward with an initial speed of 20 m/s. When the ball returns to the boy, its speed is 17 m/s. How much work does air resistance do on the ball during its flight?
32. A mouse of mass 200 g falls 100 m down a vertical mine shaft and lands at the bottom with a speed of 8.0 m/s. During its fall, how much work is done on the mouse by air resistance?
33. Using energy considerations and assuming negligible air resistance, show that a rock thrown from a bridge 20.0 m above water with an initial speed of 15.0 m/s strikes the water with a speed of 24.8 m/s independent of the direction thrown. (Hint: show that

$$K_i + U_i = K_f + U_f$$

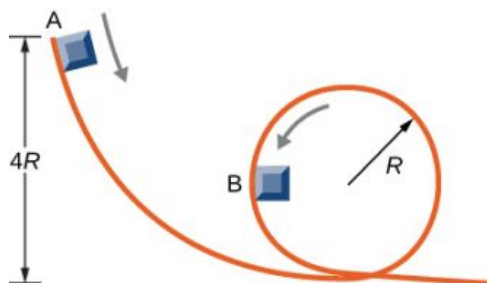
34. A 1.0-kg ball at the end of a 2.0-m string swings in a vertical plane. At its lowest point the ball is moving with a speed of 10 m/s. (a) What is its speed at the top of its path? (b) What is the tension in the string when the ball is at the bottom and at the top of its path?
35. Ignoring details associated with friction, extra forces exerted by arm and leg muscles, and other factors, we can consider a pole vault as the conversion of an athlete's running kinetic energy to gravitational potential energy. If an athlete is to lift their body 4.8 m during a vault, what speed must they have when they plant the pole?
36. Tarzan grabs a vine hanging vertically from a tall tree when he is running at 9.0 m/s. (a) How high can he swing upward? (b) Does the length of the vine affect this height?
37. Assume that the force of a bow on an arrow behaves like the spring force. In aiming the arrow, an archer pulls the bow back 50 cm and holds it in position with a force of 150 N. If the mass of the arrow is 50 g and the "spring" is massless, what is the speed of the arrow immediately after it leaves the bow?
38. A 100-kg man is skiing across level ground at a speed of 8.0 m/s when he comes to the small slope 1.8 m higher than ground level shown in the following figure. (a) If the skier coasts up the hill, what is his speed when he reaches the top plateau? Assume friction between the snow and skis is negligible. (b) What is his speed when he reaches the upper level if an 80-N frictional force acts on the skis?



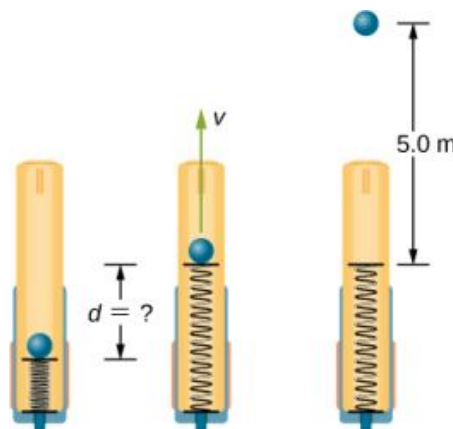
39. A sled of mass 70 kg starts from rest and slides down a 10° incline 80 m long. It then travels for 20 m horizontally before starting back up an 8° incline. It travels 80 m along this incline before coming to rest. What is the magnitude of the net work done on the sled by friction?
40. A girl on a skateboard (total mass of 40 kg) is moving at a speed of 10 m/s at the bottom of a

long ramp. The ramp is inclined at 20° with respect to the horizontal. If she travels 14.2 m upward along the ramp before stopping, what is the net frictional force on her?

41. A baseball of mass 0.25 kg is hit at home plate with a speed of 40 m/s. When it lands in a seat in the left-field bleachers a horizontal distance 120 m from home plate, it is moving at 30 m/s. If the ball lands 20 m above the spot where it was hit, how much work is done on it by air resistance?
42. A small block of mass m slides without friction around the loop-the-loop apparatus shown below. (a) If the block starts from rest at A, what is its speed at B? (b) What is the force of the track on the block at B?



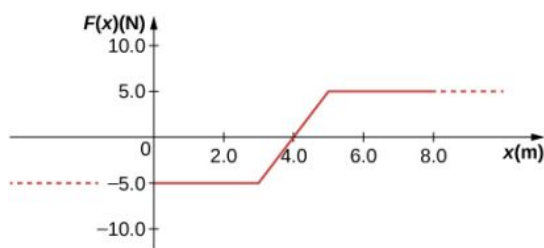
43. The massless spring of a spring gun has a force constant $k = 12 \text{ N/cm}$. When the gun is aimed vertically, a 15-g projectile is shot to a height of 5.0 m above the end of the expanded spring. (See below.) How much was the spring compressed initially?



44. A small ball is tied to a string and set rotating with negligible friction in a vertical circle. If the ball moves over the top of the circle at its slowest possible speed (so that the tension in the string is negligible), what is the tension in the string at the bottom of the circle, assuming there is no additional energy added to the ball during rotation?

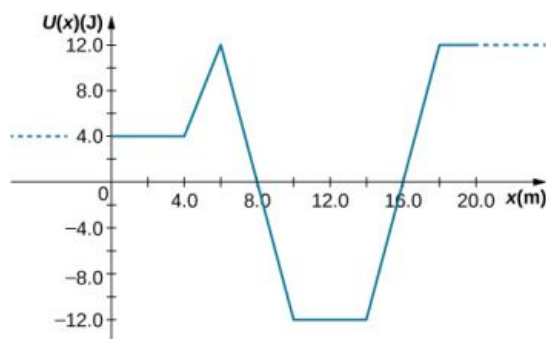
8.4 Potential Energy Diagrams and Stability

45. A mysterious constant force of 10 N acts horizontally on everything. The direction of the force is found to be always pointed toward a wall in a big hall. Find the potential energy of a particle due to this force when it is at a distance x from the wall, assuming the potential energy at the wall to be zero.
46. A single force $F(x) = -4.0x$ (in newtons) acts on a 1.0-kg body. When $x = 3.5$ m, the speed of the body is 4.0 m/s. What is its speed at $x = 2.0$ m?
47. A particle of mass 4.0 kg is constrained to move along the x -axis under a single force $F(x) = -cx^3$, where $c = 8.0 \text{ N/m}^3$. The particle's speed at A , where $x_A = 1.0$ m, is 6.0 m/s. What is its speed at B , where $x_B = -2.0$ m?
48. The force on a particle of mass 2.0 kg varies with position according to $F(x) = -3.0x^2$ (x in meters, $F(x)$ in newtons). The particle's velocity at $x = 2.0$ m is 5.0 m/s. Calculate the mechanical energy of the particle using (a) the origin as the reference point and (b) $x = 4.0$ m as the reference point. (c) Find the particle's velocity at $x = 1.0$ m. Do this part of the problem for each reference point.
49. A 4.0-kg particle moving along the x -axis is acted upon by the force whose functional form appears below. The velocity of the particle at $x = 0$ is $v = 6.0$ m/s. Find the particle's speed at $x =$ (a) 2.0 m, (b) 4.0 m, (c) 10.0 m, (d) Does the particle turn around at some point and head back toward the origin? (e) Repeat part (d) if $v = 2.0$ m/s at $x = 0$.



50. A particle of mass 0.50 kg moves along the x -axis with a potential energy whose dependence on x is shown below. (a) What is the force on the particle at $x = 2.0, 5.0, 8.0$, and 12 m? (b) If the total mechanical energy E of the particle is -6.0 J, what are the minimum and maximum positions of the particle? (c) What are these positions if $E = 2.0$ J?

- (d) If $E = 16$ J, what are the speeds of the particle at the positions listed in part (a)?



51. (a) Sketch a graph of the potential energy function $U(x) = kx^2/2 + Ae^{-\alpha x^2}$, where k , A , and α are constants. (b) What is the force corresponding to this potential energy? (c) Suppose a particle of mass m moving with this potential energy has a velocity v_a when its position is $x = a$. Show that the particle does not pass through the origin unless

$$A \leq \frac{mv_a^2 + ka^2}{2(1 - e^{-aa^2})}.$$

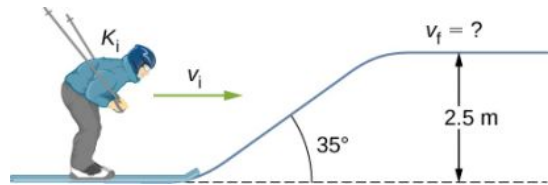
8.5 Sources of Energy

52. In the cartoon movie *Pocahontas* (<https://openstax.org/l/21pocahontclip>), Pocahontas runs to the edge of a cliff and jumps off, showcasing the fun side of her personality. (a) If she is running at 3.0 m/s before jumping off the cliff and she hits the water at the bottom of the cliff at 20.0 m/s, how high is the cliff? Assume negligible air drag in this cartoon. (b) If she jumped off the same cliff from a standstill, how fast would she be falling right before she hit the water?
53. In the reality television show “*Amazing Race*” (<https://openstax.org/l/21amazraceclip>), a contestant is firing 3.0-kg watermelons from a slingshot to hit targets down the field. The slingshot is pulled back 2.0 m and the watermelon is considered to be at ground level. The launch point is 0.75 m from the ground and the targets are 10 m horizontally away. Calculate the spring constant of the slingshot.
54. In the *Back to the Future* movies (<https://openstax.org/l/21bactofutclip>), a DeLorean car of mass 1230 kg travels at 88 miles per hour to venture back to the future. (a) What is the kinetic energy of the DeLorean? (b)

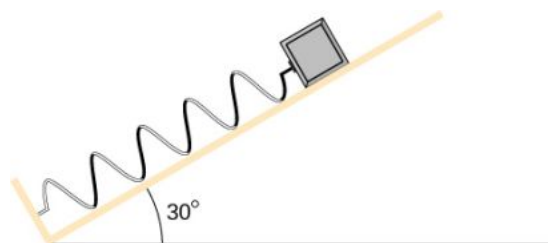
What spring constant would be needed to stop this DeLorean in a distance of 0.1 m?

55. In the *Hunger Games* movie (<https://openstax.org/l/21HungGamesclip>), Katniss Everdeen fires a 0.0200-kg arrow from ground level to pierce an apple up on a stage. The spring constant of the bow is 330 N/m and she pulls the arrow back a distance of 0.55 m. The apple on the stage is 5.00 m higher than the launching point of the arrow. At what speed does the arrow (a) leave the bow? (b) strike the apple?
56. In a “Top Fail” video (<https://openstax.org/l/21topfailvideo>), two women run at each other and collide by hitting exercise balls together. If each woman has a mass of 50 kg, which includes the exercise ball, and one woman runs to the right at 2.0 m/s and the other is running toward her at 1.0 m/s, (a) how much total kinetic energy is there in the system? (b) If energy is conserved after the collision and each exercise ball has a mass of 2.0 kg, how fast would the balls fly off toward the camera?
57. In a *Coyote/Road Runner* cartoon clip (<https://openstax.org/l/21coyroadcarcl>), a spring expands quickly and sends the coyote into a rock. If the spring extended 5 m and sent the coyote of mass 20 kg to a speed of 15 m/s, (a) what is the spring constant of this spring? (b) If the coyote were sent vertically into the air with the energy given to him by the spring, how high could he go if there were no non-conservative forces?
58. In an iconic movie scene, *Forrest Gump* (<https://openstax.org/l/21ForrGumpvid>) runs around the country. If he is running at a constant speed of 3 m/s, would it take him more or less energy to run uphill or downhill and why?
59. In the movie *Monty Python and the Holy Grail* (<https://openstax.org/l/21monpytmovcl>) a cow is catapulted from the top of a castle wall over to the people down below. The gravitational potential energy is set to zero at ground level. The cow is launched from a spring of spring constant 1.1×10^4 N/m that is expanded 0.5 m from equilibrium. If the castle is 9.1 m tall and the mass of the cow is 110 kg, (a) what is the gravitational potential energy of the cow at the top of the castle? (b) What is the elastic spring energy of the cow before the catapult is released? (c) What is the speed of the cow right before it lands on the ground?

60. A 60.0-kg skier with an initial speed of 12.0 m/s coasts up a 2.50-m high rise as shown. Find her final speed at the top, given that the coefficient of friction between her skis and the snow is 0.80.



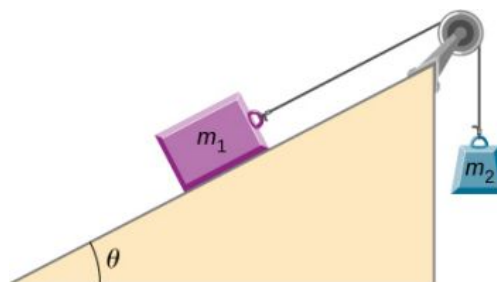
61. (a) How high a hill can a car coast up (engines disengaged) if work done by friction is negligible and its initial speed is 110 km/h? (b) If, in actuality, a 750-kg car with an initial speed of 110 km/h is observed to coast up a hill to a height 22.0 m above its starting point, how much thermal energy was generated by friction? (c) What is the average force of friction if the hill has a slope of 2.5° above the horizontal?
62. A 5.00×10^5 -kg subway train is brought to a stop from a speed of 0.500 m/s in 0.400 m by a large spring bumper at the end of its track. What is the spring constant k of the spring?
63. A pogo stick has a spring with a spring constant of 2.5×10^4 N/m, which can be compressed 12.0 cm. To what maximum height from the uncompressed spring can a child jump on the stick using only the energy in the spring, if the child and stick have a total mass of 40 kg?
64. A block of mass 500 g is attached to a spring of spring constant 80 N/m (see the following figure). The other end of the spring is attached to a support while the mass rests on a rough surface with a coefficient of friction of 0.20 that is inclined at angle of 30° . The block is pushed along the surface till the spring compresses by 10 cm and is then released from rest. (a) How much potential energy was stored in the block-spring-support system when the block was just released? (b) Determine the speed of the block when it crosses the point when the spring is neither compressed nor stretched. (c) Determine the position of the block where it just comes to rest on its way up the incline.



- 65.** A block of mass 200 g is attached at the end of a massless spring at equilibrium length of spring constant 50 N/m. The other end of the spring is attached to the ceiling and the mass is released at a height considered to be where the gravitational potential energy is zero. (a) What is the net potential energy of the block at the instant the block is at the lowest point? (b) What is the net potential energy of the block at the midpoint of its descent? (c) What is the speed of the block at the midpoint of its descent?
- 66.** A T-shirt cannon launches a shirt at 5.00 m/s from a platform height of 3.00 m from ground level. How fast will the shirt be traveling if it is caught by someone whose hands are (a) 1.00 m from ground level? (b) 4.00 m from ground level? Neglect air drag.
- 67.** A child (32 kg) jumps up and down on a trampoline. The trampoline exerts a spring restoring force on the child with a constant of 5000 N/m. At the highest point of the bounce, the child is 1.0 m above the level surface of the trampoline. What is the compression distance of

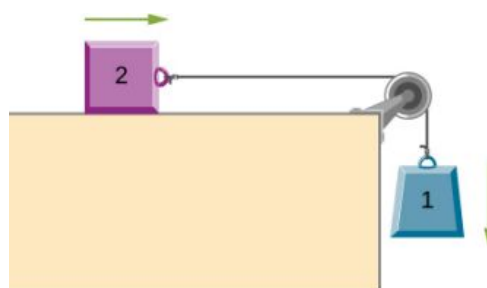
the trampoline? Neglect the bending of the legs or any transfer of energy of the child into the trampoline while jumping.

- 68.** Shown below is a box of mass m_1 that sits on a frictionless incline at an angle above the horizontal $\theta = 30^\circ$. This box is connected by a relatively massless string, over a frictionless pulley, and finally connected to a box at rest over the ledge, labeled m_2 . If m_1 and m_2 are a height h above the ground and $m_2 \gg m_1$: (a) What is the initial gravitational potential energy of the system? (b) What is the final kinetic energy of the system when m_2 hits the ground?

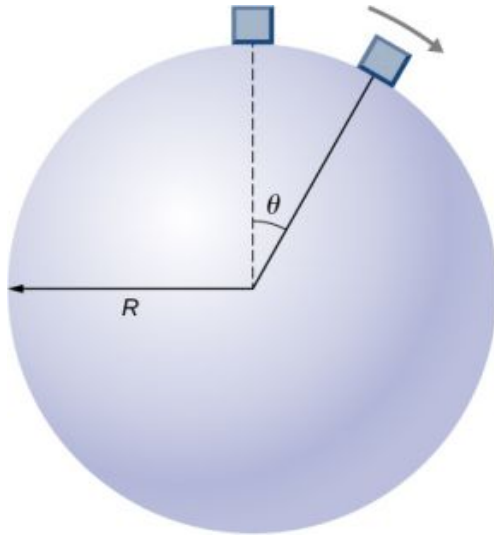


Additional Problems

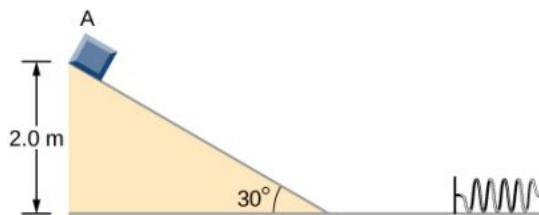
- 69.** A massless spring with force constant $k = 200$ N/m hangs from the ceiling. A 2.0-kg block is attached to the free end of the spring and released. If the block falls 17 cm before starting back upwards, how much work is done by friction during its descent?
- 70.** A particle of mass 2.0 kg moves under the influence of the force $F(x) = (-5x^2 + 7x)$ N. Suppose a frictional force also acts on the particle. If the particle's speed when it starts at $x = -4.0$ m is 0.0 m/s and when it arrives at $x = 4.0$ m is 9.0 m/s, how much work is done on it by the frictional force between $x = -4.0$ m and $x = 4.0$ m?
- 71.** Block 2 shown below slides along a frictionless table as block 1 falls. Both blocks are attached by a frictionless pulley. Find the speed of the blocks after they have each moved 2.0 m. Assume that they start at rest and that the pulley has negligible mass. Use $m_1 = 2.0$ kg and $m_2 = 4.0$ kg.



- 72.** A body of mass m and negligible size starts from rest and slides down the surface of a frictionless solid sphere of radius R . (See below.) Prove that the body leaves the sphere when $\theta = \cos^{-1}(2/3)$.

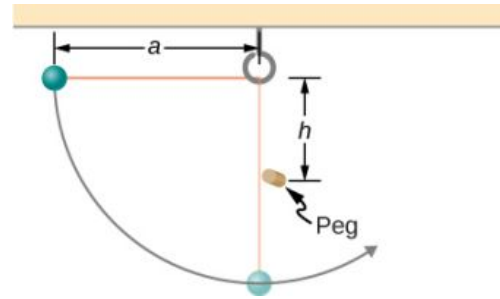


73. A mysterious force acts on all particles along a particular line and always points towards a particular point P on the line. The magnitude of the force on a particle increases as the cube of the distance from that point; that is $F \propto r^3$, if the distance from P to the position of the particle is r . Let b be the proportionality constant, and write the magnitude of the force as $F = br^3$. Find the potential energy of a particle subjected to this force when the particle is at a distance D from P , assuming the potential energy to be zero when the particle is at P .
74. An object of mass 10 kg is released at point A, slides to the bottom of the 30° incline, then collides with a horizontal massless spring, compressing it a maximum distance of 0.75 m. (See below.) The spring constant is 500 N/m, the height of the incline is 2.0 m, and the horizontal surface is frictionless. (a) What is the speed of the object at the bottom of the incline? (b) What is the work of friction on the object while it is on the incline? (c) The spring recoils and sends the object back toward the incline. What is the speed of the object when it reaches the base of the incline? (d) What vertical distance does it move back up the incline?

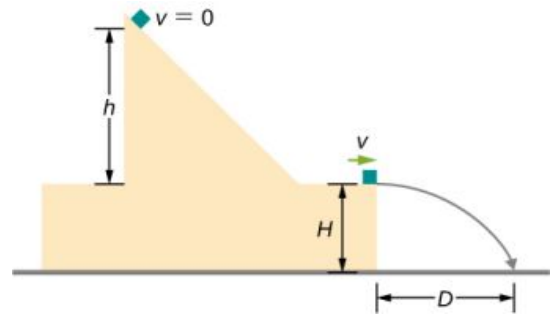


75. Shown below is a small ball of mass m attached to a string of length a . A small peg is located a

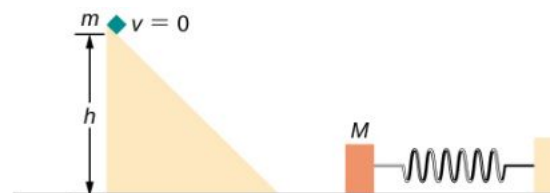
distance h below the point where the string is supported. If the ball is released when the string is horizontal, show that h must be greater than $3a/5$ if the ball is to swing completely around the peg.



76. A block leaves a frictionless inclined surface horizontally after dropping off by a height h . Find the horizontal distance D where it will land on the floor, in terms of h , H , and g .



77. A block of mass m , after sliding down a frictionless incline, strikes another block of mass M that is attached to a spring of spring constant k (see below). The blocks stick together upon impact and travel together. (a) Find the compression of the spring in terms of m , M , h , g , and k when the combination comes to rest. Hint: The speed of the combined blocks $m + M$ (v_2) is based on the speed of block m just prior to the collision with the block M (v_1) based on the equation $v_2 = (m/m) + M(v_1)$. This will be discussed further in the chapter on Linear Momentum and Collisions. (b) The loss of kinetic energy as a result of the bonding of the two masses upon impact is stored in the so-called binding energy of the two masses. Calculate the binding energy.



- 78.** A block of mass 300 g is attached to a spring of spring constant 100 N/m. The other end of the spring is attached to a support while the block rests on a smooth horizontal table and can slide freely without any friction. The block is pushed horizontally till the spring compresses by 12 cm, and then the block is released from rest. (a) How much potential energy was stored in the block-spring support system when the block was just released? (b) Determine the speed of the block when it crosses the point when the spring is neither compressed nor stretched. (c) Determine the speed of the block when it has traveled a distance of 20 cm from where it was released.

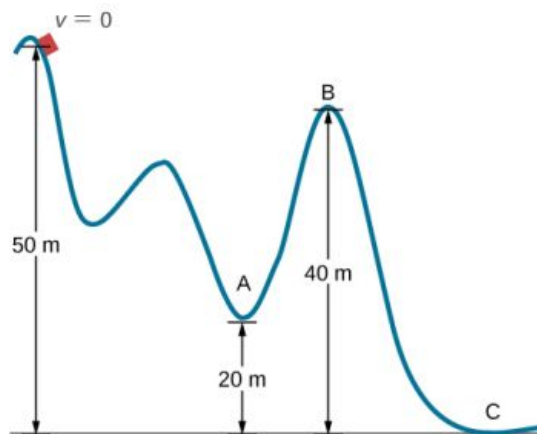
- 79.** Consider a block of mass 0.200 kg attached to a spring of spring constant 100 N/m. The block is placed on a frictionless table, and the other end of the spring is attached to the wall so that the spring is level with the table. The block is then pushed in so that the spring is compressed by 10.0 cm. Find the speed of the block as it crosses (a) the point when the spring is not stretched, (b) 5.00 cm to the left of point in (a), and (c) 5.00 cm to the right of point in (a).

- 80.** A skier starts from rest and slides downhill. What will be the speed of the skier if he drops by 20 meters in vertical height? Ignore any air resistance (which will, in reality, be quite a lot), and any friction between the skis and the snow.

- 81.** Repeat the preceding problem, but this time, suppose that the work done by air resistance cannot be ignored. Let the work done by the air resistance when the skier goes from A to B along the given hilly path be -2000 J . The work done by air resistance is negative since the air resistance acts in the opposite direction to the displacement. Supposing the mass of the skier is 50 kg, what is the speed of the skier at point B?

- 82.** Two bodies are interacting by a conservative force. Show that the mechanical energy of an isolated system consisting of two bodies interacting with a conservative force is conserved. (*Hint:* Start by using Newton's third law and the definition of work to find the work done on each body by the conservative force.)

- 83.** In an amusement park, a car rolls in a track as shown below. Find the speed of the car at A, B, and C. Note that the work done by the rolling friction is zero since the displacement of the point at which the rolling friction acts on the tires is momentarily at rest and therefore has a zero displacement.



- 84.** A 200-g steel ball is tied to a 2.00-m “massless” string and hung from the ceiling to make a pendulum, and then, the ball is brought to a position making a 30° angle with the vertical direction and released from rest. Ignoring the effects of the air resistance, find the speed of the ball when the string (a) is vertically down, (b) makes an angle of 20° with the vertical and (c) makes an angle of 10° with the vertical.

- 85.** A 300 g hockey puck is shot across an ice-covered pond. Before the hockey puck was hit, the puck was at rest. After the hit, the puck has a speed of 40 m/s. The puck comes to rest after going a distance of 30 m. (a) Describe how the energy of the puck changes over time, giving the numerical values of any work or energy involved. (b) Find the magnitude of the net friction force.

- 86.** A projectile of mass 2 kg is fired with a speed of 20 m/s at an angle of 30° with respect to the horizontal. (a) Calculate the initial total energy of the projectile given that the reference point of zero gravitational potential energy at the launch position. (b) Calculate the kinetic energy at the highest vertical position of the projectile. (c) Calculate the gravitational potential energy at the highest vertical position. (d) Calculate the maximum height that the projectile reaches. Compare this result by solving the same problem using your knowledge of projectile motion.

- 87.** An artillery shell is fired at a target 200 m above the ground. When the shell is 100 m in the air, it has a speed of 100 m/s. What is its speed when it hits its target? Neglect air friction.

- 88.** How much energy is lost to a dissipative drag force if a 60-kg person falls at a constant speed for 15 meters?

- 89.** A box slides on a frictionless surface with a total energy of 50 J. It hits a spring and compresses the spring a distance of 25 cm from equilibrium. If the same box with the same initial energy

slides on a rough surface, it only compresses the spring a distance of 15 cm, how much energy must have been lost by sliding on the rough surface?

CHAPTER 9

Linear Momentum and Collisions



FIGURE 9.1 The concepts of impulse, momentum, and center of mass are crucial for a major-league baseball player to successfully get a hit. If he misjudges these quantities, he might break his bat instead. (credit: modification of work by “Cathy T”/Flickr)

CHAPTER OUTLINE

9.1 Linear Momentum

9.2 Impulse and Collisions

9.3 Conservation of Linear Momentum

9.4 Types of Collisions

9.5 Collisions in Multiple Dimensions

9.6 Center of Mass

9.7 Rocket Propulsion

INTRODUCTION The concepts of work, energy, and the work-energy theorem are valuable for two primary reasons: First, they are powerful computational tools, making it much easier to analyze complex physical systems than is possible using Newton’s laws directly (for example, systems with nonconstant forces); and second, the observation that the total energy of a closed system is conserved means that the system can only evolve in ways that are consistent with energy conservation. In other words, a system cannot evolve randomly; it can only change in ways that conserve energy.

In this chapter, we develop and define another conserved quantity, called *linear momentum*, and another relationship (the *impulse-momentum theorem*), which will put an additional constraint on how a system evolves in time. Conservation of momentum is useful for understanding collisions, such as that shown in the above image. It is just as powerful, just as important, and just as useful as conservation of energy and the work-energy theorem.

9.1 Linear Momentum

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Explain what momentum is, physically
- Calculate the momentum of a moving object

Our study of kinetic energy showed that a complete understanding of an object's motion must include both its mass and its velocity ($K = (1/2)mv^2$). However, as powerful as this concept is, it does not include any information about the direction of the moving object's velocity vector. We'll now define a physical quantity that includes direction.

Like kinetic energy, this quantity includes both mass and velocity; like kinetic energy, it is a way of characterizing the “quantity of motion” of an object. It is given the name **momentum** (from the Latin word *movimentum*, meaning “movement”), and it is represented by the symbol p .

MOMENTUM

The momentum p of an object is the product of its mass and its velocity:

$$\vec{p} = m\vec{v}.$$

9.1



FIGURE 9.2 The velocity and momentum vectors for the ball are in the same direction. (credit: modification of work by Ben Sutherland)

As shown in [Figure 9.2](#), momentum is a vector quantity (since velocity is). This is one of the things that makes momentum useful and not a duplication of kinetic energy. It is perhaps most useful when determining whether an object's motion is difficult to change ([Figure 9.3](#)) or easy to change ([Figure 9.4](#)) over a short time interval.



FIGURE 9.3 This supertanker transports a huge mass of oil; as a consequence, it takes a long time for a force to change its (comparatively small) velocity. (credit: modification of work by “the_tahoe_guy”/Flickr)

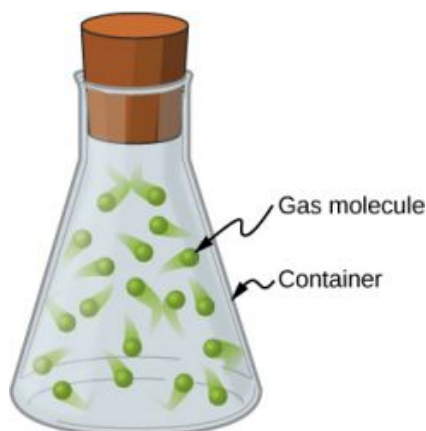


FIGURE 9.4 Gas molecules can have very large velocities, but these velocities change nearly instantaneously when they collide with the container walls or with each other. This is primarily because their masses are so tiny.

Unlike kinetic energy, momentum depends equally on an object’s mass and velocity. For example, as you will learn when you study thermodynamics, the average speed of an air molecule at room temperature is approximately 500 m/s, with an average molecular mass of 6×10^{-25} kg; its momentum is thus

$$p_{\text{molecule}} = (6 \times 10^{-25} \text{ kg}) \left(500 \frac{\text{m}}{\text{s}} \right) = 3 \times 10^{-22} \frac{\text{kg} \cdot \text{m}}{\text{s}}.$$

For comparison, a typical automobile might have a speed of only 15 m/s, but a mass of 1400 kg, giving it a momentum of

$$p_{\text{car}} = (1400 \text{ kg}) \left(15 \frac{\text{m}}{\text{s}} \right) = 21,000 \frac{\text{kg} \cdot \text{m}}{\text{s}}.$$

These momenta are different by 27 orders of magnitude, or a factor of a billion billion billion!

9.2 Impulse and Collisions

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Explain what an impulse is, physically
- Describe what an impulse does
- Relate impulses to collisions
- Apply the impulse-momentum theorem to solve problems

We have defined momentum to be the product of mass and velocity. Therefore, if an object's velocity should change (due to the application of a force on the object), then necessarily, its momentum changes as well. This indicates a connection between momentum and force. The purpose of this section is to explore and describe that connection.

Suppose you apply a force on a free object for some amount of time. Clearly, the larger the force, the larger the object's change of momentum will be. Alternatively, the more time you spend applying this force, again the larger the change of momentum will be, as depicted in [Figure 9.5](#). The amount by which the object's motion changes is therefore proportional to the magnitude of the force, and also to the time interval over which the force is applied.

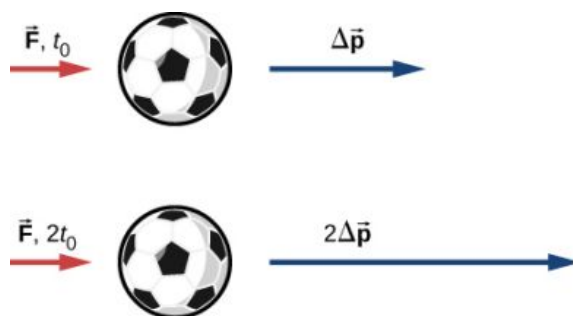


FIGURE 9.5 The change in momentum of an object is proportional to the length of time during which the force is applied. If a force is exerted on the lower ball for twice as long as on the upper ball, then the change in the momentum of the lower ball is twice that of the upper ball.

Mathematically, if a quantity is proportional to two (or more) things, then it is proportional to the product of those things. The product of a force and a time interval (over which that force acts) is called **impulse**, and is given the symbol $\vec{J}$.

IMPULSE

Let $\vec{F}(t)$ be the force applied to an object over some differential time interval dt ([Figure 9.6](#)). The resulting impulse on the object is defined as

$$d\vec{J} \equiv \vec{F}(t)dt.$$

9.2

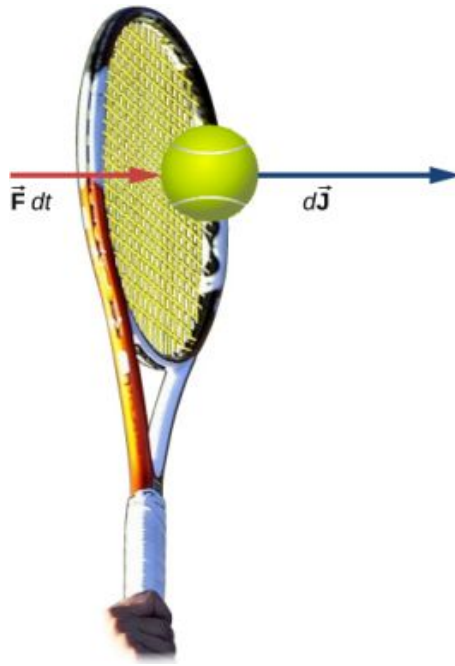


FIGURE 9.6 A force applied by a tennis racquet to a tennis ball over a time interval generates an impulse acting on the ball.

The total impulse over the interval $t_f - t_i$ is

$$\vec{J} = \int_{t_i}^{t_f} d\vec{J} \text{ or } \vec{J} \equiv \int_{t_i}^{t_f} \vec{F}(t) dt. \quad 9.3$$

[Equation 9.2](#) and [Equation 9.3](#) together say that when a force is applied for an infinitesimal time interval dt , it causes an infinitesimal impulse $d\vec{J}$, and the total impulse given to the object is defined to be the sum (integral) of all these infinitesimal impulses.

To calculate the impulse using [Equation 9.3](#), we need to know the force function $F(t)$, which we often don't. However, a result from calculus is useful here: Recall that the average value of a function over some interval is calculated by

$$f(x)_{\text{ave}} = \frac{1}{\Delta x} \int_{x_i}^{x_f} f(x) dx$$

where $\Delta x = x_f - x_i$. Applying this to the time-dependent force function, we obtain

$$\vec{F}_{\text{ave}} = \frac{1}{\Delta t} \int_{t_i}^{t_f} \vec{F}(t) dt. \quad 9.4$$

Therefore, from [Equation 9.3](#),

$$\vec{J} = \vec{F}_{\text{ave}} \Delta t. \quad 9.5$$

The idea here is that you can calculate the impulse on the object even if you don't know the details of the force as a function of time; you only need the average force. In fact, though, the process is usually reversed: You determine the impulse (by measurement or calculation) and then calculate the average force that caused that impulse.

To calculate the impulse, a useful result follows from writing the force in [Equation 9.3](#) as $\vec{F}(t) = m\vec{a}(t)$:

$$\vec{J} = \int_{t_i}^{t_f} \vec{F}(t) dt = m \int_{t_i}^{t_f} \vec{a}(t) dt = m [\vec{v}(t_f) - \vec{v}_i].$$

For a constant force $\vec{F}_{\text{ave}} = \vec{F} = m\vec{a}$, this simplifies to

$$\vec{J} = m\vec{a}\Delta t = m\vec{v}_f - m\vec{v}_i = m(\vec{v}_f - \vec{v}_i).$$

That is,

$$\vec{J} = m\Delta\vec{v}.$$

9.6

Note that the integral form, [Equation 9.3](#), applies to constant forces as well; in that case, since the force is independent of time, it comes out of the integral, which can then be trivially evaluated.



EXAMPLE 9.1

The Arizona Meteor Crater

Approximately 50,000 years ago, a large (radius of 25 m) iron-nickel meteorite collided with Earth at an estimated speed of 1.28×10^4 m/s in what is now the northern Arizona desert, in the United States. The impact produced a crater that is still visible today ([Figure 9.7](#)); it is approximately 1200 m (three-quarters of a mile) in diameter, 170 m deep, and has a rim that rises 45 m above the surrounding desert plain. Iron-nickel meteorites typically have a density of $\rho = 7970$ kg/m³. Use impulse considerations to estimate the average force and the maximum force that the meteor applied to Earth during the impact.



FIGURE 9.7 The Arizona Meteor Crater in Flagstaff, Arizona (often referred to as the Barringer Crater after the person who first suggested its origin and whose family owns the land). (credit: modification of work by “Shane.torgerson”/Wikimedia Commons)

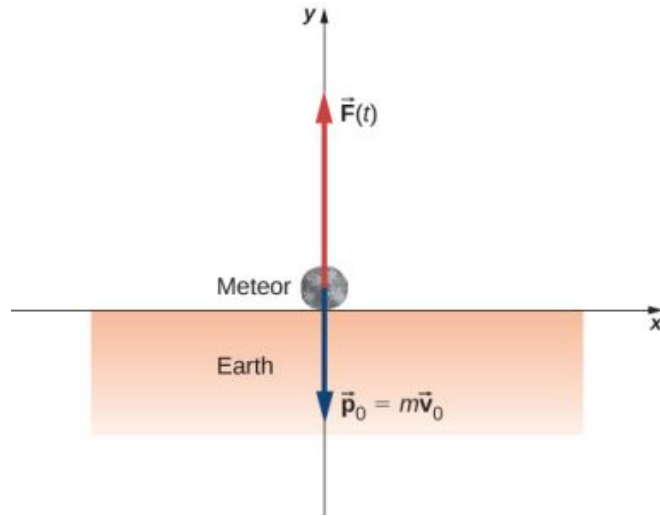
Strategy

It is conceptually easier to reverse the question and calculate the force that Earth applied on the meteor in order to stop it. Therefore, we'll calculate the force on the meteor and then use Newton's third law to argue that the force from the meteor on Earth was equal in magnitude and opposite in direction.

Using the given data about the meteor, and making reasonable guesses about the shape of the meteor and impact time, we first calculate the impulse using [Equation 9.6](#). We then use the relationship between force and impulse [Equation 9.5](#) to estimate the average force during impact. Next, we choose a reasonable force function for the impact event, calculate the average value of that function [Equation 9.4](#), and set the resulting expression equal to the calculated average force. This enables us to solve for the maximum force.

Solution

Define upward to be the $+y$ -direction. For simplicity, assume the meteor is traveling vertically downward prior to impact. In that case, its initial velocity is $\vec{v}_i = -v_i \hat{j}$, and the force Earth exerts on the meteor points upward, $\vec{F}(t) = +F(t)\hat{j}$. The situation at $t = 0$ is depicted below.



The average force during the impact is related to the impulse by

$$\vec{F}_{\text{ave}} = \frac{\vec{J}}{\Delta t}.$$

From [Equation 9.6](#), $\vec{J} = m\Delta\vec{v}$, so we have

$$\vec{F}_{\text{ave}} = \frac{m\Delta\vec{v}}{\Delta t}.$$

The mass is equal to the product of the meteor's density and its volume:

$$m = \rho V.$$

If we assume (guess) that the meteor was roughly spherical, we have

$$V = \frac{4}{3}\pi R^3.$$

Thus we obtain

$$\vec{F}_{\text{ave}} = \frac{\rho V \Delta\vec{v}}{\Delta t} = \frac{\rho \left(\frac{4}{3}\pi R^3\right) (\vec{v}_f - \vec{v}_i)}{\Delta t}.$$

The problem says the velocity at impact was $-1.28 \times 10^4 \text{ m/s} \hat{j}$ (the final velocity is zero); also, we guess that the primary impact lasted about $t_{\text{max}} = 2 \text{ s}$. Substituting these values gives

$$\begin{aligned} \vec{F}_{\text{ave}} &= \frac{\left(7970 \frac{\text{kg}}{\text{m}^3}\right) \left[\frac{4}{3}\pi (25 \text{ m})^3\right] \left[0 \frac{\text{m}}{\text{s}} - (-1.28 \times 10^4 \frac{\text{m}}{\text{s}} \hat{j})\right]}{2 \text{ s}} \\ &= + (3.33 \times 10^{12} \text{ N}) \hat{j} \end{aligned}$$

This is the average force applied during the collision. Notice that this force vector points in the same direction as the change of velocity vector $\Delta\vec{v}$.

Next, we calculate the maximum force. The impulse is related to the force function by

$$\vec{J} = \int_{t_i}^{t_{\max}} \vec{F}(t) dt.$$

We need to make a reasonable choice for the force as a function of time. We define $t = 0$ to be the moment the meteor first touches the ground. Then we assume the force is a maximum at impact, and rapidly drops to zero. A function that does this is

$$F(t) = F_{\max} e^{-t^2/(2\tau^2)}.$$

(The parameter τ represents how rapidly the force decreases to zero.) The average force is

$$F_{\text{ave}} = \frac{1}{\Delta t} \int_0^{t_{\max}} F_{\max} e^{-t^2/(2\tau^2)} dt$$

where $\Delta t = t_{\max} - 0$ s. Since we already have a numeric value for F_{ave} , we can use the result of the integral to obtain $F_{\max}$.

Choosing $\tau = \frac{1}{e} t_{\max}$ (this is a common choice, as you will see in later chapters), and guessing that $t_{\max} = 2$ s, this integral evaluates to

$$F_{\text{avg}} = 0.458 F_{\max}.$$

Thus, the maximum force has a magnitude of

$$\begin{aligned} 0.458 F_{\max} &= 3.33 \times 10^{12} \text{ N} \\ F_{\max} &= 7.27 \times 10^{12} \text{ N} \end{aligned}$$

The complete force function, including the direction, is

$$\vec{F}(t) = (7.27 \times 10^{12} \text{ N}) e^{-t^2/(8\text{s}^2)} \hat{\mathbf{j}}.$$

This is the force Earth applied to the meteor; by Newton's third law, the force the meteor applied to Earth is

$$\vec{F}(t) = -(7.27 \times 10^{12} \text{ N}) e^{-t^2/(8\text{s}^2)} \hat{\mathbf{j}}$$

which is the answer to the original question.

Significance

The graph of this function contains important information. Let's graph (the magnitude of) both this function and the average force together ([Figure 9.8](#)).

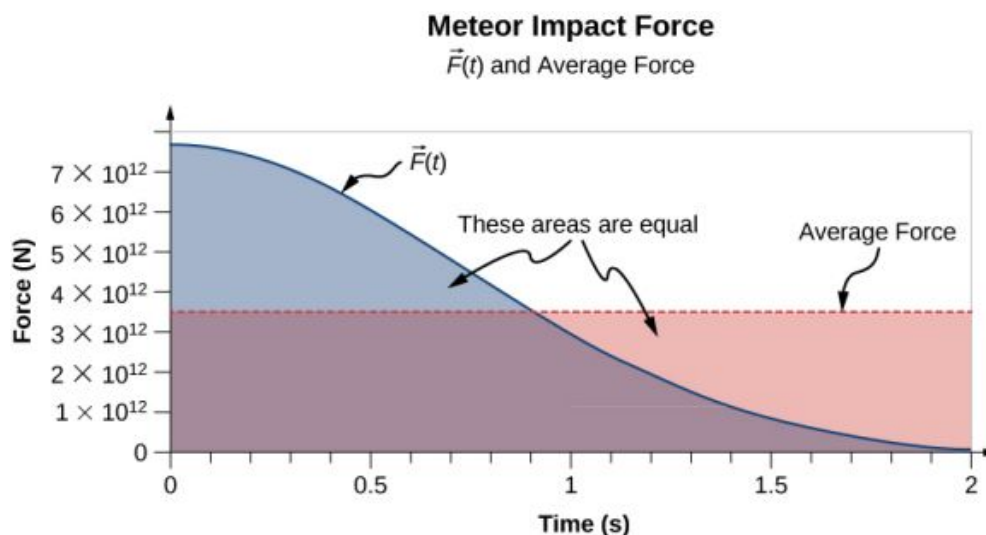


FIGURE 9.8 A graph of the average force (in red) and the force as a function of time (blue) of the meteor impact. The areas under the curves

are equal to each other, and are numerically equal to the applied impulse.

Notice that the area under each plot has been filled in. For the plot of the (constant) force F_{ave} , the area is a rectangle, corresponding to $F_{\text{ave}} \Delta t = J$. As for the plot of $F(t)$, recall from calculus that the area under the plot of a function is numerically equal to the integral of that function, over the specified interval; so here, that is

$$\int_0^{t_{\text{max}}} F(t) dt = J. \text{ Thus, the areas are equal, and both represent the impulse that the meteor applied to Earth}$$

during the two-second impact. The average force on Earth sounds like a huge force, and it is. Nevertheless, Earth barely noticed it. The acceleration Earth obtained was just

$$\vec{a} = \frac{-\vec{F}_{\text{ave}}}{M_{\text{Earth}}} = \frac{-(3.33 \times 10^{12} \text{ N}) \hat{j}}{5.97 \times 10^{24} \text{ kg}} = -\left(5.6 \times 10^{-13} \frac{\text{m}}{\text{s}^2}\right) \hat{j}$$

which is completely immeasurable. That said, the impact created seismic waves that nowadays could be detected by modern monitoring equipment.



EXAMPLE 9.2

The Benefits of Impulse

A car traveling at 27 m/s collides with a building. The collision with the building causes the car to come to a stop in approximately 1 second. The driver, who weighs 860 N, is protected by a combination of a variable-tension seatbelt and an airbag (Figure 9.9). (In effect, the driver collides with the seatbelt and airbag and *not* with the building.) The airbag and seatbelt slow his velocity, such that he comes to a stop in approximately 2.5 s.

- What average force does the driver experience during the collision?
- Without the seatbelt and airbag, his collision time (with the steering wheel) would have been approximately 0.20 s. What force would he experience in this case?

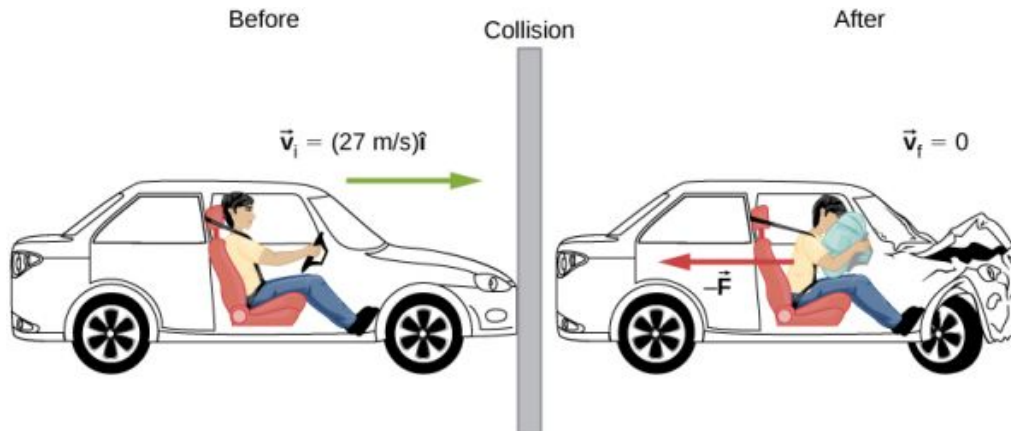


FIGURE 9.9 The motion of a car and its driver at the instant before and the instant after colliding with the wall. The restrained driver experiences a large backward force from the seatbelt and airbag, which causes his velocity to decrease to zero. (The forward force from the seatbelt is much smaller than the backward force, so we neglect it in the solution.)

Strategy

We are given the driver's weight, his initial and final velocities, and the time of collision; we are asked to calculate a force. Impulse seems the right way to tackle this; we can combine Equation 9.5 and Equation 9.6.

Solution

- Define the +x-direction to be the direction the car is initially moving. We know

$$\vec{J} = \vec{F} \Delta t$$

and

$$\vec{J} = m \Delta \vec{v}.$$

Since J is equal to both those things, they must be equal to each other:

$$\vec{F}\Delta t = m\Delta\vec{v}.$$

We need to convert this weight to the equivalent mass, expressed in SI units:

$$\frac{860 \text{ N}}{9.8 \text{ m/s}^2} = 87.8 \text{ kg}.$$

Remembering that $\Delta\vec{v} = \vec{v}_f - \vec{v}_i$, and noting that the final velocity is zero, we solve for the force:

$$\vec{F} = m \frac{0 - v_i \hat{i}}{\Delta t} = (87.8 \text{ kg}) \left(\frac{-(27 \text{ m/s}) \hat{i}}{2.5 \text{ s}} \right) = -(948 \text{ N}) \hat{i}.$$

The negative sign implies that the force slows him down. For perspective, this is about 1.1 times his own weight.

- b. Same calculation, just the different time interval:

$$\vec{F} = (87.8 \text{ kg}) \left(\frac{-(27 \text{ m/s}) \hat{i}}{0.20 \text{ s}} \right) = -(11,853 \text{ N}) \hat{i}$$

which is about 14 times his own weight. Big difference!

Significance

You see that the value of an airbag is how greatly it reduces the force on the vehicle occupants. For this reason, they have been required on all passenger vehicles in the United States since 1991, and have been commonplace throughout Europe and Asia since the mid-1990s. The change of momentum in a crash is the same, with or without an airbag; the force, however, is vastly different.

Effect of Impulse

Since an impulse is a force acting for some amount of time, it causes an object's motion to change. Recall [Equation 9.6](#):

$$\vec{J} = m\Delta\vec{v}.$$

Because $m\vec{v}$ is the momentum of a system, $m\Delta\vec{v}$ is the *change* of momentum $\Delta\vec{p}$. This gives us the following relation, called the **impulse-momentum theorem** (or relation).

IMPULSE-MOMENTUM THEOREM

An impulse applied to a system changes the system's momentum, and that change of momentum is exactly equal to the impulse that was applied:

$$\vec{J} = \Delta\vec{p}.$$

9.7

The impulse-momentum theorem is depicted graphically in [Figure 9.10](#).

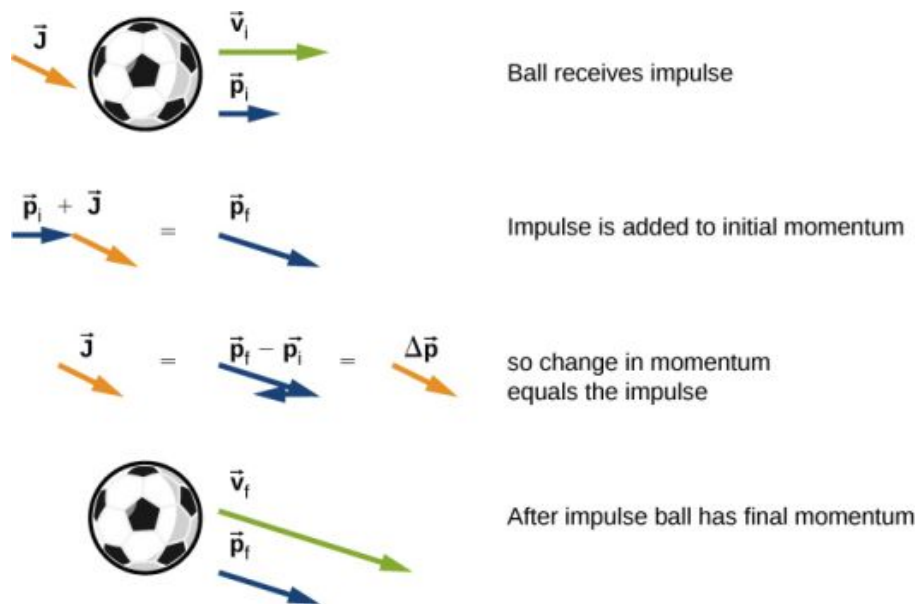


FIGURE 9.10 Illustration of impulse-momentum theorem. (a) A ball with initial velocity $\vec{v}_0$ and momentum $\vec{p}_0$ receives an impulse $\vec{J}$. (b) This impulse is added vectorially to the initial momentum. (c) Thus, the impulse equals the change in momentum, $\vec{J} = \Delta\vec{p}$. (d) After the impulse, the ball moves off with its new momentum $\vec{p}_f$.

There are two crucial concepts in the impulse-momentum theorem:

1. Impulse is a vector quantity; an impulse of, say, $-(10 \text{ N} \cdot \text{s}) \hat{i}$ is very different from an impulse of $+(10 \text{ N} \cdot \text{s}) \hat{i}$; they cause completely opposite changes of momentum.
2. An impulse does not cause momentum; rather, it causes a *change* in the momentum of an object. Thus, you must subtract the initial momentum from the final momentum, and—since momentum is also a vector quantity—you must take careful account of the signs of the momentum vectors.

The most common questions asked in relation to impulse are to calculate the applied force, or the change of velocity that occurs as a result of applying an impulse. The general approach is the same.



PROBLEM-SOLVING STRATEGY

Impulse-Momentum Theorem

1. Express the impulse as force times the relevant time interval.
2. Express the impulse as the change of momentum, usually $m\Delta v$.
3. Equate these and solve for the desired quantity.

EXAMPLE 9.3

Moving the *Enterprise*

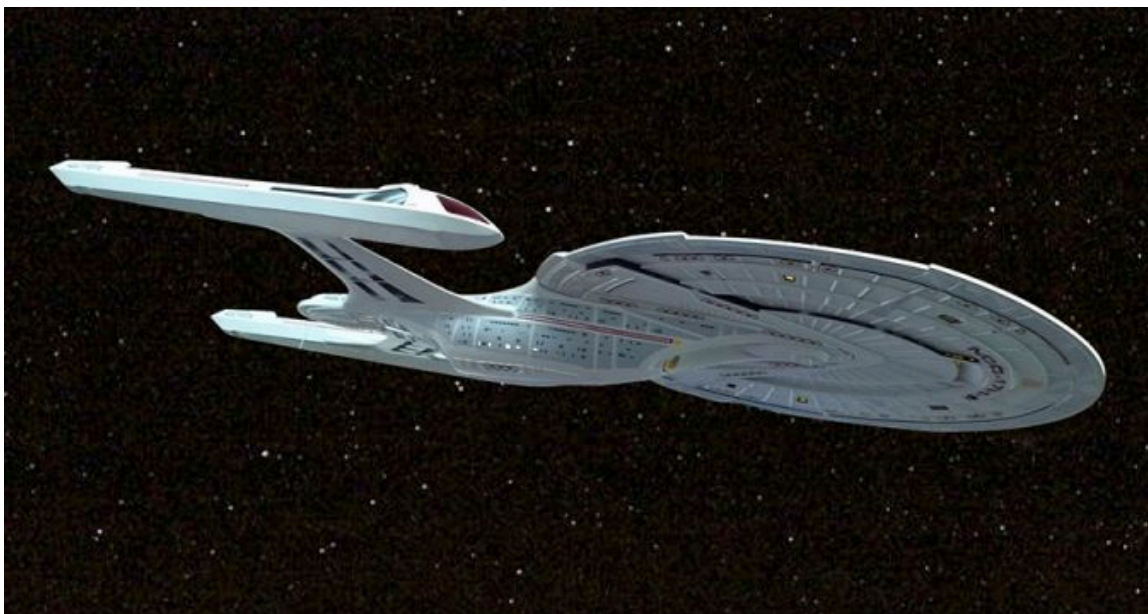


FIGURE 9.11 The fictional starship *Enterprise* from the Star Trek adventures operated on so-called “impulse engines” that combined matter with antimatter to produce energy.

When Captain Picard commands, “Take us out,” the starship *Enterprise* ([Figure 9.11](#)) starts from rest to a final speed of $v_f = 7.5 \times 10^7$ m/s. Assuming this maneuver is completed in 60 s, what average force did the impulse engines apply to the ship?

Strategy

We are asked for a force; we know the initial and final speeds (and hence the change in speed), and we know the time interval over which this all happened. In particular, we know the amount of time that the force acted. This suggests using the impulse-momentum relation. To use that, though, we need the mass of the *Enterprise*. An internet search gives a best estimate of the mass of the *Enterprise* (in the 2009 movie) as 2×10^9 kg.

Solution

Because this problem involves only one direction (i.e., the direction of the force applied by the engines), we only need the scalar form of the impulse-momentum theorem [Equation 9.7](#), which is

$$\Delta p = J$$

with

$$\Delta p = m\Delta v$$

and

$$J = F\Delta t.$$

Equating these expressions gives

$$F\Delta t = m\Delta v.$$

Solving for the magnitude of the force and inserting the given values leads to

$$F = \frac{m\Delta v}{\Delta t} = \frac{(2 \times 10^9 \text{ kg})(7.5 \times 10^7 \text{ m/s})}{60 \text{ s}} = 2.5 \times 10^{15} \text{ N}.$$

Significance

Using Newton's Second Law, this force causes an acceleration of $1.25 \times 10^6 \text{ m/s}^2$. This is 130,000 times gravity, which is unimaginably huge. It goes almost without saying that such a force would kill everyone on board instantly, as well as destroying every piece of equipment. Fortunately, the *Enterprise* has “inertial dampeners.” It is left as an exercise for the reader’s imagination to determine how these work.

✓ CHECK YOUR UNDERSTANDING 9.1

The U.S. Air Force uses “10gs” (an acceleration equal to $10 \times 9.8 \text{ m/s}^2$) as the maximum acceleration a human can withstand (but only for several seconds) and survive. How much time must the *Enterprise* spend accelerating if the humans on board are to experience an average of at most 10gs of acceleration? (Assume the inertial dampeners are offline.)

EXAMPLE 9.4

The iPhone Drop

Apple released its iPhone 6 Plus in November 2014. According to many reports, it was originally supposed to have a screen made from sapphire, but that was changed at the last minute for a hardened glass screen. Reportedly, this was because the sapphire screen cracked when the phone was dropped. What force did the iPhone 6 Plus experience as a result of being dropped?

Strategy

The force the phone experiences is due to the impulse applied to it by the floor when the phone collides with the floor. Our strategy then is to use the impulse-momentum relationship. We calculate the impulse, estimate the impact time, and use this to calculate the force.

We need to make a couple of reasonable estimates, as well as find technical data on the phone itself. First, let’s suppose that the phone is most often dropped from about chest height on an average-height person. Second, assume that it is dropped from rest, that is, with an initial vertical velocity of zero. Finally, we assume that the phone bounces very little—the height of its bounce is assumed to be negligible.

Solution

Define upward to be the $+y$ -direction. A typical height is approximately $h = 1.5 \text{ m}$ and, as stated, $\vec{v}_i = (0 \text{ m/s})\hat{i}$. The average force on the phone is related to the impulse the floor applies on it during the collision:

$$\vec{F}_{\text{ave}} = \frac{\vec{J}}{\Delta t}.$$

The impulse $\vec{J}$ equals the change in momentum,

$$\vec{J} = \Delta \vec{p}$$

so

$$\vec{F}_{\text{ave}} = \frac{\Delta \vec{p}}{\Delta t}.$$

Next, the change of momentum is

$$\Delta \vec{p} = m\Delta \vec{v}.$$

We need to be careful with the velocities here; this is the change of velocity due to the collision with the floor. But the phone also has an initial drop velocity [$\vec{v}_i = (0 \text{ m/s})\hat{j}$], so we label our velocities. Let:

- $\vec{v}_i$ = the initial velocity with which the phone was dropped (zero, in this example)

- $\vec{v}_1$ = the velocity the phone had the instant just before it hit the floor
- $\vec{v}_2$ = the final velocity of the phone as a result of hitting the floor

Figure 9.12 shows the velocities at each of these points in the phone's trajectory.

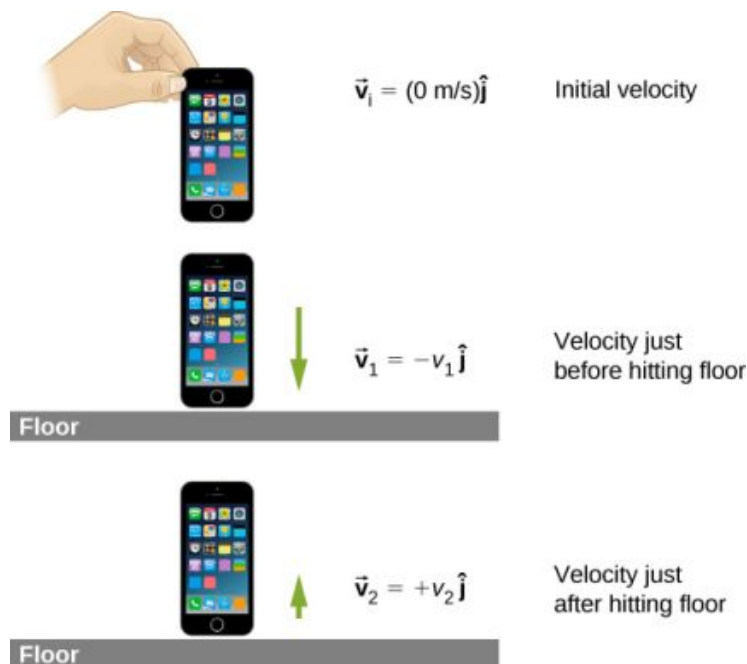


FIGURE 9.12 (a) The initial velocity of the phone is zero, just after the person drops it. (b) Just before the phone hits the floor, its velocity is $\vec{v}_1$, which is unknown at the moment, except for its direction, which is downward ($-\hat{j}$). (c) After bouncing off the floor, the phone has a velocity $\vec{v}_2$, which is also unknown, except for its direction, which is upward ($+\hat{j}$).

With these definitions, the change of momentum of the phone during the collision with the floor is

$$m\Delta\vec{v} = m(\vec{v}_2 - \vec{v}_1).$$

Since we assume the phone doesn't bounce at all when it hits the floor (or at least, the bounce height is negligible), then $\vec{v}_2$ is zero, so

$$\begin{aligned} m\Delta\vec{v} &= m[0 - (-v_1\hat{j})] \\ m\Delta\vec{v} &= +mv_1\hat{j}. \end{aligned}$$

We can get the speed of the phone just before it hits the floor using either kinematics or conservation of energy. We'll use conservation of energy here; you should re-do this part of the problem using kinematics and prove that you get the same answer.

First, define the zero of potential energy to be located at the floor. Conservation of energy then gives us:

$$\begin{aligned} E_i &= E_f \\ K_i + U_i &= K_f + U_f \\ \frac{1}{2}mv_i^2 + mgh_{\text{drop}} &= \frac{1}{2}mv_f^2 + mgh_{\text{floor}}. \end{aligned}$$

Defining $h_{\text{floor}} = 0$ and using $\vec{v}_i = (0 \text{ m/s})\hat{j}$ gives

$$\begin{aligned} \frac{1}{2}mv_1^2 &= mgh_{\text{drop}} \\ v_1 &= \pm\sqrt{2gh_{\text{drop}}}. \end{aligned}$$

Because v_1 is a vector magnitude, it must be positive. Thus, $m\Delta v = mv_1 = m\sqrt{2gh_{\text{drop}}}$. Inserting this result into the expression for force gives

$$\begin{aligned}
 \vec{\mathbf{F}} &= \frac{\Delta \vec{\mathbf{p}}}{\Delta t} \\
 &= \frac{m \Delta \vec{\mathbf{v}}}{\Delta t} \\
 &= \frac{+mv_1 \hat{\mathbf{j}}}{\Delta t} \\
 &= \frac{m \sqrt{2gh}}{\Delta t} \hat{\mathbf{j}}.
 \end{aligned}$$

Finally, we need to estimate the collision time. One common way to estimate a collision time is to calculate how long the object would take to travel its own length. The phone is moving at 5.4 m/s just before it hits the floor, and it is 0.14 m long, giving an estimated collision time of 0.026 s. Inserting the given numbers, we obtain

$$\vec{\mathbf{F}} = \frac{(0.172 \text{ kg}) \sqrt{2(9.8 \text{ m/s}^2)(1.5 \text{ m})}}{0.026 \text{ s}} \hat{\mathbf{j}} = (36 \text{ N}) \hat{\mathbf{j}}.$$

Significance

The iPhone itself weighs just $(0.172 \text{ kg})(9.81 \text{ m/s}^2) = 1.68 \text{ N}$; the force the floor applies to it is therefore over 20 times its weight.

✓ CHECK YOUR UNDERSTANDING 9.2

What if we had assumed the phone *did* bounce on impact? Would this have increased the force on the iPhone, decreased it, or made no difference?

Momentum and Force

In [Example 9.3](#), we obtained an important relationship:

$$\vec{\mathbf{F}}_{\text{ave}} = \frac{\Delta \vec{\mathbf{p}}}{\Delta t}. \quad 9.8$$

In words, the average force applied to an object is equal to the change of the momentum that the force causes, divided by the time interval over which this change of momentum occurs. This relationship is very useful in situations where the collision time Δt is small, but measureable; typical values would be 1/10th of a second, or even one thousandth of a second. Car crashes, punting a football, or collisions of subatomic particles would meet this criterion.

For a *continuously* changing momentum—due to a continuously changing force—this becomes a powerful conceptual tool. In the limit $\Delta t \rightarrow dt$, [Equation 9.2](#) becomes

$$\vec{\mathbf{F}} = \frac{d\vec{\mathbf{p}}}{dt}. \quad 9.9$$

This says that the rate of change of the system's momentum (implying that momentum is a function of time) is exactly equal to the net applied force (also, in general, a function of time). This is, in fact, Newton's second law, written in terms of momentum rather than acceleration. This is the relationship Newton himself presented in his *Principia Mathematica* (although he called it “quantity of motion” rather than “momentum”).

If the mass of the system remains constant, [Equation 9.3](#) reduces to the more familiar form of Newton's second law. We can see this by substituting the definition of momentum:

$$\vec{\mathbf{F}} = \frac{d(m\vec{\mathbf{v}})}{dt} = m \frac{d\vec{\mathbf{v}}}{dt} = m\vec{\mathbf{a}}.$$

The assumption of constant mass allowed us to pull m out of the derivative. If the mass is not constant, we cannot

use this form of the second law, but instead must start from [Equation 9.3](#). Thus, one advantage to expressing force in terms of changing momentum is that it allows for the mass of the system to change, as well as the velocity; this is a concept we'll explore when we study the motion of rockets.

NEWTON'S SECOND LAW OF MOTION IN TERMS OF MOMENTUM

The net external force on a system is equal to the rate of change of the momentum of that system caused by the force:

$$\vec{\mathbf{F}} = \frac{d\vec{\mathbf{p}}}{dt}.$$

Although [Equation 9.3](#) allows for changing mass, as we will see in [Rocket Propulsion](#), the relationship between momentum and force remains useful when the mass of the system is constant, as in the following example.



EXAMPLE 9.5

Calculating Force: Venus Williams' Tennis Serve

During the 2007 French Open, Venus Williams hit the fastest recorded serve in a premier women's match, reaching a speed of 58 m/s (209 km/h). What is the average force exerted on the 0.057-kg tennis ball by Venus Williams' racquet? Assume that the ball's speed just after impact is 58 m/s, as shown in [Figure 9.13](#), that the initial horizontal component of the velocity before impact is negligible, and that the ball remained in contact with the racquet for 5.0 ms.

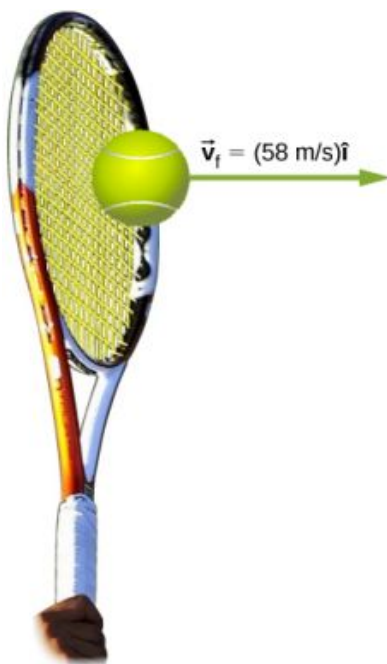


FIGURE 9.13 The final velocity of the tennis ball is $\vec{v}_f = (58 \text{ m/s})\hat{i}$.

Strategy

This problem involves only one dimension because the ball starts from having no horizontal velocity component before impact. Newton's second law stated in terms of momentum is then written as

$$\vec{\mathbf{F}} = \frac{d\vec{\mathbf{p}}}{dt}.$$

As noted above, when mass is constant, the change in momentum is given by

$$\Delta p = m\Delta v = m(v_f - v_i)$$

where we have used scalars because this problem involves only one dimension. In this example, the velocity just after impact and the time interval are given; thus, once Δp is calculated, we can use $F = \frac{\Delta p}{\Delta t}$ to find the force.

Solution

To determine the change in momentum, insert the values for the initial and final velocities into the equation above:

$$\begin{aligned}\Delta p &= m(v_f - v_i) \\ &= (0.057 \text{ kg})(58 \text{ m/s} - 0 \text{ m/s}) \\ &= 3.3 \frac{\text{kg}\cdot\text{m}}{\text{s}}.\end{aligned}$$

Now the magnitude of the net external force can be determined by using

$$F = \frac{\Delta p}{\Delta t} = \frac{3.3 \frac{\text{kg}\cdot\text{m}}{\text{s}}}{5.0 \times 10^{-3} \text{ s}} = 6.6 \times 10^2 \text{ N}.$$

where we have retained only two significant figures in the final step.

Significance

This quantity was the average force exerted by Venus Williams' racquet on the tennis ball during its brief impact (note that the ball also experienced the 0.57-N force of gravity, but that force was not due to the racquet). This problem could also be solved by first finding the acceleration and then using $F = ma$, but one additional step would be required compared with the strategy used in this example.

9.3 Conservation of Linear Momentum

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Explain the meaning of “conservation of momentum”
- Correctly identify if a system is, or is not, closed
- Define a system whose momentum is conserved
- Mathematically express conservation of momentum for a given system
- Calculate an unknown quantity using conservation of momentum

Recall Newton's third law: When two objects of masses m_1 and m_2 interact (meaning that they apply forces on each other), the force that object 2 applies to object 1 is equal in magnitude and opposite in direction to the force that object 1 applies on object 2. Let:

- $\vec{F}_{21}$ = the force on m_1 from m_2
- $\vec{F}_{12}$ = the force on m_2 from m_1

Then, in symbols, Newton's third law says

$$\begin{aligned}\vec{F}_{21} &= -\vec{F}_{12} \\ m_1 \vec{a}_1 &= -m_2 \vec{a}_2.\end{aligned}\tag{9.10}$$

(Recall that these two forces do not cancel because they are applied to different objects. F_{21} causes m_1 to accelerate, and F_{12} causes m_2 to accelerate.)

Although the magnitudes of the forces on the objects are the same, the accelerations are not, simply because the masses (in general) are different. Therefore, the changes in velocity of each object are different:

$$\frac{d\vec{v}_1}{dt} \neq \frac{d\vec{v}_2}{dt}.$$

However, the products of the mass and the change of velocity are equal (in magnitude):

$$m_1 \frac{d\vec{v}_1}{dt} = -m_2 \frac{d\vec{v}_2}{dt}. \quad 9.11$$

It's a good idea, at this point, to make sure you're clear on the physical meaning of the derivatives in [Equation 9.3](#). Because of the interaction, each object ends up getting its velocity changed, by an amount dv . Furthermore, the interaction occurs over a time interval dt , which means that the change of velocities also occurs over dt . This time interval is the same for each object.

Let's assume, for the moment, that the masses of the objects do not change during the interaction. (We'll relax this restriction later.) In that case, we can pull the masses inside the derivatives:

$$\frac{d}{dt}(m_1 \vec{v}_1) = -\frac{d}{dt}(m_2 \vec{v}_2) \quad 9.12$$

and thus

$$\frac{d\vec{p}_1}{dt} = -\frac{d\vec{p}_2}{dt}. \quad 9.13$$

This says that *the rate at which momentum changes is the same for both objects*. The masses are different, and the changes of velocity are different, but the rate of change of the product of m and $\vec{v}$ are the same.

Physically, this means that during the interaction of the two objects (m_1 and m_2), both objects have their momentum changed; but those changes are identical in magnitude, though opposite in sign. For example, the momentum of object 1 might increase, which means that the momentum of object 2 decreases by exactly the same amount.

In light of this, let's re-write [Equation 9.12](#) in a more suggestive form:

$$\frac{d\vec{p}_1}{dt} + \frac{d\vec{p}_2}{dt} = 0. \quad 9.14$$

This says that during the interaction, although object 1's momentum changes, and object 2's momentum also changes, these two changes cancel each other out, so that the total change of momentum of the two objects together is zero.

Since the total combined momentum of the two objects together never changes, then we could write

$$\frac{d}{dt}(\vec{p}_1 + \vec{p}_2) = 0 \quad 9.15$$

from which it follows that

$$\vec{p}_1 + \vec{p}_2 = \text{constant}. \quad 9.16$$

As shown in [Figure 9.14](#), the total momentum of the system before and after the collision remains the same.

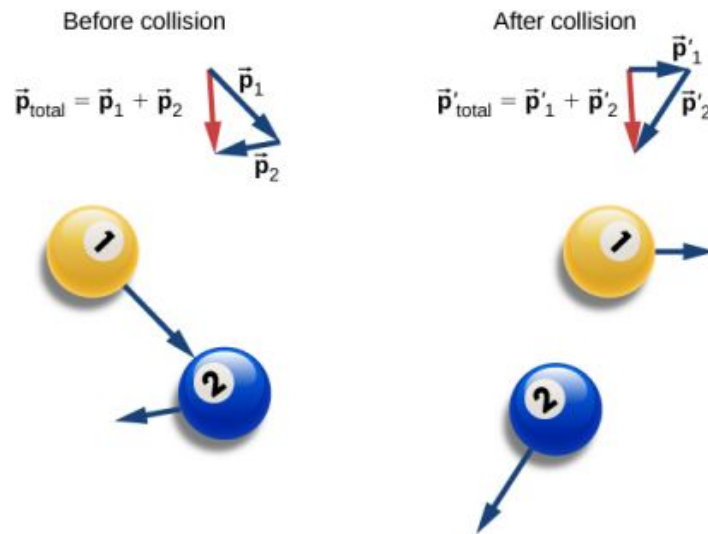


FIGURE 9.14 Before the collision, the two billiard balls travel with momenta $\vec{p}_1$ and $\vec{p}_2$. The total momentum of the system is the sum of these, as shown by the red vector labeled $\vec{p}_{\text{total}}$ on the left. After the collision, the two billiard balls travel with different momenta $\vec{p}'_1$ and $\vec{p}'_2$. The total momentum, however, has not changed, as shown by the red vector arrow $\vec{p}'_{\text{total}}$ on the right.

Generalizing this result to N objects, we obtain

$$\begin{aligned}\vec{p}_1 + \vec{p}_2 + \vec{p}_3 + \cdots + \vec{p}_N &= \text{constant} \\ \sum_{j=1}^N \vec{p}_j &= \text{constant.}\end{aligned}\tag{9.17}$$

Equation 9.17 is the definition of the total (or net) momentum of a system of N interacting objects, along with the statement that the total momentum of a system of objects is constant in time—or better, is conserved.

CONSERVATION LAWS

If the value of a physical quantity is constant in time, we say that the quantity is conserved.

Requirements for Momentum Conservation

There is a complication, however. A system must meet two requirements for its momentum to be conserved:

1. *The mass of the system must remain constant during the interaction.*

As the objects interact (apply forces on each other), they may *transfer* mass from one to another; but any mass one object gains is balanced by the loss of that mass from another. The total mass of the system of objects, therefore, remains unchanged as time passes:

$$\left[\frac{dm}{dt} \right]_{\text{system}} = 0.$$

2. *The net external force on the system must be zero.*

As the objects collide, or explode, and move around, they exert forces on each other. However, all of these forces are internal to the system, and thus each of these internal forces is balanced by another internal force that is equal in magnitude and opposite in sign. As a result, the change in momentum caused by each internal force is cancelled by another momentum change that is equal in magnitude and opposite in direction.

Therefore, internal forces cannot change the total momentum of a system because the changes sum to zero. However, if there is some external force that acts on all of the objects (gravity, for example, or friction), then this force changes the momentum of the system as a whole; that is to say, the momentum of the system is changed by the external force. Thus, for the momentum of the system to be conserved, we must have

$$\vec{F}_{\text{ext}} = \vec{0}.$$

A system of objects that meets these two requirements is said to be a **closed system** (also called an isolated system). Thus, the more compact way to express this is shown below.

LAW OF CONSERVATION OF MOMENTUM

The total momentum of a closed system is conserved:

$$\sum_{j=1}^N \vec{p}_j = \text{constant}.$$

This statement is called the **Law of Conservation of Momentum**. Along with the conservation of energy, it is one of the foundations upon which all of physics stands. All our experimental evidence supports this statement: from the motions of galactic clusters to the quarks that make up the proton and the neutron, and at every scale in between. *In a closed system, the total momentum never changes.*

Note that there absolutely *can* be external forces acting on the system; but for the system's momentum to remain constant, these external forces have to cancel, so that the *net* external force is zero. Billiard balls on a table all have a weight force acting on them, but the weights are balanced (canceled) by the normal forces, so there is no *net* force.

The Meaning of 'System'

A **system** (mechanical) is the collection of objects in whose motion (kinematics and dynamics) you are interested. If you are analyzing the bounce of a ball on the ground, you are probably only interested in the motion of the ball, and not of Earth; thus, the ball is your system. If you are analyzing a car crash, the two cars together compose your system ([Figure 9.15](#)).

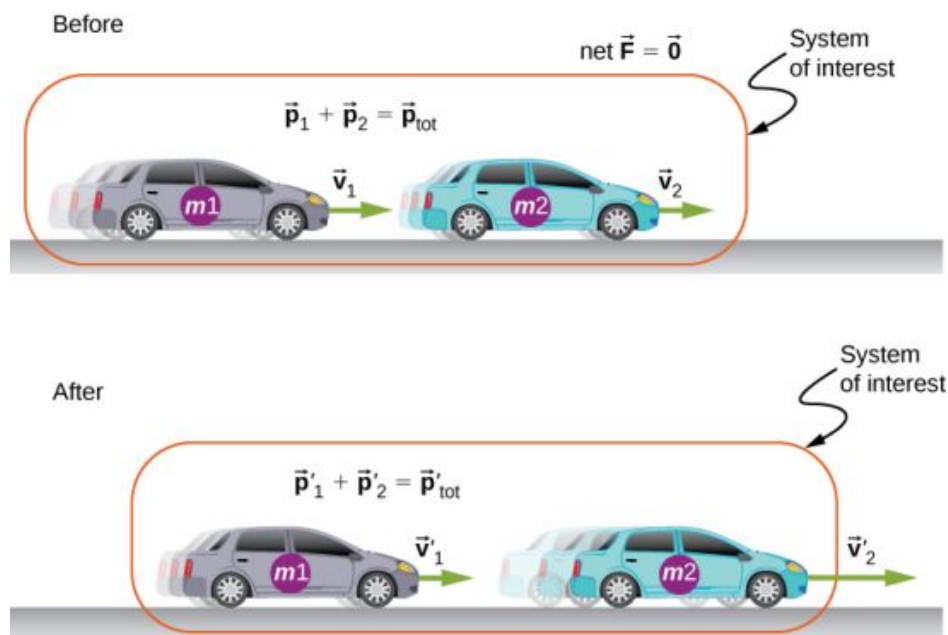


FIGURE 9.15 The two cars together form the system that is to be analyzed. It is important to remember that the contents (the mass) of the system do not change before, during, or after the objects in the system interact.



PROBLEM-SOLVING STRATEGY

Conservation of Momentum

Using conservation of momentum requires four basic steps. The first step is crucial:

1. Identify a closed system (total mass is constant, no net external force acts on the system).
2. Write down an expression representing the total momentum of the system before the “event” (explosion or collision).
3. Write down an expression representing the total momentum of the system after the “event.”
4. Set these two expressions equal to each other, and solve this equation for the desired quantity.



EXAMPLE 9.6

Colliding Carts

Two carts in a physics lab roll on a level track, with negligible friction. These carts have small magnets at their ends, so that when they collide, they stick together (Figure 9.16). The first cart has a mass of 675 grams and is rolling at 0.75 m/s to the right; the second has a mass of 500 grams and is rolling at 1.33 m/s, also to the right. After the collision, what is the velocity of the two joined carts?



FIGURE 9.16 Two lab carts collide and stick together after the collision.

Strategy

We have a collision. We’re given masses and initial velocities; we’re asked for the final velocity. This all suggests using conservation of momentum as a method of solution. However, we can only use it if we have a closed system. So we need to be sure that the system we choose has no net external force on it, and that its mass is not changed by the collision.

Defining the system to be the two carts meets the requirements for a closed system: The combined mass of the two carts certainly doesn’t change, and while the carts definitely exert forces on each other, those forces are internal to the system, so they do not change the momentum of the system as a whole. In the vertical direction, the weights of the carts are canceled by the normal forces on the carts from the track.

Solution

Conservation of momentum is

$$\vec{p}_f = \vec{p}_i.$$

Define the direction of their initial velocity vectors to be the +x-direction. The initial momentum is then

$$\vec{p}_i = m_1 v_1 \hat{i} + m_2 v_2 \hat{i}.$$

The final momentum of the now-linked carts is

$$\vec{p}_f = (m_1 + m_2) \vec{v}_f.$$

Equating:

$$\begin{aligned} (m_1 + m_2) \vec{v}_f &= m_1 v_1 \hat{i} + m_2 v_2 \hat{i} \\ \vec{v}_f &= \left(\frac{m_1 v_1 + m_2 v_2}{m_1 + m_2} \right) \hat{i}. \end{aligned}$$

Substituting the given numbers:

$$\begin{aligned} \vec{v}_f &= \left[\frac{(0.675 \text{ kg})(0.75 \text{ m/s}) + (0.5 \text{ kg})(1.33 \text{ m/s})}{1.175 \text{ kg}} \right] \hat{i} \\ &= (0.997 \text{ m/s}) \hat{i}. \end{aligned}$$

Significance

The principles that apply here to two laboratory carts apply identically to all objects of whatever type or size. Even for photons, the concepts of momentum and conservation of momentum are still crucially important even at that scale. (Since they are massless, the momentum of a photon is defined very differently from the momentum of ordinary objects. You will learn about this when you study quantum physics.)

✓ CHECK YOUR UNDERSTANDING 9.3

Suppose the second, smaller cart had been initially moving to the left. What would the sign of the final velocity have been in this case?

✳ EXAMPLE 9.7

A Bouncing Superball

A superball of mass 0.25 kg is dropped from rest from a height of $h = 1.50$ m above the floor. It bounces with no loss of energy and returns to its initial height (Figure 9.17).

- What is the superball's change of momentum during its bounce on the floor?
- What was Earth's change of momentum due to the ball colliding with the floor?
- What was Earth's change of velocity as a result of this collision?

(This example shows that you have to be careful about defining your system.)

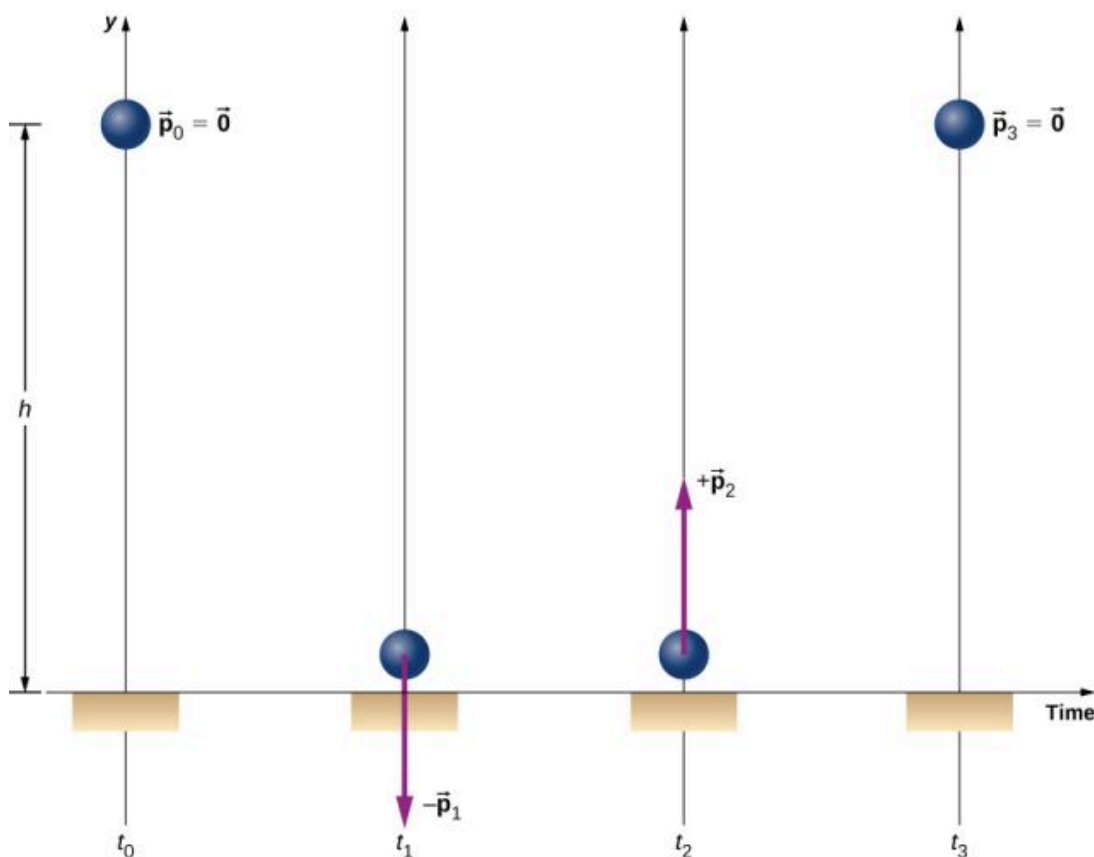


FIGURE 9.17 A superball is dropped to the floor (t_0), hits the floor (t_1), bounces (t_2), and returns to its initial height (t_3).

Strategy

Since we are asked only about the ball's change of momentum, we define our system to be the ball. But this is

clearly not a closed system; gravity applies a downward force on the ball while it is falling, and the normal force from the floor applies a force during the bounce. Thus, we cannot use conservation of momentum as a strategy. Instead, we simply determine the ball's momentum just before it collides with the floor and just after, and calculate the difference. We have the ball's mass, so we need its velocities.

Solution

- a. Since this is a one-dimensional problem, we use the scalar form of the equations. Let:
- p_0 = the magnitude of the ball's momentum at time t_0 , the moment it was released; since it was dropped from rest, this is zero.
 - p_1 = the magnitude of the ball's momentum at time t_1 , the instant just before it hits the floor.
 - p_2 = the magnitude of the ball's momentum at time t_2 , just after it loses contact with the floor after the bounce.

The ball's change of momentum is

$$\begin{aligned}\Delta\vec{p} &= \vec{p}_2 - \vec{p}_1 \\ &= p_2\hat{j} - (-p_1\hat{j}) \\ &= (p_2 + p_1)\hat{j}.\end{aligned}$$

Its velocity just before it hits the floor can be determined from either conservation of energy or kinematics. We use kinematics here; you should re-solve it using conservation of energy and confirm you get the same result. We want the velocity just before it hits the ground (at time t_1). We know its initial velocity $v_0 = 0$ (at time t_0), the height it falls, and its acceleration; we don't know the fall time. We could calculate that, but instead we use

$$\vec{v}_1 = -\hat{j}\sqrt{2gy} = -5.4 \text{ m/s}\hat{j}.$$

Thus the ball has a momentum of

$$\begin{aligned}\vec{p}_1 &= -(0.25 \text{ kg})(-5.4 \text{ m/s}\hat{j}) \\ &= -(1.4 \text{ kg} \cdot \text{m/s})\hat{j}.\end{aligned}$$

We don't have an easy way to calculate the momentum after the bounce. Instead, we reason from the symmetry of the situation.

Before the bounce, the ball starts with zero velocity and falls 1.50 m under the influence of gravity, achieving some amount of momentum just before it hits the ground. On the return trip (after the bounce), it starts with some amount of momentum, rises the same 1.50 m it fell, and ends with zero velocity. Thus, the motion after the bounce was the mirror image of the motion before the bounce. From this symmetry, it must be true that the ball's momentum after the bounce must be equal and opposite to its momentum before the bounce. (This is a subtle but crucial argument; make sure you understand it before you go on.)

Therefore,

$$\vec{p}_2 = -\vec{p}_1 = +(1.4 \text{ kg} \cdot \text{m/s})\hat{j}.$$

Thus, the ball's change of momentum during the bounce is

$$\begin{aligned}\Delta\vec{p} &= \vec{p}_2 - \vec{p}_1 \\ &= (1.4 \text{ kg} \cdot \text{m/s})\hat{j} - (-1.4 \text{ kg} \cdot \text{m/s})\hat{j} \\ &= +(2.8 \text{ kg} \cdot \text{m/s})\hat{j}.\end{aligned}$$

- b. What was Earth's change of momentum due to the ball colliding with the floor?
- Your instinctive response may well have been either "zero; the Earth is just too massive for that tiny ball to have affected it" or possibly, "more than zero, but utterly negligible." But no—if we re-define our system to be the Superball + Earth, then this system is closed (neglecting the gravitational pulls of the Sun, the Moon, and the other planets in the solar system), and therefore the total change of momentum of this new system must be zero. Therefore, Earth's change of momentum is exactly the same magnitude:

$$\Delta\vec{p}_{\text{Earth}} = -2.8 \text{ kg} \cdot \text{m/s}\hat{j}.$$

- c. What was Earth's change of velocity as a result of this collision?

This is where your instinctive feeling is probably correct:

$$\begin{aligned}\Delta \vec{v}_{\text{Earth}} &= \frac{\Delta \vec{p}_{\text{Earth}}}{M_{\text{Earth}}} \\ &= -\frac{2.8 \text{ kg}\cdot\text{m/s}}{5.97 \times 10^{24} \text{ kg}} \hat{\mathbf{j}} \\ &= -(4.7 \times 10^{-25} \text{ m/s}) \hat{\mathbf{j}}.\end{aligned}$$

This change of Earth's velocity is utterly negligible.

Significance

It is important to realize that the answer to part (c) is not a velocity; it is a change of velocity, which is a very different thing. Nevertheless, to give you a feel for just how small that change of velocity is, suppose you were moving with a velocity of $4.7 \times 10^{-25} \text{ m/s}$. At this speed, it would take you about 7 million years to travel a distance equal to the diameter of a hydrogen atom.

✓ CHECK YOUR UNDERSTANDING 9.4

Would the ball's change of momentum have been larger, smaller, or the same, if it had collided with the floor and stopped (without bouncing)?

EXAMPLE 9.8

Ice Hockey 1

Two hockey pucks of identical mass are on a flat, horizontal ice hockey rink. The red puck is motionless; the blue puck is moving at 2.5 m/s to the left (Figure 9.18). It collides with the motionless red puck. The pucks have a mass of 15 g. After the collision, the red puck is moving at 2.5 m/s, to the left. What is the final velocity of the blue puck?

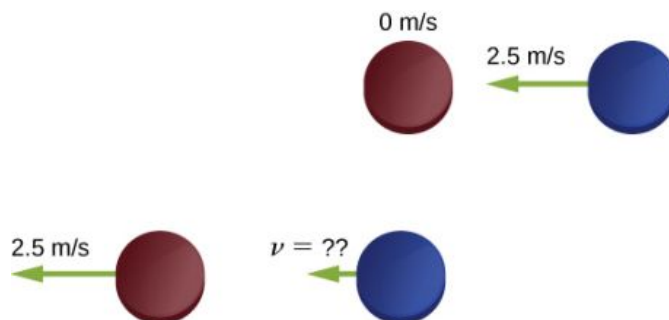


FIGURE 9.18 Two identical hockey pucks colliding. The top diagram shows the pucks the instant before the collision, and the bottom diagram shows the pucks the instant after the collision. The net external force is zero.

Strategy

We're told that we have two colliding objects, we're told the masses and initial velocities, and one final velocity; we're asked for both final velocities. Conservation of momentum seems like a good strategy. Define the system to be the two pucks; there's no friction, so we have a closed system.

Before you look at the solution, what do you think the answer will be?

The blue puck final velocity will be:

- zero
- 2.5 m/s to the left
- 2.5 m/s to the right
- 1.25 m/s to the left
- 1.25 m/s to the right

- something else

Solution

Define the $+x$ -direction to point to the right. Conservation of momentum then reads

$$\begin{aligned}\vec{p}_f &= \vec{p}_i \\ mv_{rf}\hat{i} + mv_{bf}\hat{i} &= mv_{ri}\hat{i} - mv_{bi}\hat{i}.\end{aligned}$$

Before the collision, the momentum of the system is entirely and only in the blue puck. Thus,

$$\begin{aligned}mv_{rf}\hat{i} + mv_{bf}\hat{i} &= -mv_{bi}\hat{i} \\ v_{rf}\hat{i} + v_{bf}\hat{i} &= -v_{bi}\hat{i}.\end{aligned}$$

(Remember that the masses of the pucks are equal.) Substituting numbers:

$$\begin{aligned}-(2.5 \text{ m/s})\hat{i} + \vec{v}_{bf} &= -(2.5 \text{ m/s})\hat{i} \\ \vec{v}_{bf} &= 0.\end{aligned}$$

Significance

Evidently, the two pucks simply exchanged momentum. The blue puck transferred all of its momentum to the red puck. In fact, this is what happens in similar collision where $m_1 = m_2$.

✓ CHECK YOUR UNDERSTANDING 9.5

Even if there were some friction on the ice, it is still possible to use conservation of momentum to solve this problem, but you would need to impose an additional condition on the problem. What is that additional condition?

✿ EXAMPLE 9.9

Landing of *Philae*

On November 12, 2014, the European Space Agency successfully landed a probe named *Philae* on Comet 67P/Churyumov/Gerasimenko (Figure 9.19). During the landing, however, the probe actually landed three times, because it bounced twice. Let's calculate how much the comet's speed changed as a result of the first bounce.



FIGURE 9.19 An artist's rendering of *Philae* landing on a comet. (credit: modification of work by "DLR German Aerospace Center"/Flickr)

Let's define upward to be the $+y$ -direction, perpendicular to the surface of the comet, and $y = 0$ to be at the surface of the comet. Here's what we know:

- The mass of Comet 67P: $M_c = 1.0 \times 10^{13} \text{ kg}$
- The acceleration due to the comet's gravity: $\vec{a} = -(5.0 \times 10^{-3} \text{ m/s}^2) \hat{j}$
- *Philae's* mass: $M_p = 96 \text{ kg}$
- Initial touchdown speed: $\vec{v}_1 = -(1.0 \text{ m/s}) \hat{j}$
- Initial upward speed due to first bounce: $\vec{v}_2 = (0.38 \text{ m/s}) \hat{j}$
- Landing impact time: $\Delta t = 1.3 \text{ s}$

Strategy

We're asked for how much the comet's speed changed, but we don't know much about the comet, beyond its mass and the acceleration its gravity causes. However, we *are* told that the *Philae* lander collides with (lands on) the comet, and bounces off of it. A collision suggests momentum as a strategy for solving this problem.

If we define a system that consists of both *Philae* and Comet 67/P, then there is no net external force on this system, and thus the momentum of this system is conserved. (We'll neglect the gravitational force of the sun.) Thus, if we calculate the change of momentum of the lander, we automatically have the change of momentum of the comet. Also, the comet's change of velocity is directly related to its change of momentum as a result of the lander "colliding" with it.

Solution

Let $\vec{p}_1$ be *Philae's* momentum at the moment just before touchdown, and $\vec{p}_2$ be its momentum just after the first bounce. Then its momentum just before landing was

$$\vec{p}_1 = M_p \vec{v}_1 = (96 \text{ kg}) (-1.0 \text{ m/s} \hat{j}) = -(96 \text{ kg} \cdot \text{m/s}) \hat{j}$$

and just after was

$$\vec{p}_2 = M_p \vec{v}_2 = (96 \text{ kg}) (+0.38 \text{ m/s} \hat{j}) = (36.5 \text{ kg} \cdot \text{m/s}) \hat{j}.$$

Therefore, the lander's change of momentum during the first bounce is

$$\begin{aligned} \Delta \vec{p} &= \vec{p}_2 - \vec{p}_1 \\ &= (36.5 \text{ kg} \cdot \text{m/s}) \hat{j} - (-96.0 \text{ kg} \cdot \text{m/s} \hat{j}) = (133 \text{ kg} \cdot \text{m/s}) \hat{j} \end{aligned}$$

Notice how important it is to include the negative sign of the initial momentum.

Now for the comet. Since momentum of the system must be conserved, the *comet's* momentum changed by exactly the negative of this:

$$\Delta \vec{p}_c = -\Delta \vec{p} = -(133 \text{ kg} \cdot \text{m/s}) \hat{j}.$$

Therefore, its change of velocity is

$$\Delta \vec{v}_c = \frac{\Delta \vec{p}_c}{M_c} = \frac{-(133 \text{ kg} \cdot \text{m/s}) \hat{j}}{1.0 \times 10^{13} \text{ kg}} = -(1.33 \times 10^{-11} \text{ m/s}) \hat{j}.$$

Significance

This is a very small change in velocity, about a thousandth of a billionth of a meter per second. Crucially, however, it is *not* zero.

CHECK YOUR UNDERSTANDING 9.6

The changes of momentum for *Philae* and for Comet 67/P were equal (in magnitude). Were the impulses experienced by *Philae* and the comet equal? How about the forces? How about the changes of kinetic energies?

9.4 Types of Collisions

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Identify the type of collision
- Correctly label a collision as elastic or inelastic
- Use kinetic energy along with momentum and impulse to analyze a collision

Although momentum is conserved in all interactions, not all interactions (collisions or explosions) are the same. The possibilities include:

- A single object can explode into multiple objects (explosions).
- Multiple objects can collide and bounce off each other, called an elastic collision, resulting in the same kinetic energy of the system before and after the collision.
- Multiple objects can collide and the system loses kinetic energy, called an inelastic collision. One such case is where the two objects stick together, forming a single object.

It's useful, therefore, to categorize different types of interactions, according to how the interacting objects move before and after the interaction.

Explosions

The first possibility is that a single object may break apart into two or more pieces. An example of this is a firecracker, or a bow and arrow, or a rocket rising through the air toward space. These can be difficult to analyze if the number of fragments after the collision is more than about three or four; but nevertheless, the total momentum of the system before and after the explosion is identical.

Note that if the object is initially motionless, then the system (which is just the object) has no momentum and no kinetic energy. After the explosion, the net momentum of all the pieces of the object must sum to zero (since the momentum of this closed system cannot change). However, the system *will* have a great deal of kinetic energy after the explosion, although it had none before. Thus, we see that, although the momentum of the system is conserved in an explosion, the kinetic energy of the system most definitely is not; it increases. This interaction—one object becoming many, with an increase of kinetic energy of the system—is called an **explosion**.

Where does the energy come from? Does conservation of energy still hold? Yes; some form of potential energy is converted to kinetic energy. In the case of gunpowder burning and pushing out a bullet, chemical potential energy is converted to kinetic energy of the bullet, and of the recoiling gun. For a bow and arrow, it is elastic potential energy in the bowstring.

Inelastic

The second possibility is the reverse: that two or more objects collide with each other and stick together, thus (after the collision) forming one single composite object. The total mass of this composite object is the sum of the masses of the original objects, and the new single object moves with a velocity dictated by the conservation of momentum. However, it turns out again that, although the total momentum of the system of objects remains constant, the kinetic energy doesn't; but this time, the kinetic energy decreases. This type of collision is called **inelastic**.

Any collision where the objects stick together will result in the maximum loss of kinetic energy (i.e., K_f will be a minimum).

Such a collision is called **perfectly inelastic**. In the extreme case, multiple objects collide, stick together, and remain motionless after the collision. Since the objects are all motionless after the collision, the final kinetic energy is also zero; therefore, the loss of kinetic energy is a maximum.

- If $0 < K_f < K_i$, the collision is inelastic.
- If K_f is the lowest energy, or the energy lost by both objects is the most, the collision is perfectly inelastic (objects stick together).
- If $K_f = K_i$, the collision is elastic.

Elastic

The extreme case on the other end is if two or more objects approach each other, collide, and bounce off each other, moving away from each other at the same relative speed at which they approached each other. In this case, the total kinetic energy of the system is conserved. Such an interaction is called **elastic**.



PROBLEM-SOLVING STRATEGY

Collisions

A closed system always conserves momentum; it might also conserve kinetic energy, but very often it doesn't. Energy-momentum problems confined to a plane (as ours are) usually have two unknowns. Generally, this approach works well:

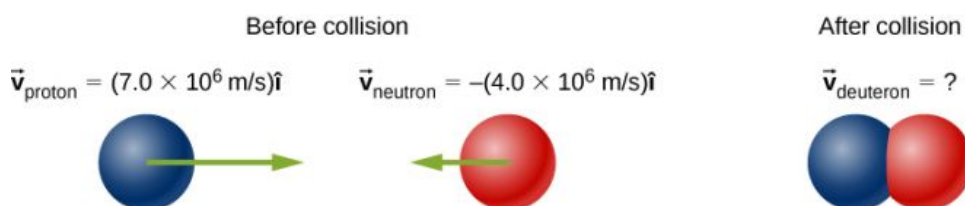
1. Define a closed system.
2. Write down the expression for conservation of momentum.
3. If kinetic energy is conserved, write down the expression for conservation of kinetic energy; if not, write down the expression for the change of kinetic energy.
4. You now have two equations in two unknowns, which you solve by standard methods.



EXAMPLE 9.10

Formation of a Deuteron

A proton (mass 1.67×10^{-27} kg) collides with a neutron (with essentially the same mass as the proton) to form a particle called a *deuteron*. What is the velocity of the deuteron if it is formed from a proton moving with velocity 7.0×10^6 m/s to the right and a neutron moving with velocity 4.0×10^6 m/s to the left?



Strategy

Define the system to be the two particles. This is a collision, so we should first identify what kind. Since we are told the two particles form a single particle after the collision, this means that the collision is perfectly inelastic. Thus, kinetic energy is not conserved, but momentum is. Thus, we use conservation of momentum to determine the final velocity of the system.

Solution

Treat the two particles as having identical masses M . Use the subscripts p, n, and d for proton, neutron, and deuteron, respectively. This is a one-dimensional problem, so we have

$$Mv_p - Mv_n = 2Mv_d.$$

The masses divide out:

$$\begin{aligned} v_p - v_n &= 2v_d \\ 7.0 \times 10^6 \text{ m/s} - 4.0 \times 10^6 \text{ m/s} &= 2v_d \\ v_d &= 1.5 \times 10^6 \text{ m/s.} \end{aligned}$$

The velocity is thus $\vec{v}_d = (1.5 \times 10^6 \text{ m/s}) \hat{i}$.

Significance

This is essentially how particle colliders like the Large Hadron Collider work: They accelerate particles up to very

high speeds (large momenta), but in opposite directions. This maximizes the creation of so-called “daughter particles.”



EXAMPLE 9.11

Ice Hockey 2

(This is a variation of an earlier example.)

Two ice hockey pucks of different masses are on a flat, horizontal hockey rink. The red puck has a mass of 15 grams, and is motionless; the blue puck has a mass of 12 grams, and is moving at 2.5 m/s to the left. It collides with the motionless red puck (Figure 9.20). If the collision is perfectly elastic, what are the final velocities of the two pucks?

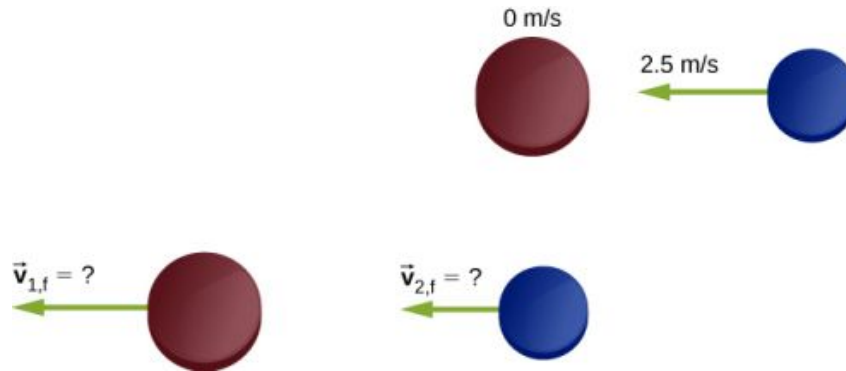


FIGURE 9.20 Two different hockey pucks colliding. The top diagram shows the pucks the instant before the collision, and the bottom diagram shows the pucks the instant after the collision. The net external force is zero.

Strategy

We’re told that we have two colliding objects, and we’re told their masses and initial velocities; we’re asked for both final velocities. Conservation of momentum seems like a good strategy; define the system to be the two pucks. There is no friction, so we have a closed system. We have two unknowns (the two final velocities), but only one equation. The comment about the collision being perfectly elastic is the clue; it suggests that kinetic energy is also conserved in this collision. That gives us our second equation.

The initial momentum and initial kinetic energy of the system resides entirely and only in the second puck (the blue one); the collision transfers some of this momentum and energy to the first puck.

Solution

Conservation of momentum, in this case, reads

$$\begin{aligned} p_i &= p_f \\ m_2 v_{2,i} &= m_1 v_{1,f} + m_2 v_{2,f}. \end{aligned}$$

Conservation of kinetic energy reads

$$\begin{aligned} K_i &= K_f \\ \frac{1}{2} m_2 v_{2,i}^2 &= \frac{1}{2} m_1 v_{1,f}^2 + \frac{1}{2} m_2 v_{2,f}^2. \end{aligned}$$

There are our two equations in two unknowns. The algebra is tedious but not terribly difficult; you definitely should work it through. The solution is

$$\begin{aligned} v_{1,f} &= \frac{(m_1 - m_2)v_{1,i} + 2m_2 v_{2,i}}{m_1 + m_2} \\ v_{2,f} &= \frac{(m_2 - m_1)v_{2,i} + 2m_1 v_{1,i}}{m_1 + m_2}. \end{aligned}$$

Substituting with the given numbers where a positive direction is to the left, we obtain

$$v_{1,f} = 2.22 \frac{\text{m}}{\text{s}}$$

$$v_{2,f} = -0.28 \frac{\text{m}}{\text{s}}.$$

Significance

Notice that after the collision, the blue puck is moving to the right; its direction of motion was reversed. The red puck is now moving to the left.

CHECK YOUR UNDERSTANDING 9.7

There is a possible mathematical second to the system of equations solved in this example (because the energy equation is quadratic): $v_{1,f} = 0$, $v_{2,f} = -2.5$ m/s. This solution is unacceptable on physical grounds; what's wrong with it?

EXAMPLE 9.12

Thor vs. Iron Man

The 2012 movie “The Avengers” has a scene where Iron Man and Thor fight. At the beginning of the fight, Thor throws his hammer at Iron Man, hitting him and throwing him slightly up into the air and against a small tree, which breaks. From the video, Iron Man is standing still when the hammer hits him. The distance between Thor and Iron Man is approximately 10 m, and the hammer takes about 1 s to reach Iron Man after Thor releases it. The tree is about 2 m behind Iron Man, which he hits in about 0.75 s. Also from the video, Iron Man's trajectory to the tree is very close to horizontal. Assuming Iron Man's total mass is 200 kg:

- Estimate the mass of Thor's hammer
- Estimate how much kinetic energy was lost in this collision

Strategy

After the collision, Thor's hammer is in contact with Iron Man for the entire time, so this is a perfectly inelastic collision. Thus, with the correct choice of a closed system, we expect momentum is conserved, but not kinetic energy. We use the given numbers to estimate the initial momentum, the initial kinetic energy, and the final kinetic energy. Because this is a one-dimensional problem, we can go directly to the scalar form of the equations.

Solution

- First, we posit conservation of momentum. For that, we need a closed system. The choice here is the system (hammer + Iron Man), from the time of collision to the moment just before Iron Man and the hammer hit the tree. Let:
 - M_H = mass of the hammer
 - M_I = mass of Iron Man
 - v_H = velocity of the hammer before hitting Iron Man
 - v = combined velocity of Iron Man + hammer after the collision

Again, Iron Man's initial velocity was zero. Conservation of momentum here reads:

$$M_H v_H = (M_H + M_I) v.$$

We are asked to find the mass of the hammer, so we have

$$\begin{aligned}
 M_H v_H &= M_H v + M_I v \\
 M_H (v_H - v) &= M_I v \\
 M_H &= \frac{M_I v}{v_H - v} \\
 &= \frac{(200 \text{ kg}) \left(\frac{2 \text{ m}}{0.75 \text{ s}} \right)}{10 \frac{\text{m}}{\text{s}} - \left(\frac{2 \text{ m}}{0.75 \text{ s}} \right)} \\
 &= 73 \text{ kg}.
 \end{aligned}$$

Considering the uncertainties in our estimates, this should be expressed with just one significant figure; thus, $M_H = 7 \times 10^1 \text{ kg}$.

- b. The initial kinetic energy of the system, like the initial momentum, is all in the hammer:

$$\begin{aligned}
 K_i &= \frac{1}{2} M_H v_H^2 \\
 &= \frac{1}{2} (70 \text{ kg}) (10 \text{ m/s})^2 \\
 &= 3500 \text{ J}.
 \end{aligned}$$

After the collision,

$$\begin{aligned}
 K_f &= \frac{1}{2} (M_H + M_I) v^2 \\
 &= \frac{1}{2} (70 \text{ kg} + 200 \text{ kg}) (2.67 \text{ m/s})^2 \\
 &= 960 \text{ J}.
 \end{aligned}$$

Thus, there was a loss of $3500 \text{ J} - 960 \text{ J} = 2540 \text{ J}$.

Significance

From other scenes in the movie, Thor apparently can control the hammer's velocity with his mind. It is possible, therefore, that he mentally causes the hammer to maintain its initial velocity of 10 m/s while Iron Man is being driven backward toward the tree. If so, this would represent an external force on our system, so it would not be closed. Thor's mental control of his hammer is beyond the scope of this book, however.



EXAMPLE 9.13

Analyzing a Car Crash

At a stoplight, a small car (1200 kg) collides with a motionless large truck (3000 kg). The car comes to an instantaneous stop; the truck slides straight ahead, coming to a stop after sliding 10 meters. The measured coefficient of friction between the truck's tires and the road was 0.62. How fast was the car moving at the moment of impact?

Strategy

At first it may seem we don't have enough information to solve this problem. Although we know the initial speed of the truck, we don't know the speed of the car (indeed, that's what we're asked to find), so we don't know the initial momentum of the system. Similarly, we know the final speed of the car, but not the speed of the truck immediately after impact. The fact that the truck eventually slid to a speed of zero doesn't help with the final momentum, since an external friction force caused that. Nor can we calculate an impulse, since we don't know the collision time, or the amount of time the truck slid before stopping. A useful strategy is to impose a restriction on the analysis.

Suppose we define a system consisting of just the car and the truck. The momentum of this system isn't conserved, because of the friction between the truck and the road. But if we *could* find the speed of the truck the instant after impact—before friction had any measurable effect on the truck—then we could consider the momentum of the system to be conserved, with that restriction.

Can we find the final speed of the truck? Yes; we invoke the work-energy theorem.

Solution

First, define some variables. Let:

- M_c and M_T be the masses of the car and truck, respectively
- $v_{T,i}$ and $v_{T,f}$ be the velocities of the truck before and after the collision, respectively
- $v_{c,i}$ and $v_{c,f}$ be the velocities of the car before and after the collision, respectively
- K_i and K_f be the kinetic energies of the truck immediately after the collision, and after the truck has stopped sliding (so $K_f = 0$).
- d be the distance the truck slides after the collision before eventually coming to a stop.

Since we actually want the initial speed of the car, and since the car is not part of the work-energy calculation, let's start with conservation of momentum. For the car + truck system, conservation of momentum reads

$$p_i = p_f$$

$$M_T v_{T,i} + M_c v_{c,i} = M_T v_{T,f} + M_c v_{c,f}.$$

Since the car's initial velocity was zero, as was the truck's final velocity, this simplifies to

$$v_{c,i} = \frac{M_T}{M_c} v_{T,f}.$$

So now we need the truck's speed immediately after impact. Recall that

$$W = \Delta K$$

where

$$\begin{aligned} \Delta K &= K_f - K_i \\ &= 0 - \frac{1}{2} M_T v_{T,f}^2. \end{aligned}$$

Also,

$$W = \vec{F} \cdot \vec{d} = Fd \cos \theta.$$

The work is done over the distance the truck slides, which we've called d . Equating:

$$Fd \cos \theta = -\frac{1}{2} M_T v_{T,f}^2.$$

Friction is the force on the truck that does the work to stop the sliding. With a level road, the friction force is

$$F = \mu_k M_T g.$$

Since the angle between the directions of the friction force vector and the displacement d is 180° , and $\cos(180^\circ) = -1$, we have

$$-(\mu_k M_T g) d = -\frac{1}{2} M_T v_{T,f}^2$$

(Notice that the truck's mass divides out; evidently the mass of the truck doesn't matter.)

Solving for the truck's speed immediately after the collision gives

$$v_{T,f} = \sqrt{2\mu_k g d}.$$

Substituting the given numbers:

$$\begin{aligned} v_{T,f} &= \sqrt{2(0.62) \left(9.81 \frac{\text{m}}{\text{s}^2} \right) (10 \text{ m})} \\ &= 11.0 \text{ m/s.} \end{aligned}$$

Now we can calculate the initial speed of the car:

$$v_{T,i} = \left(\frac{3000 \text{ kg}}{1200 \text{ kg}} \right) \left(11.0 \frac{\text{m}}{\text{s}} \right) = 27.5 \text{ m/s.}$$

Significance

This is an example of the type of analysis done by investigators of major car accidents. A great deal of legal and financial consequences depend on an accurate analysis and calculation of momentum and energy.

This collision converted some of the initial kinetic energy to other forms (presumably the deformation of the car and truck and some heat and sound):

$$KE_i = \frac{1}{2} (1200) (27.5^2) = 4.5 \times 10^5 \text{ J}$$

$$KE_f = \frac{1}{2} (3000) (11^2) = 1.8 \times 10^5 \text{ J}$$

$$\text{Change in kinetic energy} = 2.7 \times 10^5 \text{ J}$$

✓ CHECK YOUR UNDERSTANDING 9.8

Suppose there had been no friction (the collision happened on ice); that would make μ_k zero, and thus $v_{c,f} = \sqrt{2\mu_k g d} = 0$, which is obviously wrong. What is the mistake in this conclusion?

Subatomic Collisions and Momentum

Conservation of momentum is crucial to our understanding of atomic and subatomic particles because much of what we know about these particles comes from collision experiments.

At the beginning of the twentieth century, there was considerable interest in, and debate about, the structure of the atom. It was known that atoms contain two types of electrically charged particles: negatively charged electrons and positively charged protons. (The existence of an electrically neutral particle was suspected, but would not be confirmed until 1932.) The question was, how were these particles arranged in the atom? Were they distributed uniformly throughout the volume of the atom (as J.J. Thomson proposed), or arranged at the corners of regular polygons (which was Gilbert Lewis' model), or rings of negative charge that surround the positively charged nucleus—rather like the planetary rings surrounding Saturn (as suggested by Hantaro Nagaoka), or something else?

The New Zealand physicist Ernest Rutherford (along with the German physicist Hans Geiger and the British physicist Ernest Marsden) performed the crucial experiment in 1909. They bombarded a thin sheet of gold foil with a beam of high-energy (that is, high-speed) alpha-particles (the nucleus of a helium atom). The alpha-particles collided with the gold atoms, and their subsequent velocities were detected and analyzed, using conservation of momentum and conservation of energy.

If the charges of the gold atoms were distributed uniformly (per Thomson), then the alpha-particles should collide with them and nearly all would be deflected through many angles, all small; the Nagaoka model would produce a similar result. If the atoms were arranged as regular polygons (Lewis), the alpha-particles would deflect at a relatively small number of angles.

What *actually* happened is that nearly *none* of the alpha-particles were deflected. Those that were, were deflected at large angles, some close to 180° —those alpha-particles reversed direction completely ([Figure 9.21](#)). None of the existing atomic models could explain this. Eventually, Rutherford developed a model of the atom that was much closer to what we now have—again, using conservation of momentum and energy as his starting point.

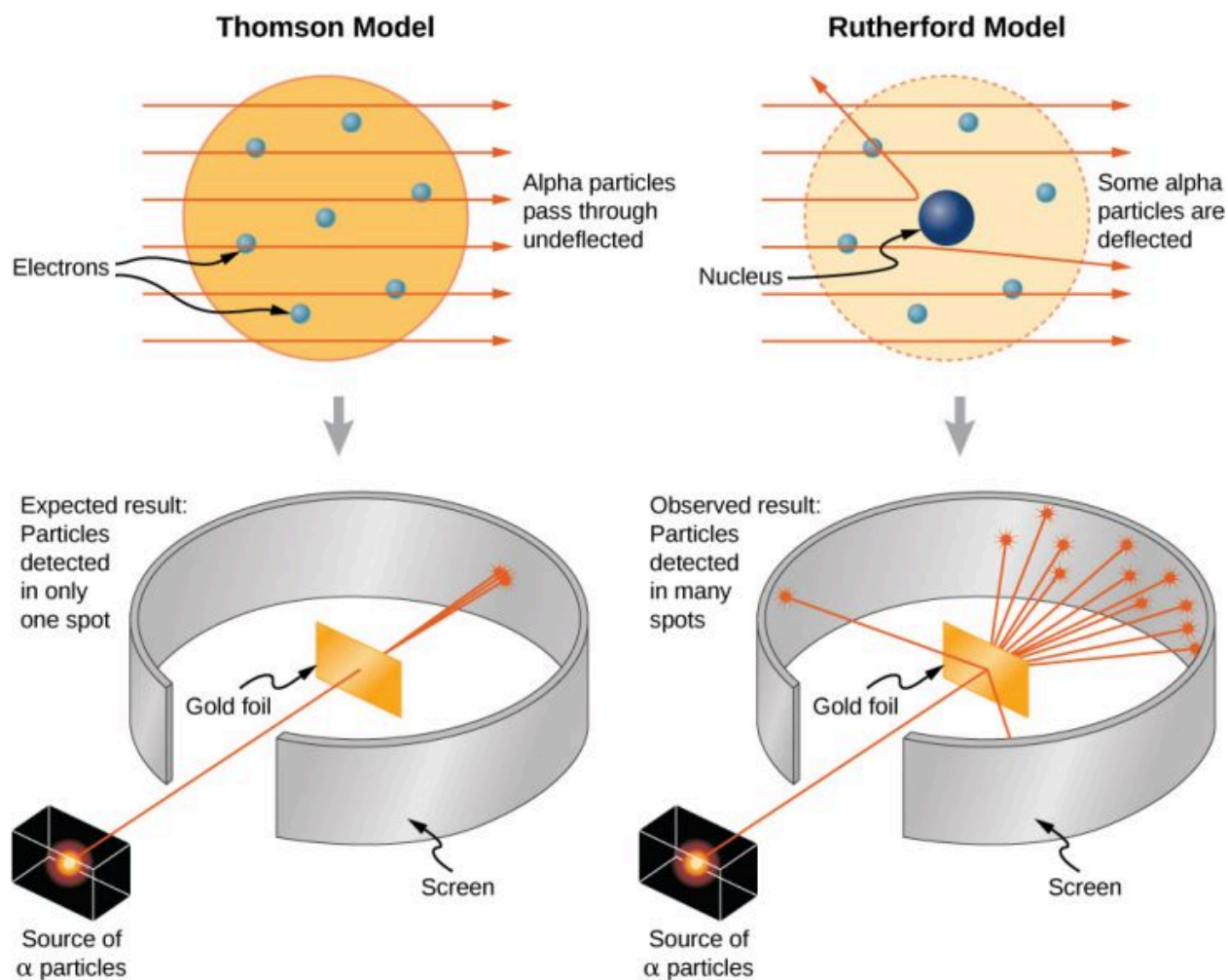


FIGURE 9.21 The Thomson and Rutherford models of the atom. The Thomson model predicted that nearly all of the incident alpha-particles would be scattered and at small angles. Rutherford and Geiger found that nearly none of the alpha particles were scattered, but those few that were deflected did so through very large angles. The results of Rutherford's experiments were inconsistent with the Thomson model. Rutherford used conservation of momentum and energy to develop a new, and better model of the atom—the nuclear model.

9.5 Collisions in Multiple Dimensions

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Express momentum as a two-dimensional vector
- Write equations for momentum conservation in component form
- Calculate momentum in two dimensions, as a vector quantity

It is far more common for collisions to occur in two dimensions; that is, the angle between the initial velocity vectors is neither zero nor 180° . Let's see what complications arise from this.

The first idea we need is that momentum is a vector; like all vectors, it can be expressed as a sum of perpendicular components (usually, though not always, an x-component and a y-component, and a z-component if necessary). Thus, when we write down the statement of conservation of momentum for a problem, our momentum vectors can be, and usually will be, expressed in component form.

The second idea we need comes from the fact that momentum is related to force:

$$\vec{\mathbf{F}} = \frac{d\vec{\mathbf{p}}}{dt}.$$

Expressing both the force and the momentum in component form,

$$F_x = \frac{dp_x}{dt}, \quad F_y = \frac{dp_y}{dt}, \quad F_z = \frac{dp_z}{dt}.$$

Remember, these equations are simply Newton's second law, in vector form and in component form. We know that Newton's second law is true in each direction, independently of the others. It follows therefore (via Newton's third law) that conservation of momentum is also true in each direction independently.

These two ideas motivate the solution to two-dimensional problems: We write down the expression for conservation of momentum twice: once in the x -direction and once in the y -direction.

$$p_{f,x} = p_{1,i,x} + p_{2,i,x}$$

9.18

$$p_{f,y} = p_{1,i,y} + p_{2,i,y}$$

This procedure is shown graphically in [Figure 9.22](#).

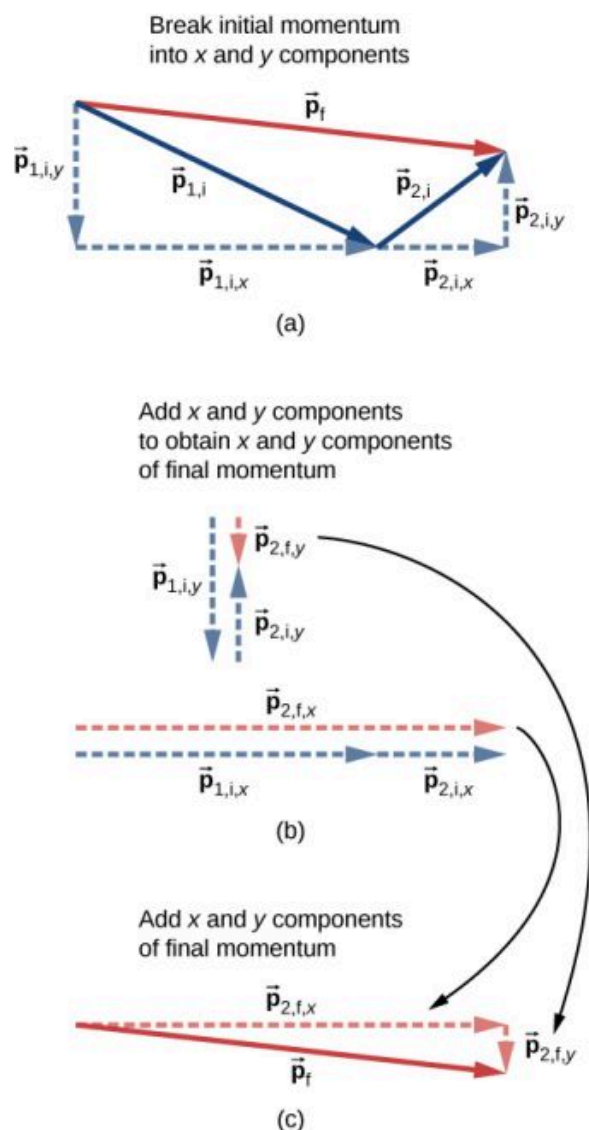


FIGURE 9.22 (a) For two-dimensional momentum problems, break the initial momentum vectors into their x - and y -components. (b) Add the x - and y -components together separately. This gives you the x - and y -components of the final momentum, which are shown as red dashed vectors. (c) Adding these components together gives the final momentum.

We solve each of these two component equations independently to obtain the x - and y -components of the desired velocity vector:

$$v_{f,x} = \frac{m_1 v_{1,i,x} + m_2 v_{2,i,x}}{m}$$

$$v_{f,y} = \frac{m_1 v_{1,i,y} + m_2 v_{2,i,y}}{m}.$$

(Here, m represents the total mass of the system.) Finally, combine these components using the Pythagorean theorem,

$$v_f = |\vec{v}_f| = \sqrt{v_{f,x}^2 + v_{f,y}^2}.$$



PROBLEM-SOLVING STRATEGY

Conservation of Momentum in Two Dimensions

The method for solving a two-dimensional (or even three-dimensional) conservation of momentum problem is generally the same as the method for solving a one-dimensional problem, except that you have to conserve momentum in both (or all three) dimensions simultaneously:

1. Identify a closed system.
2. Write down the equation that represents conservation of momentum in the x -direction, and solve it for the desired quantity. If you are calculating a vector quantity (velocity, usually), this will give you the x -component of the vector.
3. Write down the equation that represents conservation of momentum in the y -direction, and solve. This will give you the y -component of your vector quantity.
4. Assuming you are calculating a vector quantity, use the Pythagorean theorem to calculate its magnitude, using the results of steps 3 and 4.



EXAMPLE 9.14

Traffic Collision

A small car of mass 1200 kg traveling east at 60 km/hr collides at an intersection with a truck of mass 3000 kg that is traveling due north at 40 km/hr ([Figure 9.23](#)). The two vehicles are locked together. What is the velocity of the combined wreckage?

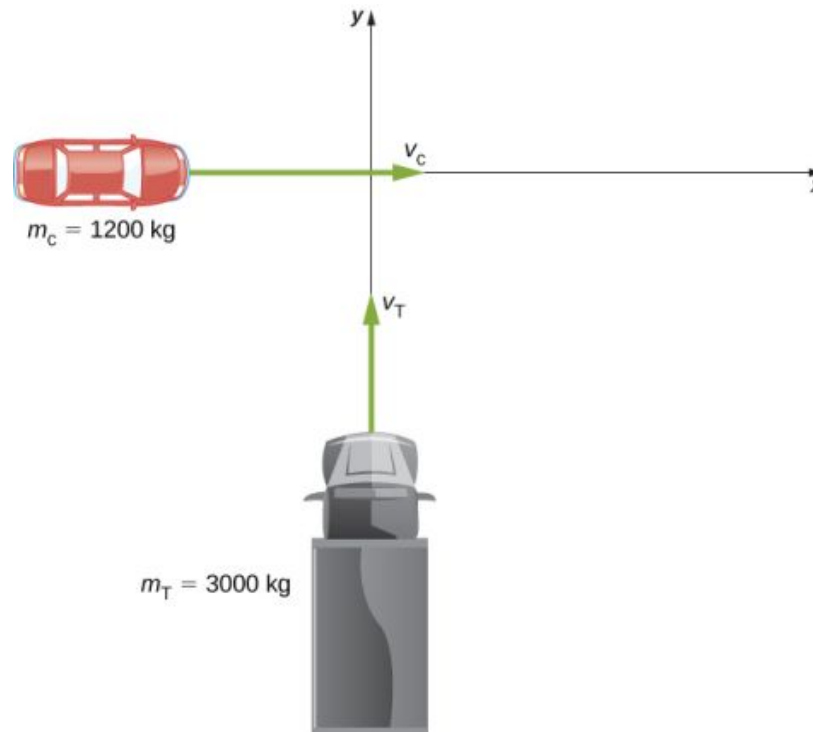


FIGURE 9.23 A large truck moving north is about to collide with a small car moving east. The final momentum vector has both x- and y-components.

Strategy

First off, we need a closed system. The natural system to choose is the (car + truck), but this system is not closed; friction from the road acts on both vehicles. We avoid this problem by restricting the question to finding the velocity at the instant just after the collision, so that friction has not yet had any effect on the system. With that restriction, momentum is conserved for this system.

Since there are two directions involved, we do conservation of momentum twice: once in the x-direction and once in the y-direction.

Solution

Before the collision the total momentum is

$$\vec{\mathbf{p}} = m_c \vec{\mathbf{v}}_c + m_T \vec{\mathbf{v}}_T.$$

After the collision, the wreckage has momentum

$$\vec{\mathbf{p}} = (m_c + m_T) \vec{\mathbf{v}}_w.$$

Since the system is closed, momentum must be conserved, so we have

$$m_c \vec{\mathbf{v}}_c + m_T \vec{\mathbf{v}}_T = (m_c + m_T) \vec{\mathbf{v}}_w.$$

We have to be careful; the two initial momenta are not parallel. We must add vectorially ([Figure 9.24](#)).

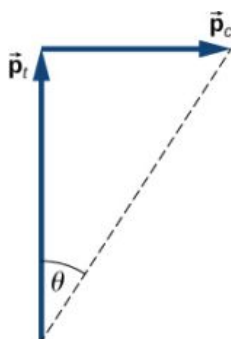


FIGURE 9.24 Graphical addition of momentum vectors. Notice that, although the car's velocity is larger than the truck's, its momentum is smaller.

If we define the $+x$ -direction to point east and the $+y$ -direction to point north, as in the figure, then (conveniently),

$$\begin{aligned}\vec{p}_c &= p_c \hat{i} = m_c v_c \hat{i} \\ \vec{p}_T &= p_T \hat{j} = m_T v_T \hat{j}.\end{aligned}$$

Therefore, in the x -direction:

$$\begin{aligned}m_c v_c &= (m_c + m_T) v_{w,x} \\ v_{w,x} &= \left(\frac{m_c}{m_c + m_T} \right) v_c\end{aligned}$$

and in the y -direction:

$$\begin{aligned}m_T v_T &= (m_c + m_T) v_{w,y} \\ v_{w,y} &= \left(\frac{m_T}{m_c + m_T} \right) v_T.\end{aligned}$$

Applying the Pythagorean theorem gives

$$\begin{aligned}|\vec{v}_w| &= \sqrt{\left[\left(\frac{m_c}{m_c + m_T} \right) v_c \right]^2 + \left[\left(\frac{m_T}{m_c + m_T} \right) v_T \right]^2} \\ &= \sqrt{\left[\left(\frac{1200 \text{ kg}}{4200 \text{ kg}} \right) (16.67 \text{ m/s}) \right]^2 + \left[\left(\frac{3000 \text{ kg}}{4200 \text{ kg}} \right) (11.1 \text{ m/s}) \right]^2} \\ &= \sqrt{\left(4.76 \text{ m/s} \right)^2 + \left(7.93 \text{ m/s} \right)^2} \\ &= 9.25 \text{ m/s} \approx 33.3 \text{ km/hr}.\end{aligned}$$

As for its direction, using the angle shown in the figure,

$$\theta = \tan^{-1} \left(\frac{v_{w,x}}{v_{w,y}} \right) = \tan^{-1} \left(\frac{4.76 \text{ m/s}}{7.93 \text{ m/s}} \right) = 31^\circ.$$

This angle is east of north, or 31° counterclockwise from the $+x$ -direction.

Significance

As a practical matter, accident investigators usually work in the “opposite direction”; they measure the distance of skid marks on the road (which gives the stopping distance) and use the work-energy theorem along with conservation of momentum to determine the speeds and directions of the cars prior to the collision. We saw that analysis in an earlier section.

CHECK YOUR UNDERSTANDING 9.9

Suppose the initial velocities were *not* at right angles to each other. How would this change both the physical result

and the mathematical analysis of the collision?

EXAMPLE 9.15

Exploding Scuba Tank

A common scuba tank is an aluminum cylinder that weighs 31.7 pounds empty (Figure 9.25). When full of compressed air, the internal pressure is between 2500 and 3000 psi (pounds per square inch). Suppose such a tank, which had been sitting motionless, suddenly explodes into three pieces. The first piece, weighing 10 pounds, shoots off horizontally at 235 miles per hour; the second piece (7 pounds) shoots off at 172 miles per hour, also in the horizontal plane, but at a 19° angle to the first piece. What is the mass and initial velocity of the third piece? (Do all work, and express your final answer, in SI units.)



FIGURE 9.25 A scuba tank explodes into three pieces.

Strategy

To use conservation of momentum, we need a closed system. If we define the system to be the scuba tank, this is not a closed system, since gravity is an external force. However, the problem asks for just the initial velocity of the third piece, so we can neglect the effect of gravity and consider the tank by itself as a closed system. Notice that, for this system, the initial momentum vector is zero.

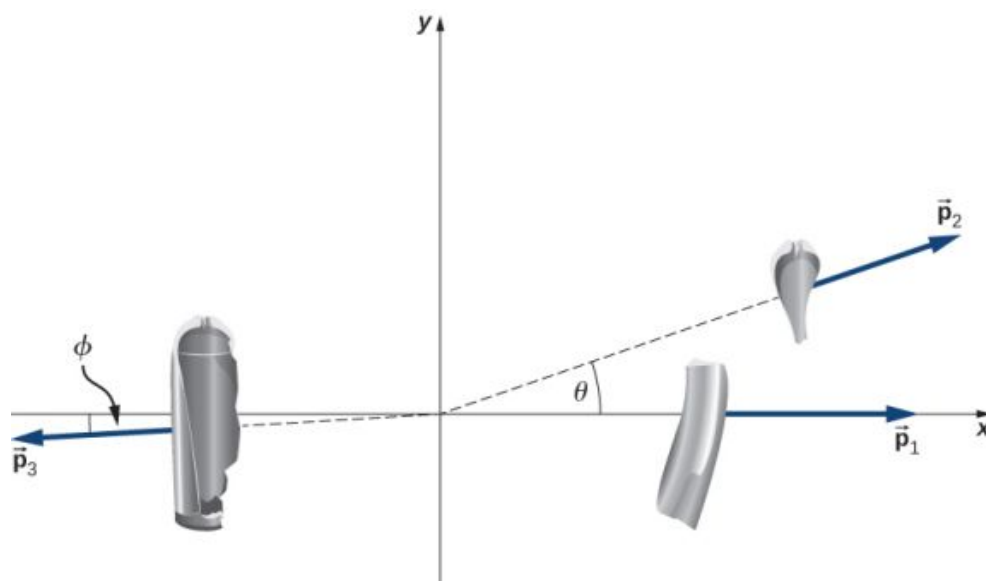
We choose a coordinate system where all the motion happens in the xy -plane. We then write down the equations for conservation of momentum in each direction, thus obtaining the x - and y -components of the momentum of the third piece, from which we obtain its magnitude (via the Pythagorean theorem) and its direction. Finally, dividing this momentum by the mass of the third piece gives us the velocity.

Solution

First, let's get all the conversions to SI units out of the way:

$$\begin{aligned}
 31.7 \text{ lb} &\times \frac{1 \text{ kg}}{2.2 \text{ lb}} \rightarrow 14.4 \text{ kg} \\
 10 \text{ lb} &\rightarrow 4.5 \text{ kg} \\
 235 \frac{\text{miles}}{\text{hour}} &\times \frac{1 \text{ hour}}{3600 \text{ s}} \times \frac{1609 \text{ m}}{\text{mile}} = 105 \frac{\text{m}}{\text{s}} \\
 7 \text{ lb} &\rightarrow 3.2 \text{ kg} \\
 172 \frac{\text{mile}}{\text{hour}} &= 77 \frac{\text{m}}{\text{s}} \\
 m_3 &= 14.4 \text{ kg} - (4.5 \text{ kg} + 3.2 \text{ kg}) = 6.7 \text{ kg}.
 \end{aligned}$$

Now apply conservation of momentum in each direction.



x-direction:

$$\begin{aligned}
 p_{f,x} &= p_{0,x} \\
 p_{1,x} + p_{2,x} + p_{3,x} &= 0 \\
 m_1 v_{1,x} + m_2 v_{2,x} + p_{3,x} &= 0 \\
 p_{3,x} &= -m_1 v_{1,x} - m_2 v_{2,x}
 \end{aligned}$$

y-direction:

$$\begin{aligned}
 p_{f,y} &= p_{0,y} \\
 p_{1,y} + p_{2,y} + p_{3,y} &= 0 \\
 m_1 v_{1,y} + m_2 v_{2,y} + p_{3,y} &= 0 \\
 p_{3,y} &= -m_1 v_{1,y} - m_2 v_{2,y}
 \end{aligned}$$

From our chosen coordinate system, we write the x-components as

$$\begin{aligned}
 p_{3,x} &= -m_1 v_1 - m_2 v_2 \cos \theta \\
 &= -(4.5 \text{ kg}) \left(105 \frac{\text{m}}{\text{s}} \right) - (3.2 \text{ kg}) \left(77 \frac{\text{m}}{\text{s}} \right) \cos(19^\circ) \\
 &= -705 \frac{\text{kg}\cdot\text{m}}{\text{s}}.
 \end{aligned}$$

For the y-direction, we have

$$\begin{aligned}
 p_{3,y} &= 0 - m_2 v_2 \sin \theta \\
 &= -(3.2 \text{ kg}) \left(77 \frac{\text{m}}{\text{s}} \right) \sin(19^\circ) \\
 &= -80.2 \frac{\text{kg}\cdot\text{m}}{\text{s}}.
 \end{aligned}$$

This gives the magnitude of p_3 :

$$\begin{aligned}
 p_3 &= \sqrt{p_{3,x}^2 + p_{3,y}^2} \\
 &= \sqrt{\left(-705 \frac{\text{kg}\cdot\text{m}}{\text{s}} \right)^2 + \left(-80.2 \frac{\text{kg}\cdot\text{m}}{\text{s}} \right)^2} \\
 &= 710 \frac{\text{kg}\cdot\text{m}}{\text{s}}.
 \end{aligned}$$

The velocity of the third piece is therefore

$$v_3 = \frac{p_3}{m_3} = \frac{710 \frac{\text{kg}\cdot\text{m}}{\text{s}}}{6.7 \text{ kg}} = 106 \frac{\text{m}}{\text{s}}.$$

The direction of its velocity vector is the same as the direction of its momentum vector:

$$\phi = \tan^{-1} \left(\frac{p_{3,y}}{p_{3,x}} \right) = \tan^{-1} \left(\frac{80.2 \frac{\text{kg}\cdot\text{m}}{\text{s}}}{705 \frac{\text{kg}\cdot\text{m}}{\text{s}}} \right) = 6.49^\circ.$$

Because ϕ is below the $-x$ -axis, the actual angle is 183.5° from the $+x$ -direction.

Significance

The enormous velocities here are typical; an exploding tank of any compressed gas can easily punch through the wall of a house and cause significant injury, or death. Fortunately, such explosions are extremely rare, on a percentage basis.

✓ CHECK YOUR UNDERSTANDING 9.10

Notice that the mass of the air in the tank was neglected in the analysis and solution. How would the solution method change if the air was included? How large a difference do you think it would make in the final answer?

9.6 Center of Mass

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Explain the meaning and usefulness of the concept of center of mass
- Calculate the center of mass of a given system
- Apply the center of mass concept in two and three dimensions
- Calculate the velocity and acceleration of the center of mass

We have been avoiding an important issue up to now: When we say that an object moves (more correctly, accelerates) in a way that obeys Newton's second law, we have been ignoring the fact that all objects are actually made of many constituent particles. A car has an engine, steering wheel, seats, passengers; a football is leather and rubber surrounding air; a brick is made of atoms. There are many different types of particles, and they are generally not distributed uniformly in the object. How do we include these facts into our calculations?

Then too, an extended object might change shape as it moves, such as a water balloon or a cat falling ([Figure 9.26](#)). This implies that the constituent particles are applying internal forces on each other, in addition to the external force that is acting on the object as a whole. We want to be able to handle this, as well.

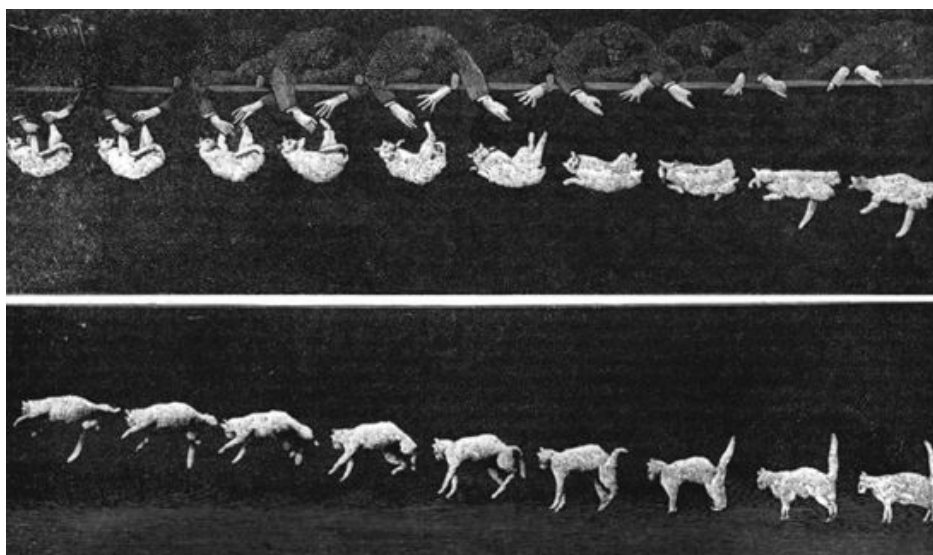


FIGURE 9.26 As the cat falls, its body performs complicated motions so it can land on its feet, but one point in the system moves with the simple uniform acceleration of gravity.

The problem before us, then, is to determine what part of an extended object is obeying Newton's second law when an external force is applied and to determine how the motion of the object as a whole is affected by both the internal and external forces.

Be warned: To treat this new situation correctly, we must be rigorous and completely general. We won't make any assumptions about the nature of the object, or of its constituent particles, or either the internal or external forces. Thus, the arguments will be complex.

Internal and External Forces

Suppose we have an extended object of mass M , made of N interacting particles. Let's label their masses as m_j , where $j = 1, 2, 3, \dots, N$. Note that

$$M = \sum_{j=1}^N m_j. \quad 9.19$$

If we apply some net **external force** $\vec{F}_{\text{ext}}$ on the object, every particle experiences some "share" or some fraction of that external force. Let:

$\vec{f}_j^{\text{ext}}$ = the fraction of the external force that the j th particle experiences.

Notice that these fractions of the total force are not necessarily equal; indeed, they virtually never are. (They *can* be, but they usually aren't.) In general, therefore,

$$\vec{f}_1^{\text{ext}} \neq \vec{f}_2^{\text{ext}} \neq \dots \neq \vec{f}_N^{\text{ext}}.$$

Next, we assume that each of the particles making up our object can interact with (apply forces on) every other particle of the object. We won't try to guess what kind of forces they are; but since these forces are the result of particles of the object acting on other particles of the same object, we refer to them as **internal forces** $\vec{f}_j^{\text{int}}$; thus:

$\vec{f}_j^{\text{int}}$ = the net internal force that the j th particle experiences from all the other particles that make up the object.

Now, the *net* force, internal plus external, on the j th particle is the vector sum of these:

$$\vec{f}_j = \vec{f}_j^{\text{int}} + \vec{f}_j^{\text{ext}}. \quad 9.20$$

where again, this is for all N particles; $j = 1, 2, 3, \dots, N$.

As a result of this fractional force, the momentum of each particle gets changed:

$$\begin{aligned}\vec{\mathbf{f}}_j &= \frac{d\vec{\mathbf{p}}_j}{dt} \\ \vec{\mathbf{f}}_j^{\text{int}} + \vec{\mathbf{f}}_j^{\text{ext}} &= \frac{d\vec{\mathbf{p}}_j}{dt}.\end{aligned}\tag{9.21}$$

The net force $\vec{\mathbf{F}}$ on the *object* is the vector sum of these forces:

$$\begin{aligned}\vec{\mathbf{F}}_{\text{net}} &= \sum_{j=1}^N \left(\vec{\mathbf{f}}_j^{\text{int}} + \vec{\mathbf{f}}_j^{\text{ext}} \right) \\ &= \sum_{j=1}^N \vec{\mathbf{f}}_j^{\text{int}} + \sum_{j=1}^N \vec{\mathbf{f}}_j^{\text{ext}}.\end{aligned}\tag{9.22}$$

This net force changes the momentum of the object as a whole, and the net change of momentum of the object must be the vector sum of all the individual changes of momentum of all of the particles:

$$\vec{\mathbf{F}}_{\text{net}} = \sum_{j=1}^N \frac{d\vec{\mathbf{p}}_j}{dt}.\tag{9.23}$$

Combining [Equation 9.22](#) and [Equation 9.23](#) gives

$$\sum_{j=1}^N \vec{\mathbf{f}}_j^{\text{int}} + \sum_{j=1}^N \vec{\mathbf{f}}_j^{\text{ext}} = \sum_{j=1}^N \frac{d\vec{\mathbf{p}}_j}{dt}.\tag{9.24}$$

Let's now think about these summations. First consider the internal forces term; remember that each $\vec{\mathbf{f}}_j^{\text{int}}$ is the force on the j th particle from the other particles in the object. But by Newton's third law, for every one of these forces, there must be another force that has the same magnitude, but the opposite sign (points in the opposite direction). These forces do not cancel; however, that's not what we're doing in the summation. Rather, we're simply *mathematically adding up* all the internal force vectors. That is, in general, the internal forces for any individual part of the object won't cancel, but when all the internal forces are added up, the internal forces must cancel in pairs. It follows, therefore, that the sum of all the internal forces must be zero:

$$\sum_{j=1}^N \vec{\mathbf{f}}_j^{\text{int}} = 0.$$

(This argument is subtle, but crucial; take plenty of time to completely understand it.)

For the external forces, this summation is simply the total external force that was applied to the whole object:

$$\sum_{j=1}^N \vec{\mathbf{f}}_j^{\text{ext}} = \vec{\mathbf{F}}_{\text{ext}}.$$

As a result,

$$\vec{\mathbf{F}}_{\text{ext}} = \sum_{j=1}^N \frac{d\vec{\mathbf{p}}_j}{dt}.\tag{9.25}$$

This is an important result. [Equation 9.25](#) tells us that the total change of momentum of the entire object (all N particles) is due only to the external forces; the internal forces do not change the momentum of the object as a whole. This is why you can't lift yourself in the air by standing in a basket and pulling up on the handles: For the system of you + basket, your upward pulling force is an internal force.

Force and Momentum

Remember that our actual goal is to determine the equation of motion for the entire object (the entire system of particles). To that end, let's define:

$\vec{p}_{\text{CM}}$ = the total momentum of the system of N particles (the reason for the subscript will become clear shortly)

Then we have

$$\vec{p}_{\text{CM}} \equiv \sum_{j=1}^N \vec{p}_j,$$

and therefore [Equation 9.25](#) can be written simply as

$$\vec{F} = \frac{d\vec{p}_{\text{CM}}}{dt}. \quad 9.26$$

Since this change of momentum is caused by only the net external force, we have dropped the “ext” subscript.

This is Newton's second law, but now for the entire extended object. If this feels a bit anticlimactic, remember what is hiding inside it: $\vec{p}_{\text{CM}}$ is the vector sum of the momentum of (in principle) hundreds of thousands of billions of billions of particles (6.02×10^{23}), all caused by one simple net external force—a force that you can calculate.

Center of Mass

Our next task is to determine what part of the extended object, if any, is obeying [Equation 9.26](#).

It's tempting to take the next step; does the following equation mean anything?

$$\vec{F} = M\vec{a} \quad 9.27$$

If it *does* mean something (acceleration of what, exactly?), then we could write

$$M\vec{a} = \frac{d\vec{p}_{\text{CM}}}{dt}$$

and thus

$$M\vec{a} = \sum_{j=1}^N \frac{d\vec{p}_j}{dt} = \frac{d}{dt} \sum_{j=1}^N \vec{p}_j.$$

which follows because the derivative of a sum is equal to the sum of the derivatives.

Now, $\vec{p}_j$ is the momentum of the j th particle. Defining the positions of the constituent particles (relative to some coordinate system) as $\vec{r}_j = (x_j, y_j, z_j)$, we thus have

$$\vec{p}_j = m_j \vec{v}_j = m_j \frac{d\vec{r}_j}{dt}.$$

Substituting back, we obtain

$$\begin{aligned} M\vec{a} &= \frac{d}{dt} \sum_{j=1}^N m_j \frac{d\vec{r}_j}{dt} \\ &= \frac{d^2}{dt^2} \sum_{j=1}^N m_j \vec{r}_j. \end{aligned}$$

Dividing both sides by M (the total mass of the extended object) gives us

$$\vec{\mathbf{a}} = \frac{d^2}{dt^2} \left(\frac{1}{M} \sum_{j=1}^N m_j \vec{\mathbf{r}}_j \right). \quad 9.28$$

Thus, the point in the object that traces out the trajectory dictated by the applied force in [Equation 9.27](#) is inside the parentheses in [Equation 9.28](#).

Looking at this calculation, notice that (inside the parentheses) we are calculating the product of each particle's mass with its position, adding all N of these up, and dividing this sum by the total mass of particles we summed. This is reminiscent of an average; inspired by this, we'll (loosely) interpret it to be the weighted average position of the mass of the extended object. It's actually called the **center of mass** of the object. Notice that the position of the center of mass has units of meters; that suggests a definition:

$$\vec{\mathbf{r}}_{\text{CM}} \equiv \frac{1}{M} \sum_{j=1}^N m_j \vec{\mathbf{r}}_j. \quad 9.29$$

So, the point that obeys [Equation 9.26](#) (and therefore [Equation 9.27](#) as well) is the center of mass of the object, which is located at the position vector $\vec{\mathbf{r}}_{\text{CM}}$.

It may surprise you to learn that there does not have to be any actual mass at the center of mass of an object. For example, a hollow steel sphere with a vacuum inside it is spherically symmetrical (meaning its mass is uniformly distributed about the center of the sphere); all of the sphere's mass is out on its surface, with no mass inside. But it can be shown that the center of mass of the sphere is at its geometric center, which seems reasonable. Thus, there is no mass at the position of the center of mass of the sphere. (Another example is a doughnut.) The procedure to find the center of mass is illustrated in [Figure 9.27](#).

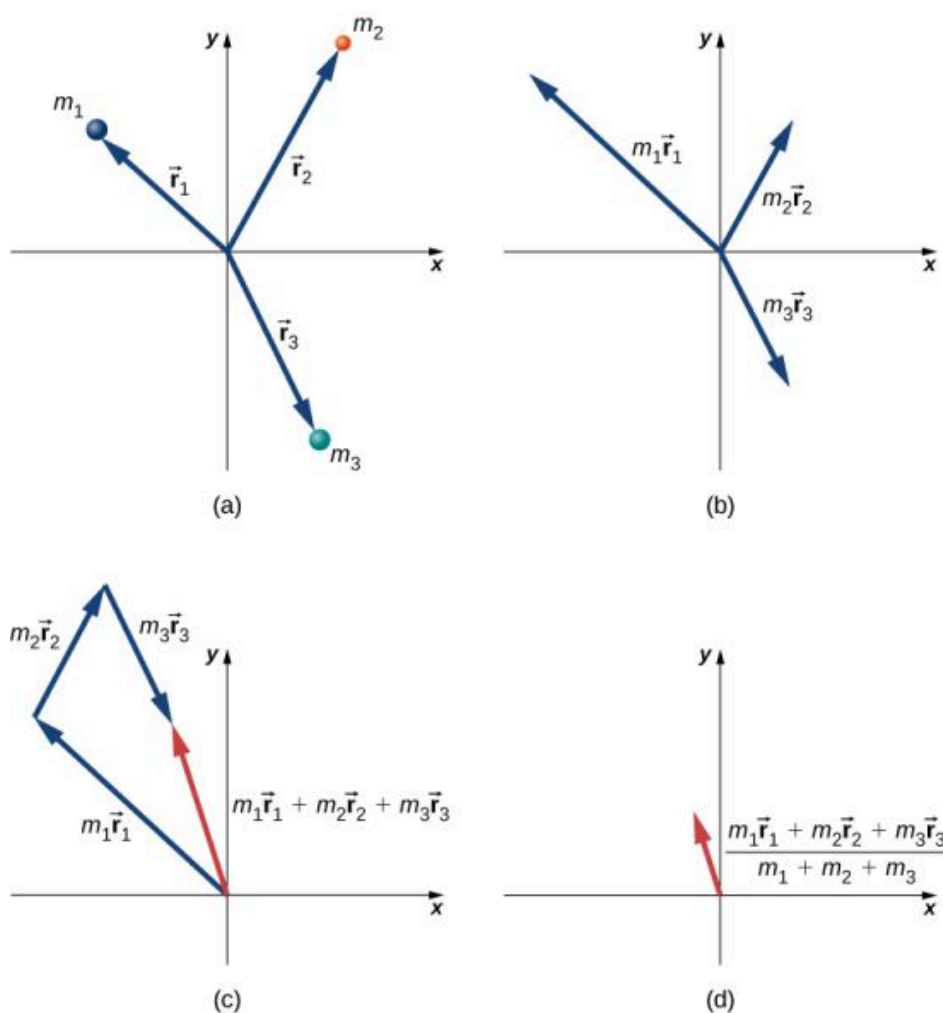


FIGURE 9.27 Finding the center of mass of a system of three different particles. (a) Position vectors are created for each object. (b) The position vectors are multiplied by the mass of the corresponding object. (c) The scaled vectors from part (b) are added together. (d) The final vector is divided by the total mass. This vector points to the center of mass of the system. Note that no mass is actually present at the center of mass of this system.

Since $\vec{r}_j = x_j\hat{i} + y_j\hat{j} + z_j\hat{k}$, it follows that:

$$r_{\text{CM},x} = \frac{1}{M} \sum_{j=1}^N m_j x_j \quad 9.30$$

$$r_{\text{CM},y} = \frac{1}{M} \sum_{j=1}^N m_j y_j \quad 9.31$$

$$r_{\text{CM},z} = \frac{1}{M} \sum_{j=1}^N m_j z_j \quad 9.32$$

and thus

$$\begin{aligned} \vec{r}_{\text{CM}} &= r_{\text{CM},x}\hat{i} + r_{\text{CM},y}\hat{j} + r_{\text{CM},z}\hat{k} \\ r_{\text{CM}} = |\vec{r}_{\text{CM}}| &= \left(r_{\text{CM},x}^2 + r_{\text{CM},y}^2 + r_{\text{CM},z}^2 \right)^{1/2}. \end{aligned}$$

Therefore, you can calculate the components of the center of mass vector individually.

Finally, to complete the kinematics, the instantaneous velocity of the center of mass is calculated exactly as you might suspect:

$$\vec{v}_{\text{CM}} = \frac{d}{dt} \left(\frac{1}{M} \sum_{j=1}^N m_j \vec{r}_j \right) = \frac{1}{M} \sum_{j=1}^N m_j \vec{v}_j \quad 9.33$$

and this, like the position, has x -, y -, and z -components.

To calculate the center of mass in actual situations, we recommend the following procedure:



PROBLEM-SOLVING STRATEGY

Calculating the Center of Mass

The center of mass of an object is a position vector. Thus, to calculate it, do these steps:

1. Define your coordinate system. Typically, the origin is placed at the location of one of the particles. This is not required, however.
2. Determine the x , y , z -coordinates of each particle that makes up the object.
3. Determine the mass of each particle, and sum them to obtain the total mass of the object. Note that the mass of the object at the origin *must* be included in the total mass.
4. Calculate the x -, y -, and z -components of the center of mass vector, using [Equation 9.30](#), [Equation 9.31](#), and [Equation 9.32](#).
5. If required, use the Pythagorean theorem to determine its magnitude.

Here are two examples that will give you a feel for what the center of mass is.



EXAMPLE 9.16

Center of Mass of the Earth-Moon System

Using data from text appendix, determine how far the center of mass of the Earth-moon system is from the center of Earth. Compare this distance to the radius of Earth, and comment on the result. Ignore the other objects in the solar system.

Strategy

We get the masses and separation distance of the Earth and moon, impose a coordinate system, and use [Equation 9.29](#) with just $N = 2$ objects. We use a subscript “e” to refer to Earth, and subscript “m” to refer to the moon.

Solution

Define the origin of the coordinate system as the center of Earth. Then, with just two objects, [Equation 9.29](#) becomes

$$R = \frac{m_e r_e + m_m r_m}{m_e + m_m}.$$

From [Appendix D](#),

$$m_e = 5.97 \times 10^{24} \text{ kg}$$

$$m_m = 7.36 \times 10^{22} \text{ kg}$$

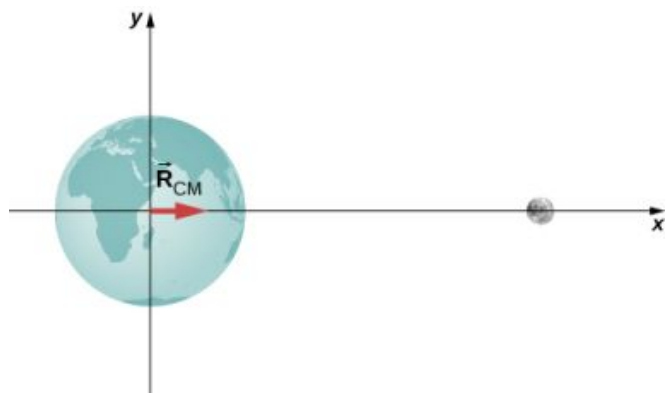
$$r_m = 3.82 \times 10^8 \text{ m}.$$

We defined the center of Earth as the origin, so $r_e = 0 \text{ m}$. Inserting these into the equation for R gives

$$\begin{aligned}
 R &= \frac{(5.97 \times 10^{24} \text{ kg})(0 \text{ m}) + (7.36 \times 10^{22} \text{ kg})(3.82 \times 10^8 \text{ m})}{5.97 \times 10^{24} \text{ kg} + 7.36 \times 10^{22} \text{ kg}} \\
 &= 4.64 \times 10^6 \text{ m}.
 \end{aligned}$$

Significance

The radius of Earth is $6.37 \times 10^6 \text{ m}$, so the center of mass of the Earth-moon system is $(6.37 - 4.64) \times 10^6 \text{ m} = 1.73 \times 10^6 \text{ m} = 1730 \text{ km}$ (roughly 1080 miles) *below* the surface of Earth. The location of the center of mass is shown (not to scale).



✓ CHECK YOUR UNDERSTANDING 9.11

Suppose we included the sun in the system. Approximately where would the center of mass of the Earth-moon-sun system be located? (Feel free to actually calculate it.)

✿ EXAMPLE 9.17

Center of Mass of a Salt Crystal

Figure 9.28 shows a single crystal of sodium chloride—ordinary table salt. The sodium and chloride ions form a single unit, NaCl. When multiple NaCl units group together, they form a cubic lattice. The smallest possible cube (called the *unit cell*) consists of four sodium ions and four chloride ions, alternating. The length of one edge of this cube (i.e., the bond length) is $2.36 \times 10^{-10} \text{ m}$. Find the location of the center of mass of the unit cell. Specify it either by its coordinates $(r_{\text{CM},x}, r_{\text{CM},y}, r_{\text{CM},z})$, or by r_{CM} and two angles.

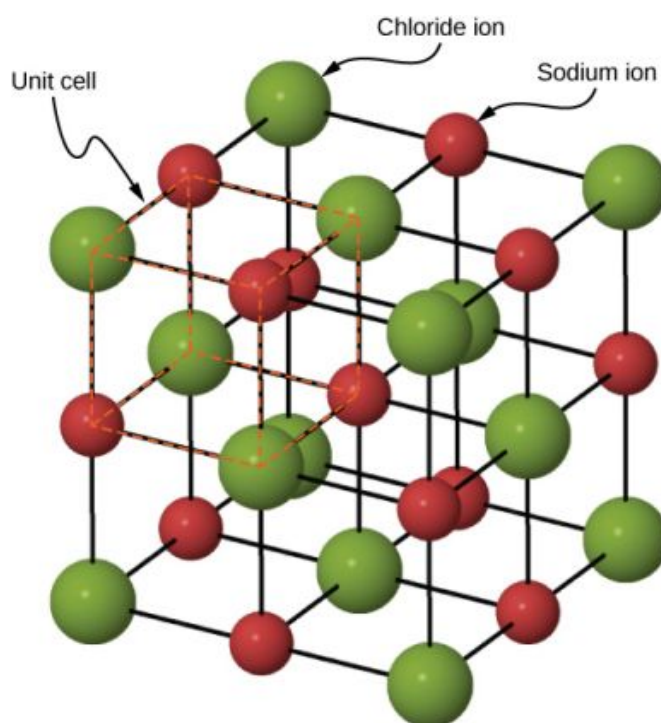


FIGURE 9.28 A drawing of a sodium chloride (NaCl) crystal.

Strategy

We can look up all the ion masses. If we impose a coordinate system on the unit cell, this will give us the positions of the ions. We can then apply [Equation 9.30](#), [Equation 9.31](#), and [Equation 9.32](#) (along with the Pythagorean theorem).

Solution

Define the origin to be at the location of the chloride ion at the bottom left of the unit cell. [Figure 9.29](#) shows the coordinate system.

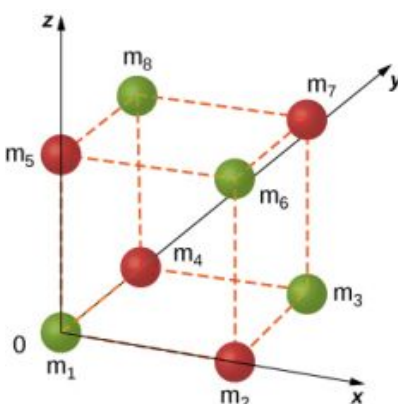


FIGURE 9.29 A single unit cell of a NaCl crystal.

There are eight ions in this crystal, so $N = 8$:

$$\vec{r}_{\text{CM}} = \frac{1}{M} \sum_{j=1}^8 m_j \vec{r}_j.$$

The mass of each of the chloride ions is

$$35.453\text{u} \times \frac{1.660 \times 10^{-27} \text{ kg}}{\text{u}} = 5.885 \times 10^{-26} \text{ kg}$$

so we have

$$m_1 = m_3 = m_6 = m_8 = 5.885 \times 10^{-26} \text{ kg}.$$

For the sodium ions,

$$m_2 = m_4 = m_5 = m_7 = 3.816 \times 10^{-26} \text{ kg}.$$

The total mass of the unit cell is therefore

$$M = (4)(5.885 \times 10^{-26} \text{ kg}) + (4)(3.816 \times 10^{-26} \text{ kg}) = 3.880 \times 10^{-25} \text{ kg}.$$

From the geometry, the locations are

$$\vec{r}_1 = 0$$

$$\vec{r}_2 = (2.36 \times 10^{-10} \text{ m}) \hat{i}$$

$$\vec{r}_3 = r_{3x} \hat{i} + r_{3y} \hat{j} = (2.36 \times 10^{-10} \text{ m}) \hat{i} + (2.36 \times 10^{-10} \text{ m}) \hat{j}$$

$$\vec{r}_4 = (2.36 \times 10^{-10} \text{ m}) \hat{j}$$

$$\vec{r}_5 = (2.36 \times 10^{-10} \text{ m}) \hat{k}$$

$$\vec{r}_6 = r_{6x} \hat{i} + r_{6z} \hat{k} = (2.36 \times 10^{-10} \text{ m}) \hat{i} + (2.36 \times 10^{-10} \text{ m}) \hat{k}$$

$$\vec{r}_7 = r_{7x} \hat{i} + r_{7y} \hat{j} + r_{7z} \hat{k} = (2.36 \times 10^{-10} \text{ m}) \hat{i} + (2.36 \times 10^{-10} \text{ m}) \hat{j} + (2.36 \times 10^{-10} \text{ m}) \hat{k}$$

$$\vec{r}_8 = r_{8y} \hat{j} + r_{8z} \hat{k} = (2.36 \times 10^{-10} \text{ m}) \hat{j} + (2.36 \times 10^{-10} \text{ m}) \hat{k}.$$

Substituting:

$$\begin{aligned} |\vec{r}_{\text{CM},x}| &= \sqrt{r_{\text{CM},x}^2 + r_{\text{CM},y}^2 + r_{\text{CM},z}^2} \\ &= \frac{1}{M} \sum_{j=1}^8 m_j (r_x)_j \\ &= \frac{1}{M} (m_1 r_{1x} + m_2 r_{2x} + m_3 r_{3x} + m_4 r_{4x} + m_5 r_{5x} + m_6 r_{6x} + m_7 r_{7x} + m_8 r_{8x}) \\ &= \frac{1}{3.8804 \times 10^{-25} \text{ kg}} [(5.885 \times 10^{-26} \text{ kg})(0 \text{ m}) + (3.816 \times 10^{-26} \text{ kg})(2.36 \times 10^{-10} \text{ m}) \\ &\quad + (5.885 \times 10^{-26} \text{ kg})(2.36 \times 10^{-10} \text{ m}) \\ &\quad + (3.816 \times 10^{-26} \text{ kg})(2.36 \times 10^{-10} \text{ m}) + 0 + 0 \\ &\quad + (3.816 \times 10^{-26} \text{ kg})(2.36 \times 10^{-10} \text{ m}) + 0] \\ &= 1.18 \times 10^{-10} \text{ m}. \end{aligned}$$

Similar calculations give $r_{\text{CM},y} = r_{\text{CM},z} = 1.18 \times 10^{-10} \text{ m}$ (you could argue that this must be true, by symmetry, but it's a good idea to check).

Significance

Although this is a great exercise to determine the center of mass given a Chloride ion at the origin, in fact the origin could be chosen at any location. Therefore, there is no meaningful application of the center of mass of a unit cell beyond as an exercise.

CHECK YOUR UNDERSTANDING 9.12

Suppose you have a macroscopic salt crystal (that is, a crystal that is large enough to be visible with your unaided eye). It is made up of a *huge* number of unit cells. Is the center of mass of this crystal necessarily at the geometric center of the crystal?

Two crucial concepts come out of these examples:

1. As with all problems, you must define your coordinate system and origin. For center-of-mass calculations, it often makes sense to choose your origin to be located at one of the masses of your system. That choice automatically defines its distance in [Equation 9.29](#) to be zero. However, you must still include the mass of the object at your origin in your calculation of M , the total mass [Equation 9.19](#). In the Earth-moon system example, this means including the mass of Earth. If you hadn't, you'd have ended up with the center of mass of the system being at the center of the moon, which is clearly wrong.
2. In the second example (the salt crystal), notice that there is no mass at all at the location of the center of mass. This is an example of what we stated above, that there does not have to be any actual mass at the center of mass of an object.

Center of Mass of Continuous Objects

If the object in question has its mass distributed uniformly in space, rather than as a collection of discrete particles, then $m_j \rightarrow dm$, and the summation becomes an integral:

$$\vec{r}_{\text{CM}} = \frac{1}{M} \int \vec{r} dm. \quad 9.34$$

In this context, r is a characteristic dimension of the object (the radius of a sphere, the length of a long rod). To generate an integrand that can actually be calculated, you need to express the differential mass element dm as a function of the mass density of the continuous object, and the dimension r . An example will clarify this.



EXAMPLE 9.18

CM of a Uniform Thin Hoop

Find the center of mass of a uniform thin hoop (or ring) of mass M and radius r .

Strategy

First, the hoop's symmetry suggests the center of mass should be at its geometric center. If we define our coordinate system such that the origin is located at the center of the hoop, the integral should evaluate to zero.

We replace dm with an expression involving the density of the hoop and the radius of the hoop. We then have an expression we can actually integrate. Since the hoop is described as “thin,” we treat it as a one-dimensional object, neglecting the thickness of the hoop. Therefore, its density is expressed as the number of kilograms of material per meter. Such a density is called a **linear mass density**, and is given the symbol λ ; this is the Greek letter “lambda,” which is the equivalent of the English letter “l” (for “linear”).

Since the hoop is described as uniform, this means that the linear mass density λ is constant. Thus, to get our expression for the differential mass element dm , we multiply λ by a differential length of the hoop, substitute, and integrate (with appropriate limits for the definite integral).

Solution

First, define our coordinate system and the relevant variables ([Figure 9.30](#)).

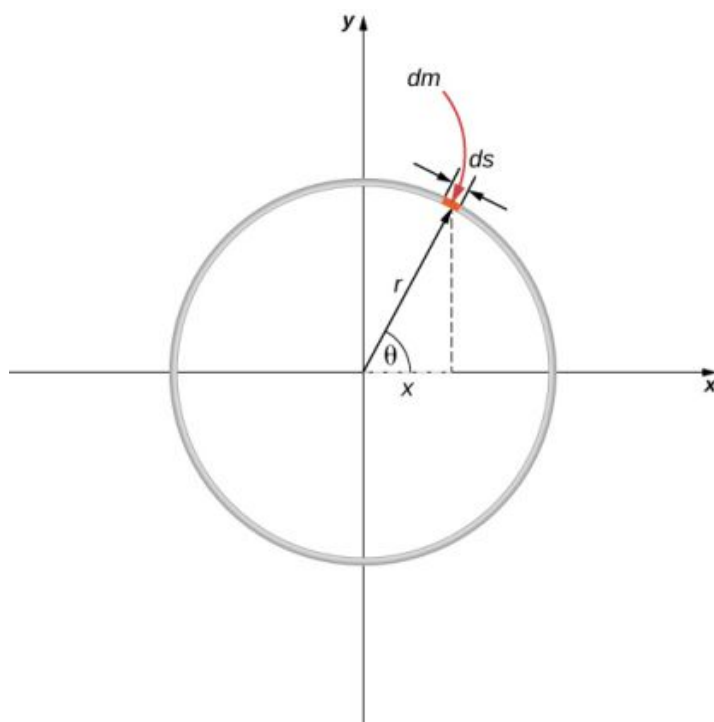


FIGURE 9.30 Finding the center of mass of a uniform hoop. We express the coordinates of a differential piece of the hoop, and then integrate around the hoop.

The center of mass is calculated with [Equation 9.34](#):

$$\vec{r}_{\text{CM}} = \frac{1}{M} \int_a^b \vec{r} dm.$$

We have to determine the limits of integration a and b . Expressing $\vec{r}$ in component form gives us

$$\vec{r}_{\text{CM}} = \frac{1}{M} \int_a^b [(r \cos \theta) \hat{i} + (r \sin \theta) \hat{j}] dm.$$

In the diagram, we highlighted a piece of the hoop that is of differential length ds ; it therefore has a differential mass $dm = \lambda ds$. Substituting:

$$\vec{r}_{\text{CM}} = \frac{1}{M} \int_a^b [(r \cos \theta) \hat{i} + (r \sin \theta) \hat{j}] \lambda ds.$$

However, the arc length ds subtends a differential angle $d\theta$, so we have

$$ds = r d\theta$$

and thus

$$\vec{r}_{\text{CM}} = \frac{1}{M} \int_a^b [(r \cos \theta) \hat{i} + (r \sin \theta) \hat{j}] \lambda r d\theta.$$

One more step: Since λ is the linear mass density, it is computed by dividing the total mass by the length of the hoop:

$$\lambda = \frac{M}{2\pi r}$$

giving us

$$\begin{aligned}
\vec{r}_{\text{CM}} &= \frac{1}{M} \int_a^b [(r \cos \theta) \hat{i} + (r \sin \theta) \hat{j}] \left(\frac{M}{2\pi r} \right) r d\theta \\
&= \frac{1}{2\pi} \int_a^b [(r \cos \theta) \hat{i} + (r \sin \theta) \hat{j}] d\theta.
\end{aligned}$$

Notice that the variable of integration is now the angle θ . This tells us that the limits of integration (around the circular hoop) are $\theta = 0$ to $\theta = 2\pi$, so $a = 0$ and $b = 2\pi$. Also, for convenience, we separate the integral into the x - and y -components of $\vec{r}_{\text{CM}}$. The final integral expression is

$$\begin{aligned}
\vec{r}_{\text{CM}} &= r_{\text{CM},x} \hat{i} + r_{\text{CM},y} \hat{j} \\
&= \left[\frac{1}{2\pi} \int_0^{2\pi} (r \cos \theta) d\theta \right] \hat{i} + \left[\frac{1}{2\pi} \int_0^{2\pi} (r \sin \theta) d\theta \right] \hat{j} \\
&= 0\hat{i} + 0\hat{j} = \vec{0}
\end{aligned}$$

as expected.

Center of Mass and Conservation of Momentum

How does all this connect to conservation of momentum?

Suppose you have N objects with masses $m_1, m_2, m_3, \dots, m_N$ and initial velocities $\vec{v}_1, \vec{v}_2, \vec{v}_3, \dots, \vec{v}_N$. The center of mass of the objects is

$$\vec{r}_{\text{CM}} = \frac{1}{M} \sum_{j=1}^N m_j \vec{r}_j.$$

Its velocity is

$$\vec{v}_{\text{CM}} = \frac{d\vec{r}_{\text{CM}}}{dt} = \frac{1}{M} \sum_{j=1}^N m_j \frac{d\vec{r}_j}{dt} \quad 9.35$$

and thus the initial momentum of the center of mass is

$$\begin{aligned}
\left[M \frac{d\vec{r}_{\text{CM}}}{dt} \right]_i &= \sum_{j=1}^N m_j \frac{d\vec{r}_{j,i}}{dt} \\
M \vec{v}_{\text{CM},i} &= \sum_{j=1}^N m_j \vec{v}_{j,i}.
\end{aligned}$$

After these masses move and interact with each other, the momentum of the center of mass is

$$M \vec{v}_{\text{CM},f} = \sum_{j=1}^N m_j \vec{v}_{j,f}.$$

But conservation of momentum tells us that the right-hand side of both equations must be equal, which says

$$M \vec{v}_{\text{CM},f} = M \vec{v}_{\text{CM},i}. \quad 9.36$$

This result implies that conservation of momentum is expressed in terms of the center of mass of the system. Notice that as an object moves through space with no net external force acting on it, an individual particle of the object may accelerate in various directions, with various magnitudes, depending on the net internal force acting on that object at any time. (Remember, it is only the vector sum of all the internal forces that vanishes, not the internal force on a single particle.) Thus, such a particle's momentum will not be constant—but the momentum of the entire extended object will be, in accord with [Equation 9.36](#).

[Equation 9.36](#) implies another important result: Since M represents the mass of the entire system of particles, it is necessarily constant. (If it isn't, we don't have a closed system, so we can't expect the system's momentum to be conserved.) As a result, [Equation 9.36](#) implies that, for a closed system,

$$\vec{v}_{\text{CM},f} = \vec{v}_{\text{CM},i}. \quad 9.37$$

That is to say, *in the absence of an external force, the velocity of the center of mass never changes.*

You might be tempted to shrug and say, “Well yes, that’s just Newton’s first law,” but remember that Newton’s first law discusses the constant velocity of a particle, whereas [Equation 9.37](#) applies to the center of mass of a (possibly vast) collection of interacting particles, and that there may not be any particle at the center of mass at all! So, this really is a remarkable result.



EXAMPLE 9.19

Fireworks Display

When a fireworks rocket explodes, thousands of glowing fragments fly outward in all directions, and fall to Earth in an elegant and beautiful display ([Figure 9.31](#)). Describe what happens, in terms of conservation of momentum and center of mass.



FIGURE 9.31 These exploding fireworks are a vivid example of conservation of momentum and the motion of the center of mass.

The picture shows radial symmetry about the central points of the explosions; this suggests the idea of center of mass. We can also see the parabolic motion of the glowing particles; this brings to mind projectile motion ideas.

Solution

Initially, the fireworks rocket is launched and flies more or less straight upward; this is the cause of the more-or-less-straight, white trail going high into the sky below the explosion in the upper-right of the picture (the yellow explosion). This trail is not parabolic because the explosive shell, during its launch phase, is actually a rocket; the impulse applied to it by the ejection of the burning fuel applies a force on the shell during the rise-time interval. (This is a phenomenon we will study in the next section.) The shell has multiple forces on it; thus, it is not in free-fall prior to the explosion.

At the instant of the explosion, the thousands of glowing fragments fly outward in a radially symmetrical pattern.

The symmetry of the explosion is the result of all the internal forces summing to zero $\left(\sum_j \vec{f}_j^{\text{int}} = 0\right)$; for every internal force, there is another that is equal in magnitude and opposite in direction.

However, as we learned above, these internal forces cannot change the momentum of the center of mass of the (now exploded) shell. Since the rocket force has now vanished, the center of mass of the shell is now a projectile (the only force on it is gravity), so its trajectory does become parabolic. The two red explosions on the left show the path of their centers of mass at a slightly longer time after explosion compared to the yellow explosion on the upper right.

In fact, if you look carefully at all three explosions, you can see that the glowing trails are not truly radially symmetric; rather, they are somewhat denser on one side than the other. Specifically, the yellow explosion and the lower middle explosion are slightly denser on their right sides, and the upper-left explosion is denser on its left side. This is because of the momentum of their centers of mass; the differing trail densities are due to the momentum each piece of the shell had at the moment of its explosion. The fragment for the explosion on the upper left of the picture had a momentum that pointed upward and to the left; the middle fragment's momentum pointed upward and slightly to the right; and the right-side explosion clearly upward and to the right (as evidenced by the white rocket exhaust trail visible below the yellow explosion).

Finally, each fragment is a projectile on its own, thus tracing out thousands of glowing parabolas.

Significance

In the discussion above, we said, "...the center of mass of the shell is now a projectile (the only force on it is gravity)...." This is not quite accurate, for there may not be any mass at all at the center of mass; in which case, there could not be a force acting on it. This is actually just verbal shorthand for describing the fact that the gravitational forces on all the particles act so that the center of mass changes position exactly as if all the mass of the shell were always located at the position of the center of mass.

✓ CHECK YOUR UNDERSTANDING 9.13

How would the firework display change in deep space, far away from any source of gravity?

You may sometimes hear someone describe an explosion by saying something like, "the fragments of the exploded object always move in a way that makes sure that the center of mass continues to move on its original trajectory." This makes it sound as if the process is somewhat magical: how can it be that, in *every* explosion, it *always* works out that the fragments move in just the right way so that the center of mass' motion is unchanged? Phrased this way, it would be hard to believe no explosion ever does anything differently.

The explanation of this apparently astonishing coincidence is: We defined the center of mass precisely so this is exactly what we would get. Recall that first we defined the momentum of the system:

$$\vec{p}_{\text{CM}} = \sum_{j=1}^N \frac{d\vec{p}_j}{dt}.$$

We then concluded that the net external force on the system (if any) changed this momentum:

$$\vec{F} = \frac{d\vec{p}_{\text{CM}}}{dt}$$

and then—and here's the point—we defined an acceleration that would obey Newton's second law. That is, we demanded that we should be able to write

$$\vec{a} = \frac{\vec{F}}{M}$$

which requires that

$$\vec{a} = \frac{d^2}{dt^2} \left(\frac{1}{M} \sum_{j=1}^N m_j \vec{r}_j \right).$$

where the quantity inside the parentheses is the center of mass of our system. So, it's not astonishing that the center of mass obeys Newton's second law; we defined it so that it would.

9.7 Rocket Propulsion

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Describe the application of conservation of momentum when the mass changes with time, as well as the velocity
- Calculate the speed of a rocket in empty space, at some time, given initial conditions
- Calculate the speed of a rocket in Earth's gravity field, at some time, given initial conditions

Now we deal with the case where the mass of an object is changing. We analyze the motion of a rocket, which changes its velocity (and hence its momentum) by ejecting burned fuel gases, thus causing it to accelerate in the opposite direction of the velocity of the ejected fuel (see [Figure 9.32](#)). Specifically: A fully fueled rocket ship in deep space has a total mass m_0 (this mass includes the initial mass of the fuel). At some moment in time, the rocket has a velocity $\vec{v}$ and mass m ; this mass is a combination of the mass of the empty rocket and the mass of the remaining unburned fuel it contains. (We refer to m as the “instantaneous mass” and $\vec{v}$ as the “instantaneous velocity.”) The rocket accelerates by burning the fuel it carries and ejecting the burned exhaust gases. If the burn rate of the fuel is constant, and the velocity at which the exhaust is ejected is also constant, what is the change of velocity of the rocket as a result of burning all of its fuel?



FIGURE 9.32 The space shuttle had a number of reusable parts. Solid fuel boosters on either side were recovered and refueled after each flight, and the entire orbiter returned to Earth for use in subsequent flights. The large liquid fuel tank was expended. The space shuttle was a complex assemblage of technologies, employing both solid and liquid fuel, and pioneering ceramic tiles as reentry heat shields. As a result, it permitted multiple launches as opposed to single-use rockets. (credit: modification of work by NASA)

Physical Analysis

Here's a description of what happens, so that you get a feel for the physics involved.

- As the rocket engines operate, they are continuously ejecting burned fuel gases, which have both mass and velocity, and therefore some momentum. By conservation of momentum, the rocket's momentum changes by this same amount (with the opposite sign). We will assume the burned fuel is being ejected at a constant rate, which means the rate of change of the rocket's momentum is also constant. By [Equation 9.9](#), this represents a constant force on the rocket.
- However, as time goes on, the mass of the rocket (which includes the mass of the remaining fuel) continuously decreases. Thus, even though the force on the rocket is constant, the resulting acceleration is not; it is continuously increasing.
- So, the total change of the rocket's velocity will depend on the amount of mass of fuel that is burned, and that dependence is not linear.

The problem has the mass and velocity of the rocket changing; also, the total mass of ejected gases is changing. If we define our system to be the rocket + fuel, then this is a closed system (since the rocket is in deep space, there are no external forces acting on this system); as a result, momentum is conserved for this system. Thus, we can apply conservation of momentum to answer the question ([Figure 9.33](#)).

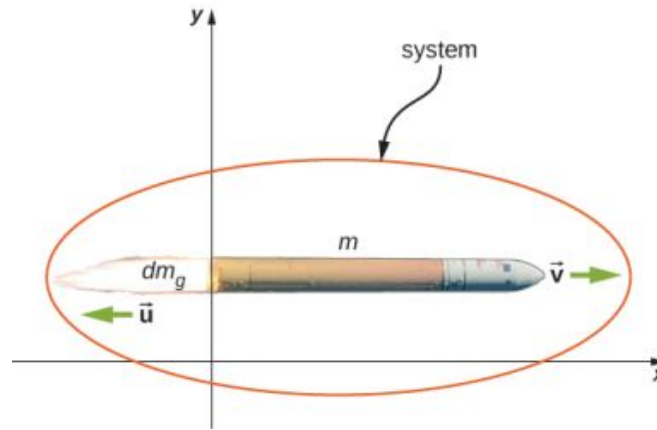


FIGURE 9.33 The rocket accelerates to the right due to the expulsion of some of its fuel mass to the left. Conservation of momentum enables us to determine the resulting change of velocity. The mass m is the instantaneous total mass of the rocket (i.e., mass of rocket body plus mass of fuel at that point in time). (credit: modification of work by NASA/Bill Ingalls)

At the same moment that the total instantaneous rocket mass is m (i.e., m is the mass of the rocket body plus the mass of the fuel at that point in time), we define the rocket's instantaneous velocity to be $\vec{v} = v\hat{i}$ (in the $+x$ -direction); this velocity is measured relative to an inertial reference system (the Earth, for example). Thus, the initial momentum of the system is

$$\vec{p}_i = mv\hat{i}.$$

The rocket's engines are burning fuel at a constant rate and ejecting the exhaust gases in the $-x$ -direction. During an infinitesimal time interval dt , the engines eject a (positive) infinitesimal mass of gas dm_g at velocity $\vec{u} = -u\hat{i}$; note that although the rocket velocity $v\hat{i}$ is measured with respect to Earth, the exhaust gas velocity is measured with respect to the (moving) rocket. Measured with respect to the Earth, therefore, the exhaust gas has velocity $(v - u)\hat{i}$.

As a consequence of the ejection of the fuel gas, the rocket's mass decreases by dm_g , and its velocity increases by $dv\hat{i}$. Therefore, including both the change for the rocket and the change for the exhaust gas, the final momentum of the system is

$$\begin{aligned}\vec{p}_f &= \vec{p}_{\text{rocket}} + \vec{p}_{\text{gas}} \\ &= (m - dm_g)(v + dv)\hat{i} + dm_g(v - u)\hat{i}\end{aligned}$$

Since all vectors are in the x -direction, we drop the vector notation. Applying conservation of momentum, we obtain

$$\begin{aligned}
 p_i &= p_f \\
 mv &= (m - dm_g)(v + dv) + dm_g(v - u) \\
 mv &= mv + mdv - dm_g v - dm_g dv + dm_g v - dm_g u \\
 mdv &= dm_g dv + dm_g u.
 \end{aligned}$$

Now, dm_g and dv are each very small; thus, their product $dm_g dv$ is very, very small, much smaller than the other two terms in this expression. We neglect this term, therefore, and obtain:

$$mdv = dm_g u.$$

Our next step is to remember that, since dm_g represents an increase in the mass of ejected gases, it must also represent a decrease of mass of the rocket:

$$dm_g = -dm.$$

Replacing this, we have

$$mdv = -dm u$$

or

$$dv = -u \frac{dm}{m}.$$

Integrating from the initial mass m_0 to the final mass m of the rocket gives us the result we are after:

$$\begin{aligned}
 \int_{v_i}^v dv &= -u \int_{m_0}^m \frac{1}{m} dm \\
 v - v_i &= u \ln \left(\frac{m_0}{m} \right)
 \end{aligned}$$

and thus our final answer is

$$\Delta v = u \ln \left(\frac{m_0}{m} \right). \quad 9.38$$

This result is called the **rocket equation**. It was originally derived by the Russian rocket scientist Konstantin Tsiolkovsky in 1897. It gives us the change of velocity that the rocket obtains from burning a mass of fuel that decreases the total rocket mass from m_0 down to m . As expected, the relationship between Δv and the change of mass of the rocket is nonlinear.



PROBLEM-SOLVING STRATEGY

Rocket Propulsion

In rocket problems, the most common questions are finding the change of velocity due to burning some amount of fuel for some amount of time; or to determine the acceleration that results from burning fuel.

1. To determine the change of velocity, use the rocket equation [Equation 9.38](#).
2. To determine the acceleration, determine the force by using the impulse-momentum theorem, using the rocket equation to determine the change of velocity.



EXAMPLE 9.20

Thrust on a Spacecraft

A spacecraft is moving in gravity-free space along a straight path when its pilot decides to accelerate forward. He turns on the thrusters, and burned fuel is ejected at a constant rate of 2.0×10^2 kg/s, at a speed (relative to the rocket) of 2.5×10^2 m/s. The initial mass of the spacecraft and its unburned fuel is 2.0×10^4 kg, and the

thrusters are on for 30 s.

- What is the thrust (the force applied to the rocket by the ejected fuel) on the spacecraft?
- What is the spacecraft's acceleration as a function of time?
- What are the spacecraft's accelerations at $t = 0, 15, 30$, and 35 s?

Strategy

- The force on the spacecraft is equal to the rate of change of the momentum of the fuel.
- Knowing the force from part (a), we can use Newton's second law to calculate the consequent acceleration. The key here is that, although the force applied to the spacecraft is constant (the fuel is being ejected at a constant rate), the mass of the spacecraft isn't; thus, the acceleration caused by the force won't be constant. We expect to get a function $a(t)$, therefore.
- We'll use the function we obtain in part (b), and just substitute the numbers given. Important: We expect that the acceleration will get larger as time goes on, since the mass being accelerated is continuously decreasing (fuel is being ejected from the rocket).

Solution

- The momentum of the ejected fuel gas is

$$p = m_g v.$$

The ejection velocity $v = 2.5 \times 10^2$ m/s is constant, and therefore the force is

$$F = \frac{dp}{dt} = v \frac{dm_g}{dt} = -v \frac{dm}{dt}.$$

Now, $\frac{dm_g}{dt}$ is the rate of change of the mass of the fuel; the problem states that this is 2.0×10^2 kg/s.

Substituting, we get

$$\begin{aligned} F &= v \frac{dm_g}{dt} \\ &= (2.5 \times 10^2 \frac{\text{m}}{\text{s}}) (2.0 \times 10^2 \frac{\text{kg}}{\text{s}}) \\ &= 5 \times 10^4 \text{ N}. \end{aligned}$$

- Above, we defined m to be the combined mass of the empty rocket plus however much unburned fuel it contained: $m = m_R + m_g$. From Newton's second law,

$$a = \frac{F}{m} = \frac{F}{m_R + m_g}.$$

The force is constant and the empty rocket mass m_R is constant, but the fuel mass m_g is decreasing at a uniform rate; specifically:

$$m_g = m_g(t) = m_{g0} - \left(\frac{dm_g}{dt} \right) t.$$

This gives us

$$a(t) = \frac{F}{m_{g1} - \left(\frac{dm_g}{dt} \right) t} = \frac{F}{M - \left(\frac{dm_g}{dt} \right) t}.$$

Notice that, as expected, the acceleration is a function of time. Substituting the given numbers:

$$a(t) = \frac{5 \times 10^4 \text{ N}}{2.0 \times 10^4 \text{ kg} - \left(2.0 \times 10^2 \frac{\text{kg}}{\text{s}} \right) t}.$$

- At $t = 0$ s:

$$a(0 \text{ s}) = \frac{5 \times 10^4 \text{ N}}{2.0 \times 10^4 \text{ kg} - \left(2.0 \times 10^2 \frac{\text{kg}}{\text{s}} \right) (0 \text{ s})} = 2.5 \frac{\text{m}}{\text{s}^2}.$$

At $t = 15$ s, $a(15 \text{ s}) = 2.9 \text{ m/s}^2$.

At $t = 30$ s, $a(30 \text{ s}) = 3.6 \text{ m/s}^2$.

Acceleration is increasing, as we expected.

At $t = 35$ s (and at any time after $t = 30$ s), the acceleration is 0 m/s^2 since the thrusters stopped working and thus no external forces are acting on the spacecraft.

Significance

Notice that the acceleration is not constant; as a result, any dynamical quantities must be calculated either using integrals, or (more easily) conservation of total energy.

✓ CHECK YOUR UNDERSTANDING 9.14

What is the physical difference (or relationship) between $\frac{dm}{dt}$ and $\frac{dm_g}{dt}$ in this example?

Rocket in a Gravitational Field

Let's now analyze the velocity change of the rocket during the launch phase, from the surface of Earth. To keep the math manageable, we'll restrict our attention to distances for which the acceleration caused by gravity can be treated as a constant g .

The analysis is similar, except that now there is an external force of $\vec{F} = -mg\hat{j}$ acting on our system. This force applies an impulse $d\vec{J} = \vec{F}dt = -mgdt\hat{j}$, which is equal to the change of momentum. This gives us

$$\begin{aligned} d\vec{p} &= d\vec{J} \\ \vec{p}_f - \vec{p}_i &= -mgdt\hat{j} \\ [(m - dm_g)(v + dv) + dm_g(v - u) - mv]\hat{j} &= -mgdt\hat{j} \end{aligned}$$

and so

$$mdv - dm_g u = -mgdt$$

where we have again neglected the term $dm_g dv$ and dropped the vector notation. Next we replace dm_g with $-dm$:

$$\begin{aligned} mdv + dm u &= -mgdt \\ mdv &= -dm u - mgdt. \end{aligned}$$

Dividing through by m gives

$$dv = -u \frac{dm}{m} - gdt$$

and integrating, we have

$$\Delta v = u \ln\left(\frac{m_0}{m}\right) - g\Delta t. \quad 9.39$$

Unsurprisingly, the rocket's velocity is affected by the (constant) acceleration of gravity.

Remember that Δt is the burn time of the fuel. Now, in the absence of gravity, [Equation 9.38](#) implies that it makes no difference how much time it takes to burn the entire mass of fuel; the change of velocity does not depend on Δt . However, in the presence of gravity, it matters a lot. The $-g\Delta t$ term in [Equation 9.39](#) tells us that the *longer* the burn time is, the *smaller* the rocket's change of velocity will be. This is the reason that the launch of a rocket is so spectacular at the first moment of liftoff: It's essential to burn the fuel as quickly as possible, to get as large a Δv as possible.

Chapter Review

Key Terms

center of mass weighted average position of the mass

closed system system for which the mass is constant and the net external force on the system is zero

elastic collision that conserves kinetic energy

explosion single object breaks up into multiple objects; kinetic energy is not conserved in explosions

external force force applied to an extended object that changes the momentum of the extended object as a whole

impulse effect of applying a force on a system for a time interval; this time interval is usually small, but does not have to be

impulse-momentum theorem change of momentum of a system is equal to the impulse applied to the system

inelastic collision that does not conserve kinetic energy

internal force force that the simple particles that make up an extended object exert on each other. Internal forces can be attractive or repulsive

Law of Conservation of Momentum total momentum of a closed system cannot change

linear mass density λ , expressed as the number of kilograms of material per meter

momentum measure of the quantity of motion that an object has; it takes into account both how fast the object is moving, and its mass; specifically, it is the product of mass and velocity; it is a vector quantity

perfectly inelastic collision after which all objects are motionless, the final kinetic energy is zero, and the loss of kinetic energy is a maximum

rocket equation derived by the Soviet physicist Konstantin Tsiolkovsky in 1897, it gives us the change of velocity that the rocket obtains from burning a mass of fuel that decreases the total rocket mass from m_i down to m

system object or collection of objects whose motion is currently under investigation; however, your system is defined at the start of the problem, you must keep that definition for the entire problem

Key Equations

Definition of momentum

$$\vec{p} = m\vec{v}$$

Impulse

$$\vec{J} \equiv \int_{t_i}^{t_f} \vec{F}(t)dt \text{ or } \vec{J} = \vec{F}_{\text{ave}} \Delta t$$

Impulse-momentum theorem

$$\vec{J} = \Delta \vec{p}$$

Average force from momentum

$$\vec{F} = \frac{\Delta \vec{p}}{\Delta t}$$

Instantaneous force from momentum (Newton's second law)

$$\vec{F}(t) = \frac{d\vec{p}}{dt}$$

Conservation of momentum

$$\frac{d\vec{p}_1}{dt} + \frac{d\vec{p}_2}{dt} = 0 \text{ or } \vec{p}_1 + \vec{p}_2 = \text{constant}$$

Generalized conservation of momentum

$$\sum_{j=1}^N \vec{p}_j = \text{constant}$$

Conservation of momentum in two dimensions

$$p_{f,x} = p_{1,i,x} + p_{2,i,x}$$

$$p_{f,y} = p_{1,i,y} + p_{2,i,y}$$

External forces

$$\vec{F}_{\text{ext}} = \sum_{j=1}^N \frac{d\vec{p}_j}{dt}$$

Newton's second law for an extended object

$$\vec{F} = \frac{d\vec{p}_{\text{CM}}}{dt}$$

Acceleration of the center of mass

$$\vec{a}_{\text{CM}} = \frac{d^2}{dt^2} \left(\frac{1}{M} \sum_{j=1}^N m_j \vec{r}_j \right) = \frac{1}{M} \sum_{j=1}^N m_j \vec{a}_j$$

Position of the center of mass for a system of particles

$$\vec{r}_{\text{CM}} \equiv \frac{1}{M} \sum_{j=1}^N m_j \vec{r}_j$$

Velocity of the center of mass

$$\vec{v}_{\text{CM}} = \frac{d}{dt} \left(\frac{1}{M} \sum_{j=1}^N m_j \vec{r}_j \right) = \frac{1}{M} \sum_{j=1}^N m_j \vec{v}_j$$

Position of the center of mass of a continuous object

$$\vec{r}_{\text{CM}} \equiv \frac{1}{M} \int \vec{r} \, dm$$

Rocket equation

$$\Delta v = u \ln \left(\frac{m_1}{m} \right)$$

Summary

9.1 Linear Momentum

- The motion of an object depends on its mass as well as its velocity. Momentum is a concept that describes this. It is a useful and powerful concept, both computationally and theoretically. The SI unit for momentum is kg·m/s.

9.2 Impulse and Collisions

- When a force is applied on an object for some amount of time, the object experiences an impulse.
- This impulse is equal to the object's change of momentum.
- Newton's second law in terms of momentum states that the net force applied to a system equals the rate of change of the momentum that the force causes.

9.3 Conservation of Linear Momentum

- The law of conservation of momentum says that the momentum of a closed system is constant in time (conserved).
- A closed (or isolated) system is defined to be one for which the mass remains constant, and the net external force is zero.
- The total momentum of a system is conserved *only* when the system is closed.

9.4 Types of Collisions

- An elastic collision is one that conserves kinetic energy.
- An inelastic collision does not conserve kinetic energy.
- Momentum is conserved regardless of whether or not kinetic energy is conserved.
- Analysis of kinetic energy changes and conservation of momentum together allow the final velocities to be calculated in terms of initial

velocities and masses in one-dimensional, two-body collisions.

9.5 Collisions in Multiple Dimensions

- The approach to two-dimensional collisions is to choose a convenient coordinate system and break the motion into components along perpendicular axes.
- Momentum is conserved in both directions simultaneously and independently.
- The Pythagorean theorem gives the magnitude of the momentum vector using the x- and y-components, calculated using conservation of momentum in each direction.

9.6 Center of Mass

- An extended object (made up of many objects) has a defined position vector called the center of mass.
- The center of mass can be thought of, loosely, as the average location of the total mass of the object.
- The center of mass of an object traces out the trajectory dictated by Newton's second law, due to the net external force.
- The internal forces within an extended object cannot alter the momentum of the extended object as a whole.

9.7 Rocket Propulsion

- A rocket is an example of conservation of momentum where the mass of the system is not constant, since the rocket ejects fuel to provide thrust.
- The rocket equation gives us the change of velocity that the rocket obtains from burning a mass of fuel that decreases the total rocket mass.

Conceptual Questions

9.1 Linear Momentum

1. An object that has a small mass and an object that has a large mass have the same

momentum. Which object has the largest kinetic energy?

2. An object that has a small mass and an object

that has a large mass have the same kinetic energy. Which mass has the largest momentum?

9.2 Impulse and Collisions

- Is it possible for a small force to produce a larger impulse on a given object than a large force? Explain.
- Why is a 10-m fall onto concrete far more dangerous than a 10-m fall onto water?
- What external force is responsible for changing the momentum of a car moving along a horizontal road?
- A piece of putty and a tennis ball with the same mass are thrown against a wall with the same velocity. Which object experiences a greater force from the wall or are the forces equal? Explain.

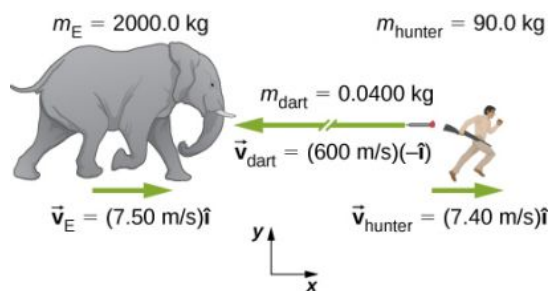
9.3 Conservation of Linear Momentum

- Under what circumstances is momentum conserved?
- Can momentum be conserved for a system if there are external forces acting on the system? If so, under what conditions? If not, why not?
- Explain in terms of momentum and Newton's laws how a car's air resistance is due in part to the fact that it pushes air in its direction of motion.
- Can objects in a system have momentum while the momentum of the system is zero? Explain your answer.
- A sprinter accelerates out of the starting blocks. Can you consider him as a closed system?

Problems

9.1 Linear Momentum

- An elephant and a hunter are having a confrontation.



- Calculate the momentum of the 2000.0-kg elephant charging the hunter at a speed of

Explain.

- A rocket in deep space (zero gravity) accelerates by firing hot gas out of its thrusters. Does the rocket constitute a closed system? Explain.

9.4 Types of Collisions

- Two objects of equal mass are moving with equal and opposite velocities when they collide. Can all the kinetic energy be lost in the collision?
- Describe a system for which momentum is conserved but mechanical energy is not. Now the reverse: Describe a system for which kinetic energy is conserved but momentum is not.

9.5 Collisions in Multiple Dimensions

- Momentum for a system can be conserved in one direction while not being conserved in another. What is the angle between the directions? Give an example.

9.6 Center of Mass

- Suppose a fireworks shell explodes, breaking into three large pieces for which air resistance is negligible. How does the explosion affect the motion of the center of mass? How would it be affected if the pieces experienced significantly more air resistance than the intact shell?

9.7 Rocket Propulsion

- It is possible for the velocity of a rocket to be greater than the exhaust velocity of the gases it ejects. When that is the case, the gas velocity and gas momentum are in the same direction as that of the rocket. How is the rocket still able to obtain thrust by ejecting the gases?

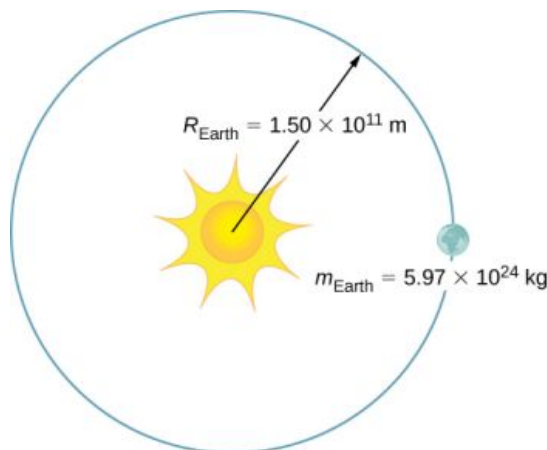
7.50 m/s.

- Calculate the ratio of the elephant's momentum to the momentum of a 0.0400-kg tranquilizer dart fired at a speed of 600 m/s.
- What is the momentum of the 90.0-kg hunter running at 7.40 m/s after missing the elephant?

- A skater of mass 40 kg is carrying a box of mass 5 kg. The skater has a speed of 5 m/s with respect to the floor and is gliding without any friction on a smooth surface.

- Find the momentum of the box with respect to the floor.

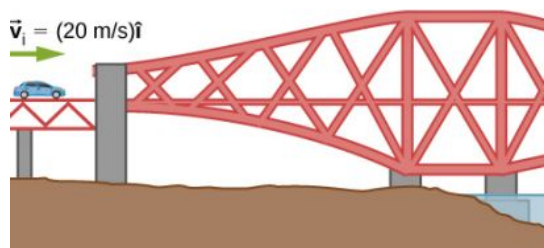
- b. Find the momentum of the box with respect to the floor after she puts the box down on the frictionless skating surface.
20. A car of mass 2000 kg is moving with a constant velocity of 10 m/s due east. What is the momentum of the car?
21. The mass of Earth is 5.97×10^{24} kg and its orbital radius is an average of 1.50×10^{11} m. Calculate the magnitude of its linear momentum at the location in the diagram.



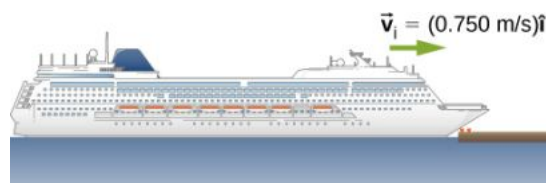
22. If a rainstorm drops 1 cm of rain over an area of 10 km^2 in the period of 1 hour, what is the momentum of the rain that falls in one second? Assume the terminal velocity of a raindrop is 10 m/s.
23. What is the average momentum of an avalanche that moves a 40-cm-thick layer of snow over an area of 100 m by 500 m over a distance of 1 km down a hill in 5.5 s? Assume a density of 350 kg/m^3 for the snow.
24. What is the average momentum of a 70.0-kg sprinter who runs the 100-m dash in 9.65 s?

9.2 Impulse and Collisions

25. A 75.0-kg person is riding in a car moving at 20.0 m/s when the car runs into a bridge abutment (see the following figure).



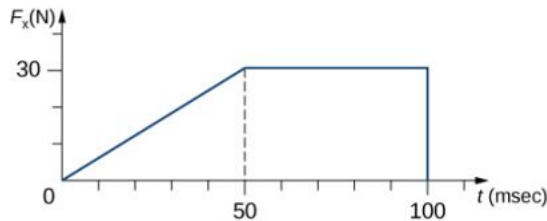
- a. Calculate the average force on the person if he is stopped by a padded dashboard that compresses an average of 1.00 cm.
- b. Calculate the average force on the person if he is stopped by an air bag that compresses an average of 15.0 cm.
26. One hazard of space travel is debris left by previous missions. There are several thousand objects orbiting Earth that are large enough to be detected by radar, but there are far greater numbers of very small objects, such as flakes of paint. Calculate the force exerted by a 0.100-mg chip of paint that strikes a spacecraft window at a relative speed of 4.00×10^3 m/s, given the collision lasts 6.00×10^{-8} s.
27. A cruise ship with a mass of 1.00×10^7 kg strikes a pier at a speed of 0.750 m/s. It comes to rest after traveling 6.00 m, damaging the ship, the pier, and the tugboat captain's finances. Calculate the average force exerted on the pier using the concept of impulse. (*Hint:* First calculate the time it took to bring the ship to rest, assuming a constant force.)



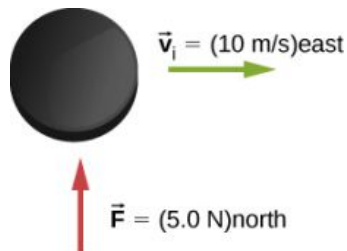
28. Calculate the final speed of a 110-kg rugby player who is initially running at 8.00 m/s but collides head-on with a padded goalpost and experiences a backward force of 1.76×10^4 N for 5.50×10^{-2} s.
29. Water from a fire hose is directed horizontally against a wall at a rate of 50.0 kg/s and a speed of 42.0 m/s. Calculate the force exerted on the wall, assuming the water's horizontal momentum is reduced to zero.
30. A 0.450-kg hammer is moving horizontally at 7.00 m/s when it strikes a nail and comes to rest after driving the nail 1.00 cm into a board. Assume constant acceleration of the hammer-nail pair.
- a. Calculate the duration of the impact.
- b. What was the average force exerted on the nail?
31. What is the momentum (as a function of time) of a 5.0-kg particle moving with a velocity $\vec{v}(t) = (2.0\hat{i} + 4.0t\hat{j})$ m/s? What is the net

force acting on this particle?

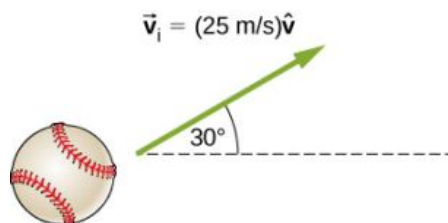
32. The x -component of a force on a 46-g golf ball by a 7-iron versus time is plotted in the following figure:



- Find the x -component of the impulse during the intervals
 - $[0, 50 \text{ ms}]$, and
 - $[50 \text{ ms}, 100 \text{ ms}]$
 - Find the change in the x -component of the momentum during the intervals
 - $[0, 50 \text{ ms}]$, and
 - $[50 \text{ ms}, 100 \text{ ms}]$
33. A hockey puck of mass 150 g is sliding due east on a frictionless table with a speed of 10 m/s. Suddenly, a constant force of magnitude 5 N and direction due north is applied to the puck for 1.5 s. Find the north and east components of the momentum at the end of the 1.5-s interval.



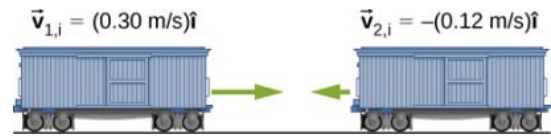
34. A ball of mass 250 g is thrown with an initial velocity of 25 m/s at an angle of 30° with the horizontal direction. Ignore air resistance. What is the momentum of the ball after 0.2 s? (Do this problem by finding the components of the momentum first, and then constructing the magnitude and direction of the momentum vector from the components.)



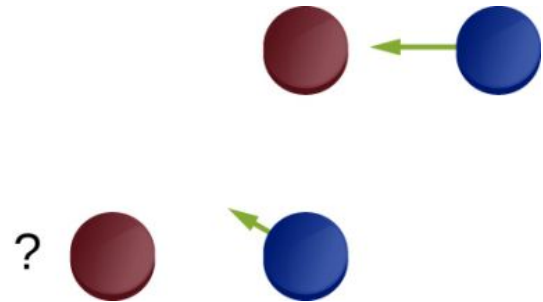
9.3 Conservation of Linear Momentum

35. Train cars are coupled together by being bumped

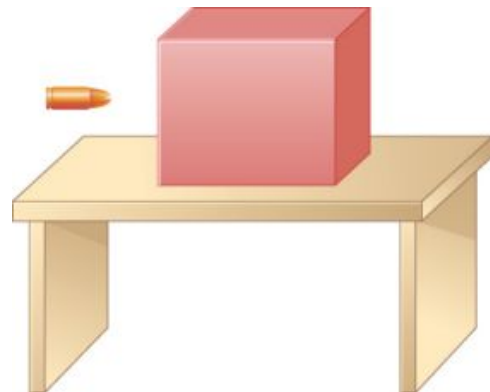
into one another. Suppose two loaded train cars are moving toward one another, the first having a mass of $1.50 \times 10^5 \text{ kg}$ and a velocity of $(0.30 \text{ m/s})\hat{i}$, and the second having a mass of $1.10 \times 10^5 \text{ kg}$ and a velocity of $-(0.12 \text{ m/s})\hat{i}$. What is their final velocity?



36. Two identical pucks collide elastically on an air hockey table. Puck 1 was originally at rest; puck 2 has an incoming speed of 6.00 m/s and scatters at an angle of 30° with respect to its incoming direction. What is the velocity (magnitude and direction) of puck 1 after the collision?



37. The figure below shows a bullet of mass 200 g traveling horizontally towards the east with speed 400 m/s, which strikes a block of mass 1.5 kg that is initially at rest on a frictionless table.



After striking the block, the bullet is embedded in the block and the block and the bullet move together as one unit.

- What is the magnitude and direction of the velocity of the block/bullet combination immediately after the impact?
- What is the magnitude and direction of the impulse by the block on the bullet?
- What is the magnitude and direction of the

- impulse from the bullet on the block?
- d. If it took 3 ms for the bullet to change the speed from 400 m/s to the final speed after impact, what is the average force between the block and the bullet during this time?
38. A 20-kg child is coasting at 3.3 m/s over flat ground in a 4.0-kg wagon.
- If the child drops a 1-kg ball out the back of the wagon, what is the final speed of the child and wagon?
 - If the child drops a 1-kg ball out the back of the wagon so that the ball is at rest relative to the ground, what is the final speed of the child and wagon?
39. A 4.5 kg puffer fish expands to 40% of its mass by taking in water. When the puffer fish is threatened, it releases the water toward the threat to move quickly forward. What is the ratio of the speed of the puffer fish forward to the speed of the expelled water backwards?
40. Explain why a cannon recoils when it fires a shell.
41. Two figure skaters are coasting in the same direction, with the leading skater moving at 5.5 m/s and the trailing skater moving at 6.2 m/s. When the trailing skater catches up with the leading skater, he picks her up without applying any horizontal forces on his skates. If the trailing skater is 50% heavier than the 50-kg leading skater, what is their speed after he picks her up?
42. A 2000-kg railway freight car coasts at 4.4 m/s underneath a grain terminal, which dumps grain directly down into the freight car. If the speed of the loaded freight car must not go below 3.0 m/s, what is the maximum mass of grain that it can accept?

9.4 Types of Collisions

43. A 5.50-kg bowling ball moving at 9.00 m/s collides with a 0.850-kg bowling pin, which is scattered at an angle of 15.8° to the initial direction of the bowling ball and with a speed of 15.0 m/s.
- Calculate the final velocity (magnitude and direction) of the bowling ball.
 - Is the collision elastic?
44. Ernest Rutherford (the first New Zealander to be awarded the Nobel Prize in Chemistry)

demonstrated that nuclei were very small and dense by scattering helium-4 nuclei from gold-197 nuclei. The energy of the incoming helium nucleus was 8.00×10^{-13} J, and the masses of the helium and gold nuclei were 6.68×10^{-27} kg and 3.29×10^{-25} kg, respectively (note that their mass ratio is 4 to 197).

a. If a helium nucleus scatters to an angle of 120° during an elastic collision with a gold nucleus, calculate the helium nucleus's final speed and the final velocity (magnitude and direction) of the gold nucleus.



- b. What is the final kinetic energy of the helium nucleus?
45. A 90.0-kg ice hockey player hits a 0.150-kg puck, giving the puck a velocity of 45.0 m/s. If both are initially at rest and if the ice is frictionless, how far does the player recoil in the time it takes the puck to reach the goal 15.0 m away?
46. A 100-g firecracker is launched vertically into the air and explodes into two pieces at the peak of its trajectory. If a 72-g piece is projected horizontally to the left at 20 m/s, what is the speed and direction of the other piece?
47. In an elastic collision, a 400-kg bumper car collides directly from behind with a second, identical bumper car that is traveling in the same direction. The initial speed of the leading bumper car is 5.60 m/s and that of the trailing car is 6.00 m/s. Assuming that the mass of the drivers is much, much less than that of the bumper cars, what are their final speeds?
48. Repeat the preceding problem if the mass of the leading bumper car is 30.0% greater than that of the trailing bumper car.
49. An alpha particle (^4He) undergoes an elastic collision with a stationary uranium nucleus (^{235}U). What percent of the kinetic energy of the alpha particle is transferred to the uranium nucleus? Assume the collision is one-dimensional.
50. You are standing on a very slippery icy surface

and throw a 1-kg football horizontally at a speed of 6.7 m/s. What is your velocity when you release the football? Assume your mass is 65 kg.

51. A 35-kg child rides a relatively massless sled down a hill and then coasts along the flat section at the bottom, where a second 35-kg child jumps on the sled as it passes by her. If the speed of the sled is 3.5 m/s before the second child jumps on, what is its speed after she jumps on?
52. A boy sleds down a hill and onto a frictionless ice-covered lake at 10.0 m/s. In the middle of the lake is a 1000-kg boulder. When the sled crashes into the boulder, he is propelled backwards from the boulder. The collision is an elastic collision. If the boy's mass is 40.0 kg and the sled's mass is 2.50 kg, what is the speed of the sled and the boulder after the collision?

9.5 Collisions in Multiple Dimensions

53. A 0.90-kg falcon is diving at 28.0 m/s at a downward angle of 35° . It catches a 0.325-kg pigeon from behind in midair. What is their combined velocity after impact if the pigeon's initial velocity was 7.00 m/s directed horizontally? Note that $\hat{v}_{1,i}$ is a unit vector pointing in the direction in which the falcon is initially flying.

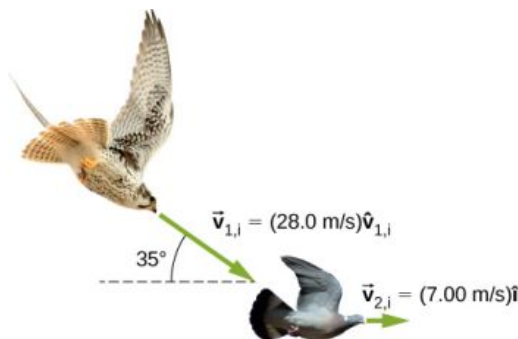
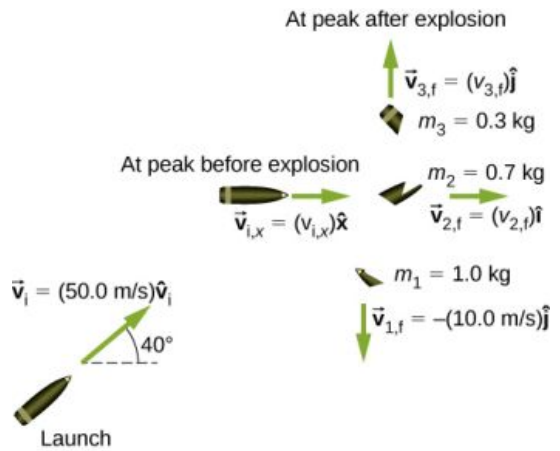


FIGURE 9.34 (credit “falcon”: modification of work by “USFWS Mountain-Prairie”/Flickr; credit “pigeon”: modification of work by Jacob Spinks)

54. A billiard ball, labeled 1, moving horizontally strikes another billiard ball, labeled 2, at rest. Before impact, ball 1 was moving at a speed of 3.00 m/s, and after impact it is moving at 0.50 m/s at 50° from the original direction. If the two balls have equal masses of 300 g, what is the velocity of the ball 2 after the impact?
55. A projectile of mass 2.0 kg is fired in the air at an angle of 40.0° to the horizon at a speed of 50.0 m/s. At the highest point in its flight, the projectile breaks into three parts of mass 1.0 kg, 0.7 kg, and 0.3 kg.

The 1.0-kg part falls straight down after breakup with an initial speed of 10.0 m/s, the 0.7-kg part moves in the original forward direction, and the 0.3-kg part goes straight up.



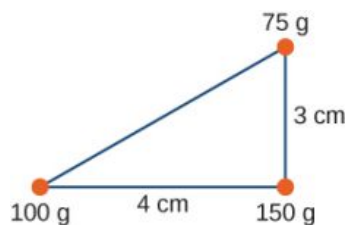
- Find the speeds of the 0.3-kg and 0.7-kg pieces immediately after the break-up.
 - How high from the break-up point does the 0.3-kg piece go before coming to rest?
 - Where does the 0.7-kg piece land relative to where it was fired from?
56. Two asteroids collide and stick together. The first asteroid has mass of 15×10^3 kg and is initially moving at 770 m/s. The second asteroid has mass of 20×10^3 kg and is moving at 1020 m/s. Their initial velocities made an angle of 20° with respect to each other. What is the final speed and direction with respect to the velocity of the first asteroid?
57. A 200-kg rocket in deep space moves with a velocity of $(121 \text{ m/s})\hat{i} + (38.0 \text{ m/s})\hat{j}$. Suddenly, it explodes into three pieces, with the first (78 kg) moving at $-(321 \text{ m/s})\hat{i} + (228 \text{ m/s})\hat{j}$ and the second (56 kg) moving at $(16.0 \text{ m/s})\hat{i} - (88.0 \text{ m/s})\hat{j}$. Find the velocity of the third piece.
58. A proton traveling at 3.0×10^6 m/s scatters elastically from an initially stationary alpha particle and is deflected at an angle of 85° with respect to its initial velocity. Given that the alpha particle has four times the mass of the proton, what percent of its initial kinetic energy does the proton retain after the collision?
59. Three 70-kg deer are standing on a flat 200-kg rock that is on an ice-covered pond. A gunshot goes off and the deer scatter, with deer A running at $(15 \text{ m/s})\hat{i} + (5.0 \text{ m/s})\hat{j}$, deer B

running at $(-12 \text{ m/s})\hat{i} + (8.0 \text{ m/s})\hat{j}$, and deer C running at $(1.2 \text{ m/s})\hat{i} - (18.0 \text{ m/s})\hat{j}$. What is the velocity of the rock on which they were standing?

60. A family is skating. The father (75 kg) skates at 8.2 m/s and collides and sticks to the mother (50 kg), who was initially moving at 3.3 m/s and at 45° with respect to the father's velocity. The pair then collides with their daughter (30 kg), who was stationary, and the three slide off together. What is their final velocity?
61. An oxygen atom (mass 16 u) moving at 733 m/s at 15.0° with respect to the $\hat{i}$ direction collides and sticks to an oxygen molecule (mass 32 u) moving at 528 m/s at 128° with respect to the $\hat{i}$ direction. The two stick together to form ozone. What is the final velocity of the ozone molecule?
62. Two cars of the same mass approach an extremely icy four-way perpendicular intersection. Car A travels northward at 30 m/s and car B is travelling eastward. They collide and stick together, traveling at 28° north of east. What was the initial velocity of car B?

9.6 Center of Mass

63. Three point masses are placed at the corners of a triangle as shown in the figure below.



Find the center of mass of the three-mass system.

64. Two particles of masses m_1 and m_2 separated by a horizontal distance D are released from the same height h at the same time. Find the vertical position of the center of mass of these two particles at a time before the two particles strike the ground. Assume no air resistance.
65. Two particles of masses m_1 and m_2 separated by a horizontal distance D are let go from the same height h at different times. Particle 1 starts at $t = 0$, and particle 2 is let go at $t = T$. Find the vertical position of the center of mass at a time before the first particle strikes the ground. Assume no air resistance.
66. Two particles of masses m_1 and m_2 move

uniformly in different circles of radii R_1 and R_2 about origin in the x, y -plane. The x - and y -coordinates of the center of mass and that of particle 1 are given as follows (where length is in meters and t in seconds):

$$x_1(t) = 4\cos(2t), y_1(t) = 4\sin(2t)$$

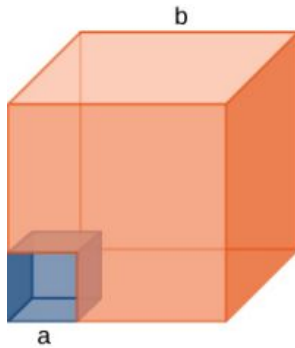
and:

$$x_{CM}(t) = 3\cos(2t), y_{CM}(t) = 3\sin(2t).$$

- Find the radius of the circle in which particle 1 moves.
 - Find the x - and y -coordinates of particle 2 and the radius of the circle this particle moves.
67. Two particles of masses m_1 and m_2 move uniformly in different circles of radii R_1 and R_2 about the origin in the x, y -plane. The coordinates of the two particles in meters are given as follows ($z = 0$ for both). Here t is in seconds:
- $$x_1(t) = 4\cos(2t)$$
- $$y_1(t) = 4\sin(2t)$$
- $$x_2(t) = 2\cos\left(3t - \frac{\pi}{2}\right)$$
- $$y_2(t) = 2\sin\left(3t - \frac{\pi}{2}\right)$$
- Find the radii of the circles of motion of both particles.
 - Find the x - and y -coordinates of the center of mass.
 - Decide if the center of mass moves in a circle by plotting its trajectory.
68. Find the center of mass of a one-meter long rod, made of 50 cm of iron (density $8 \frac{\text{g}}{\text{cm}^3}$) and 50 cm of aluminum (density $2.7 \frac{\text{g}}{\text{cm}^3}$).
69. Find the center of mass of a rod of length L whose mass density changes from one end to the other quadratically. That is, if the rod is laid out along the x -axis with one end at the origin and the other end at $x = L$, the density is given by $\rho(x) = \rho_0 + (\rho_1 - \rho_0)\left(\frac{x}{L}\right)^2$, where ρ_0 and ρ_1 are constant values.
70. Find the center of mass of a rectangular block of length a and width b that has a nonuniform density such that when the rectangle is placed in the x, y -plane with one corner at the origin and the block placed in the first quadrant with the two edges along the x - and y -axes, the density is given by $\rho(x, y) = \rho_0 x$, where ρ_0 is a constant.
71. Find the center of mass of a rectangular material

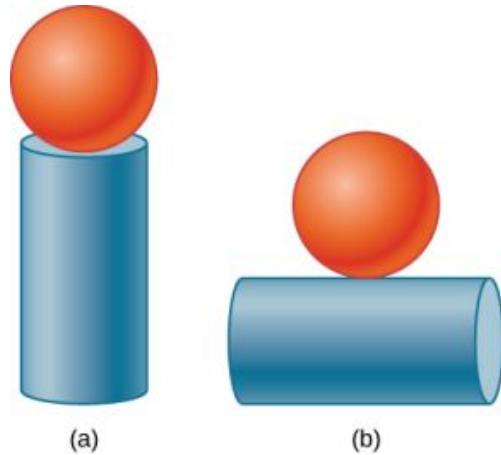
of length a and width b made up of a material of nonuniform density. The density is such that when the rectangle is placed in the xy -plane, the density is given by $\rho(x, y) = \rho_0 xy$.

72. A cube of side a is cut out of another cube of side b as shown in the figure below.



Find the location of the center of mass of the structure. (*Hint:* Think of the missing part as a negative mass overlapping a positive mass.)

73. Find the center of mass of a cone of uniform density that has a radius R at the base, height h , and mass M . Let the origin be at the center of the base of the cone and have $+z$ going through the cone vertex.
74. Find the center of mass of a thin wire of mass m and length L bent in a semicircular shape. Let the origin be at the center of the semicircle and have the wire arc from the $+x$ axis, cross the $+y$ axis, and terminate at the $-x$ axis.
75. Find the center of mass of a uniform thin semicircular plate of radius R . Let the origin be at the center of the semicircle, the plate arc from the $+x$ axis to the $-x$ axis, and the z axis be perpendicular to the plate.
76. Find the center of mass of a sphere of mass M and radius R and a cylinder of mass m , radius r , and height h arranged as shown below.



Express your answers in a coordinate system that has the origin at the center of the cylinder.

9.7 Rocket Propulsion

77. (a) A 5.00-kg squid initially at rest ejects 0.250 kg of fluid with a velocity of 10.0 m/s. What is the recoil velocity of the squid if the ejection is done in 0.100 s and there is a 5.00- N frictional force opposing the squid's movement?
(b) How much energy is lost to work done against friction?
78. A rocket takes off from Earth and reaches a speed of 100 m/s in 10.0 s. If the exhaust speed is 1500 m/s and the mass of fuel burned is 100 kg, what was the initial mass of the rocket?
79. Repeat the preceding problem but for a rocket that takes off from a space station, where there is no gravity other than the negligible gravity due to the space station.
80. How much fuel would be needed for a 1000-kg rocket (this is its mass with no fuel) to take off from Earth and reach 1000 m/s in 30 s? The exhaust speed is 1000 m/s.
81. What exhaust speed is required to accelerate a rocket in deep space from 800 m/s to 1000 m/s in 5.0 s if the total rocket mass is 1200 kg and the rocket only has 50 kg of fuel left?
82. **Unreasonable Results** Squids have been reported to jump from the ocean and travel 30.0 m (measured horizontally) before re-entering the water.
(a) Calculate the initial speed of the squid if it leaves the water at an angle of 20.0° , assuming negligible lift from the air and negligible air resistance.
(b) The squid propels itself by squirting water.

What fraction of its mass would it have to eject in order to achieve the speed found in the previous part? The water is ejected at 12.0 m/s; gravitational force and friction are neglected.

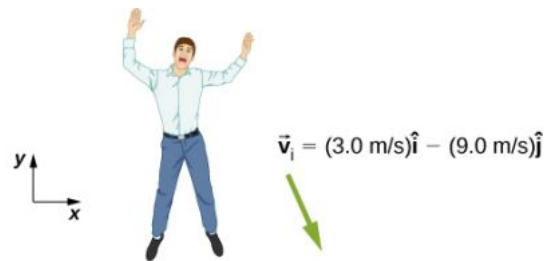
Additional Problems

83. Two 70-kg canoers paddle in a single, 50-kg canoe. Their paddling moves the canoe at 1.2 m/s with respect to the water, and the river they're in flows at 4 m/s with respect to the land. What is their momentum with respect to the land?
84. Which has a larger magnitude of momentum: a 3000-kg elephant moving at 40 km/h or a 60-kg cheetah moving at 112 km/h?
85. A driver applies the brakes and reduces the speed of her car by 20%, without changing the direction in which the car is moving. By how much does the car's momentum change?
86. Your friend claims that momentum is mass multiplied by velocity, so things with more mass have more momentum. Do you agree? Explain.
87. Dropping a glass on a cement floor is more likely to break the glass than if it is dropped from the same height on a grass lawn. Explain in terms of the impulse.
88. Your 1500-kg sports car accelerates from 0 to 30 m/s in 10 s. What average force is exerted on it during this acceleration?
89. A ball of mass m is dropped. What is the formula for the impulse exerted on the ball from the instant it is dropped to an arbitrary time τ later? Ignore air resistance.
90. Repeat the preceding problem, but including a drag force due to air of $f_{\text{drag}} = -b\vec{v}$.
91. A 5.0-g egg falls from a 90-cm-high counter onto the floor and breaks. What impulse is exerted by the floor on the egg?
92. A car crashes into a large tree that does not move. The car goes from 30 m/s to 0 in 1.3 m. (a) What impulse is applied to the 70 kg driver by the seatbelt, assuming he follows the same motion as the car? (b) What is the average force applied to the driver by the seatbelt?
93. Two hockey players approach each other head on, each traveling at the same speed v_i . They collide and get tangled together, falling down and moving off at a speed $v_i/5$. What is the ratio of their masses?
94. You are coasting on your 10-kg bicycle at 15 m/s and a 5.0-g bug splatters on your helmet. The bug was initially moving at 2.0 m/s in the same direction as you. If your mass is 60 kg, (a) what is the initial momentum of you plus your bicycle? (b) What is the initial momentum of the bug? (c) What is your change in velocity due to the collision with the bug? (d) What would the change in velocity have been if the bug were traveling in the opposite direction?
95. A load of gravel is dumped straight down into a 30 000-kg freight car coasting at 2.2 m/s on a straight section of a railroad. If the freight car's speed after receiving the gravel is 1.5 m/s, what mass of gravel did it receive?
96. Two carts on a straight track collide head on. The first cart was moving at 3.6 m/s in the positive x direction and the second was moving at 2.4 m/s in the opposite direction. After the collision, the second car continues moving in its initial direction of motion at 0.24 m/s. If the mass of the second car is 5.0 times that of the first, what is the final velocity of the first car?
97. A 100-kg astronaut finds himself separated from his spaceship by 10 m and moving away from the spaceship at 0.1 m/s. To get back to the spaceship, he throws a 10-kg tool bag away from the spaceship at 5.0 m/s. How long will he take to return to the spaceship?
98. Derive the equations giving the final speeds for two objects that collide elastically, with the mass of the objects being m_1 and m_2 and the initial speeds being $v_{1,i}$ and $v_{2,i} = 0$ (i.e., second object is initially stationary).
99. Repeat the preceding problem for the case when the initial speed of the second object is nonzero.
100. A child sleds down a hill and collides at 5.6 m/s into a stationary sled that is identical to his. The child is launched forward at the same speed, leaving behind the two sleds that lock together and slide forward more slowly. What is the speed of the two sleds after this collision?

- 101.** For the preceding problem, find the final speed of each sled for the case of an elastic collision.
- 102.** A 90-kg football player jumps vertically into the air to catch a 0.50-kg football that is thrown essentially horizontally at him at 17 m/s. What is his horizontal speed after catching the ball?
- 103.** Three skydivers are plummeting earthward. They are initially holding onto each other, but then push apart. Two skydivers of mass 70 and 80 kg gain horizontal velocities of 1.2 m/s east and 1.4 m/s southeast, respectively. What is the horizontal velocity of the third skydiver, whose mass is 55 kg?
- 104.** Two billiard balls are at rest and touching each other on a pool table. The cue ball travels at 3.8 m/s along the line of symmetry between these balls and strikes them simultaneously. If the collision is elastic, what is the velocity of the three balls after the collision?
- 105.** A billiard ball traveling at $(2.2 \text{ m/s})\hat{i} - (0.4 \text{ m/s})\hat{j}$ collides with a wall that is aligned in the $\hat{j}$ direction. Assuming the collision is elastic, what is the final velocity of the ball?
- 106.** Two identical billiard balls collide. The first one is initially traveling at $(2.2 \text{ m/s})\hat{i} - (0.4 \text{ m/s})\hat{j}$ and the second one at $-(1.4 \text{ m/s})\hat{i} + (2.4 \text{ m/s})\hat{j}$. Suppose they collide when the center of ball 1 is at the origin and the center of ball 2 is at the point $(2R, 0)$ where R is the radius of the balls. What is the final velocity of each ball?
- 107.** Repeat the preceding problem if the balls collide when the center of ball 1 is at the origin and the center of ball 2 is at the point $(0, 2R)$.
- 108.** Repeat the preceding problem if the balls collide when the center of ball 1 is at the origin and the center of ball 2 is at the point $(\sqrt{3}R/2, R/2)$.
- 109.** Where is the center of mass of a semicircular wire of radius R that is centered on the origin, begins and ends on the x axis, and lies in the x, y plane?
- 110.** Where is the center of mass of a slice of pizza that was cut into eight equal slices? Assume the origin is at the apex of the slice and measure angles with respect to an edge of the slice. The radius of the pizza is R .
- 111.** If 1% of the Earth's mass were transferred to the Moon, how far would the center of mass of the Earth-Moon-population system move? The mass of the Earth is $5.97 \times 10^{24} \text{ kg}$ and that of the Moon is $7.34 \times 10^{22} \text{ kg}$. The radius of the Moon's orbit is about $3.84 \times 10^5 \text{ m}$.
- 112.** You friend wonders how a rocket continues to climb into the sky once it is sufficiently high above the surface of Earth so that its expelled gasses no longer push on the surface. How do you respond?
- 113.** To increase the acceleration of a rocket, should you throw rocks out of the front window of the rocket or out of the back window?

Challenge Problems

- 114.** A 65-kg person jumps from the first floor window of a burning building and lands almost vertically on the ground with a horizontal velocity of 3 m/s and vertical velocity of -9 m/s . Upon impact with the ground he is brought to rest in a short time. The force experienced by his feet depends on whether he keeps his knees stiff or bends them. Find the force on his feet in each case.

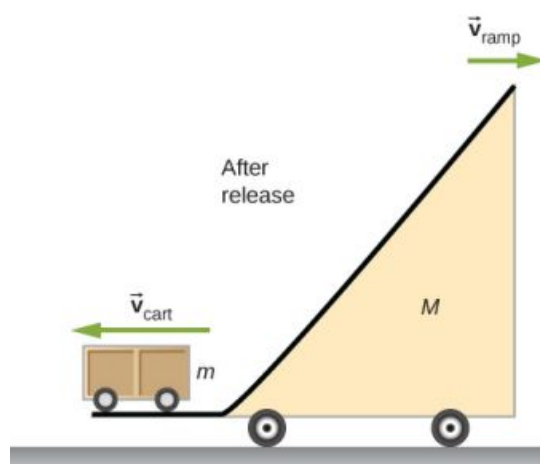
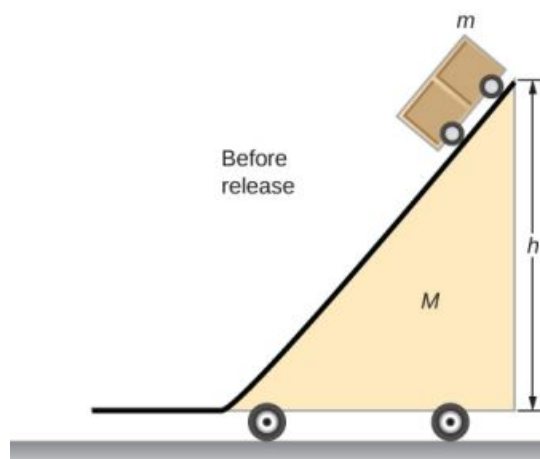


- First find the impulse on the person from the impact on the ground. Calculate both its magnitude and direction.
- Find the average force on the feet if the person keeps his leg stiff and straight and his center of mass drops by only 1 cm vertically and 1 cm horizontally during the impact.

- c. Find the average force on the feet if the person bends his legs throughout the impact so that his center of mass drops by 50 cm vertically and 5 cm horizontally during the impact.
- d. Compare the results of part (b) and (c), and draw conclusions about which way is better.

You will need to find the time the impact lasts by making reasonable assumptions about the acceleration opposite to the motion. Although the force is not constant during the impact, working with constant average force for this problem is acceptable.

115. Two projectiles of mass m_1 and m_2 are fired at the same speed but in opposite directions from two launch sites separated by a distance D . They both reach the same spot in their highest point and strike there. As a result of the impact they stick together and move as a single body afterwards. Find the place they will land.
116. Two identical objects (such as billiard balls) have a one-dimensional collision in which one is initially motionless. After the collision, the moving object is stationary and the other moves with the same speed as the other originally had. Show that both momentum and kinetic energy are conserved.
117. A ramp of mass M is at rest on a horizontal surface. A small cart of mass m is placed at the top of the ramp and released.



What are the velocities of the ramp and the cart relative to the ground at the instant the cart leaves the ramp?

118. Find the center of mass of the structure given in the figure below. Assume a uniform thickness of 20 cm, and a uniform density of 1 g/cm^3 .

